Z + b-tagged and Z + c-tagged jet: kinematics in Summer22

All POG corrections and SFs, plus BTV b- and c-tagging SFs on the leading jet

ParticleNet Loose (asked for), Medium and Tight, from one run

5 November 2026

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Selection: Z→ee or Z→μμ + at least 1 jet. Every region has the same 32 plots in the same order.

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What goes into these plots

Corrections and SFs (POG recipes)

Jets: JEC, JER, jet veto map, type-1 PUPPI MET. Leptons: muon and electron scale and smearing, ID, isolation and reco SFs, trigger SFs. Events: golden JSON, pileup weights, MET filters (new: they were missing before). The DY Z pT weights are now off.

working point	b-tag: PNet B >	c-tag: CvL >	and CvB >
Loose	0.047	0.054	0.181
Medium	0.245	0.160	0.306
Tight	0.673	0.492	0.259

Tagging is applied to the leading jet only

In MC each tagged jet gets the BTV SF for its true flavour. b-tag: particleNet_comb (b and c jets), particleNet_light. c-tag: particleNet_wc (c jets), particleNet_tnp (b jets), particleNet_light. b-tag and c-tag are separate selections, so a jet can pass both.

Yields in each region

Flavour shares are of all MC (DY + tt + VV). 'Tagged' is the fraction of Z + jet events that pass the tag.

region	data/MC Z→ee	data/MC Z→μμ	DY+b	DY+c	DY+light	tagged MC	tagged data
inclusive	0.95 ± 0.01	0.97 ± 0.01	4%	8%	87%	–	–
b-tag Loose	0.95 ± 0.03	0.95 ± 0.02	22%	21%	56%	17.9%	17.6%
c-tag Loose	0.93 ± 0.01	0.95 ± 0.01	1%	9%	90%	89.5%	88.2%
b-tag Medium	0.96 ± 0.05	0.94 ± 0.04	59%	17%	19%	6.0%	5.8%
c-tag Medium	0.94 ± 0.02	0.95 ± 0.02	3%	20%	77%	28.6%	28.2%
b-tag Tight	1.00 ± 0.07	0.90 ± 0.05	82%	5%	7%	3.5%	3.3%
c-tag Tight	0.92 ± 0.04	0.93 ± 0.03	10%	50%	39%	7.3%	7.0%

Flavour shares and tagged fractions are for Z→μμ; Z→ee is within a few percent.

The tagging SFs work: data and MC tag the same fraction of jets (within 1–2%).

Data/MC is about 0.95 in every region, the same as without tagging. The Loose c-tag keeps 89% of the events and they are 90% light, so it is not a charm selection. Medium (20% c) and Tight (50% c) are.

Tagging SFs and the overlap of the two tags

Mean SF of the tagged leading jet in DY MC, per true flavour. Overlap: tagged jets that also pass the other tag at the same working point.

region	SF b jets	SF c jets	SF light jets	overlap, DY MC	overlap, data Z→μμ
b-tag Loose	0.95	0.94	1.21	77%	83%
b-tag Medium	0.94	0.93	1.29	33%	40%
b-tag Tight	0.92	0.92	1.27	4%	3%
c-tag Loose	0.78	1.00	1.06	13%	17%
c-tag Medium	0.77	0.97	1.09	7%	8%
c-tag Tight	0.79	0.83	1.16	2%	2%

Light jets are mistagged more in data than in MC (SF 1.06–1.29).

At Loose most b-tagged jets also pass the c-tag. At Tight the two selections barely overlap.

No tag requirement

Z + at least 1 jet (pT > 30 GeV, |η| < 2.5)

	data/MC	DY+b	DY+c	DY+light	tt	tagged MC	tagged data
Z→ee	0.949 ± 0.012	3.8%	8.0%	87.4%	0.4%	–	–
Z→μμ	0.966 ± 0.009	4.2%	8.1%	87.0%	0.4%	–	–

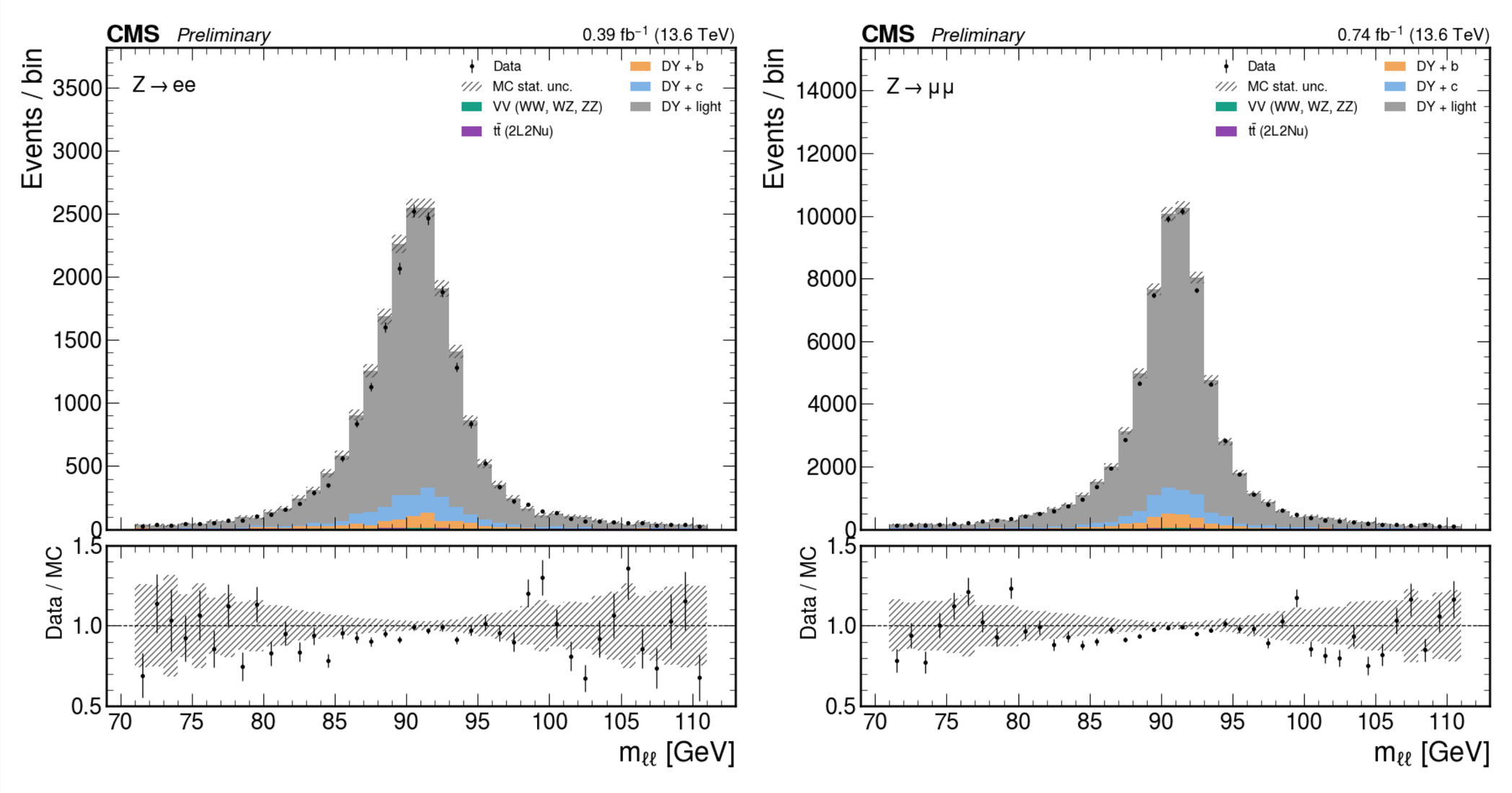
Flavour shares are of all MC. 'Tagged' = fraction of Z + jet events passing the tag. Errors are statistical only.

32 plots follow: Z and leptons, jets, Z–jet, MET and vertices, tagger scores, 2D maps. Left Z→ee, right Z→μμ.

No tag requirement

Dilepton mass

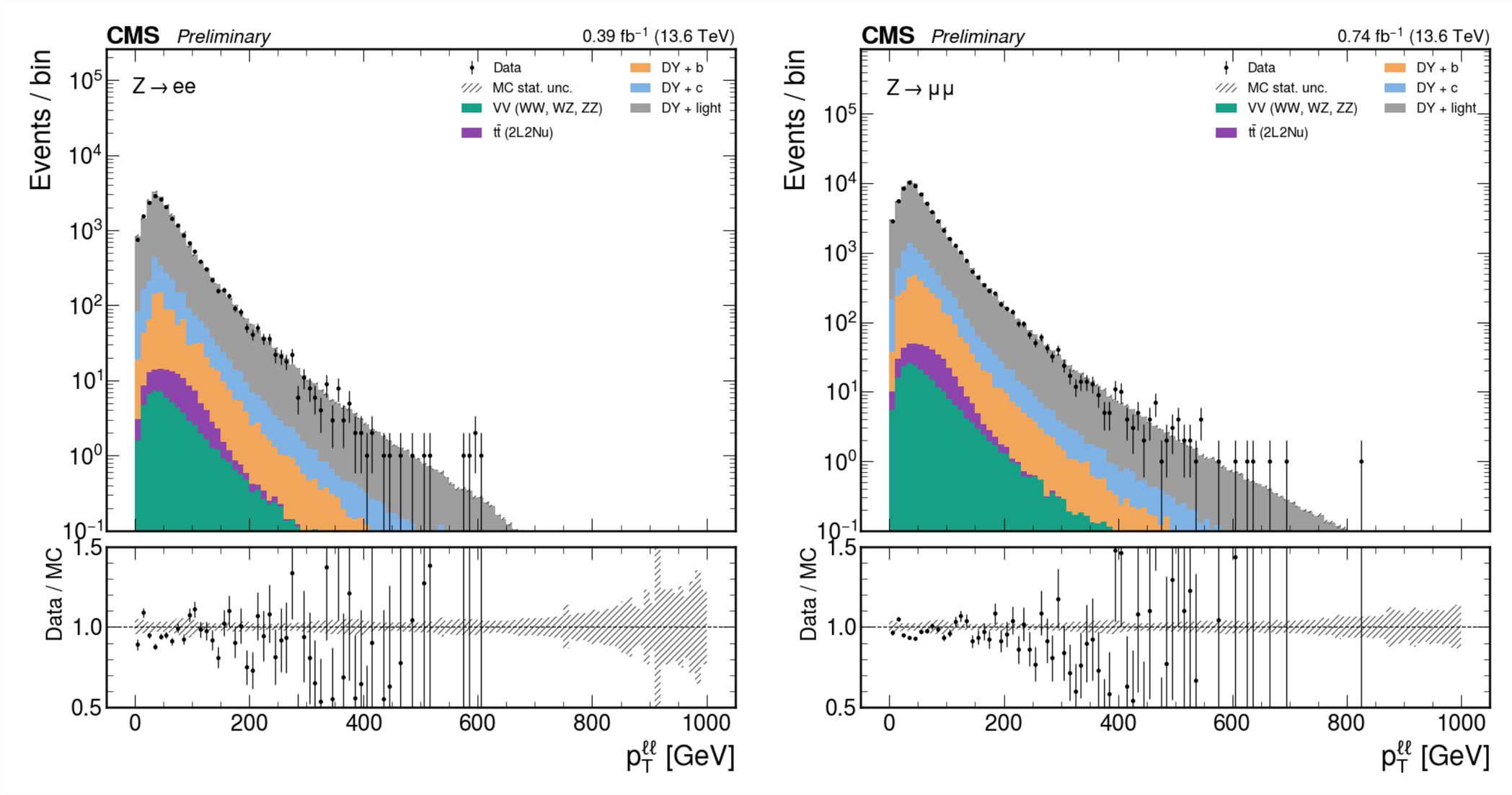
m(ℓℓ), 71–111 GeV window, linear scale.



No tag requirement

Z pT

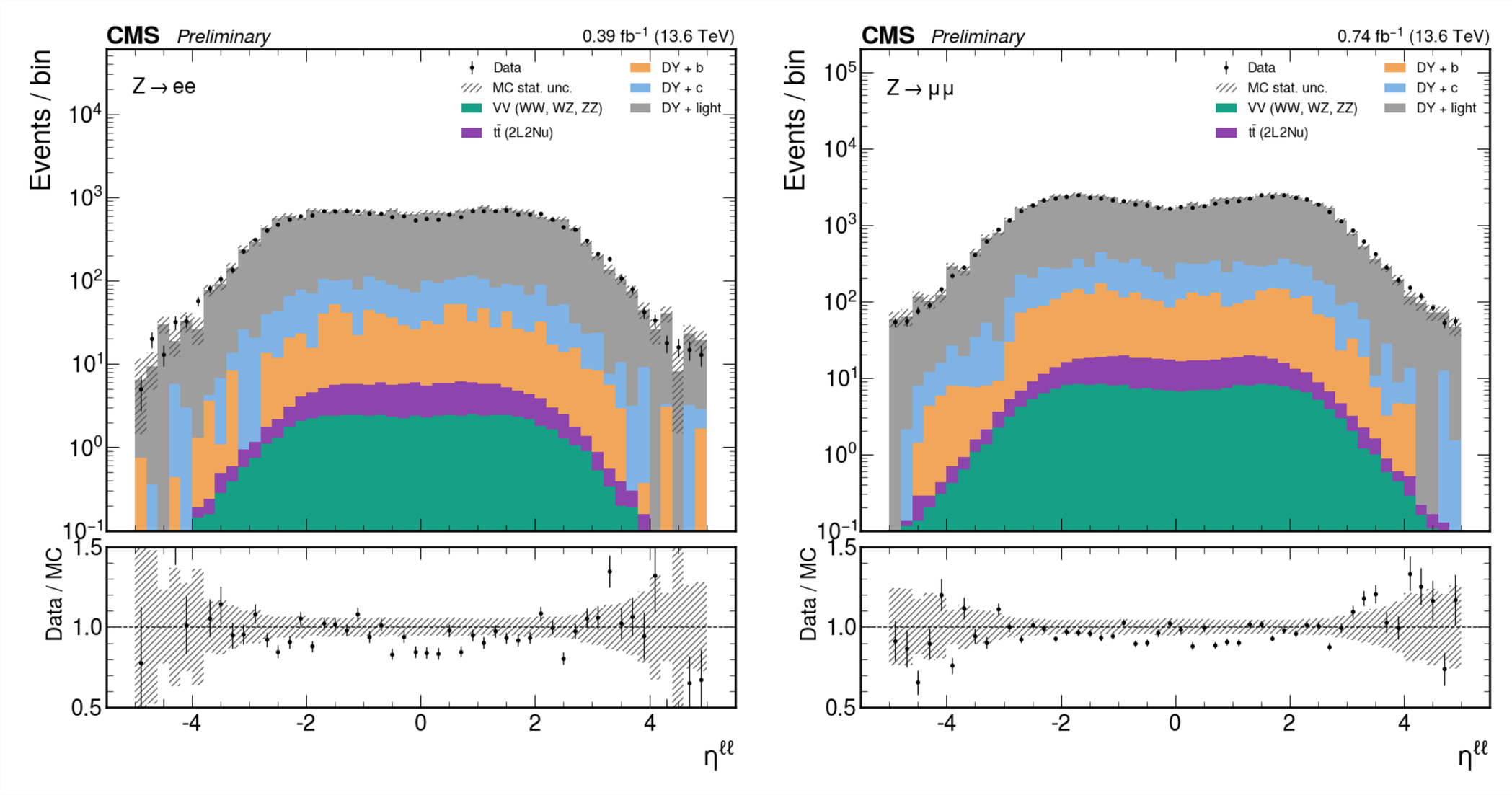
pT of the lepton pair.



No tag requirement

Z η

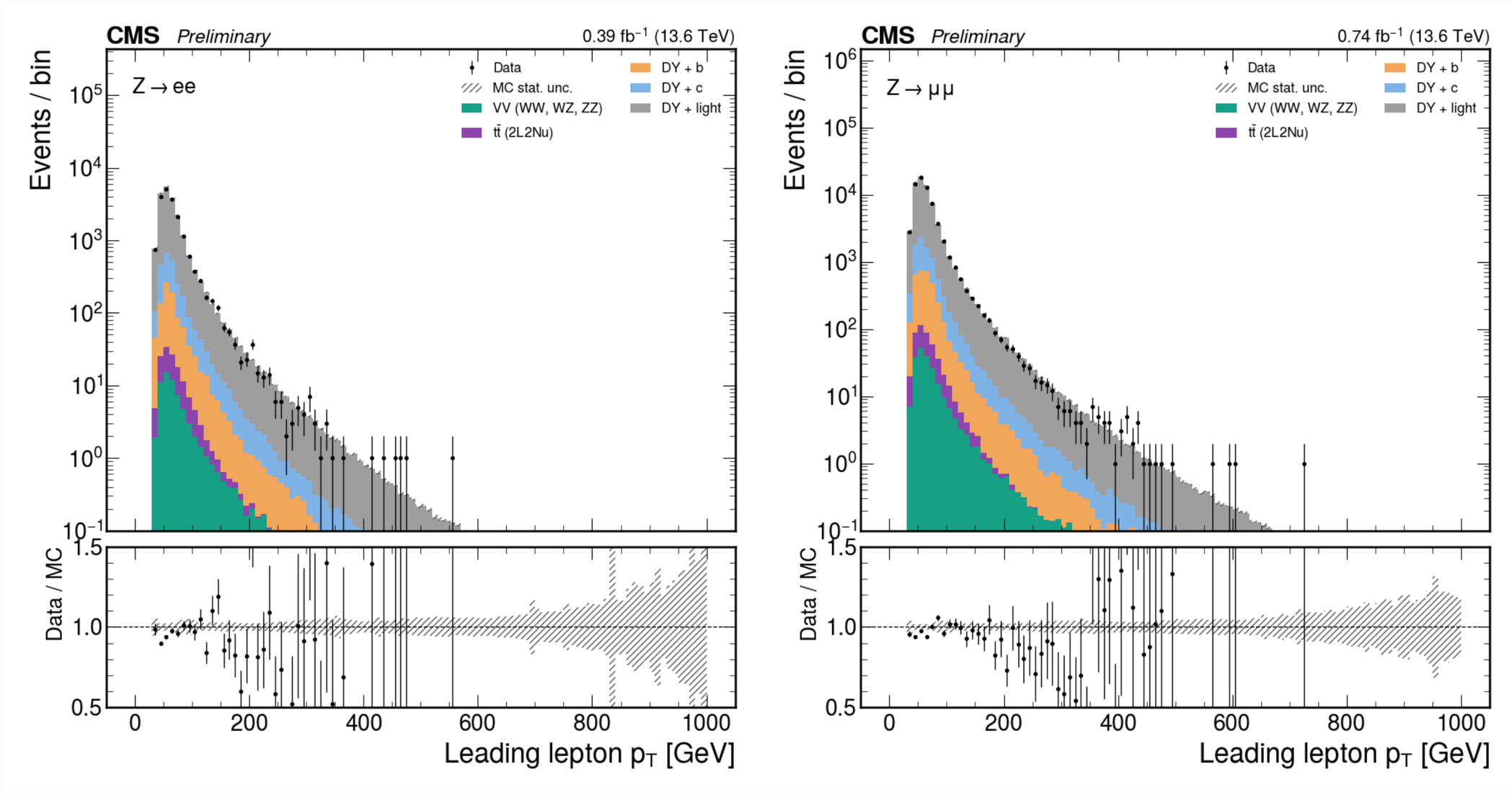
η of the lepton pair.



No tag requirement

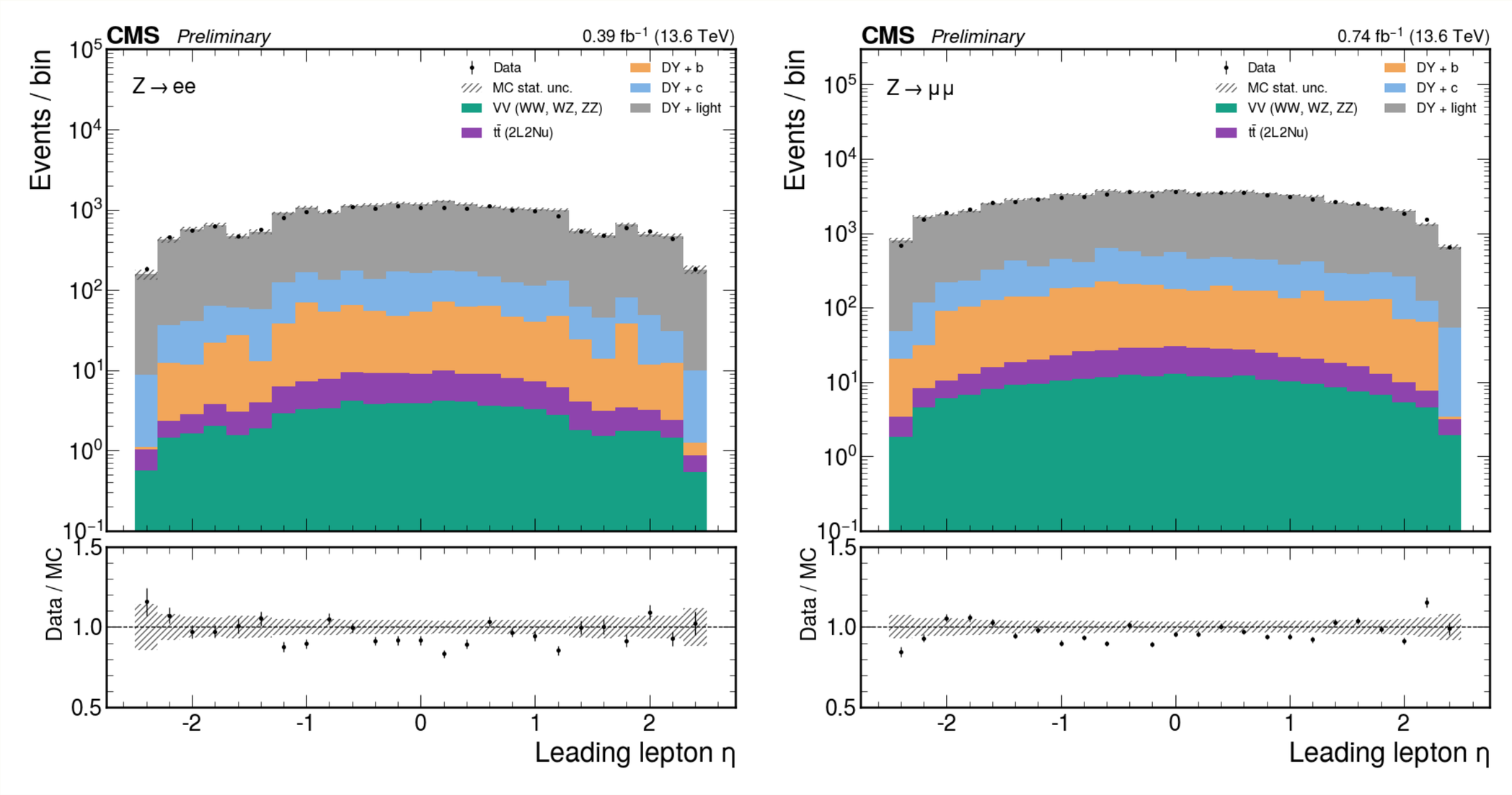
Leading-lepton pT

Leading lepton pT > 35 GeV (trigger plateau).



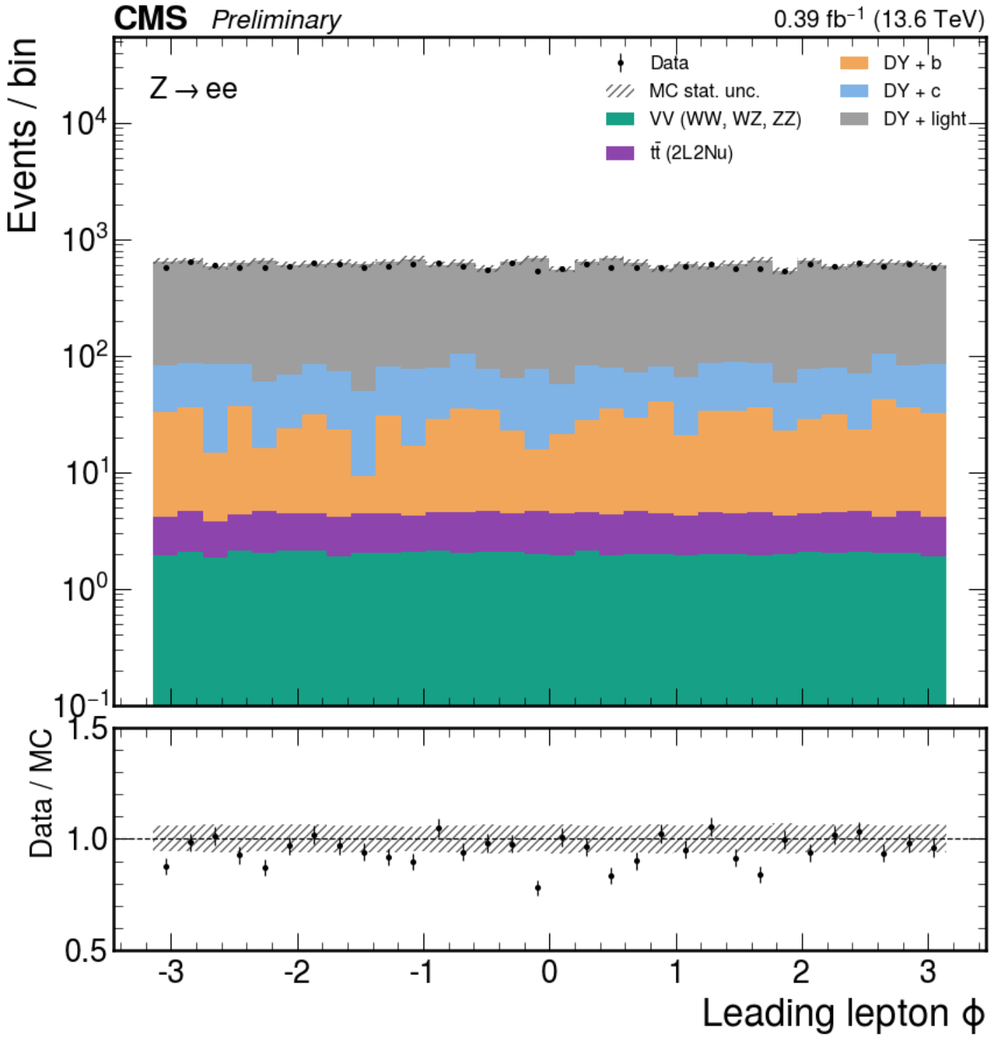
No tag requirement

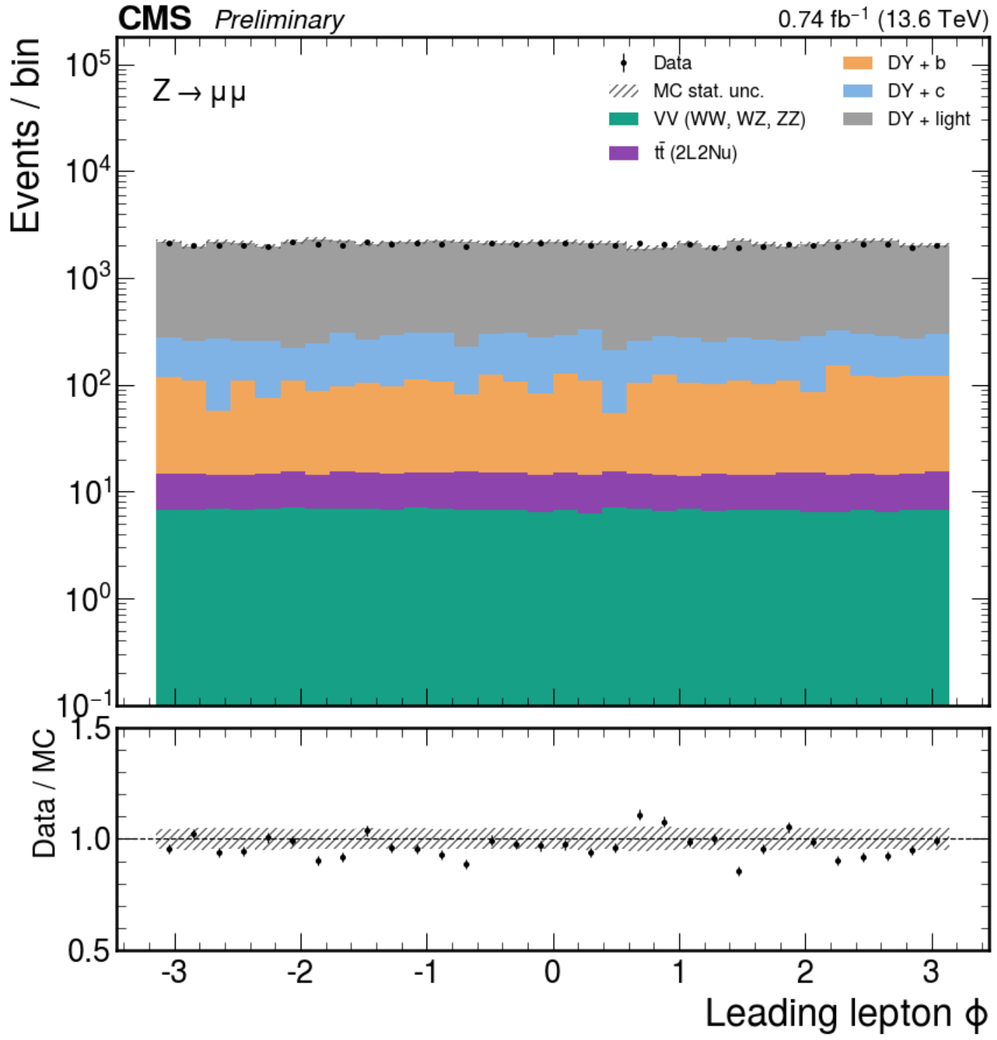
Leading-lepton η



No tag requirement

Leading-lepton φ

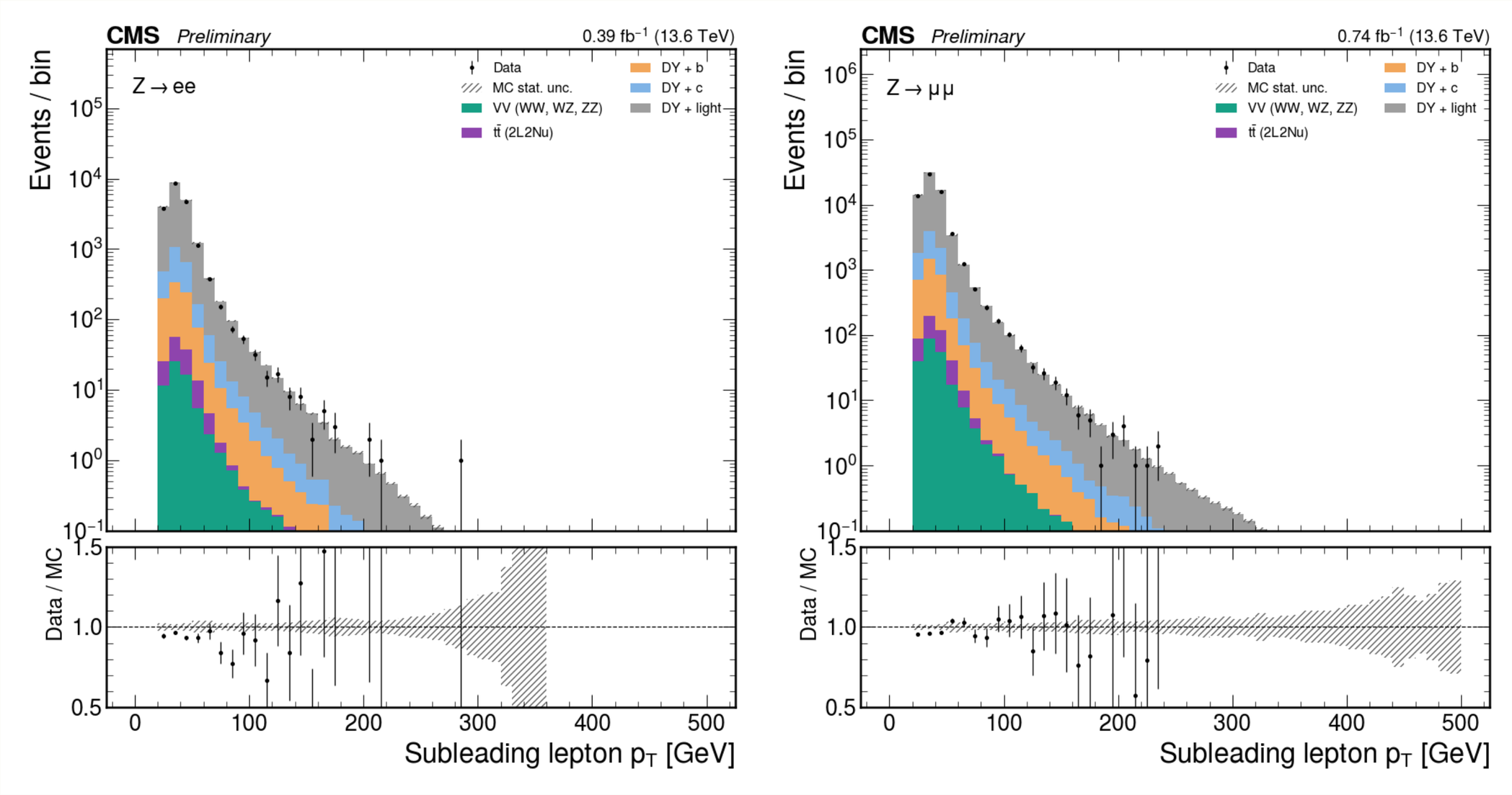




No tag requirement

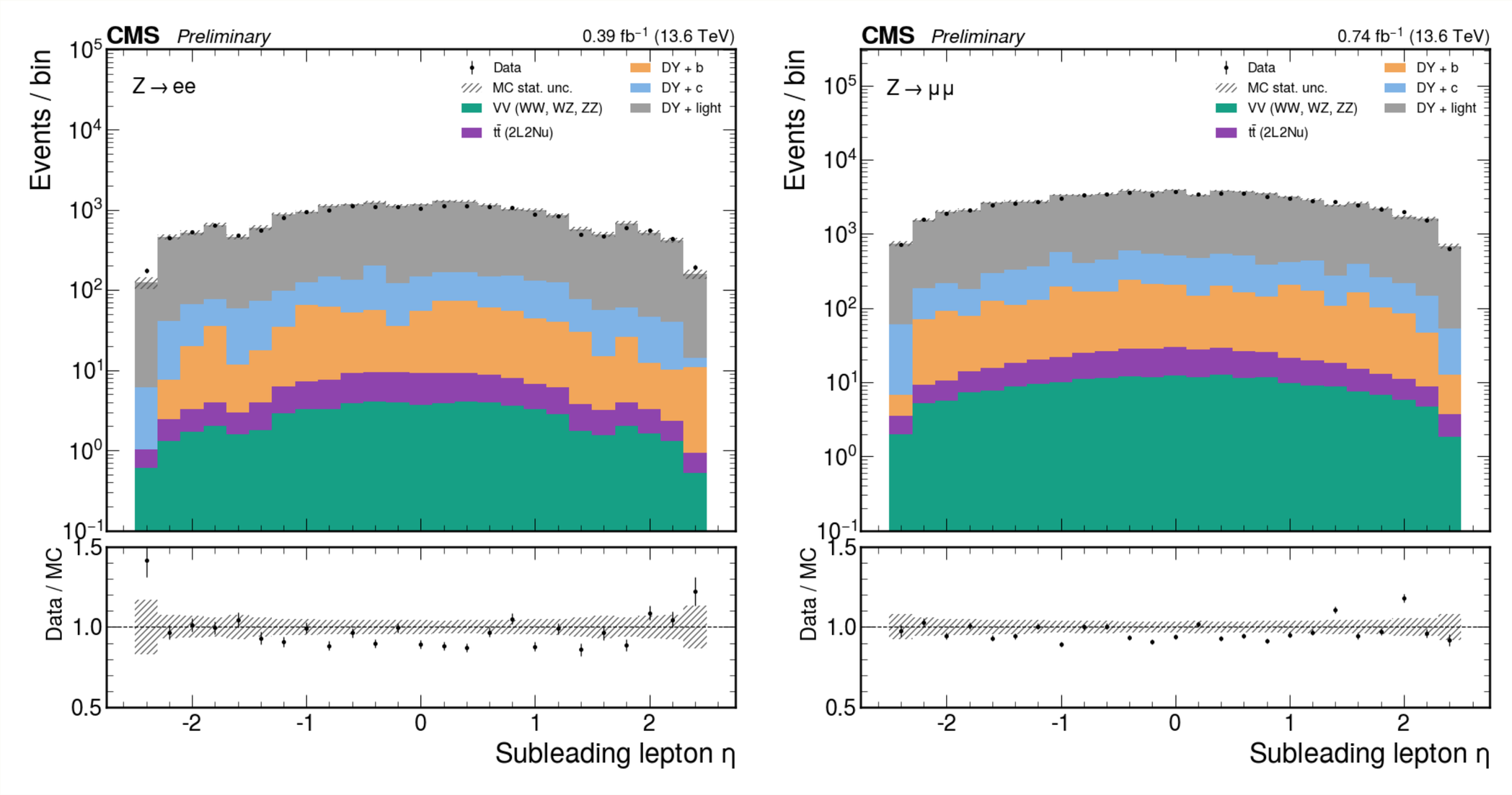
Second-lepton pT

Second lepton pT > 25 GeV.



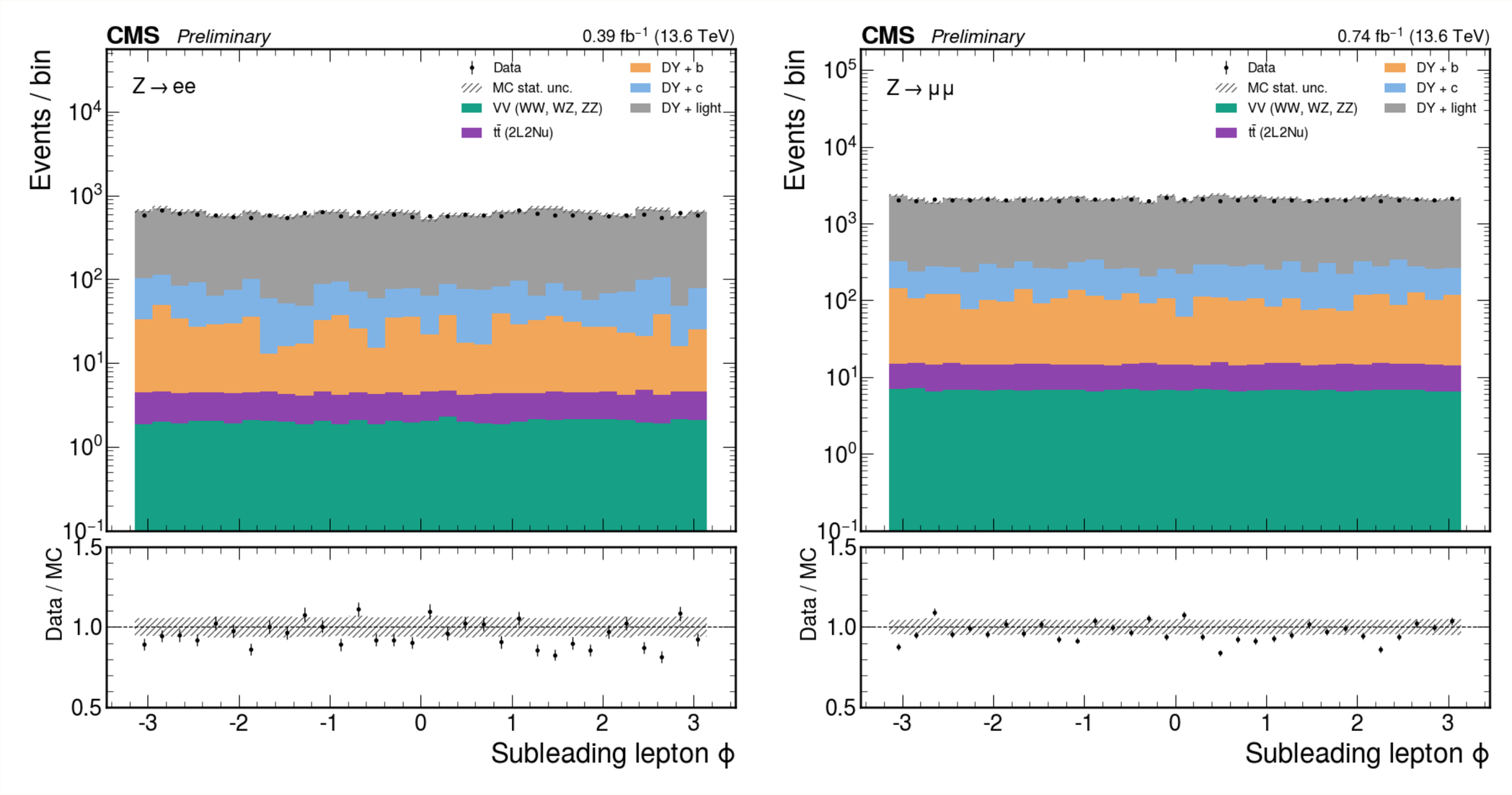
No tag requirement

Second-lepton η



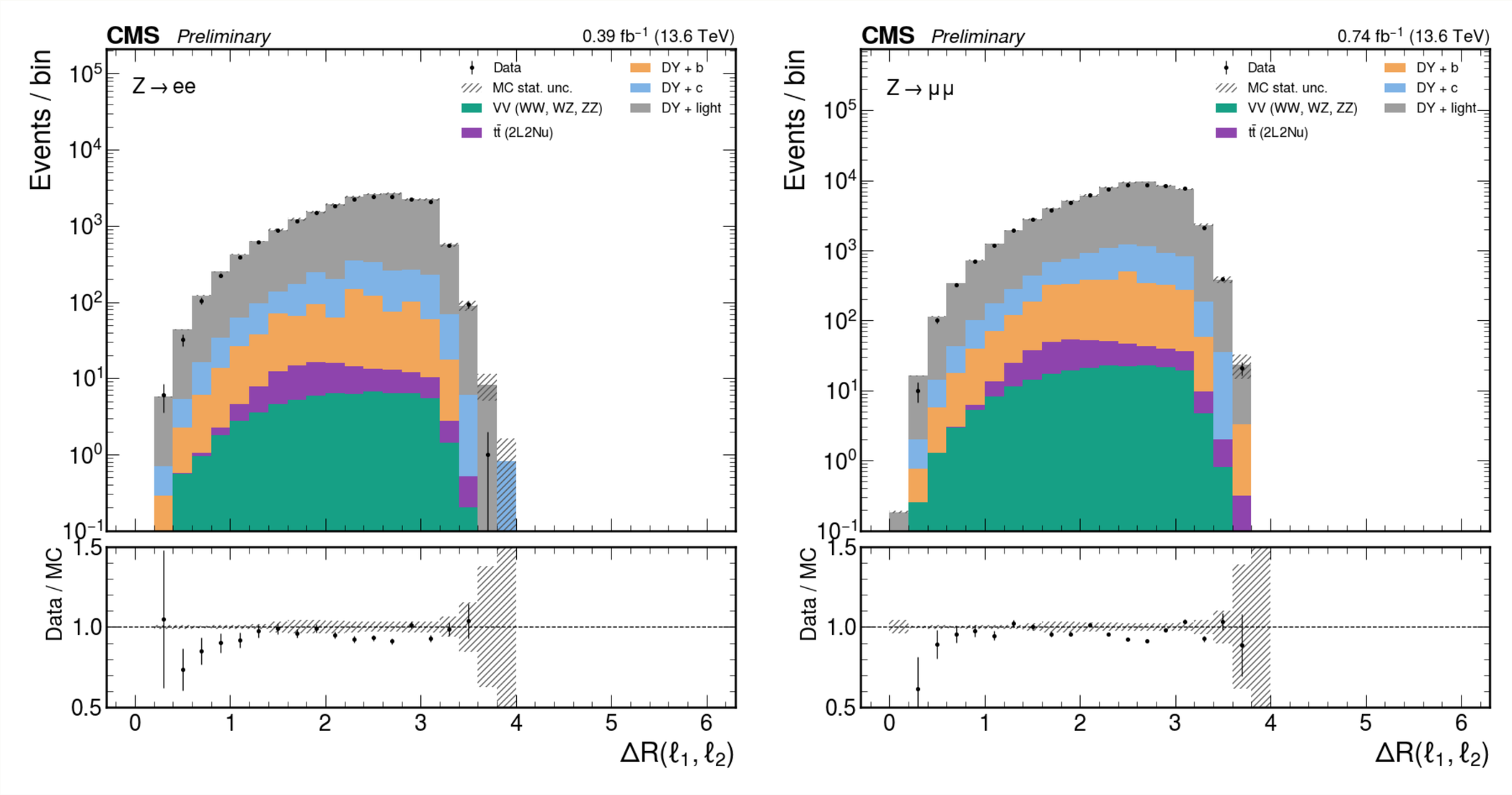
No tag requirement

Second-lepton φ



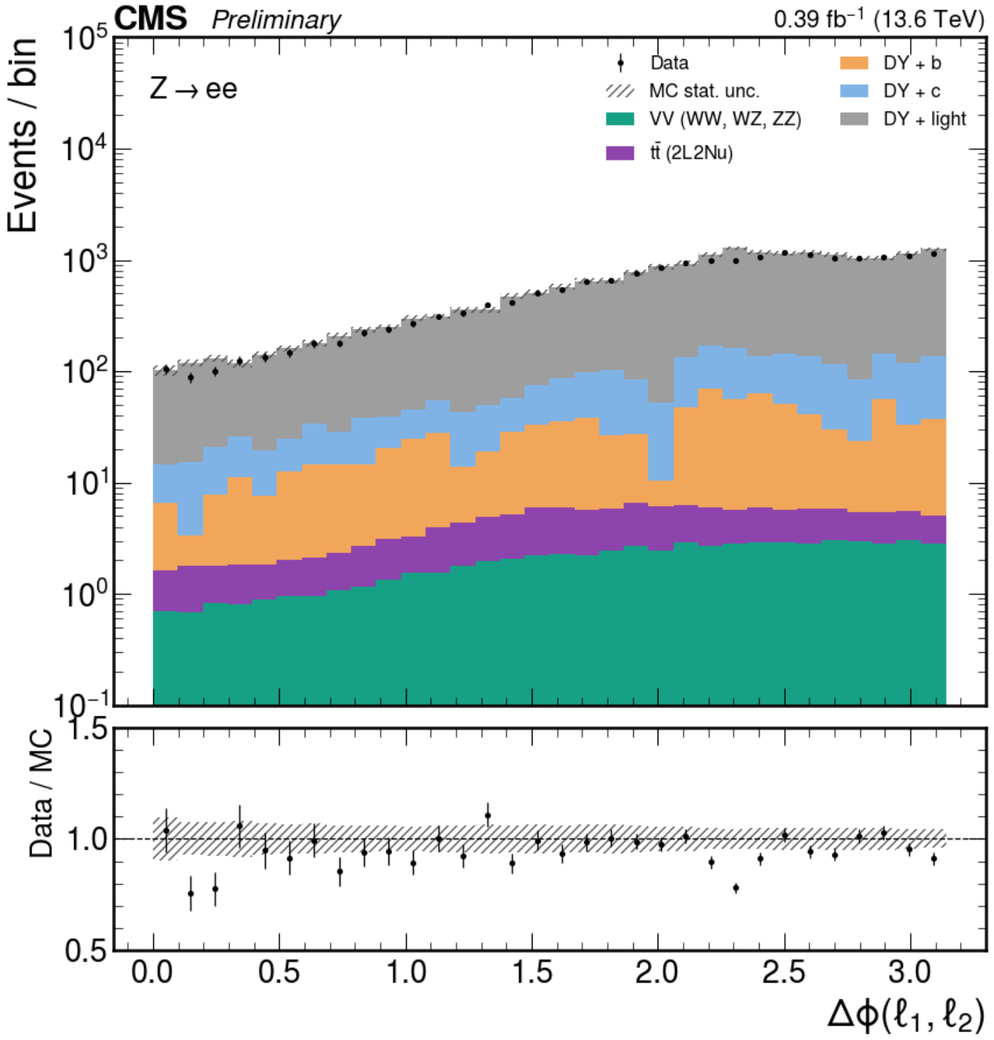
No tag requirement

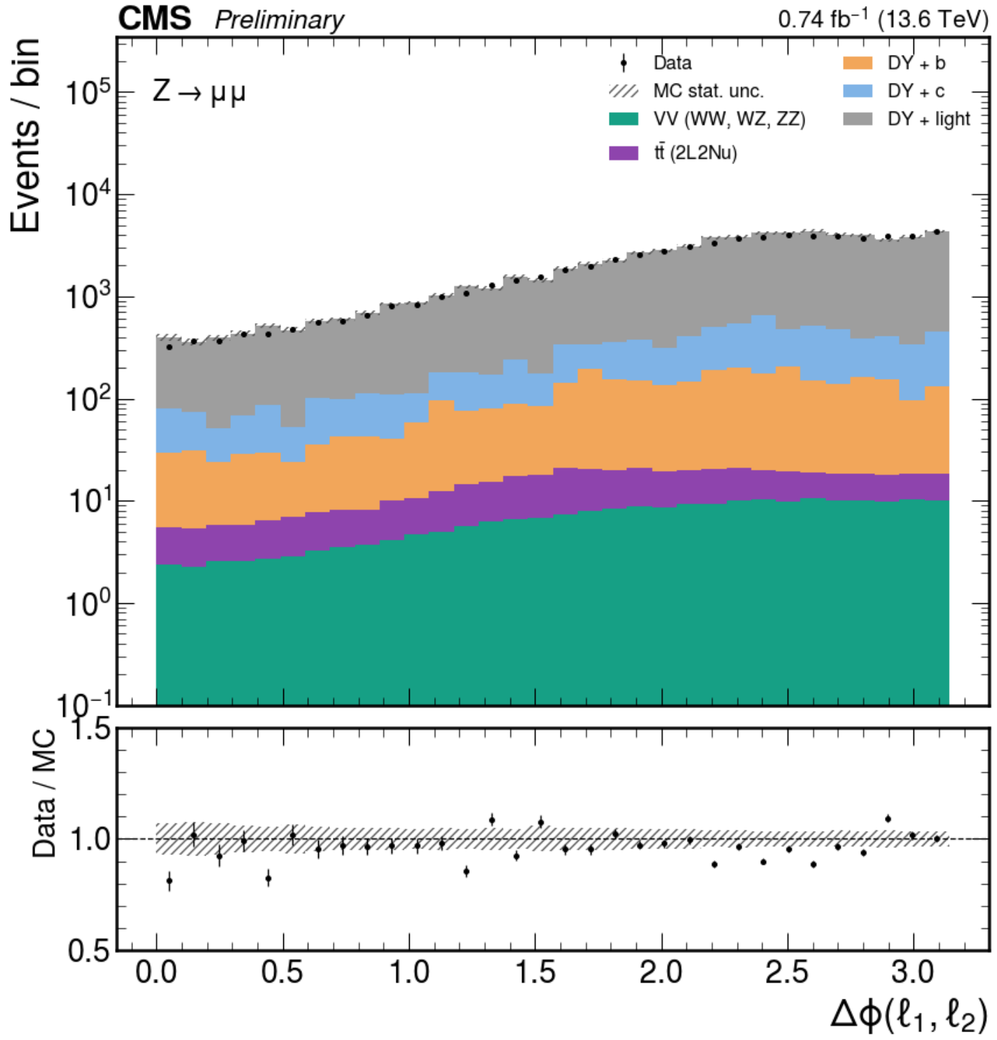
ΔR between the leptons



No tag requirement

Δφ between the leptons

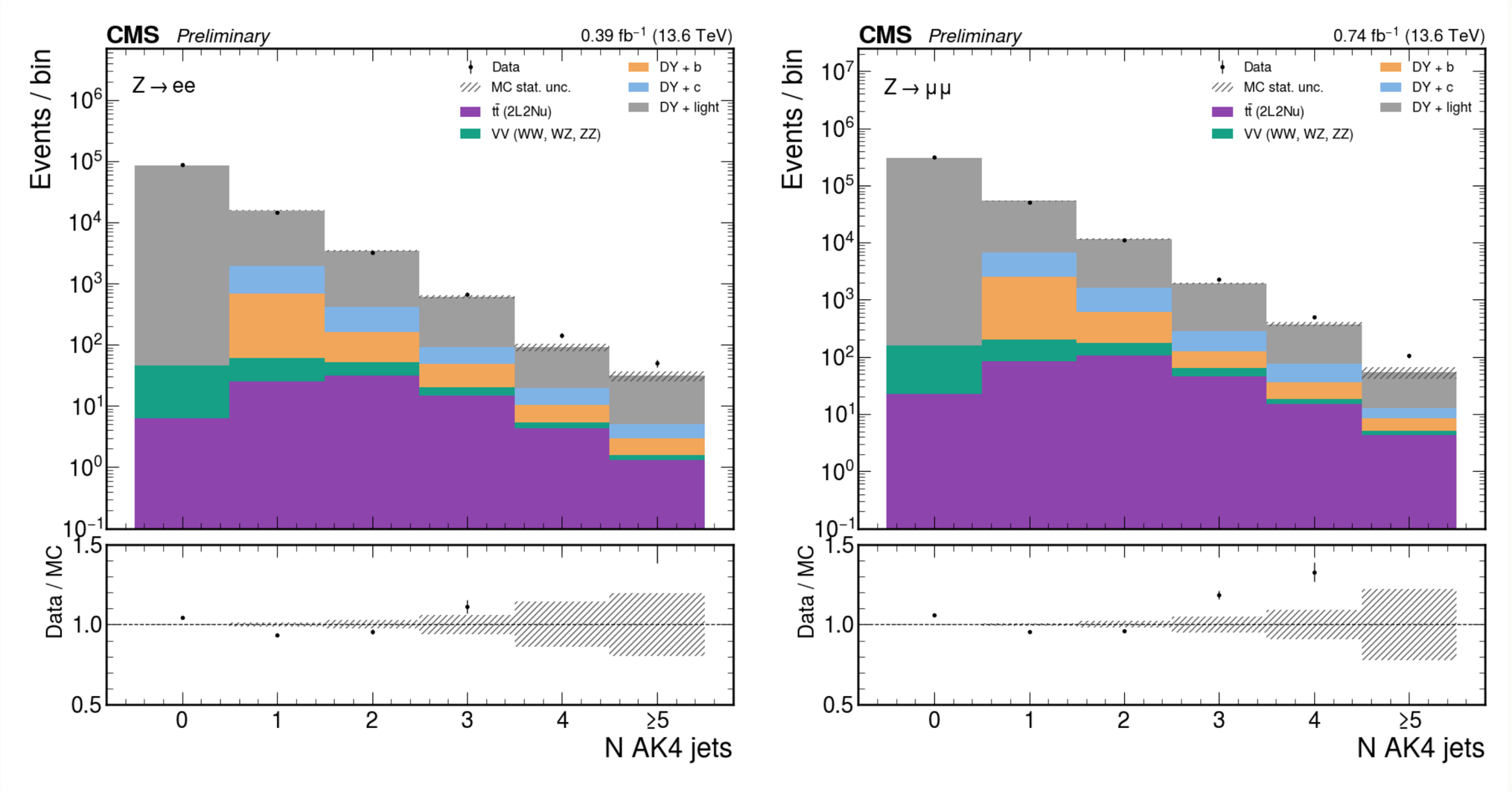




No tag requirement

Number of jets

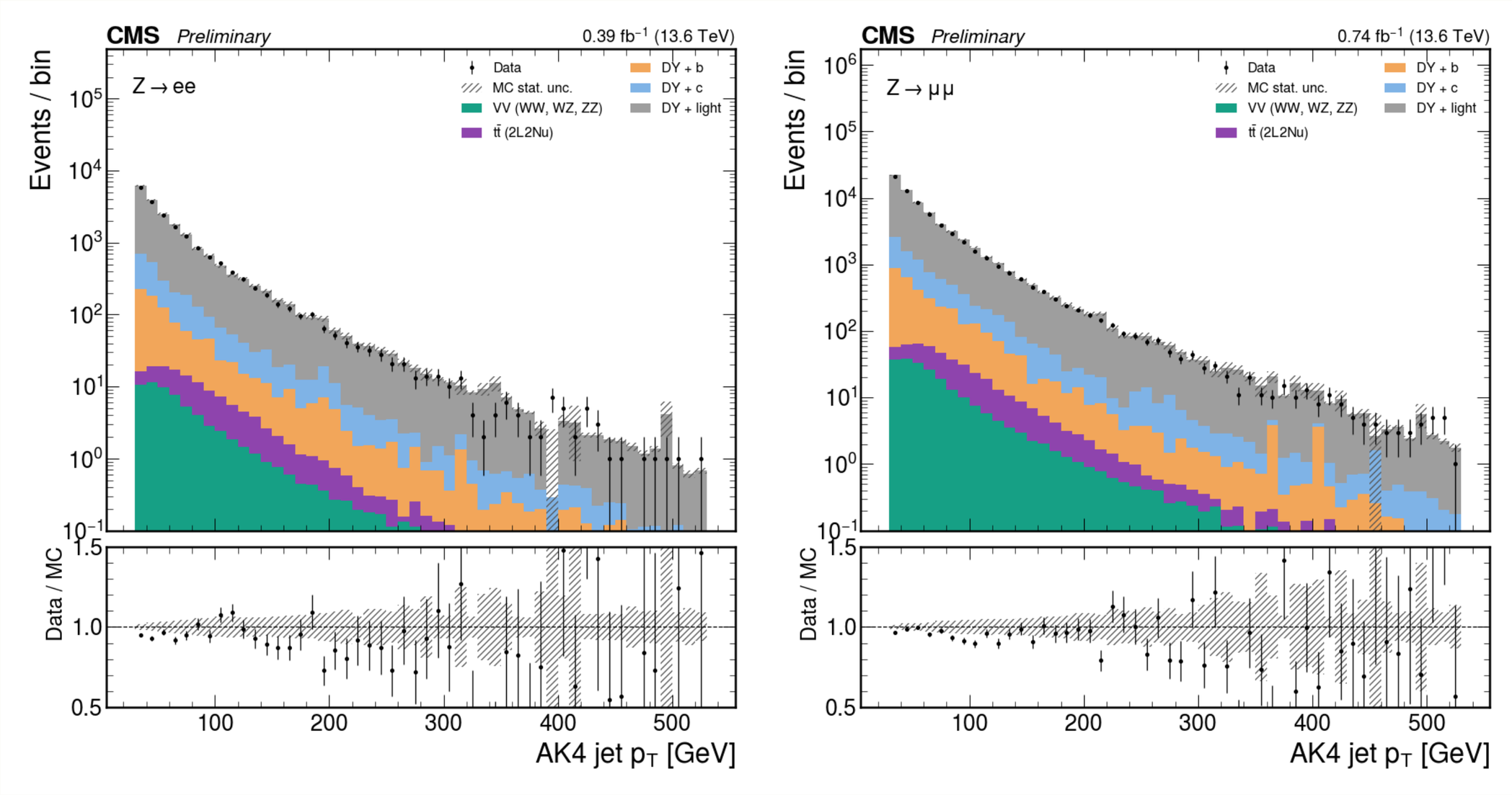
AK4 jets with pT > 30 GeV, |η| < 2.5, away from the leptons. Untagged plot: Z events before the jet cut, so the 0-jet bin is filled.



No tag requirement

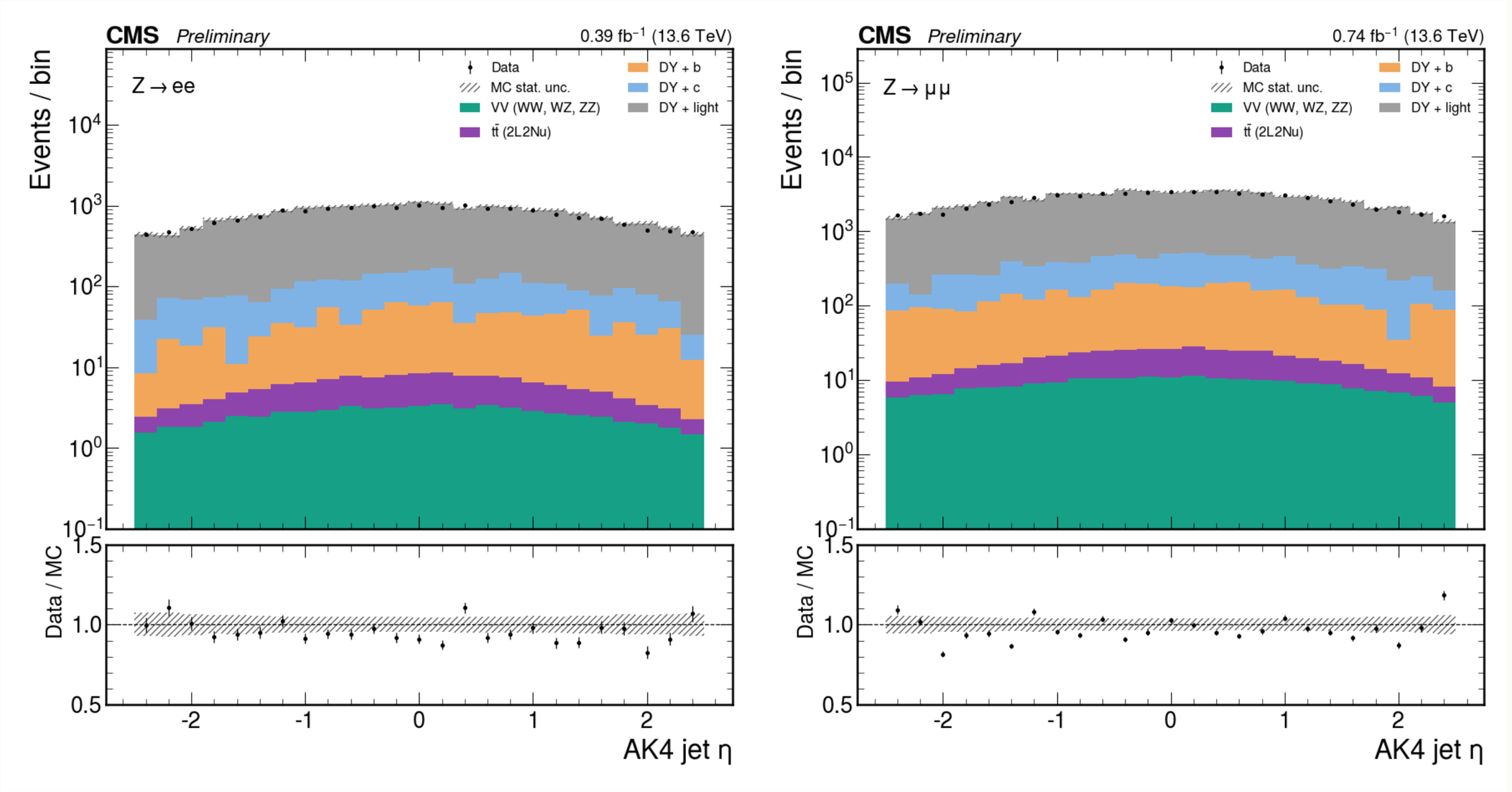
Leading-jet pT

The tag (if any) is on this jet.



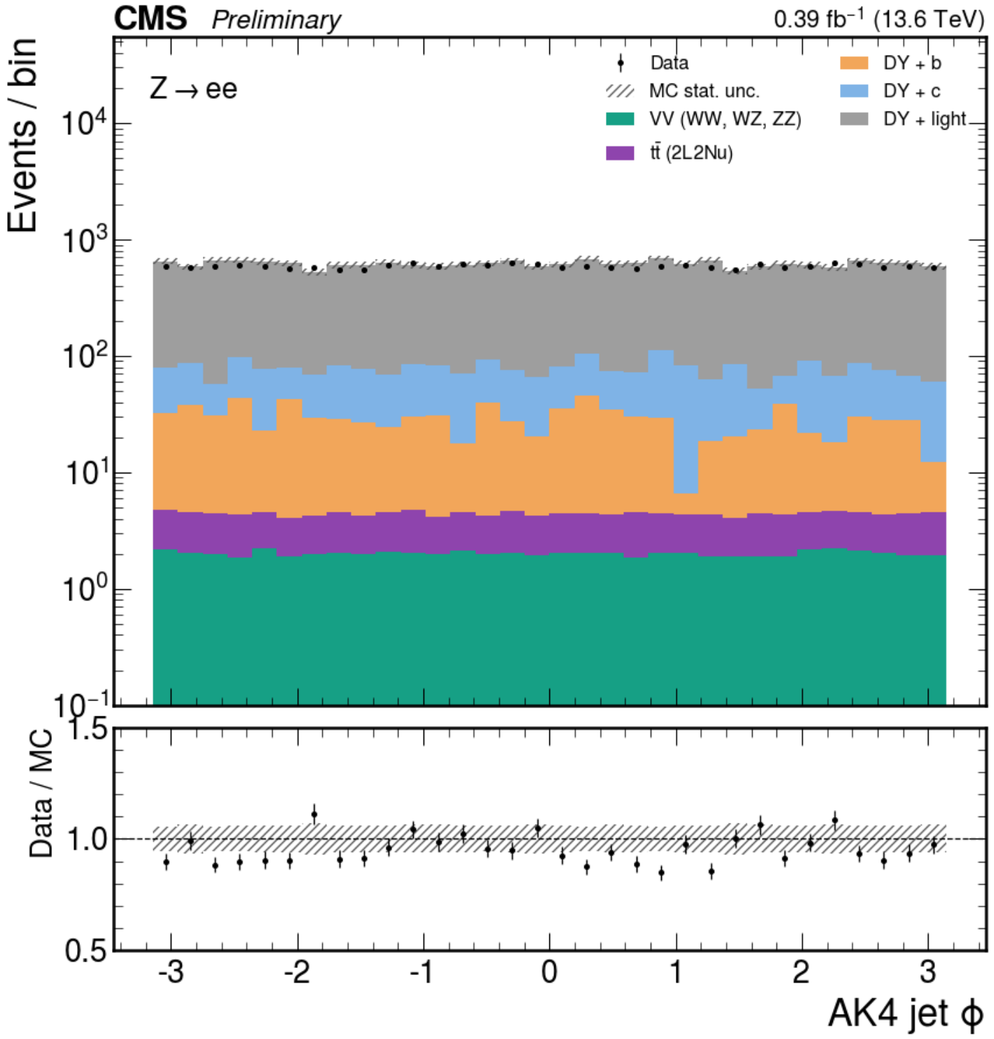
No tag requirement

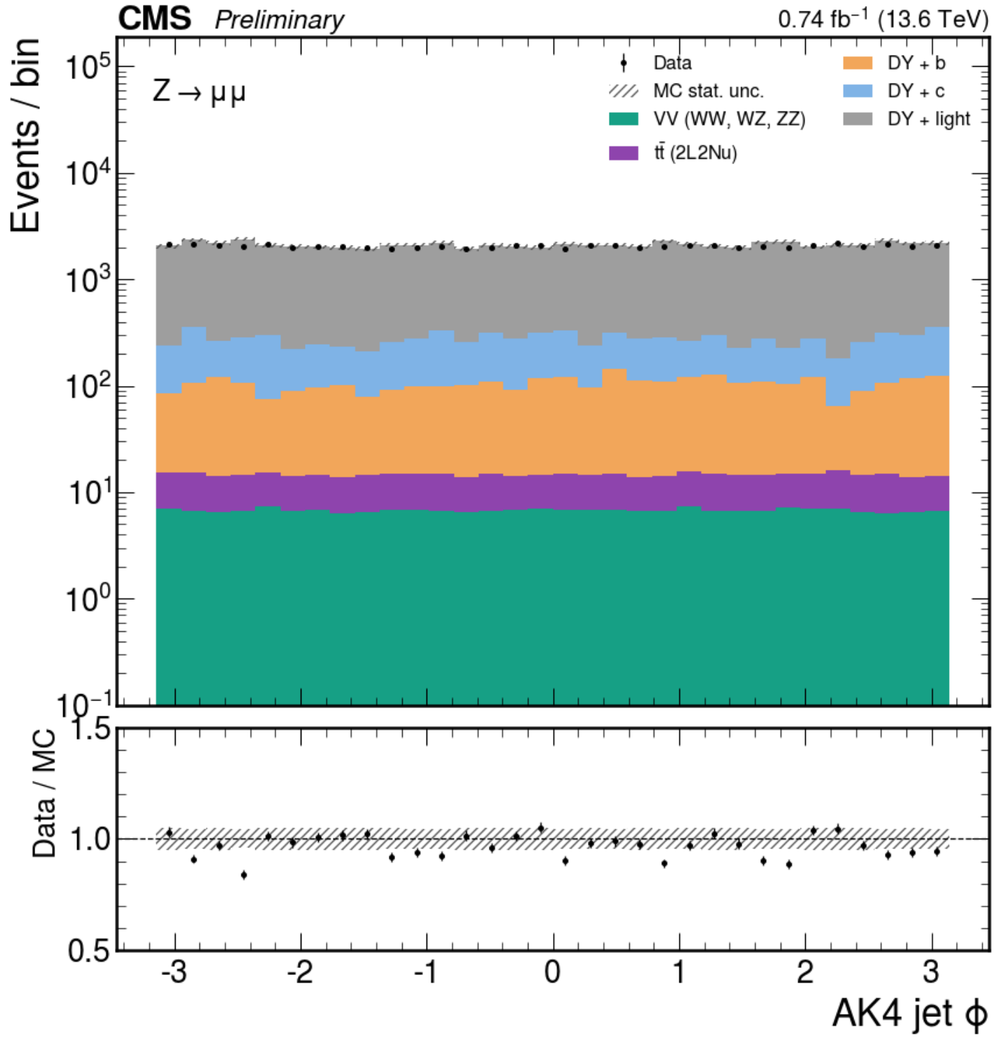
Leading-jet η



No tag requirement

Leading-jet φ

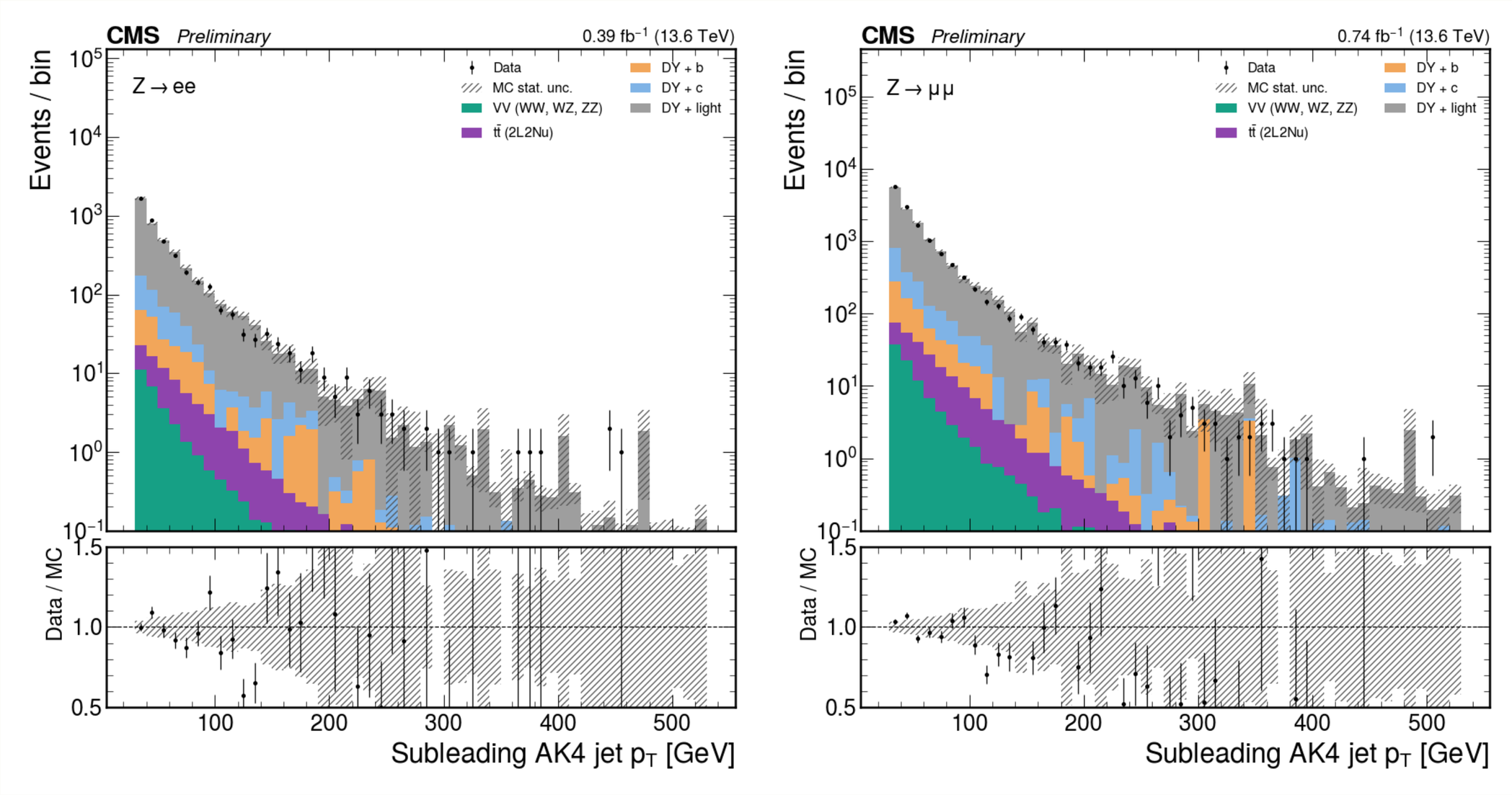




No tag requirement

Second-jet pT

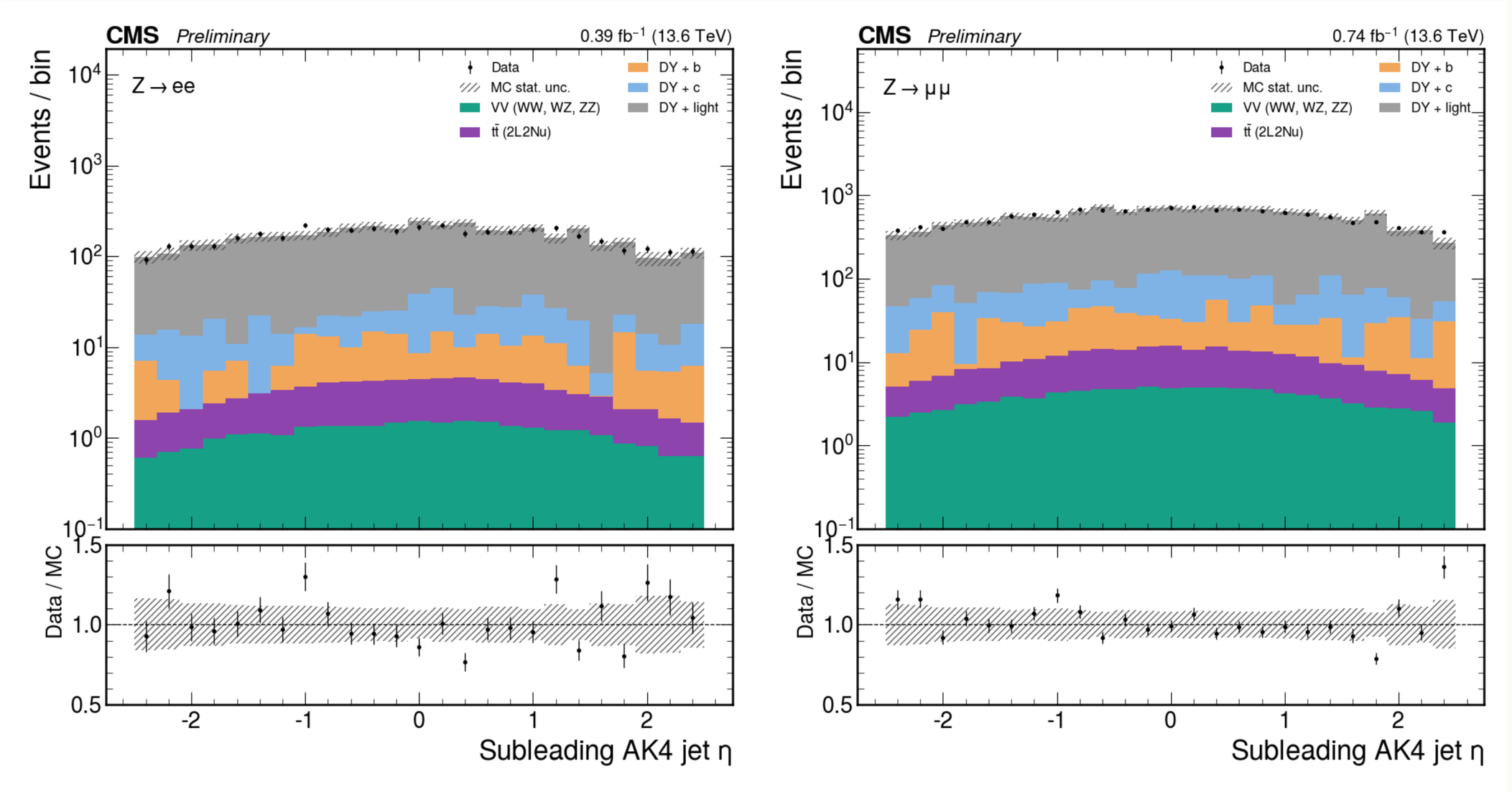
Events with at least 2 jets.



No tag requirement

Second-jet η

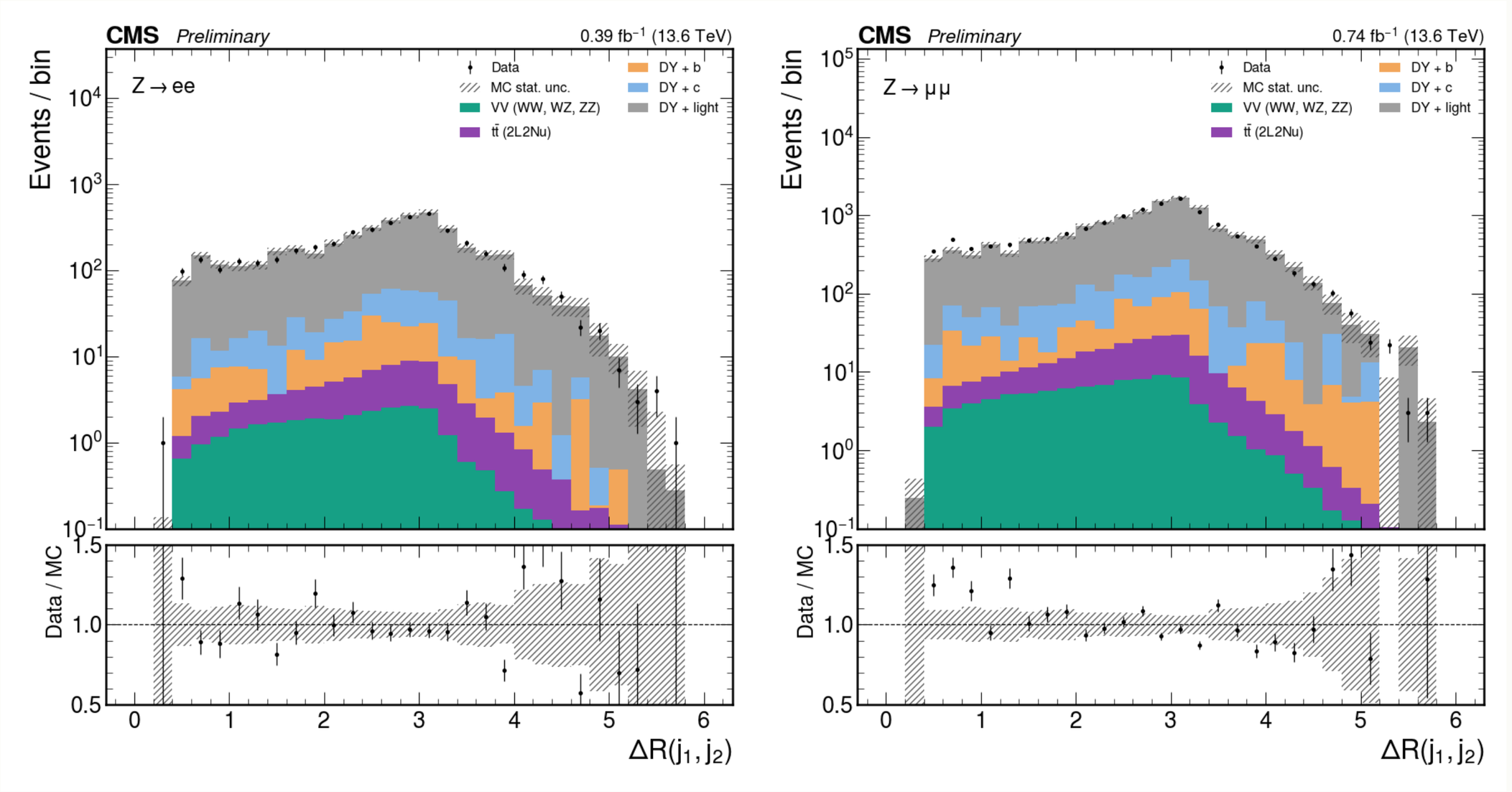
Events with at least 2 jets.



No tag requirement

ΔR between the two leading jets

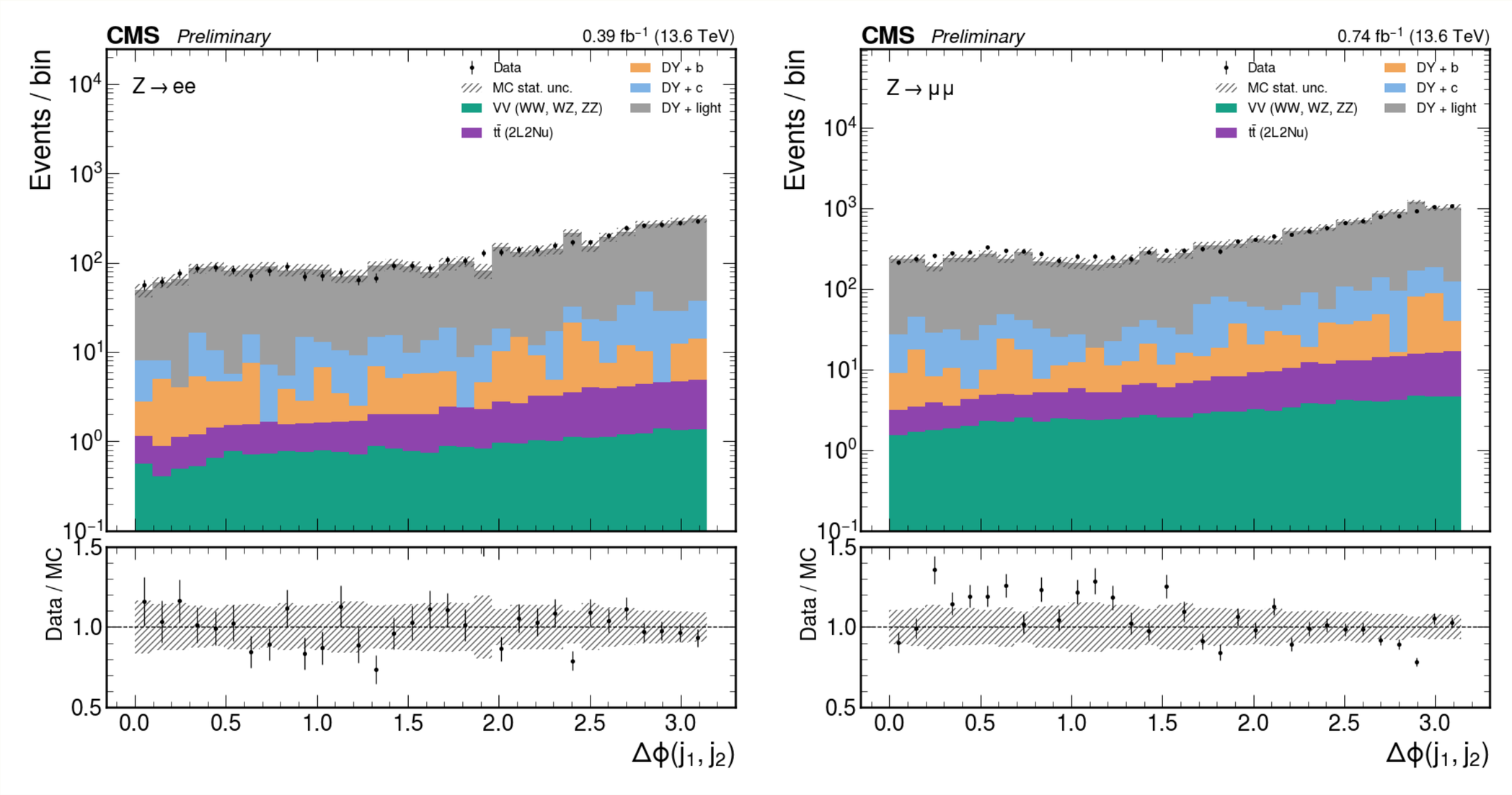
Events with at least 2 jets.



No tag requirement

Δφ between the two leading jets

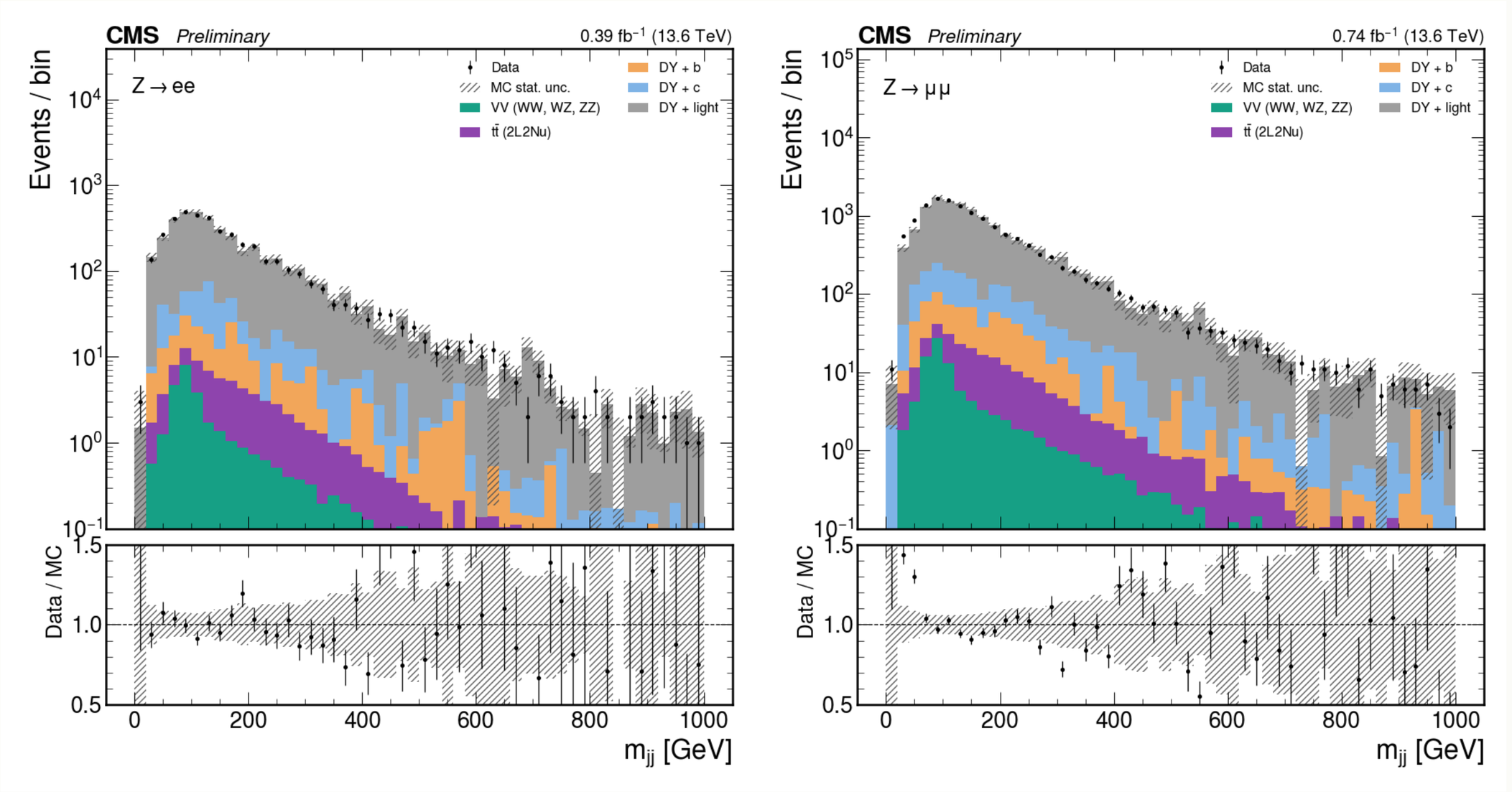
Events with at least 2 jets.



No tag requirement

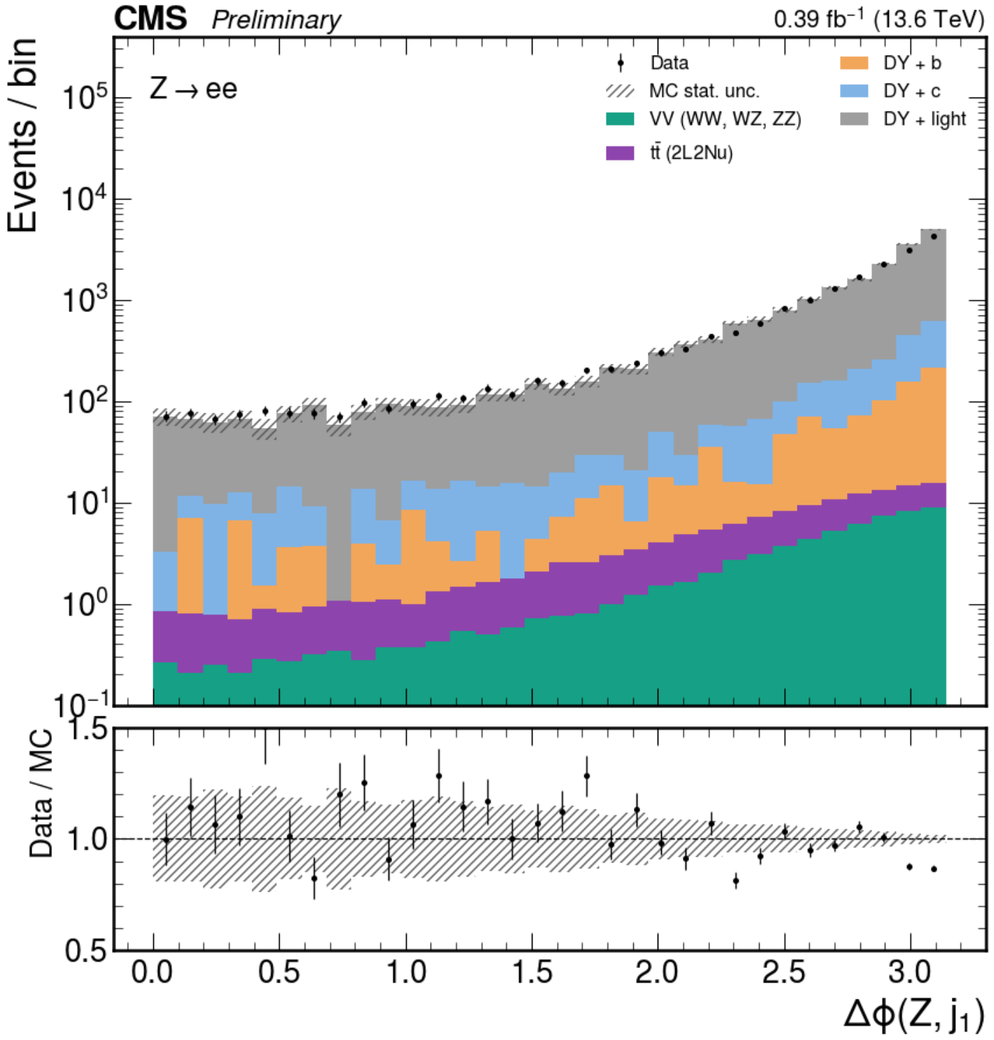
Dijet mass

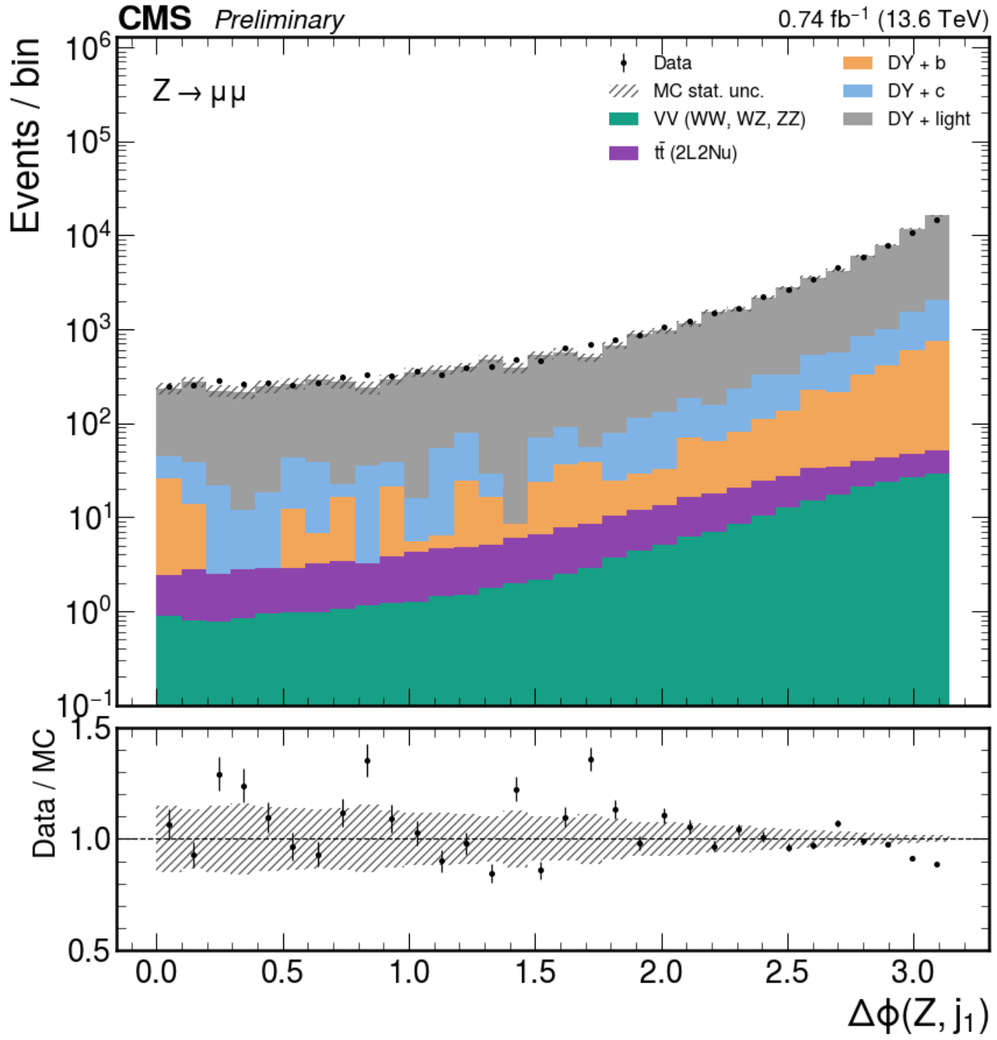
Events with at least 2 jets.



No tag requirement

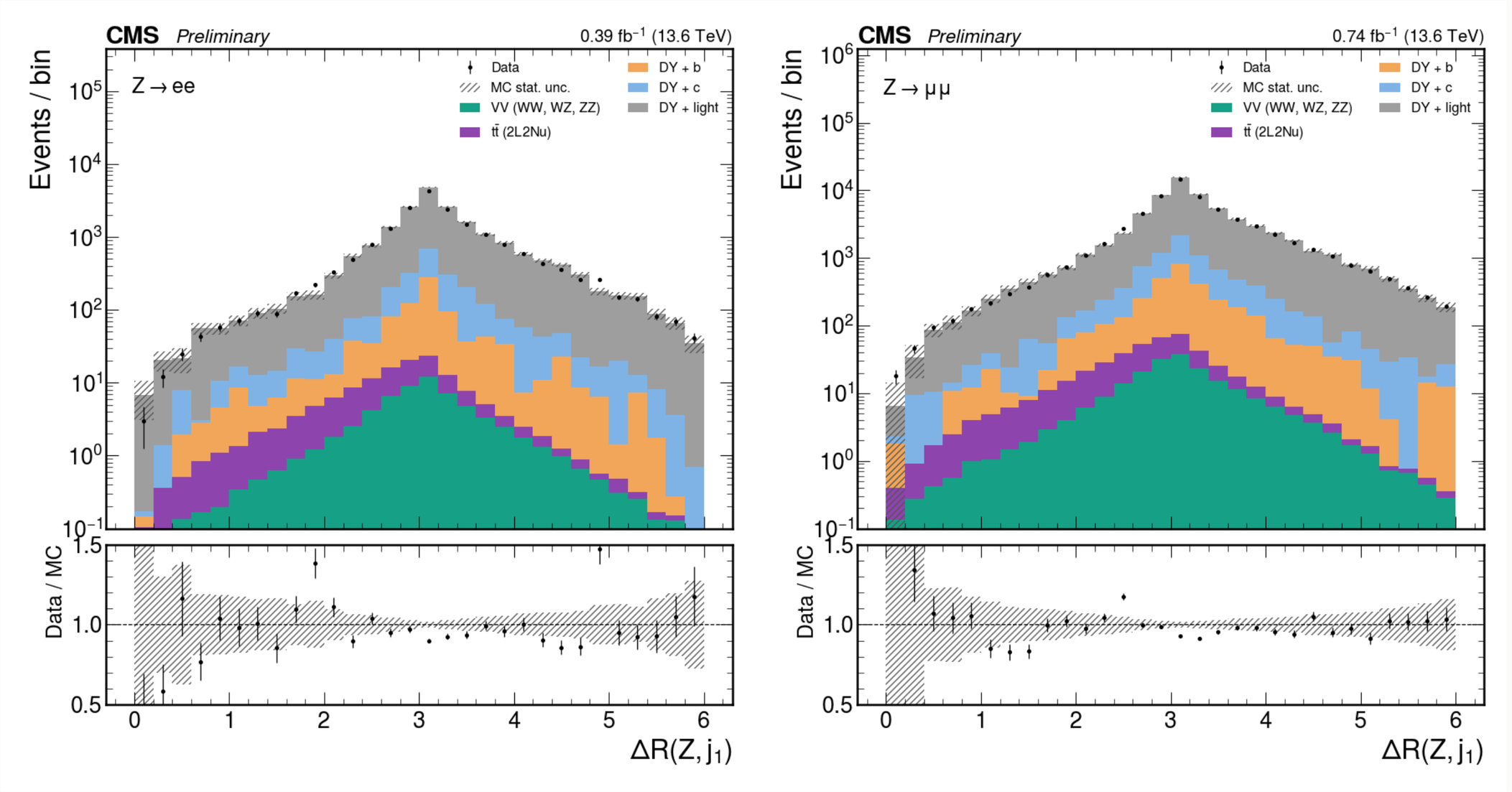
Δφ(Z, leading jet)





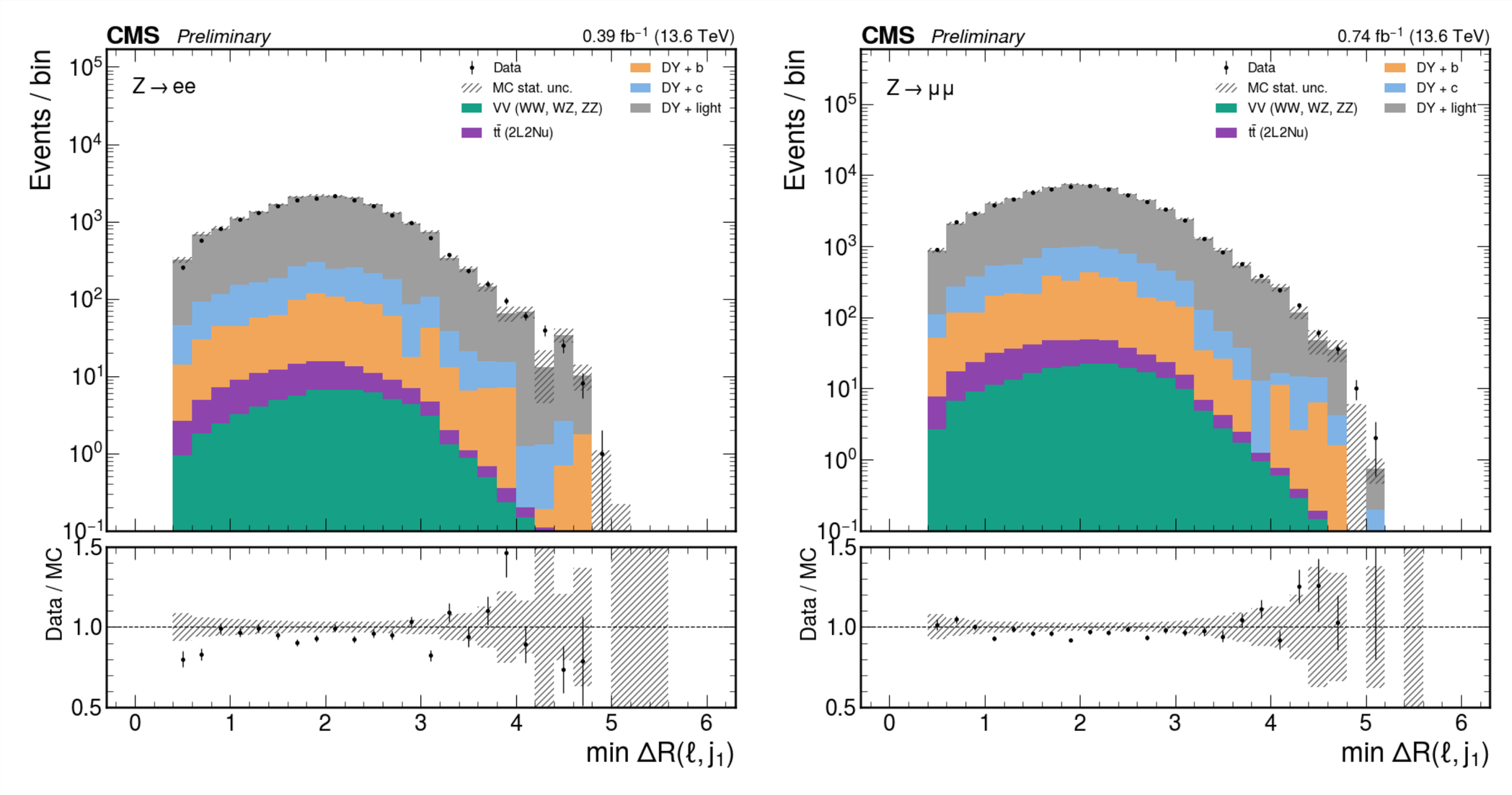
No tag requirement

ΔR(Z, leading jet)



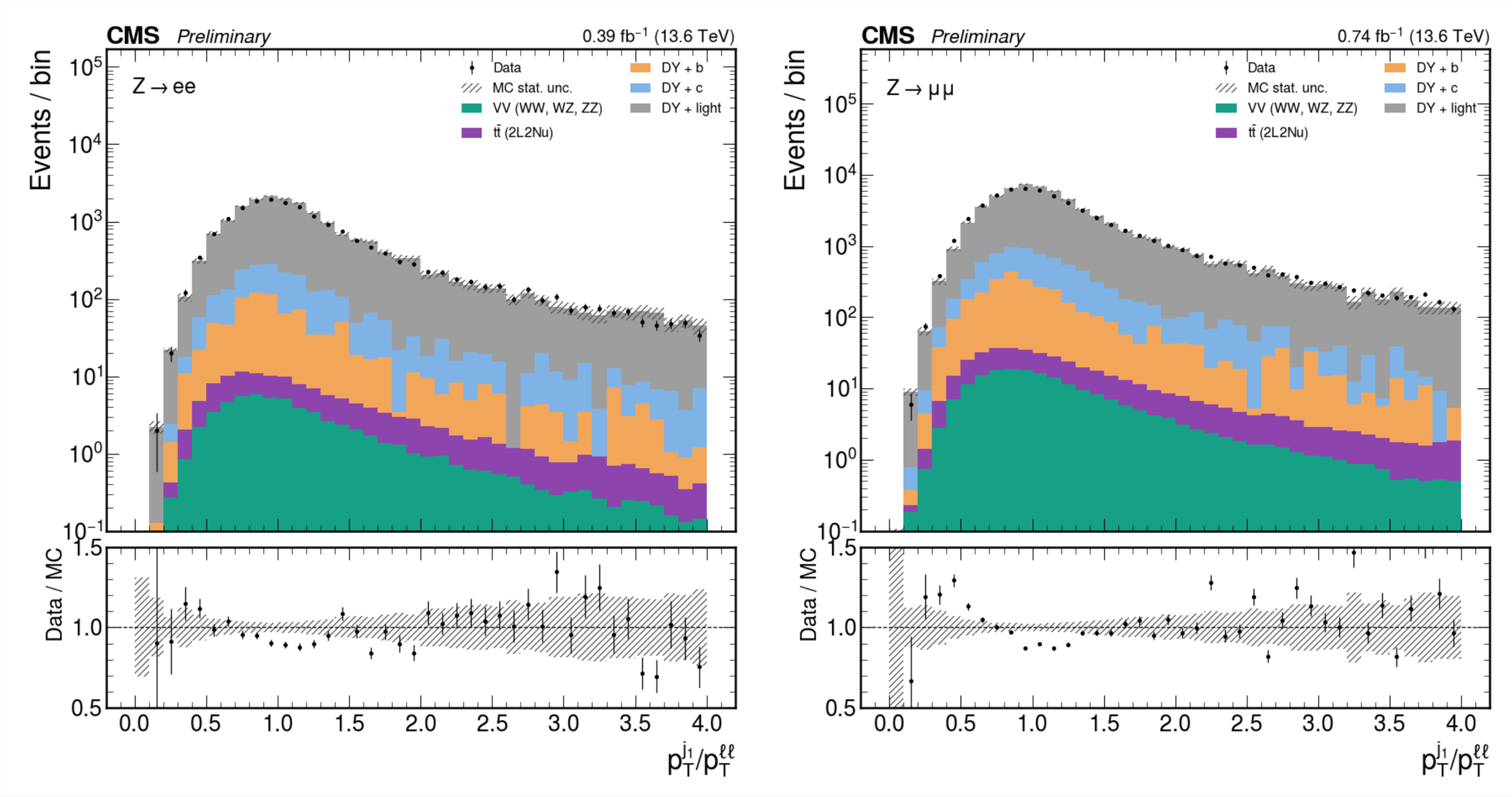
No tag requirement

Smallest ΔR(lepton, leading jet)



No tag requirement

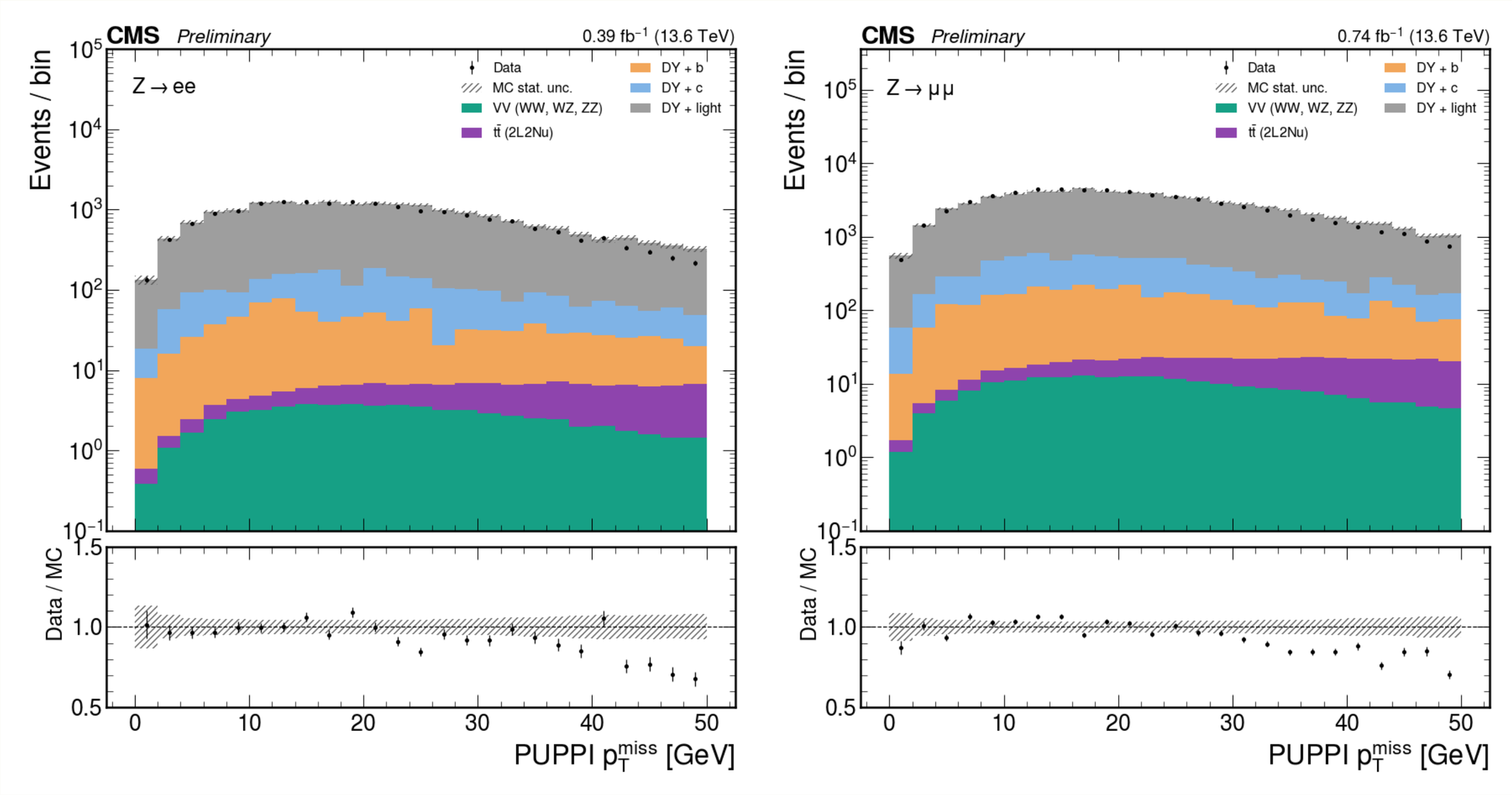
pT balance: leading-jet pT / Z pT



No tag requirement

PUPPI MET

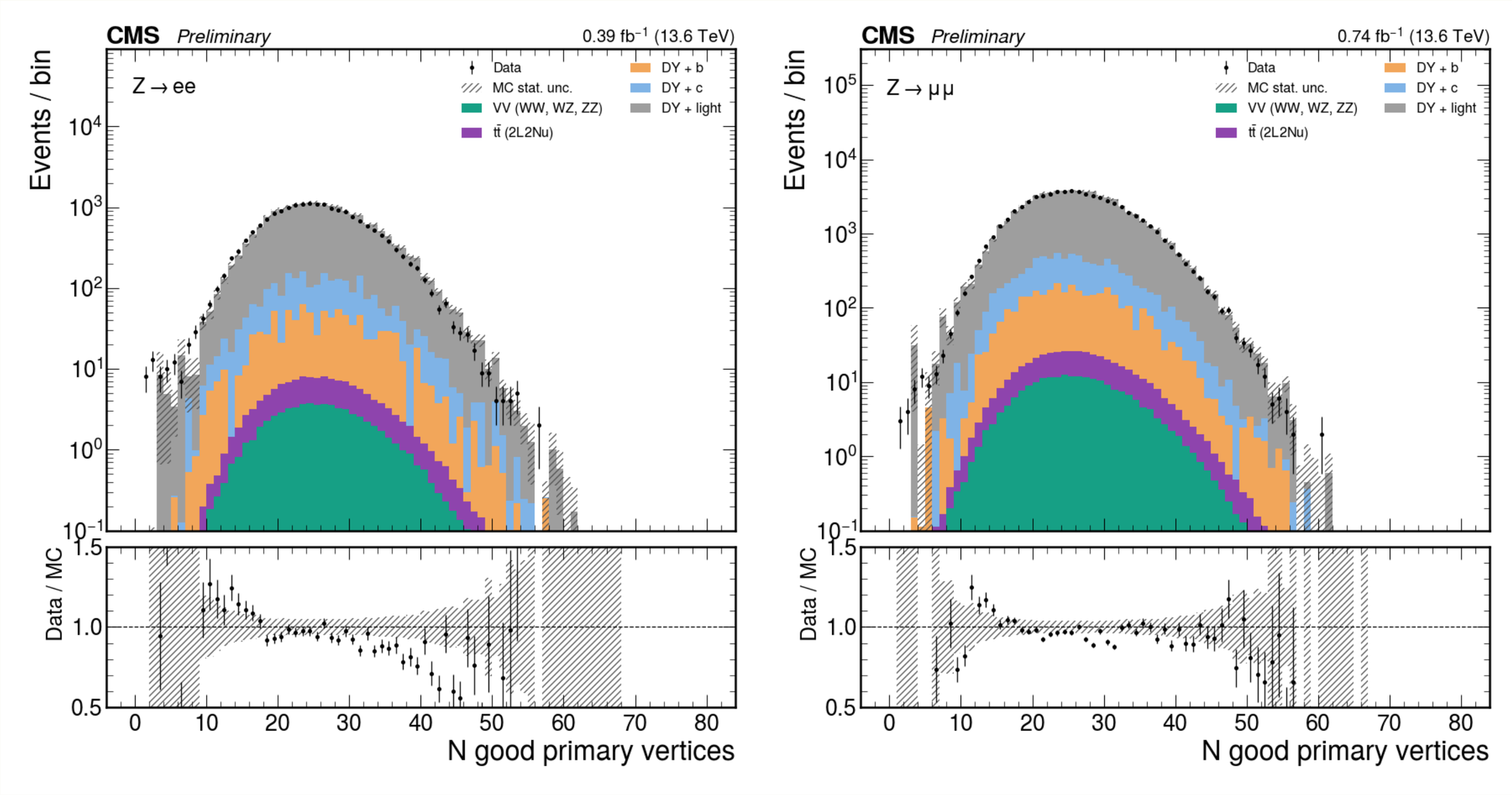
Type-1 corrected; the selection cuts at 50 GeV.



No tag requirement

Number of good primary vertices

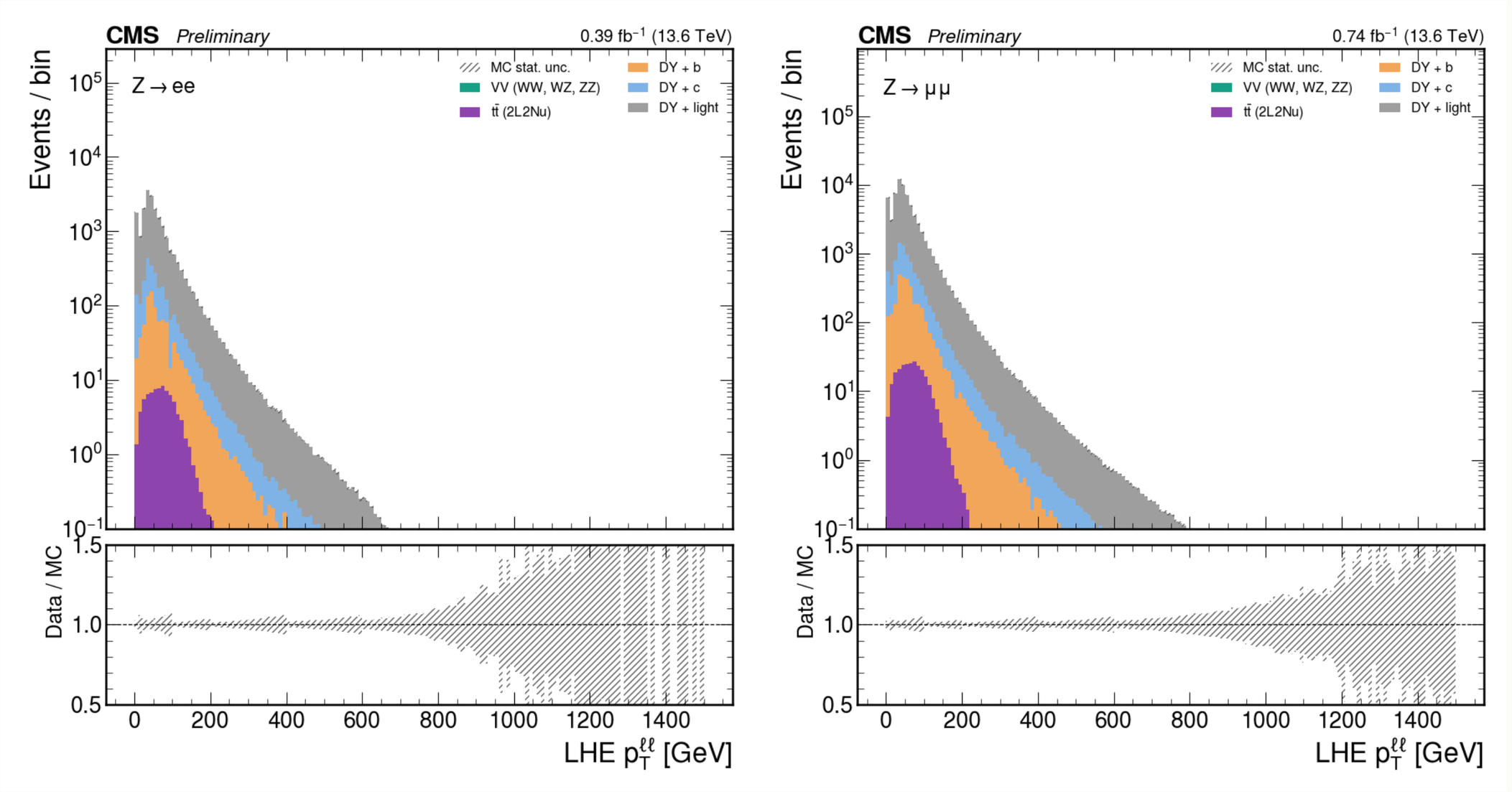
After pileup weights.



No tag requirement

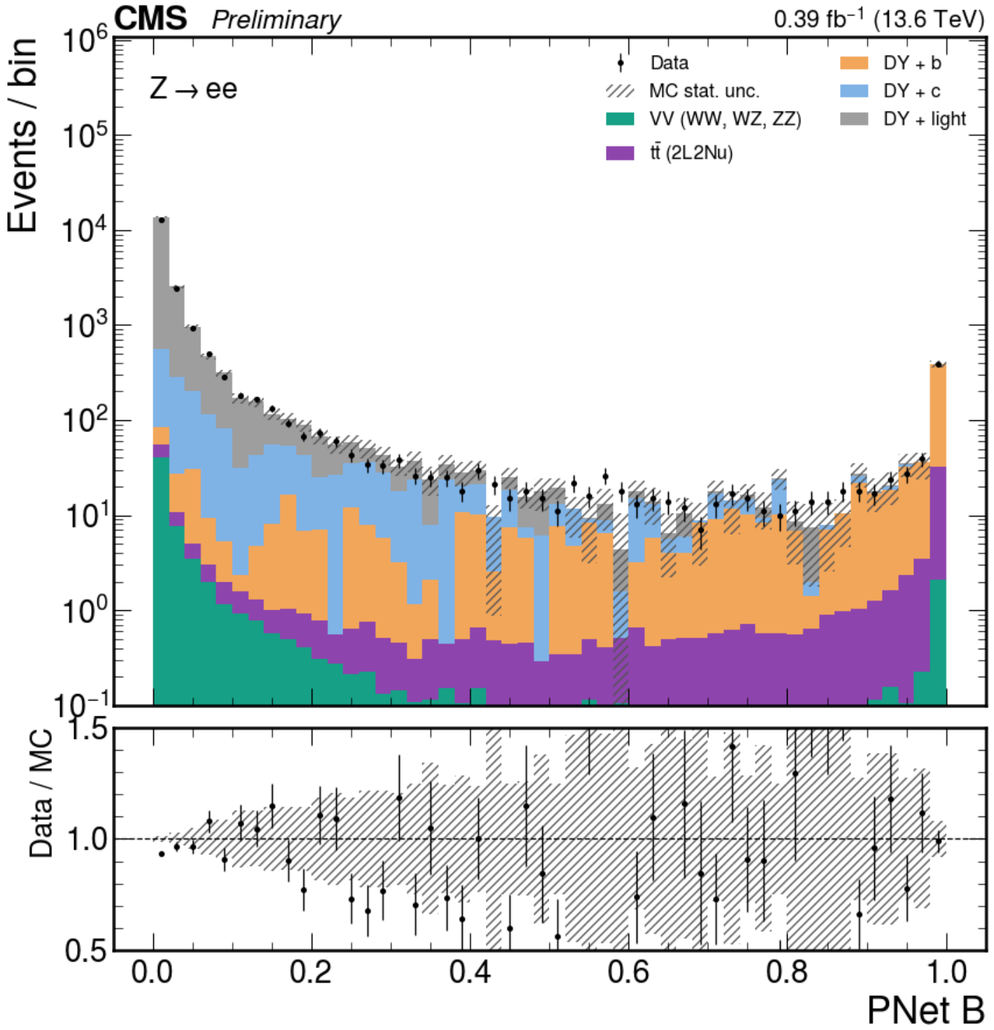
Generator Z pT (LHE)

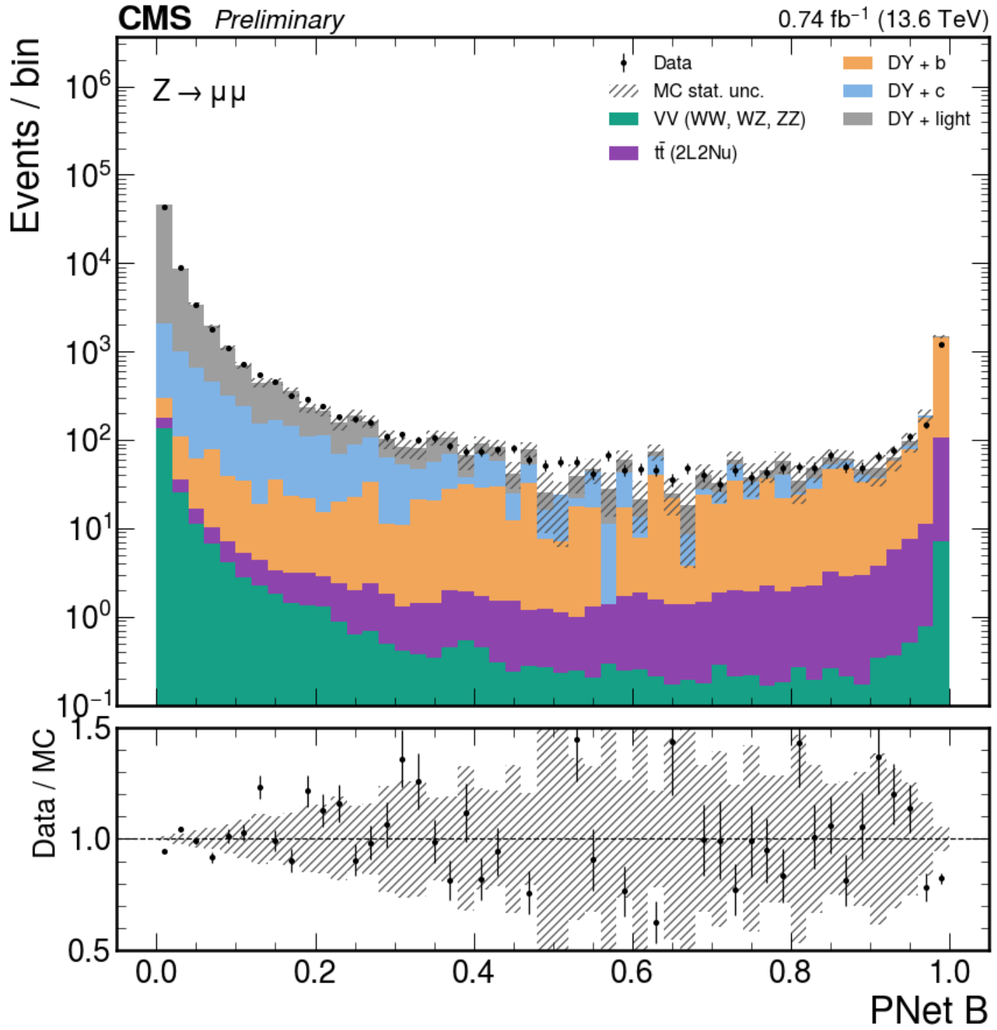
MC only, no data.



No tag requirement

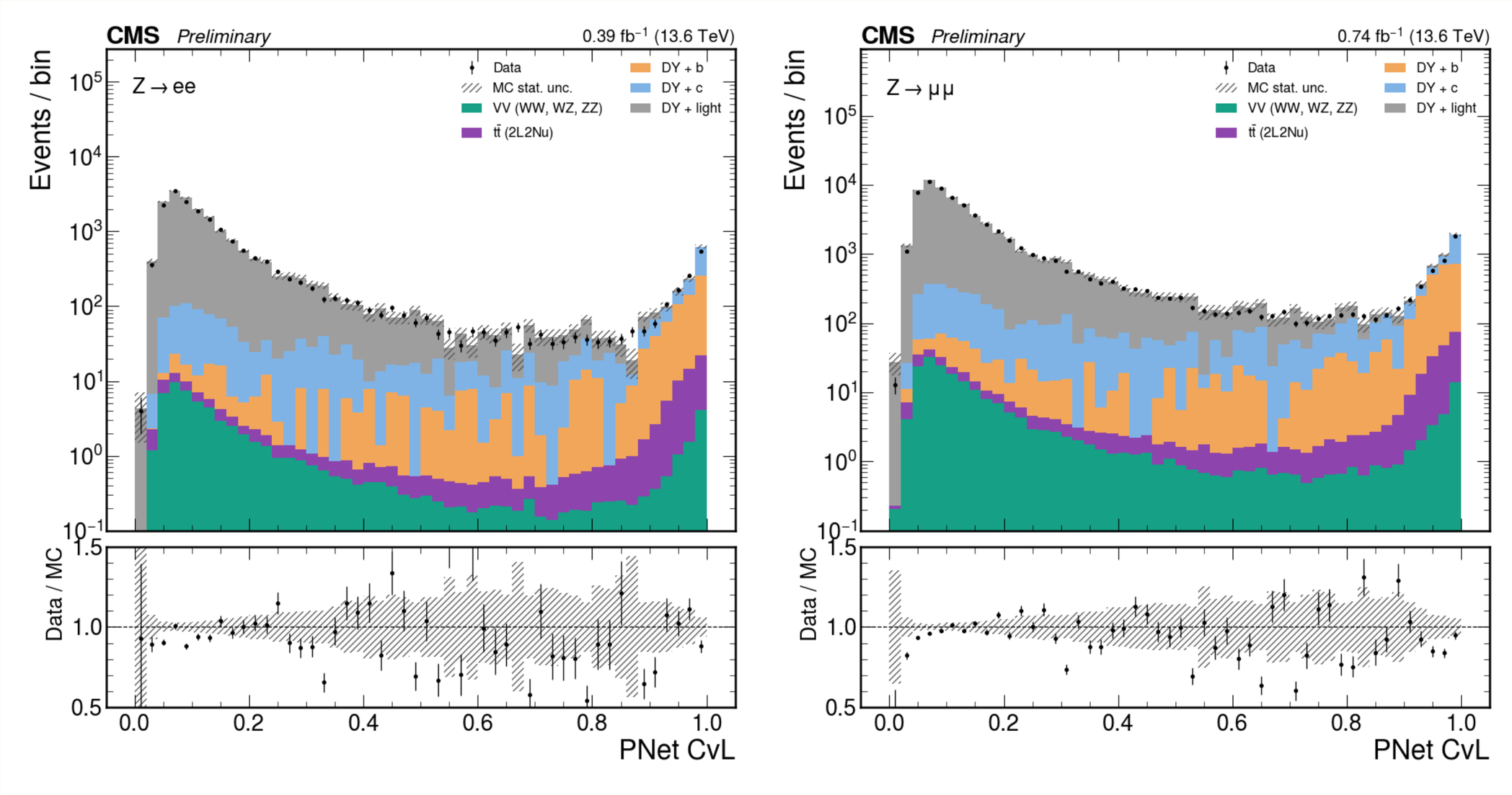
Leading-jet PNet B score





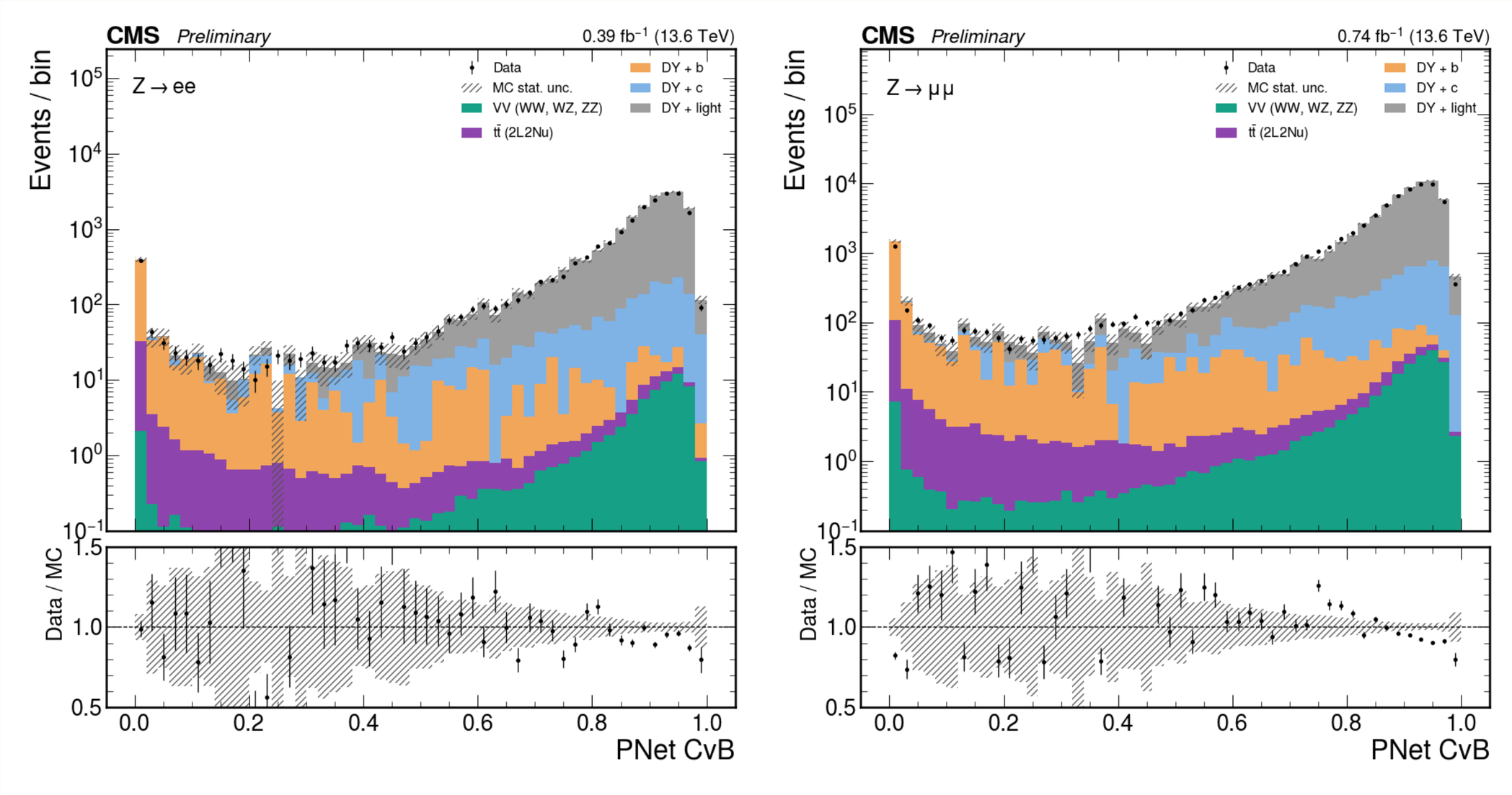
No tag requirement

Leading-jet PNet CvL score



No tag requirement

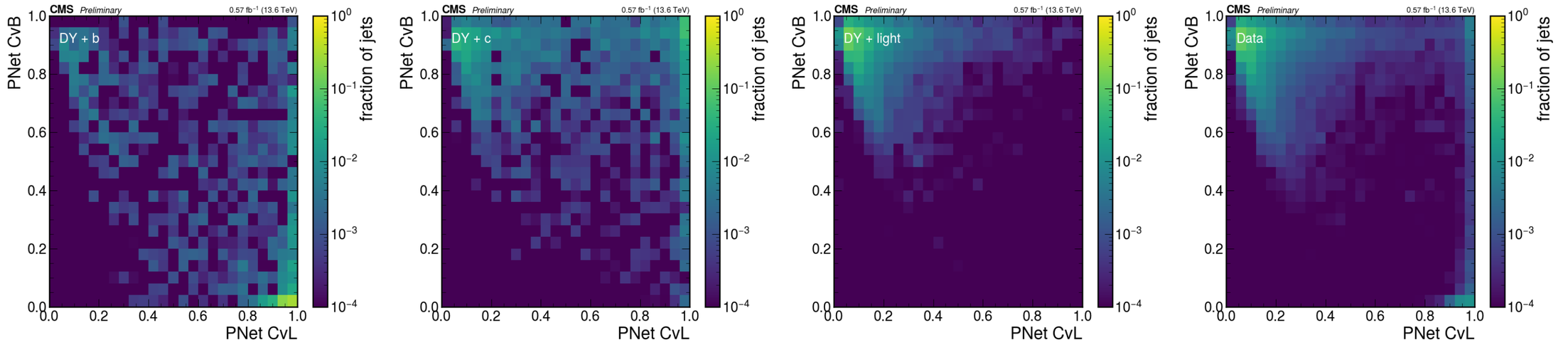
Leading-jet PNet CvB score



No tag requirement

PNet CvL vs CvB of the leading jet

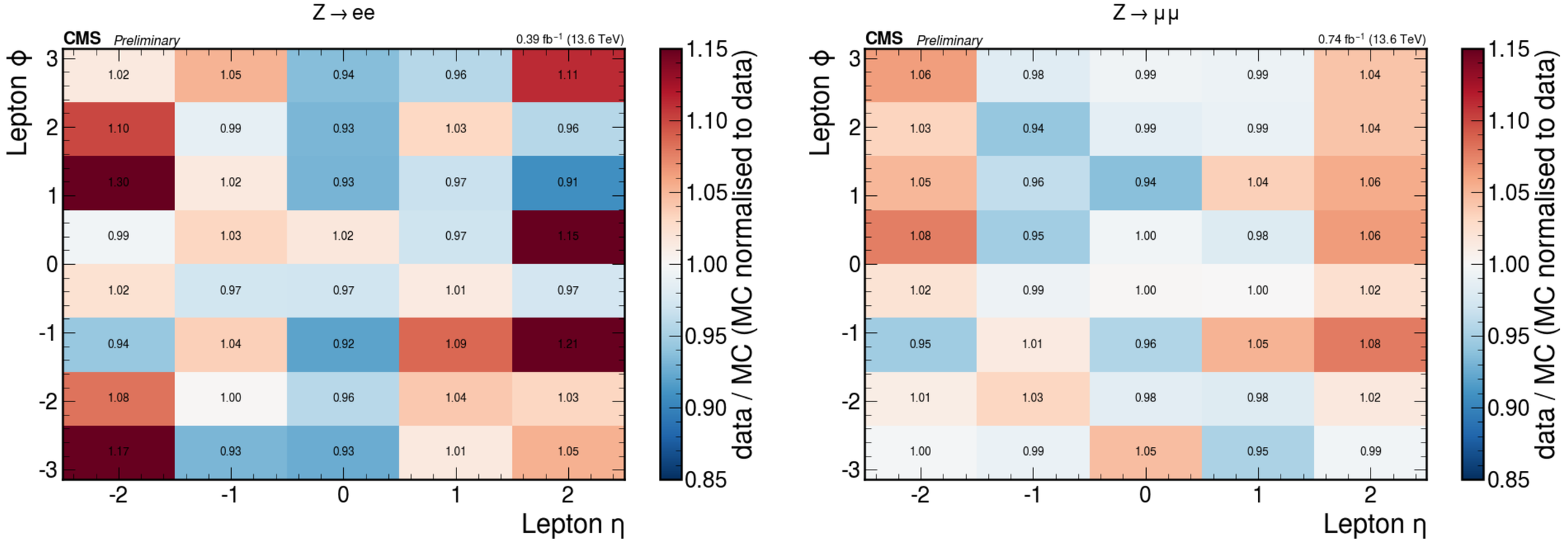
Each panel normalised to 1. Z→ee and Z→μμ added.



No tag requirement

Lepton η–φ, data / MC

Both Z leptons. MC normalised to data in each map. Colour scale stops at ±15%; in the tagged regions there are few events per cell, so read the numbers.



b-tag Loose

Leading jet PNet B > 0.047

	data/MC	DY+b	DY+c	DY+light	tt	tagged MC	tagged data
Z→ee	0.947 ± 0.031	20.7%	23.0%	54.1%	1.7%	16.9%	16.9%
Z→μμ	0.951 ± 0.021	21.5%	20.9%	55.6%	1.6%	17.9%	17.6%

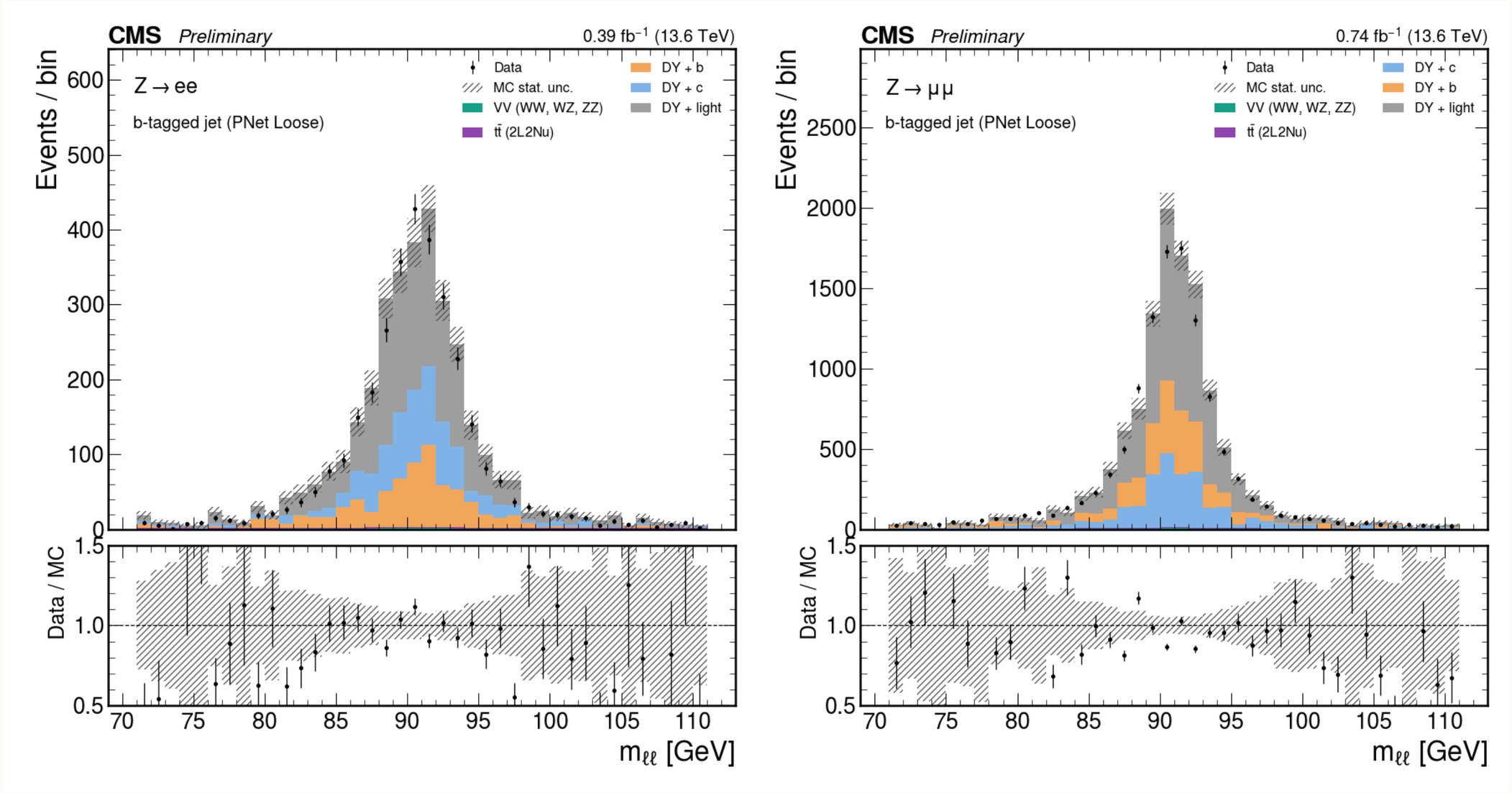
Flavour shares are of all MC. 'Tagged' = fraction of Z + jet events passing the tag. Errors are statistical only.

32 plots follow: Z and leptons, jets, Z–jet, MET and vertices, tagger scores, 2D maps. Left Z→ee, right Z→μμ.

b-tag Loose

Dilepton mass

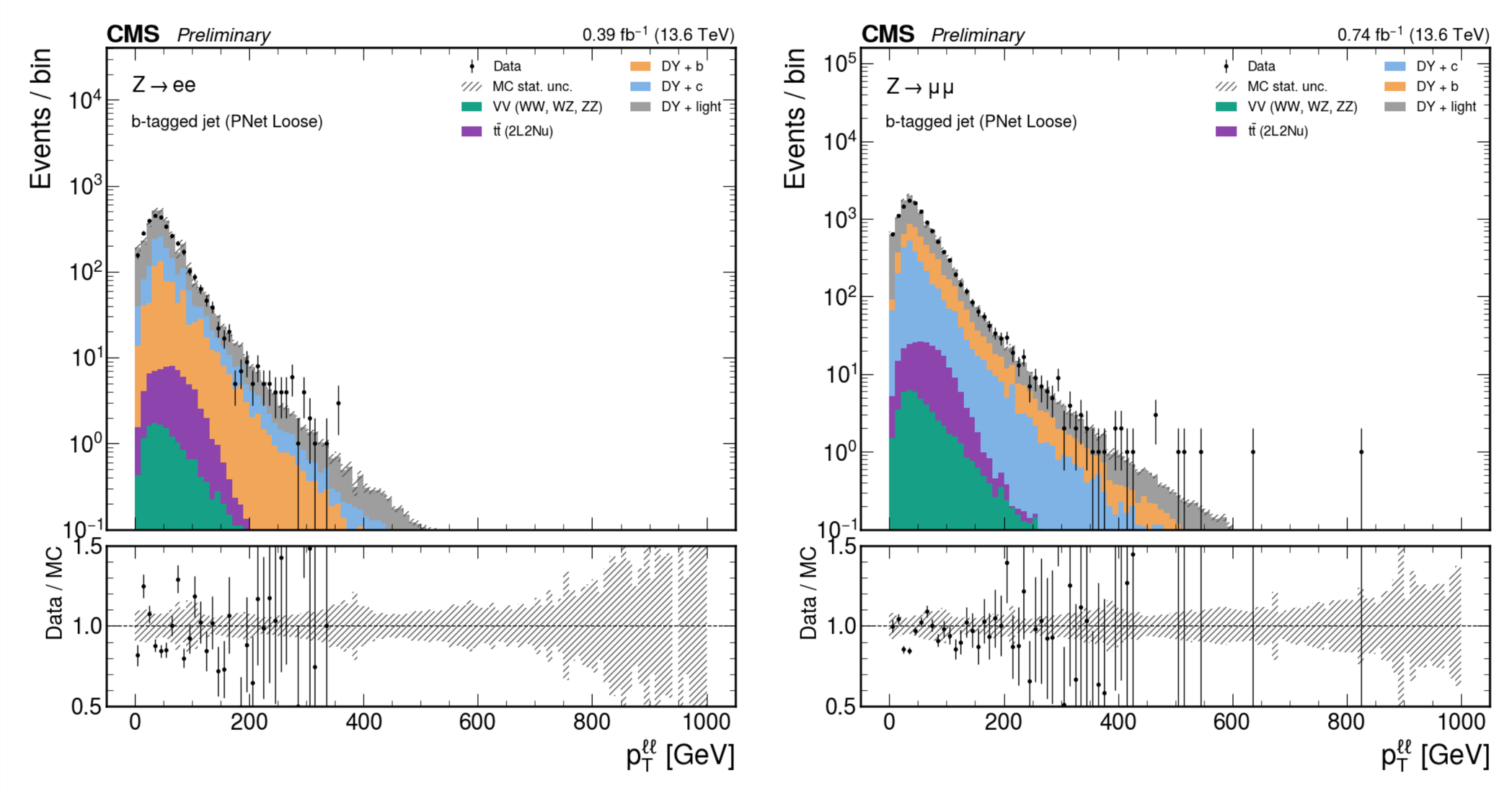
m(ℓℓ), 71–111 GeV window, linear scale.



b-tag Loose

Z pT

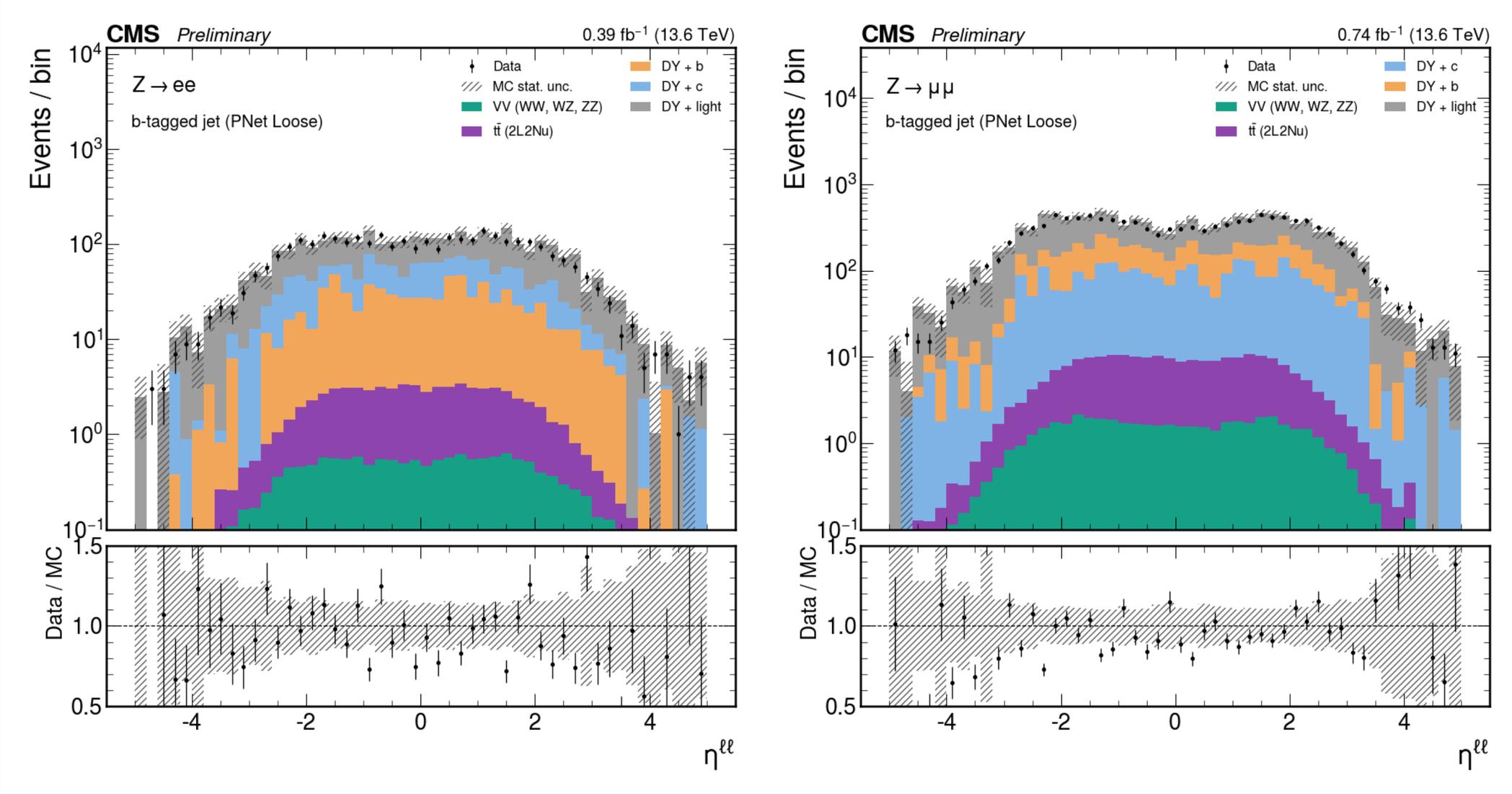
pT of the lepton pair.



b-tag Loose

Z η

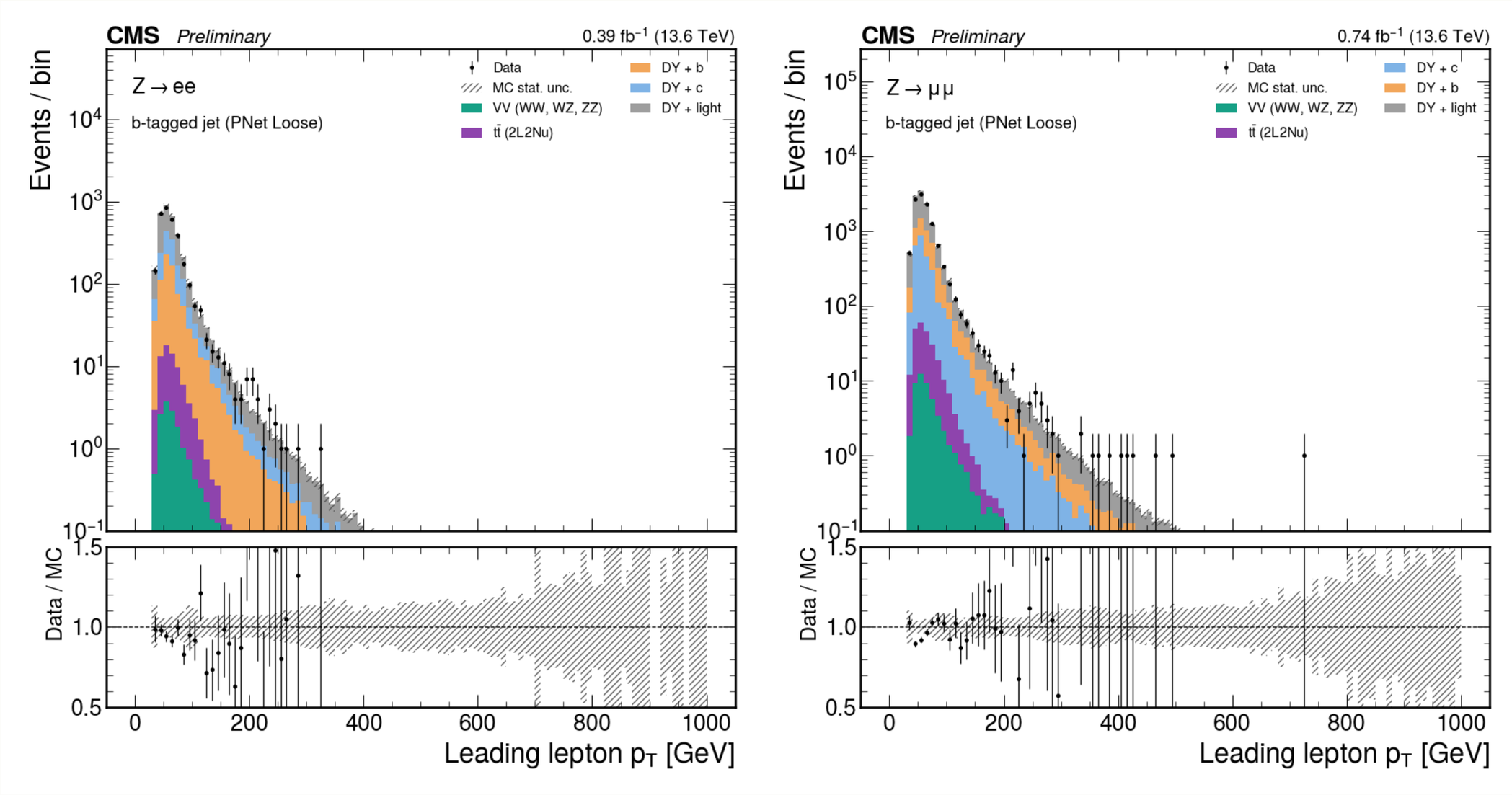
η of the lepton pair.



b-tag Loose

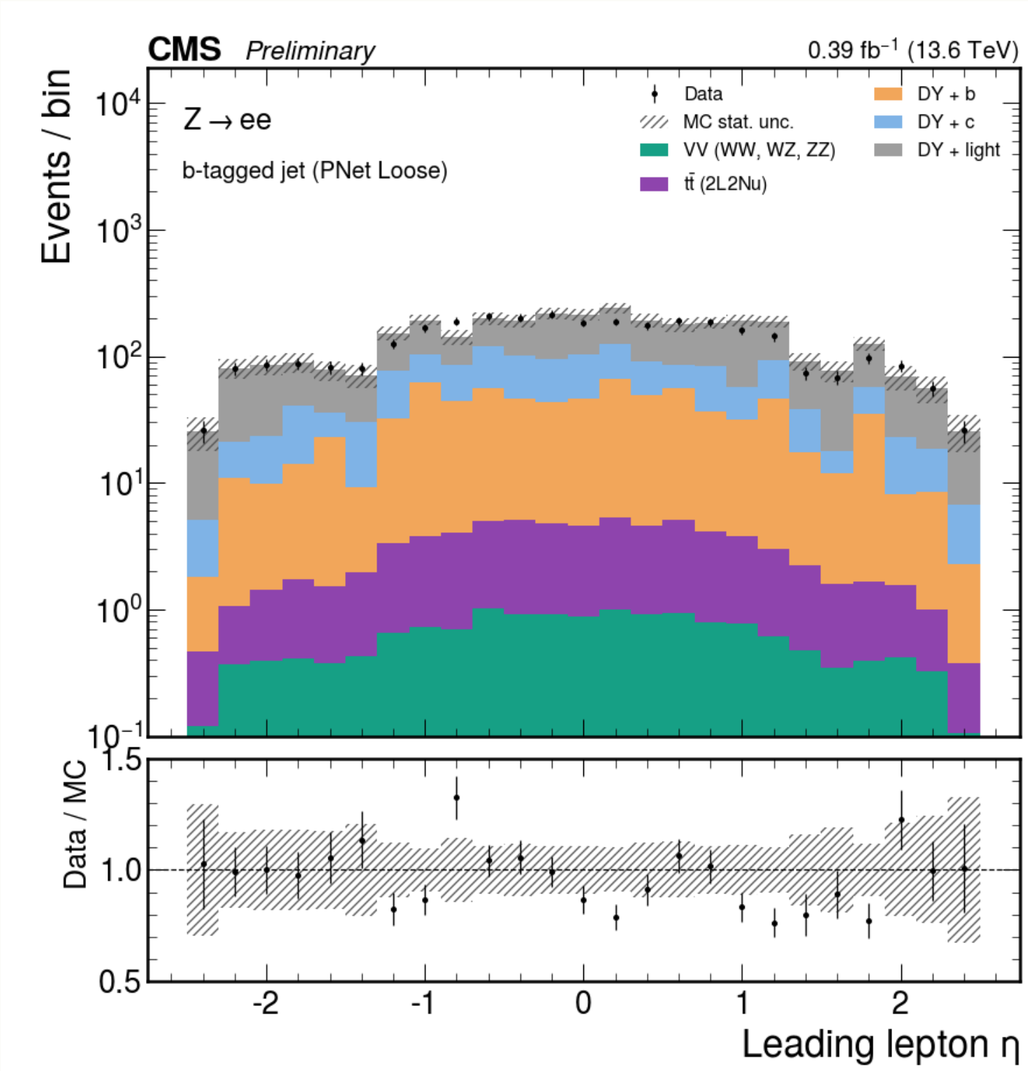
Leading-lepton pT

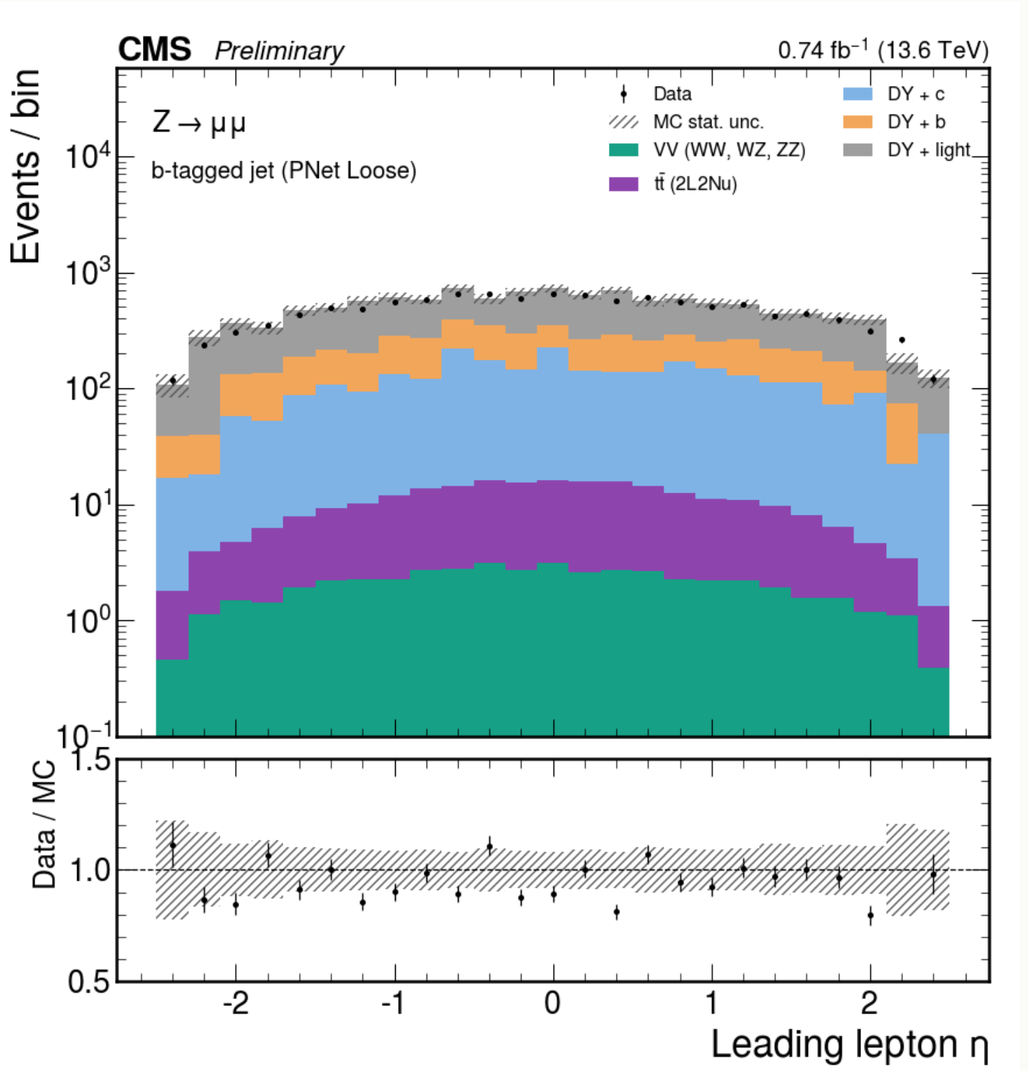
Leading lepton pT > 35 GeV (trigger plateau).



b-tag Loose

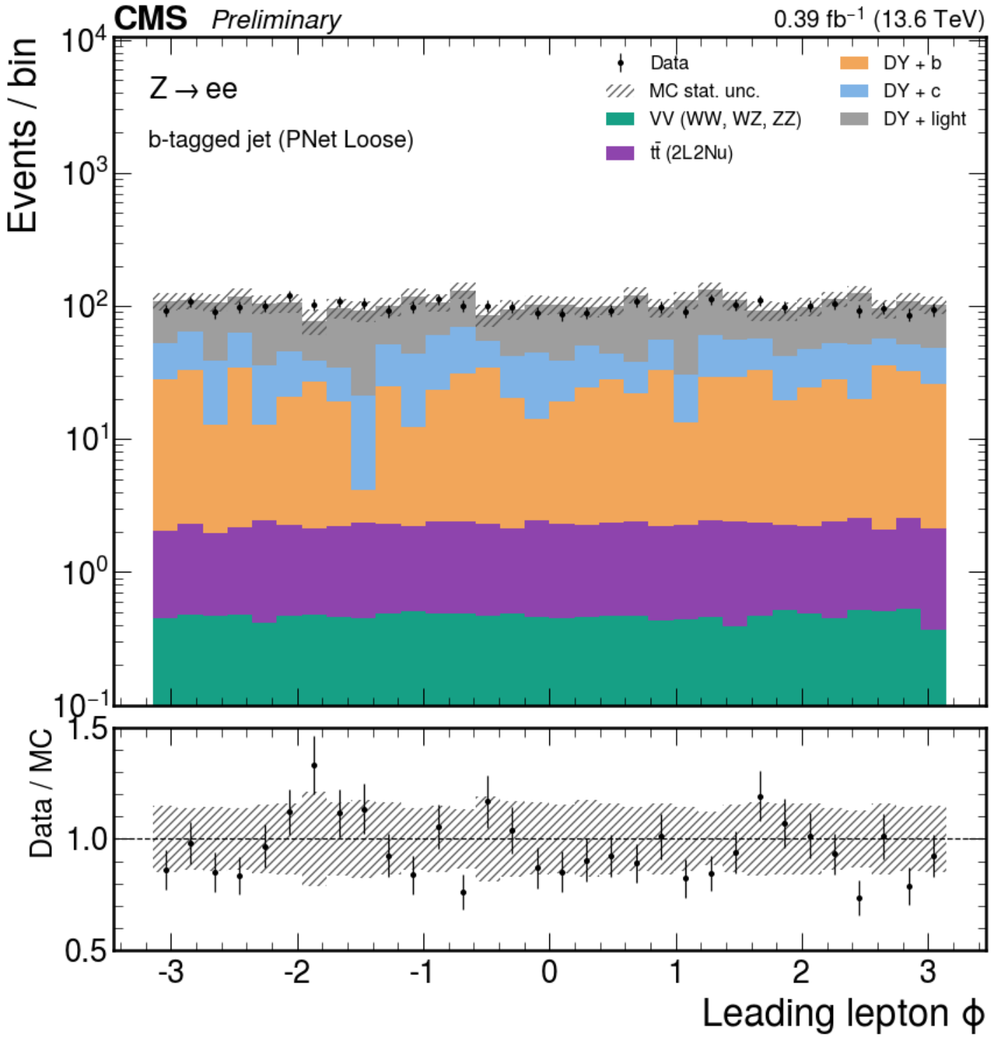
Leading-lepton η

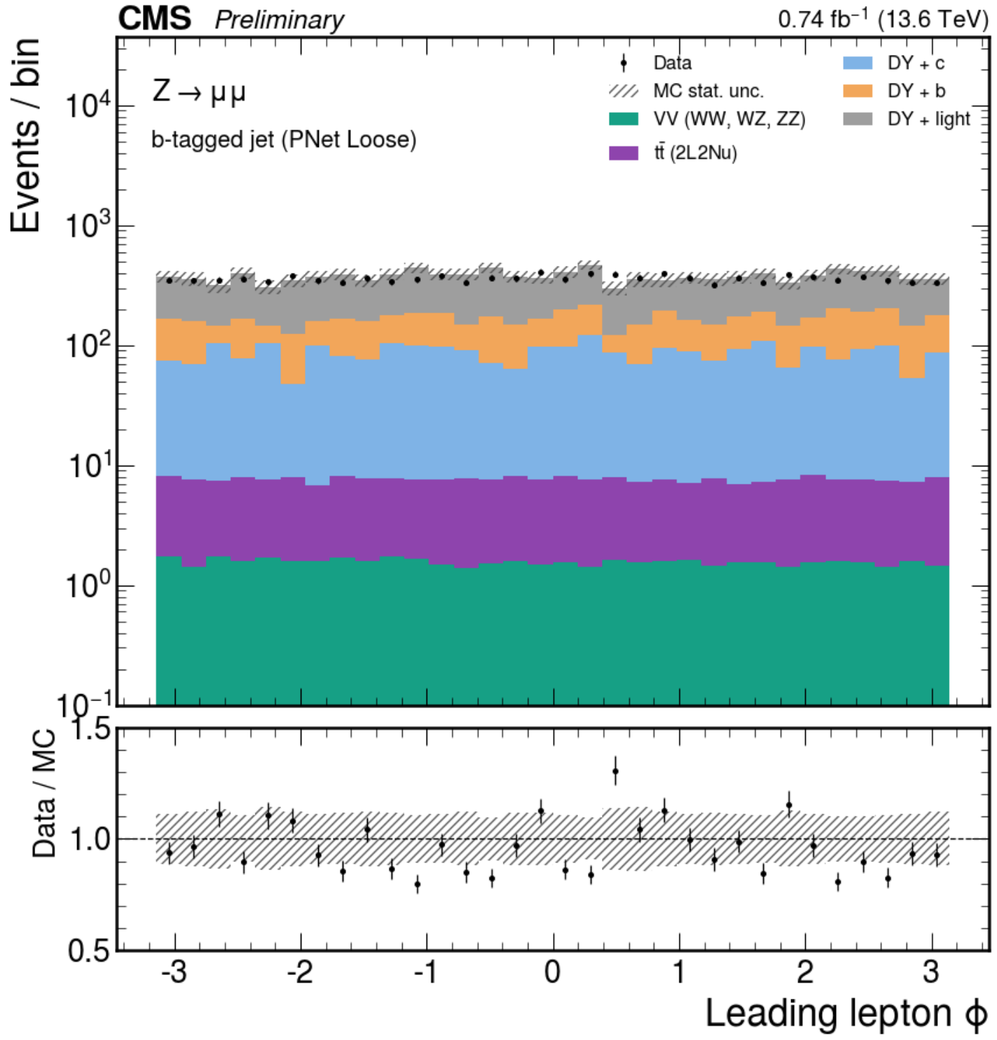




b-tag Loose

Leading-lepton φ

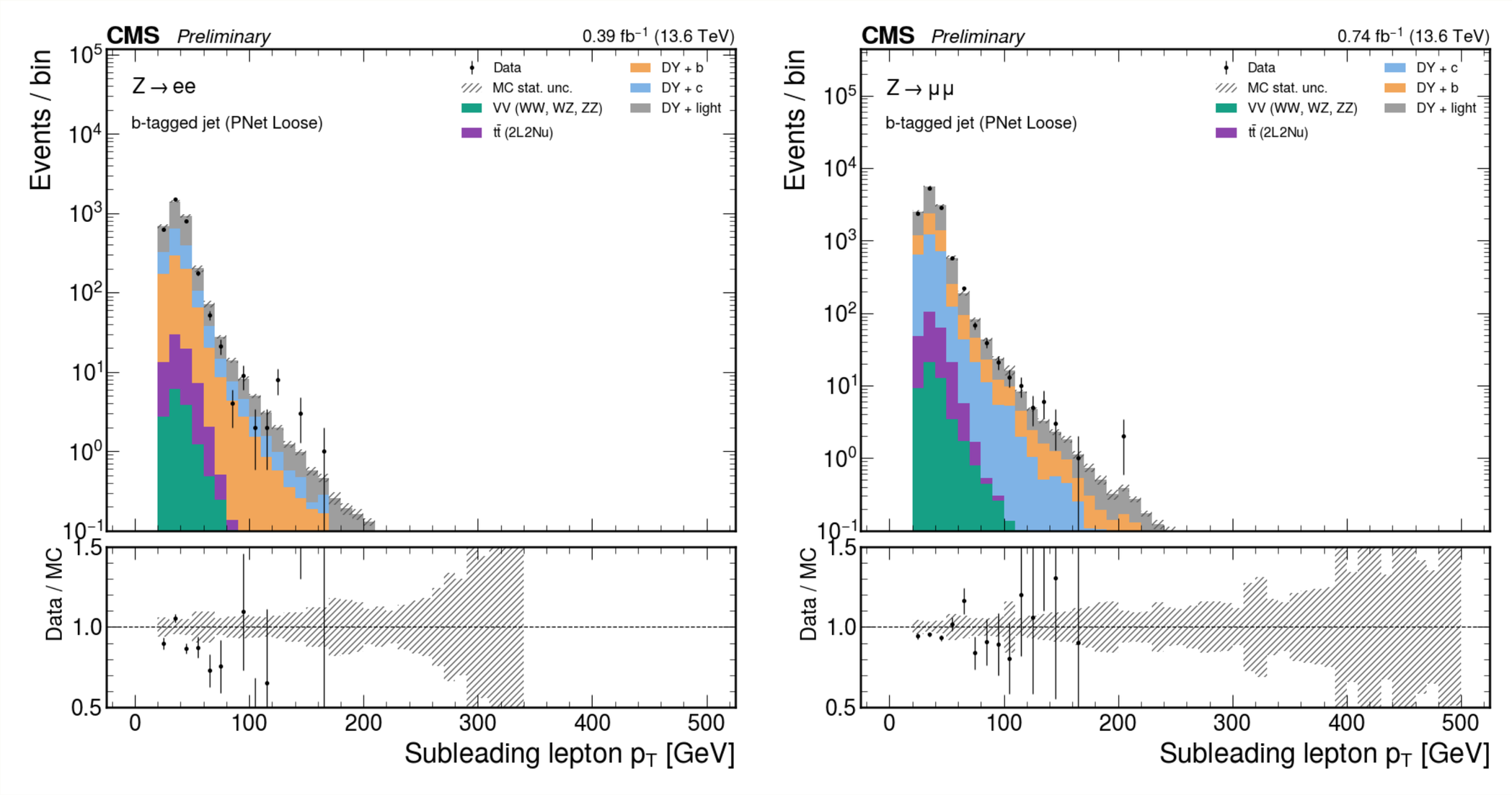




b-tag Loose

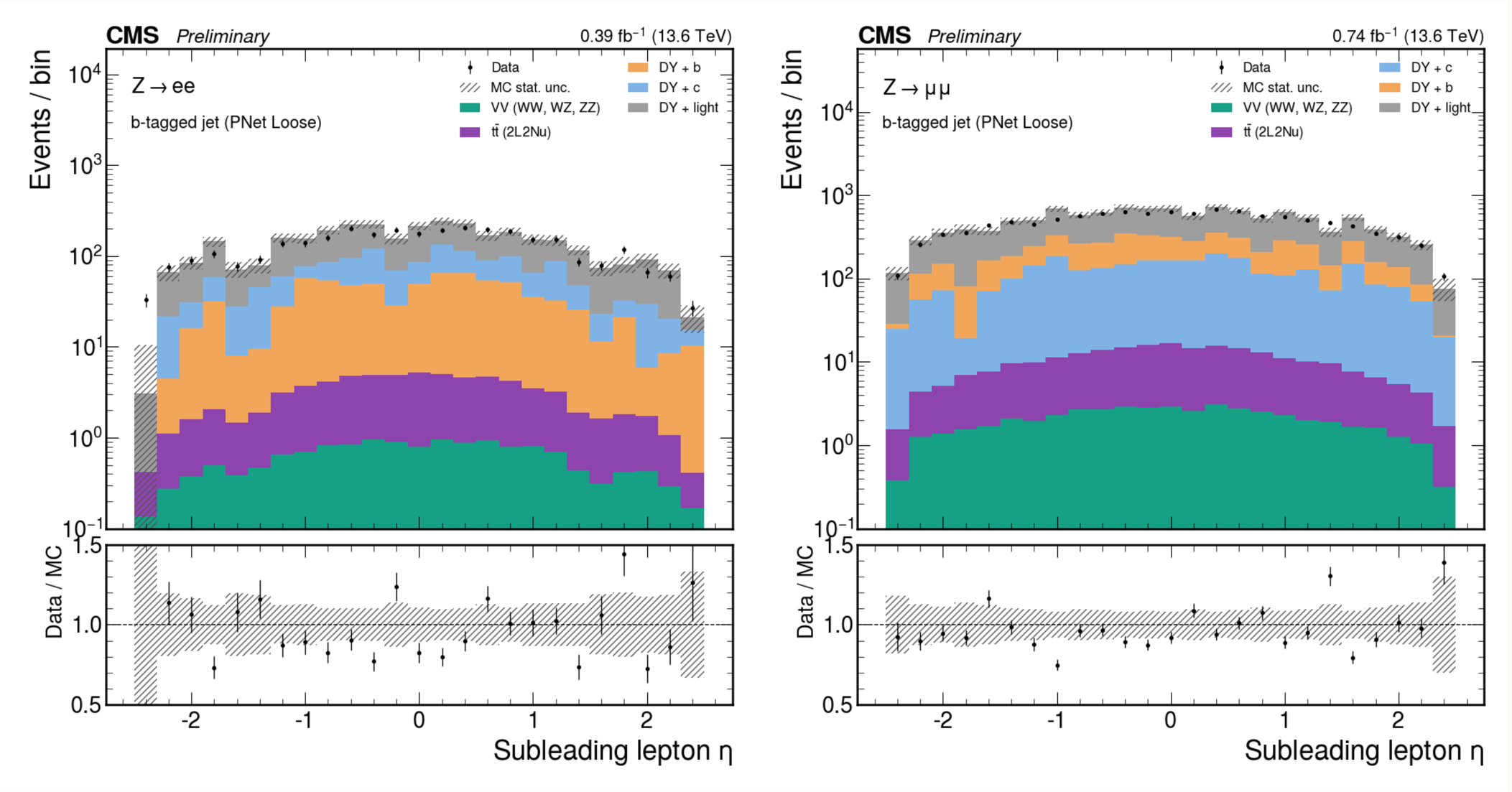
Second-lepton pT

Second lepton pT > 25 GeV.



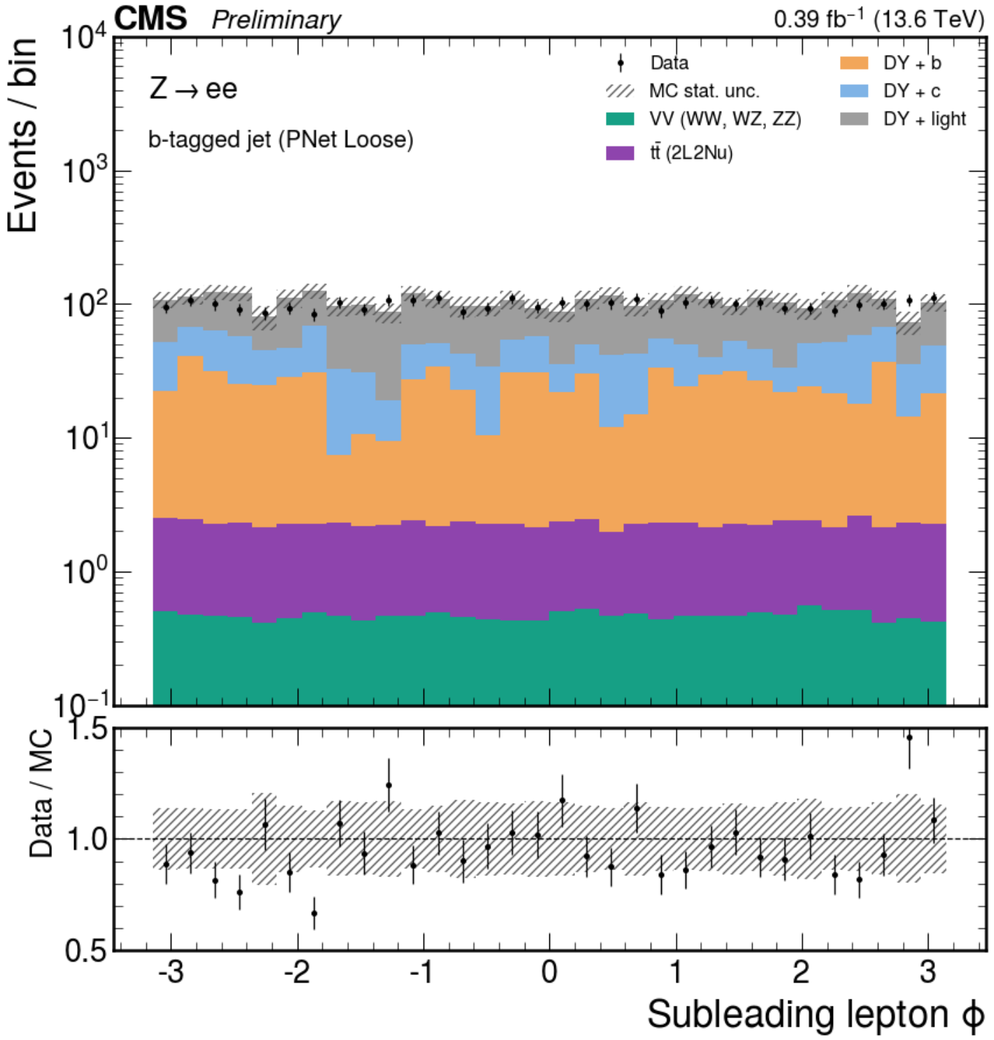
b-tag Loose

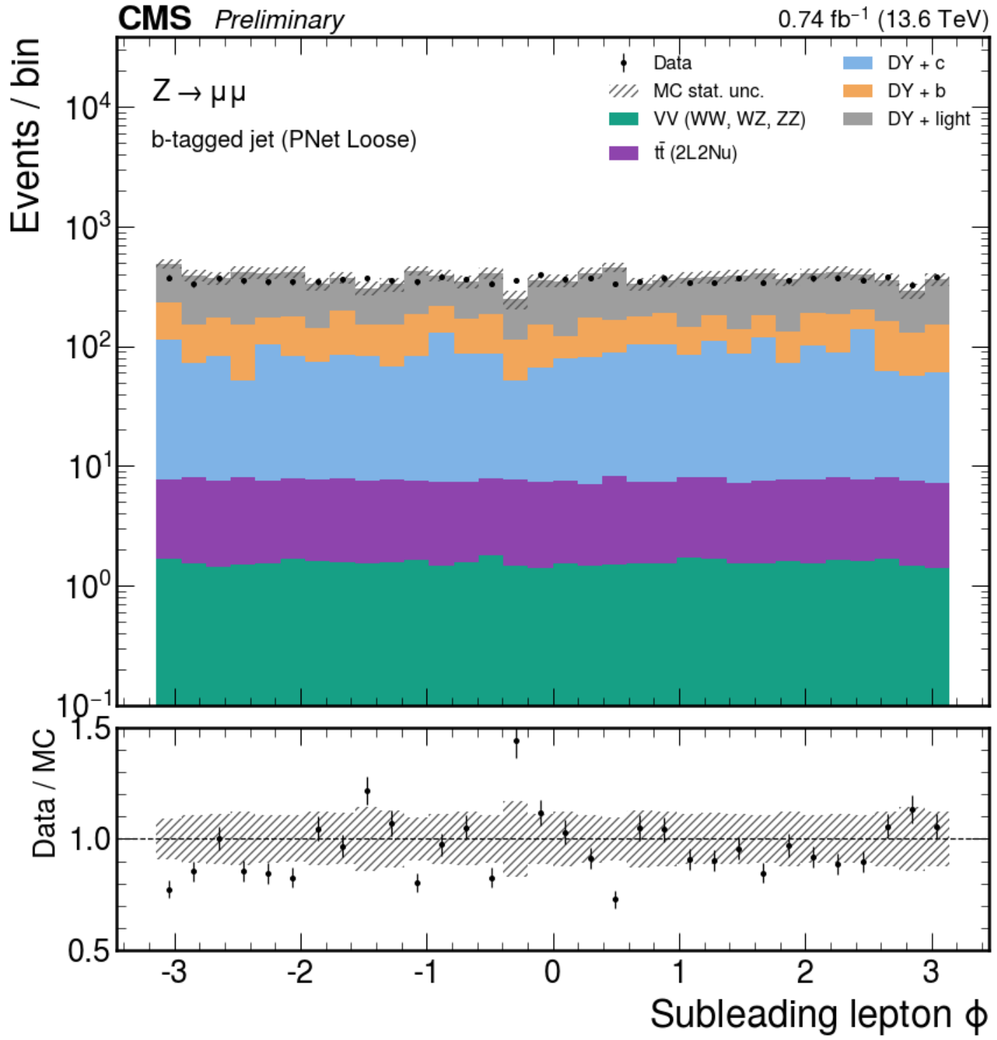
Second-lepton η



b-tag Loose

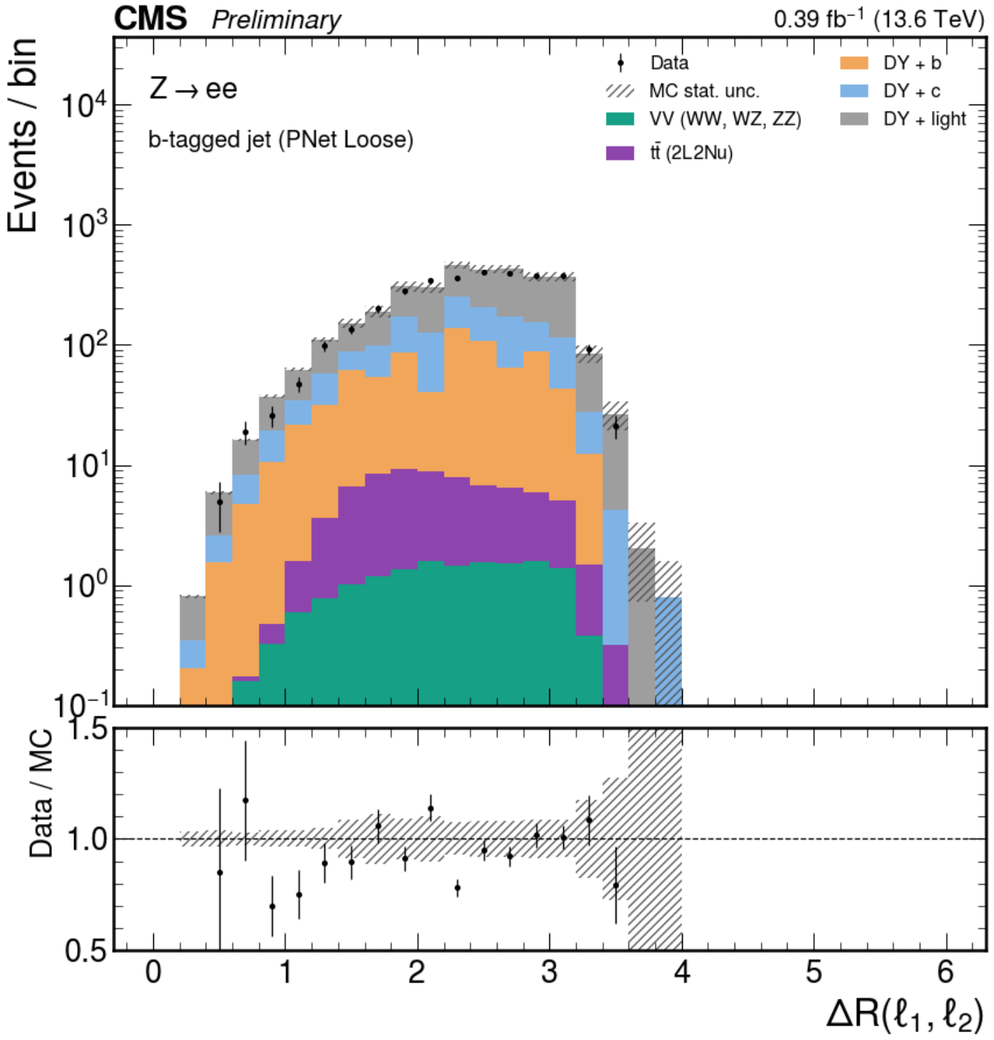
Second-lepton φ

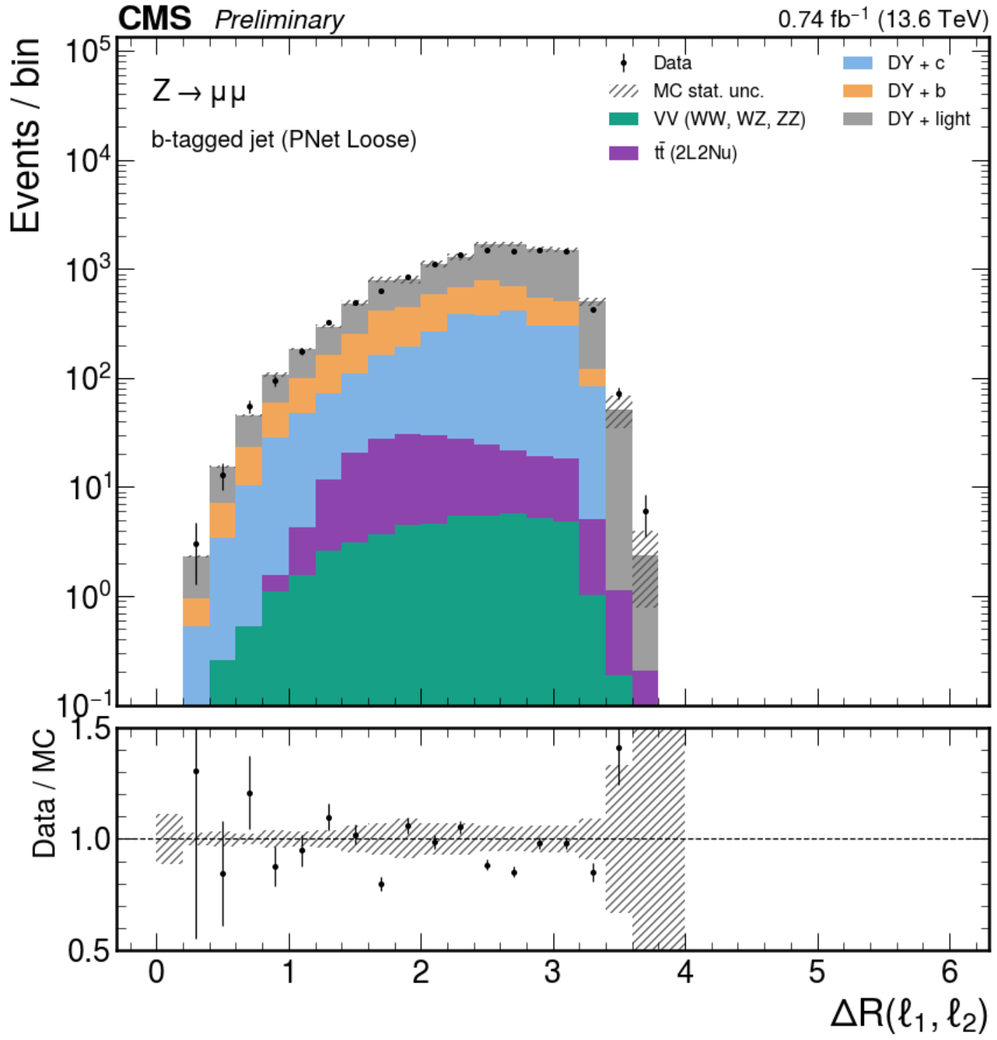




b-tag Loose

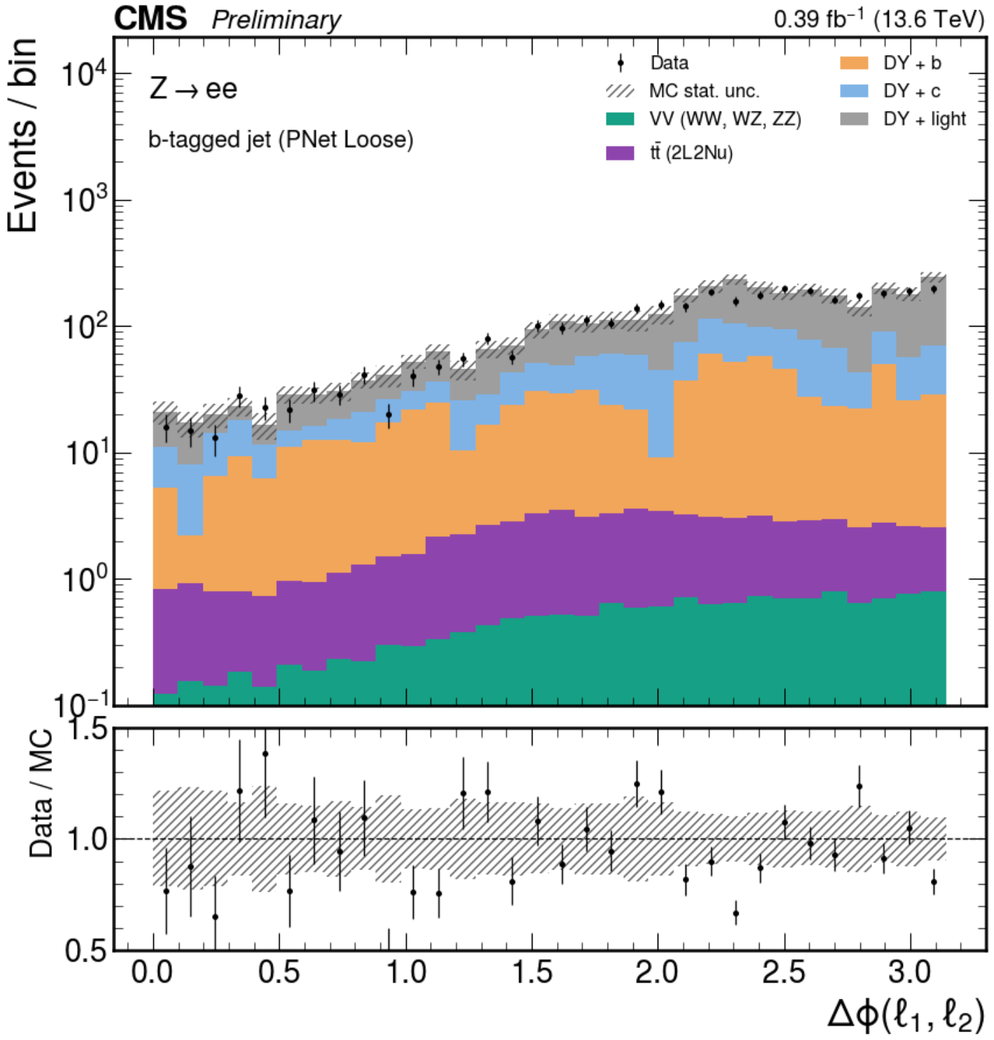
ΔR between the leptons

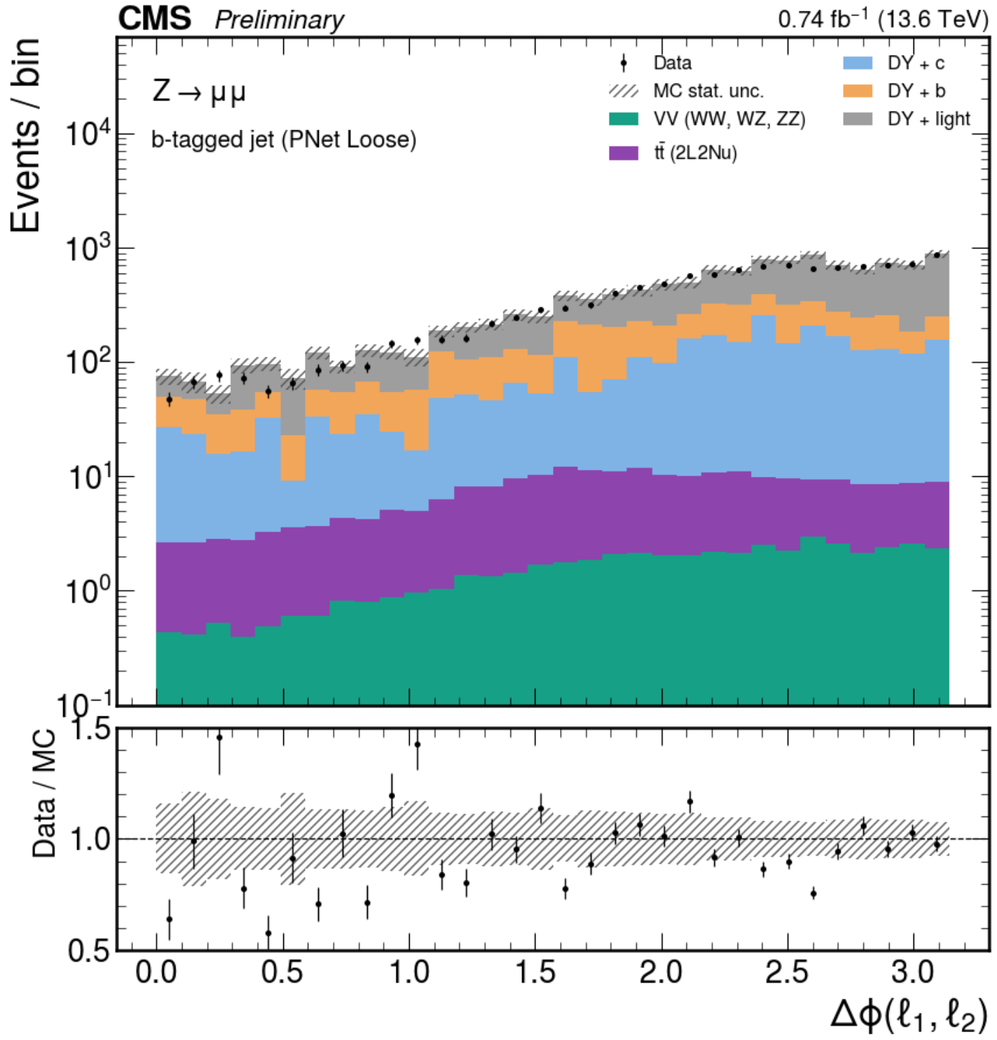




b-tag Loose

Δφ between the leptons

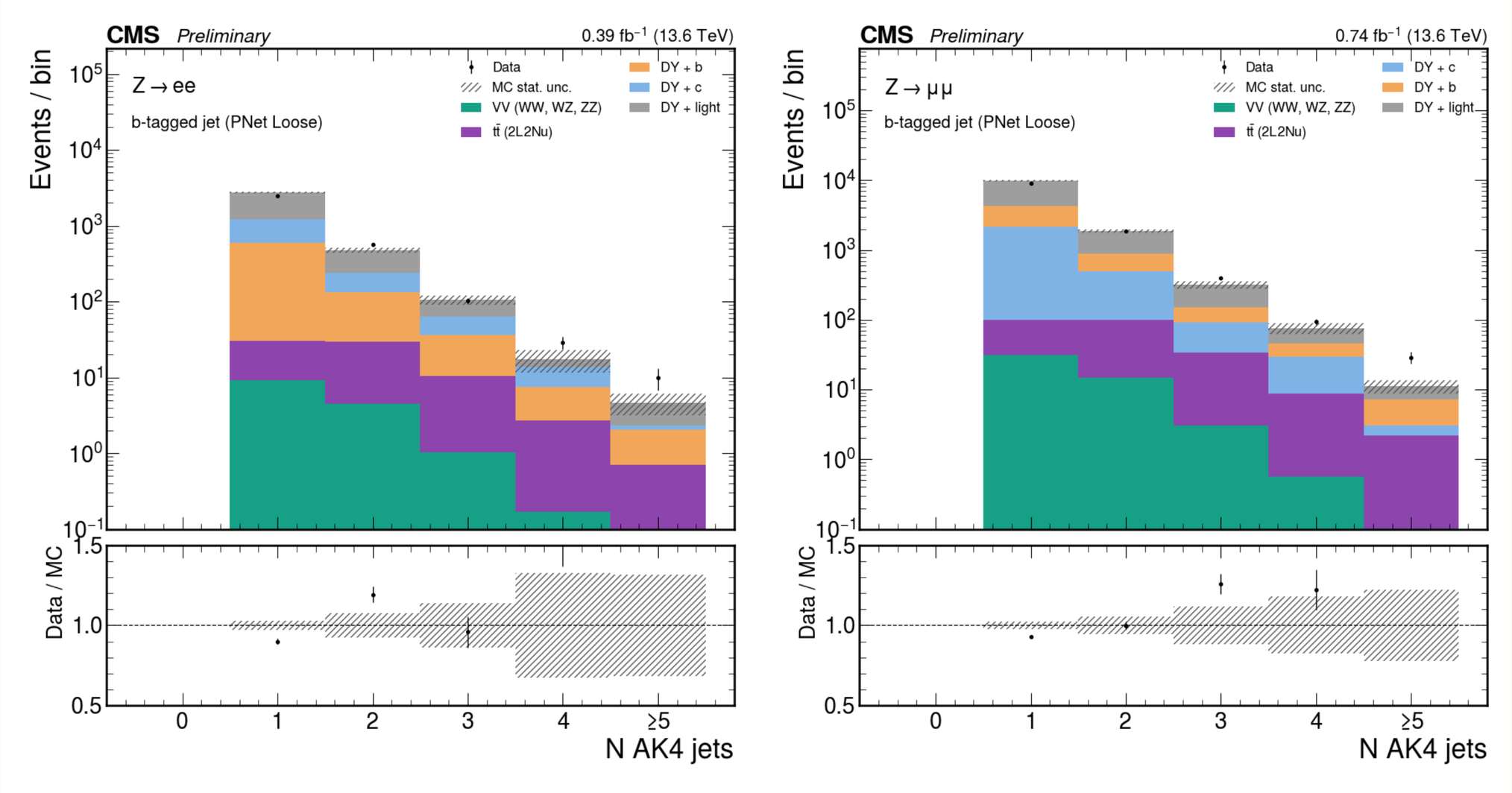




b-tag Loose

Number of jets

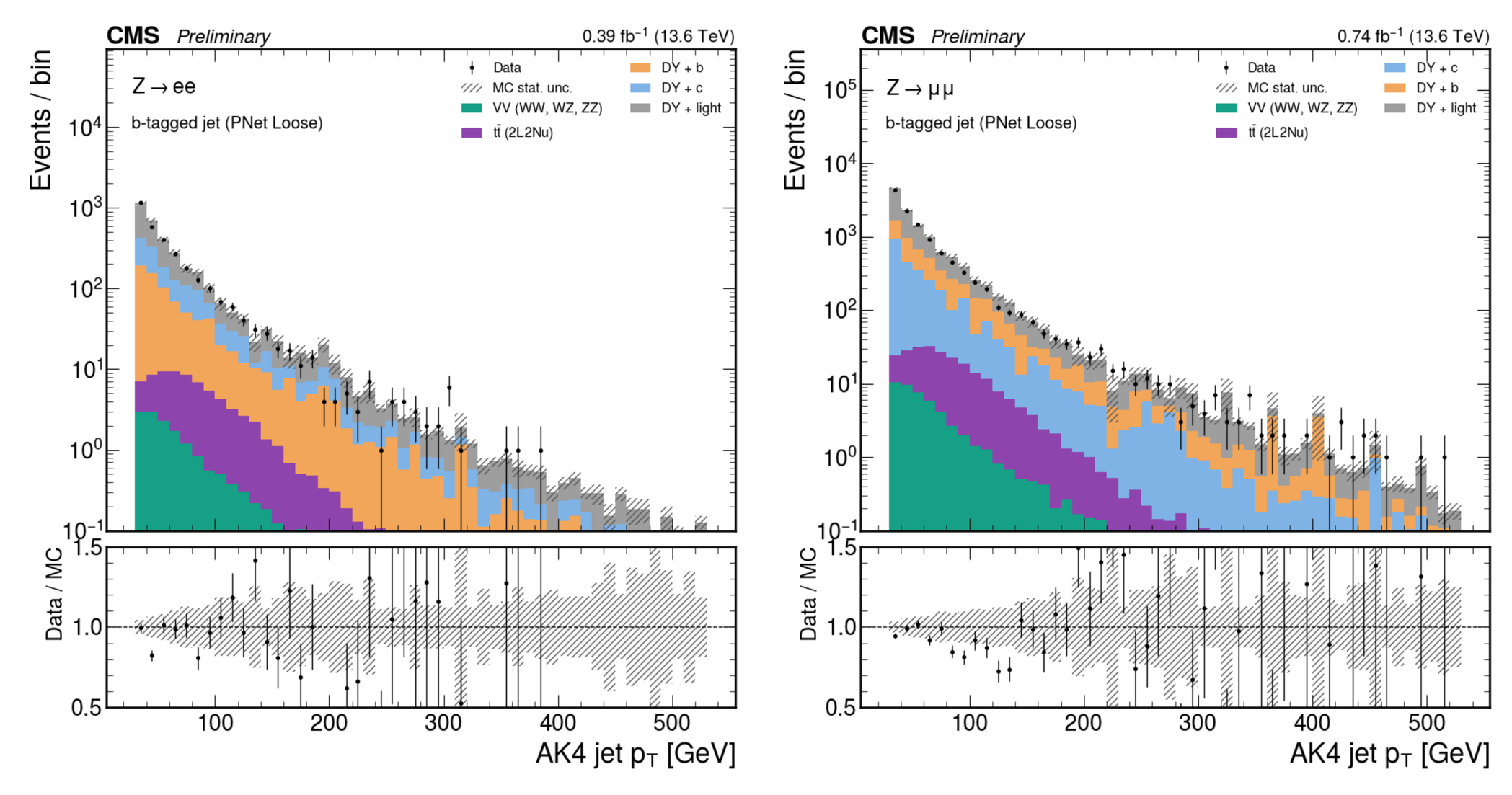
AK4 jets with pT > 30 GeV, |η| < 2.5, away from the leptons. Untagged plot: Z events before the jet cut, so the 0-jet bin is filled.



b-tag Loose

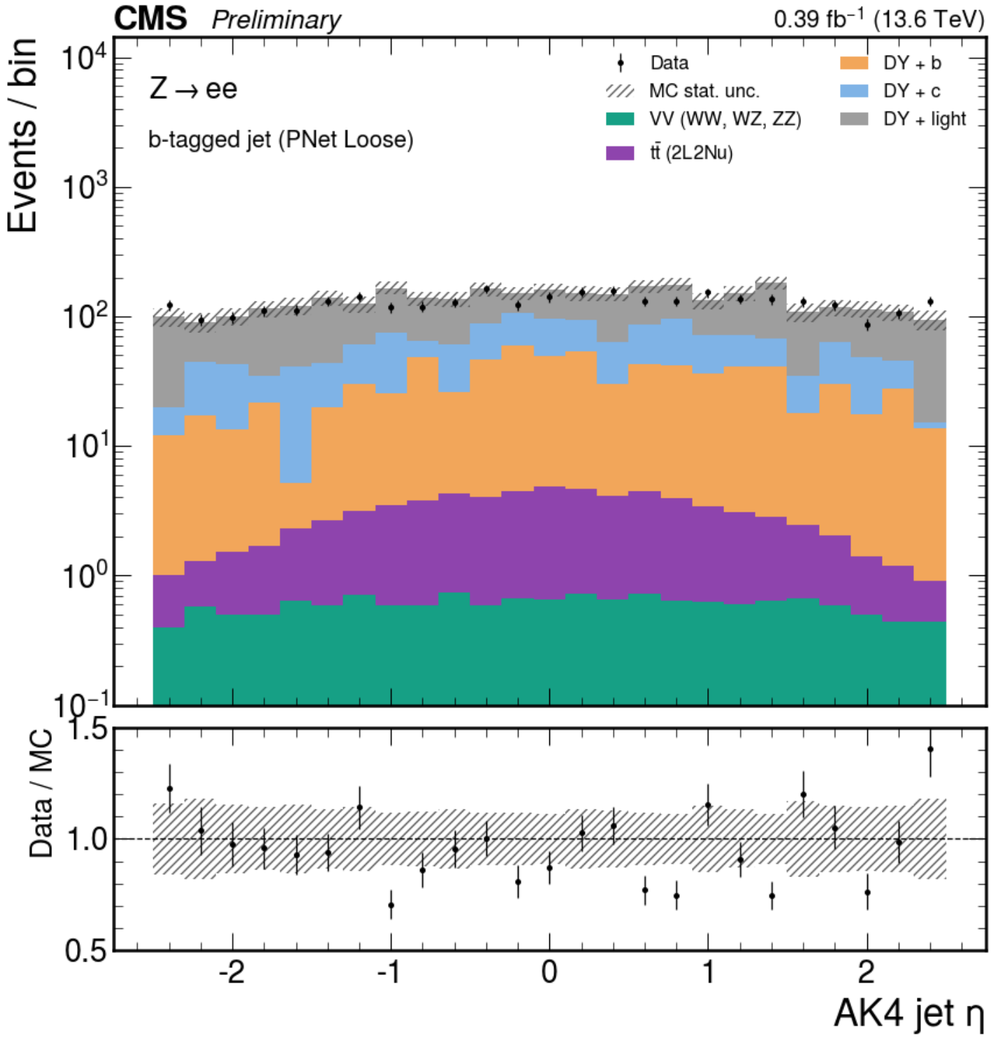
Leading-jet pT

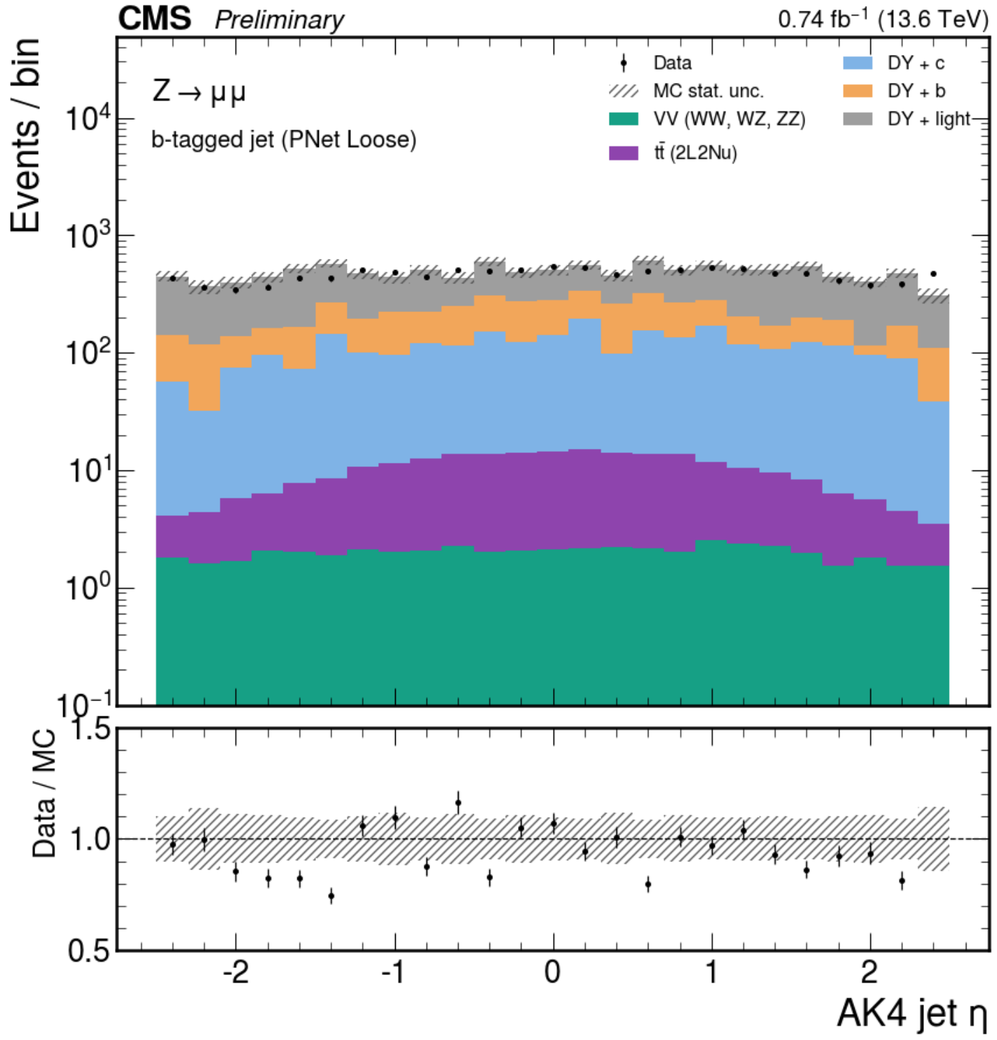
The tag (if any) is on this jet.



b-tag Loose

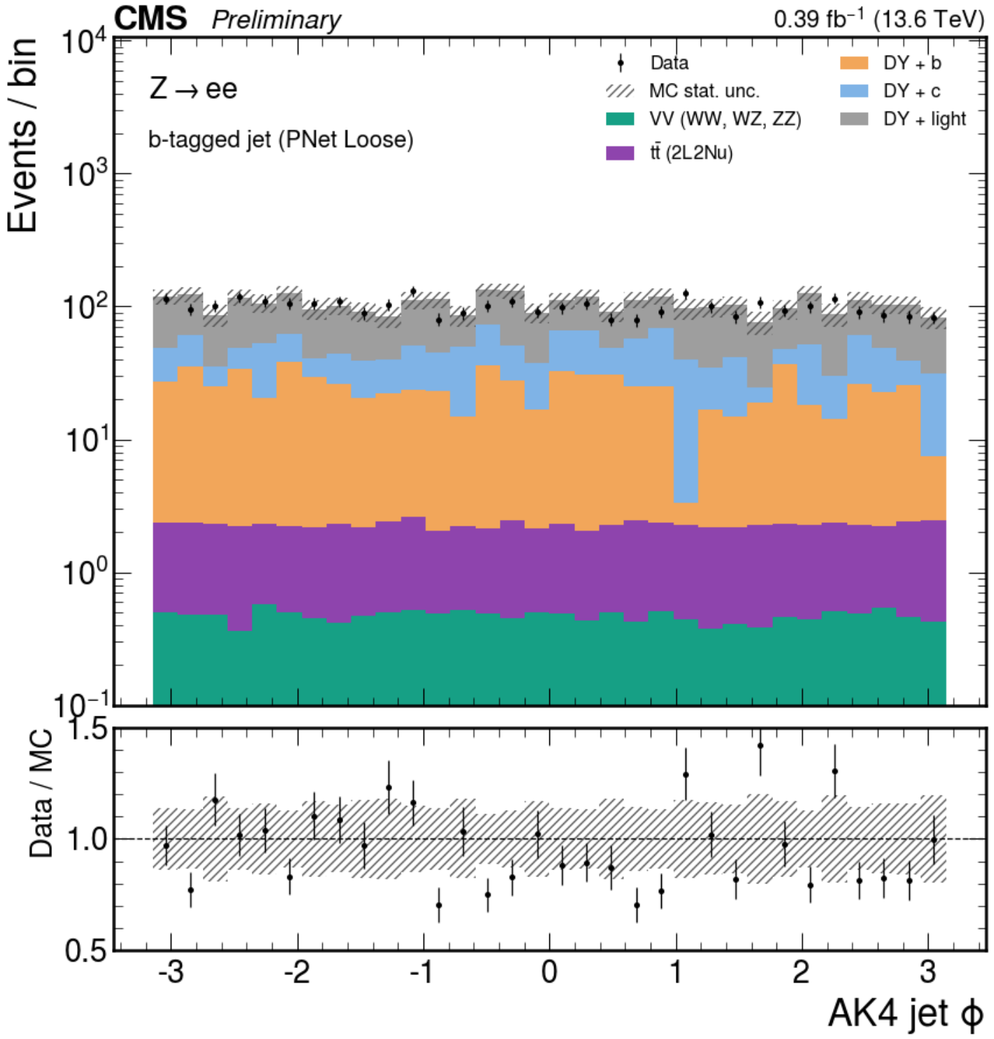
Leading-jet η

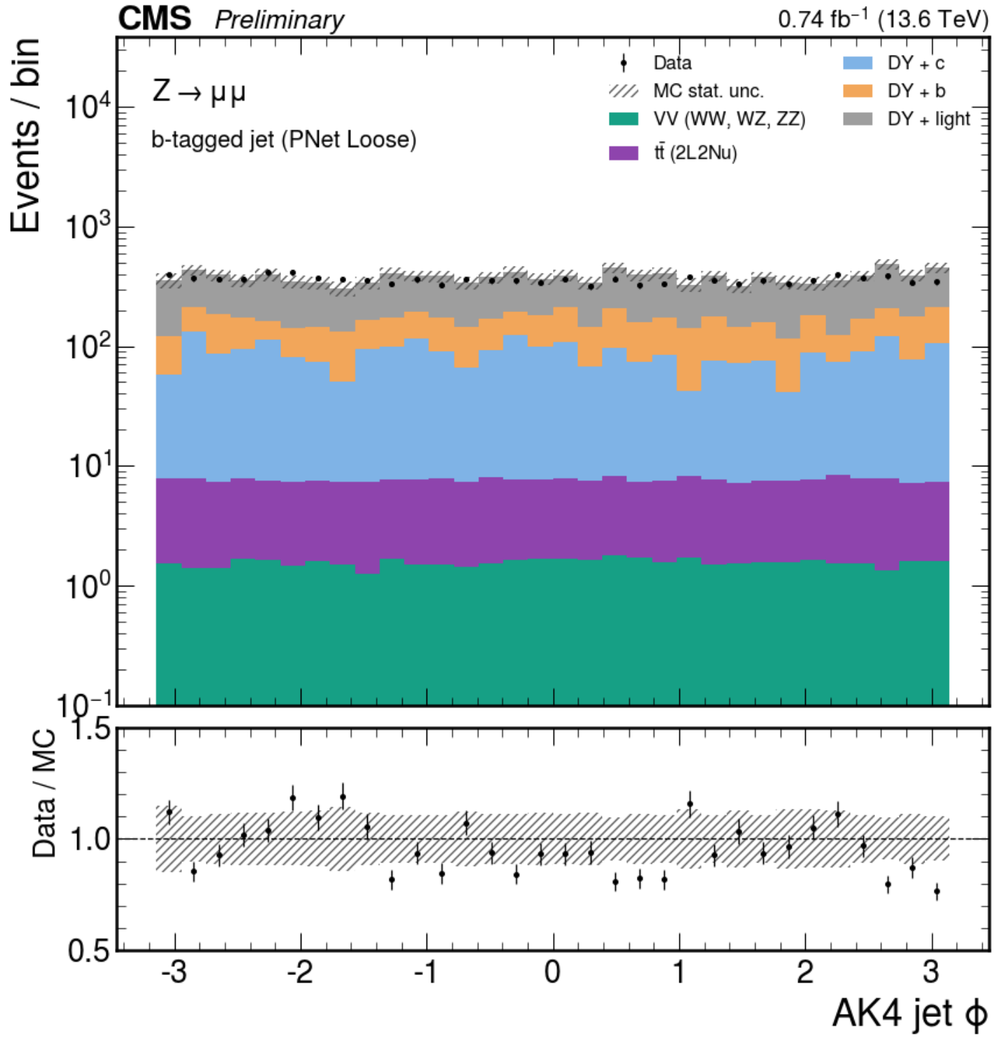




b-tag Loose

Leading-jet φ

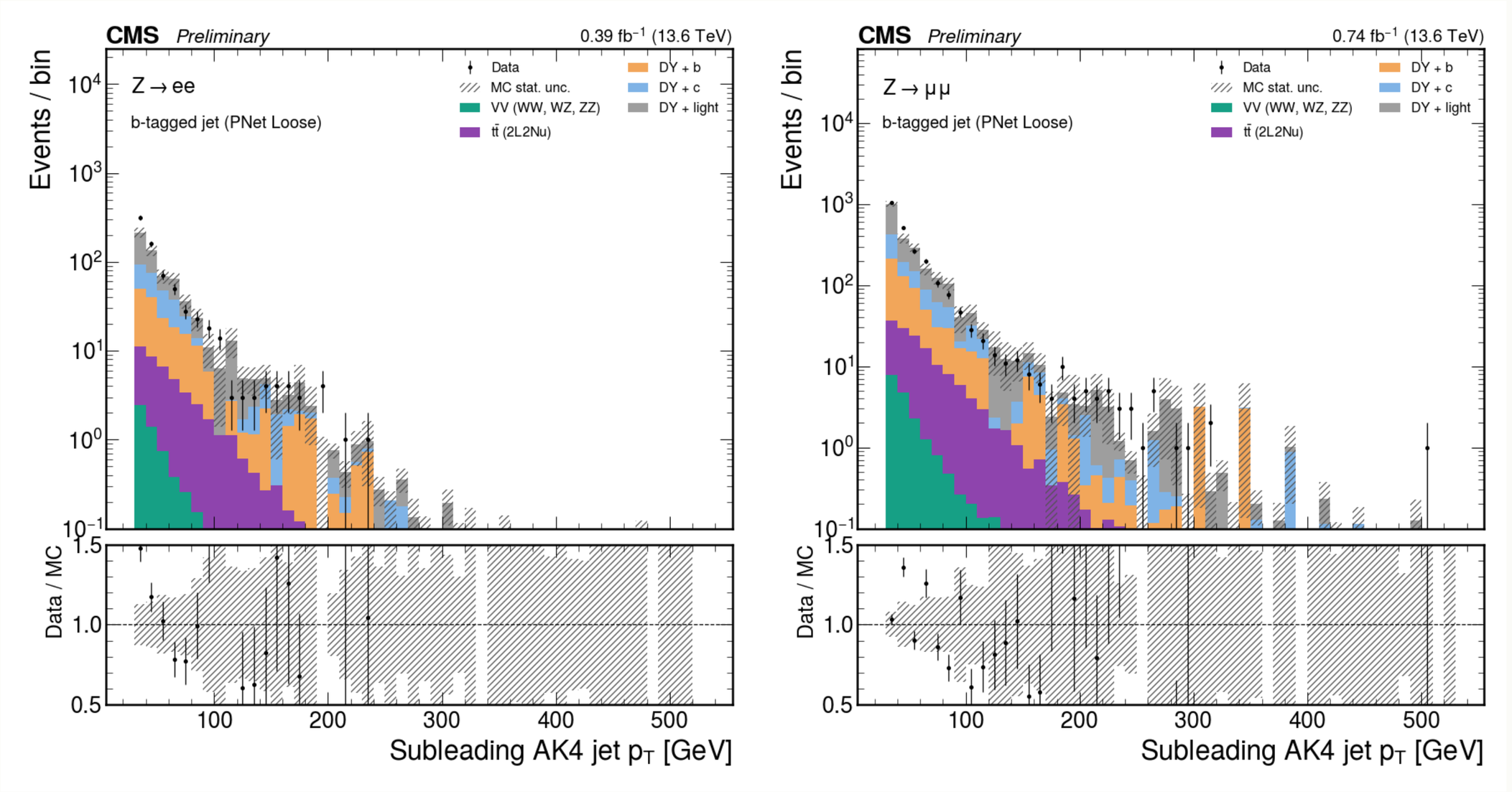




b-tag Loose

Second-jet pT

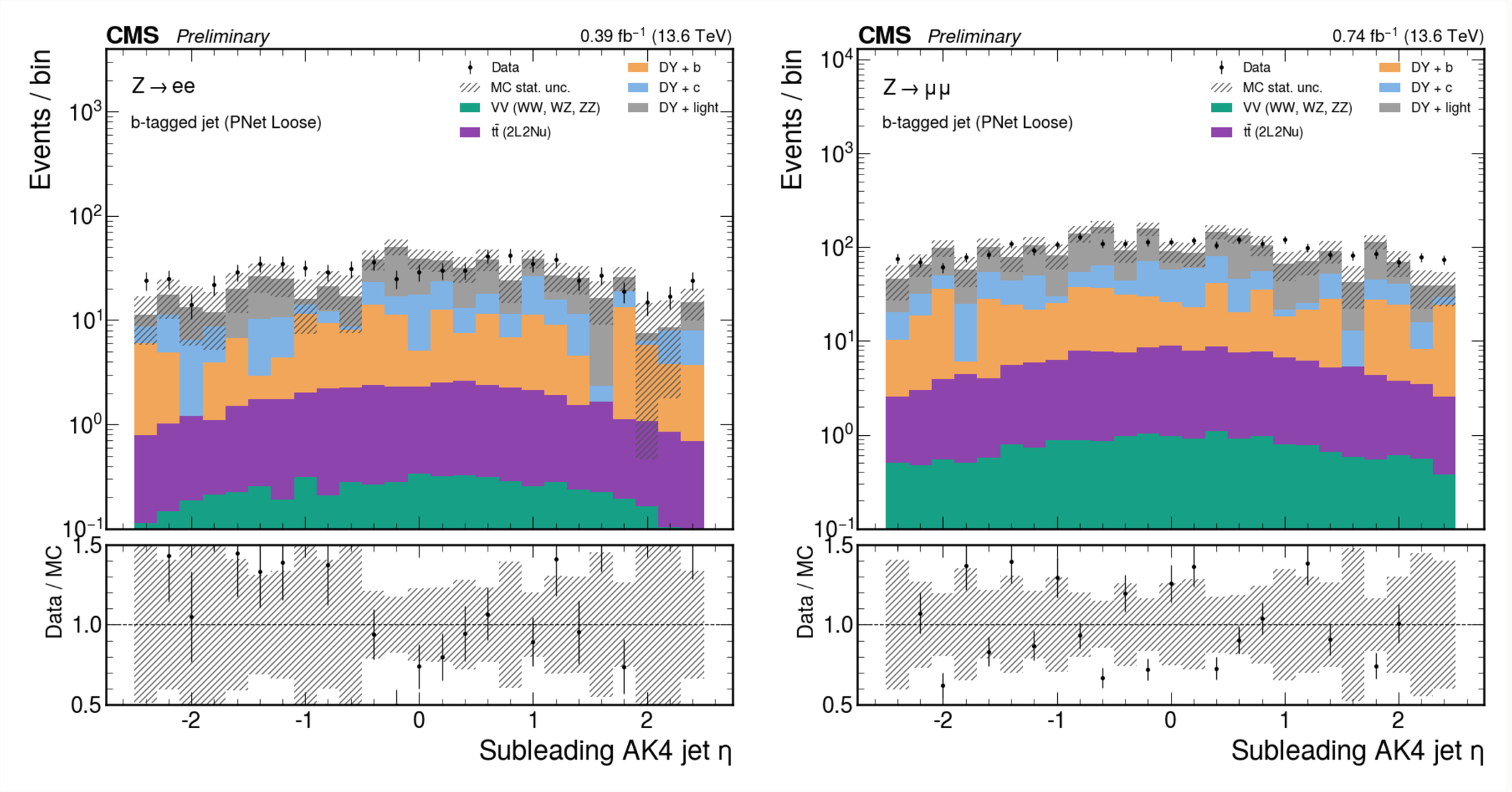
Events with at least 2 jets.



b-tag Loose

Second-jet η

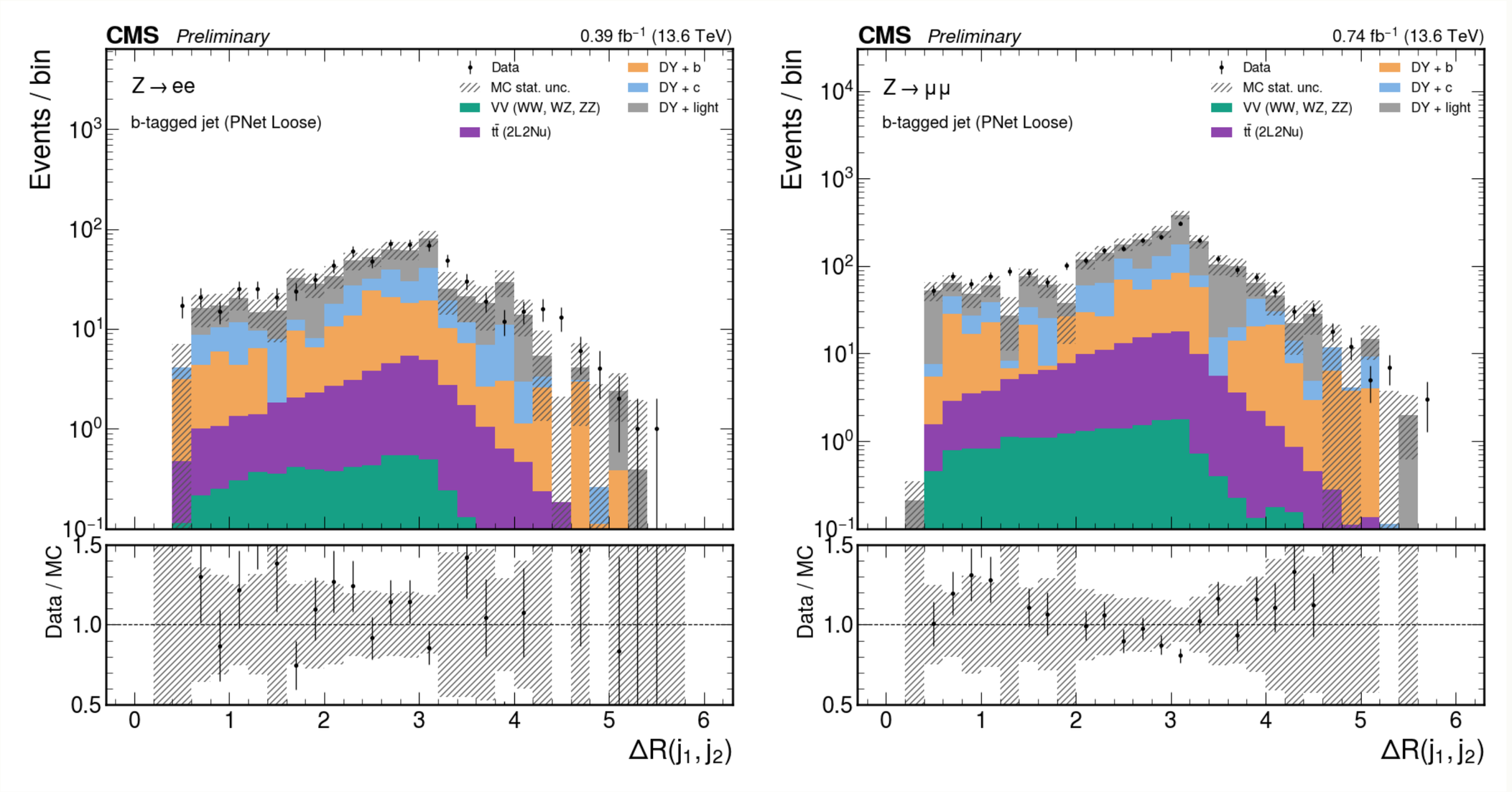
Events with at least 2 jets.



b-tag Loose

ΔR between the two leading jets

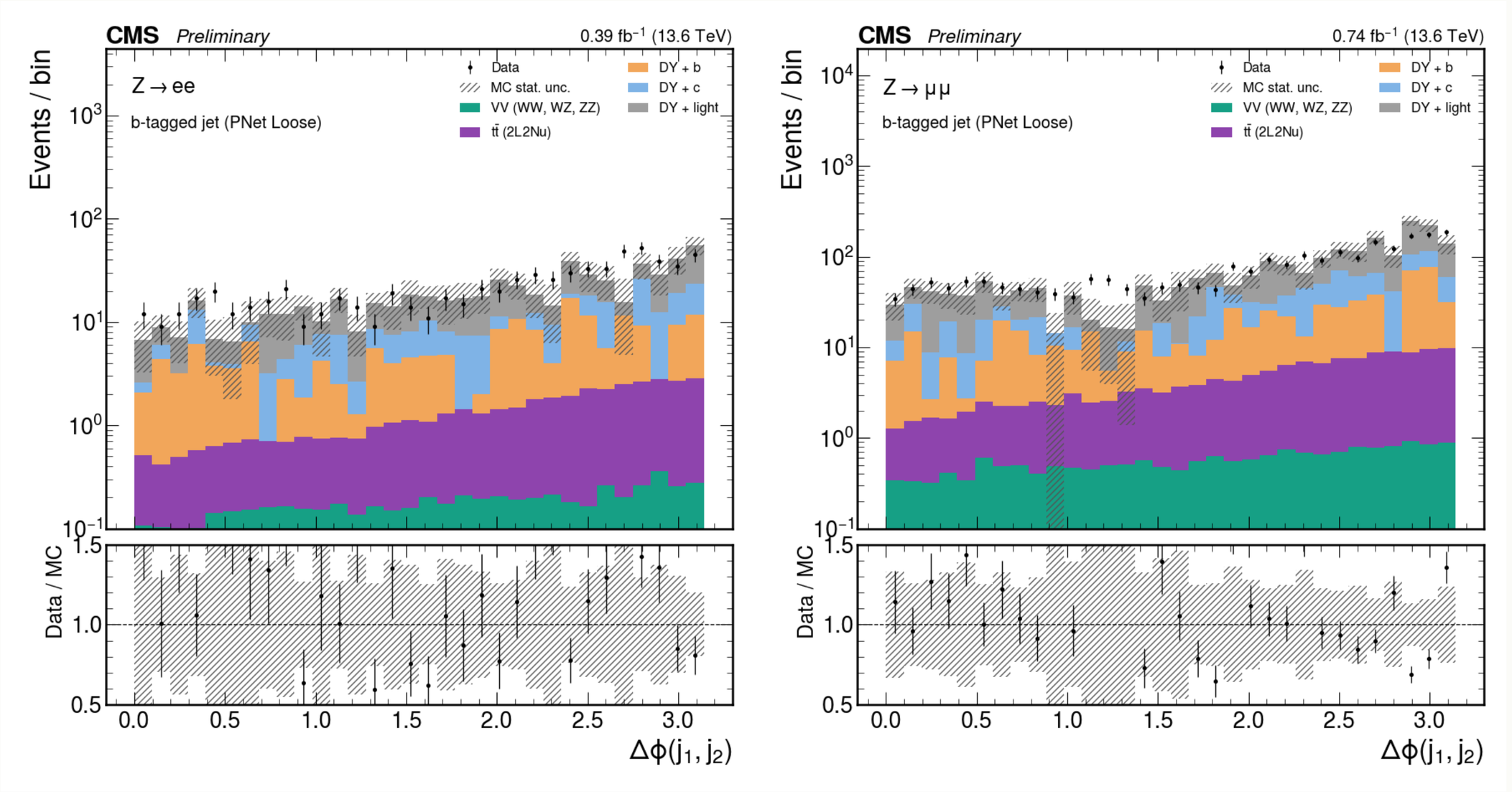
Events with at least 2 jets.



b-tag Loose

Δφ between the two leading jets

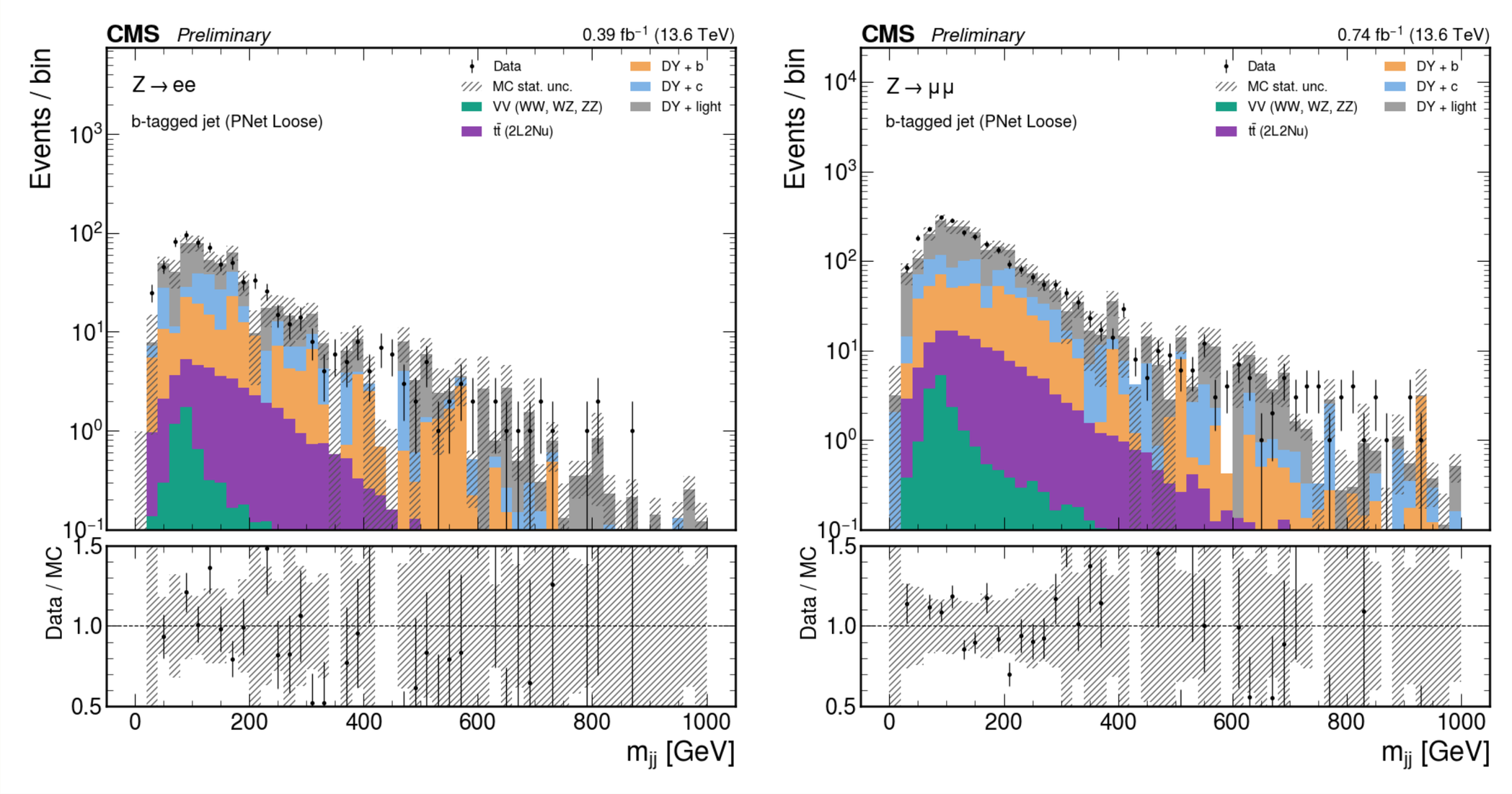
Events with at least 2 jets.



b-tag Loose

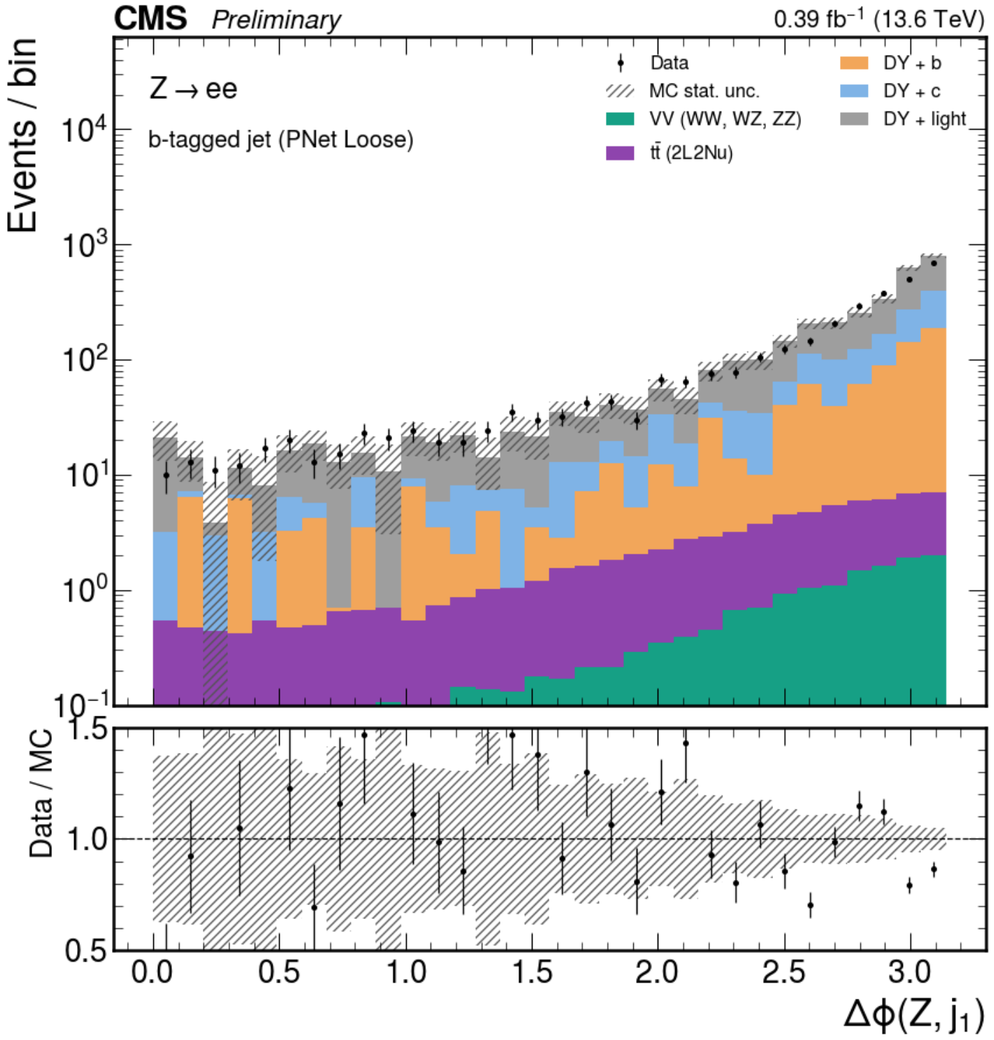
Dijet mass

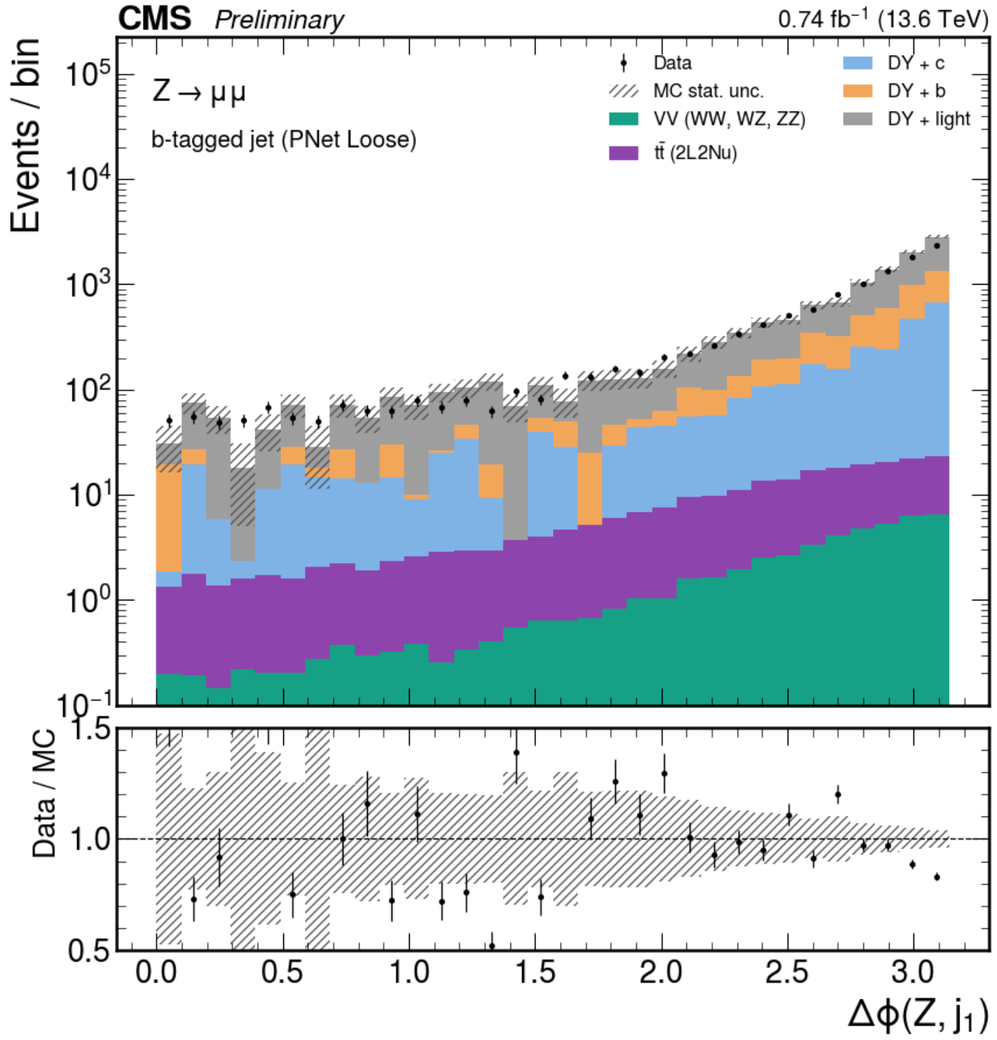
Events with at least 2 jets.



b-tag Loose

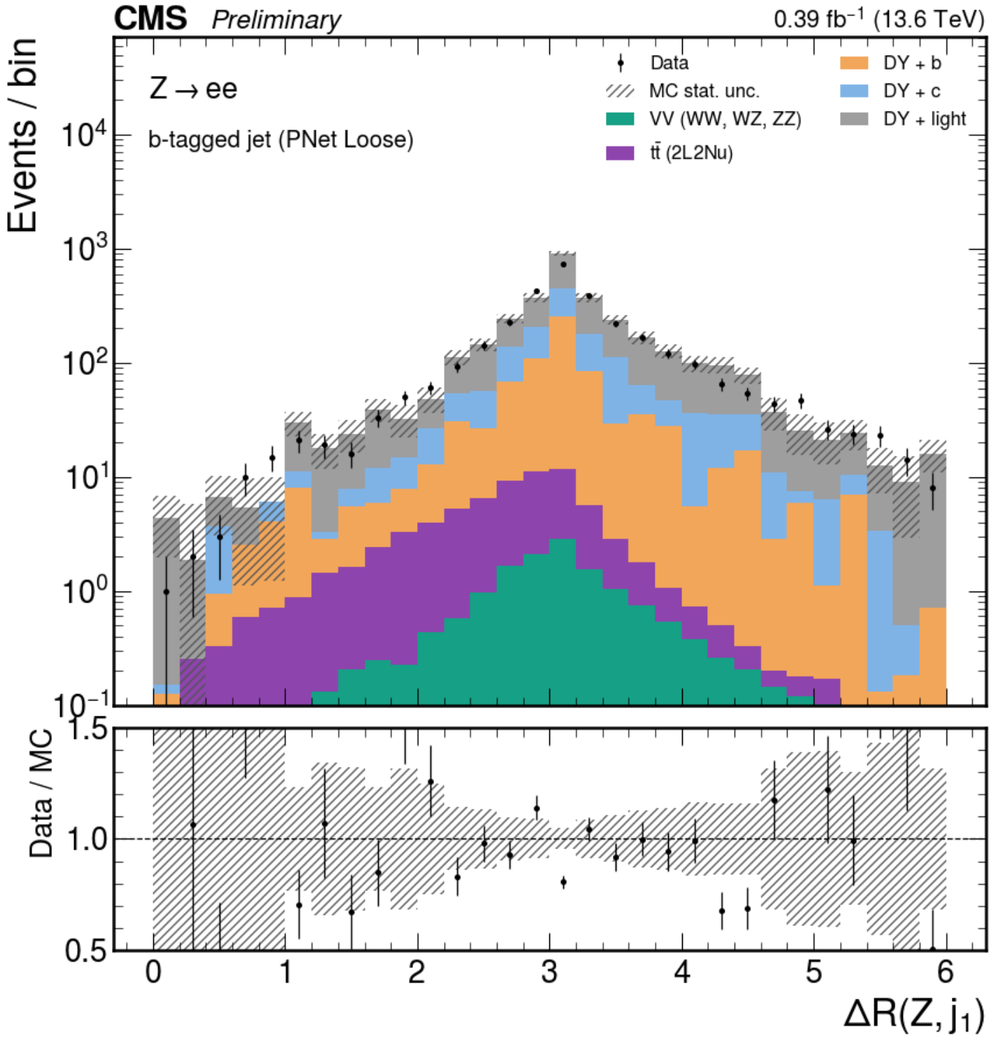
Δφ(Z, leading jet)

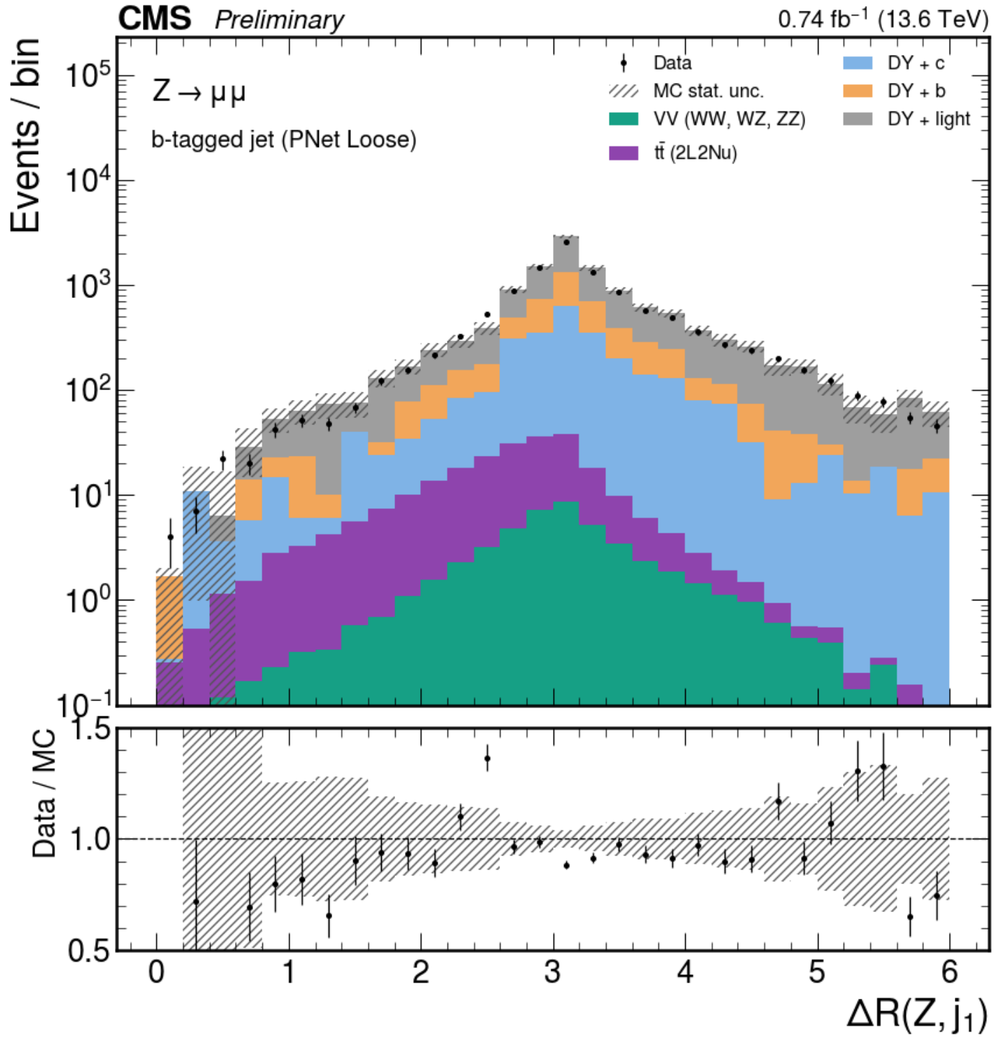




b-tag Loose

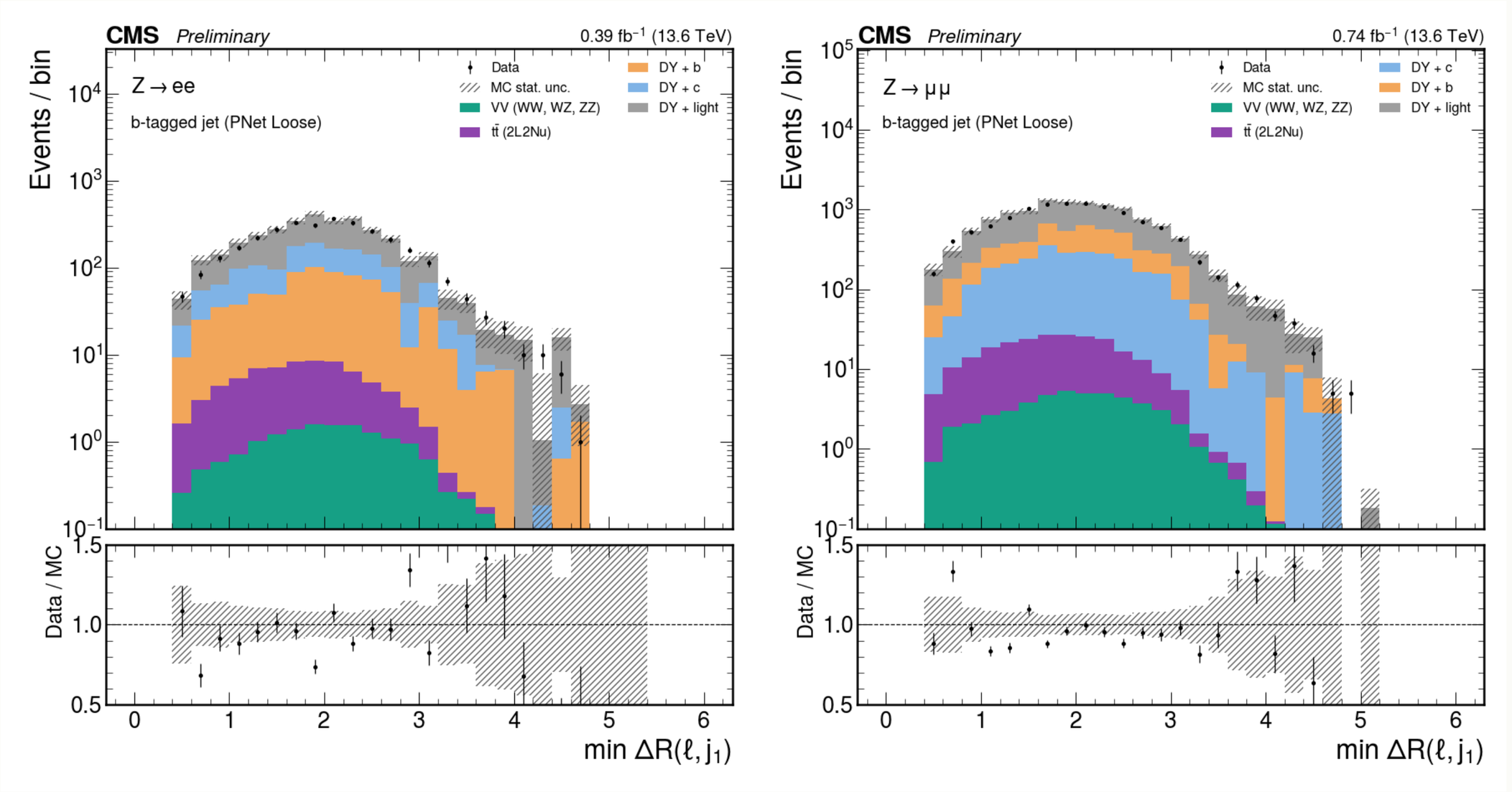
ΔR(Z, leading jet)





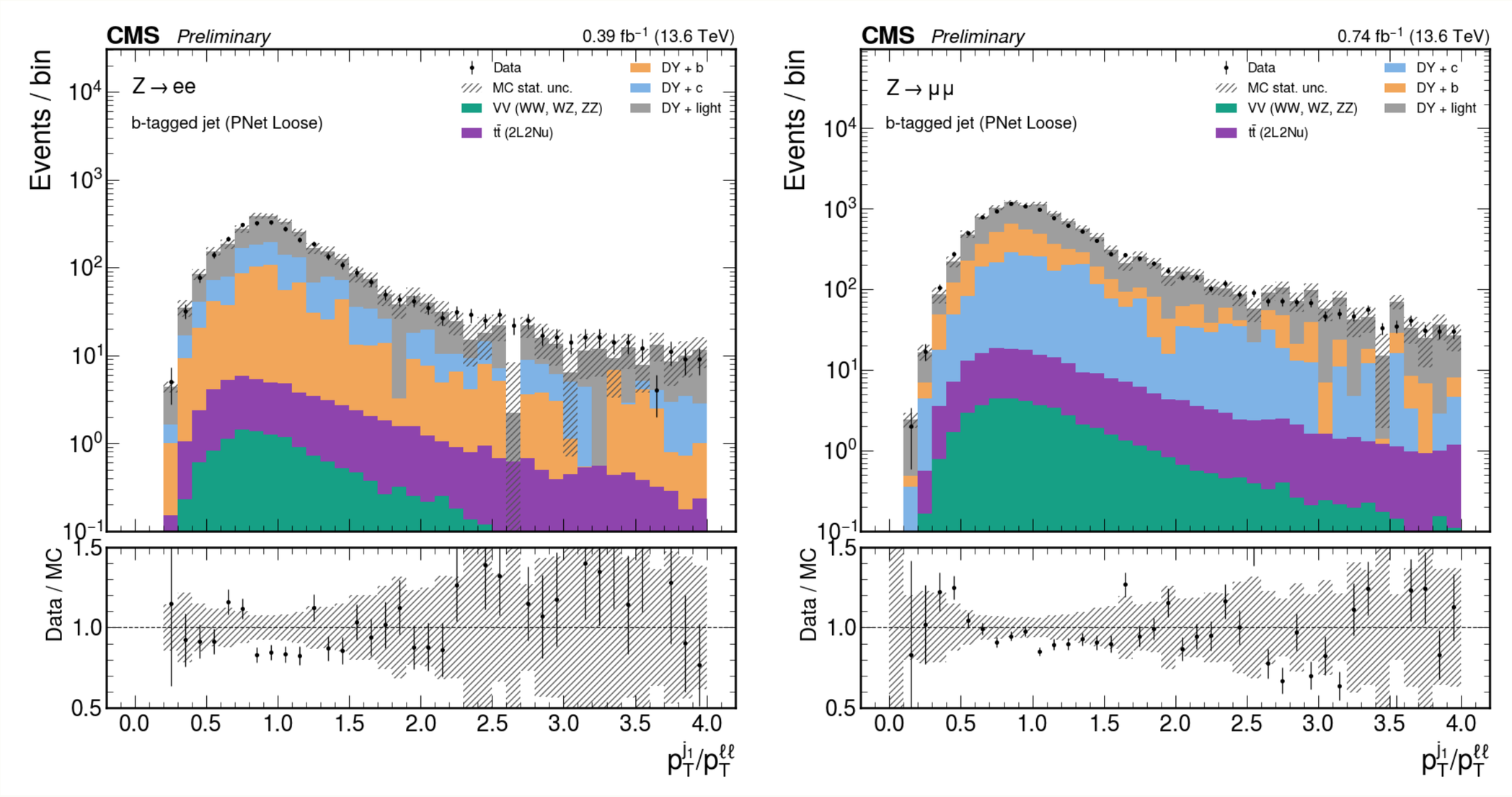
b-tag Loose

Smallest ΔR(lepton, leading jet)



b-tag Loose

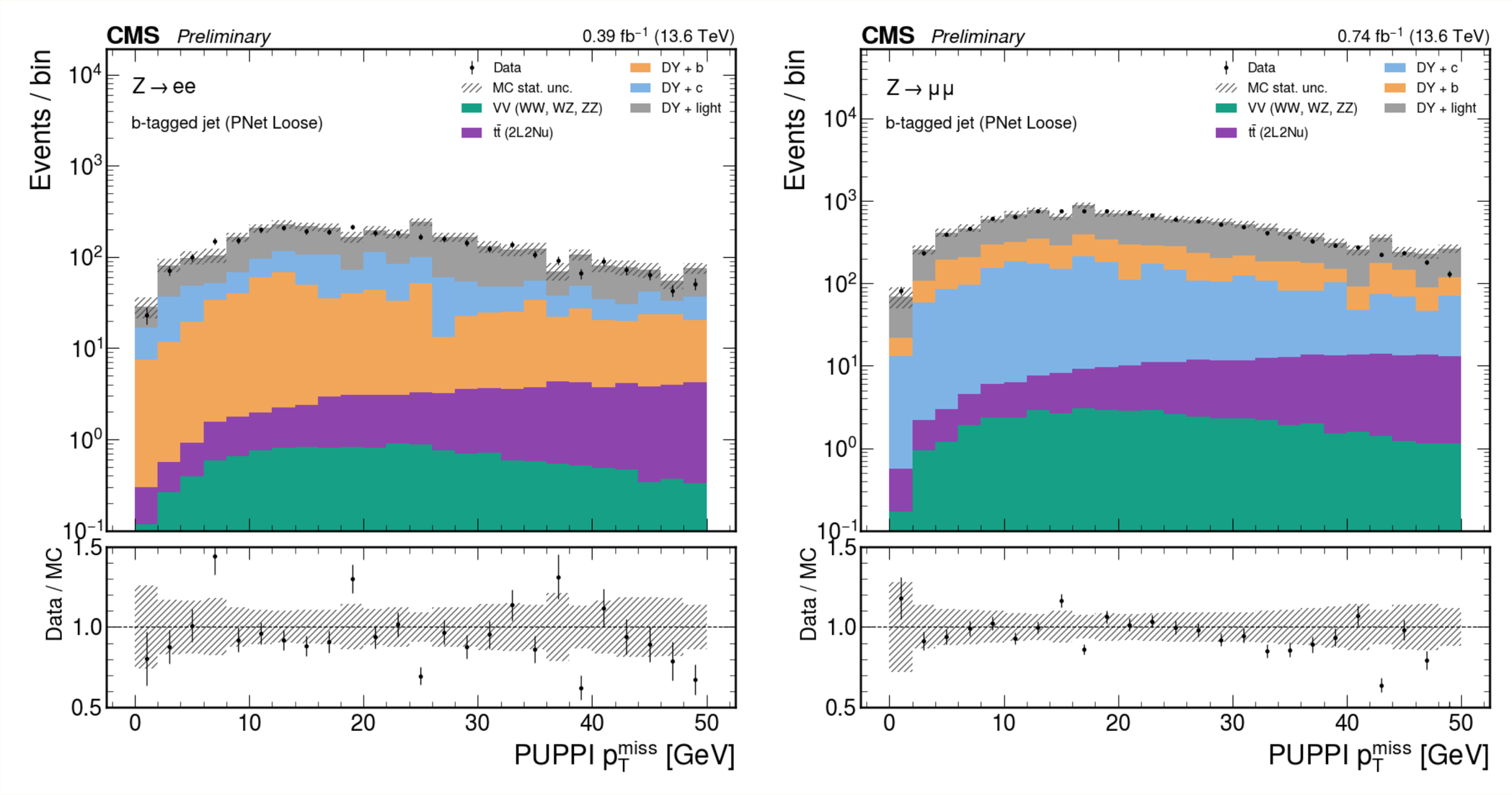
pT balance: leading-jet pT / Z pT



b-tag Loose

PUPPI MET

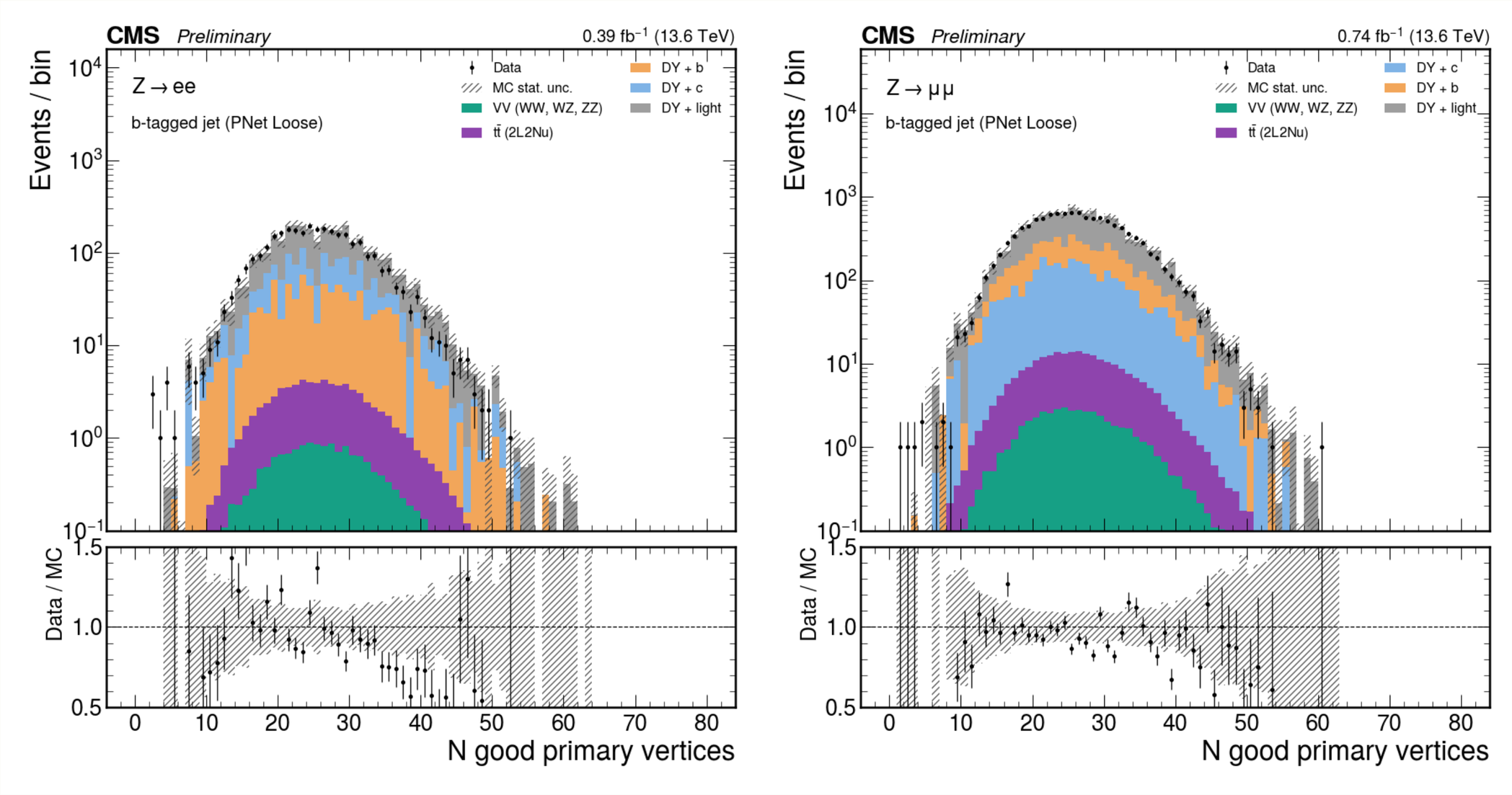
Type-1 corrected; the selection cuts at 50 GeV.



b-tag Loose

Number of good primary vertices

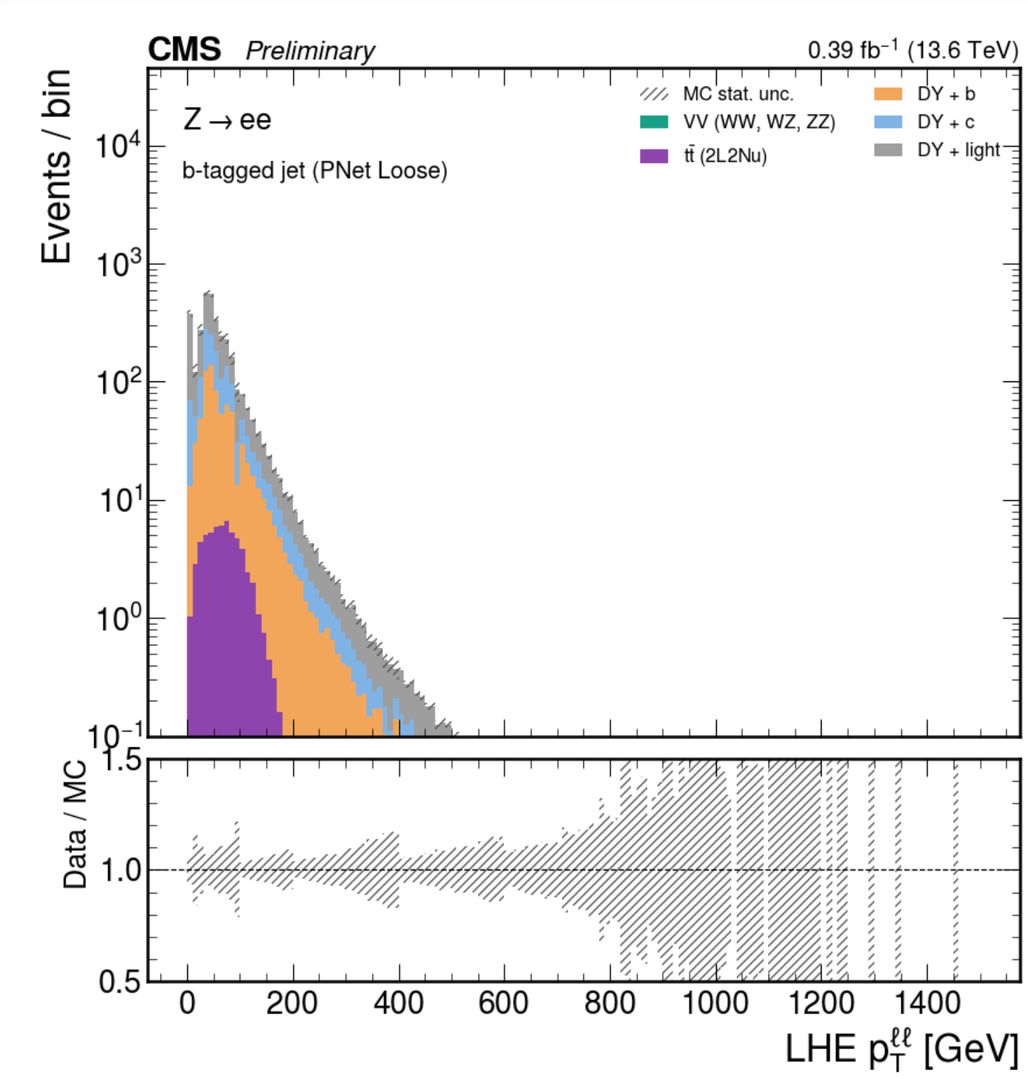
After pileup weights.

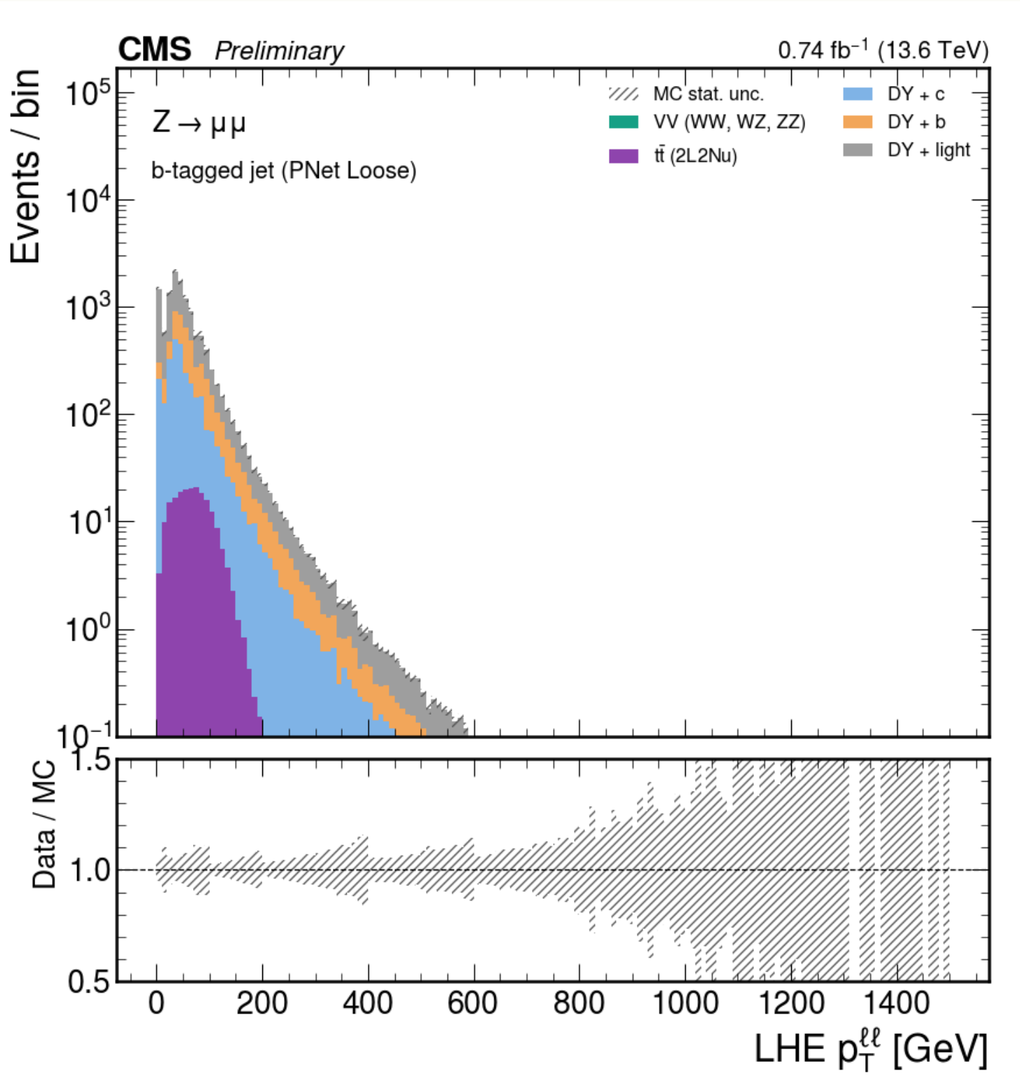


b-tag Loose

Generator Z pT (LHE)

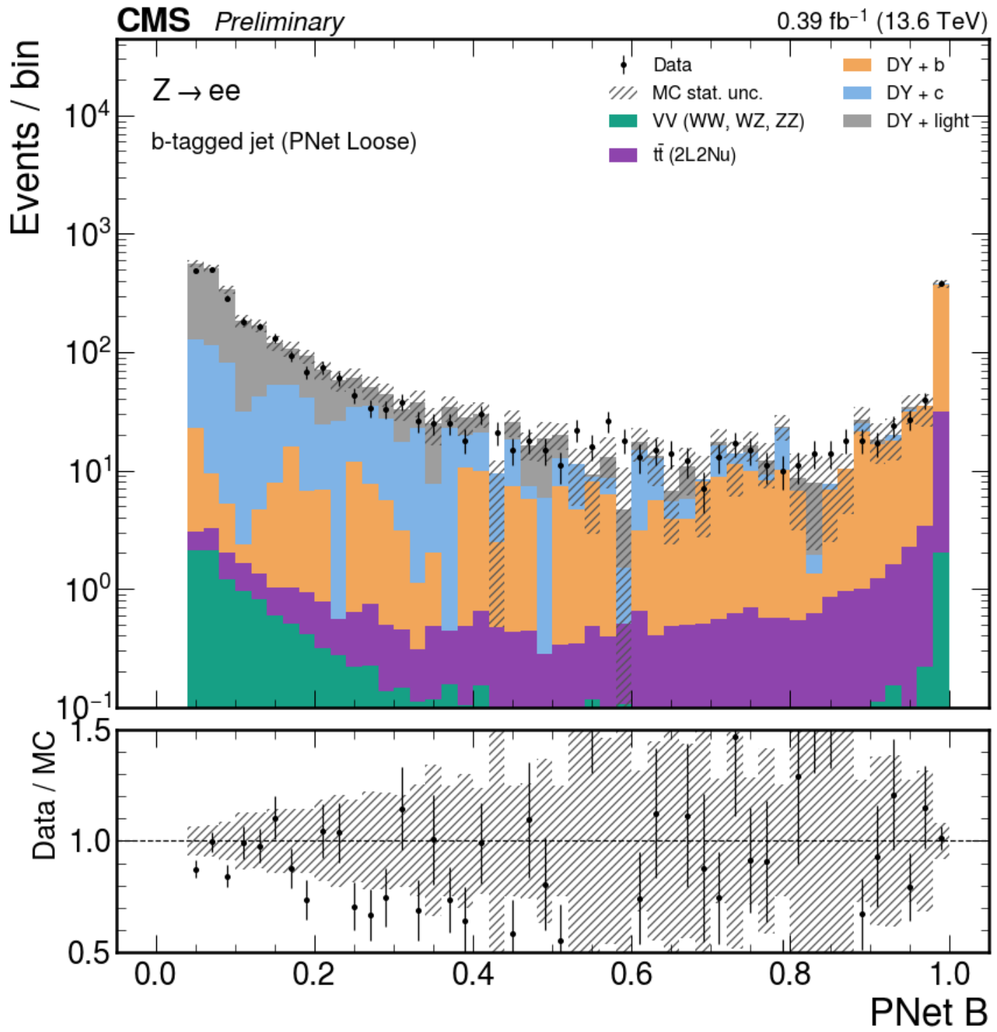
MC only, no data.

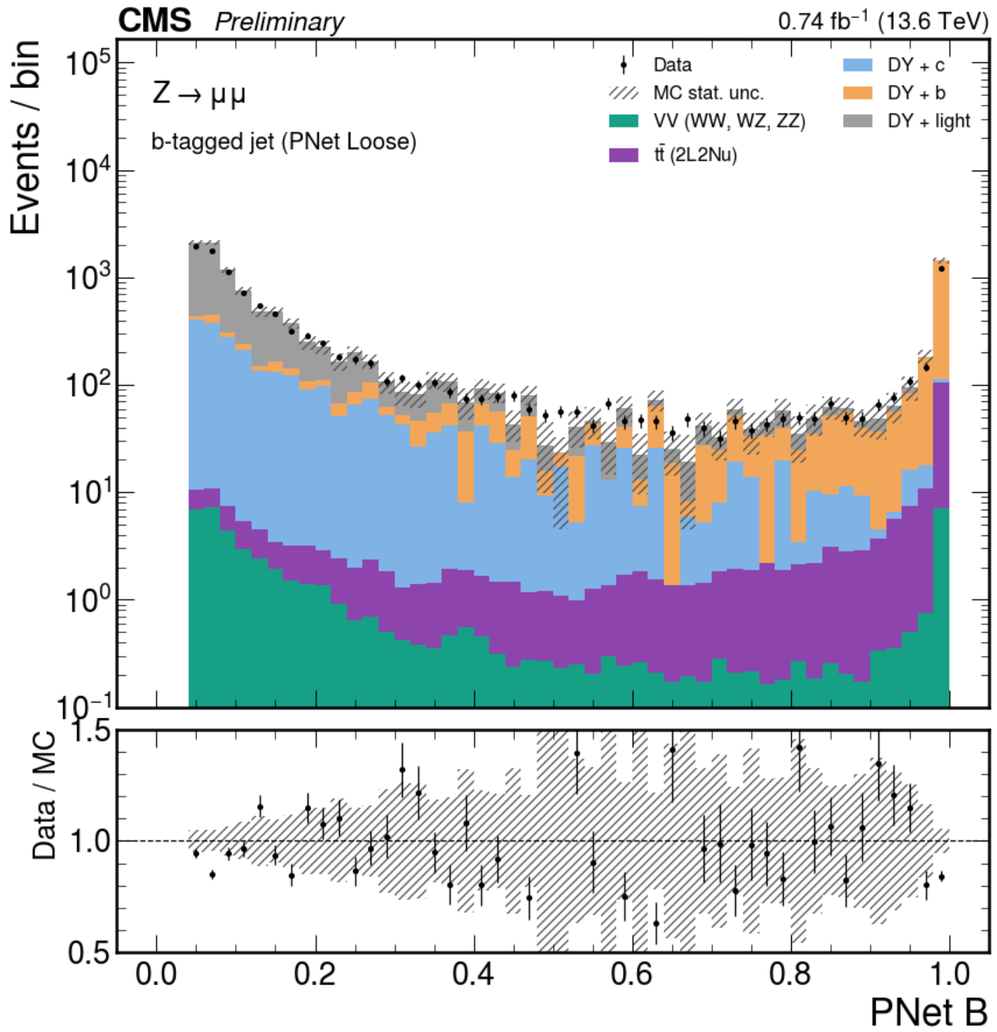




b-tag Loose

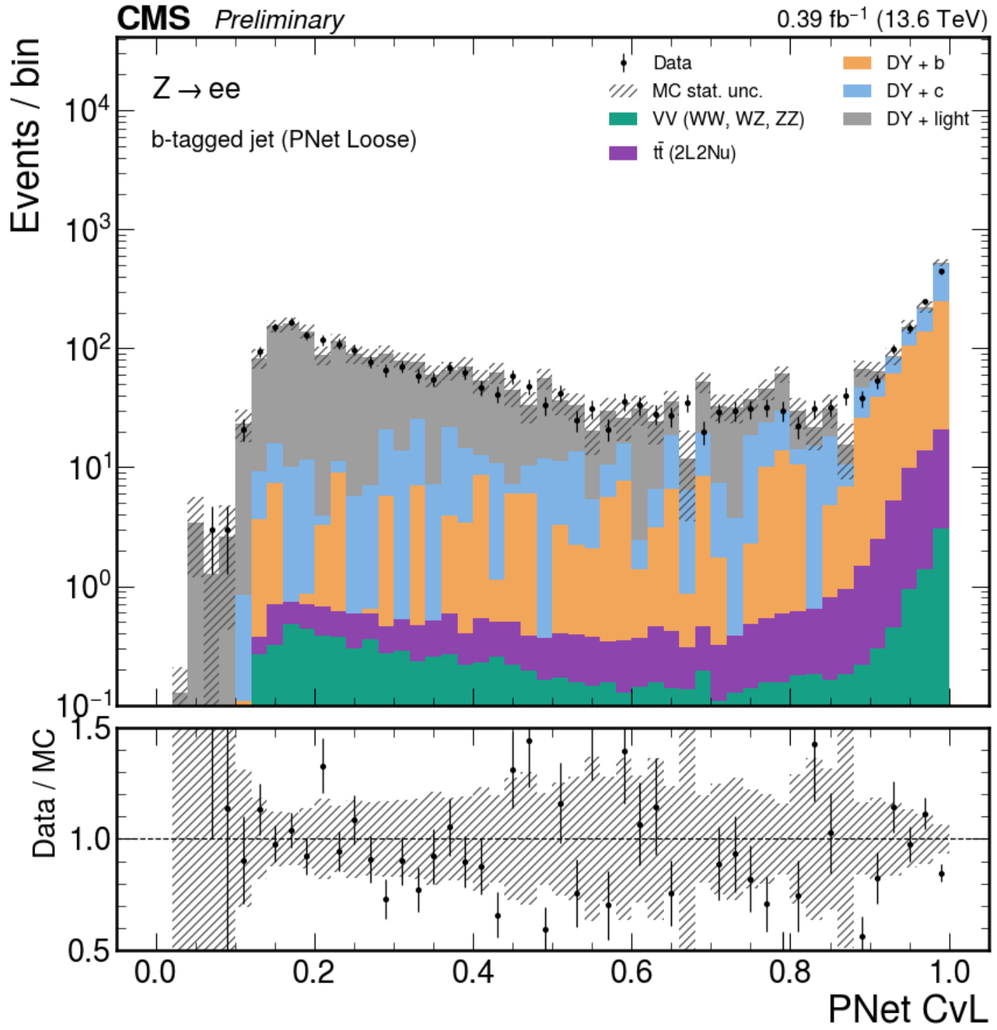
Leading-jet PNet B score

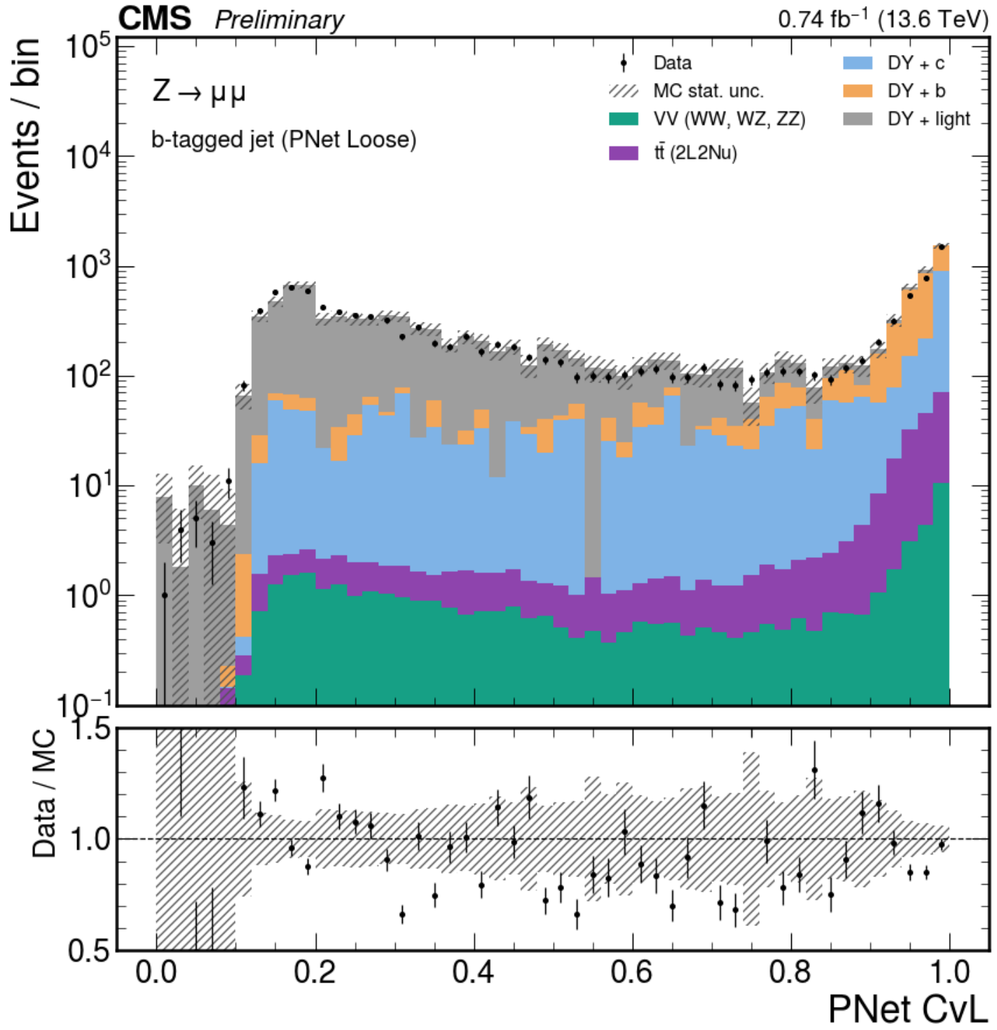




b-tag Loose

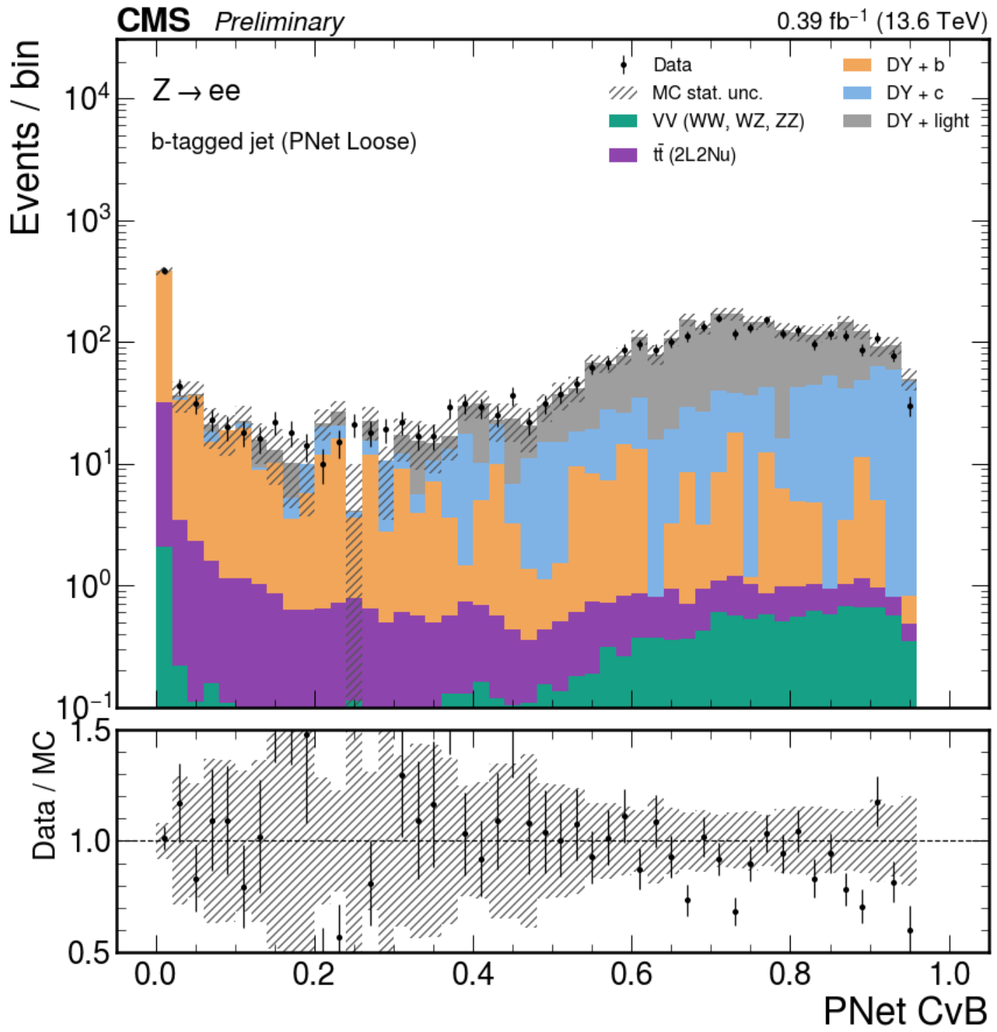
Leading-jet PNet CvL score

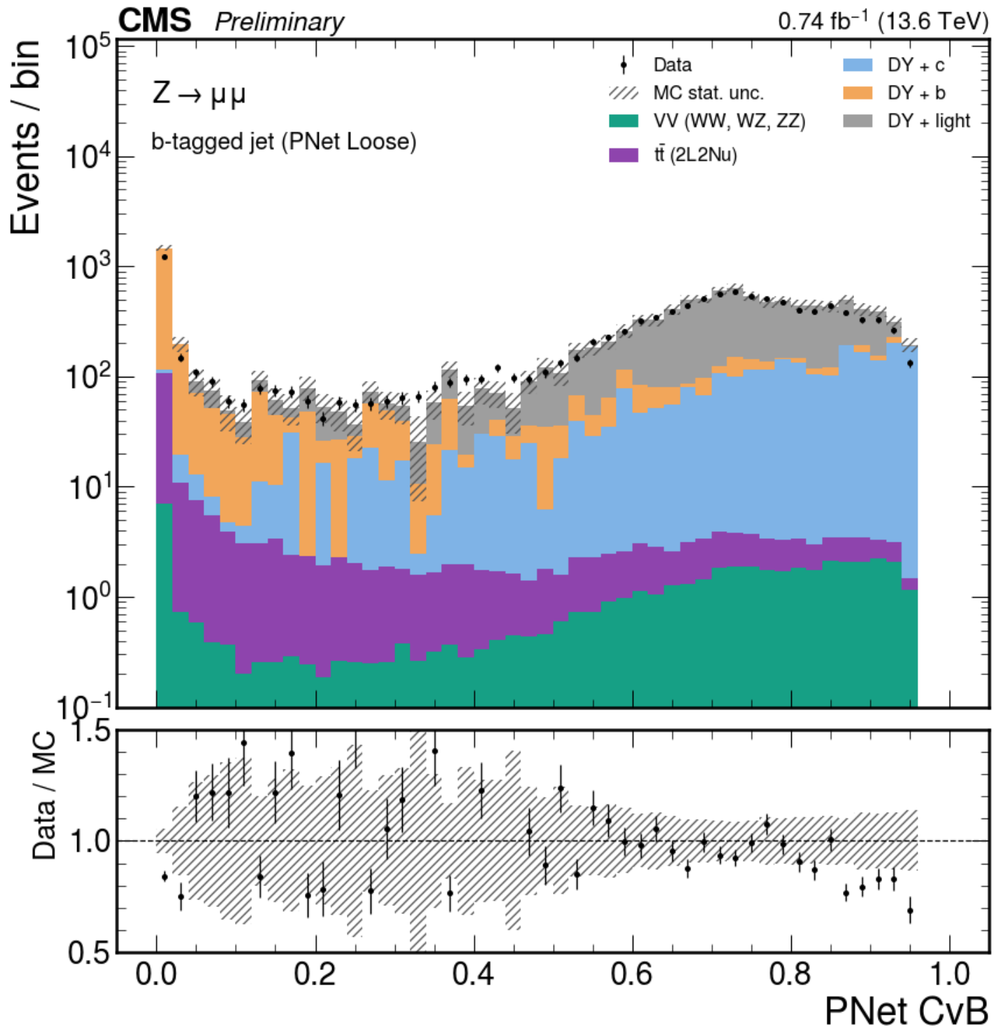




b-tag Loose

Leading-jet PNet CvB score

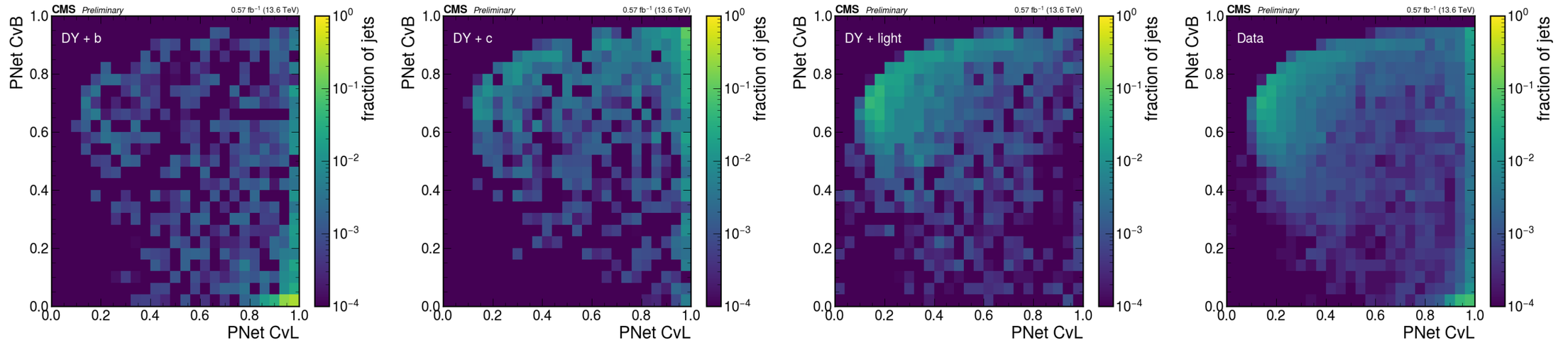




b-tag Loose

PNet CvL vs CvB of the leading jet

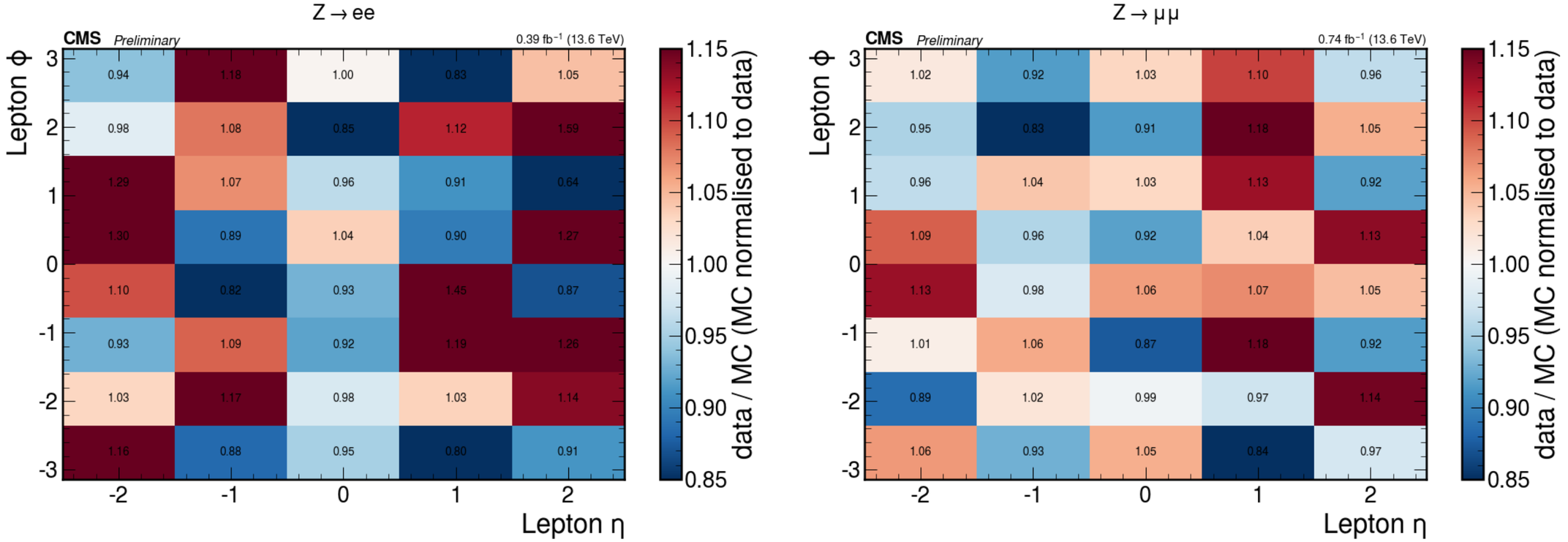
Each panel normalised to 1. Z→ee and Z→μμ added.



b-tag Loose

Lepton η–φ, data / MC

Both Z leptons. MC normalised to data in each map. Colour scale stops at ±15%; in the tagged regions there are few events per cell, so read the numbers.



c-tag Loose

Leading jet PNet CvL > 0.054 and CvB > 0.181

	data/MC	DY+b	DY+c	DY+light	tt	tagged MC	tagged data
Z→ee	0.932 ± 0.013	1.1%	8.7%	89.7%	0.2%	89.4%	87.7%
Z→μμ	0.951 ± 0.009	1.3%	8.7%	89.6%	0.2%	89.5%	88.2%

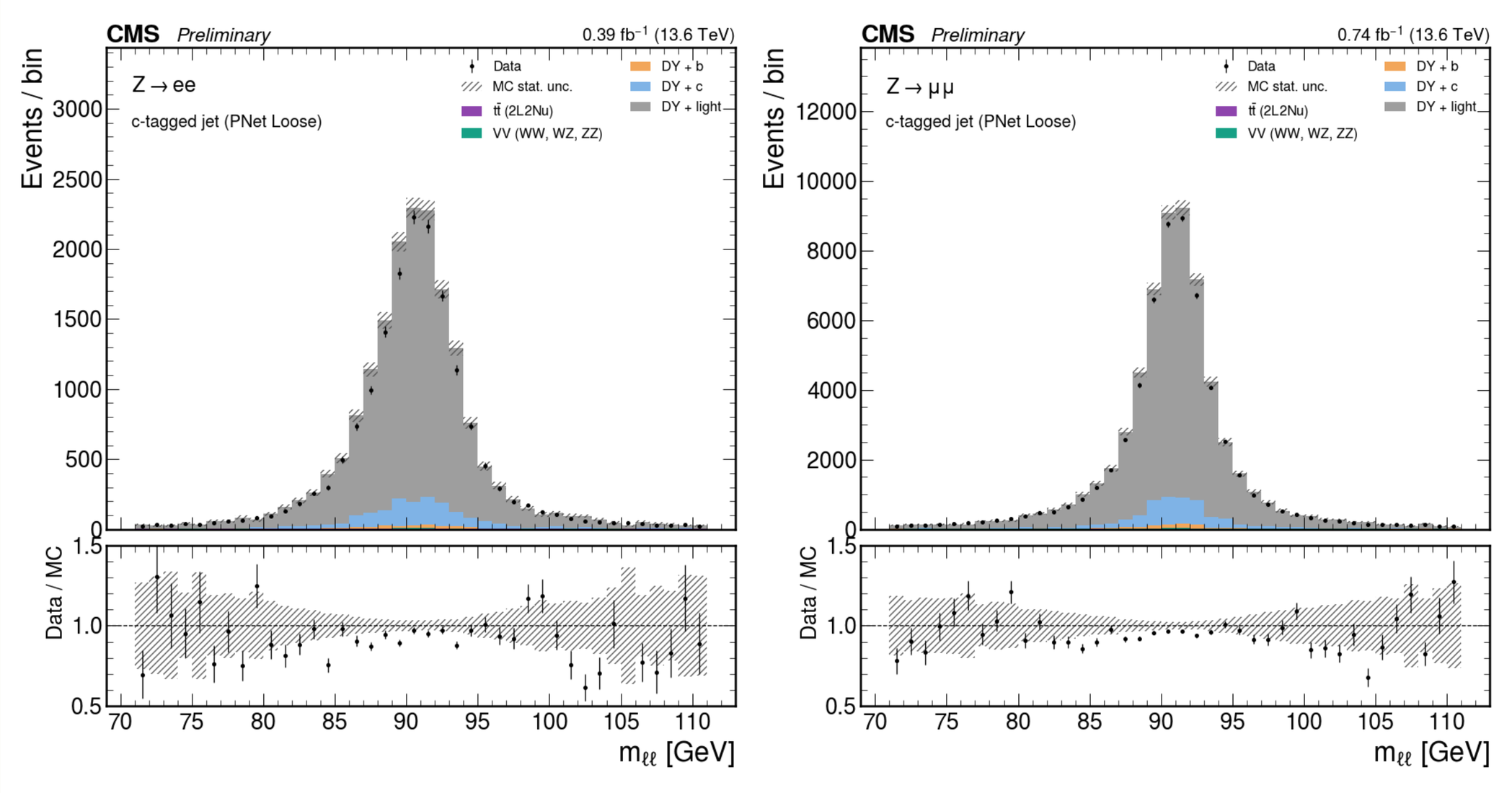
Flavour shares are of all MC. 'Tagged' = fraction of Z + jet events passing the tag. Errors are statistical only.

32 plots follow: Z and leptons, jets, Z–jet, MET and vertices, tagger scores, 2D maps. Left Z→ee, right Z→μμ.

c-tag Loose

Dilepton mass

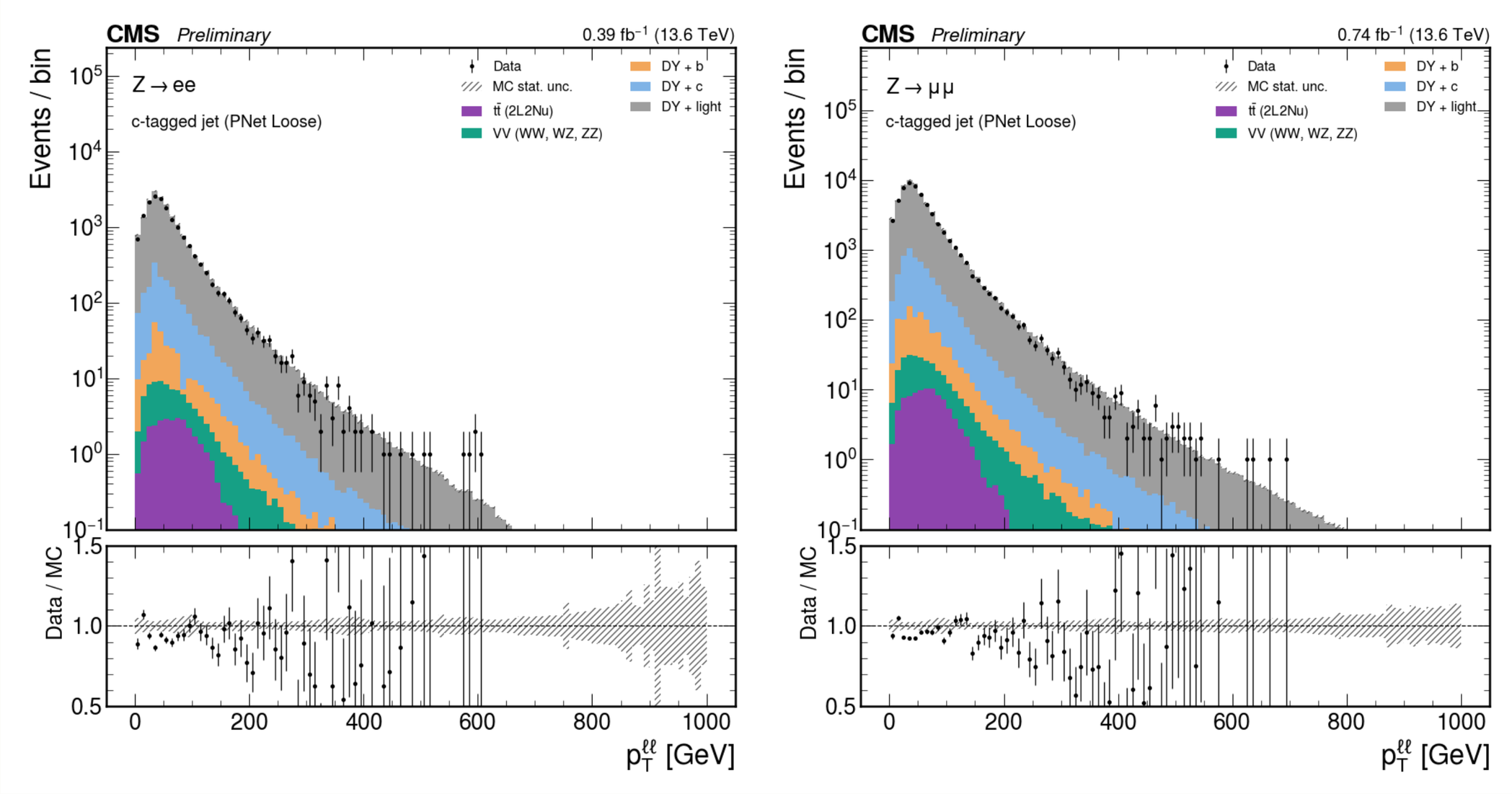
m(ℓℓ), 71–111 GeV window, linear scale.



c-tag Loose

Z pT

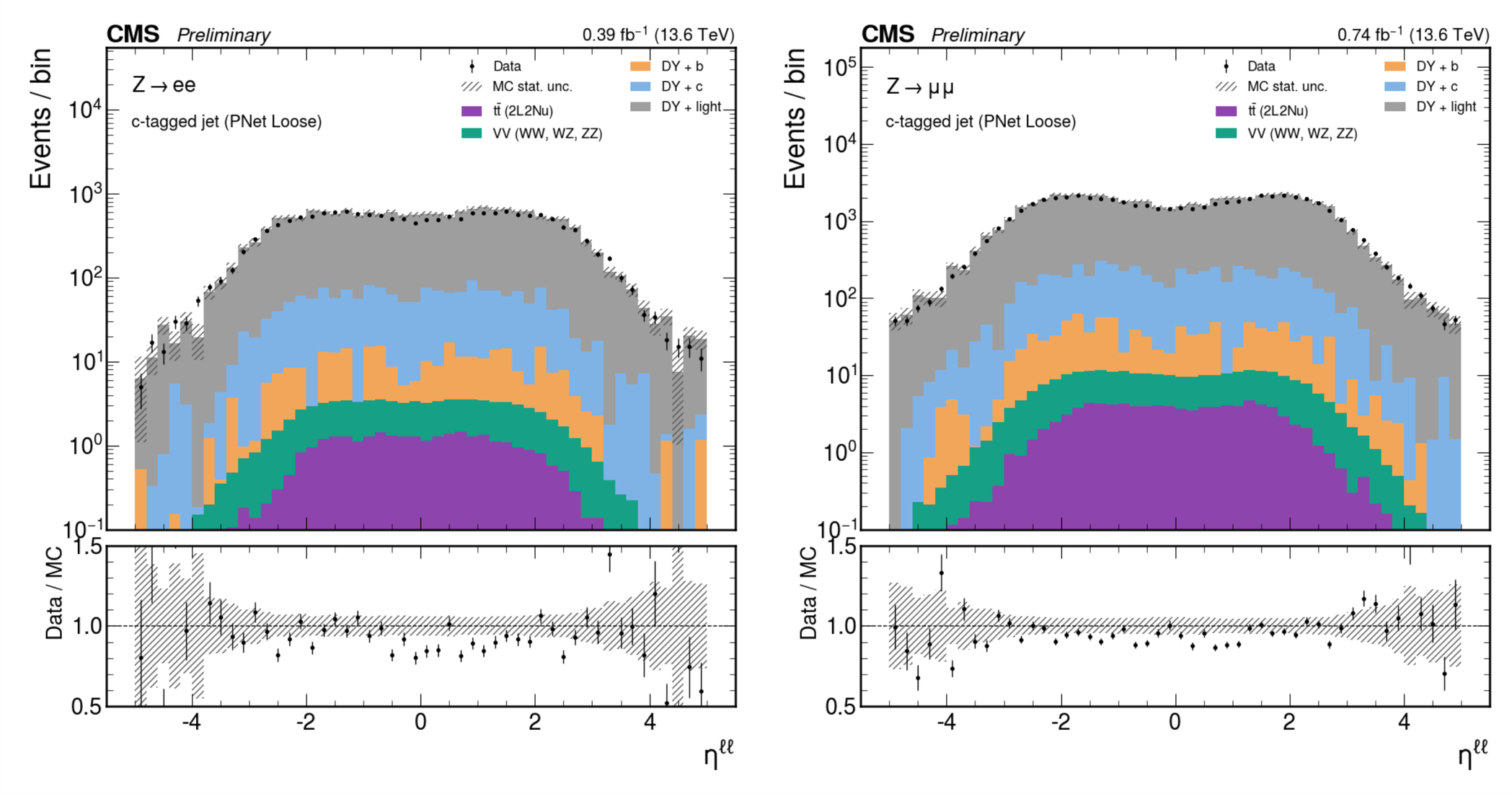
pT of the lepton pair.



c-tag Loose

Z η

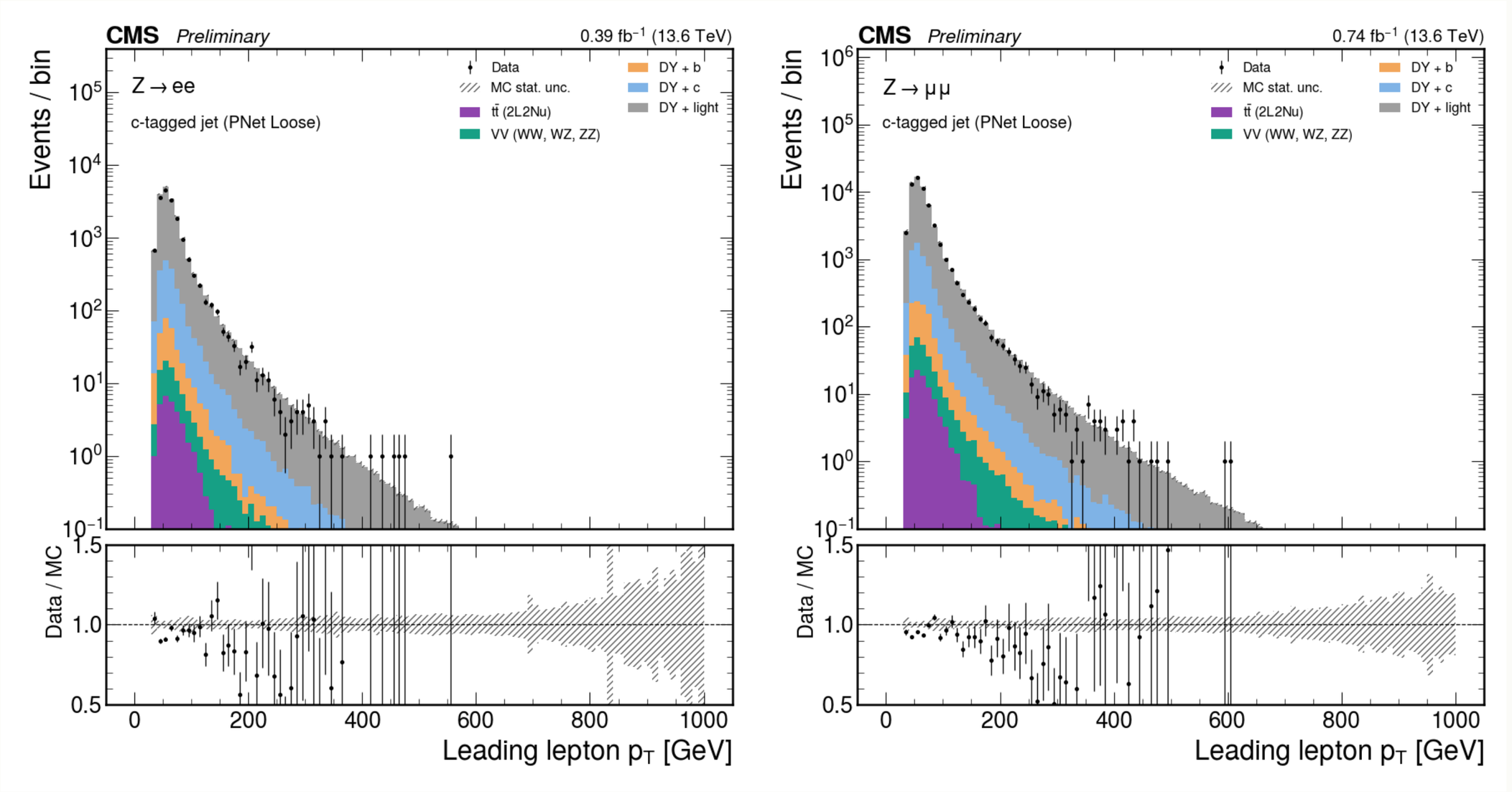
η of the lepton pair.



c-tag Loose

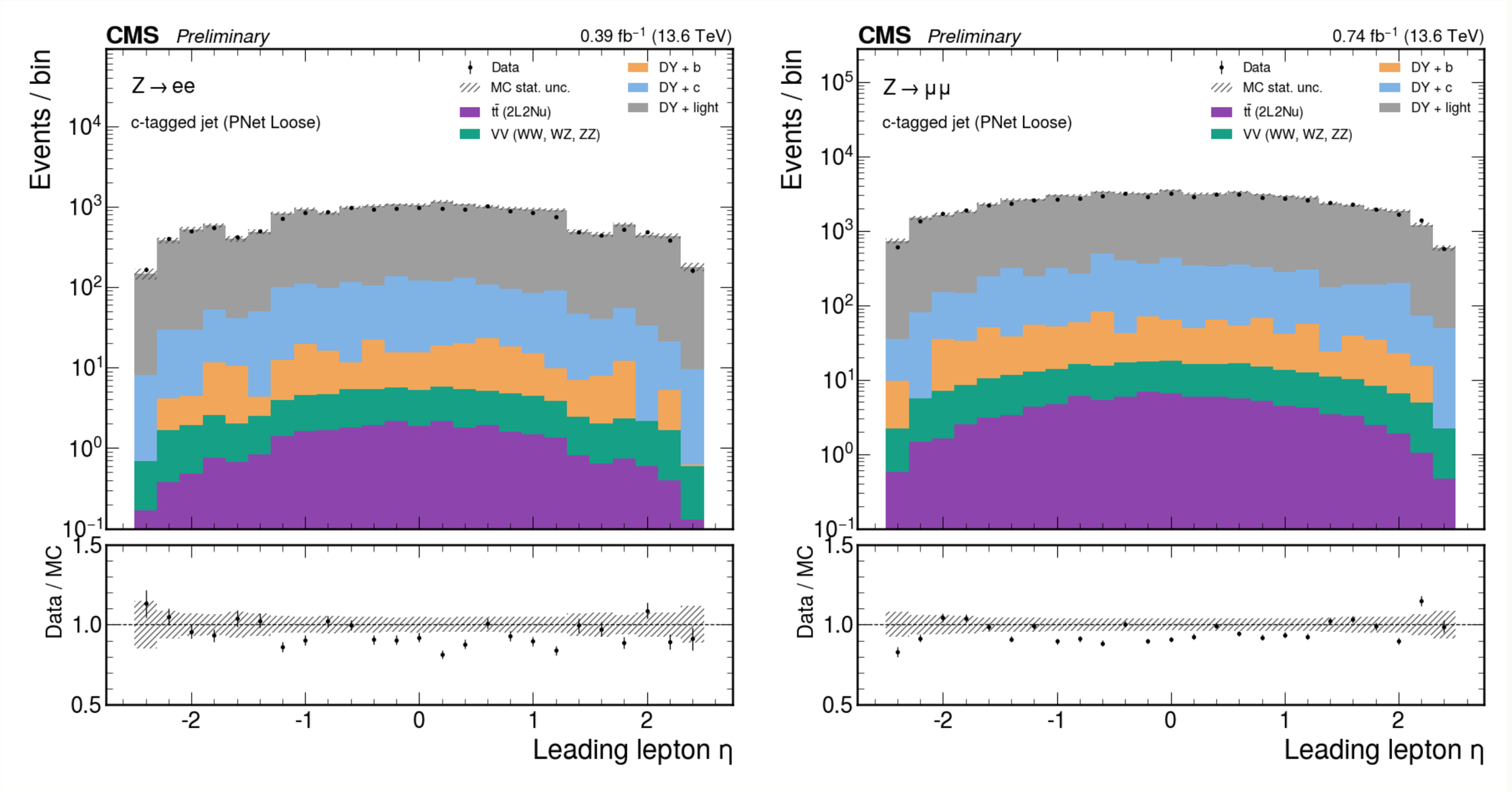
Leading-lepton pT

Leading lepton pT > 35 GeV (trigger plateau).



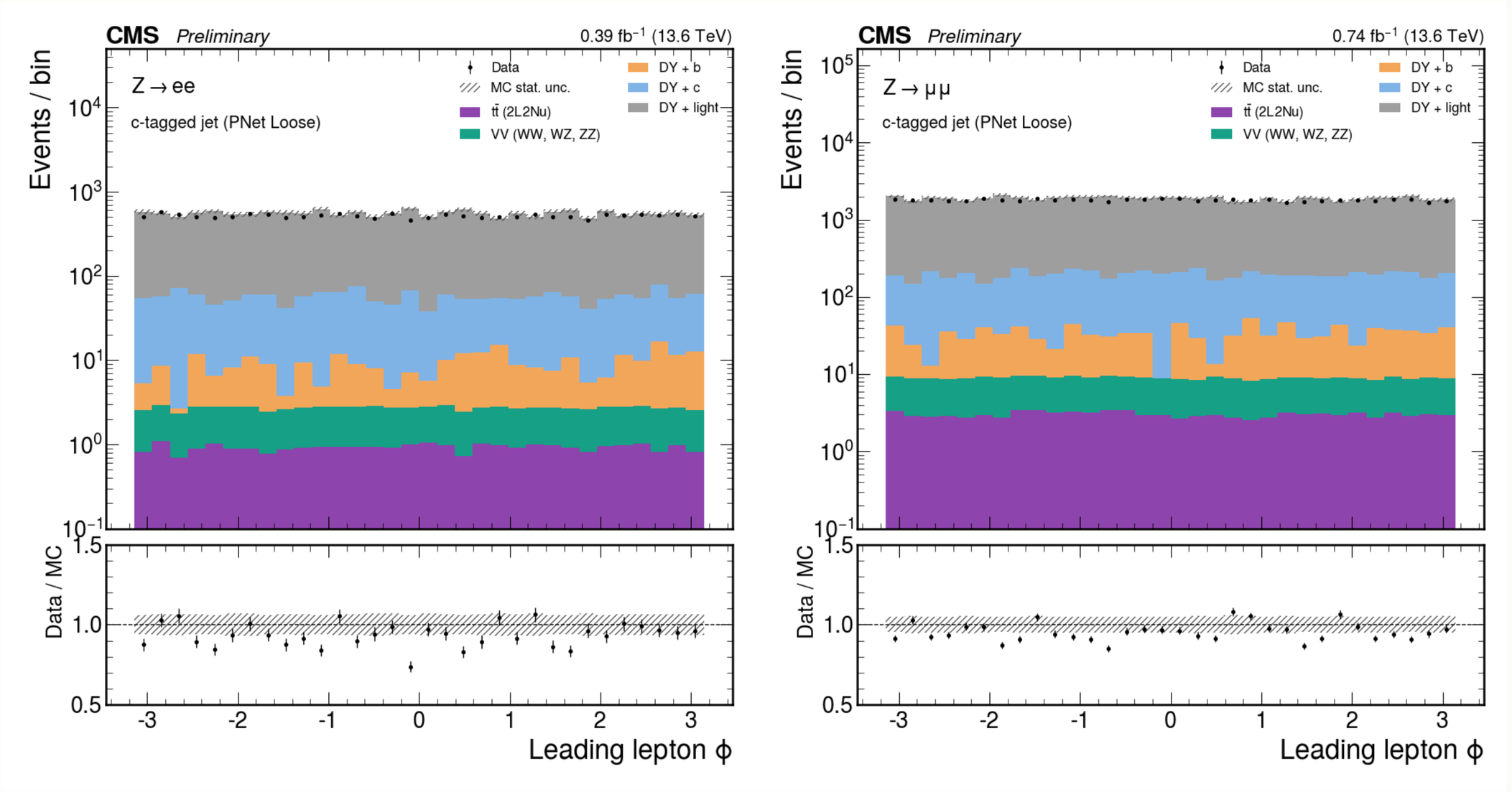
c-tag Loose

Leading-lepton η



c-tag Loose

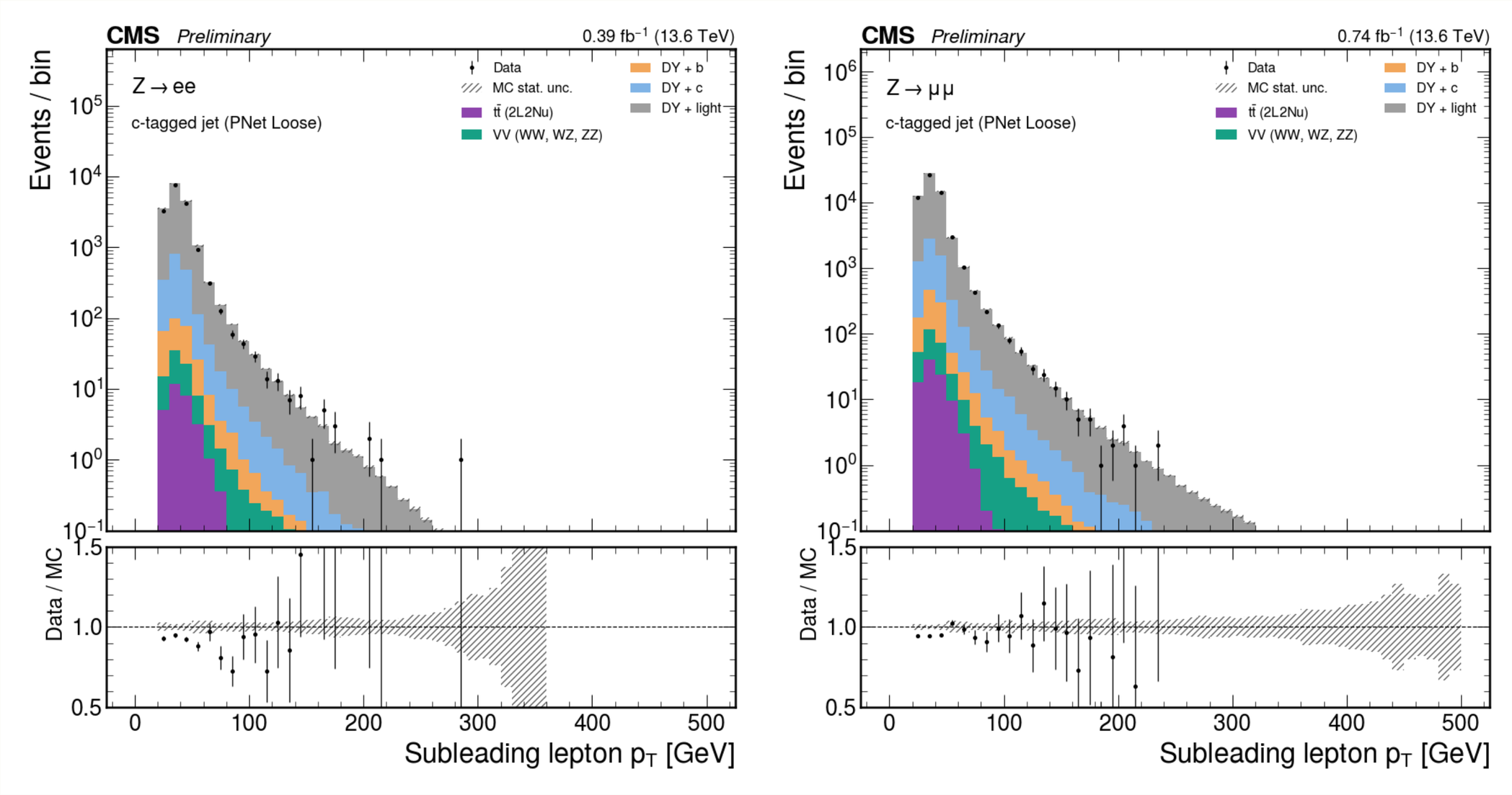
Leading-lepton φ



c-tag Loose

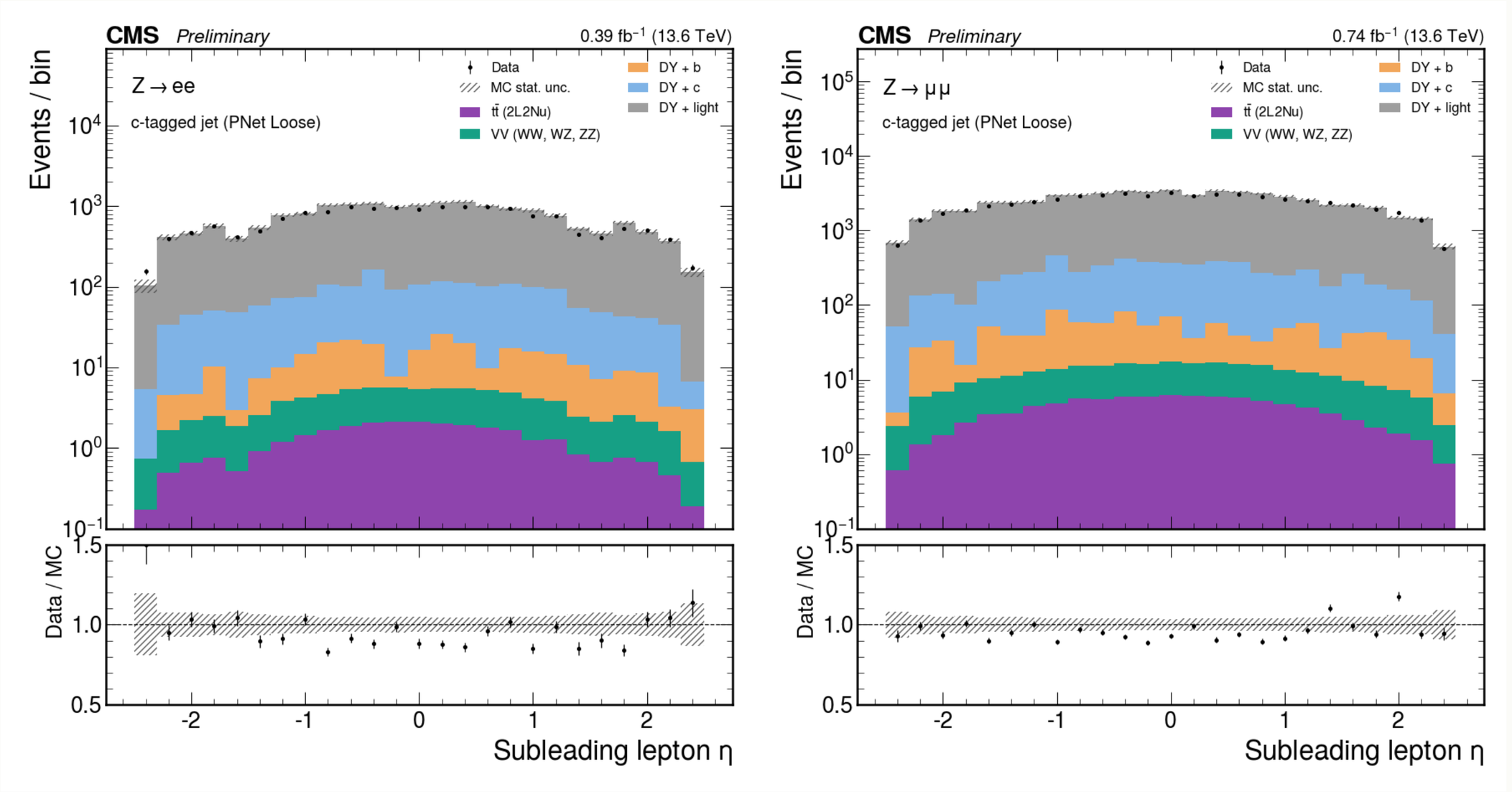
Second-lepton pT

Second lepton pT > 25 GeV.



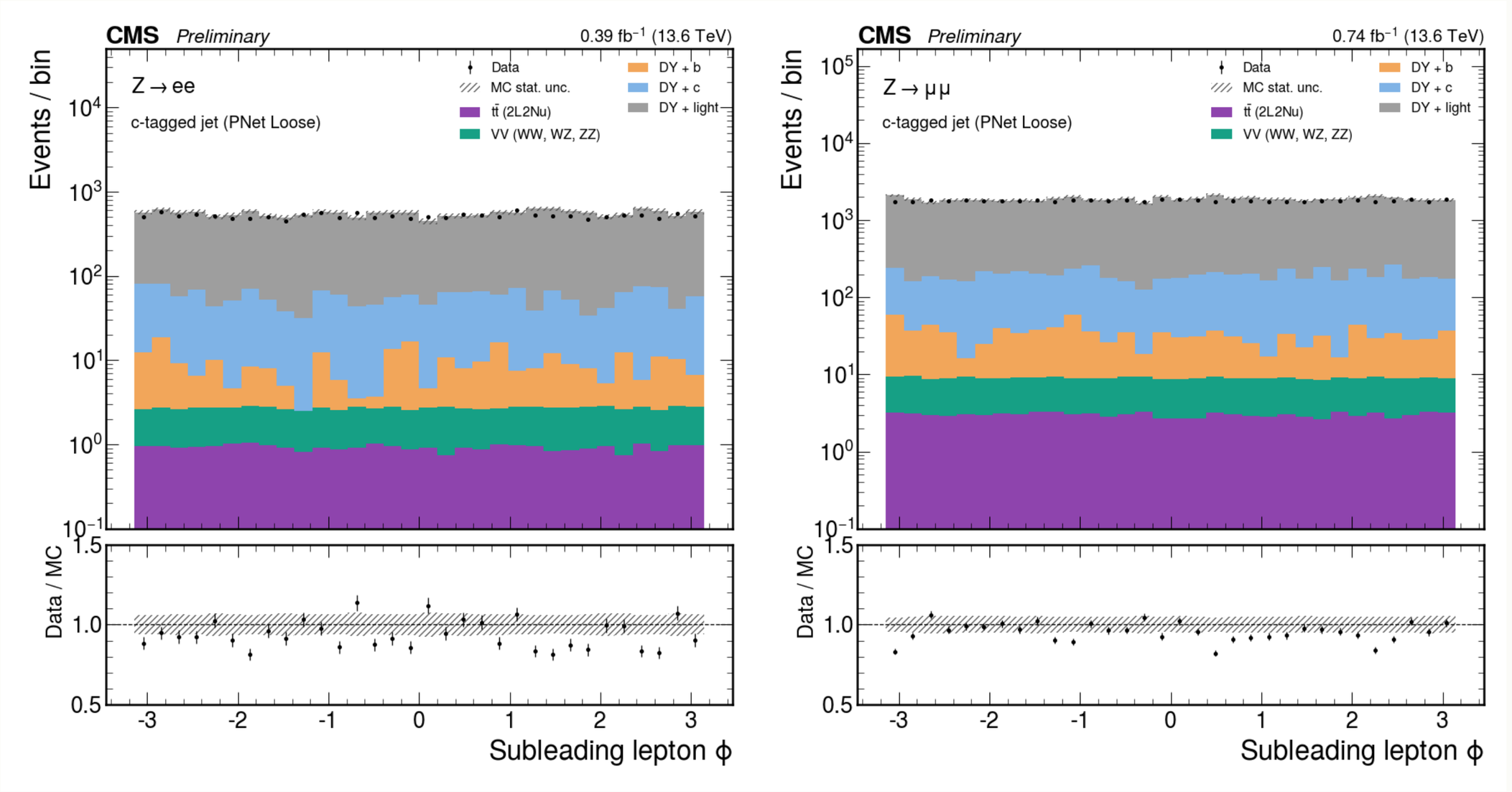
c-tag Loose

Second-lepton η



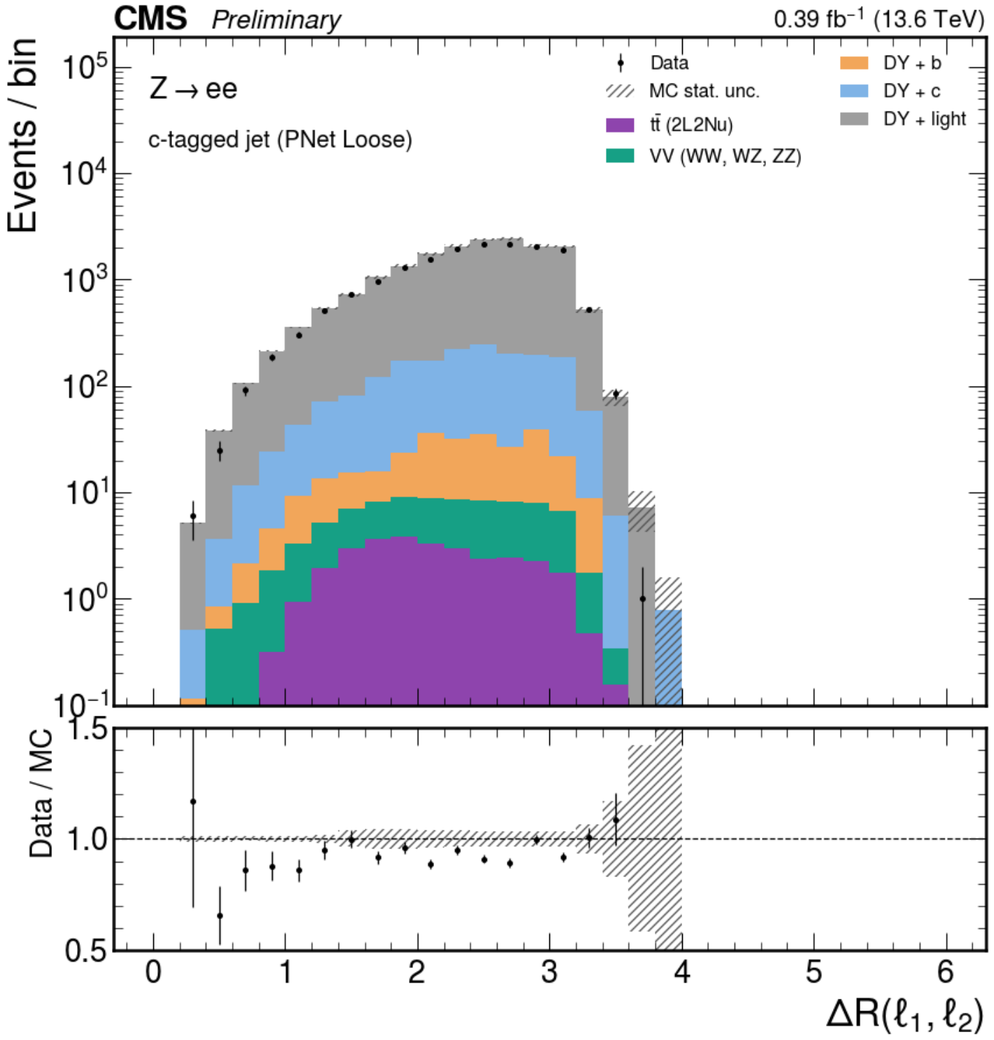
c-tag Loose

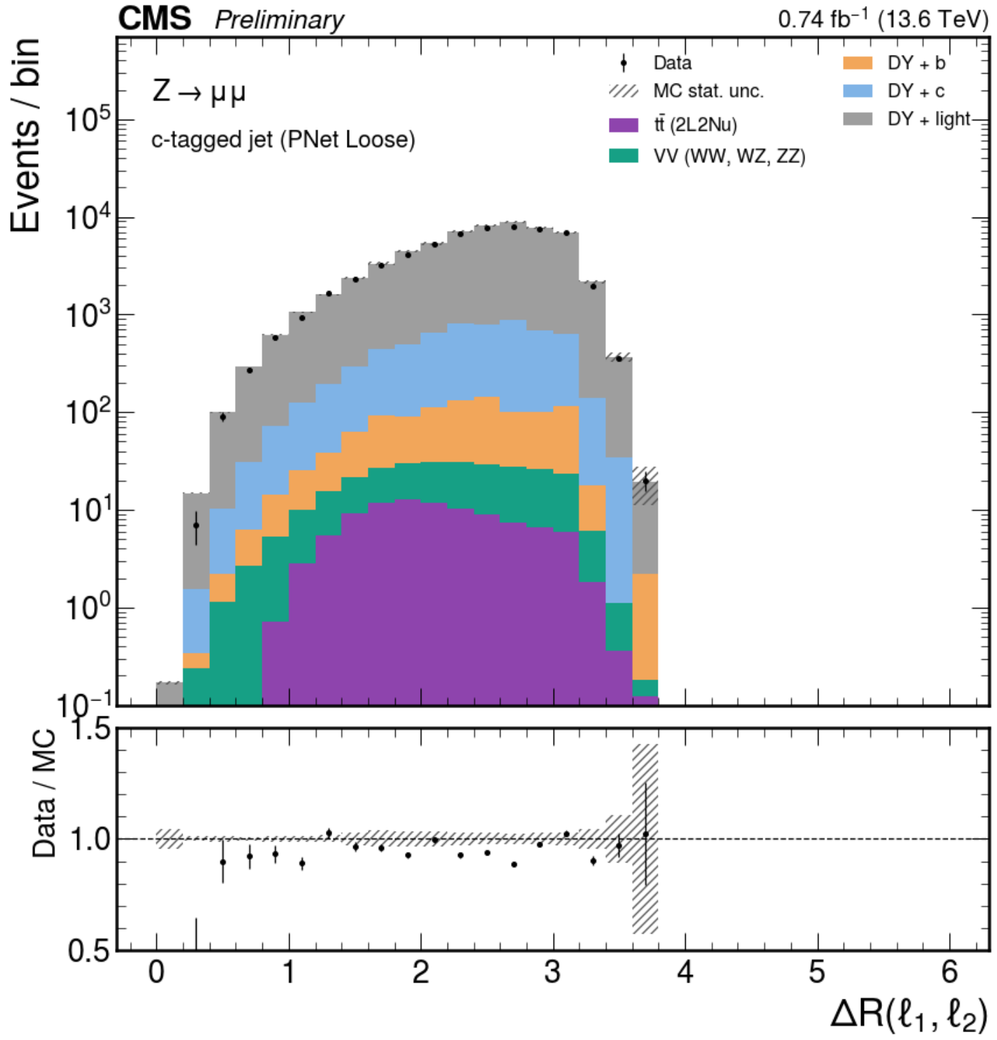
Second-lepton φ



c-tag Loose

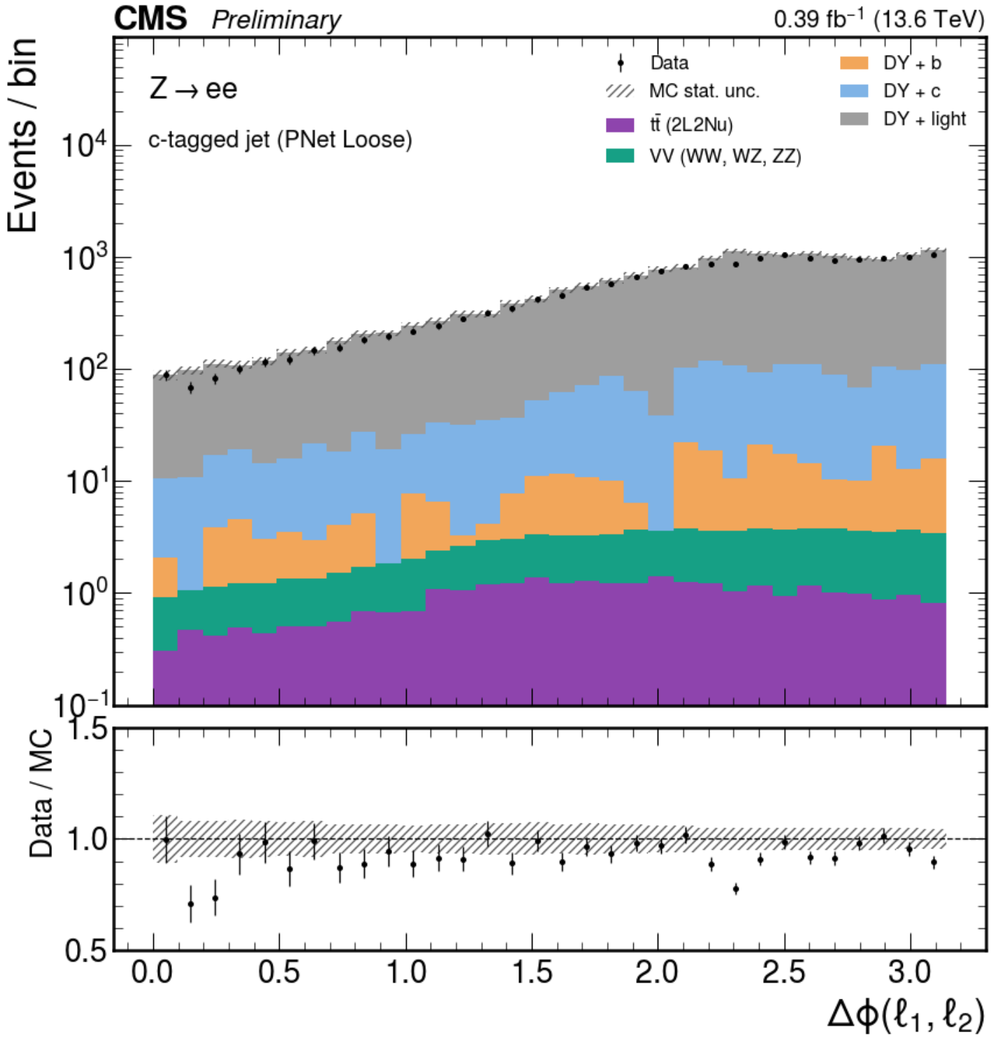
ΔR between the leptons

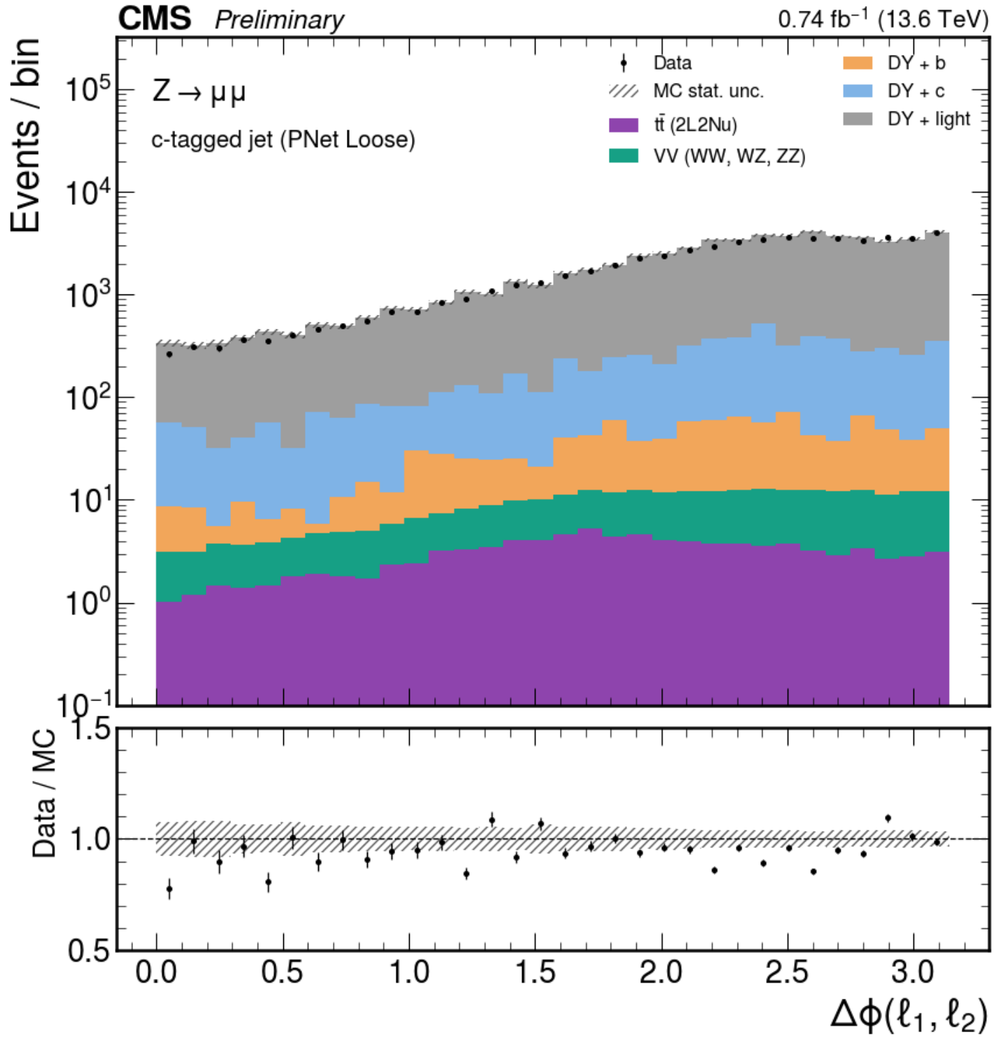




c-tag Loose

Δφ between the leptons

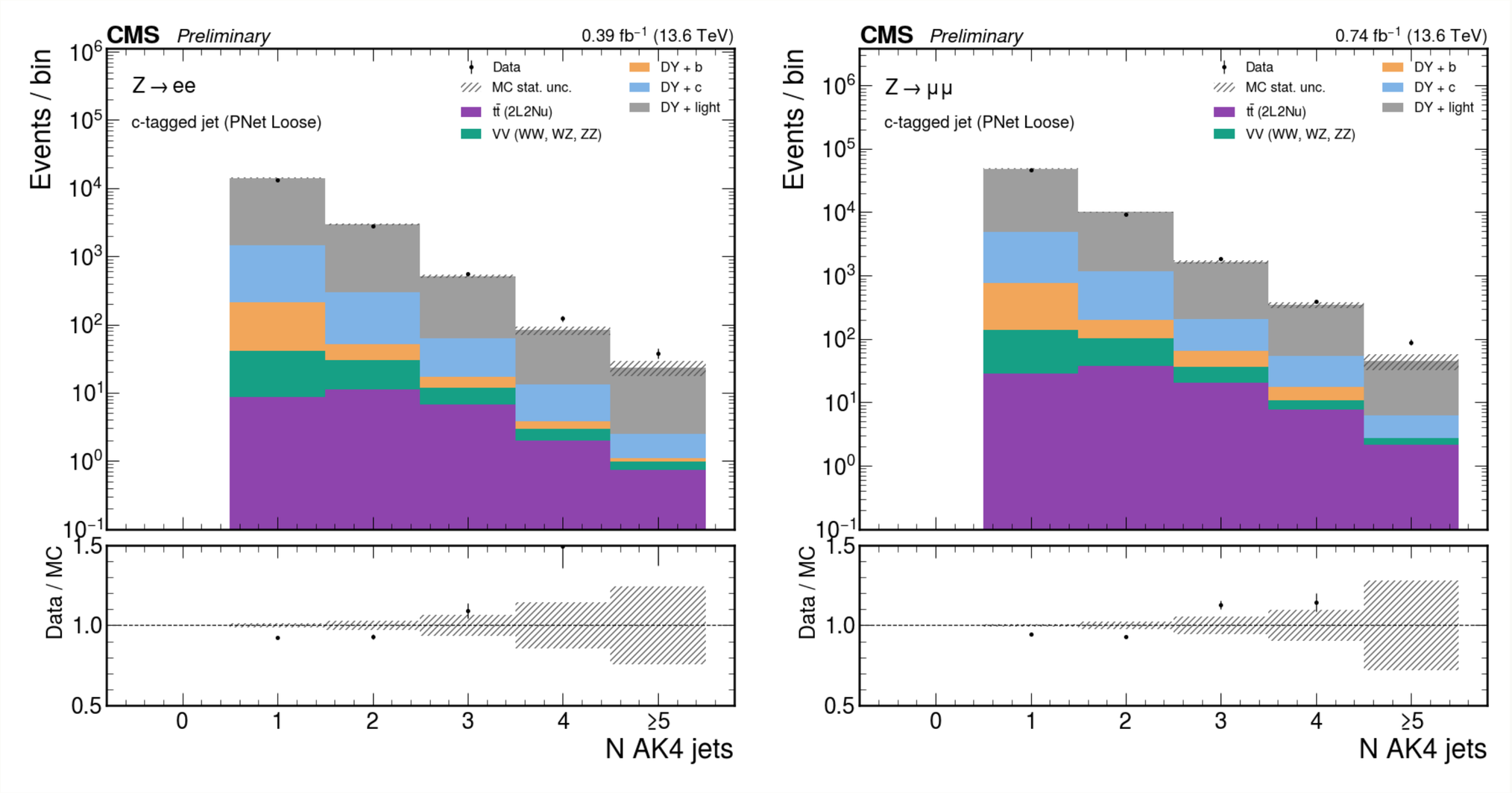




c-tag Loose

Number of jets

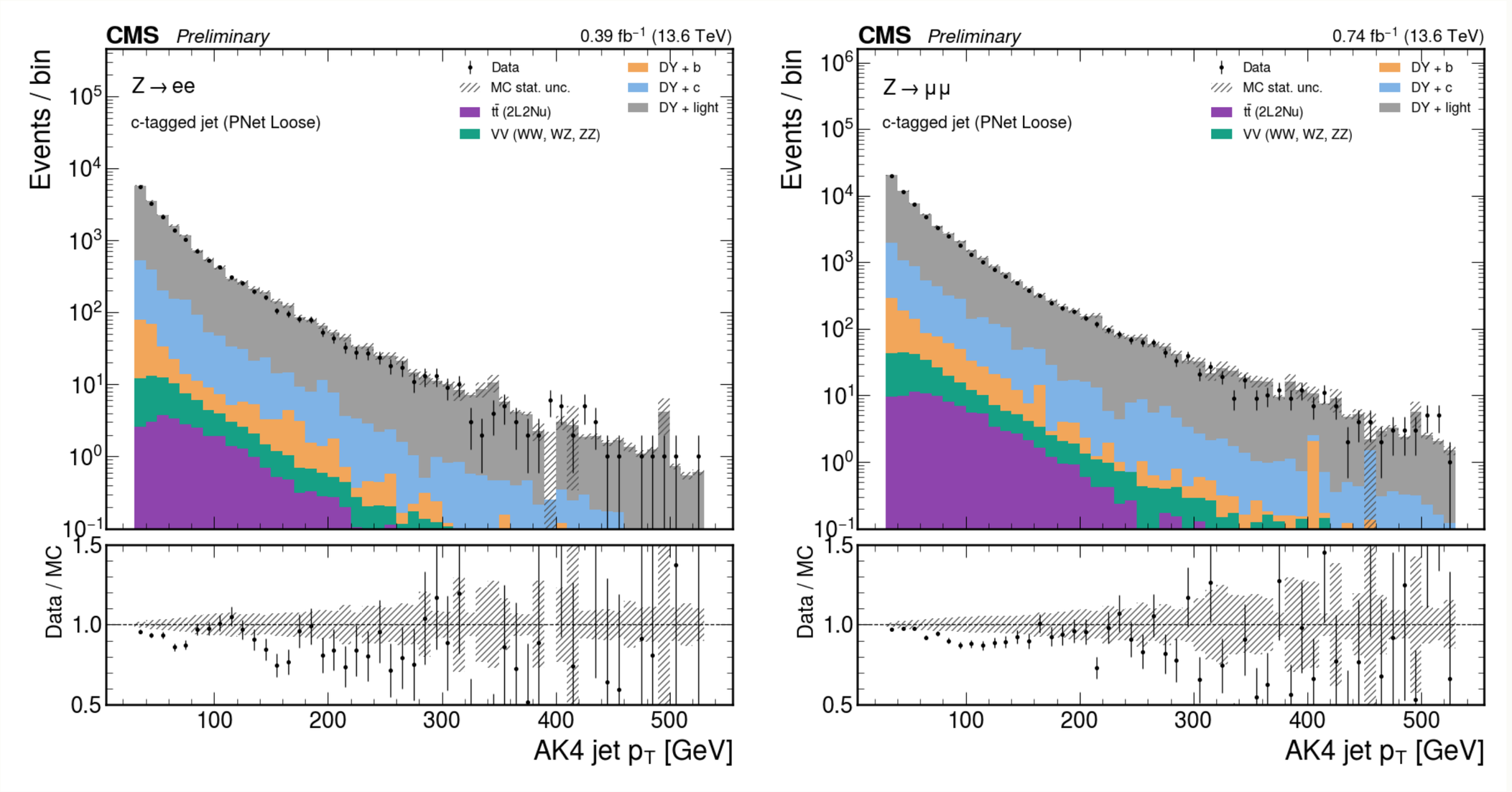
AK4 jets with pT > 30 GeV, |η| < 2.5, away from the leptons. Untagged plot: Z events before the jet cut, so the 0-jet bin is filled.



c-tag Loose

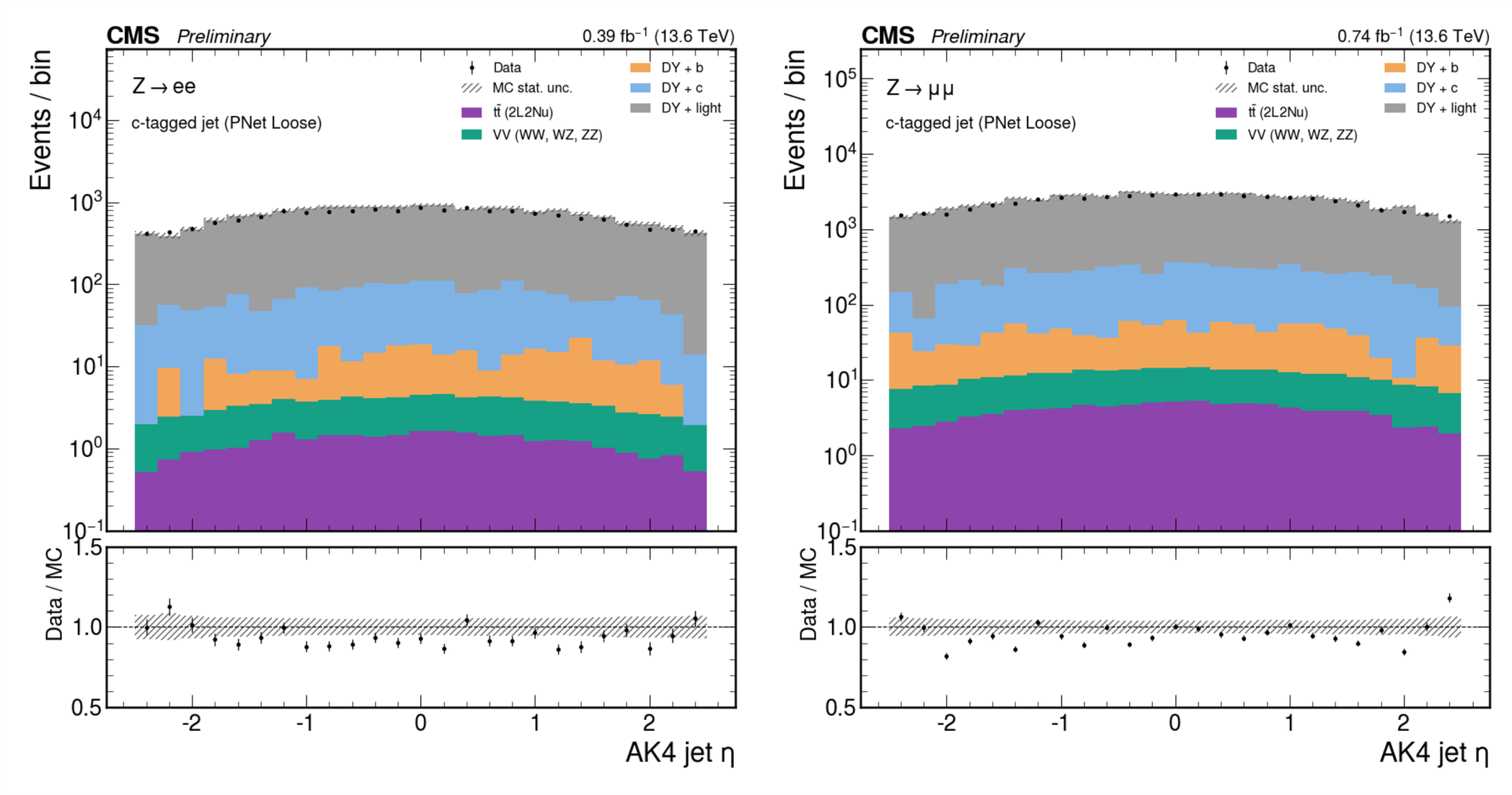
Leading-jet pT

The tag (if any) is on this jet.



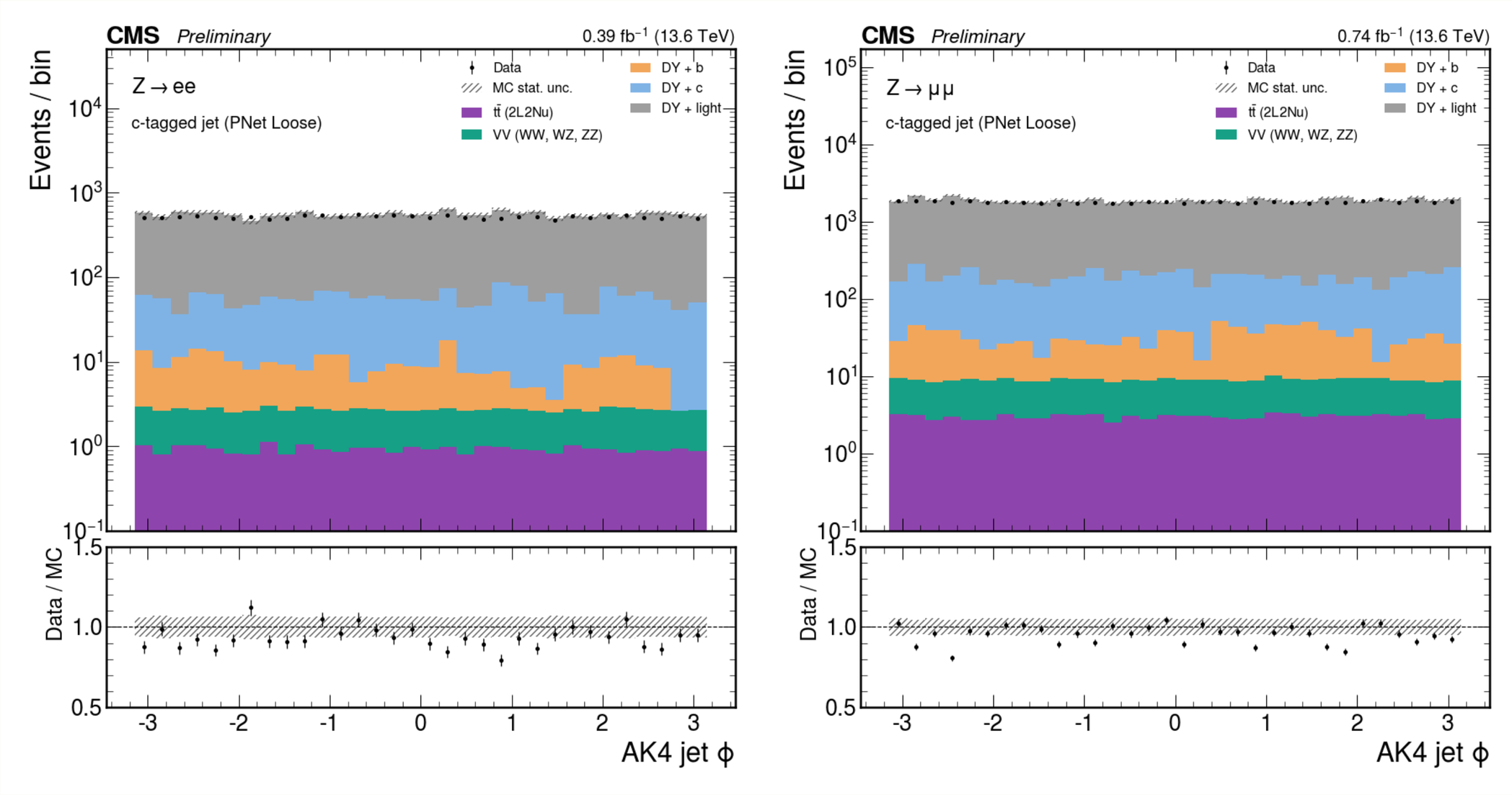
c-tag Loose

Leading-jet η



c-tag Loose

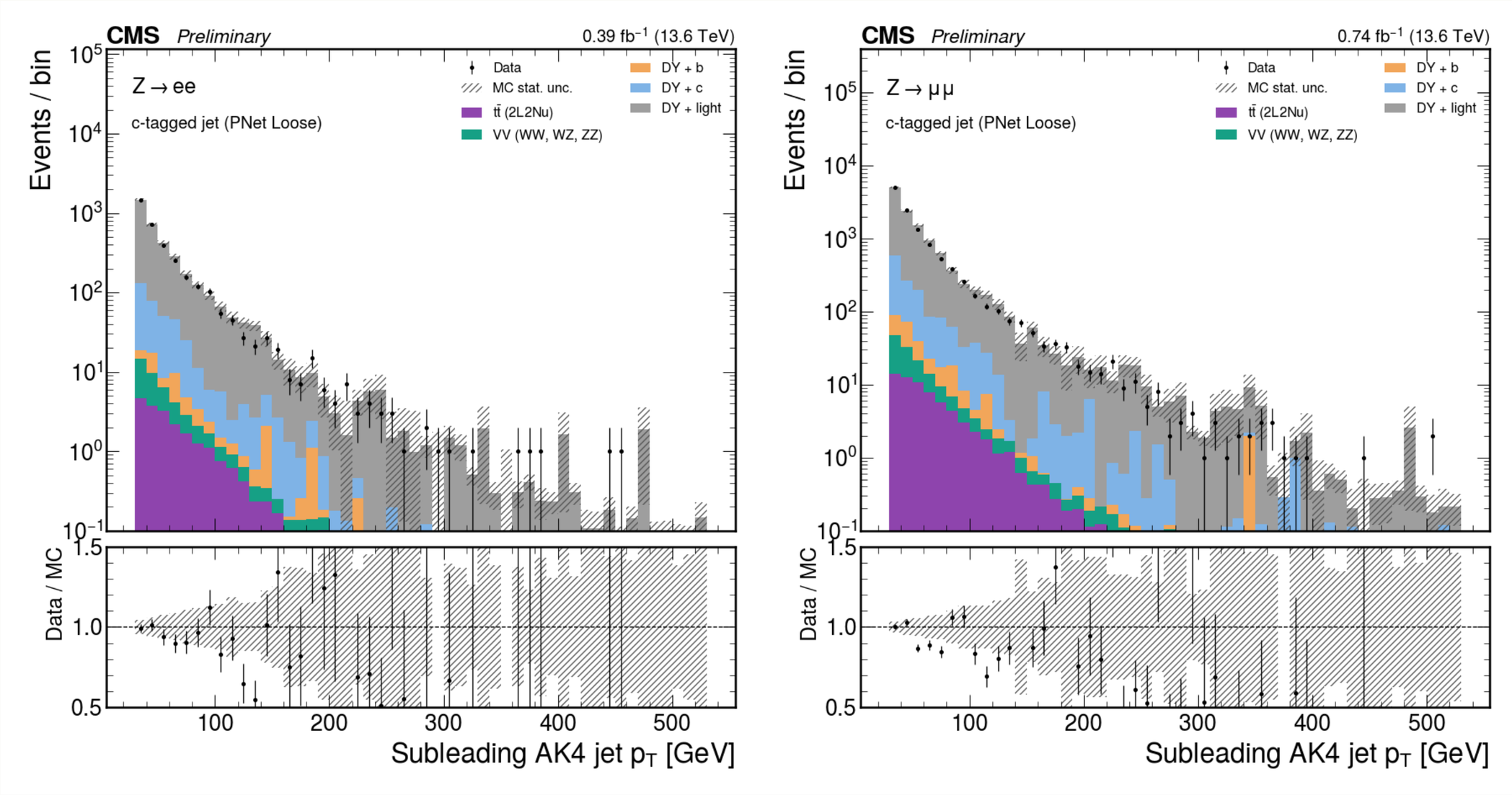
Leading-jet φ



c-tag Loose

Second-jet pT

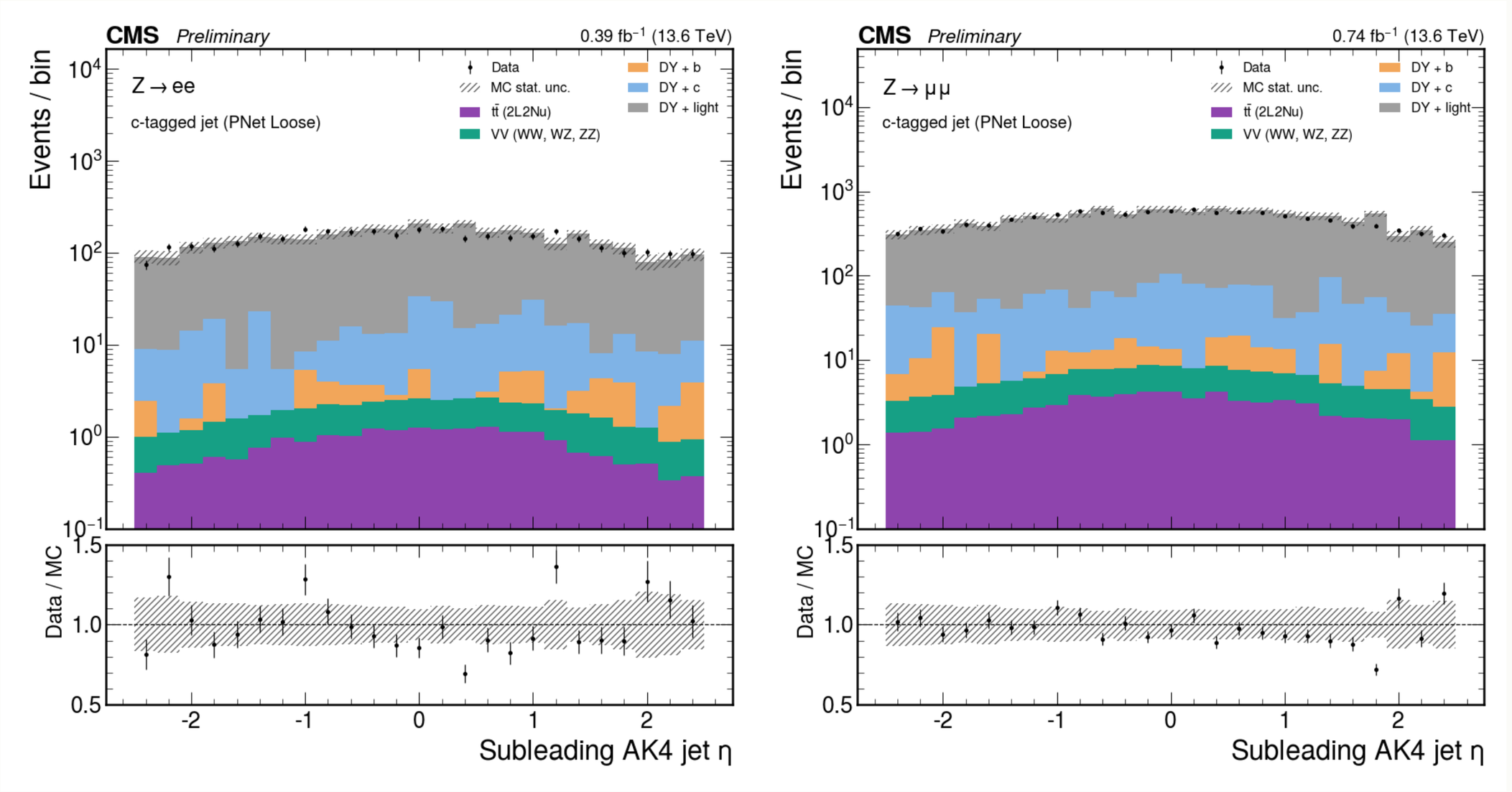
Events with at least 2 jets.



c-tag Loose

Second-jet η

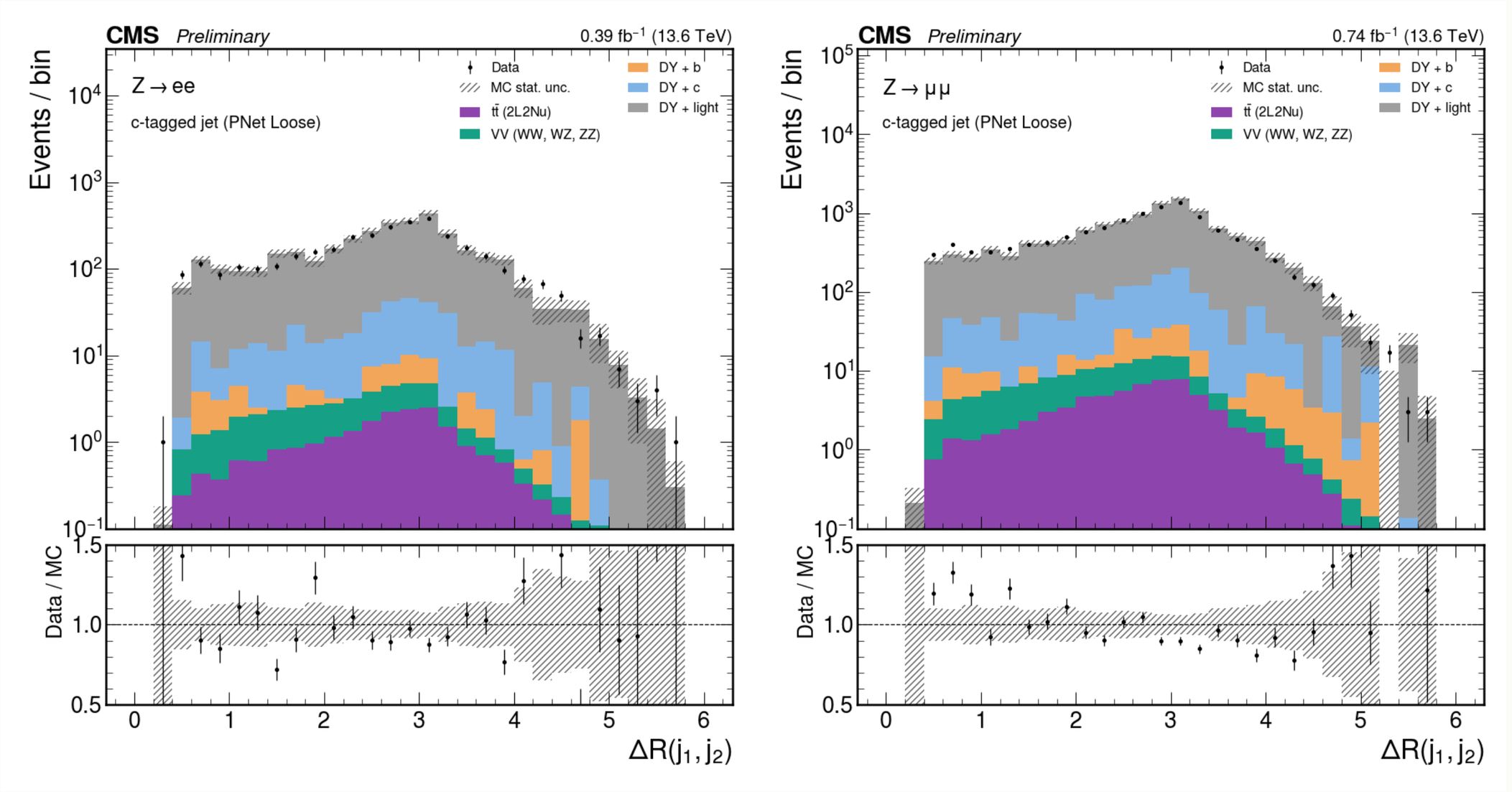
Events with at least 2 jets.



c-tag Loose

ΔR between the two leading jets

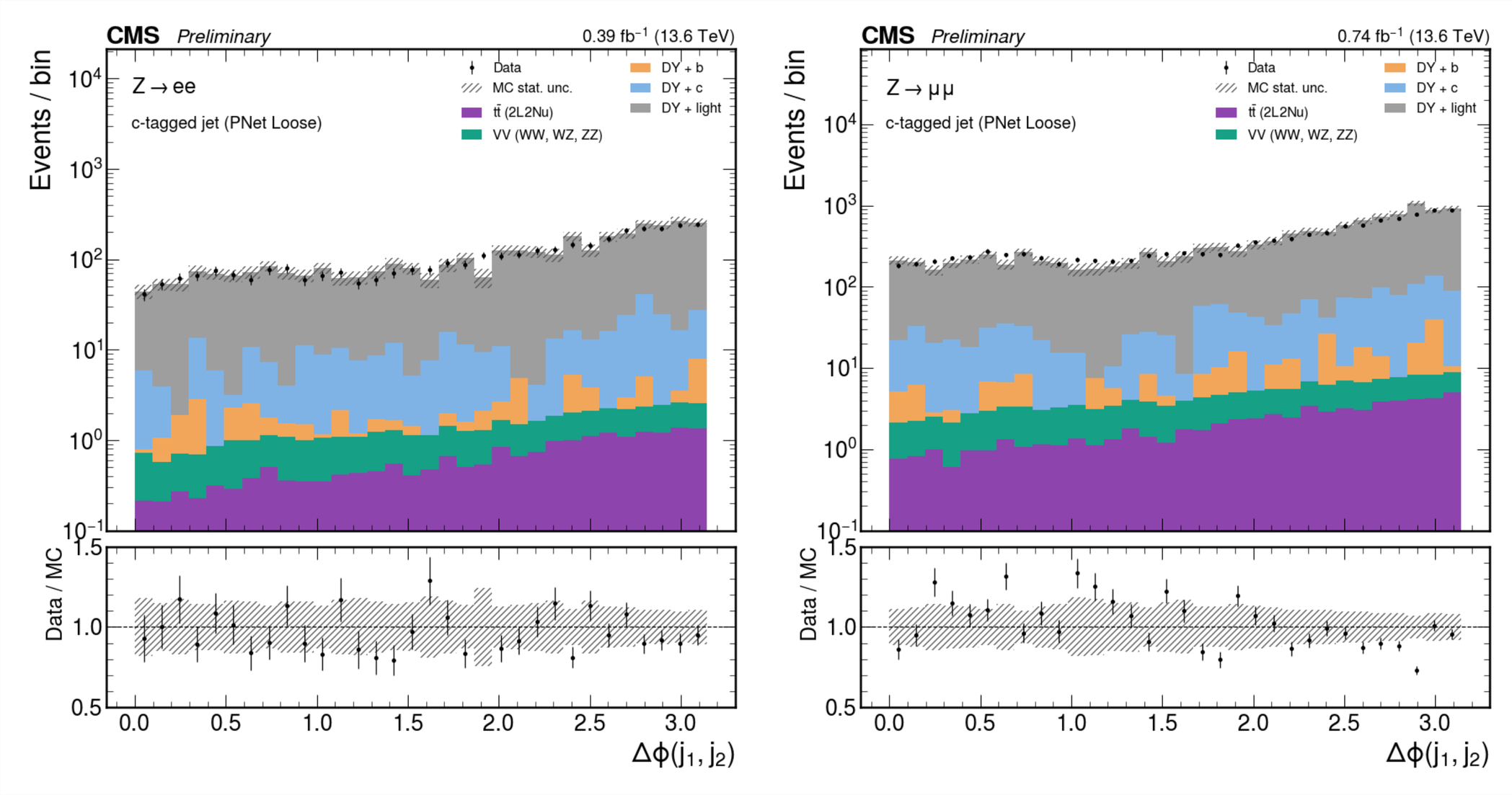
Events with at least 2 jets.



c-tag Loose

Δφ between the two leading jets

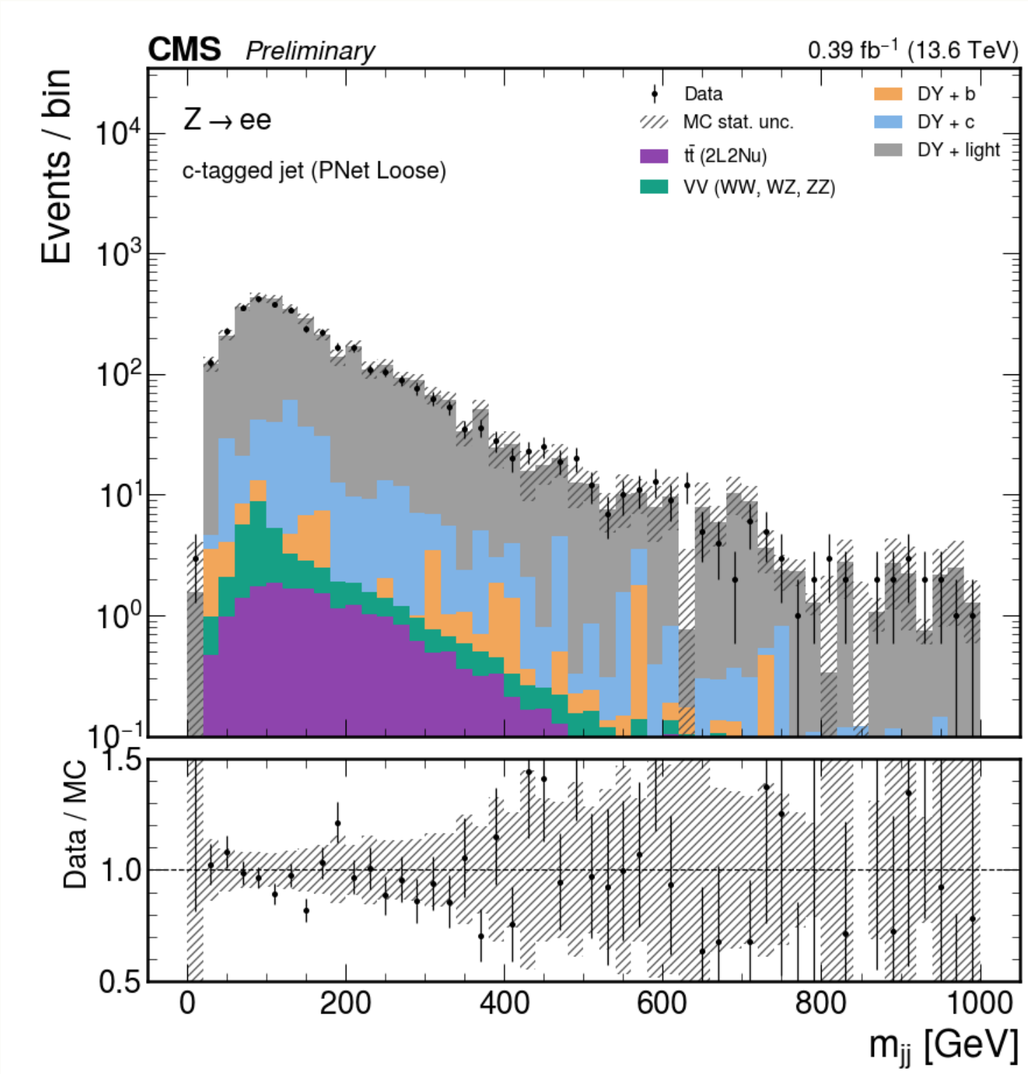
Events with at least 2 jets.

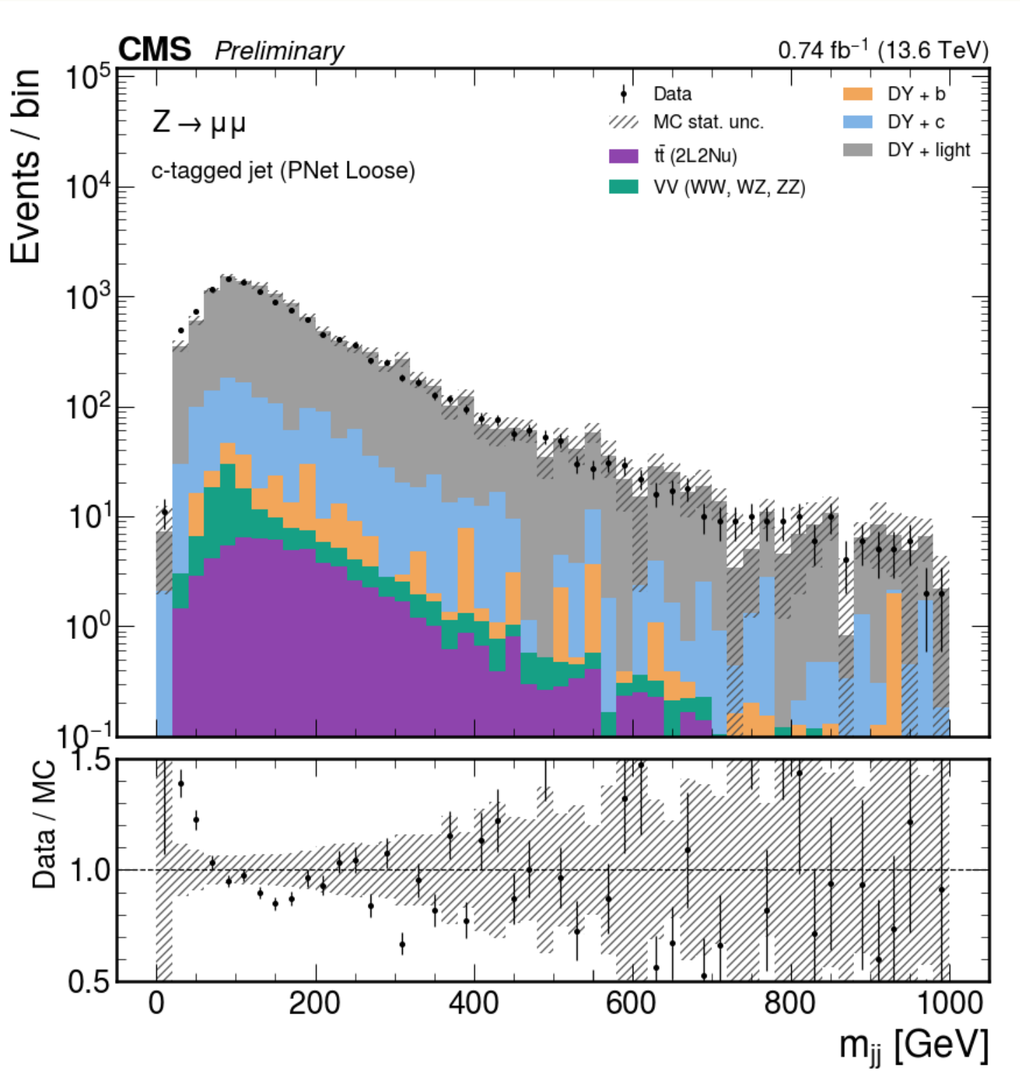


c-tag Loose

Dijet mass

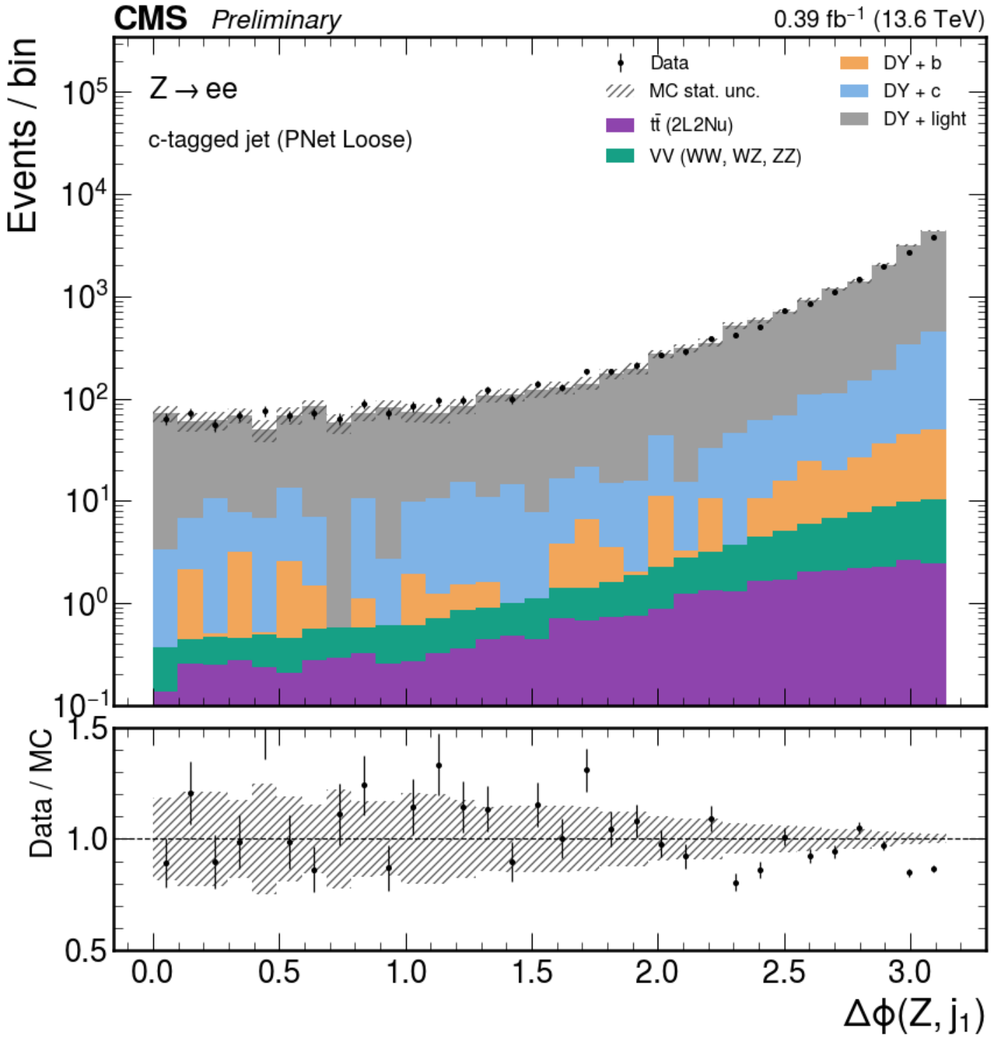
Events with at least 2 jets.

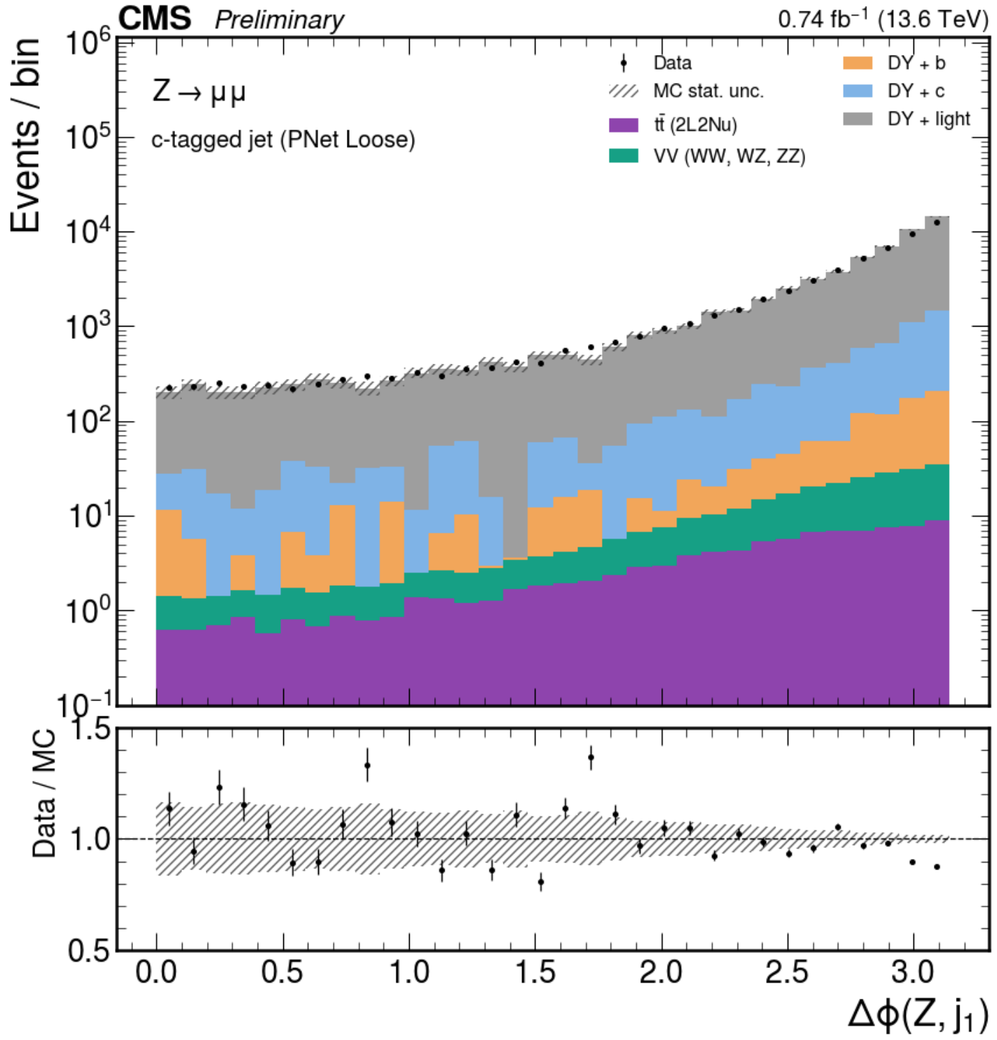




c-tag Loose

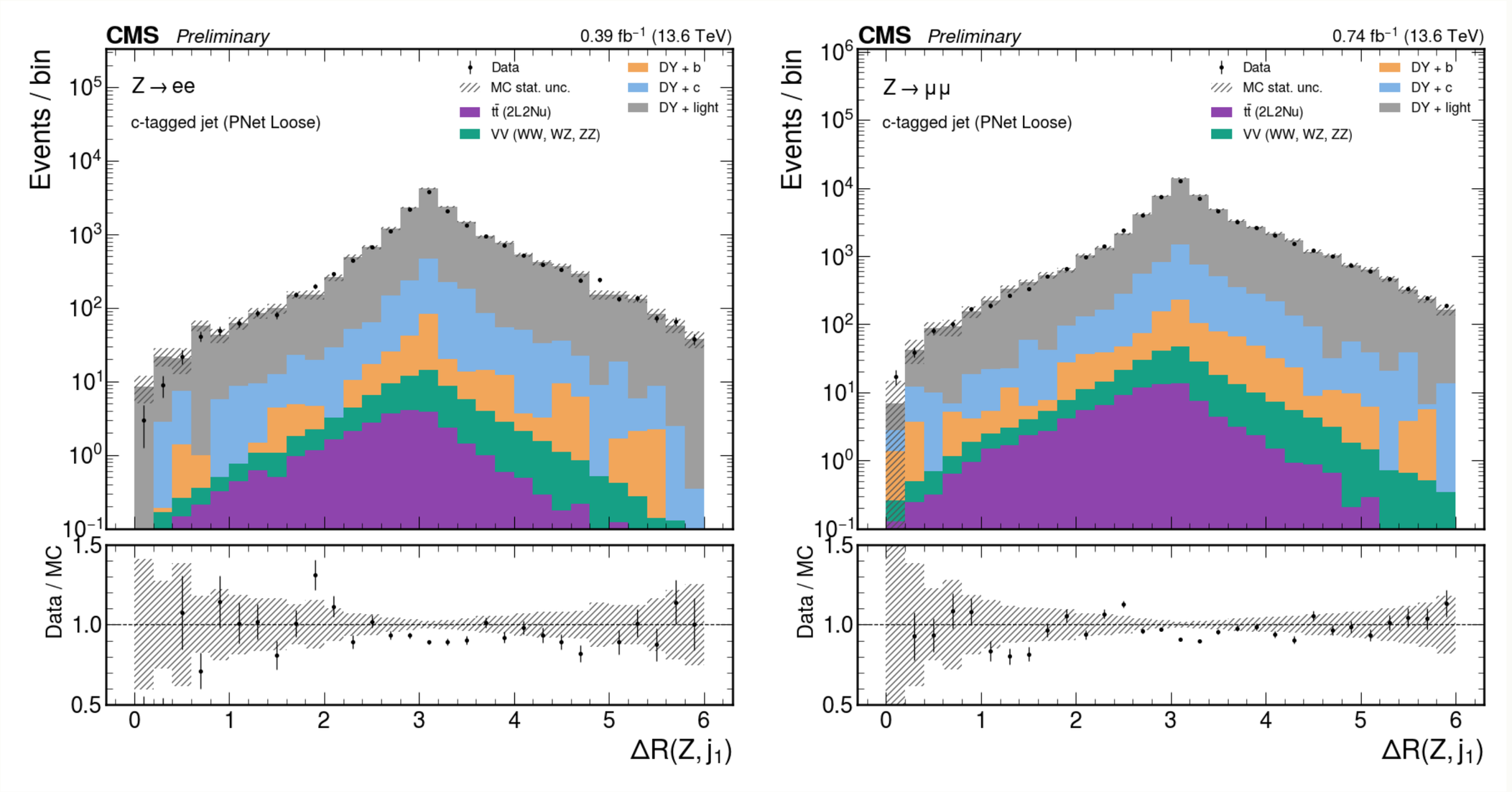
Δφ(Z, leading jet)





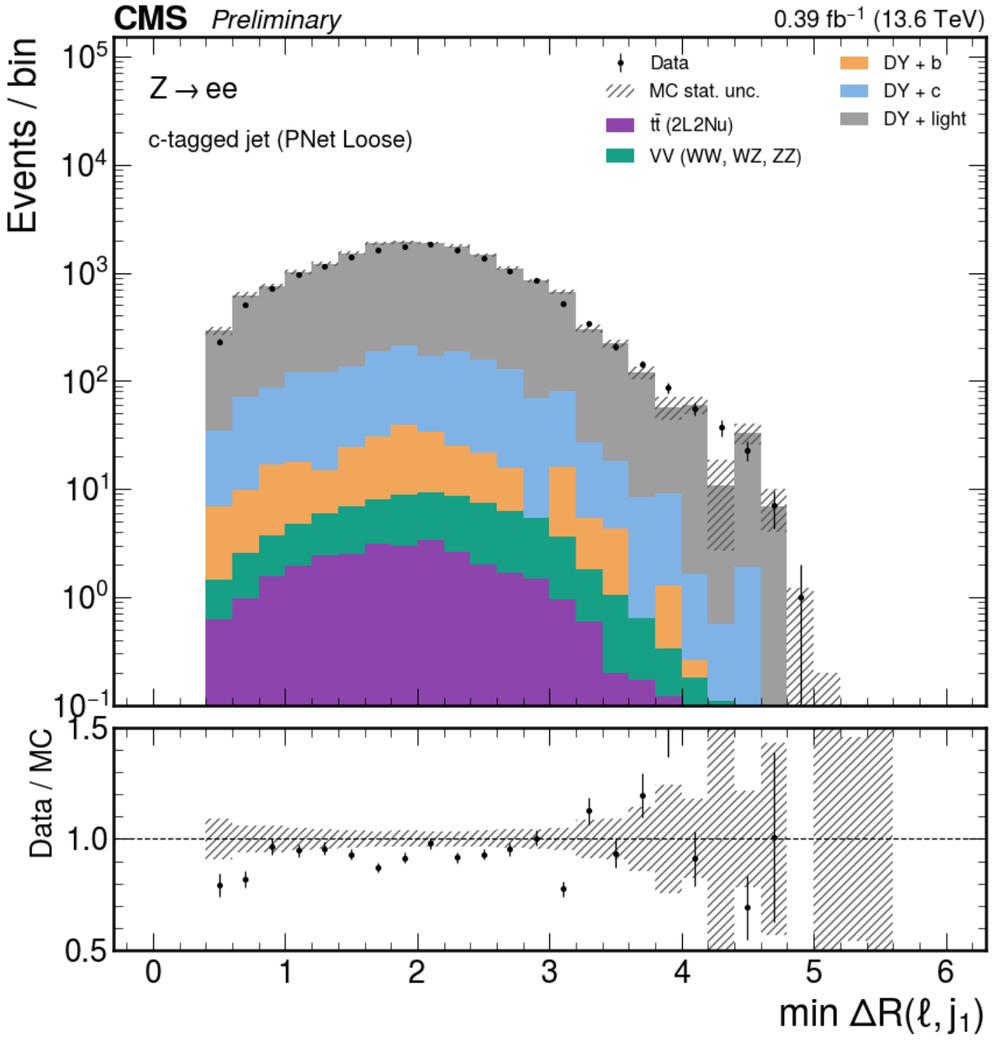
c-tag Loose

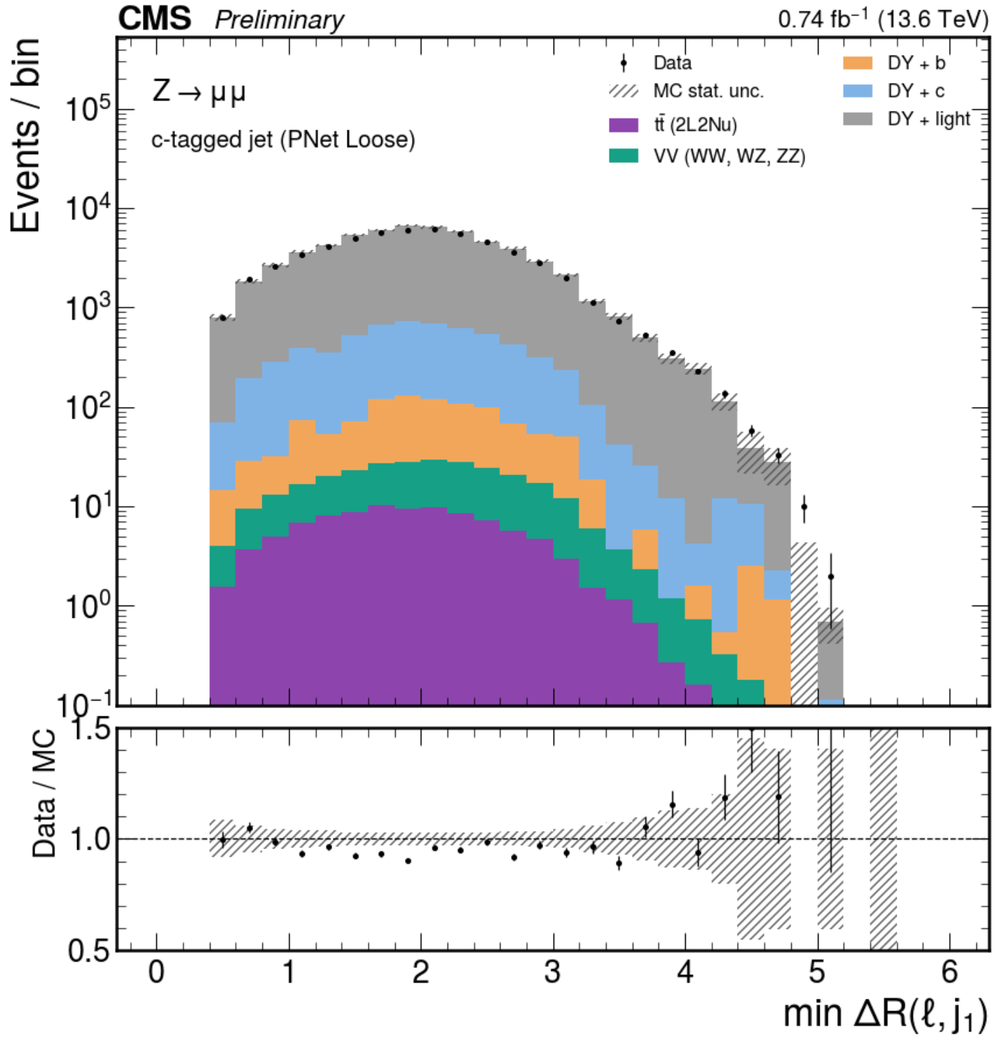
ΔR(Z, leading jet)



c-tag Loose

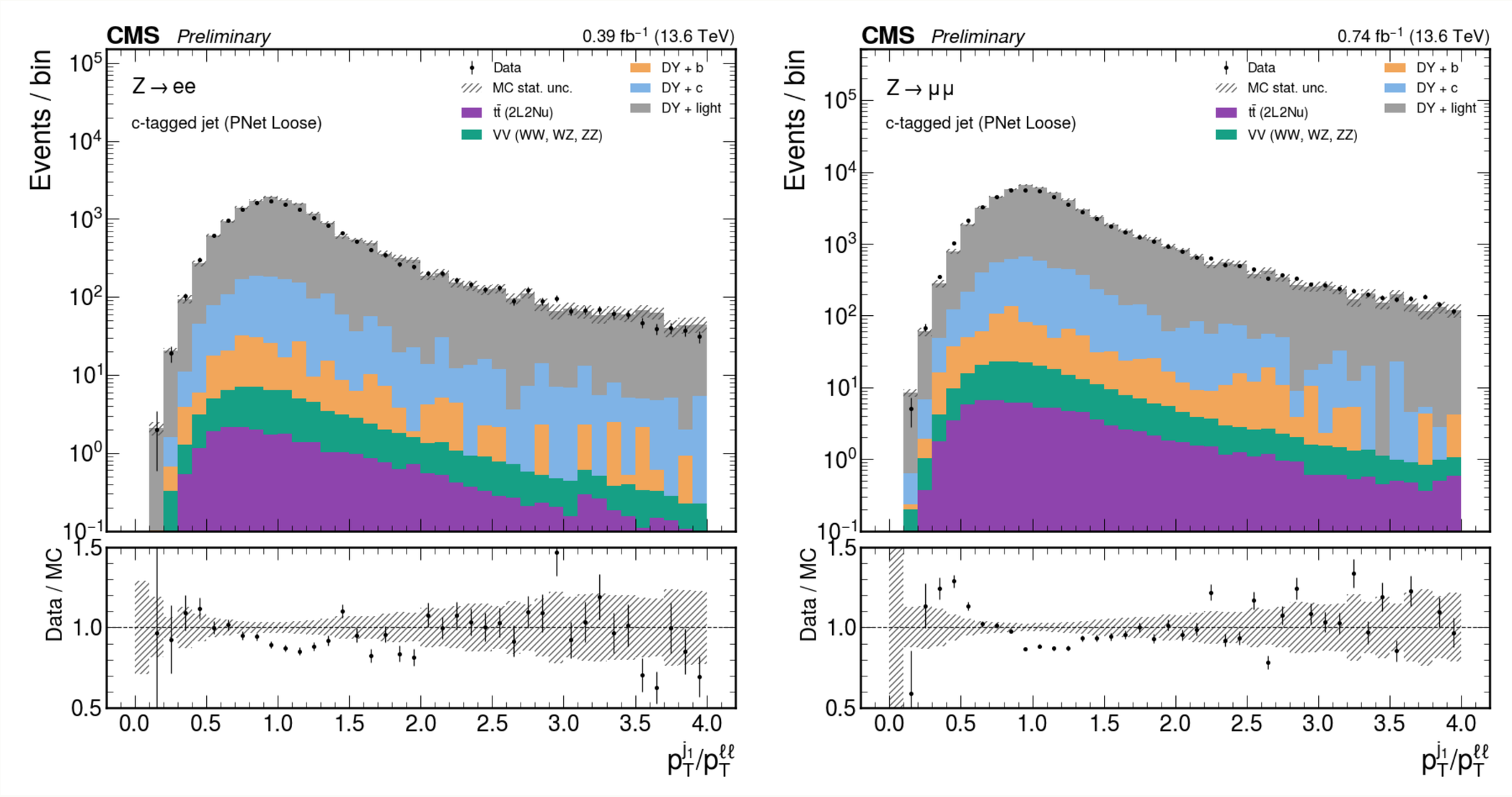
Smallest ΔR(lepton, leading jet)





c-tag Loose

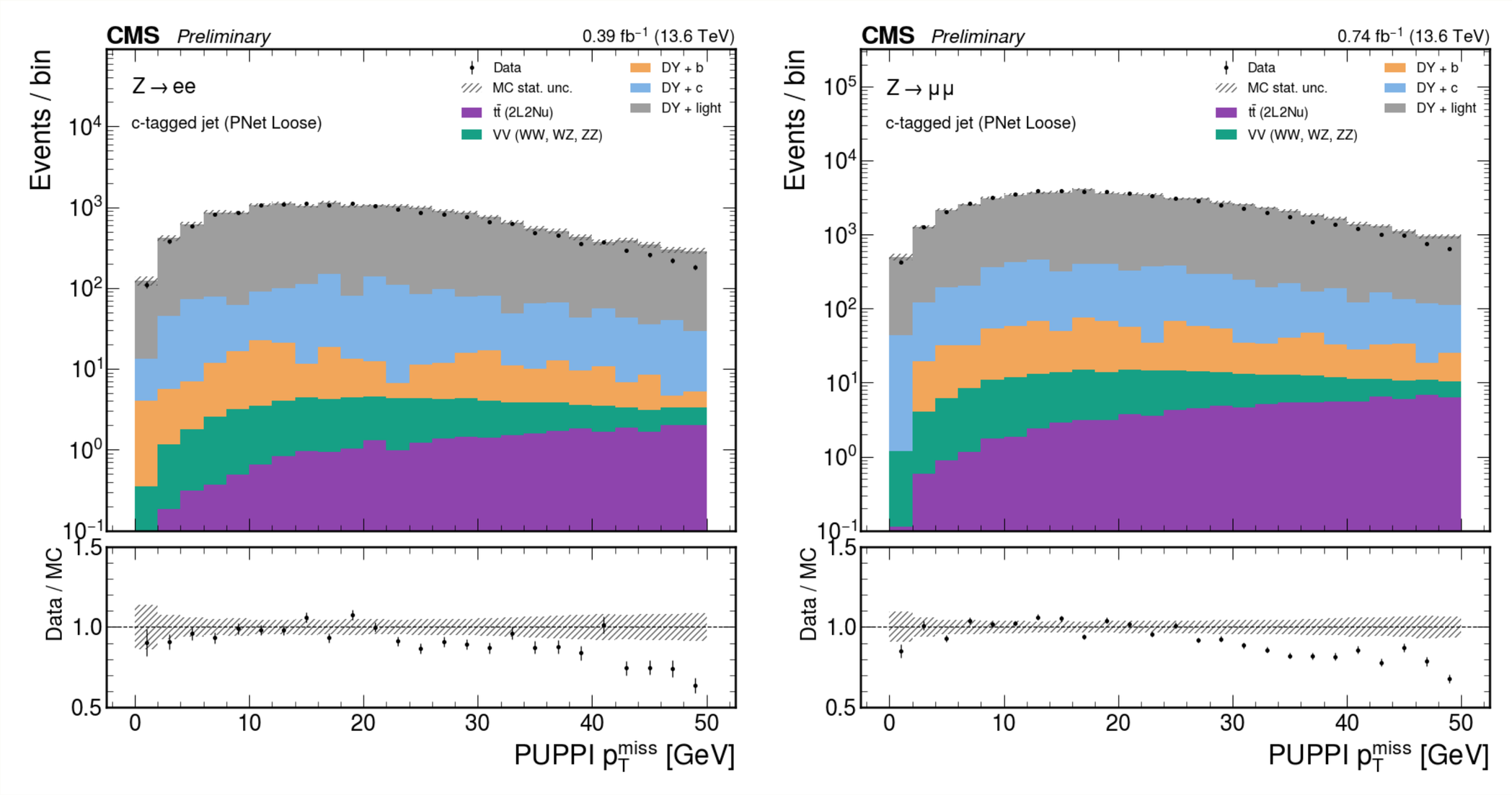
pT balance: leading-jet pT / Z pT



c-tag Loose

PUPPI MET

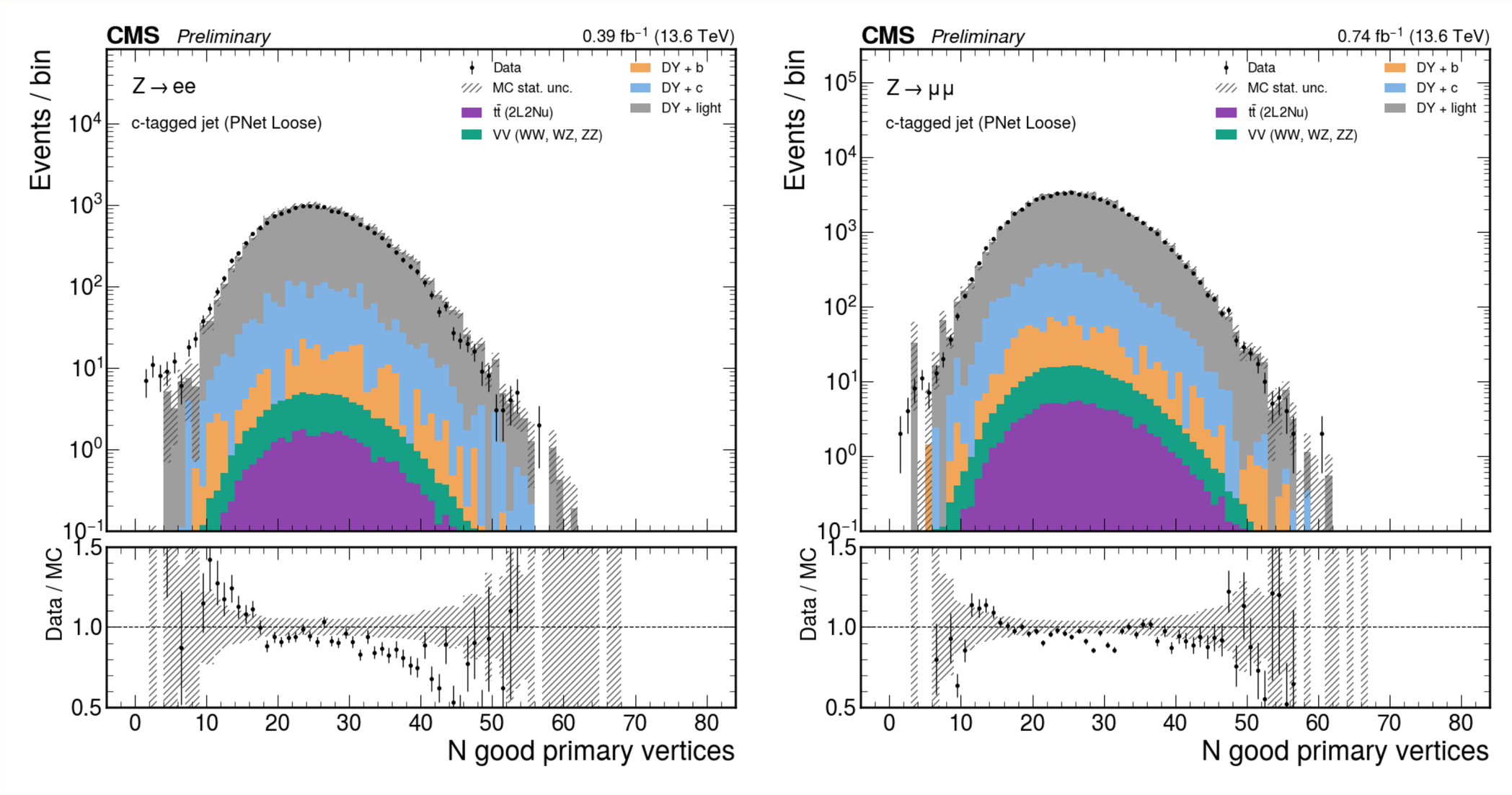
Type-1 corrected; the selection cuts at 50 GeV.



c-tag Loose

Number of good primary vertices

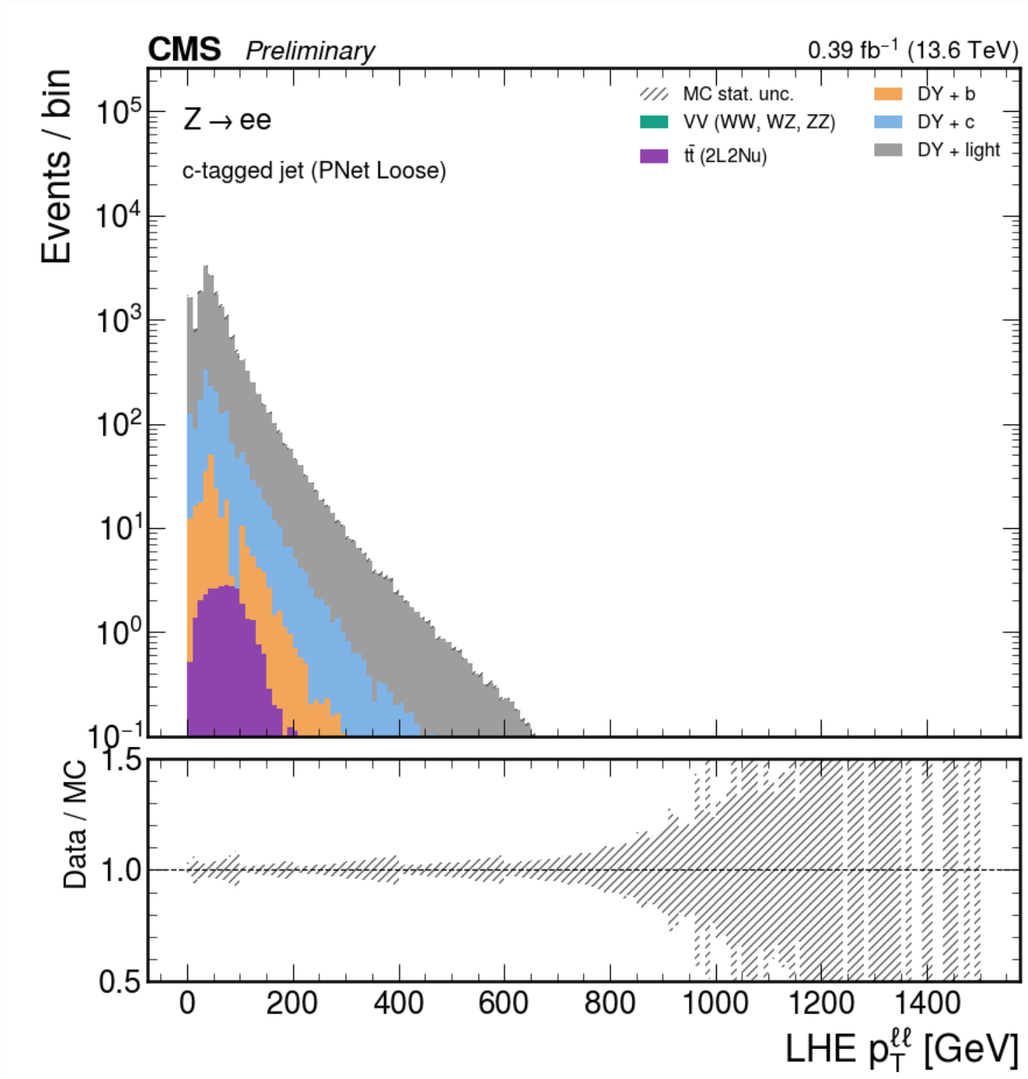
After pileup weights.

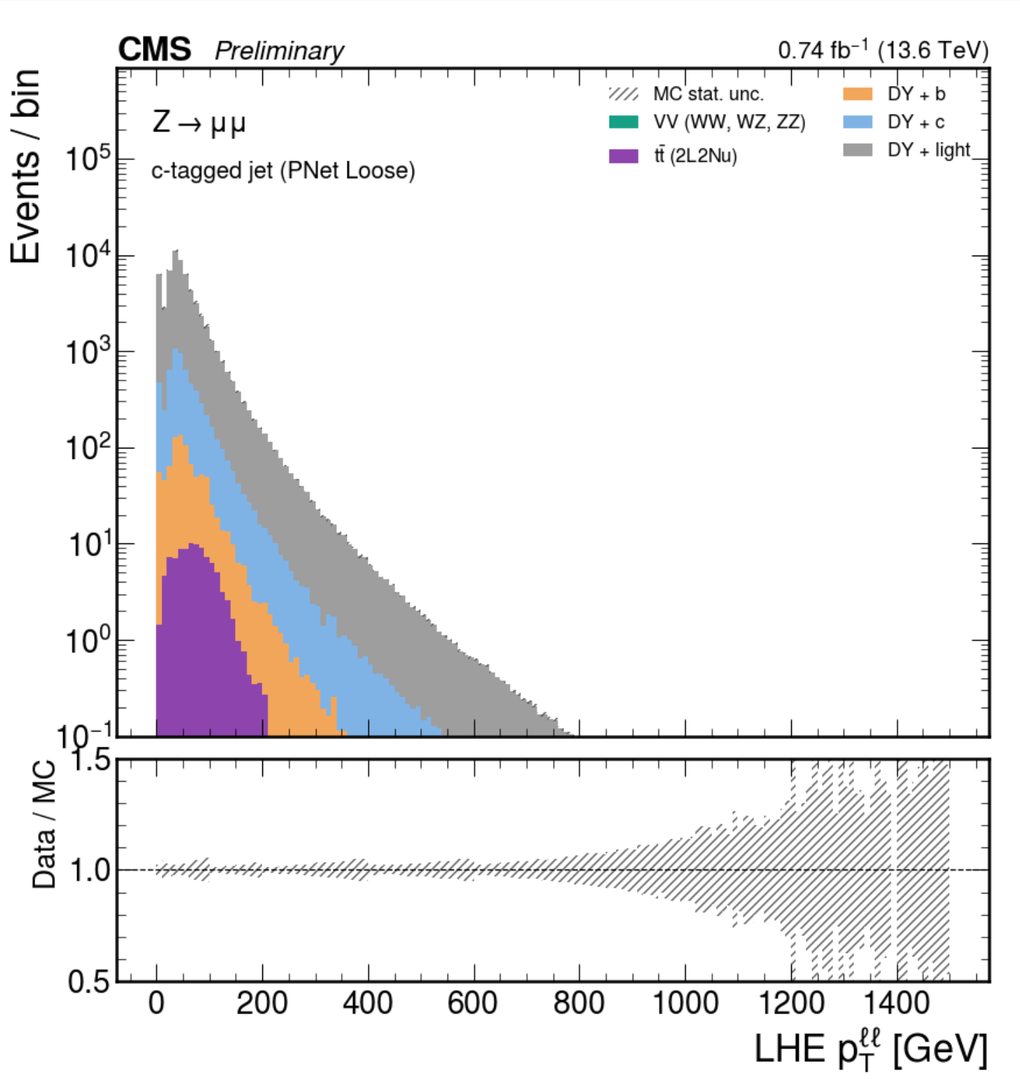


c-tag Loose

Generator Z pT (LHE)

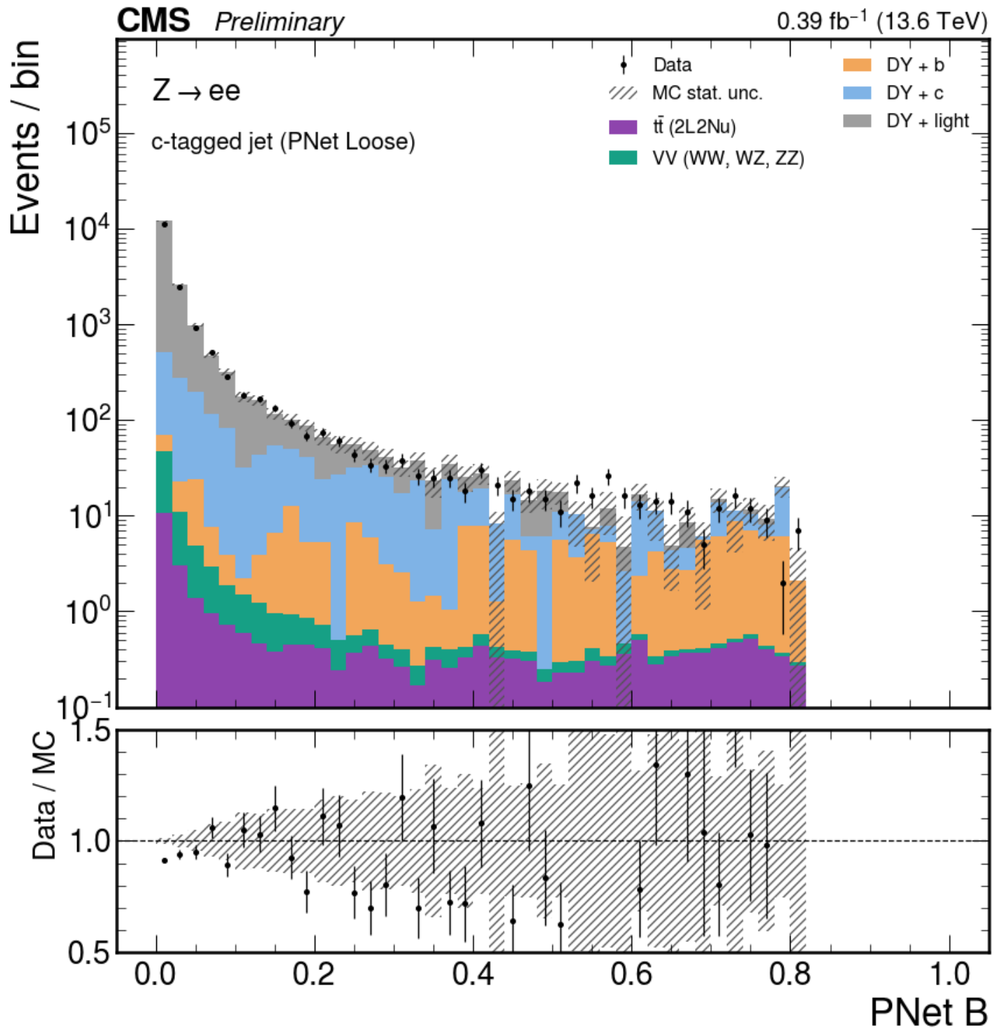
MC only, no data.

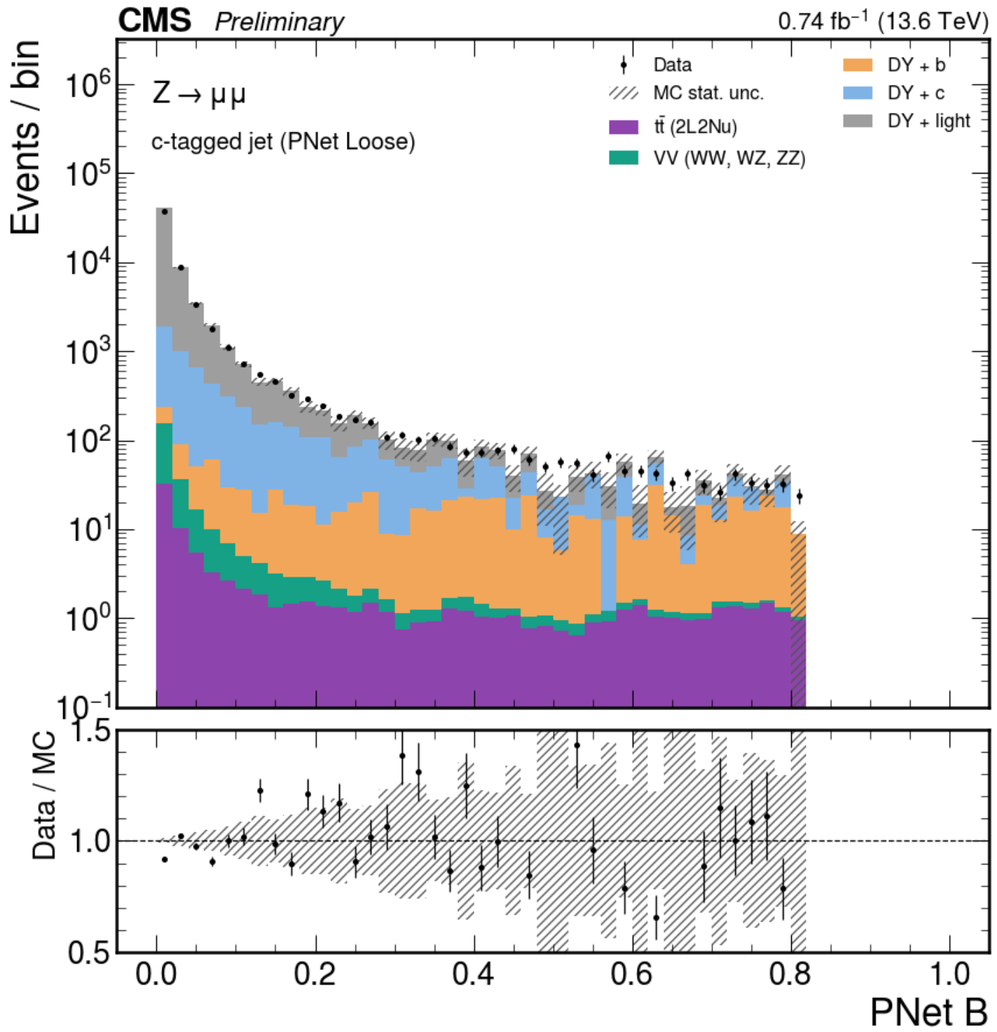




c-tag Loose

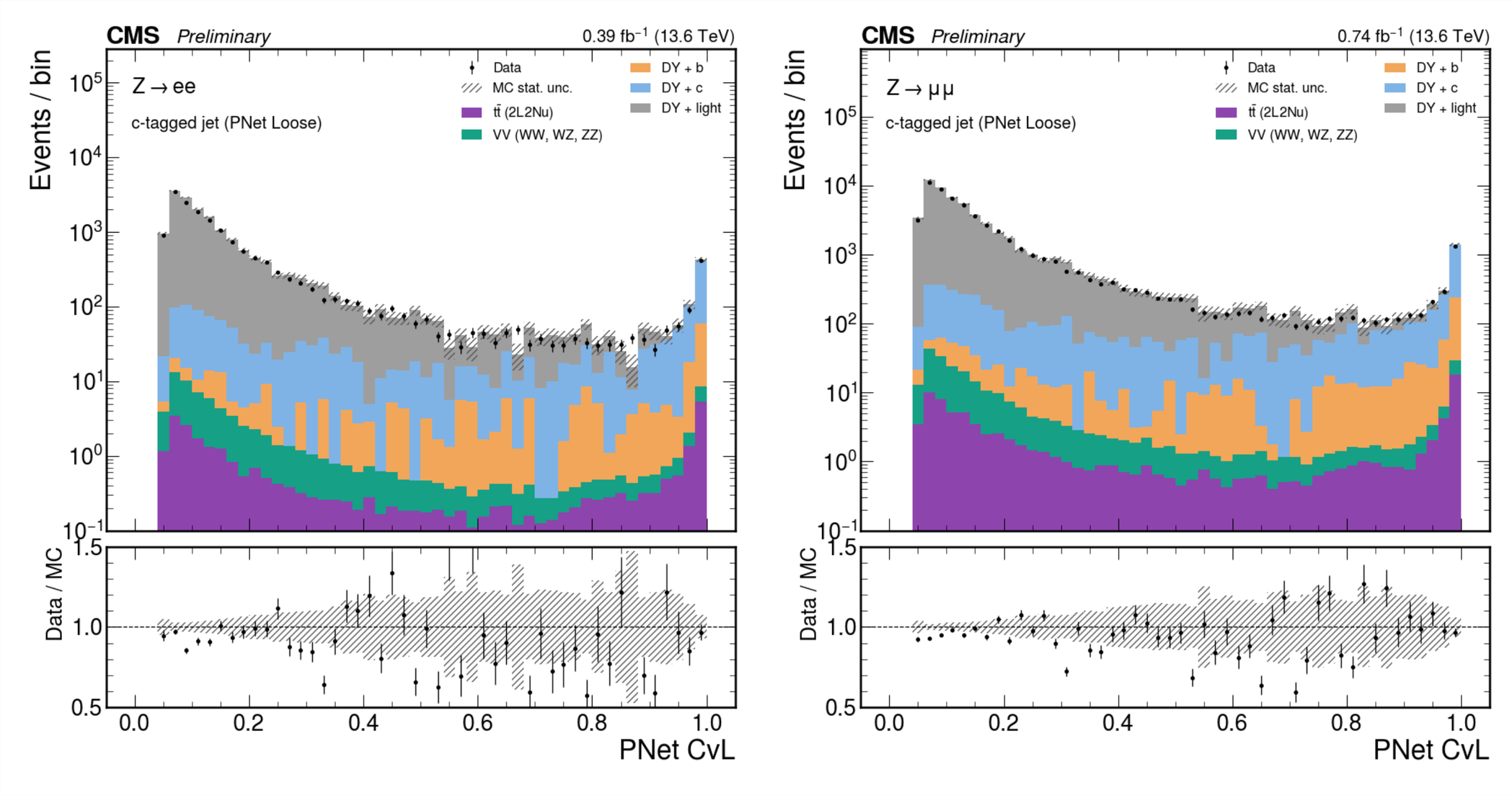
Leading-jet PNet B score





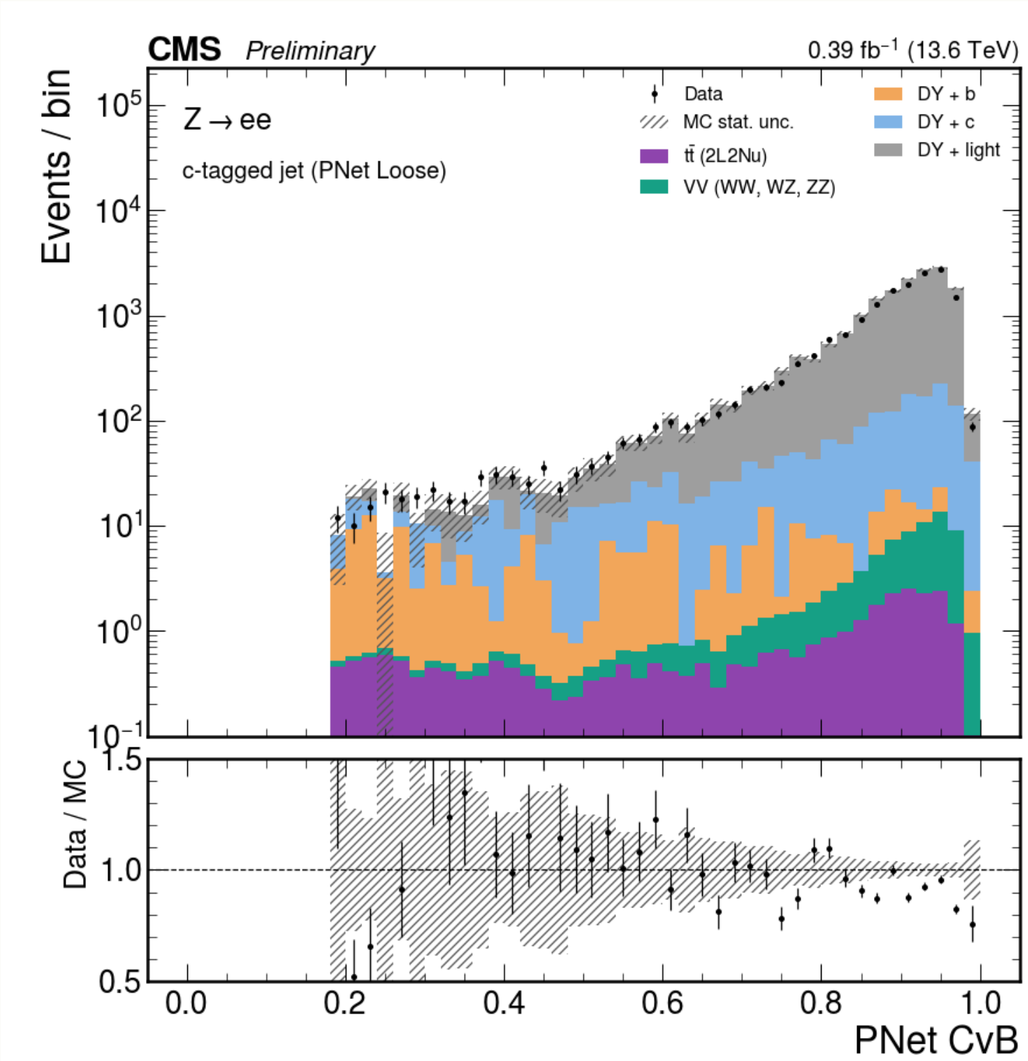
c-tag Loose

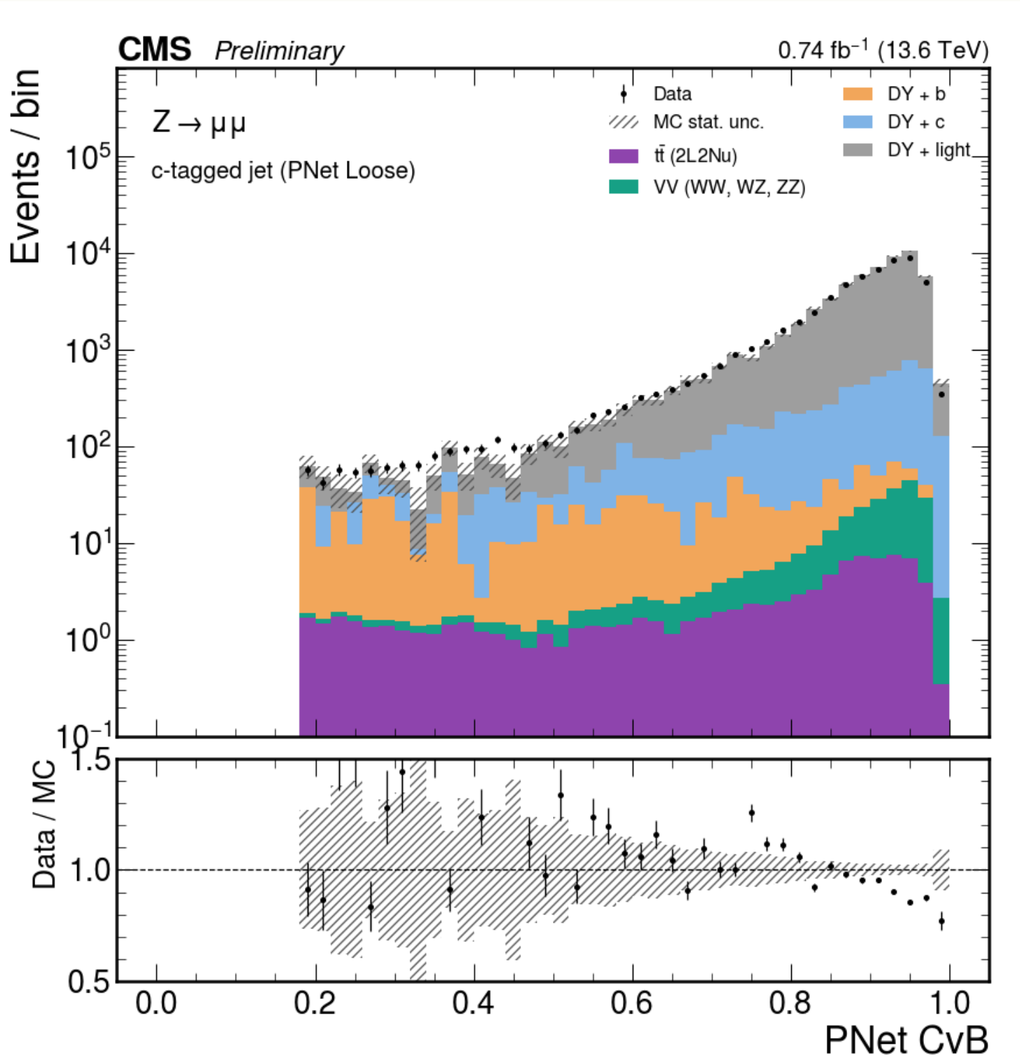
Leading-jet PNet CvL score



c-tag Loose

Leading-jet PNet CvB score

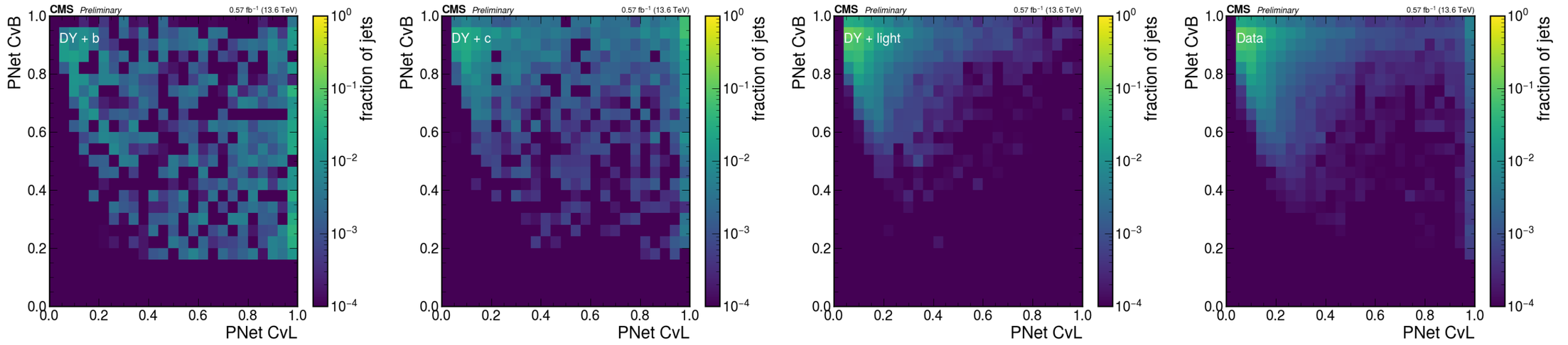




c-tag Loose

PNet CvL vs CvB of the leading jet

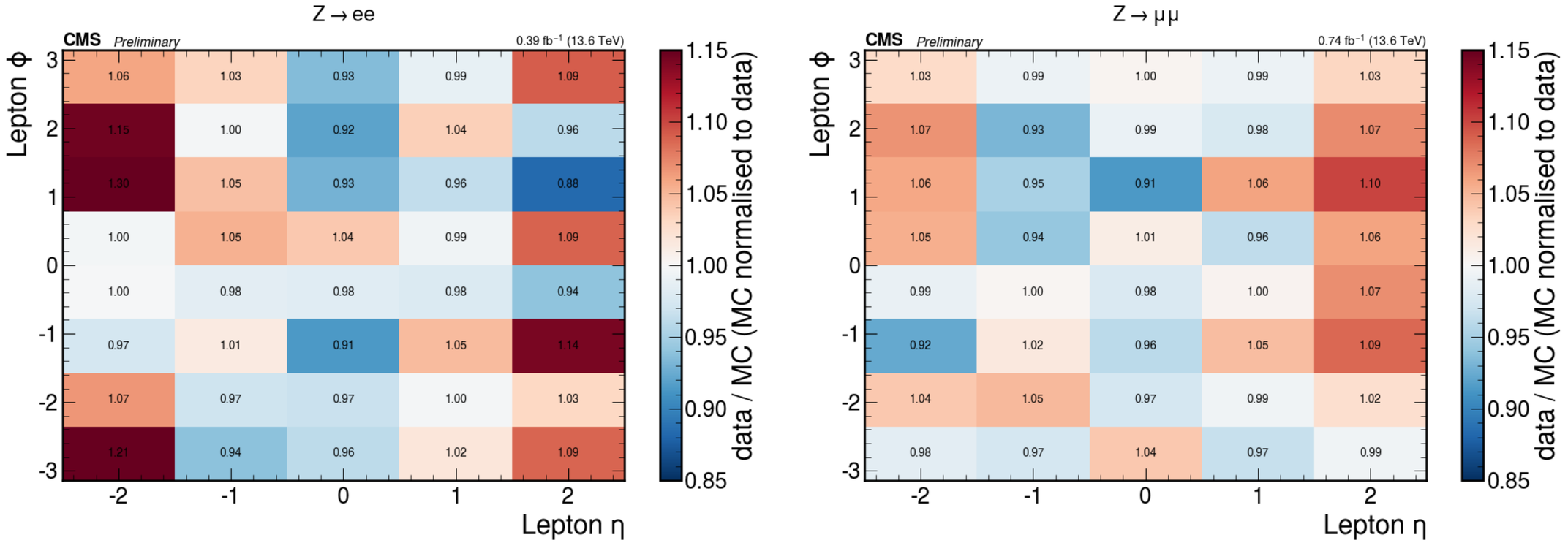
Each panel normalised to 1. Z→ee and Z→μμ added.



c-tag Loose

Lepton η–φ, data / MC

Both Z leptons. MC normalised to data in each map. Colour scale stops at ±15%; in the tagged regions there are few events per cell, so read the numbers.



b-tag Medium

Leading jet PNet B > 0.245

	data/MC	DY+b	DY+c	DY+light	tt	tagged MC	tagged data
Z→ee	0.960 ± 0.053	55.1%	22.1%	17.8%	4.5%	5.9%	5.9%
Z→μμ	0.939 ± 0.036	59.0%	16.9%	19.4%	4.3%	6.0%	5.8%

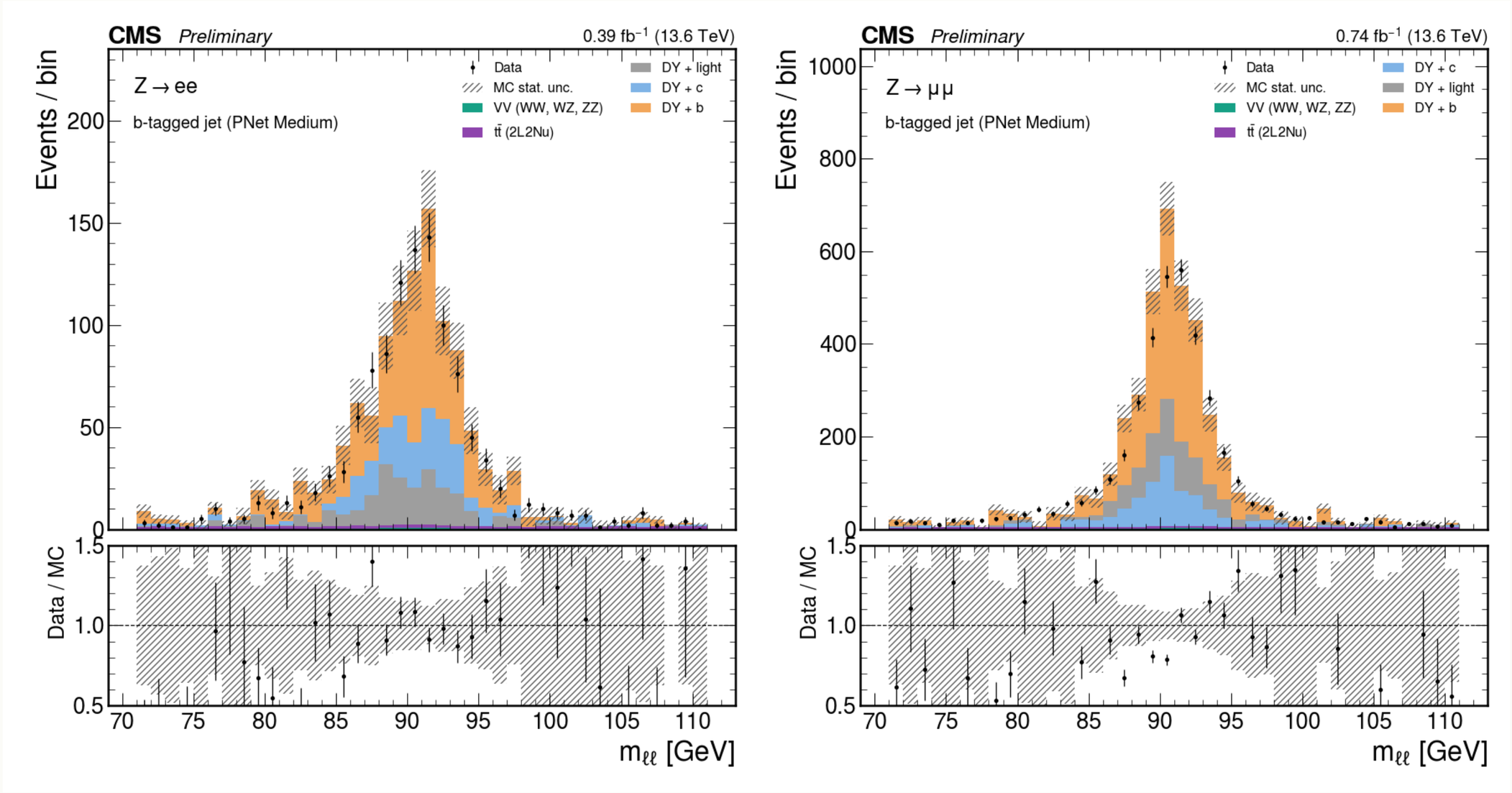
Flavour shares are of all MC. 'Tagged' = fraction of Z + jet events passing the tag. Errors are statistical only.

32 plots follow: Z and leptons, jets, Z–jet, MET and vertices, tagger scores, 2D maps. Left Z→ee, right Z→μμ.

b-tag Medium

Dilepton mass

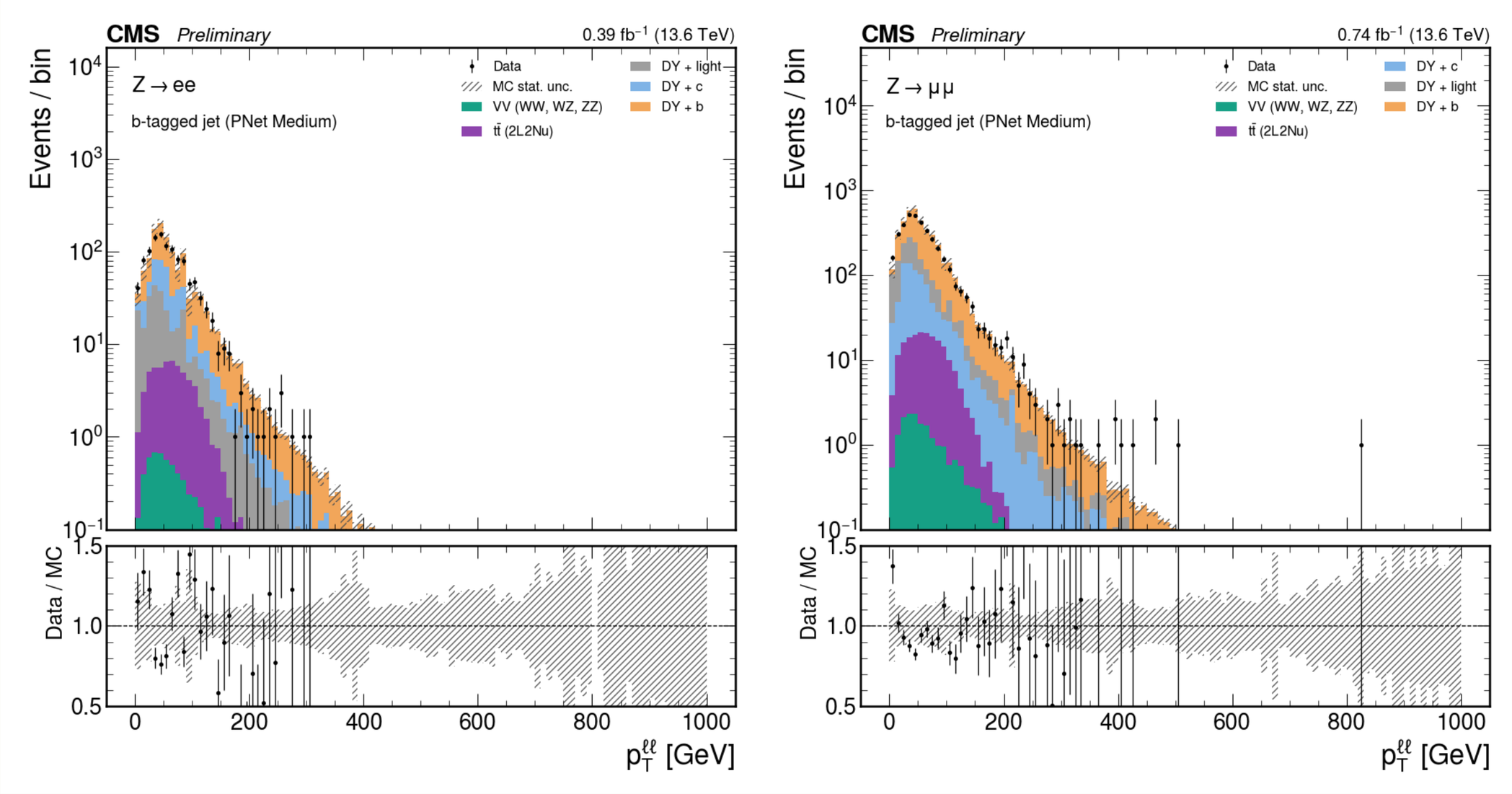
m(ℓℓ), 71–111 GeV window, linear scale.



b-tag Medium

Z pT

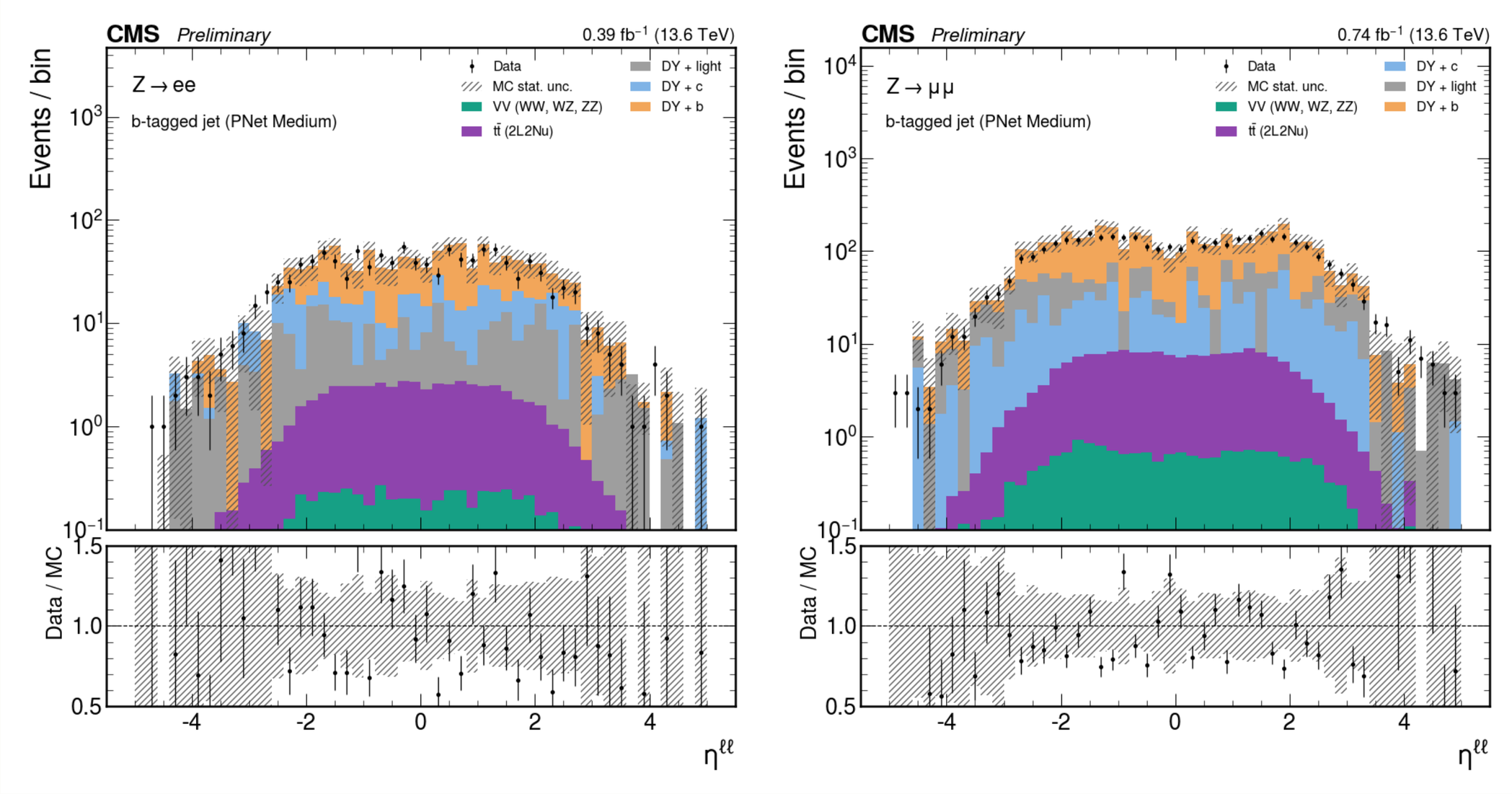
pT of the lepton pair.



b-tag Medium

Z η

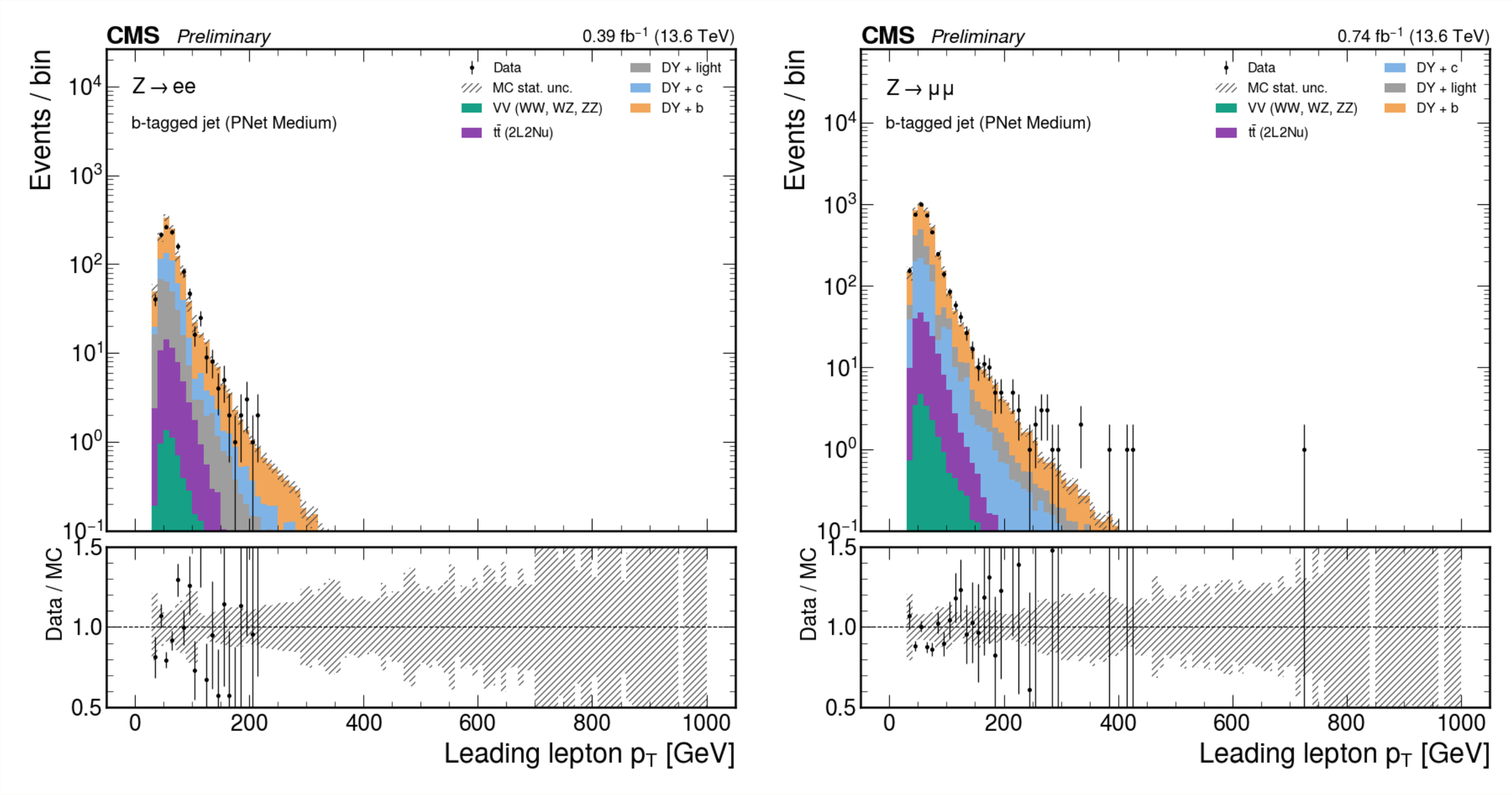
η of the lepton pair.



b-tag Medium

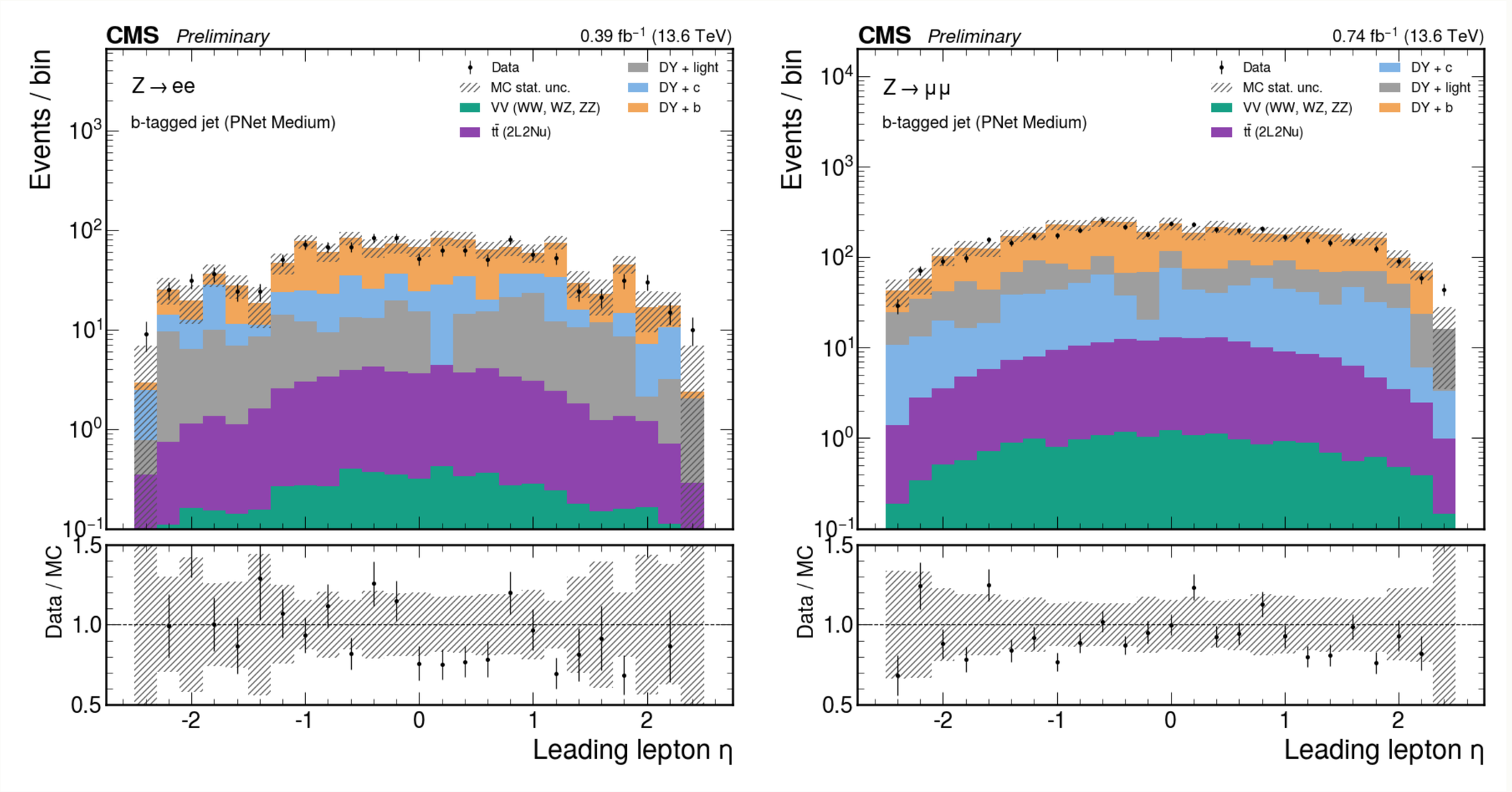
Leading-lepton pT

Leading lepton pT > 35 GeV (trigger plateau).



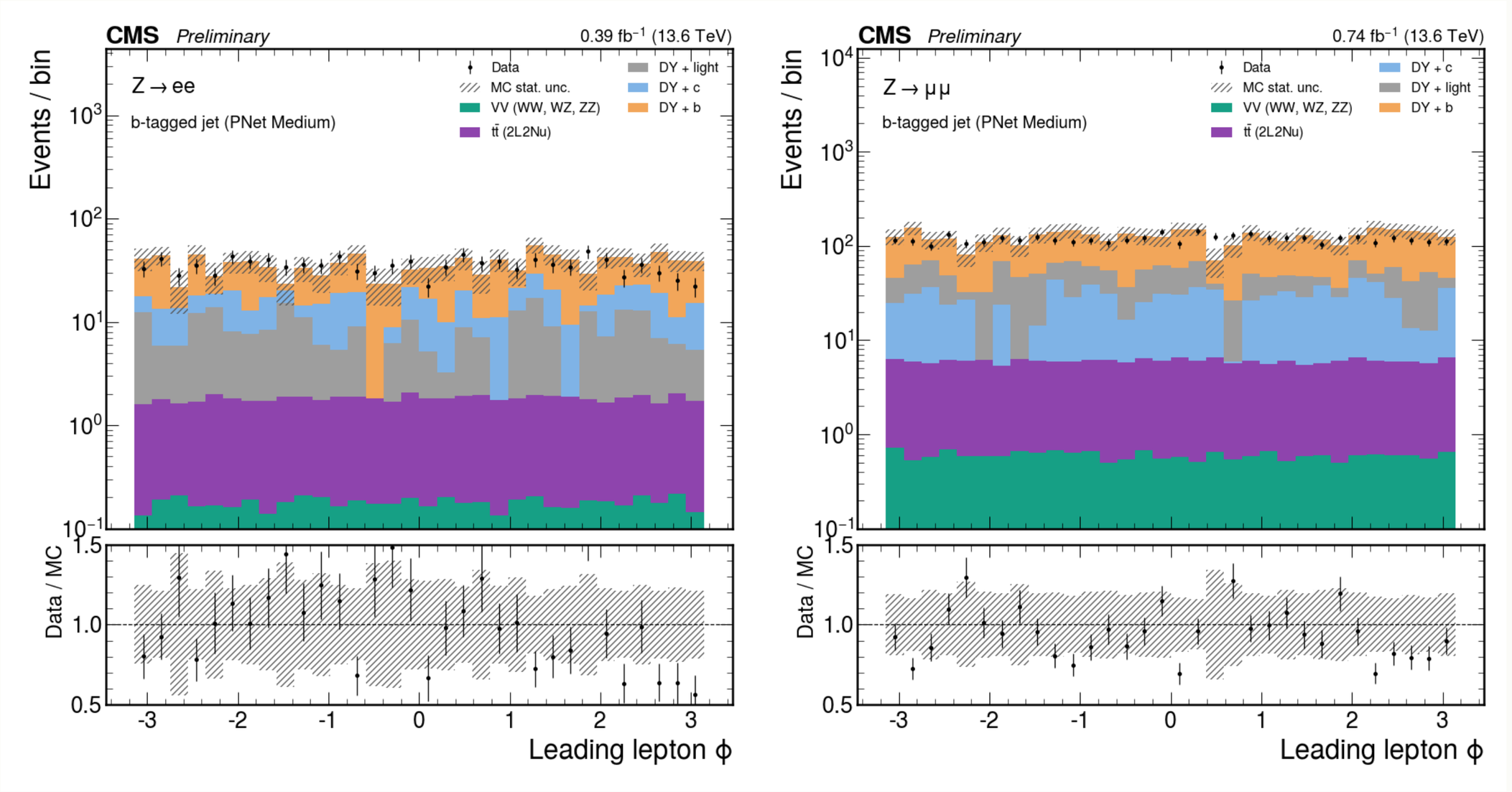
b-tag Medium

Leading-lepton η



b-tag Medium

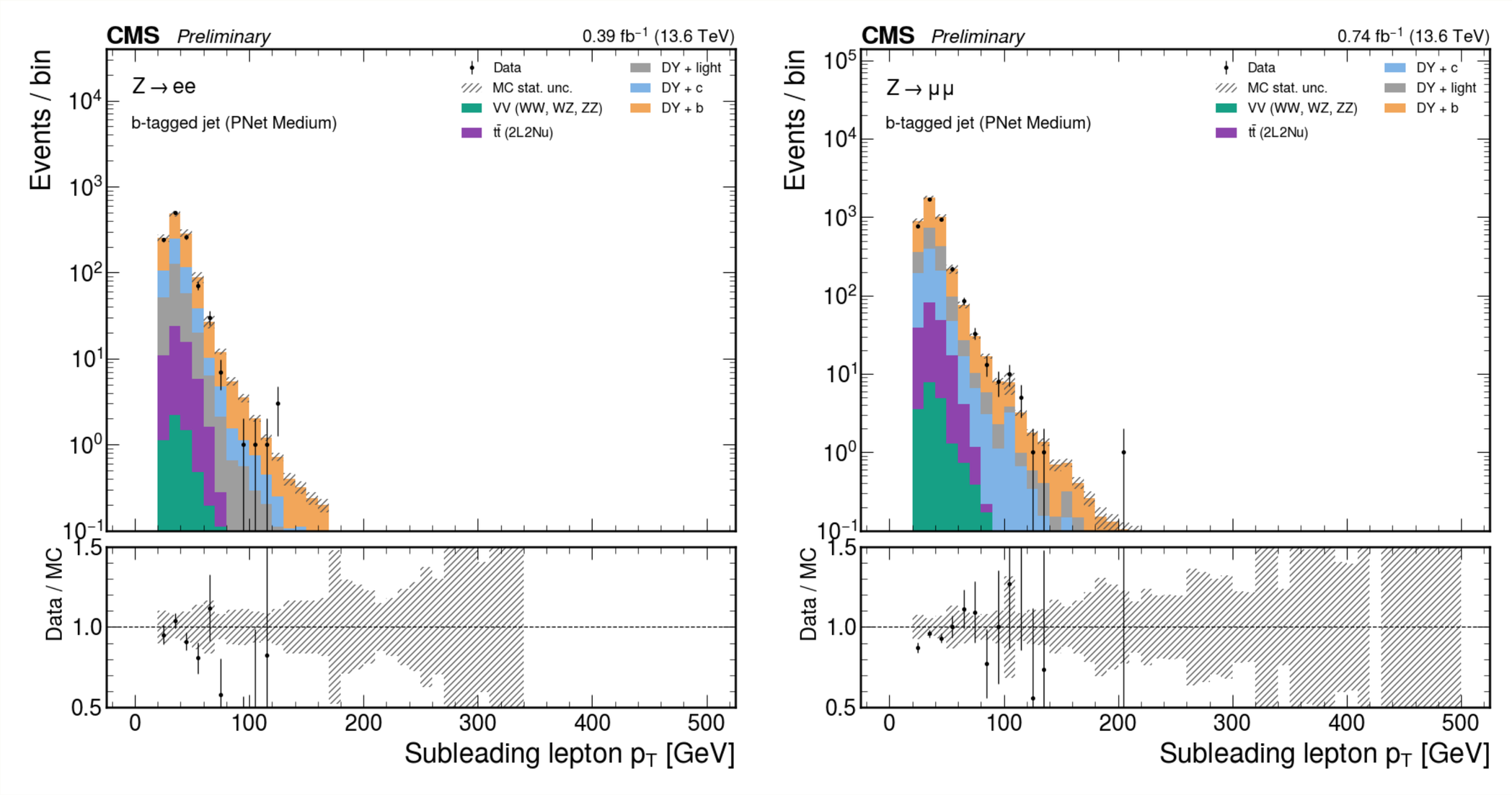
Leading-lepton φ



b-tag Medium

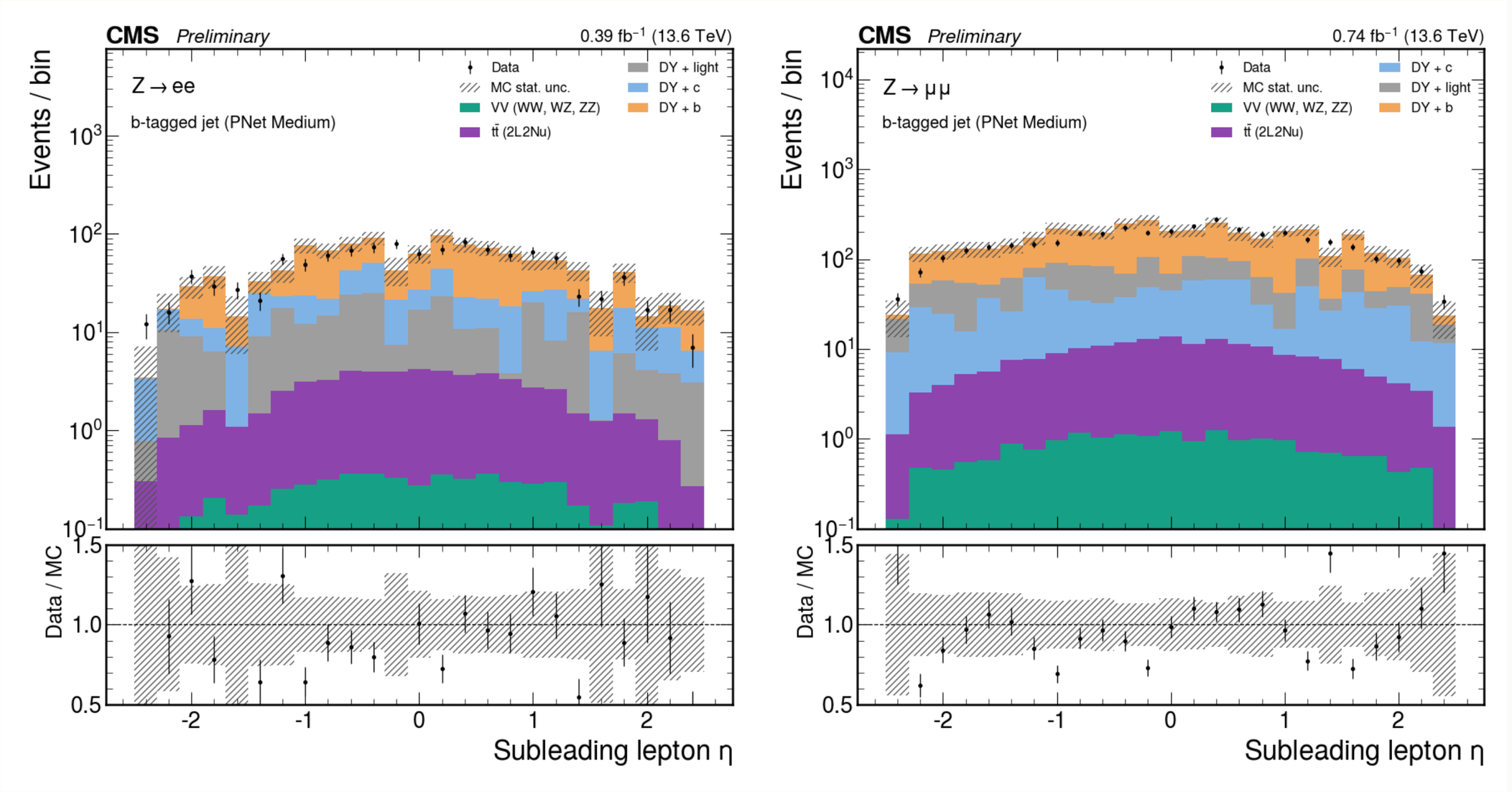
Second-lepton pT

Second lepton pT > 25 GeV.



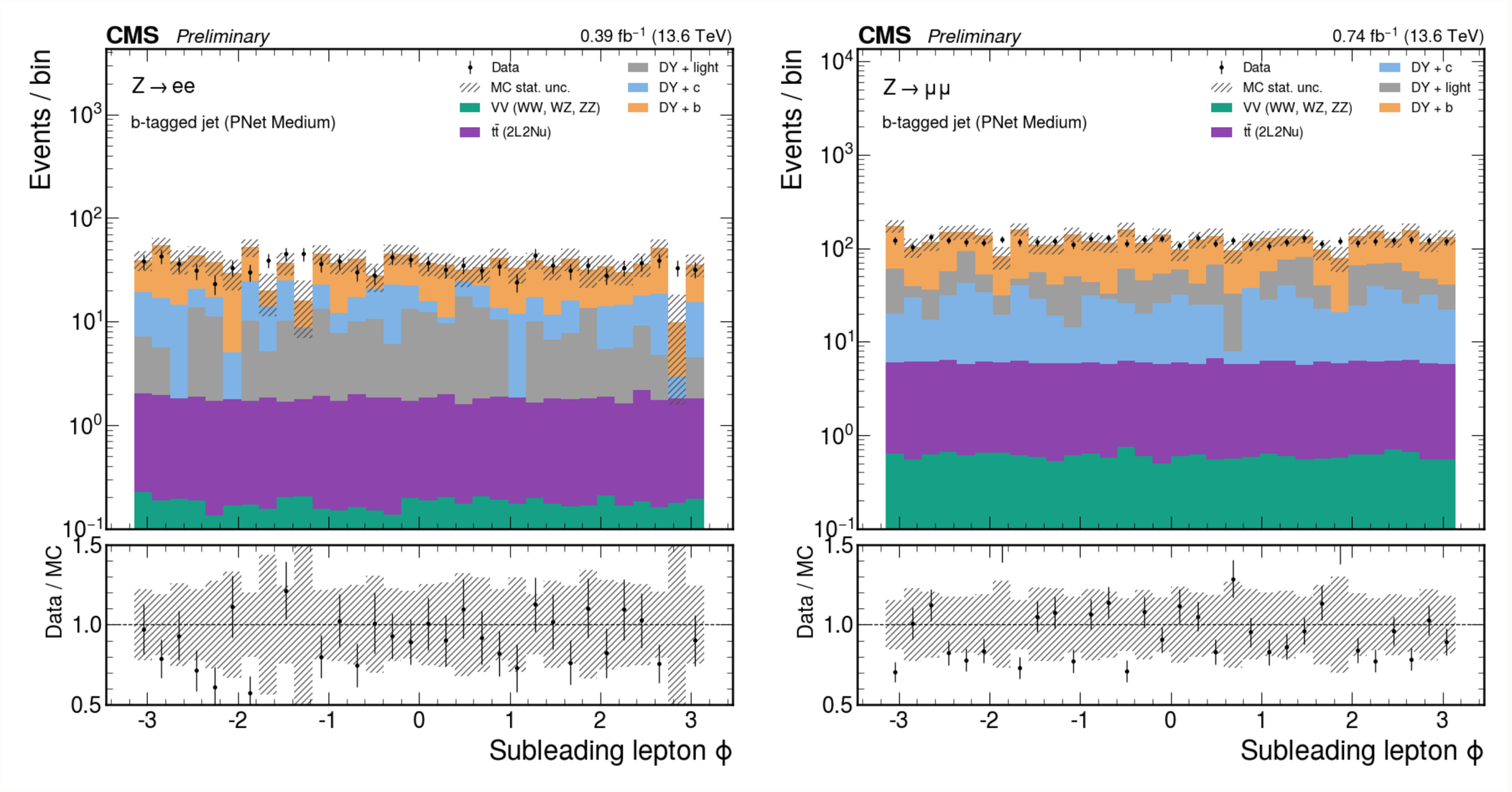
b-tag Medium

Second-lepton η



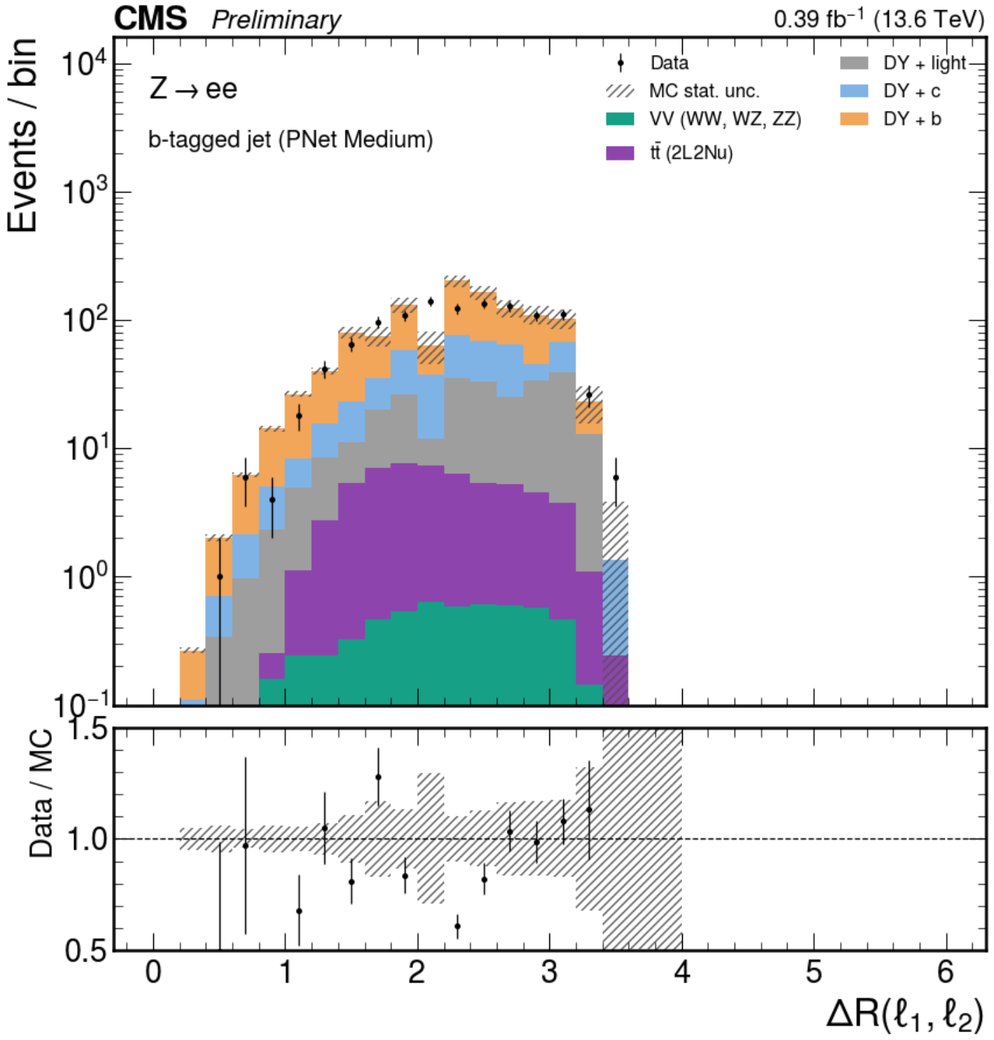
b-tag Medium

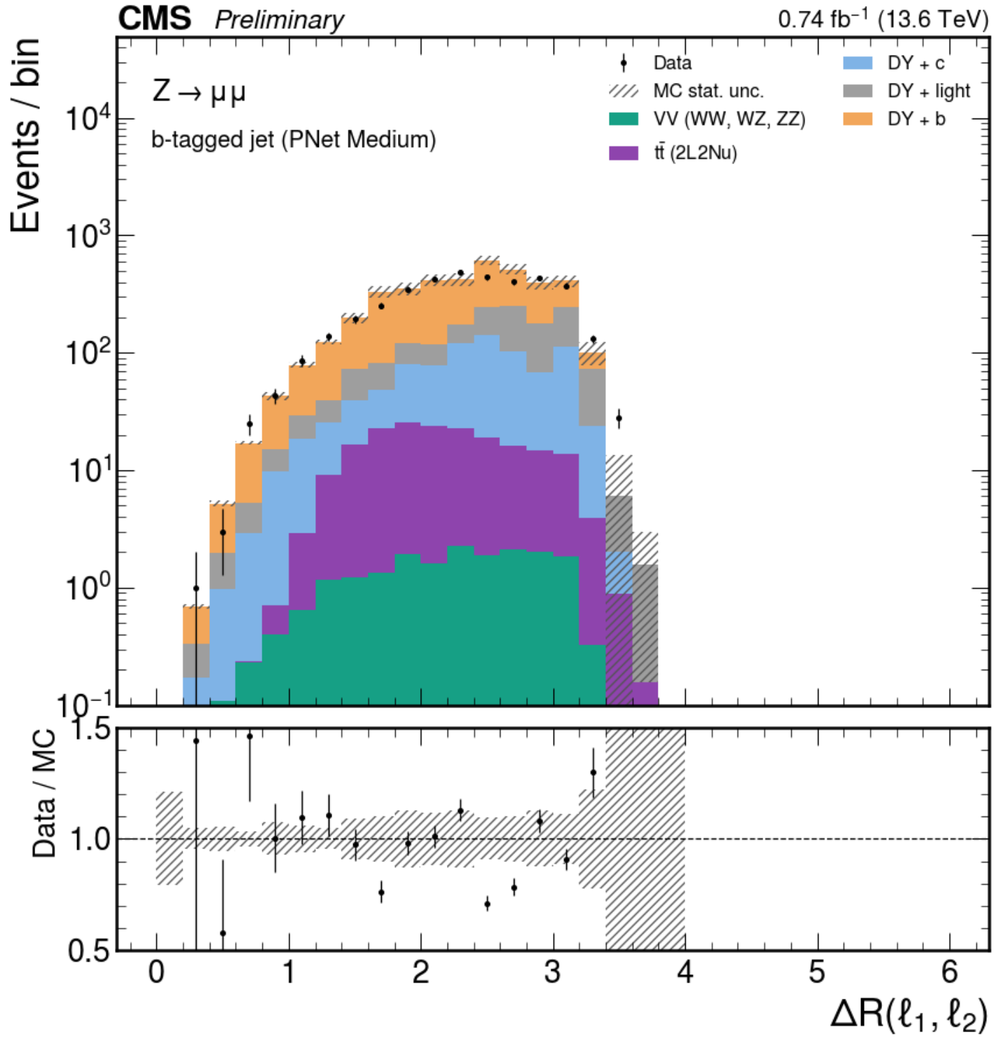
Second-lepton φ



b-tag Medium

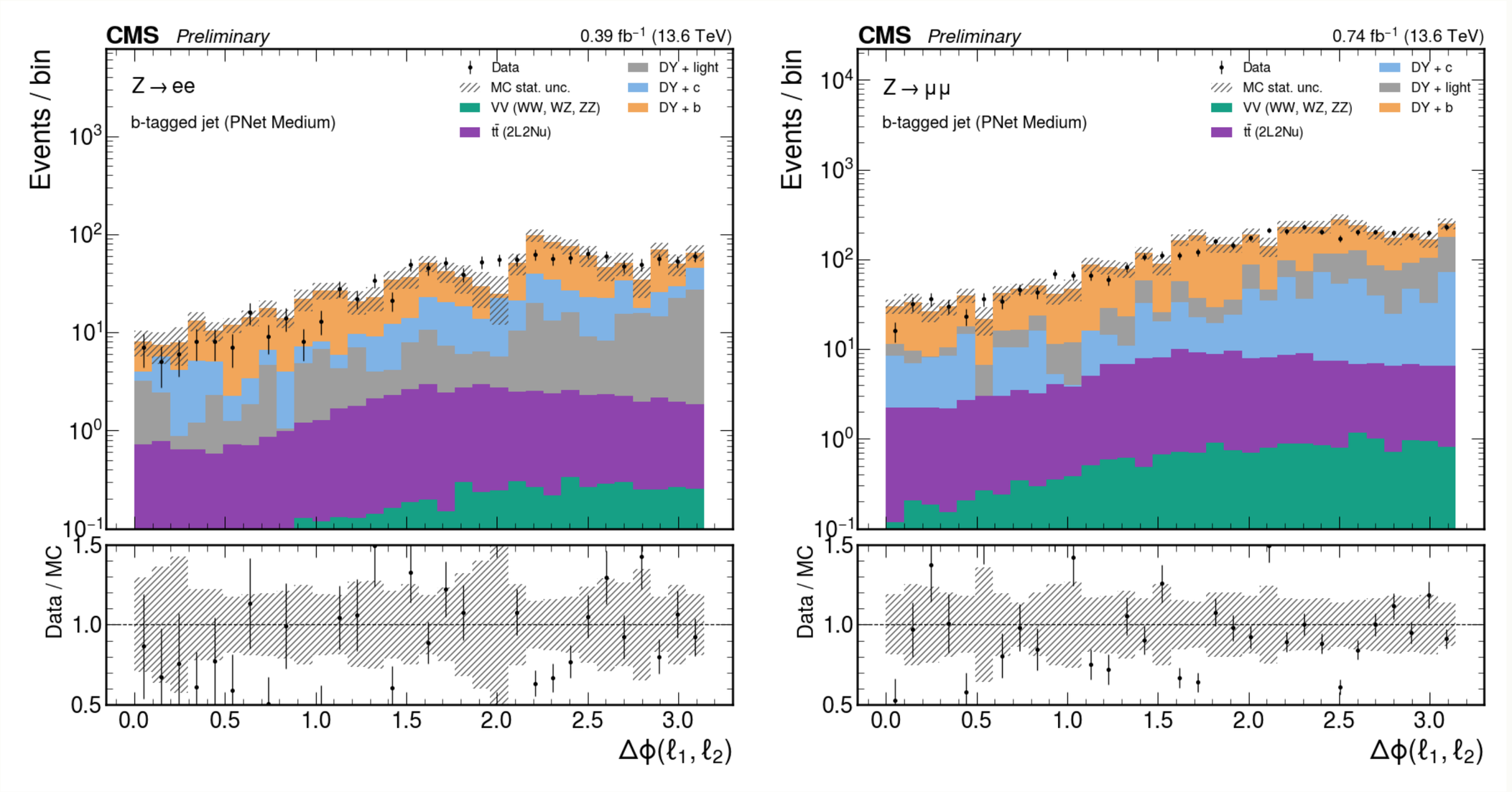
ΔR between the leptons





b-tag Medium

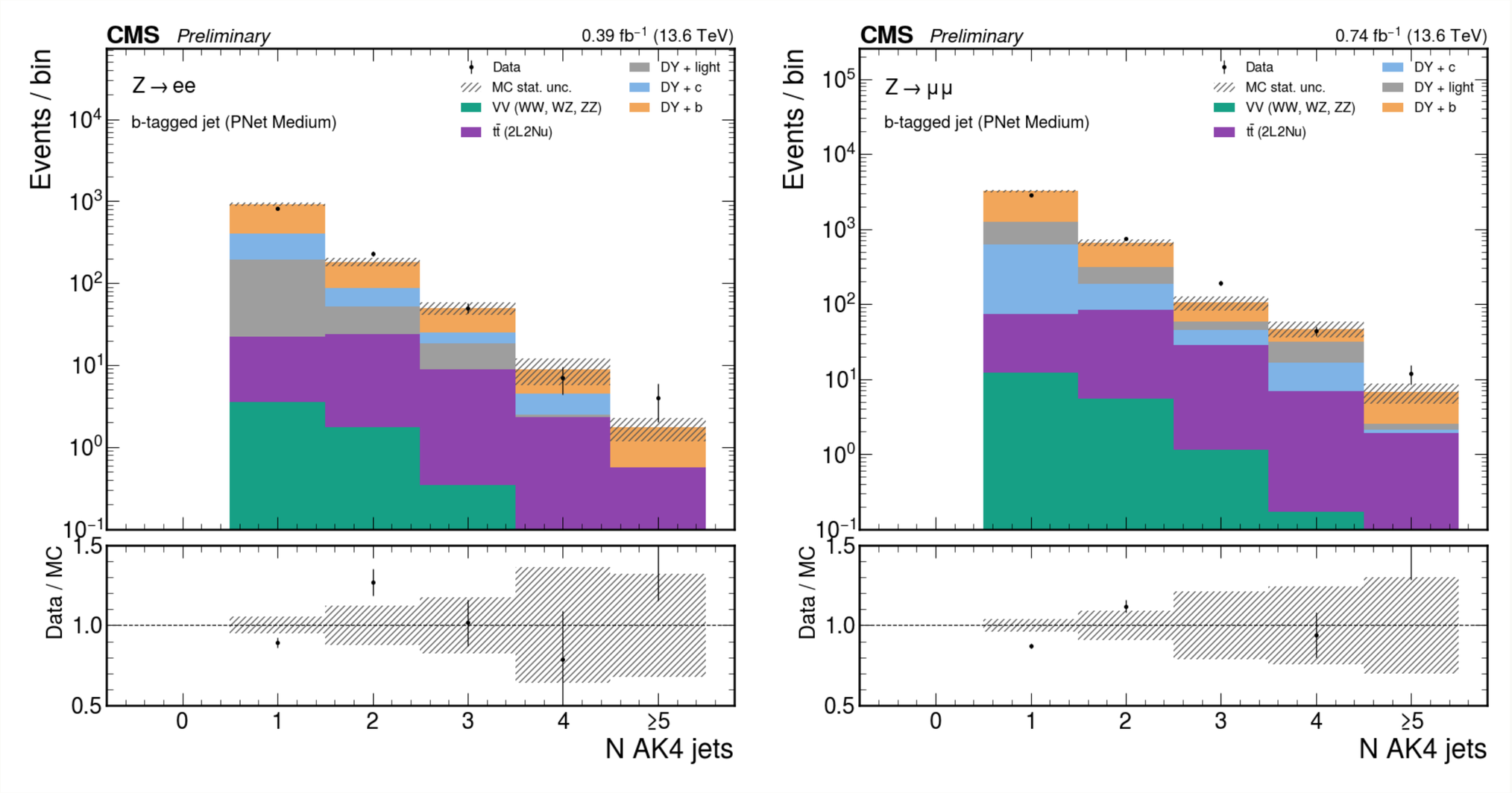
Δφ between the leptons



b-tag Medium

Number of jets

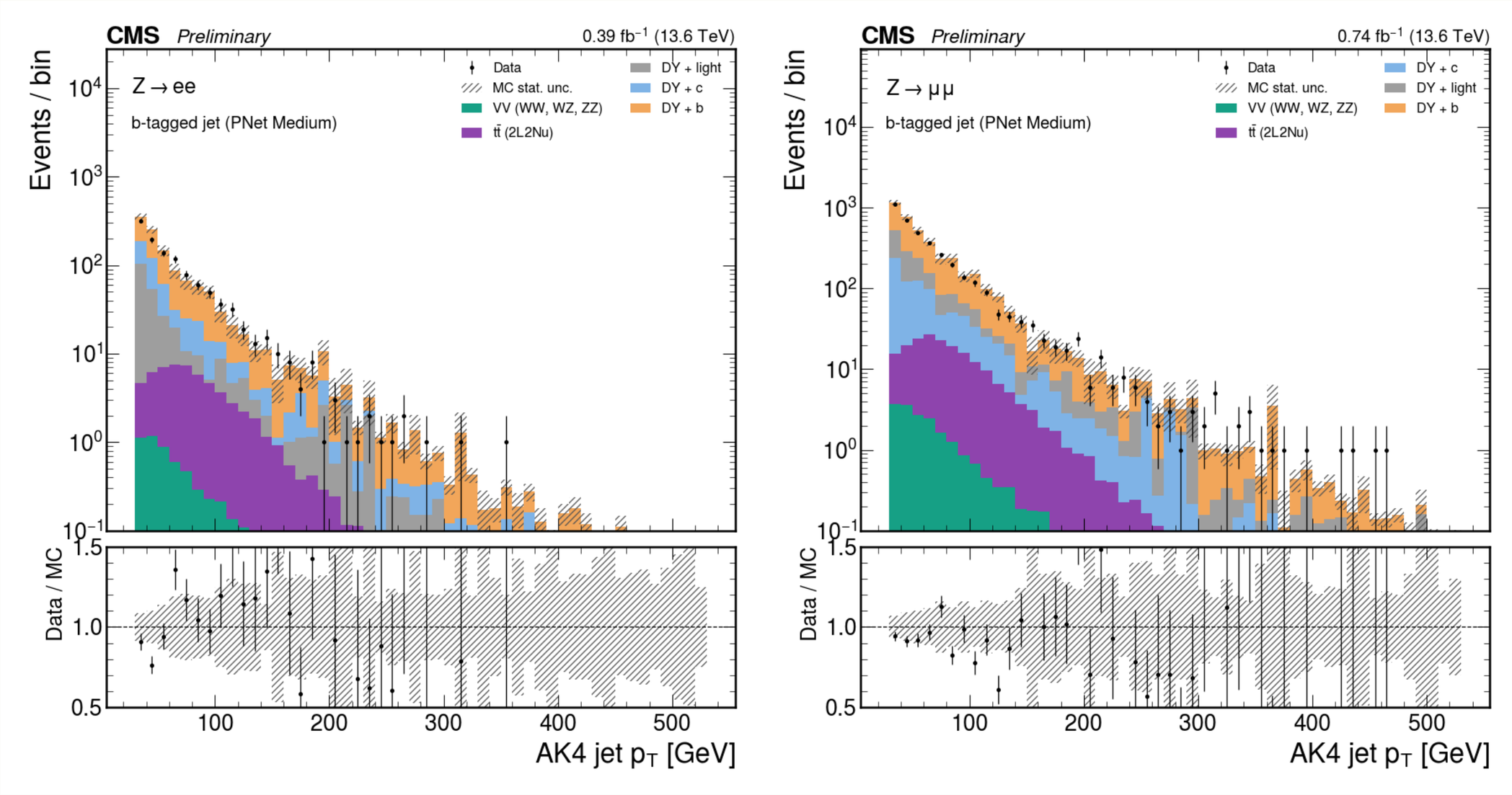
AK4 jets with pT > 30 GeV, |η| < 2.5, away from the leptons. Untagged plot: Z events before the jet cut, so the 0-jet bin is filled.



b-tag Medium

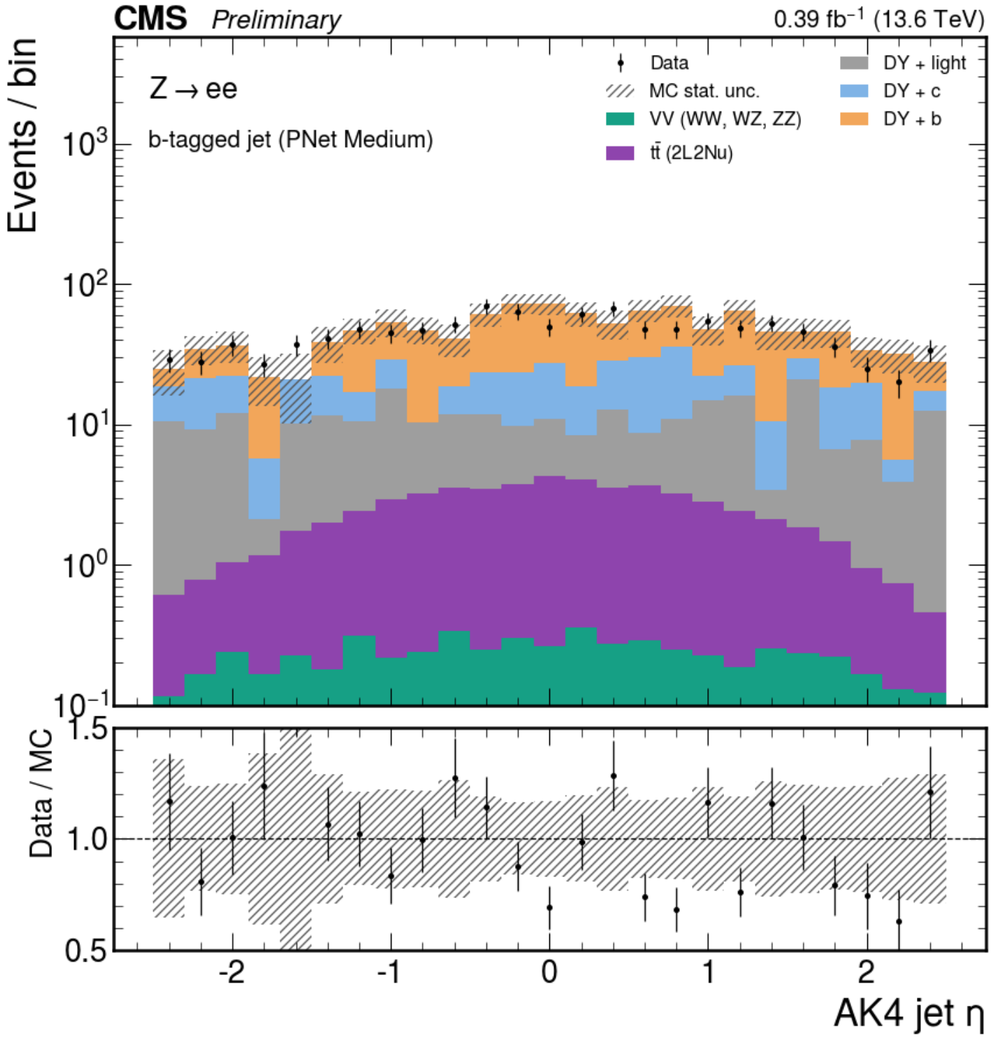
Leading-jet pT

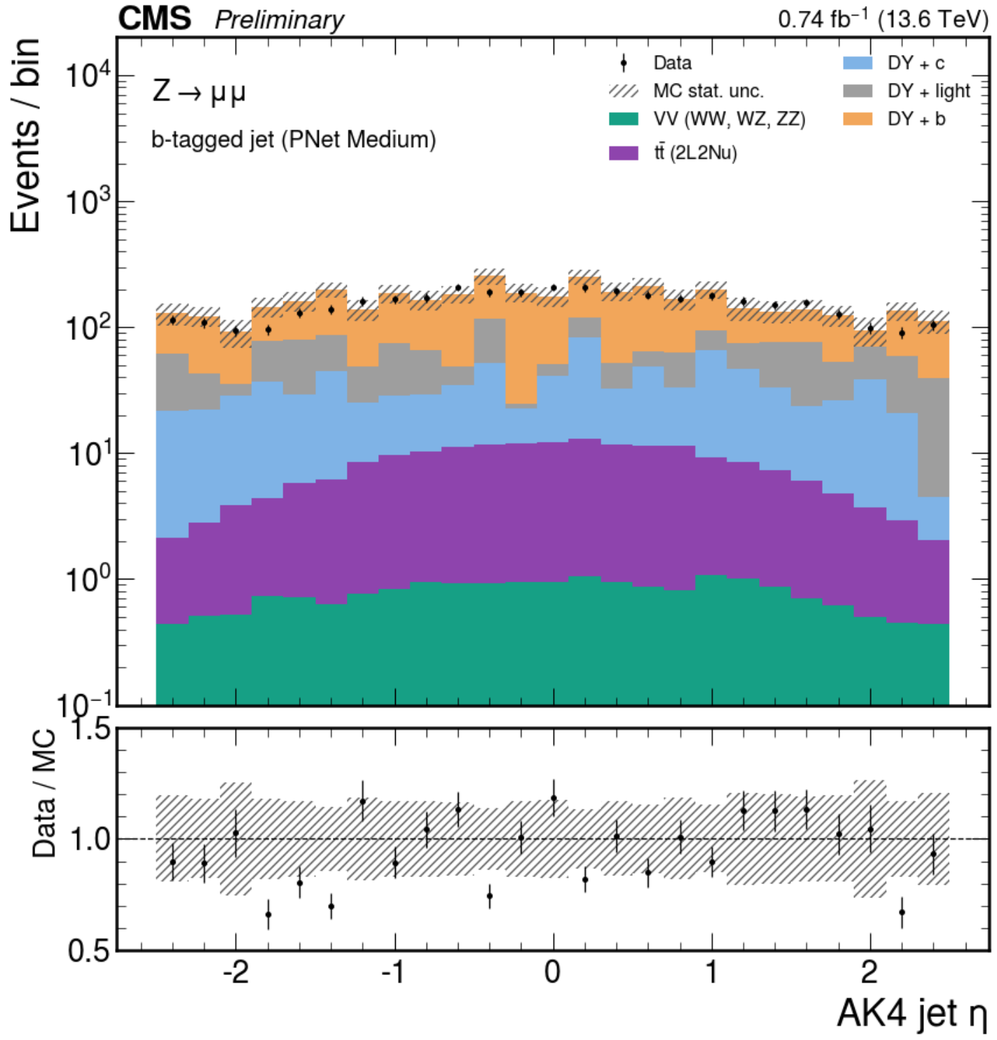
The tag (if any) is on this jet.



b-tag Medium

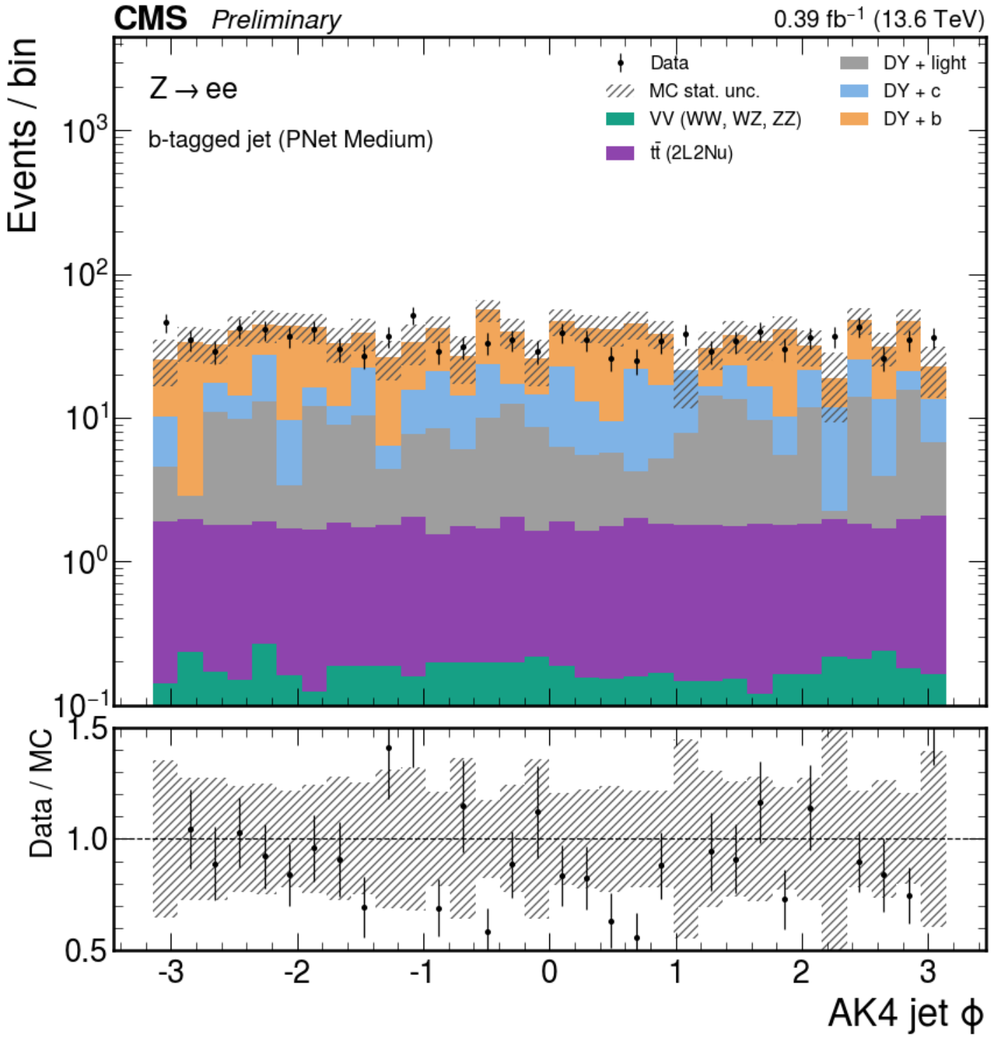
Leading-jet η

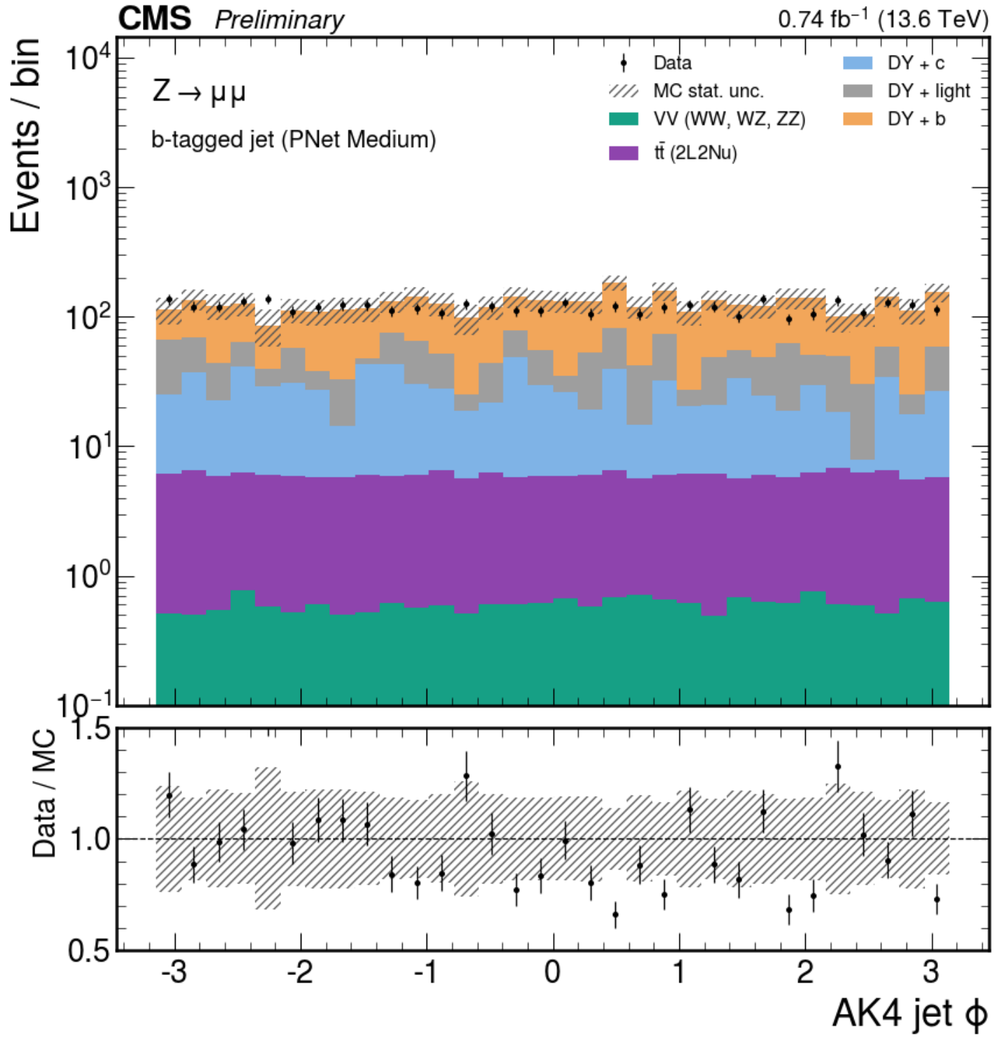




b-tag Medium

Leading-jet φ



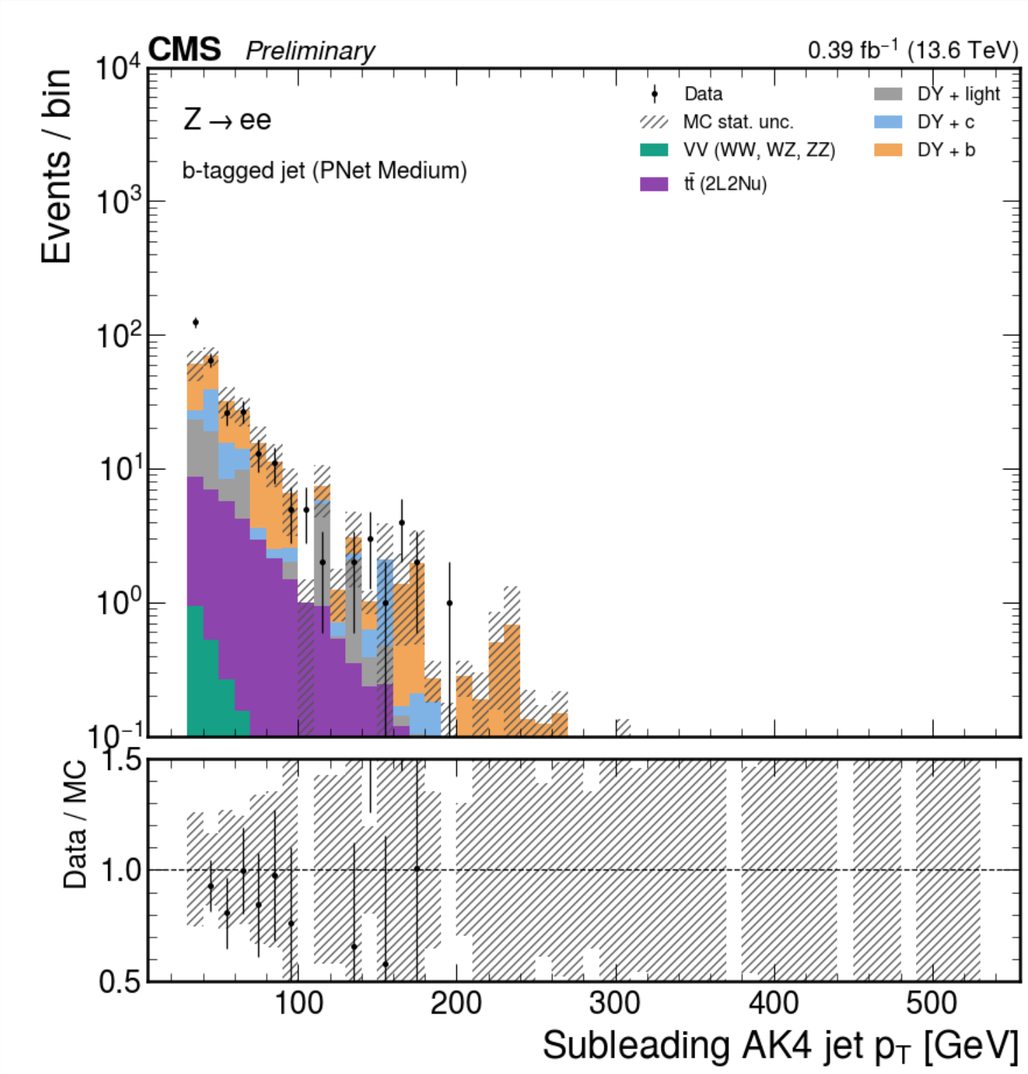


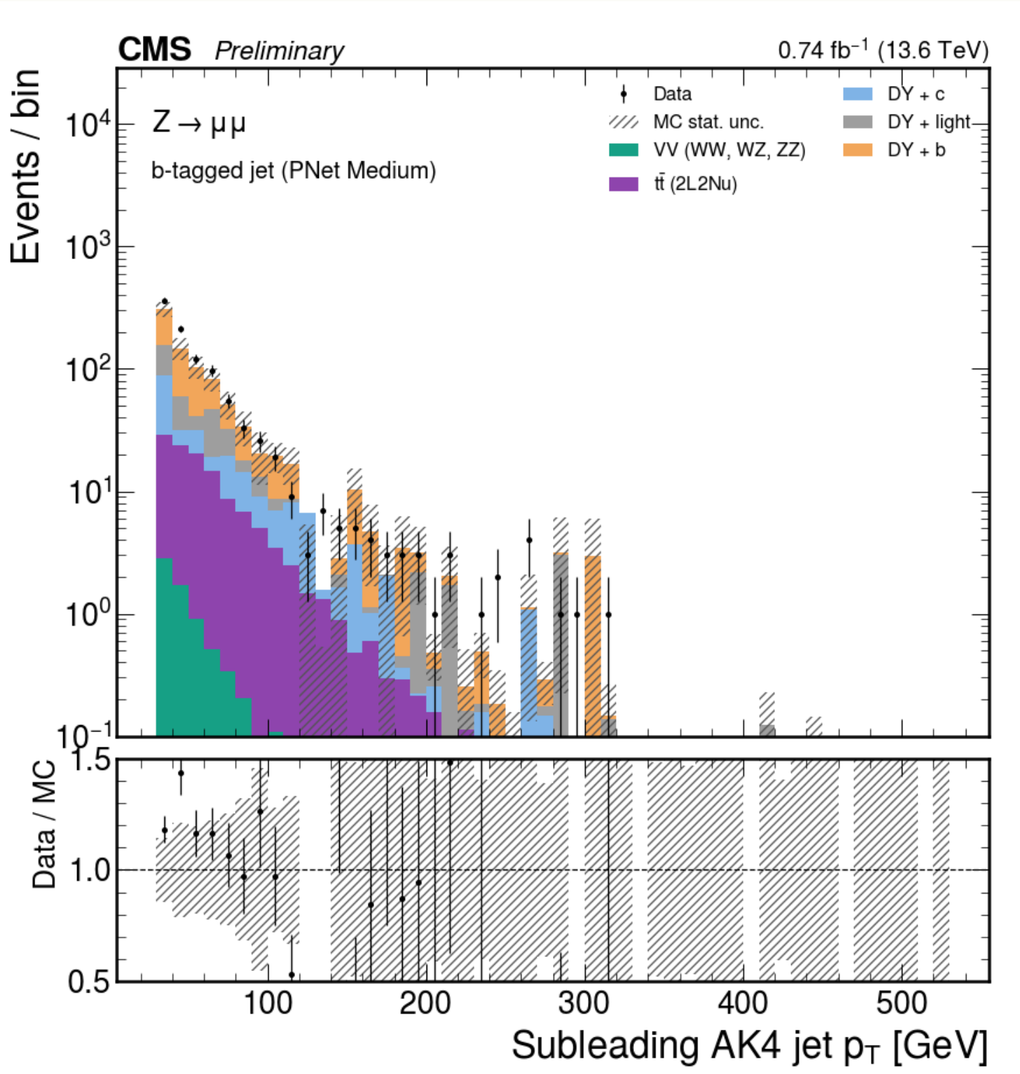
Summer22 (2022 C–D), 13.6 TeV · Z→ℓℓ + jets · 20 files per MC sample and per data channel · AK4 jets · stat. errors only · 120

b-tag Medium

Second-jet pT

Events with at least 2 jets.

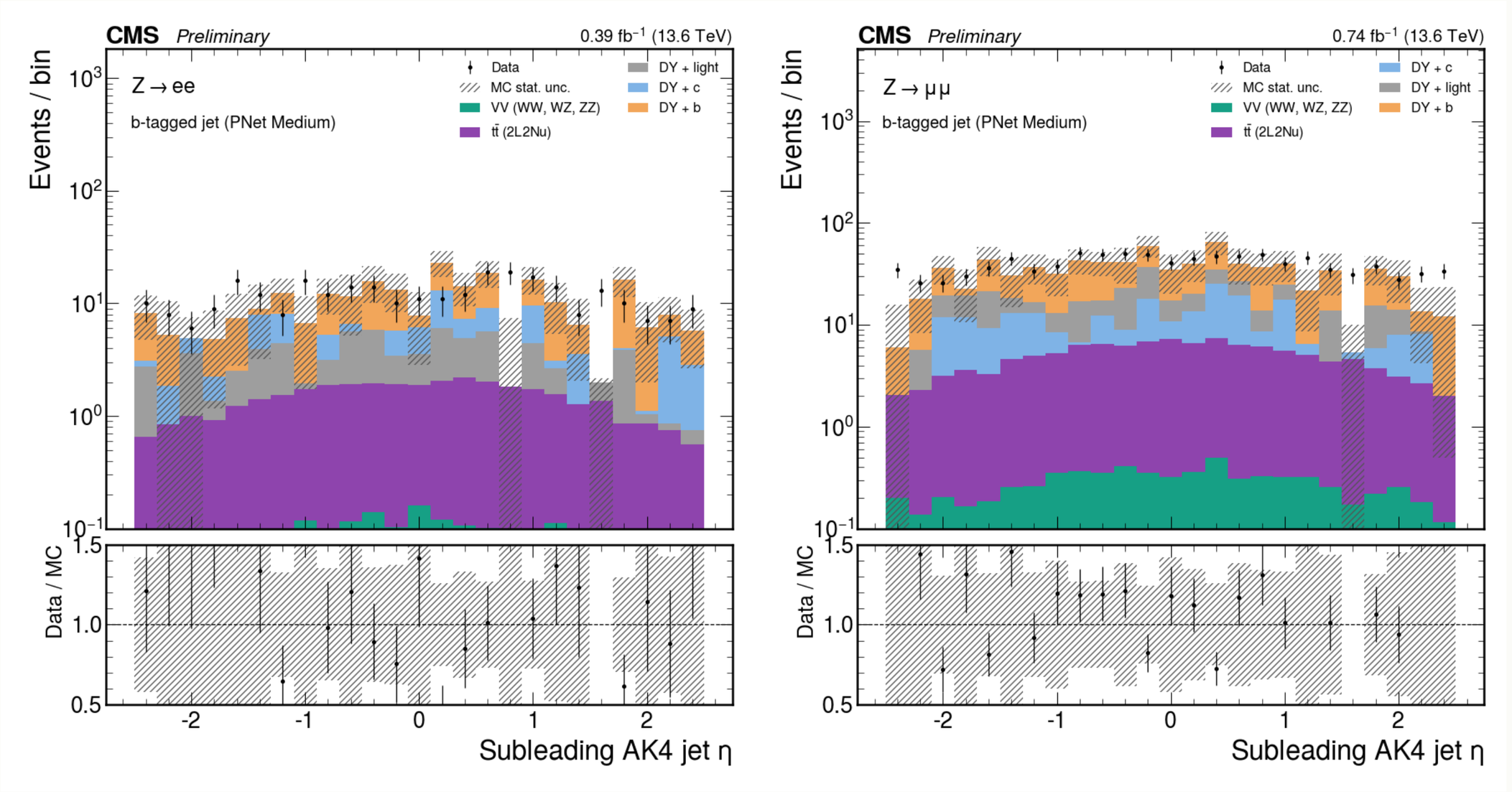




b-tag Medium

Second-jet η

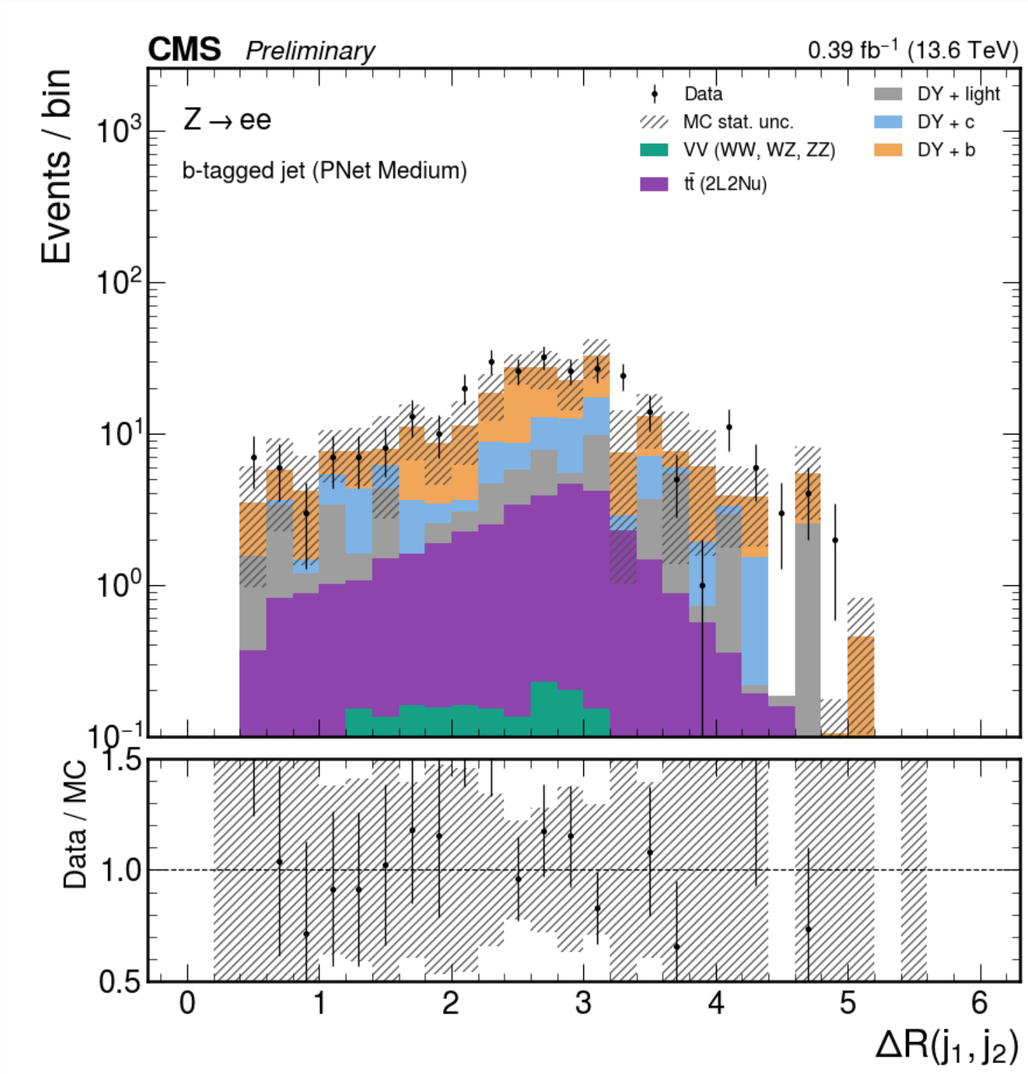
Events with at least 2 jets.

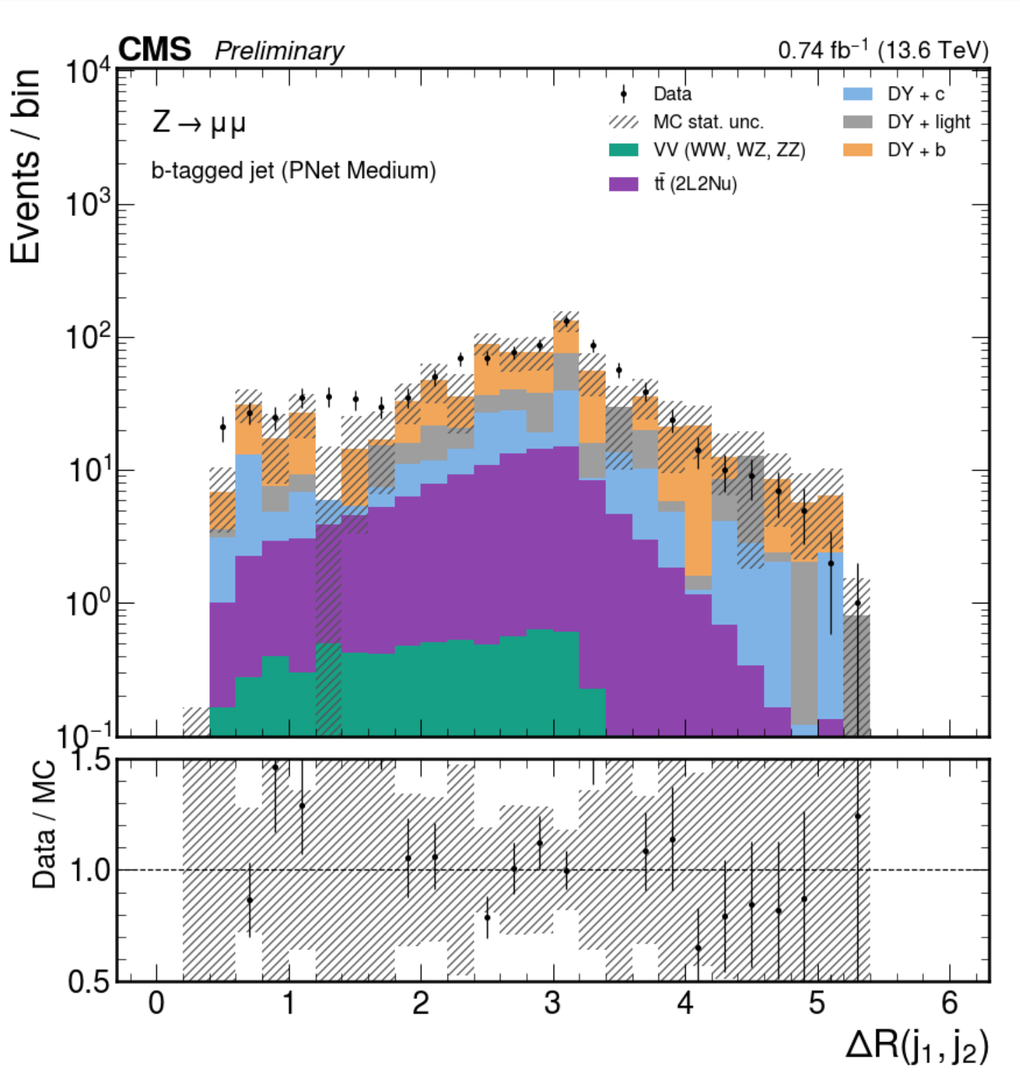


b-tag Medium

ΔR between the two leading jets

Events with at least 2 jets.

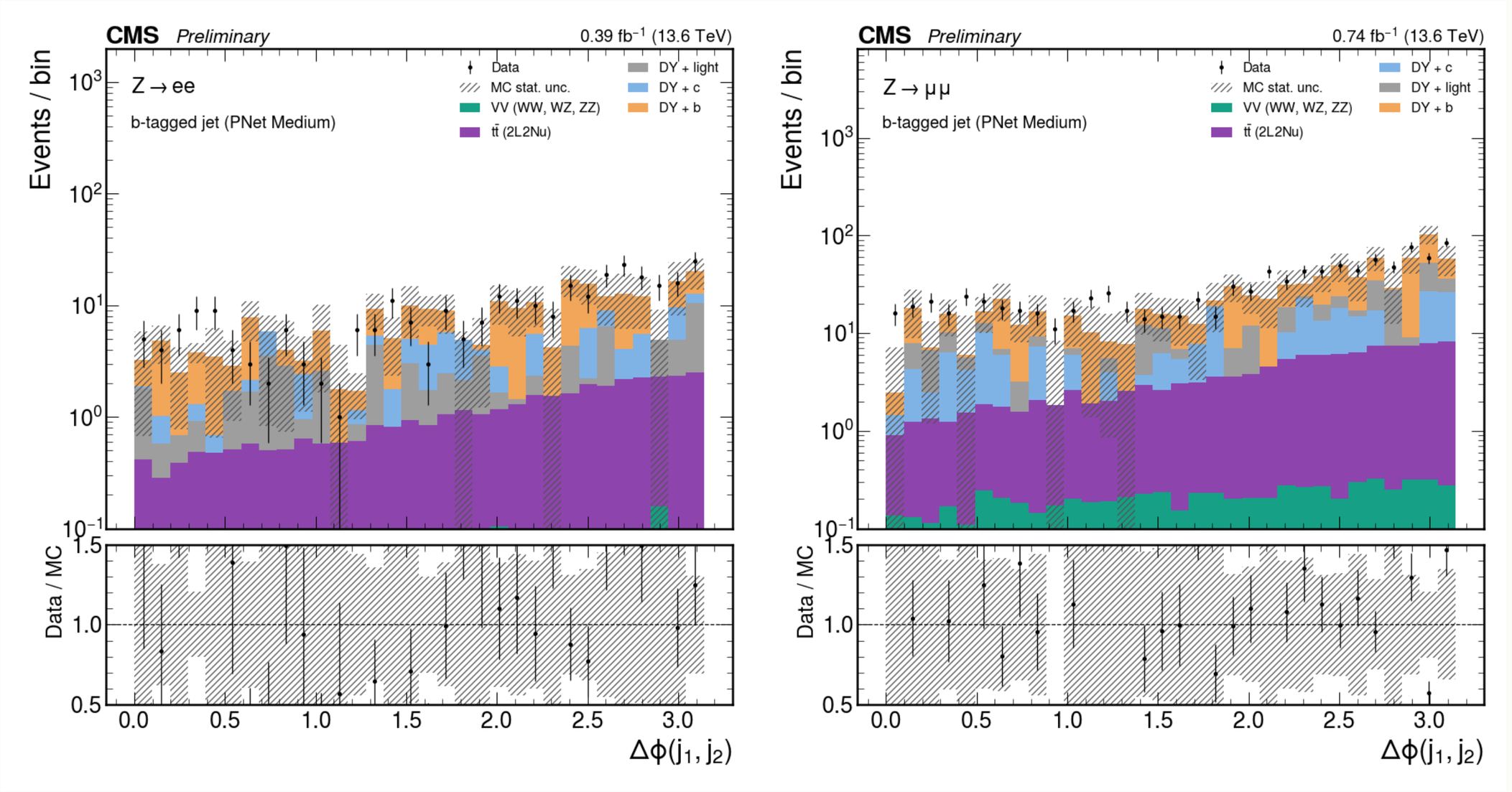




b-tag Medium

Δφ between the two leading jets

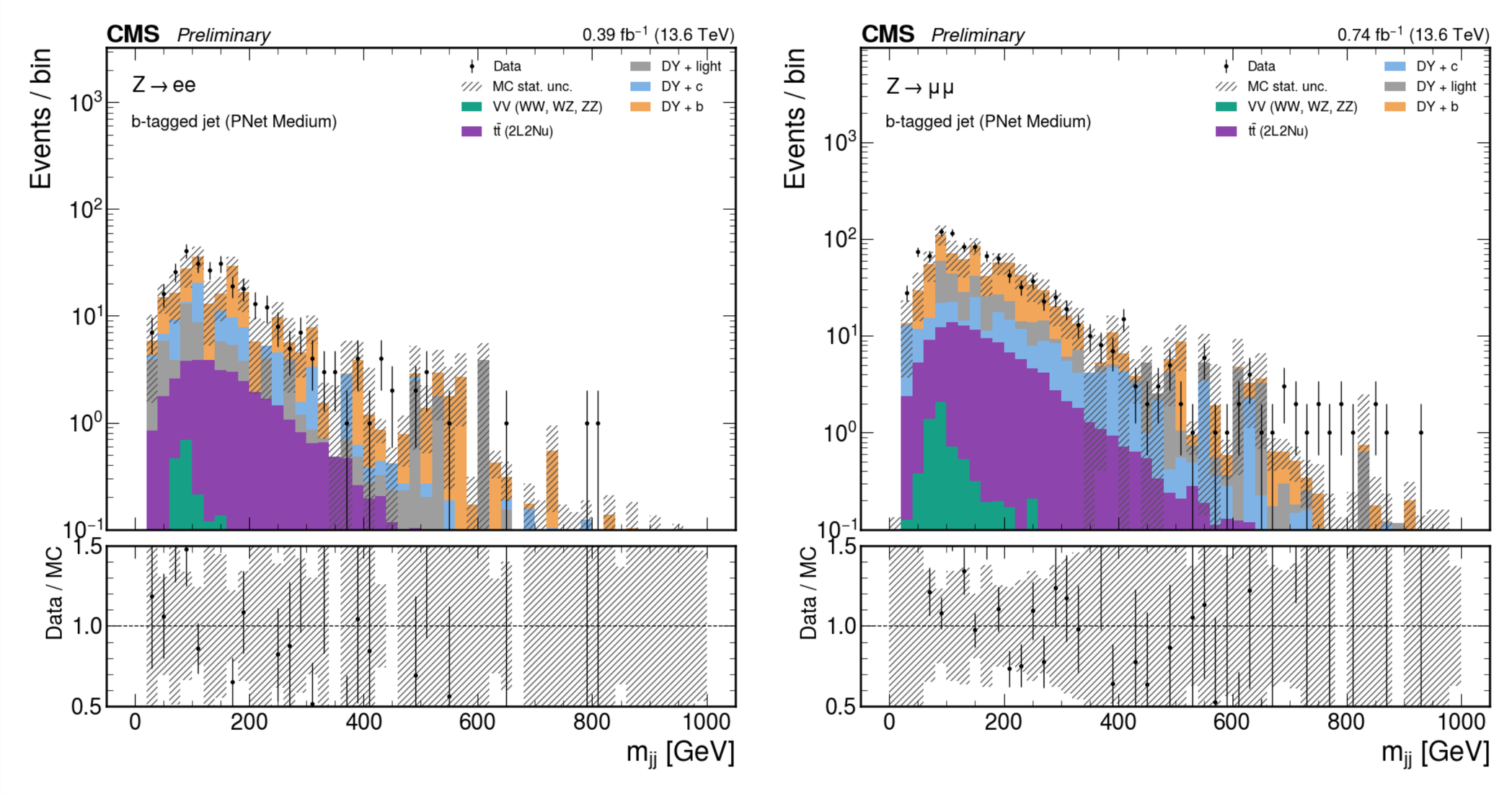
Events with at least 2 jets.



b-tag Medium

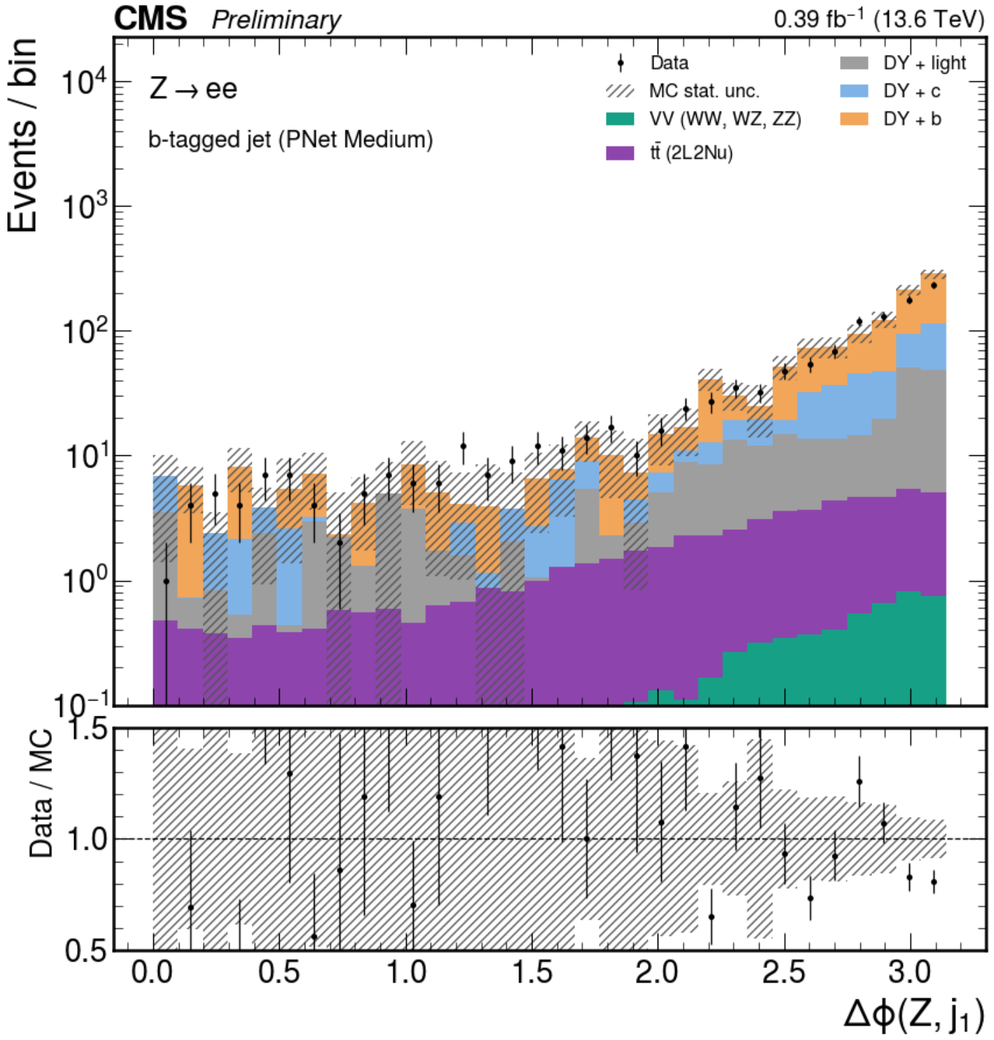
Dijet mass

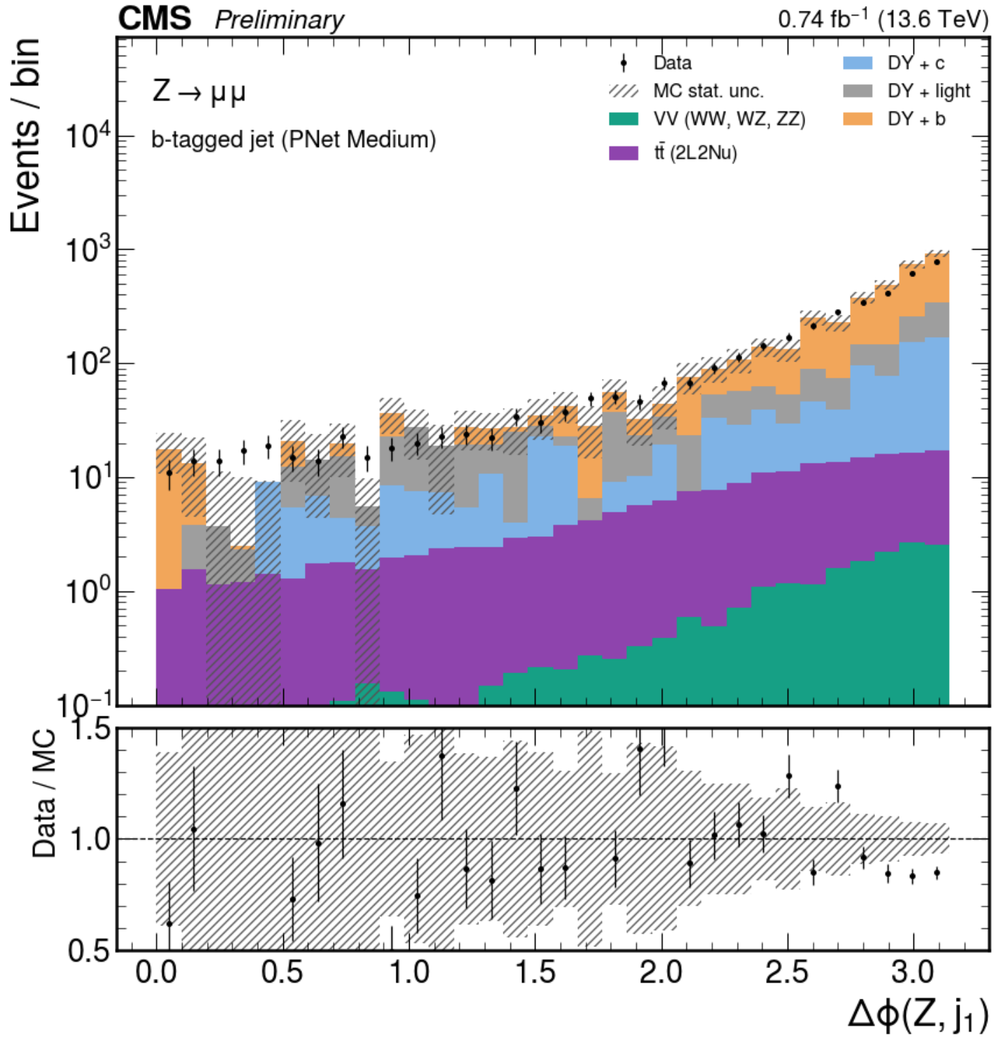
Events with at least 2 jets.



b-tag Medium

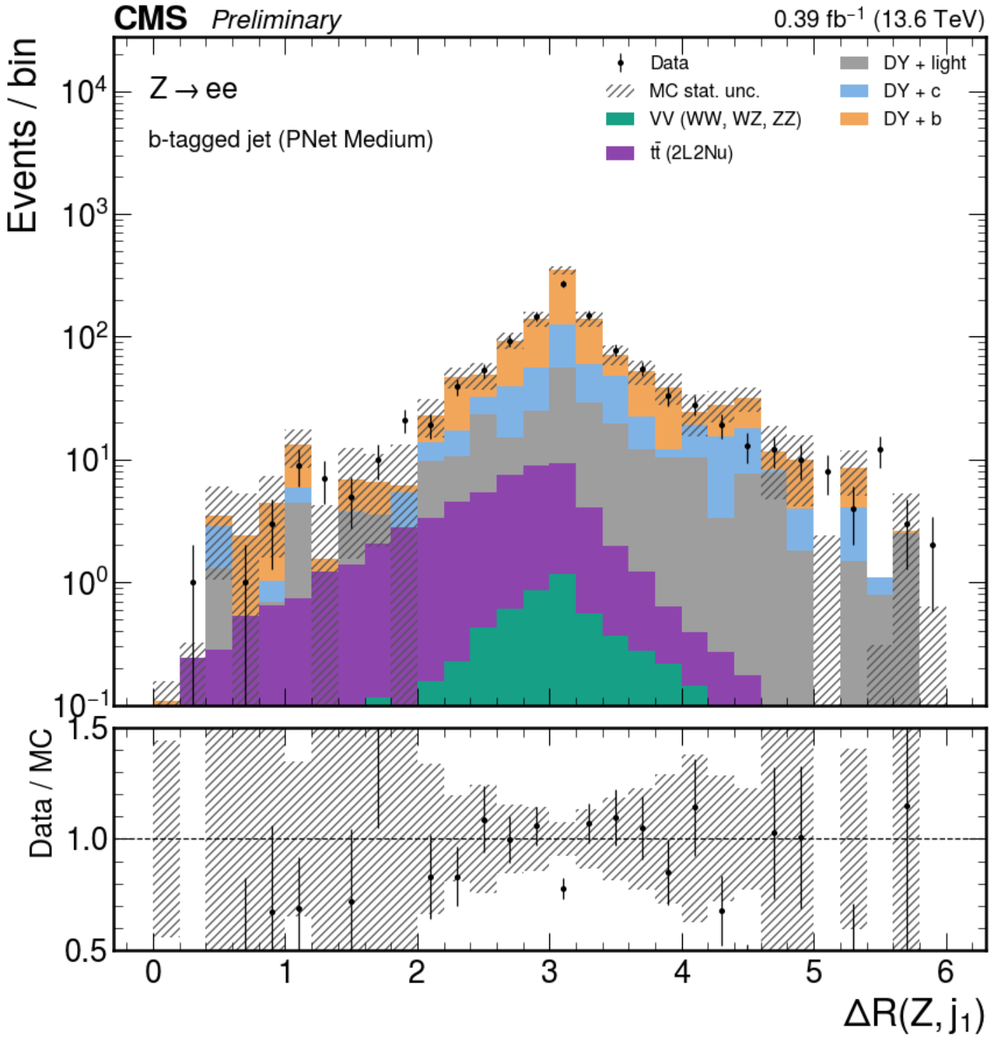
Δφ(Z, leading jet)

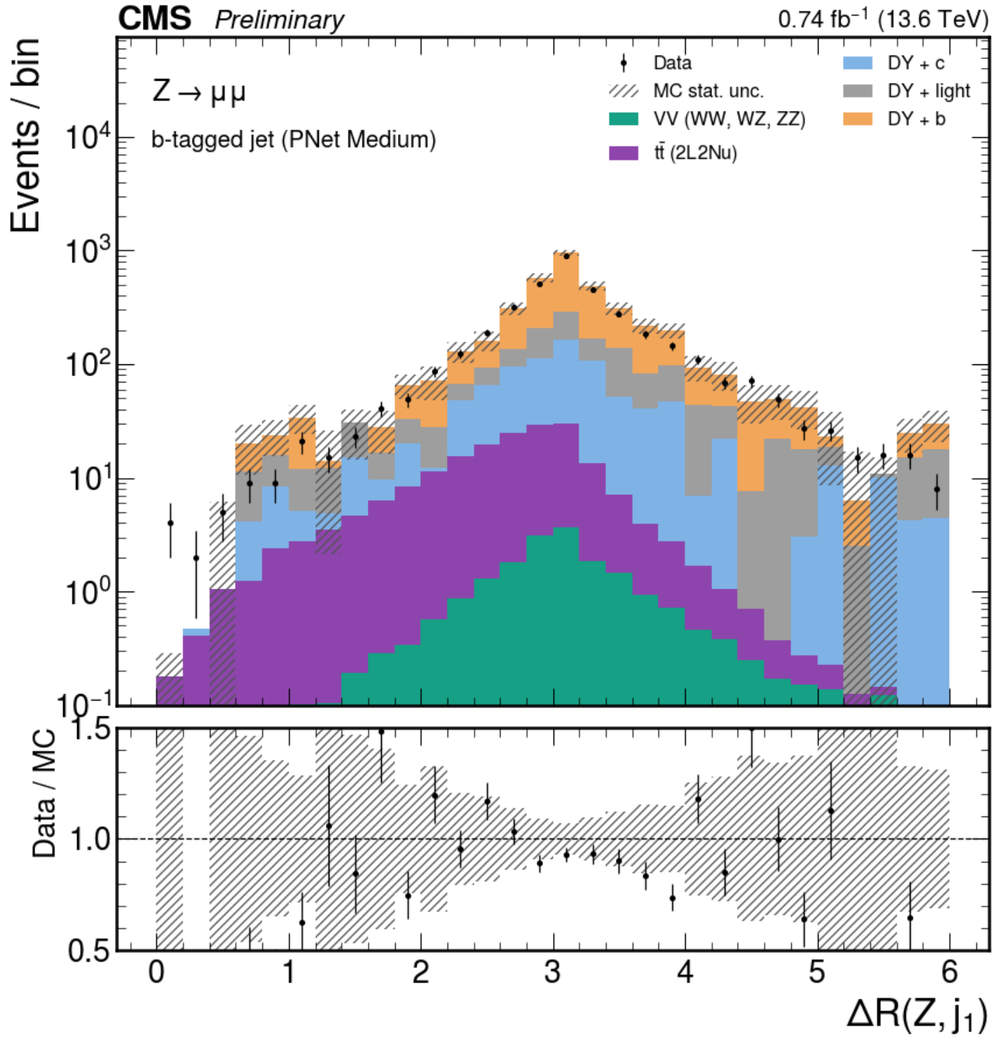




b-tag Medium

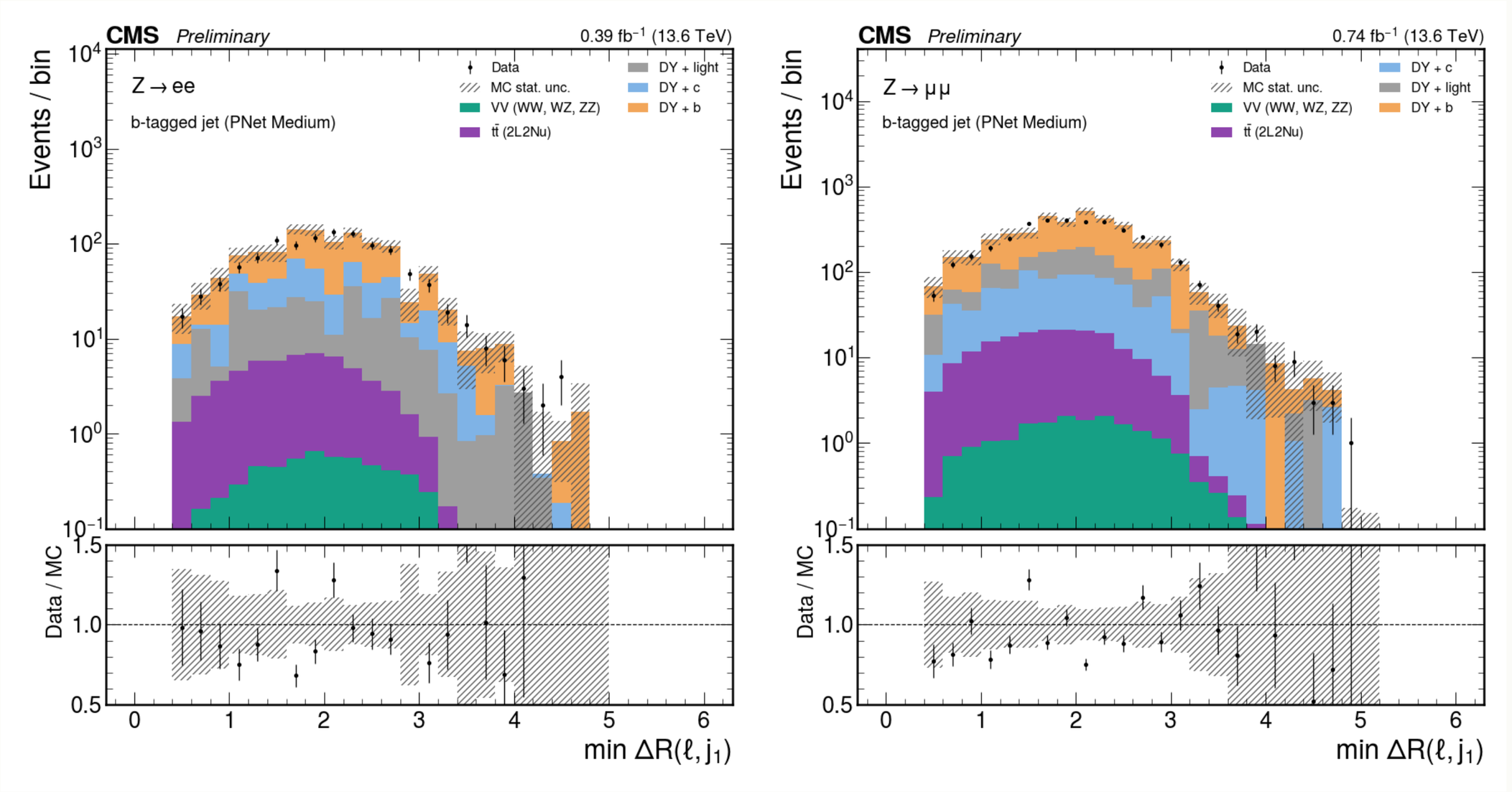
ΔR(Z, leading jet)





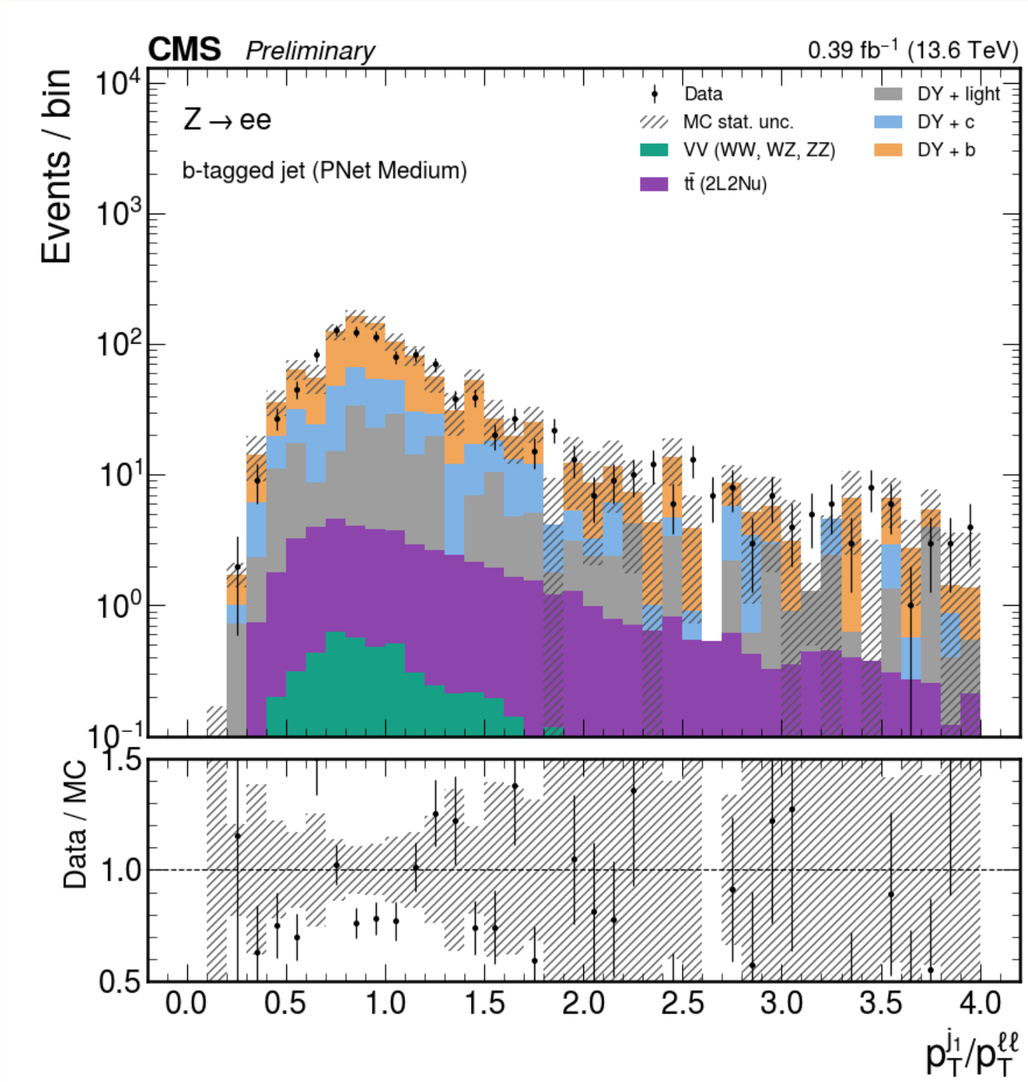
b-tag Medium

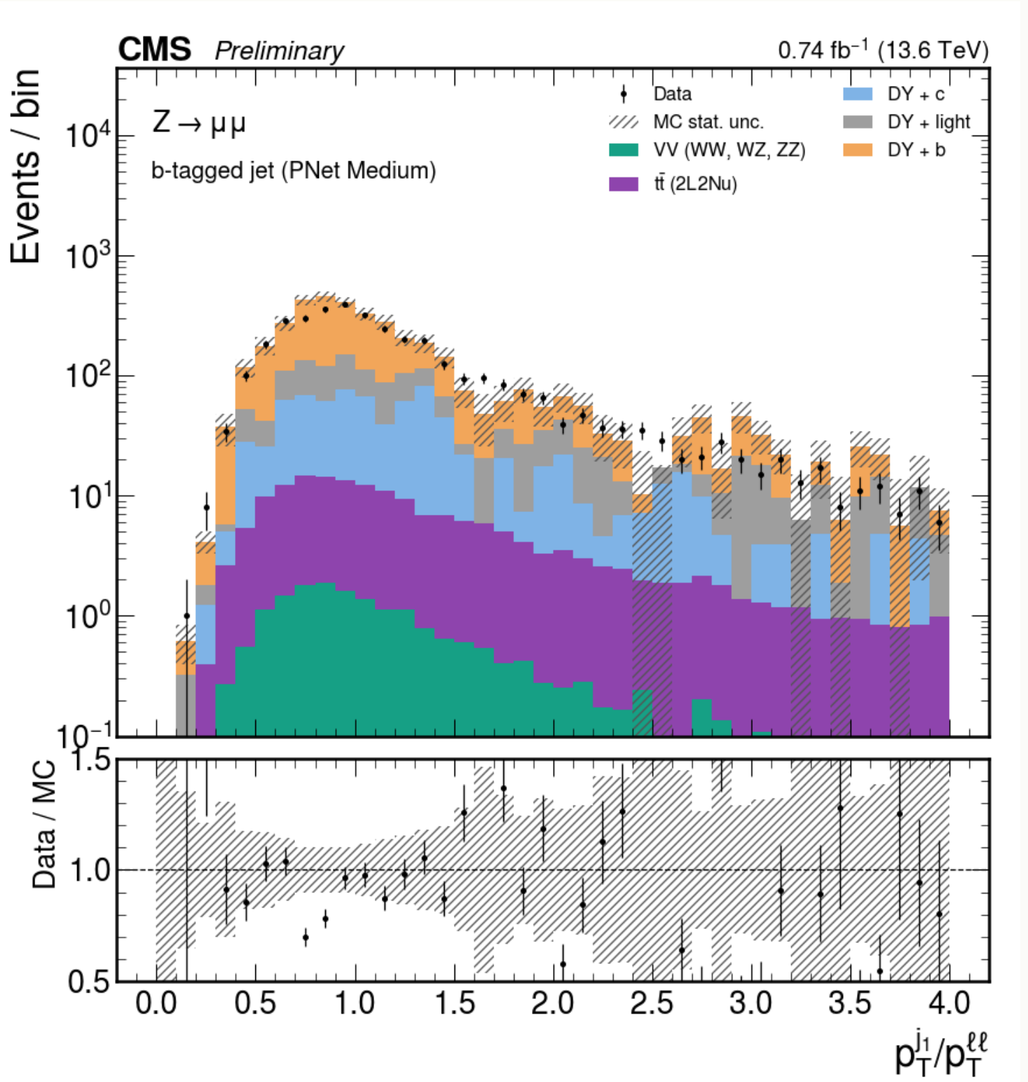
Smallest ΔR(lepton, leading jet)



b-tag Medium

pT balance: leading-jet pT / Z pT

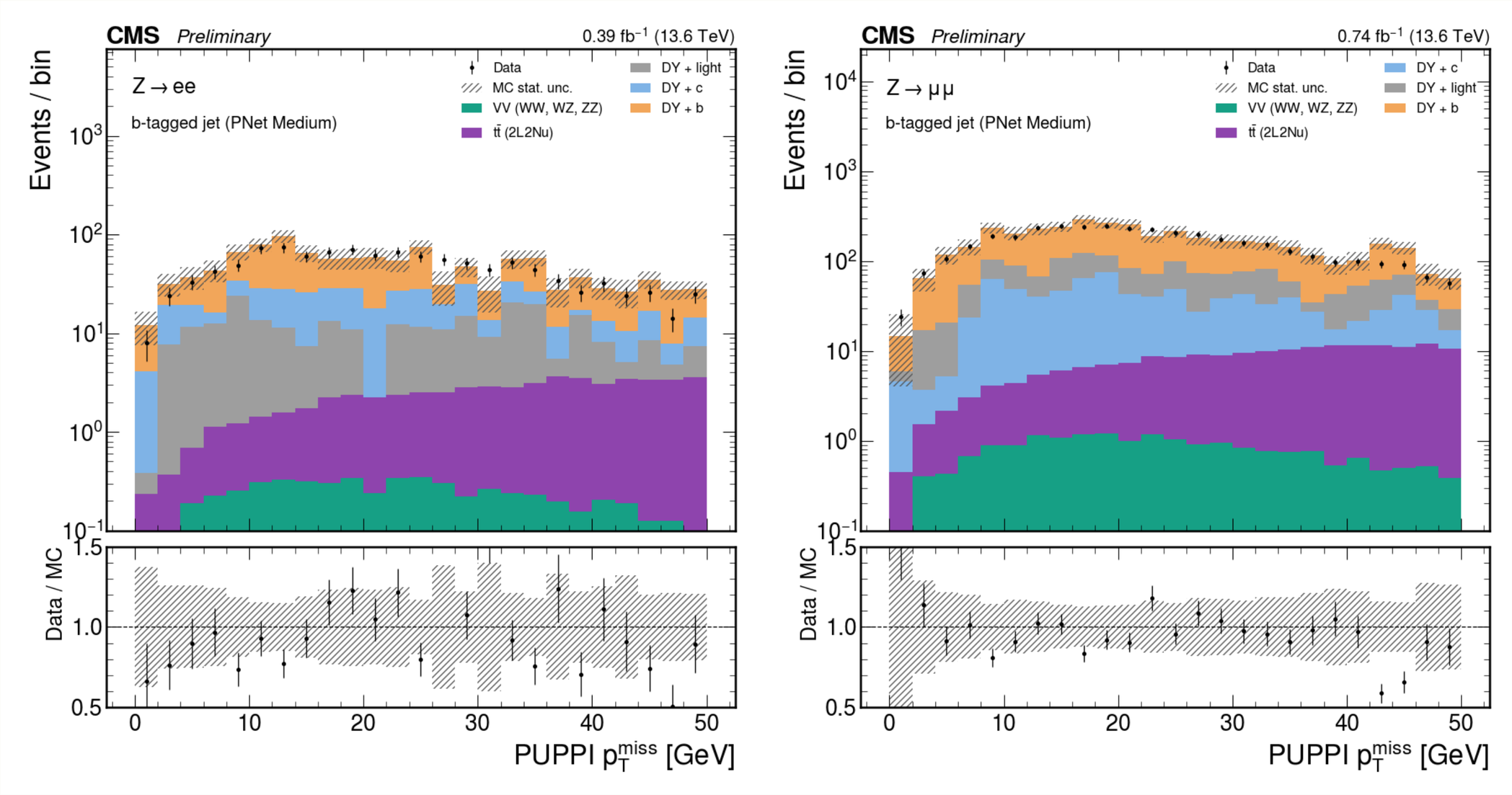




b-tag Medium

PUPPI MET

Type-1 corrected; the selection cuts at 50 GeV.

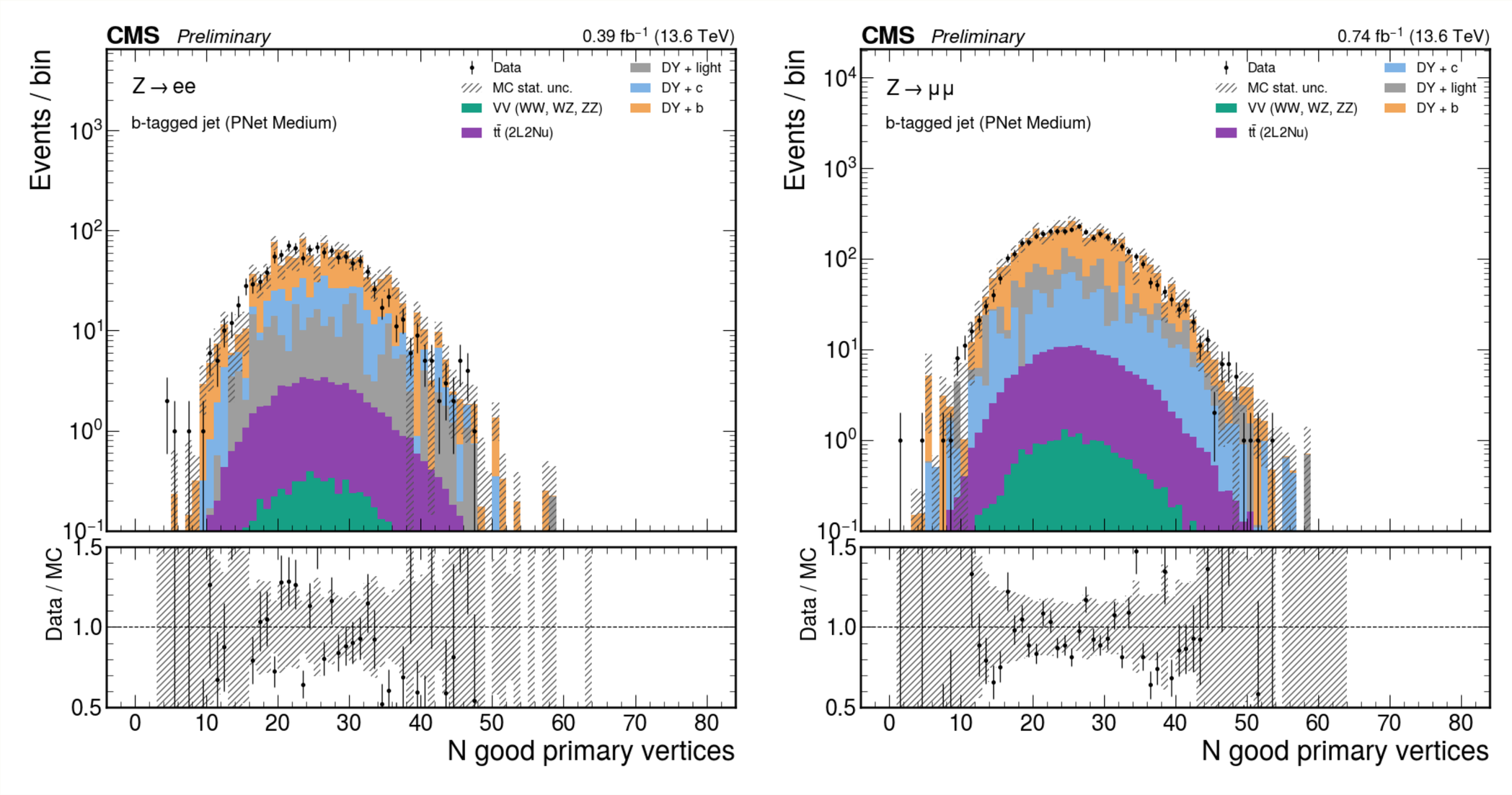


Summer22 (2022 C–D), 13.6 TeV · Z→ℓℓ + jets · 20 files per MC sample and per data channel · AK4 jets · stat. errors only · 130

b-tag Medium

Number of good primary vertices

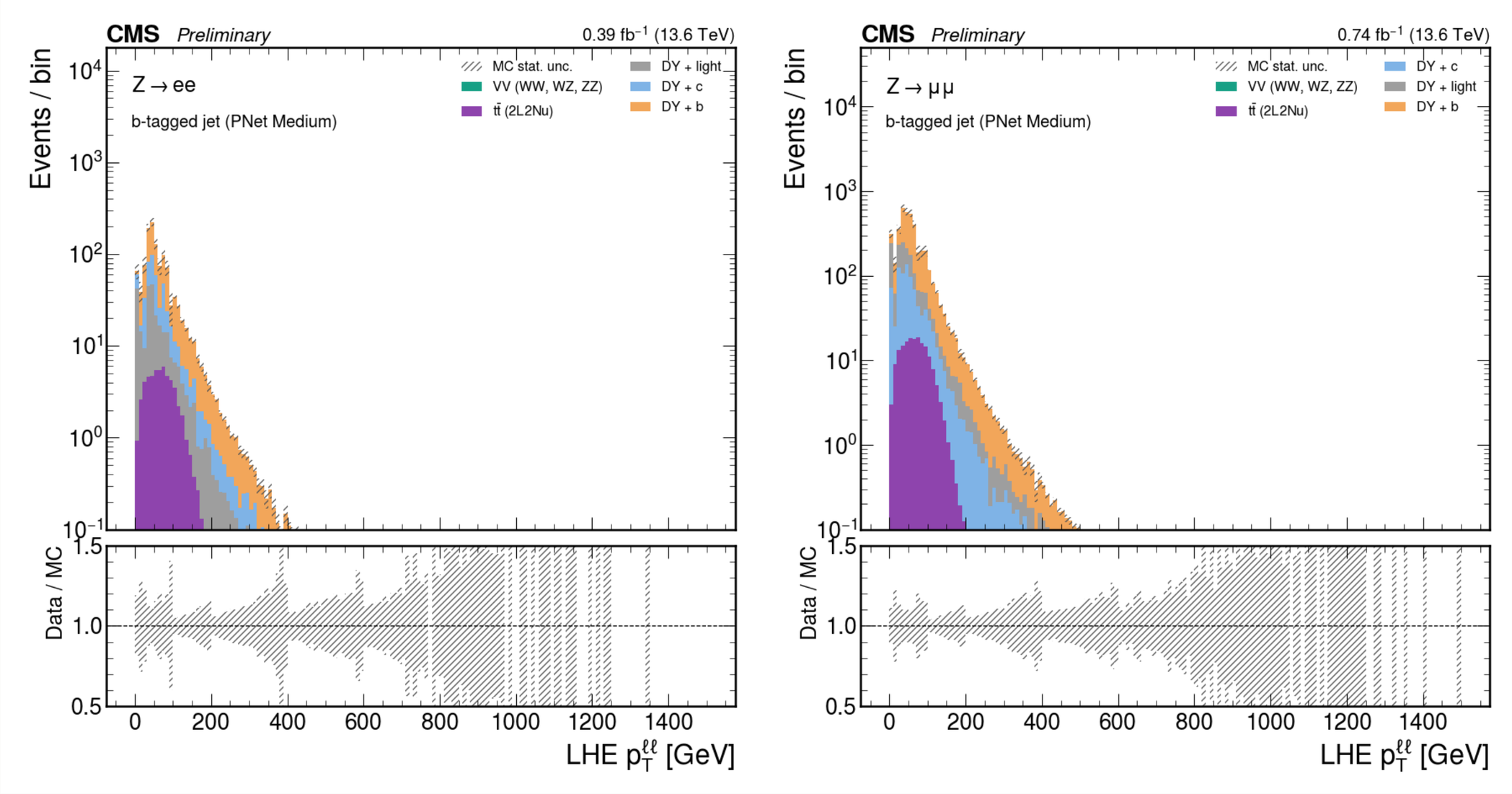
After pileup weights.



b-tag Medium

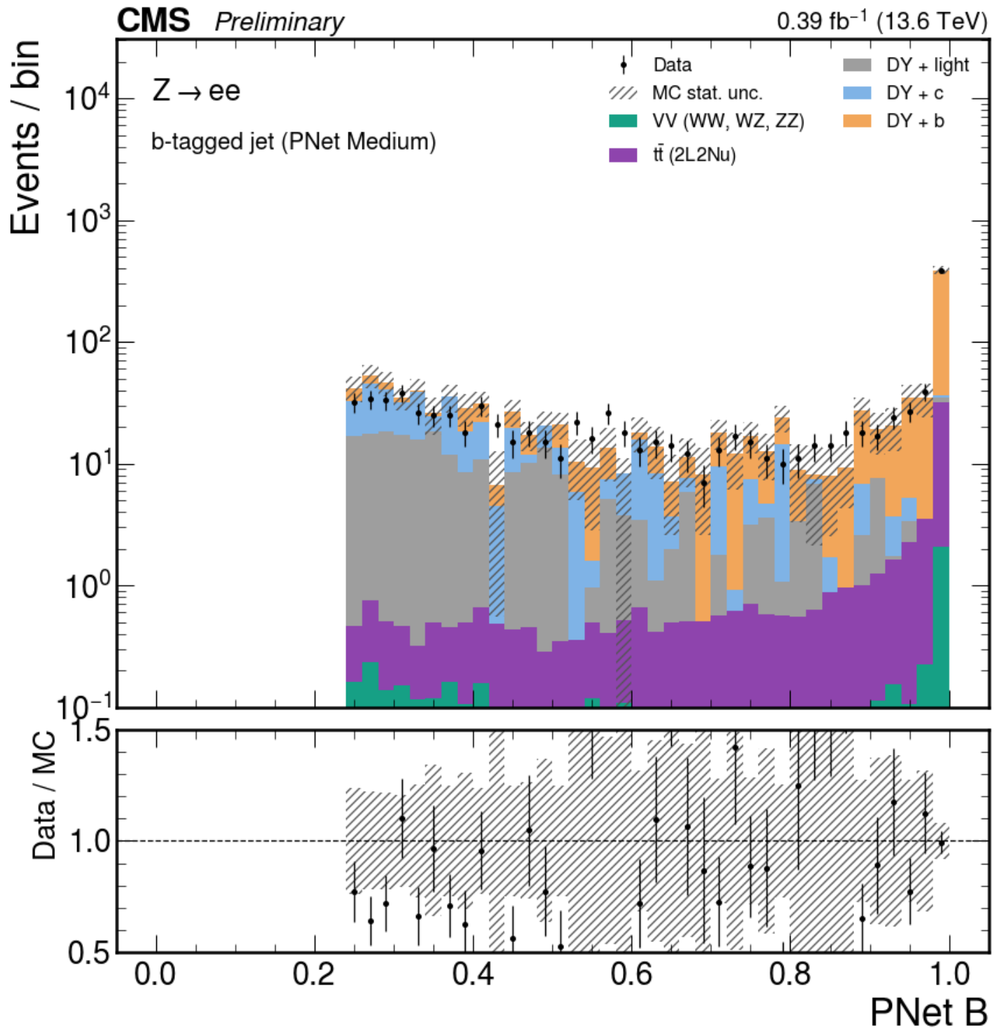
Generator Z pT (LHE)

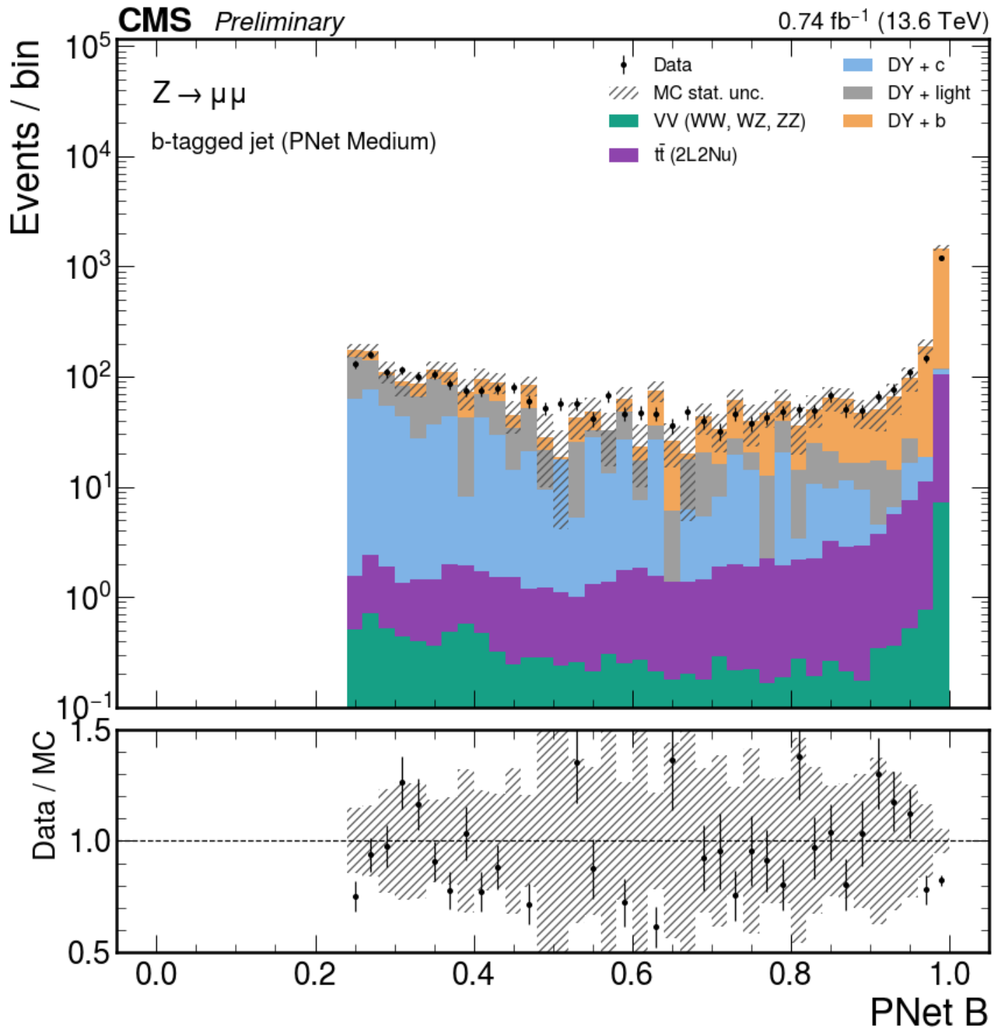
MC only, no data.



b-tag Medium

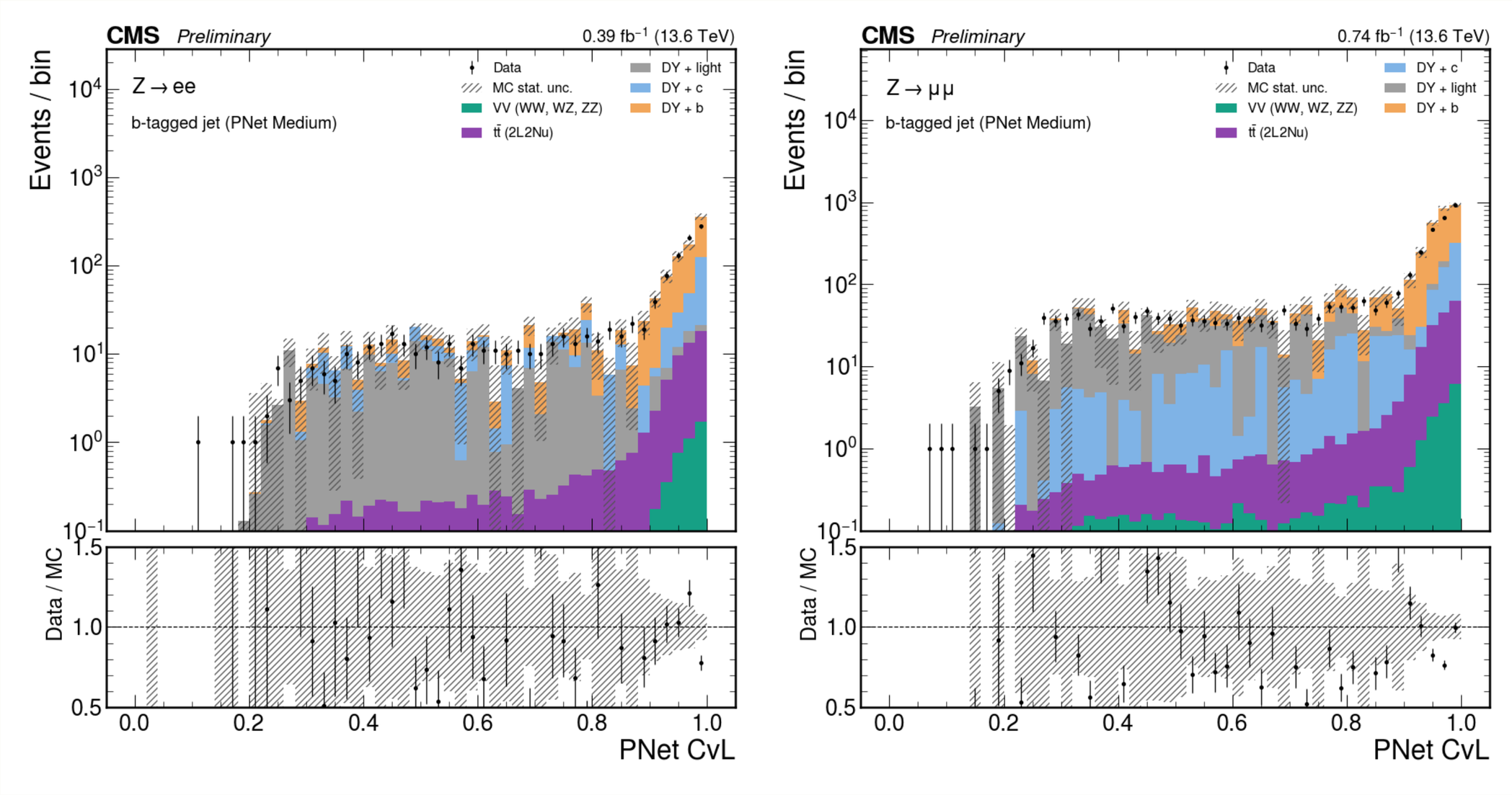
Leading-jet PNet B score





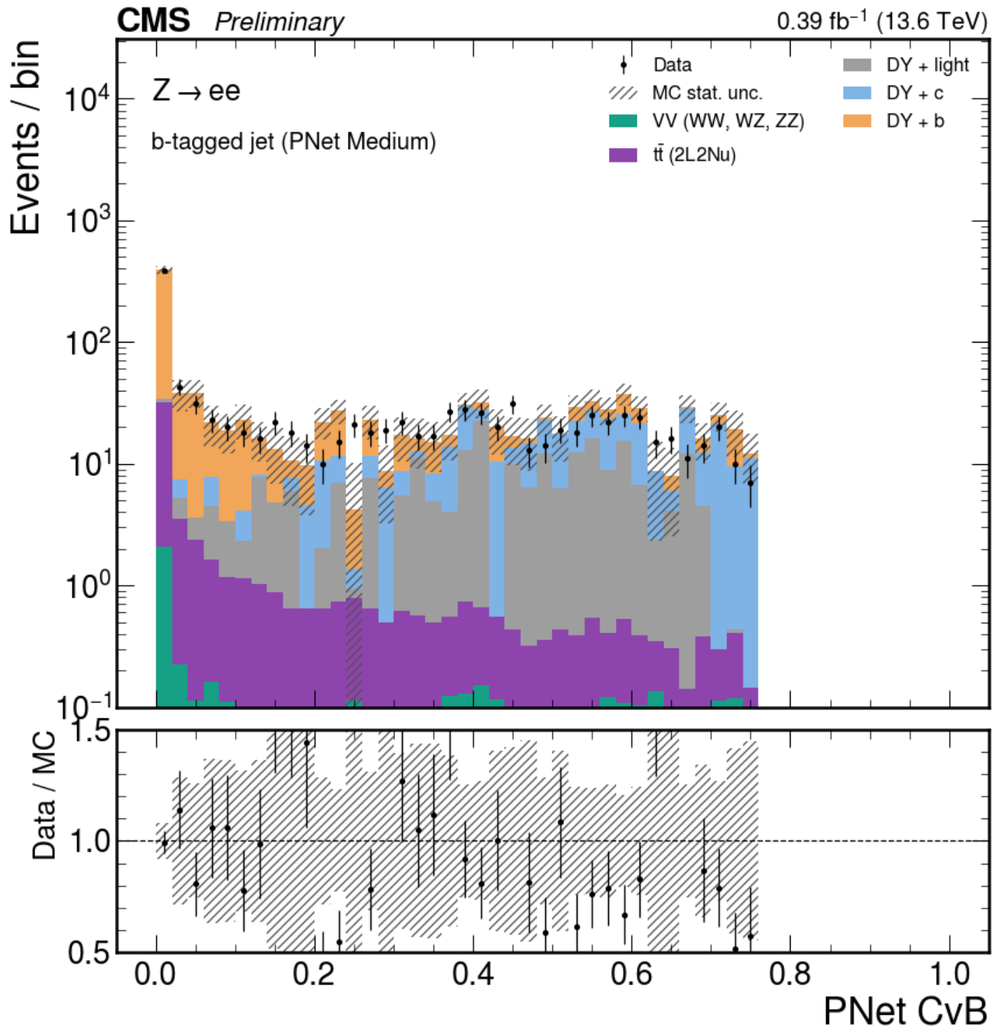
b-tag Medium

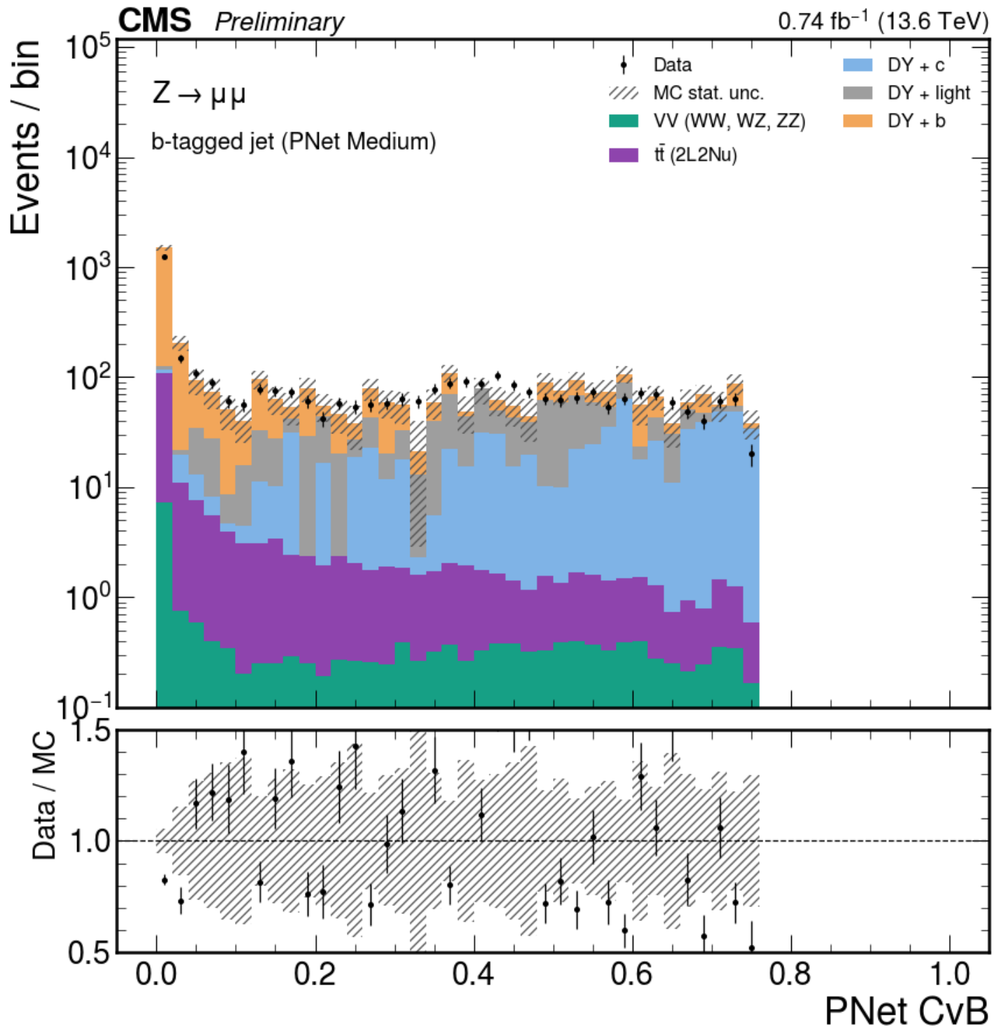
Leading-jet PNet CvL score



b-tag Medium

Leading-jet PNet CvB score

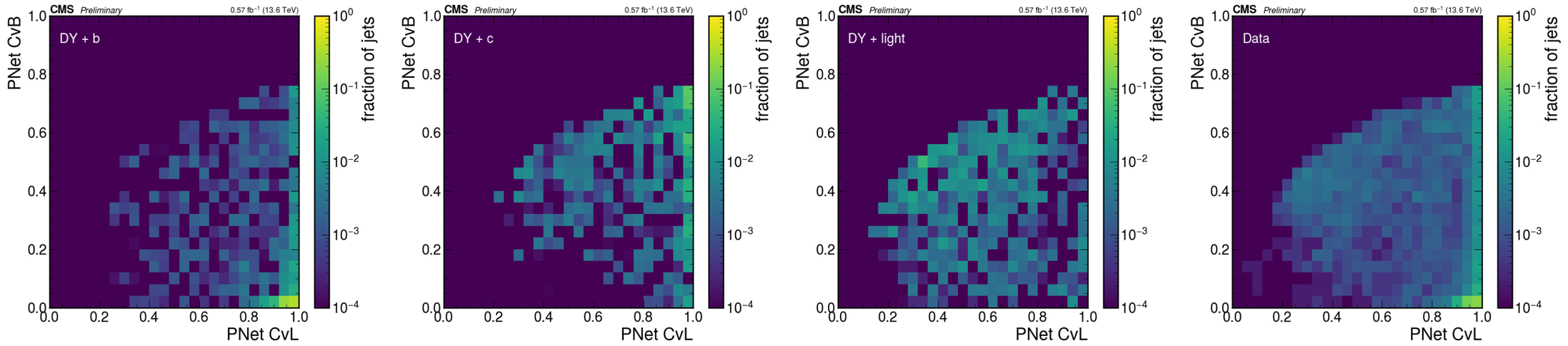




b-tag Medium

PNet CvL vs CvB of the leading jet

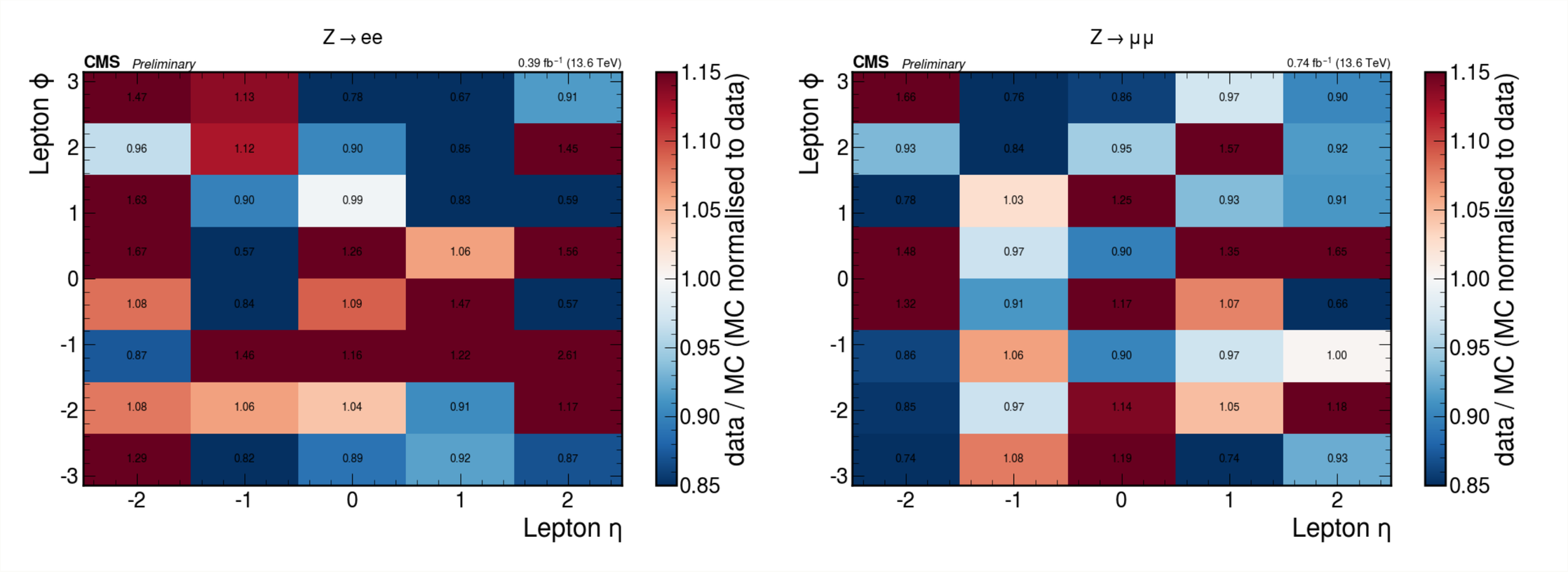
Each panel normalised to 1. Z→ee and Z→μμ added.



b-tag Medium

Lepton η–φ, data / MC

Both Z leptons. MC normalised to data in each map. Colour scale stops at ±15%; in the tagged regions there are few events per cell, so read the numbers.



c-tag Medium

Leading jet PNet CvL > 0.160 and CvB > 0.306

	data/MC	DY+b	DY+c	DY+light	tt	tagged MC	tagged data
Z→ee	0.938 ± 0.023	2.9%	20.2%	76.2%	0.3%	27.9%	27.6%
Z→μμ	0.951 ± 0.017	3.1%	19.5%	76.7%	0.3%	28.6%	28.2%

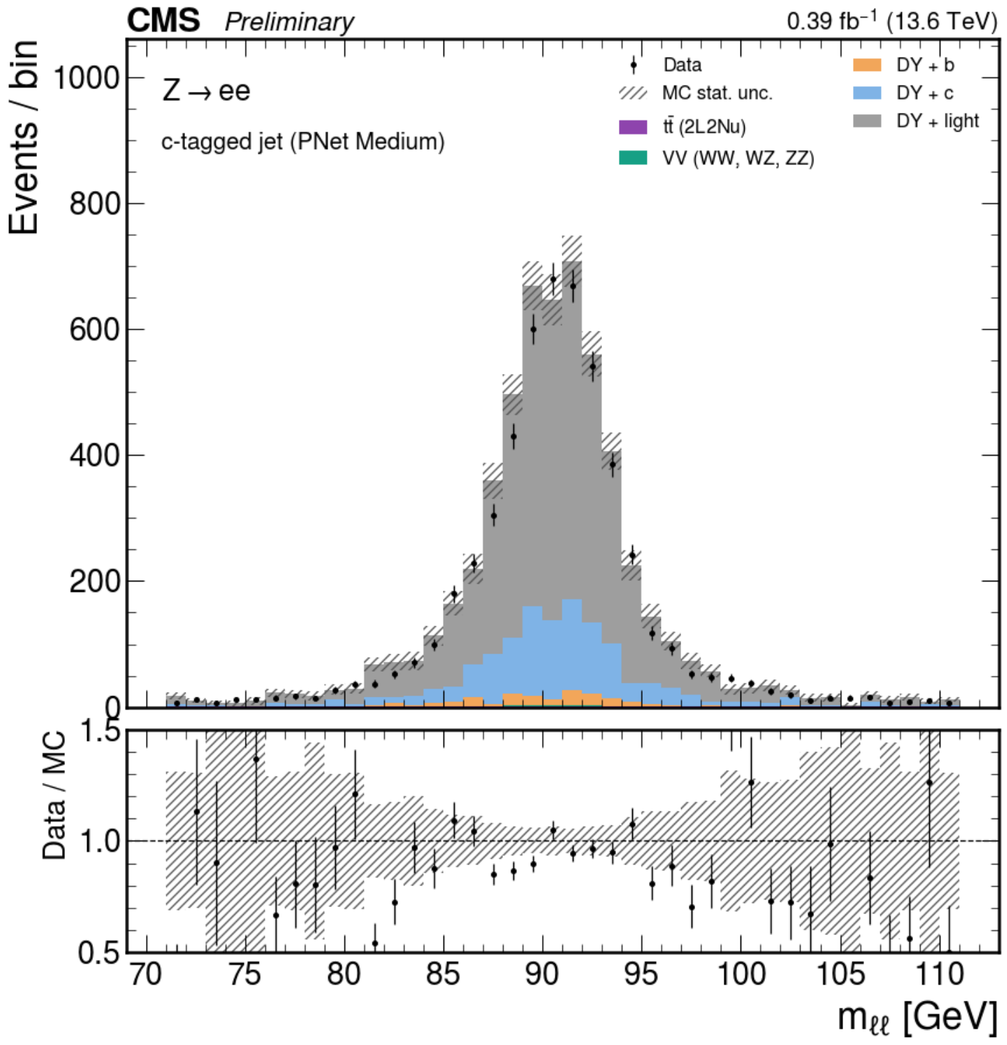
Flavour shares are of all MC. 'Tagged' = fraction of Z + jet events passing the tag. Errors are statistical only.

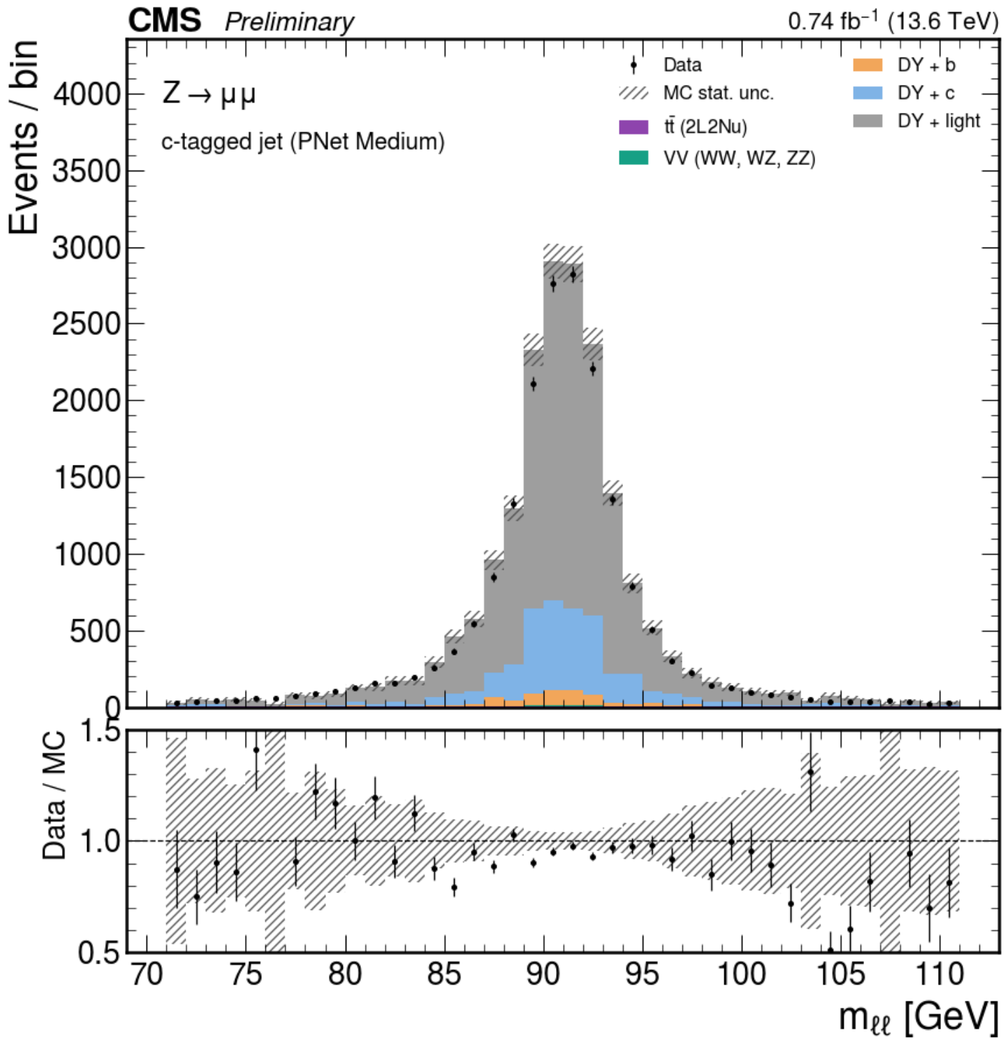
32 plots follow: Z and leptons, jets, Z–jet, MET and vertices, tagger scores, 2D maps. Left Z→ee, right Z→μμ.

c-tag Medium

Dilepton mass

m(ℓℓ), 71–111 GeV window, linear scale.

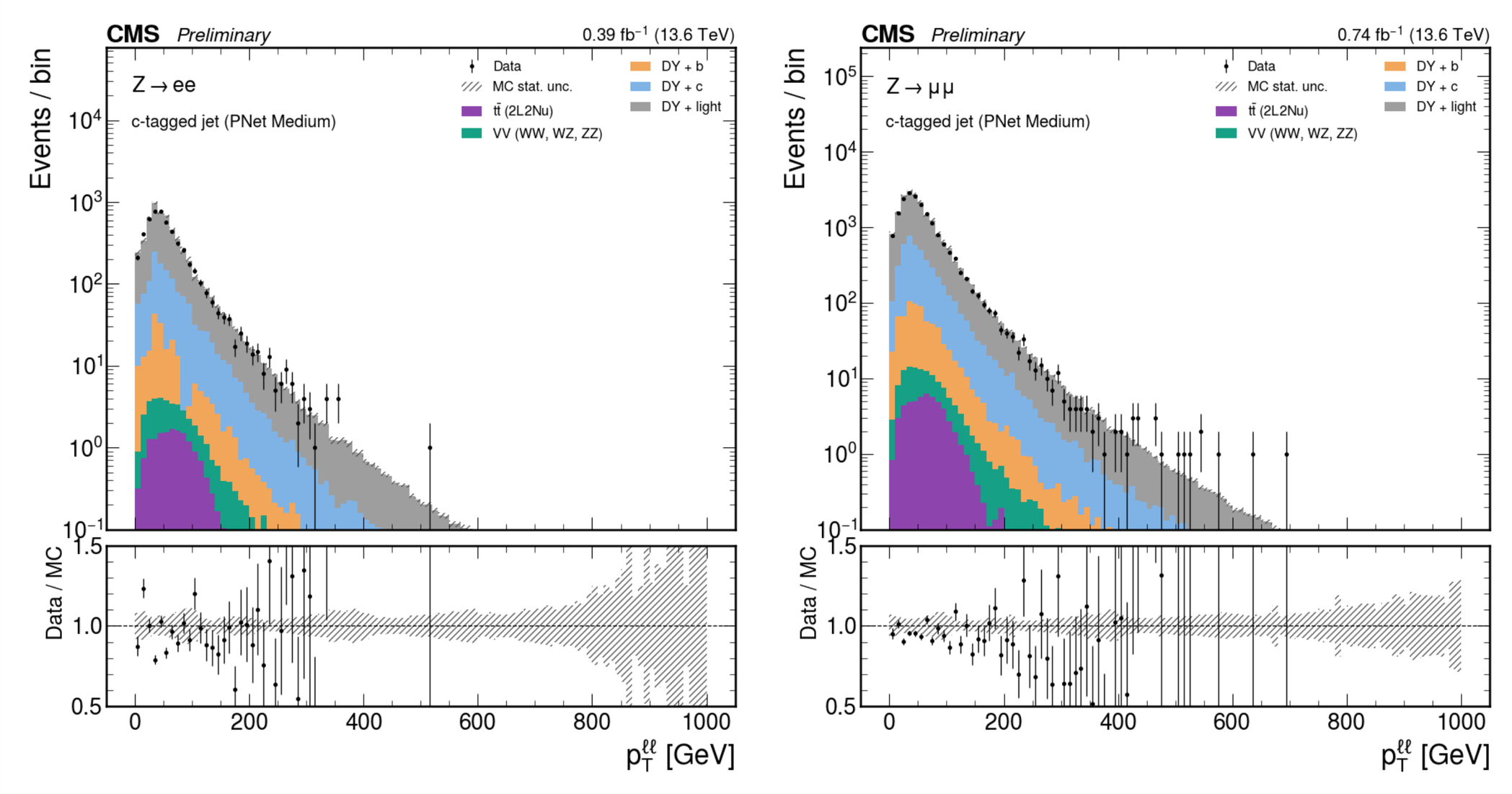




c-tag Medium

Z pT

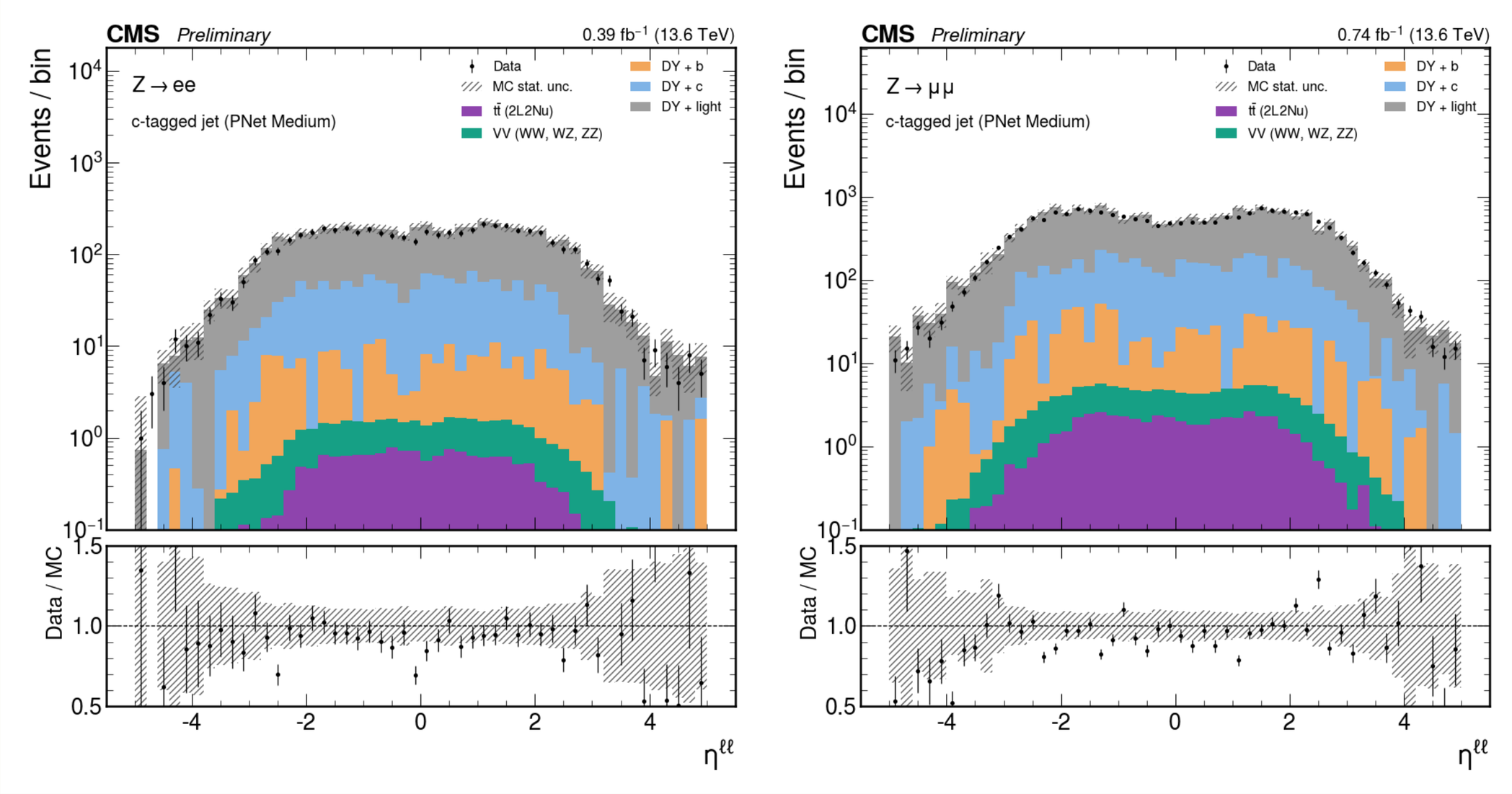
pT of the lepton pair.



c-tag Medium

Z η

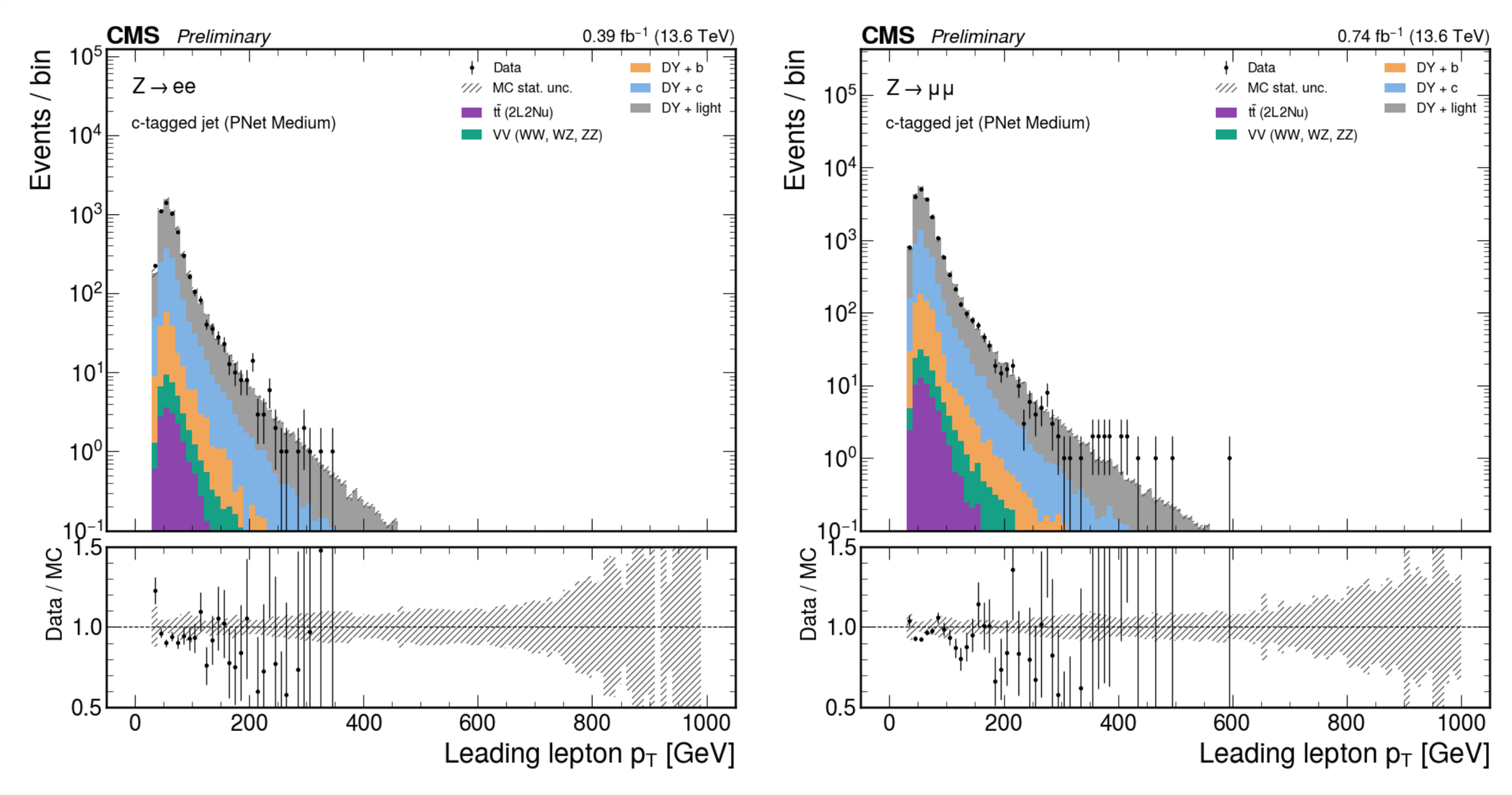
η of the lepton pair.



c-tag Medium

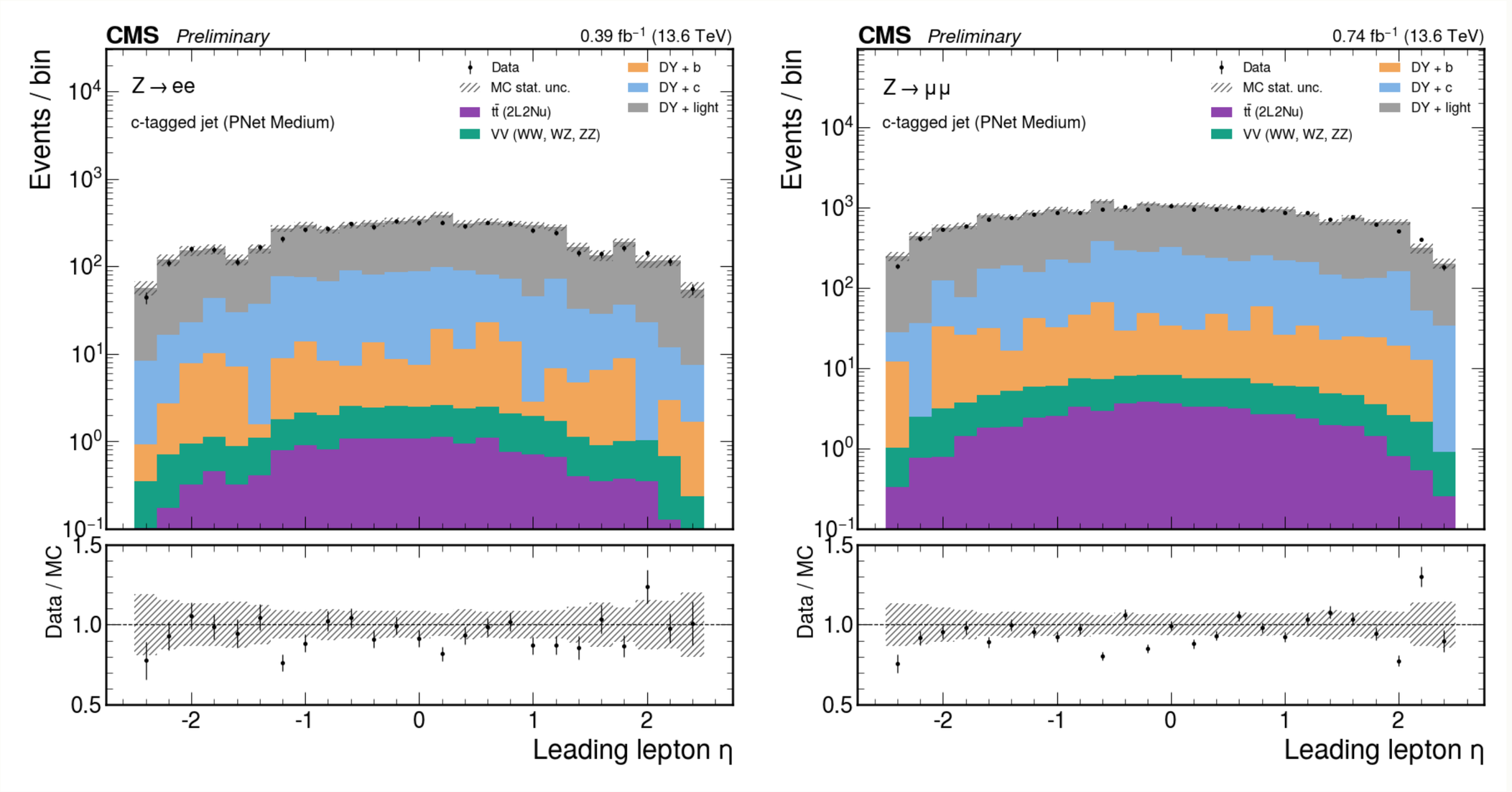
Leading-lepton pT

Leading lepton pT > 35 GeV (trigger plateau).



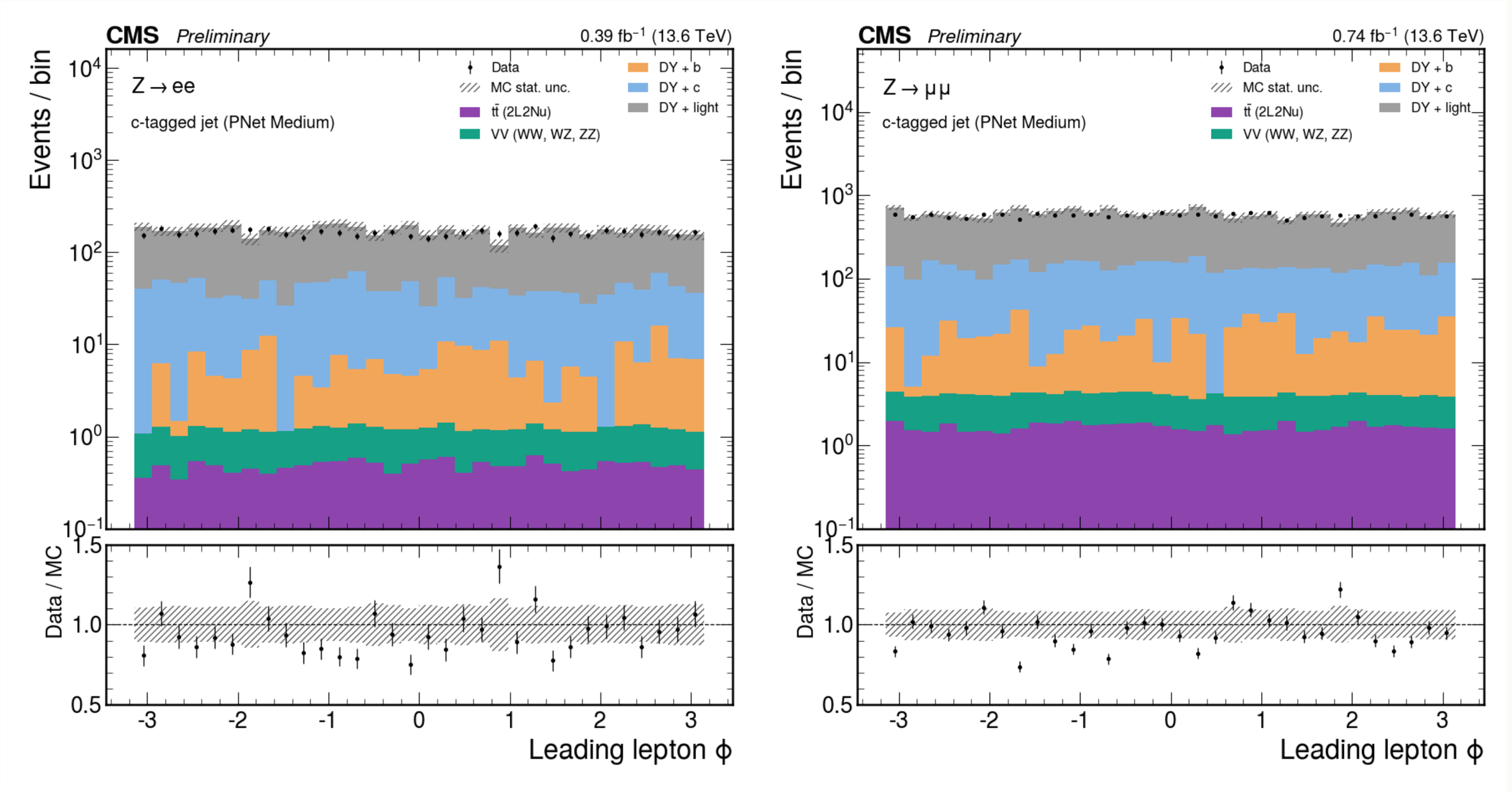
c-tag Medium

Leading-lepton η



c-tag Medium

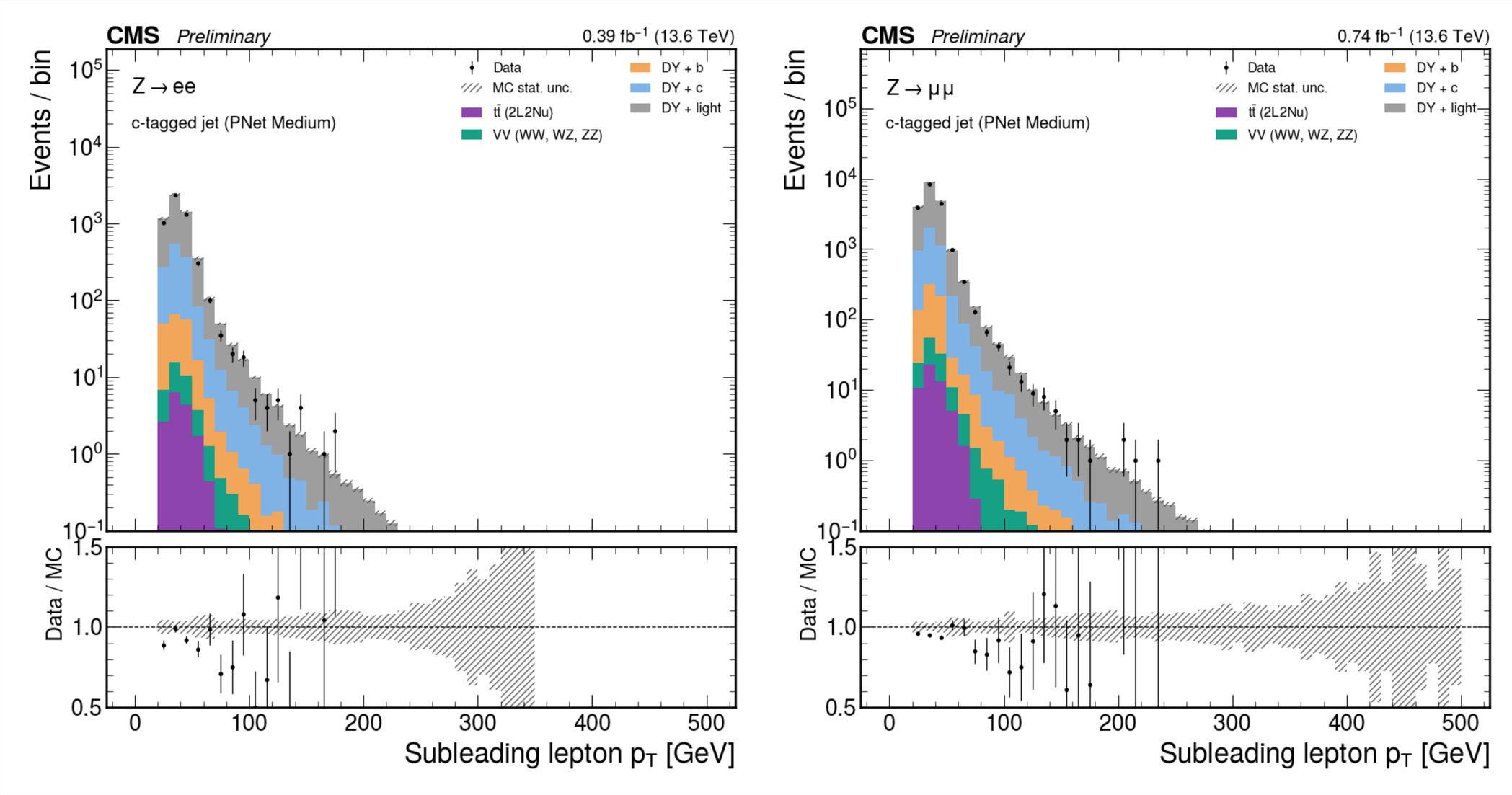
Leading-lepton φ



c-tag Medium

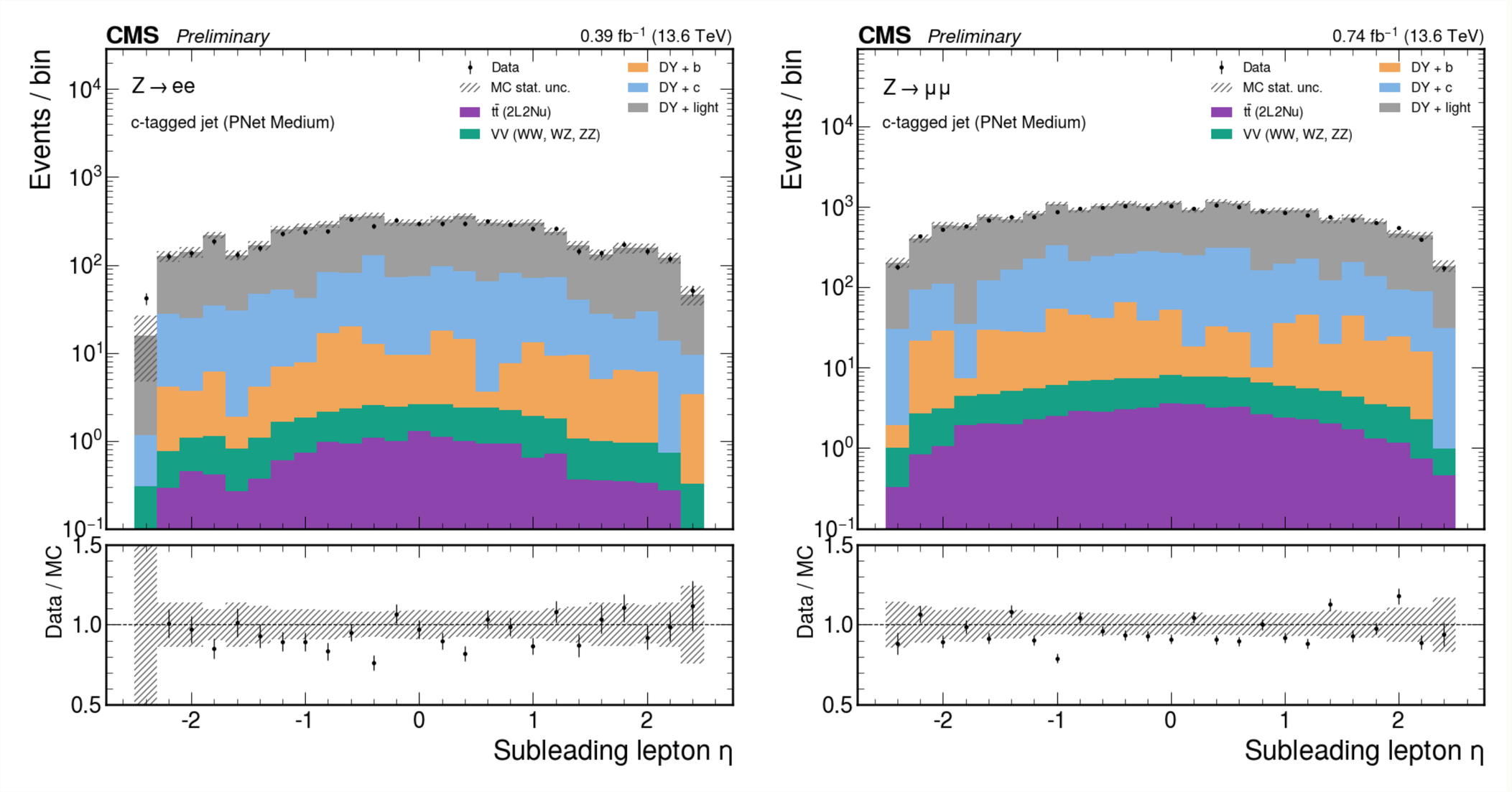
Second-lepton pT

Second lepton pT > 25 GeV.



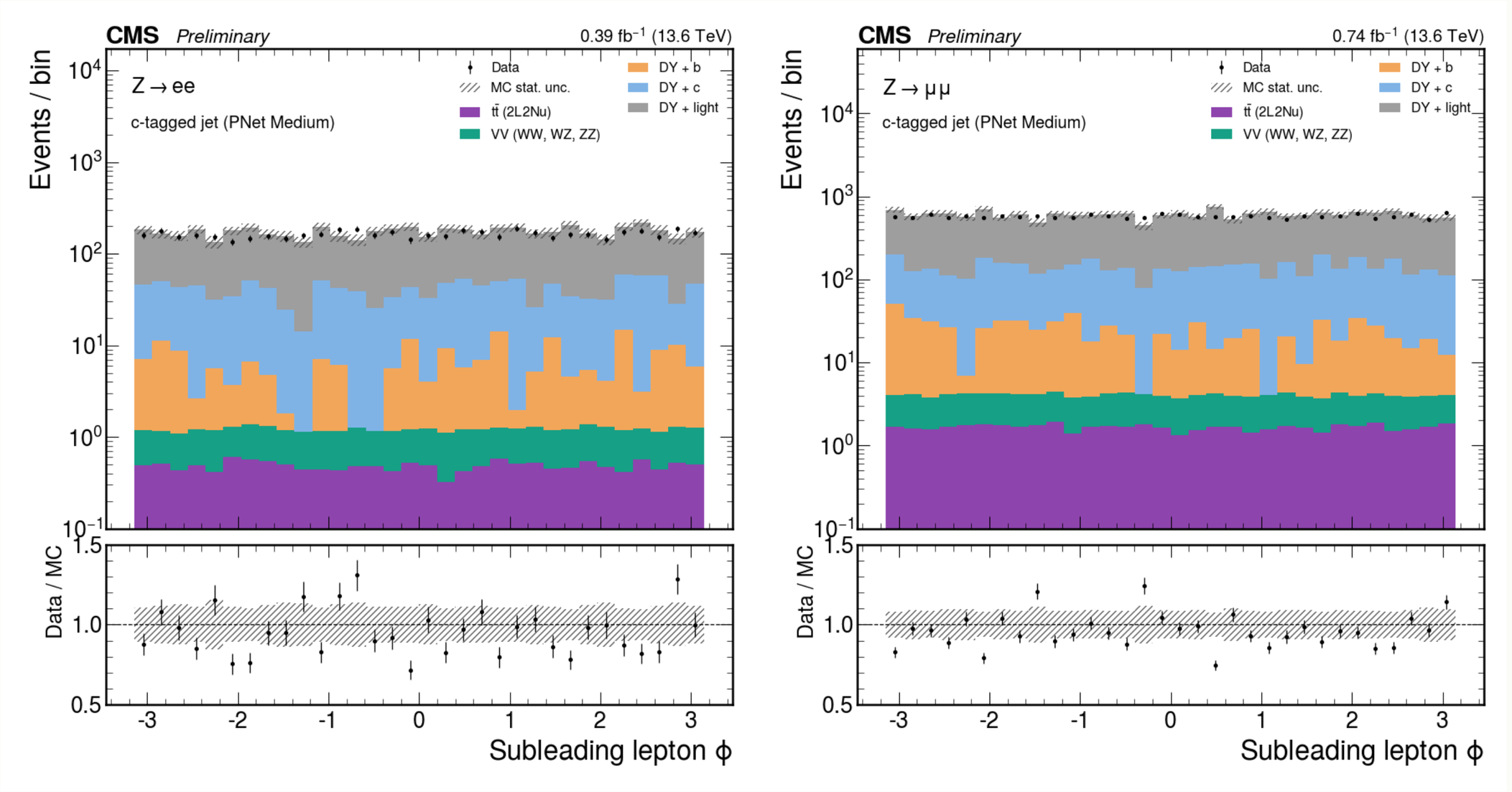
c-tag Medium

Second-lepton η



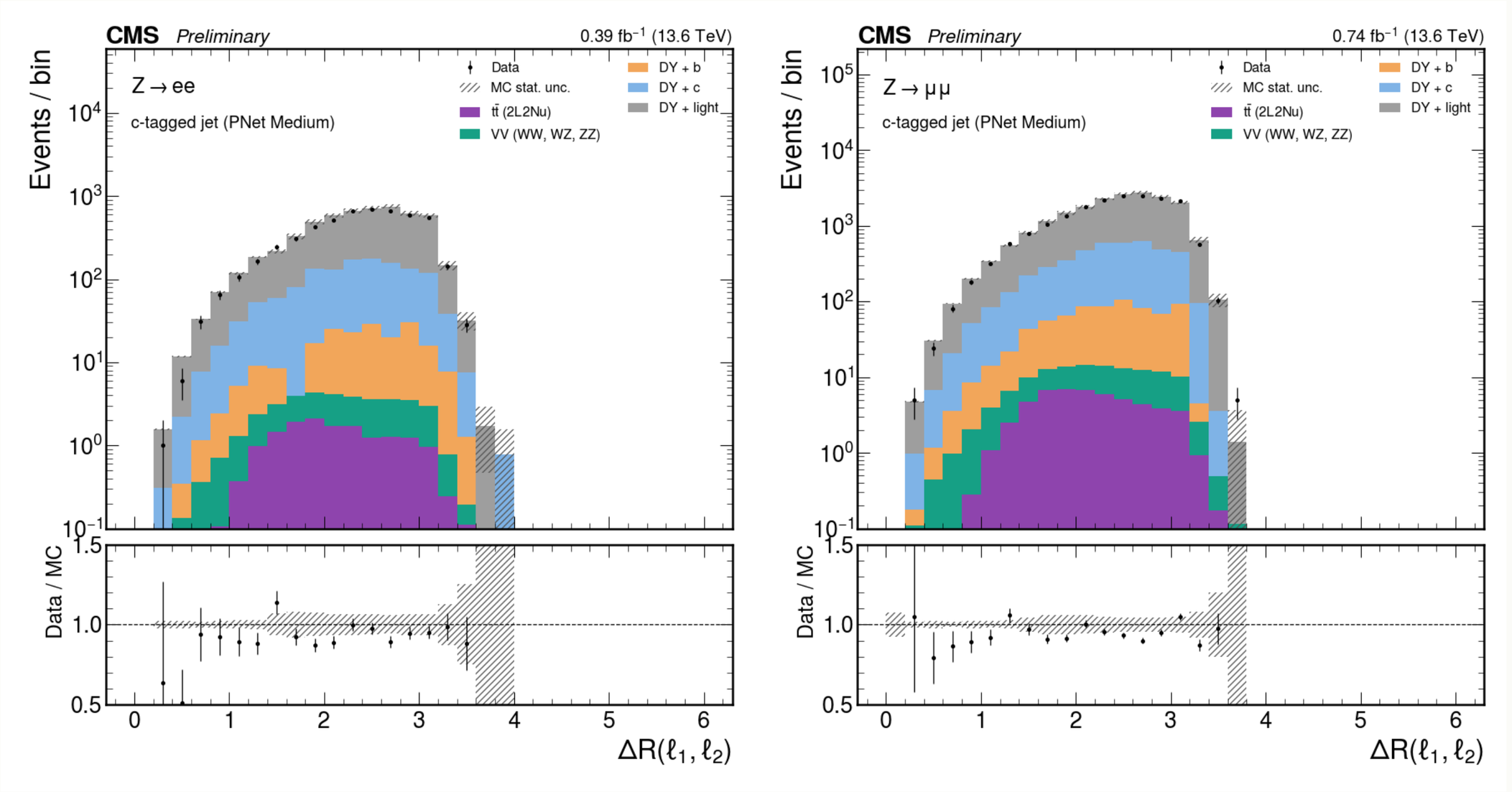
c-tag Medium

Second-lepton φ



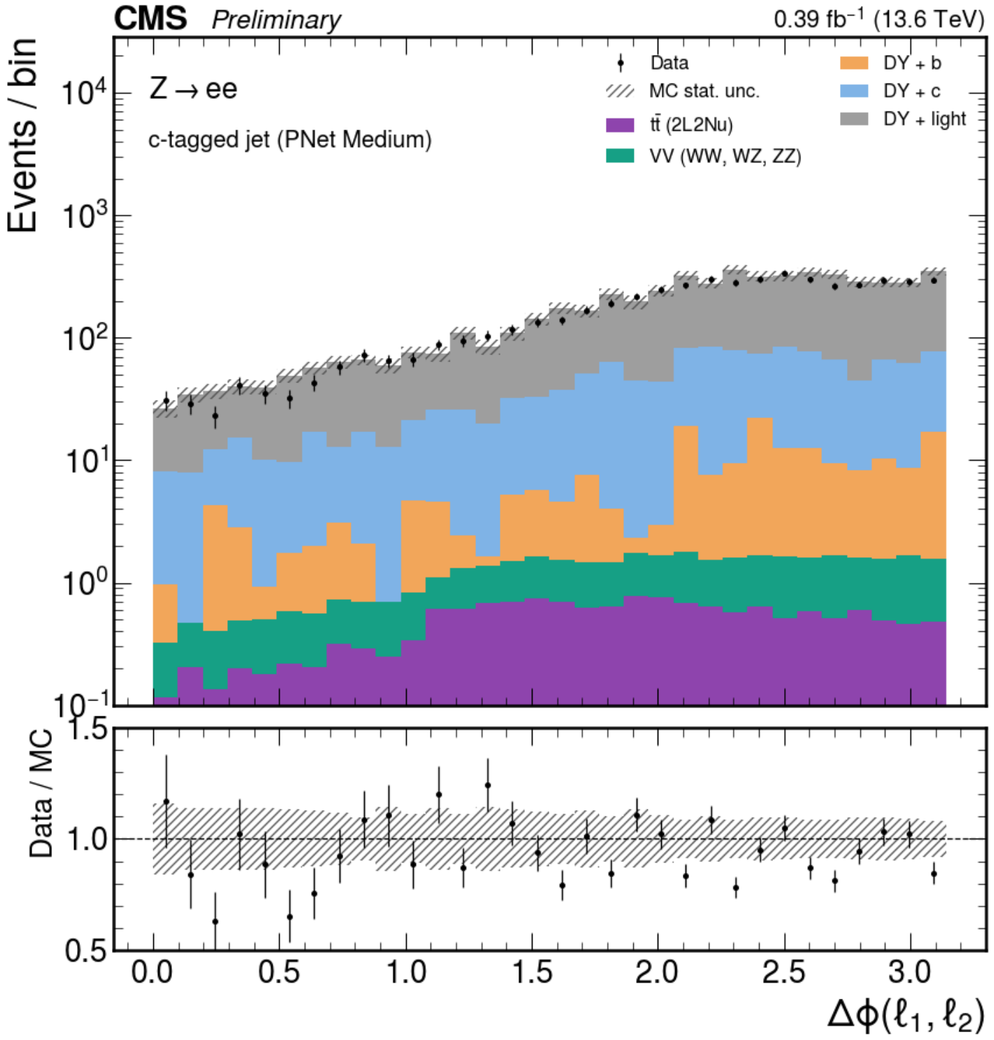
c-tag Medium

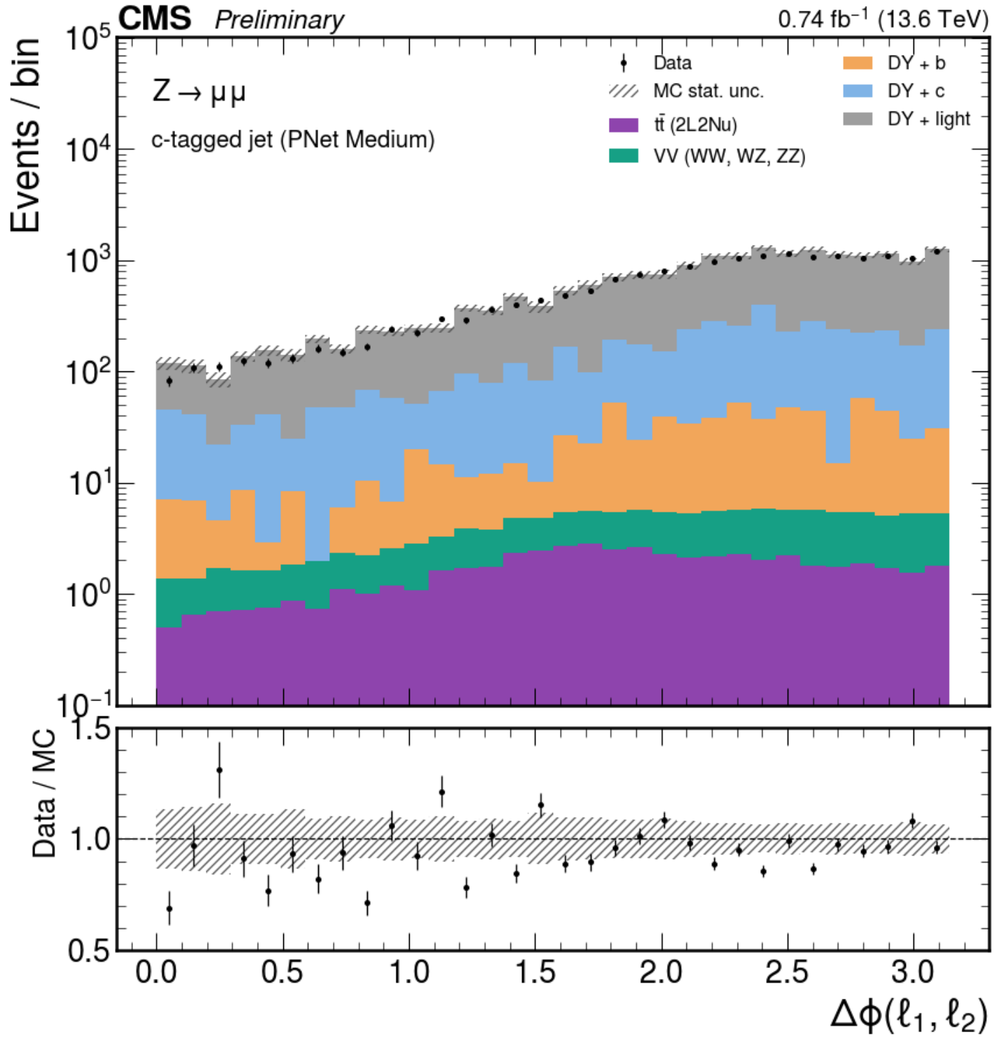
ΔR between the leptons



c-tag Medium

Δφ between the leptons

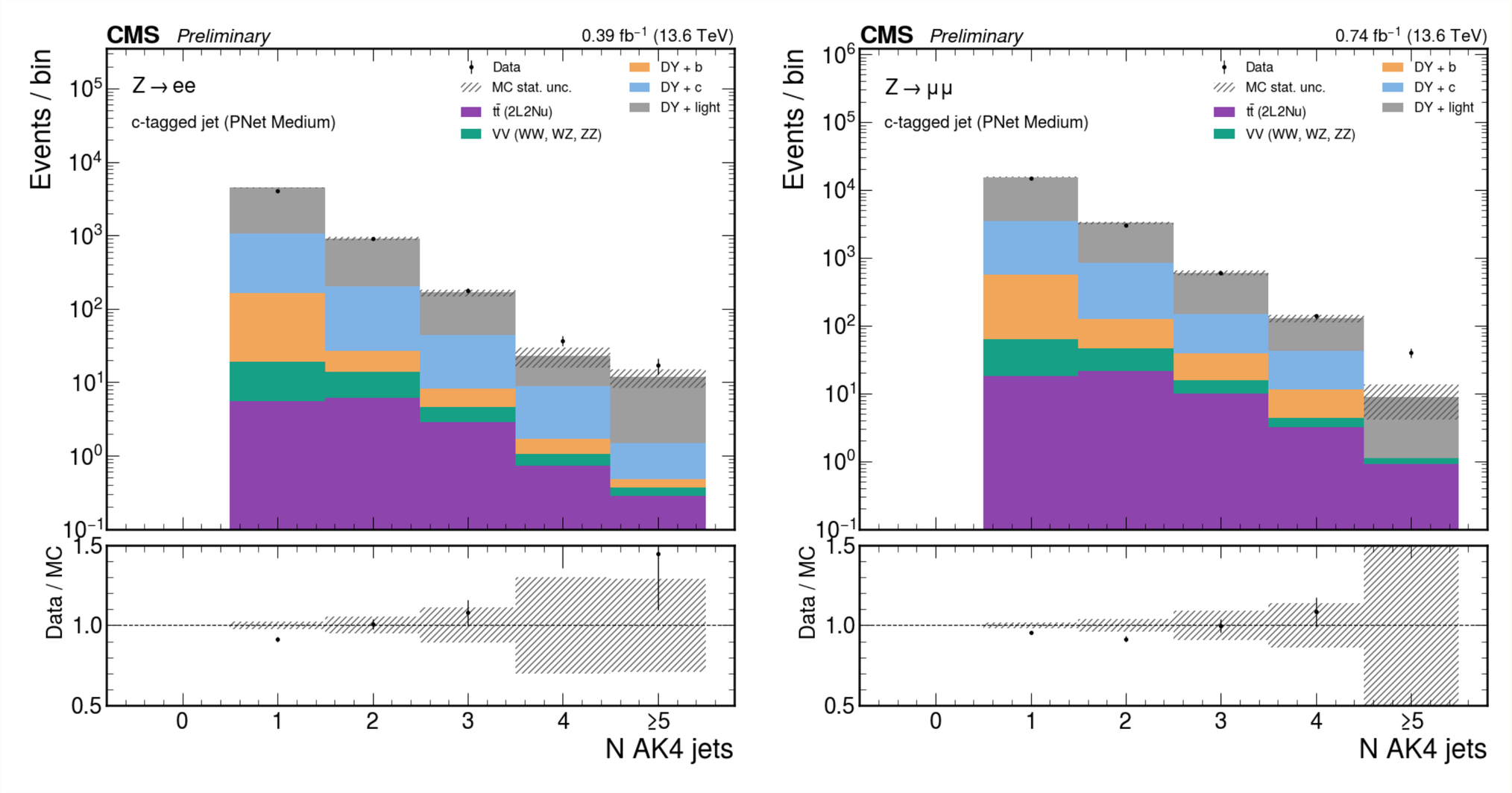




c-tag Medium

Number of jets

AK4 jets with pT > 30 GeV, |η| < 2.5, away from the leptons. Untagged plot: Z events before the jet cut, so the 0-jet bin is filled.

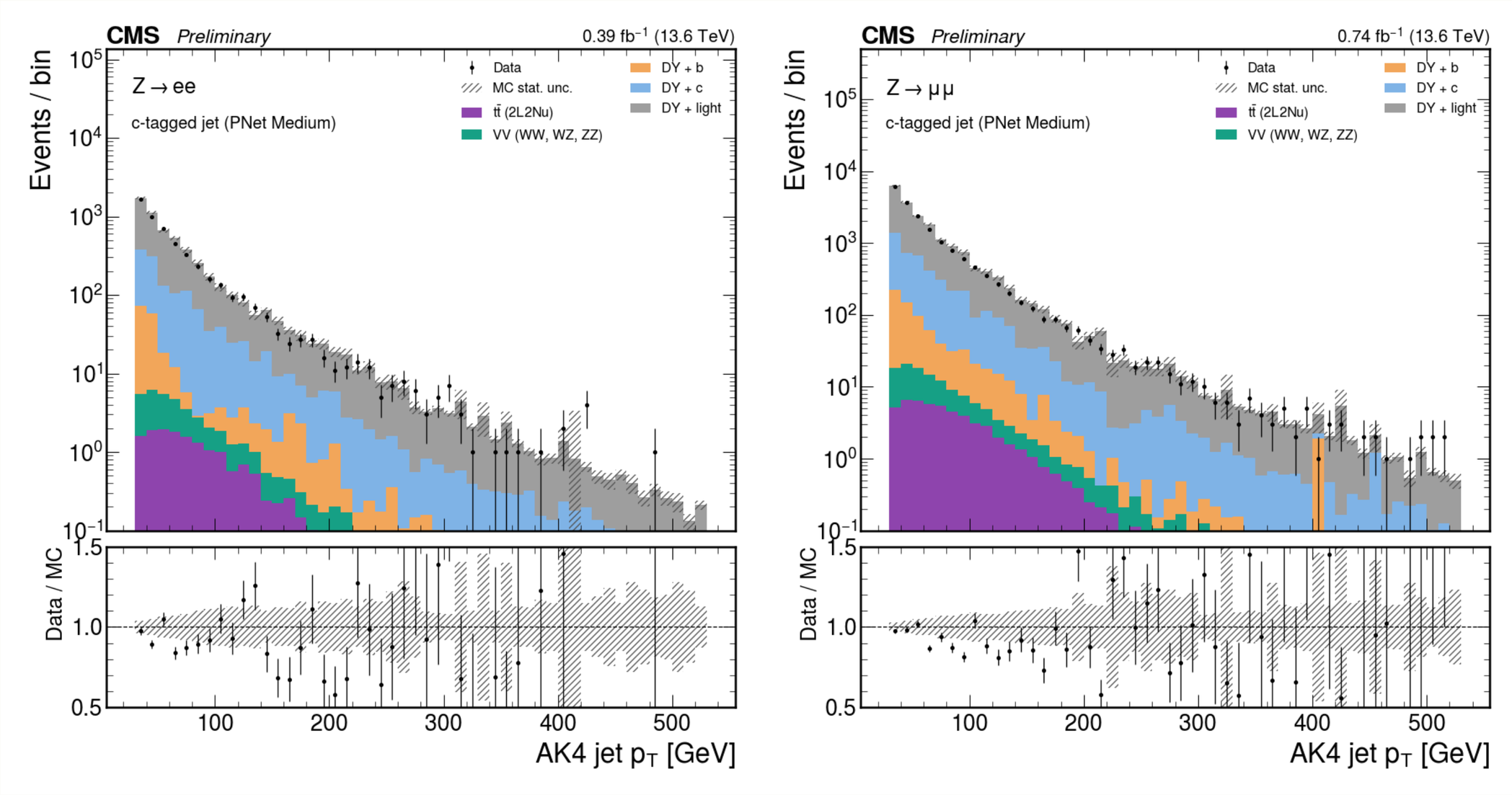


Summer22 (2022 C–D), 13.6 TeV · Z→ℓℓ + jets · 20 files per MC sample and per data channel · AK4 jets · stat. errors only · 150

c-tag Medium

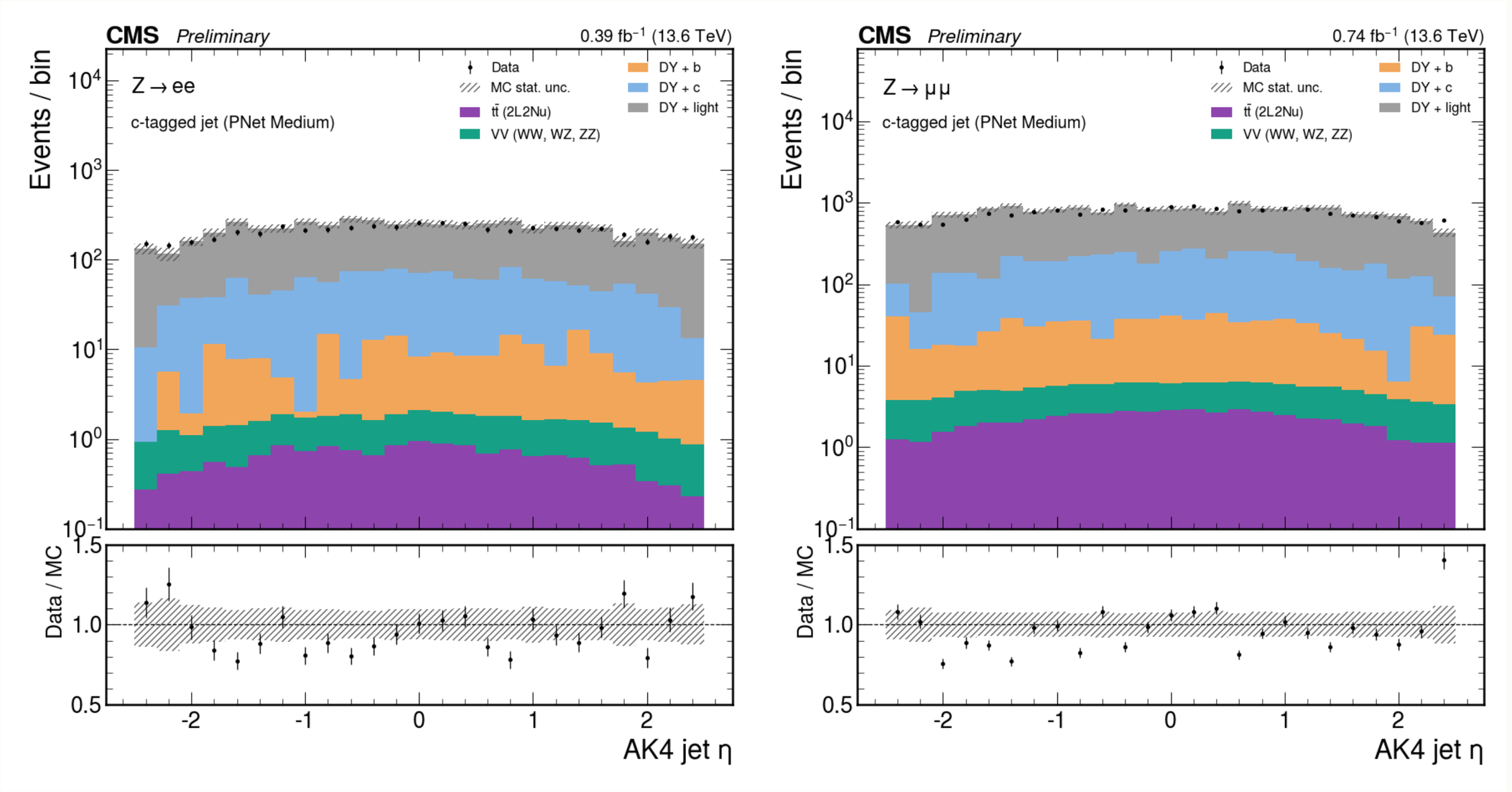
Leading-jet pT

The tag (if any) is on this jet.



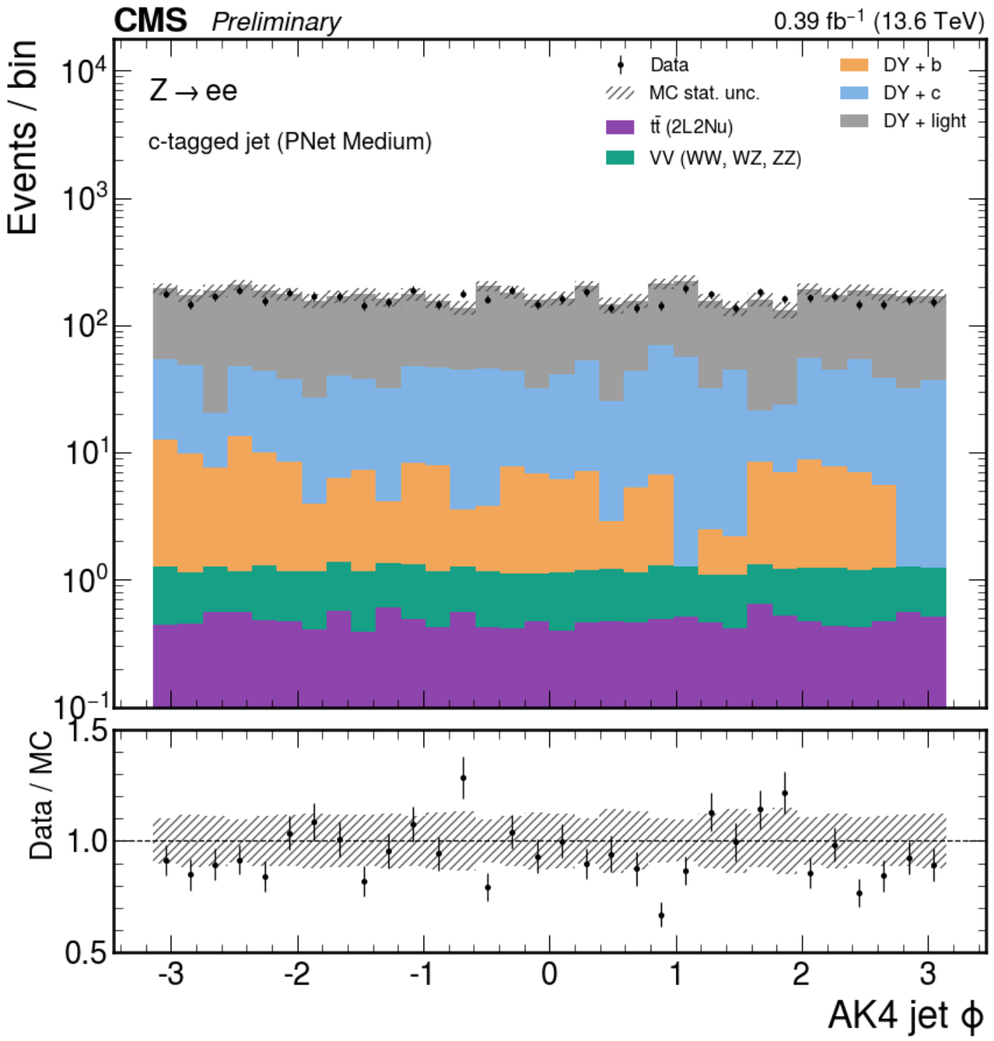
c-tag Medium

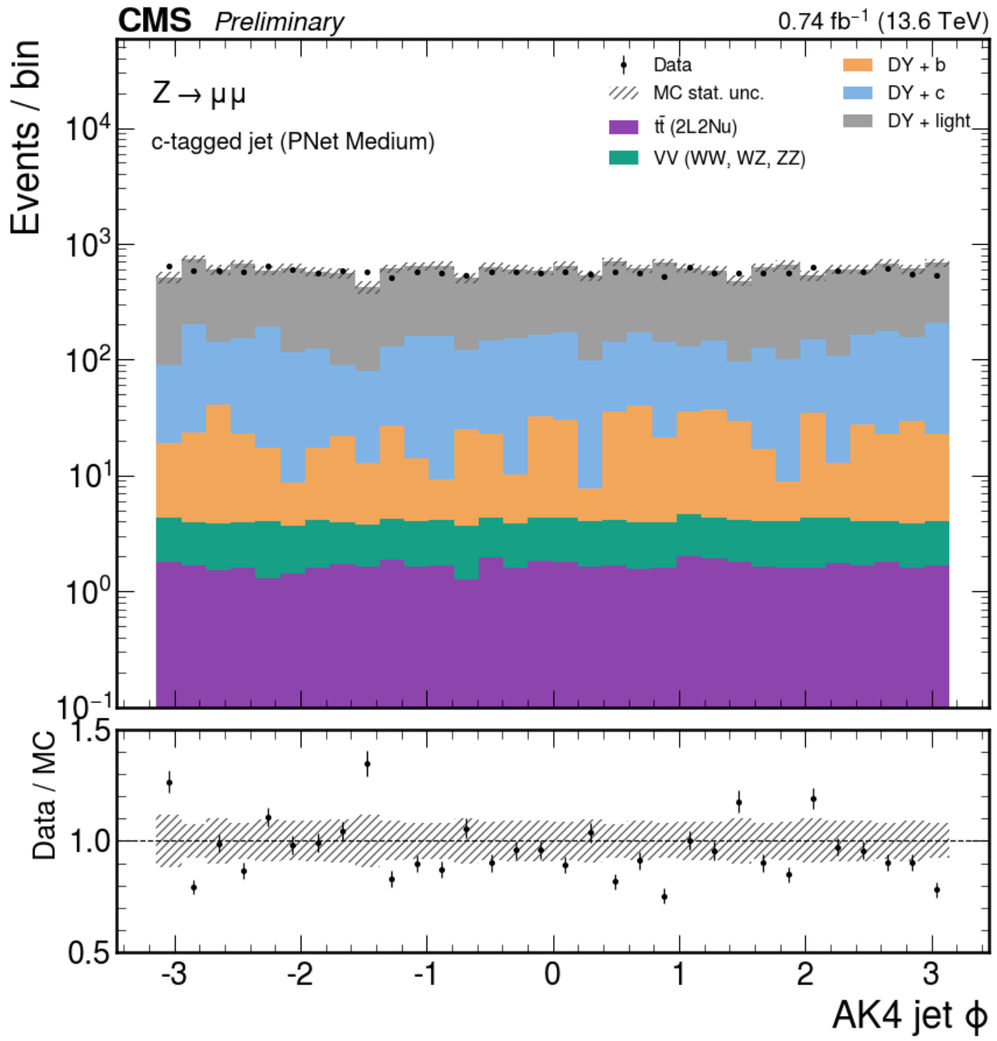
Leading-jet η



c-tag Medium

Leading-jet φ

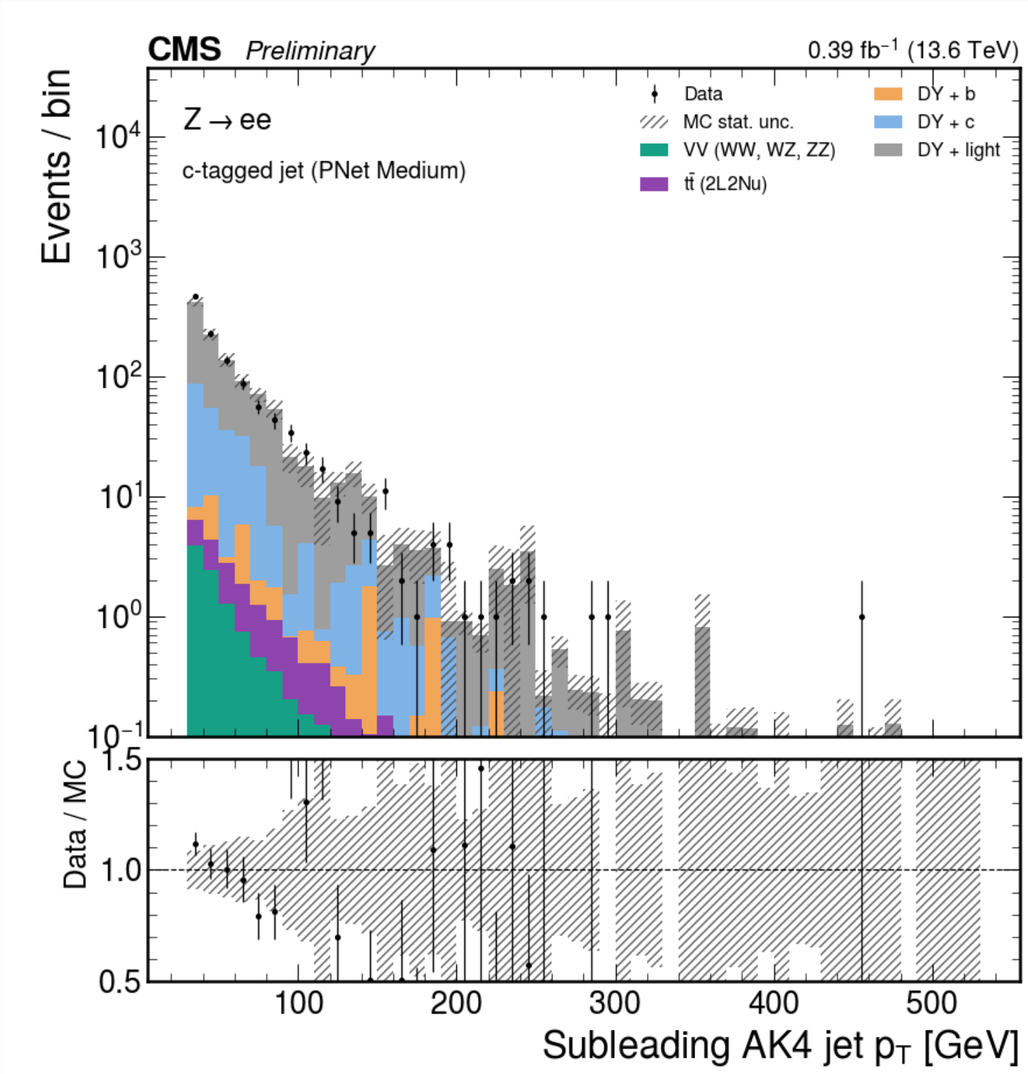


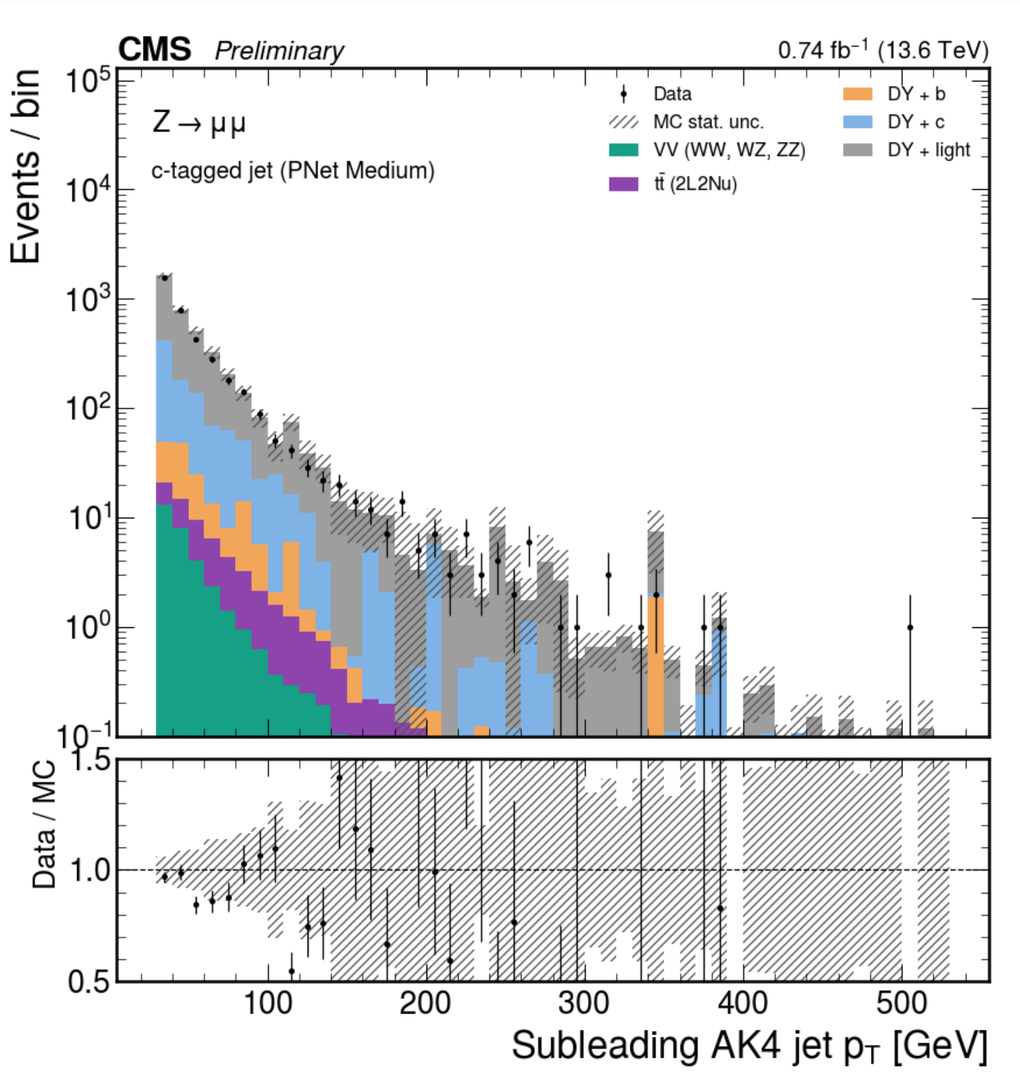


c-tag Medium

Second-jet pT

Events with at least 2 jets.

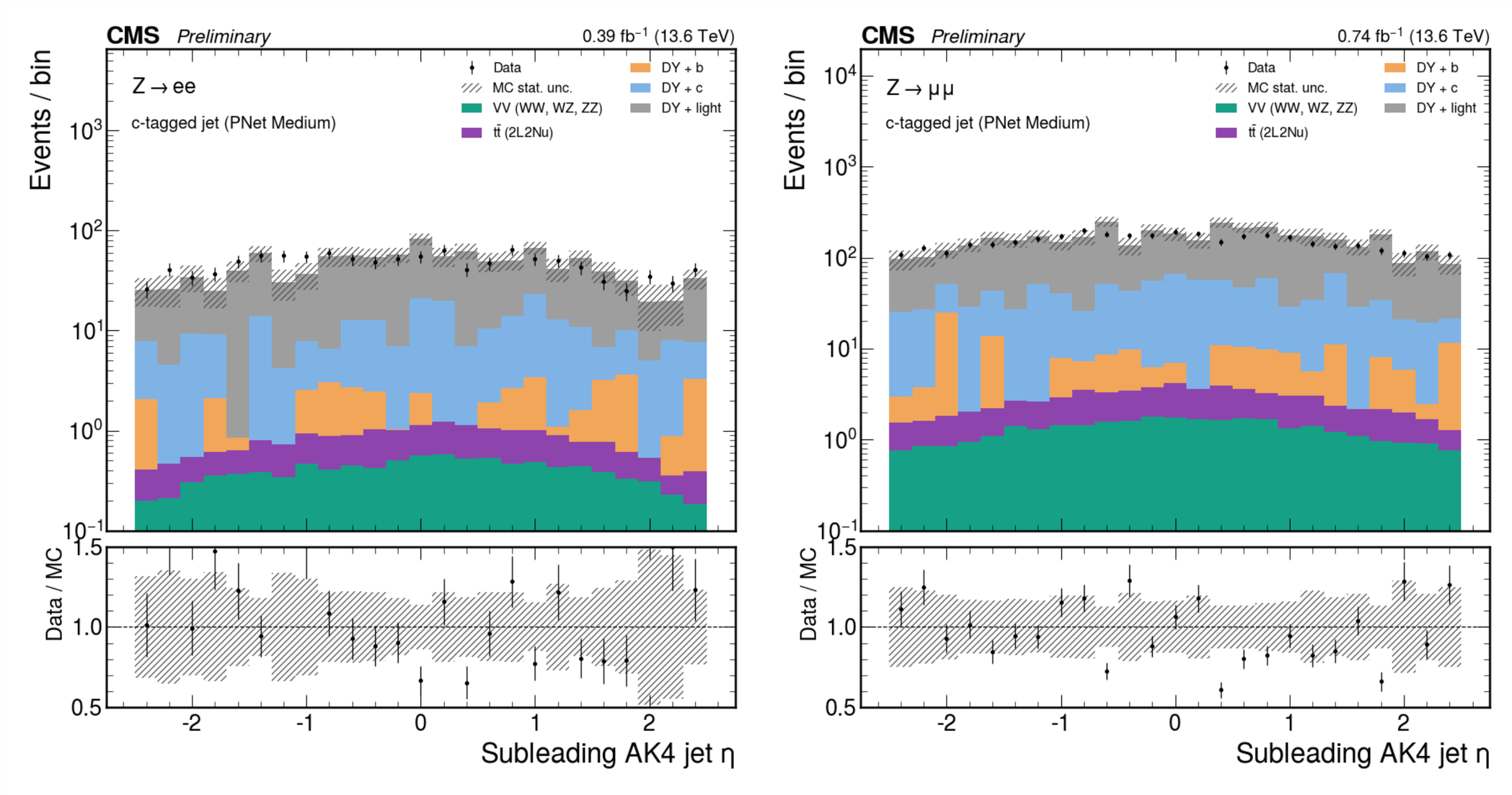




c-tag Medium

Second-jet η

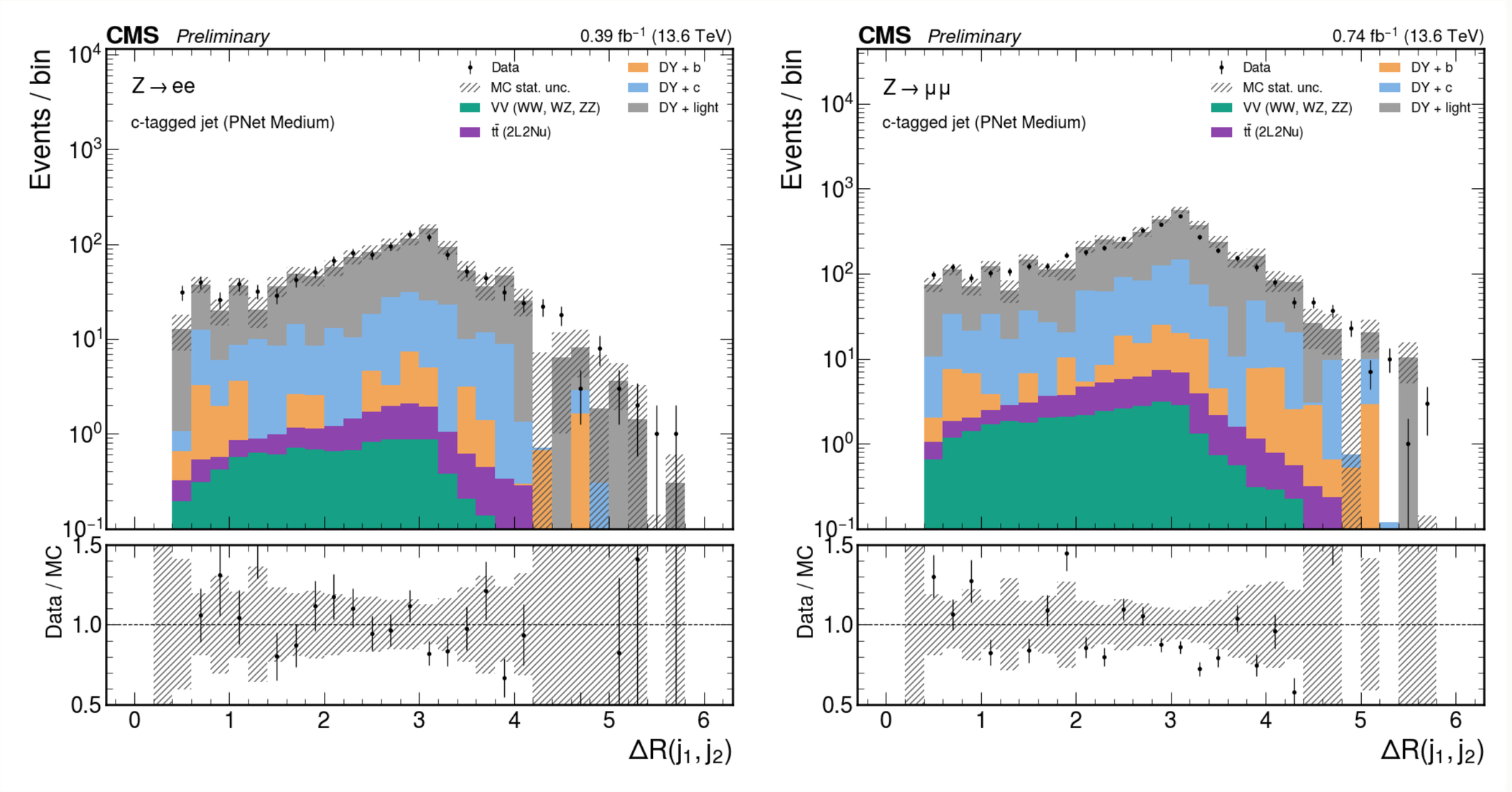
Events with at least 2 jets.



c-tag Medium

ΔR between the two leading jets

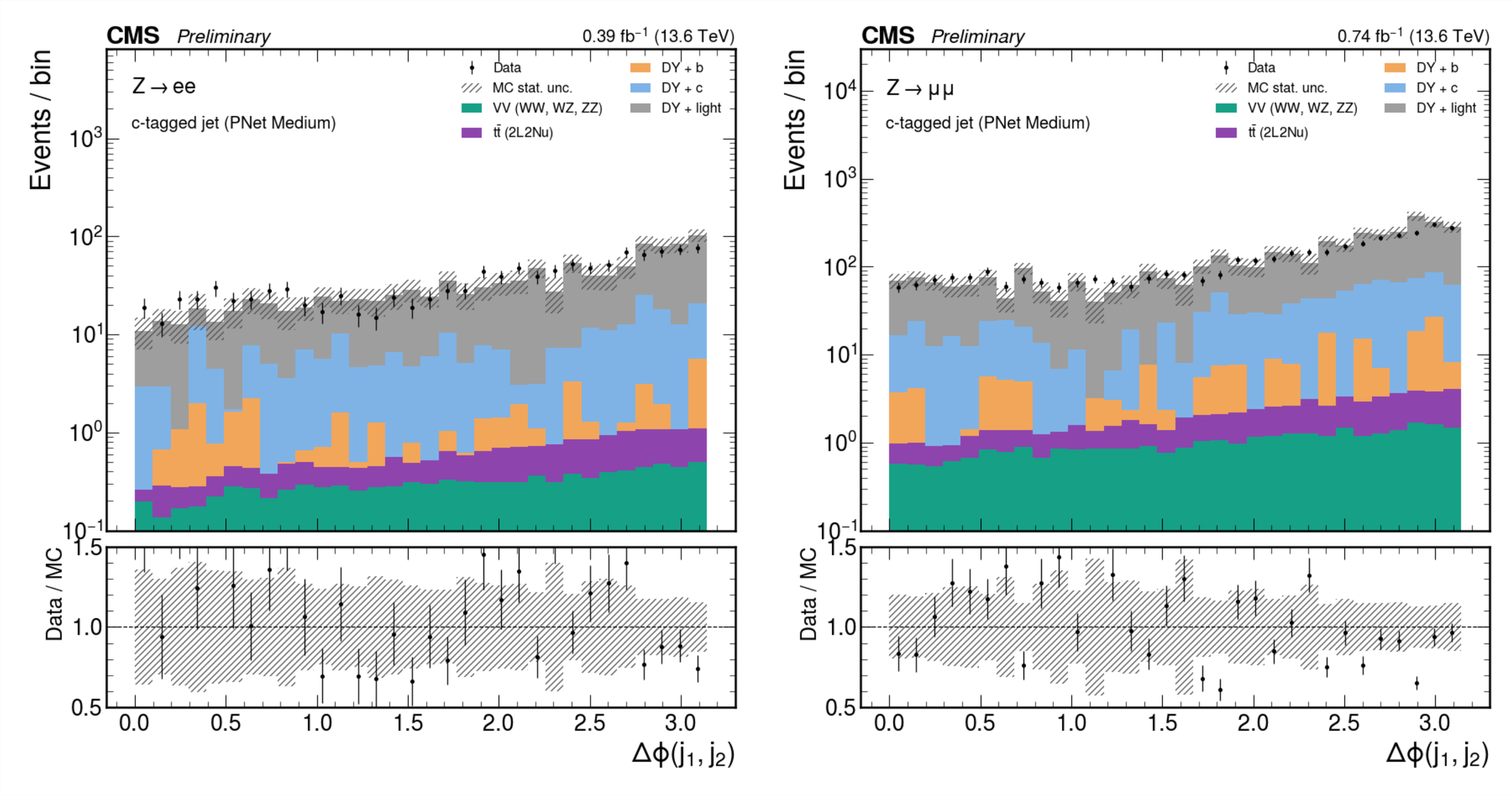
Events with at least 2 jets.



c-tag Medium

Δφ between the two leading jets

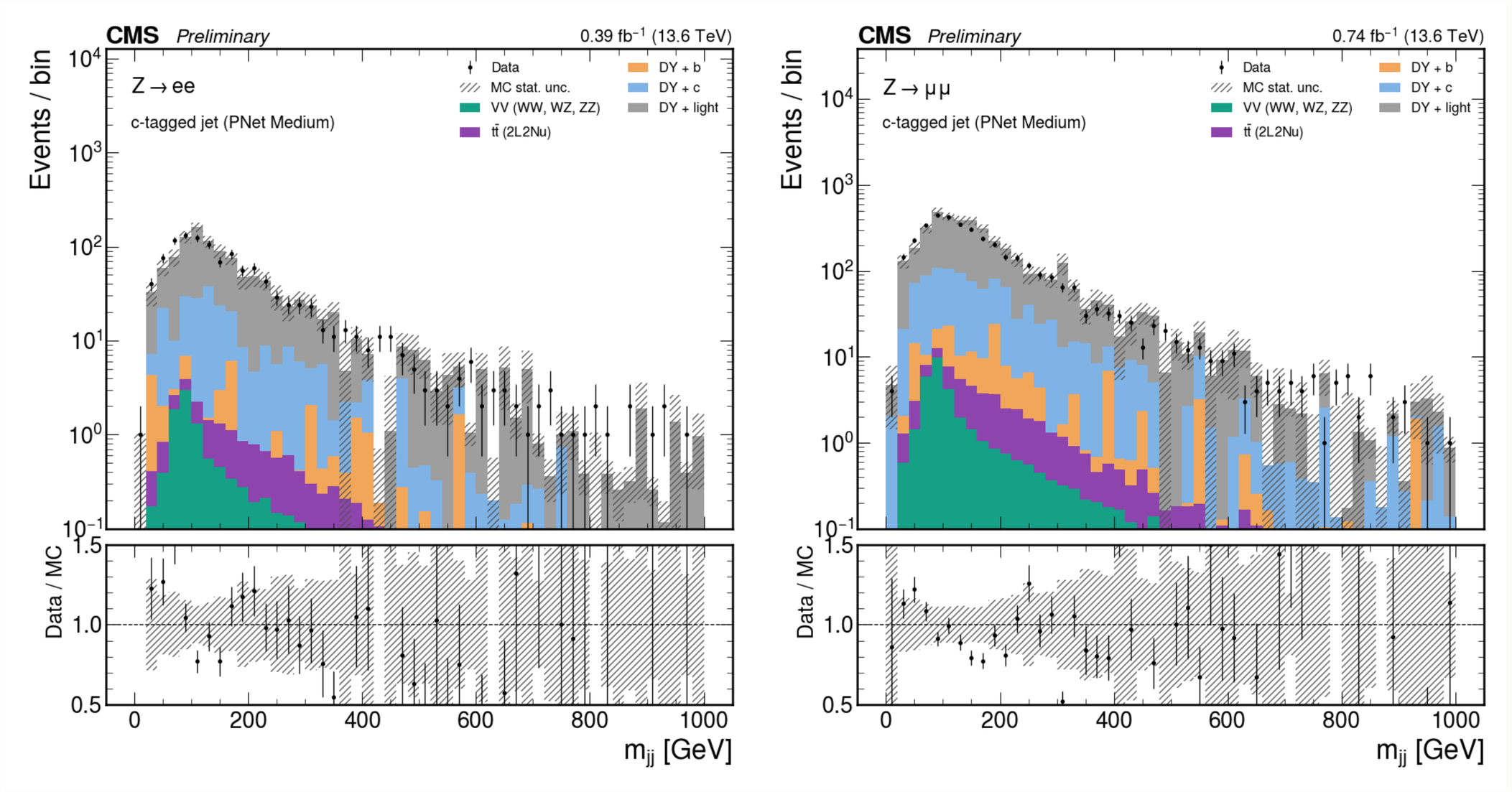
Events with at least 2 jets.



c-tag Medium

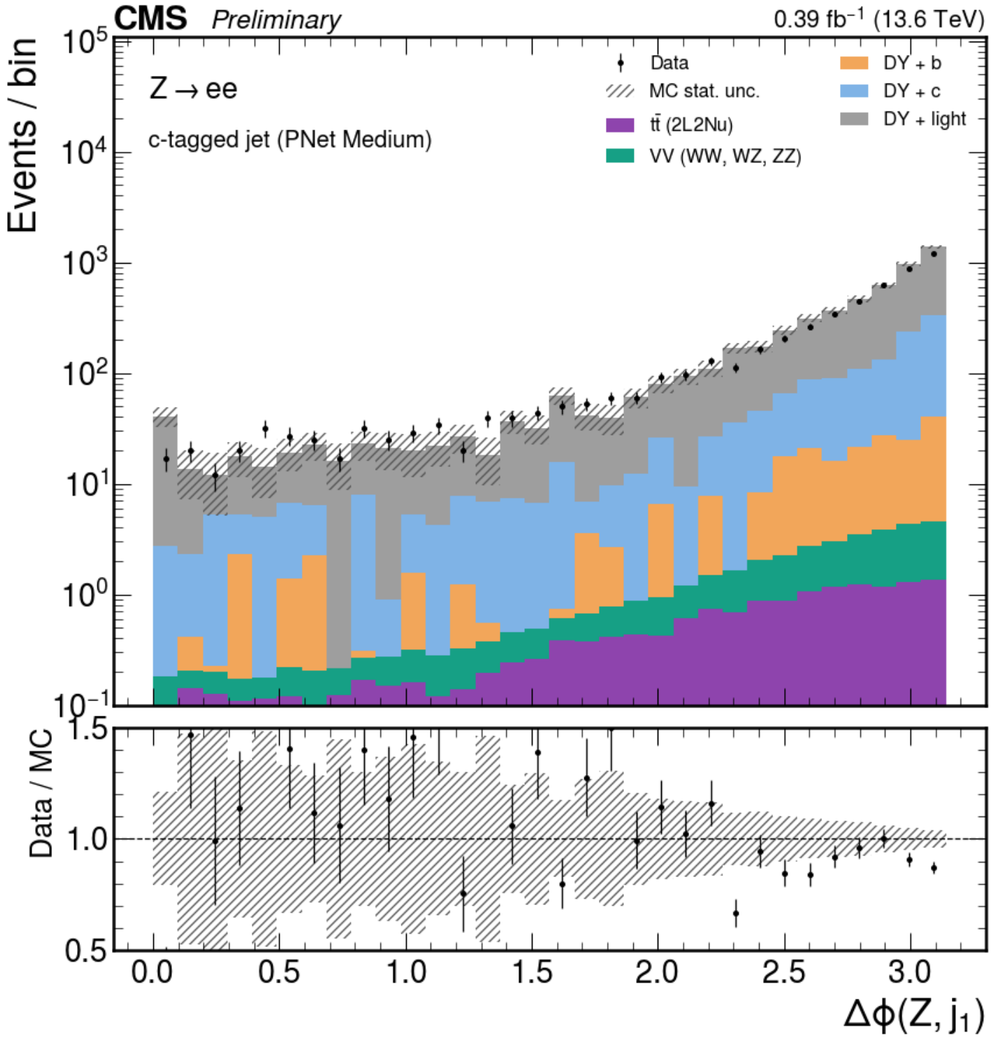
Dijet mass

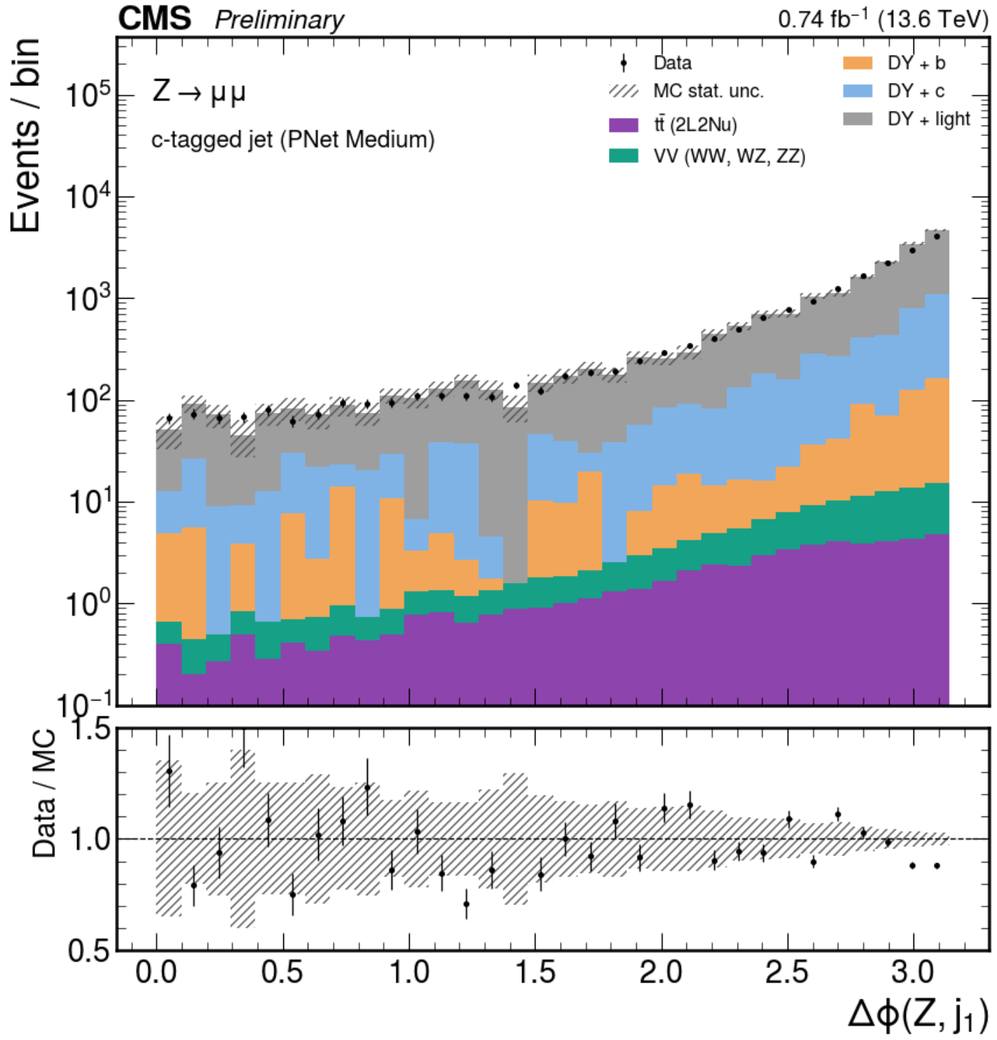
Events with at least 2 jets.



c-tag Medium

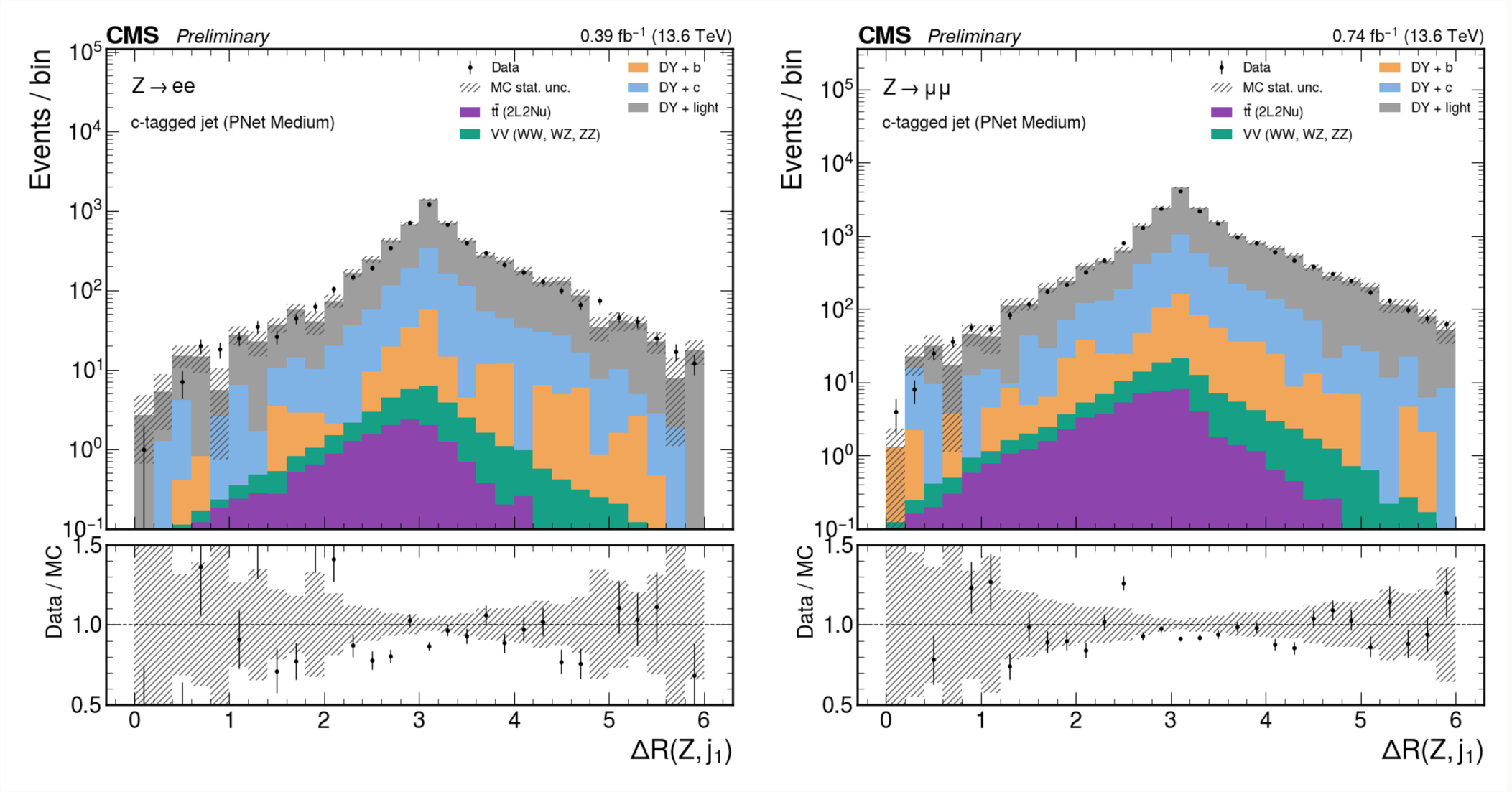
Δφ(Z, leading jet)





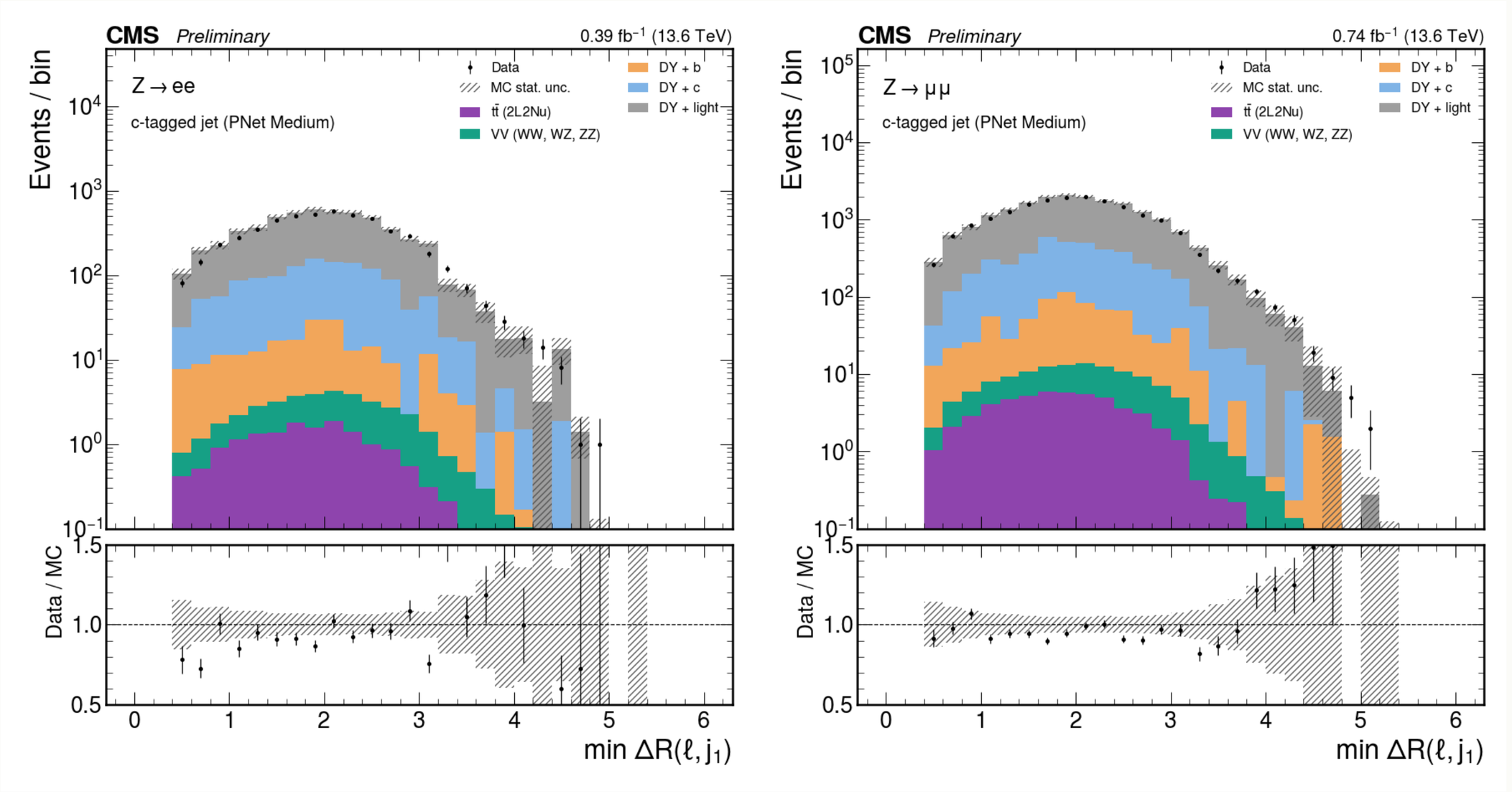
c-tag Medium

ΔR(Z, leading jet)



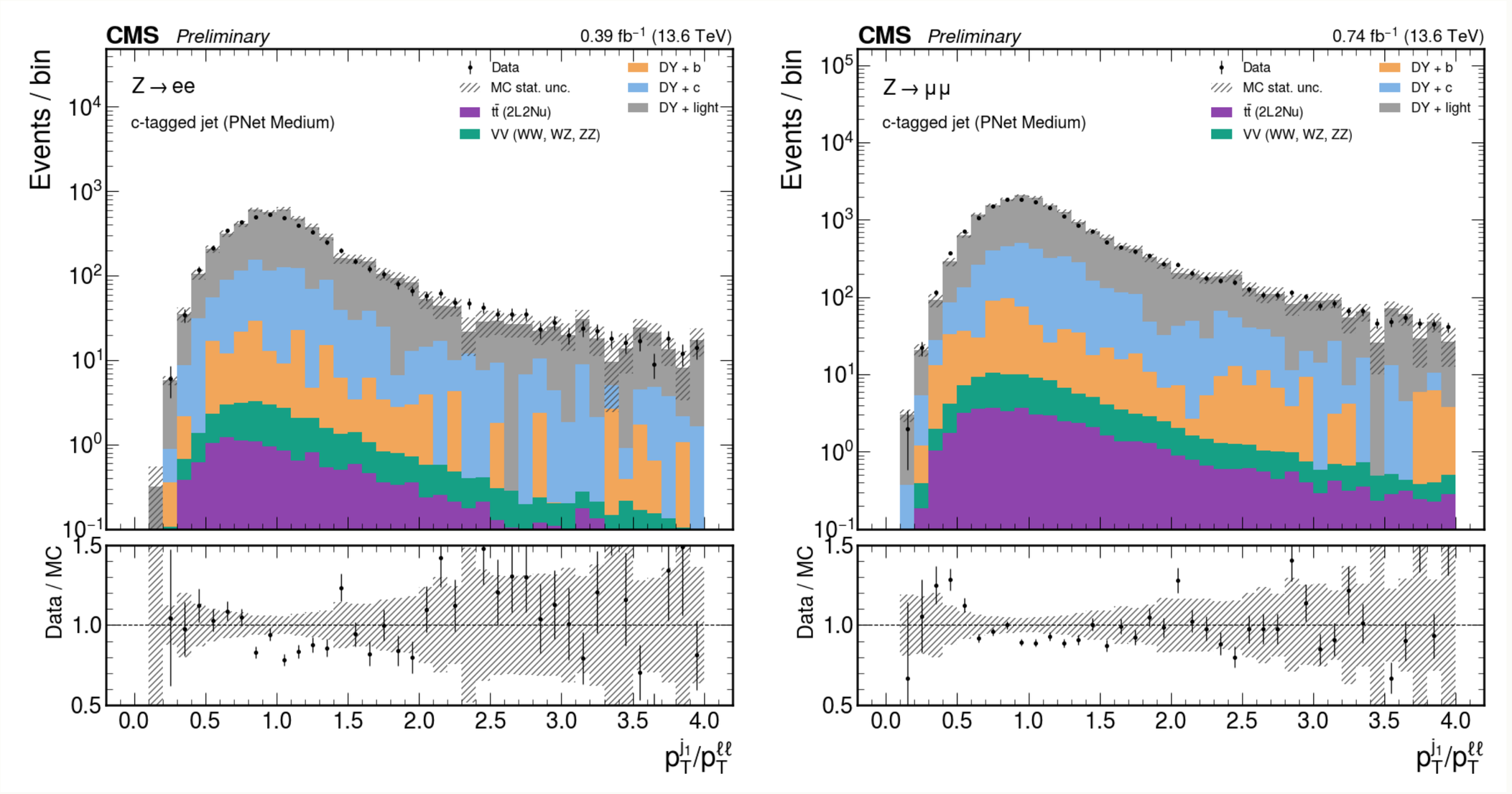
c-tag Medium

Smallest ΔR(lepton, leading jet)



c-tag Medium

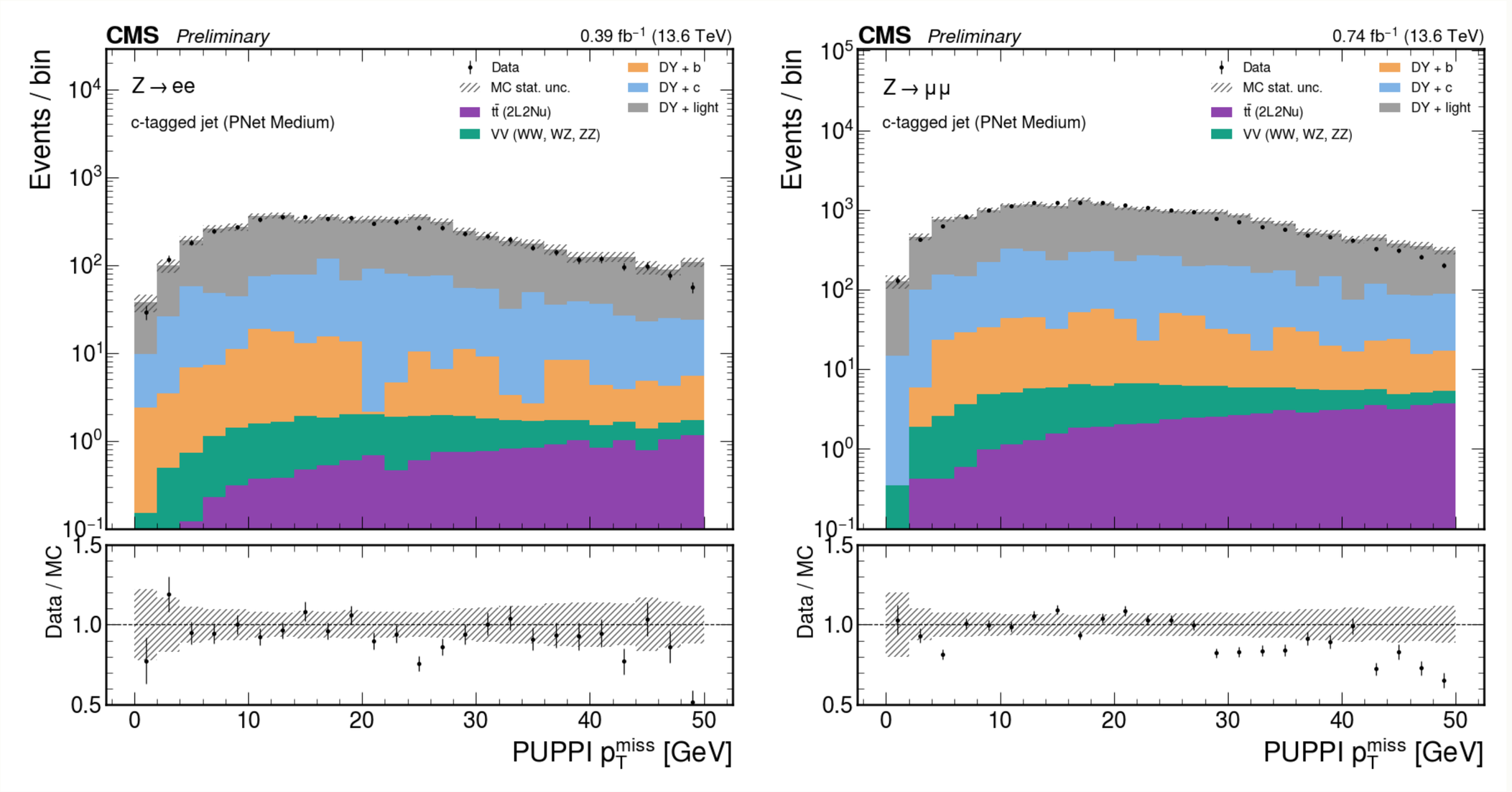
pT balance: leading-jet pT / Z pT



c-tag Medium

PUPPI MET

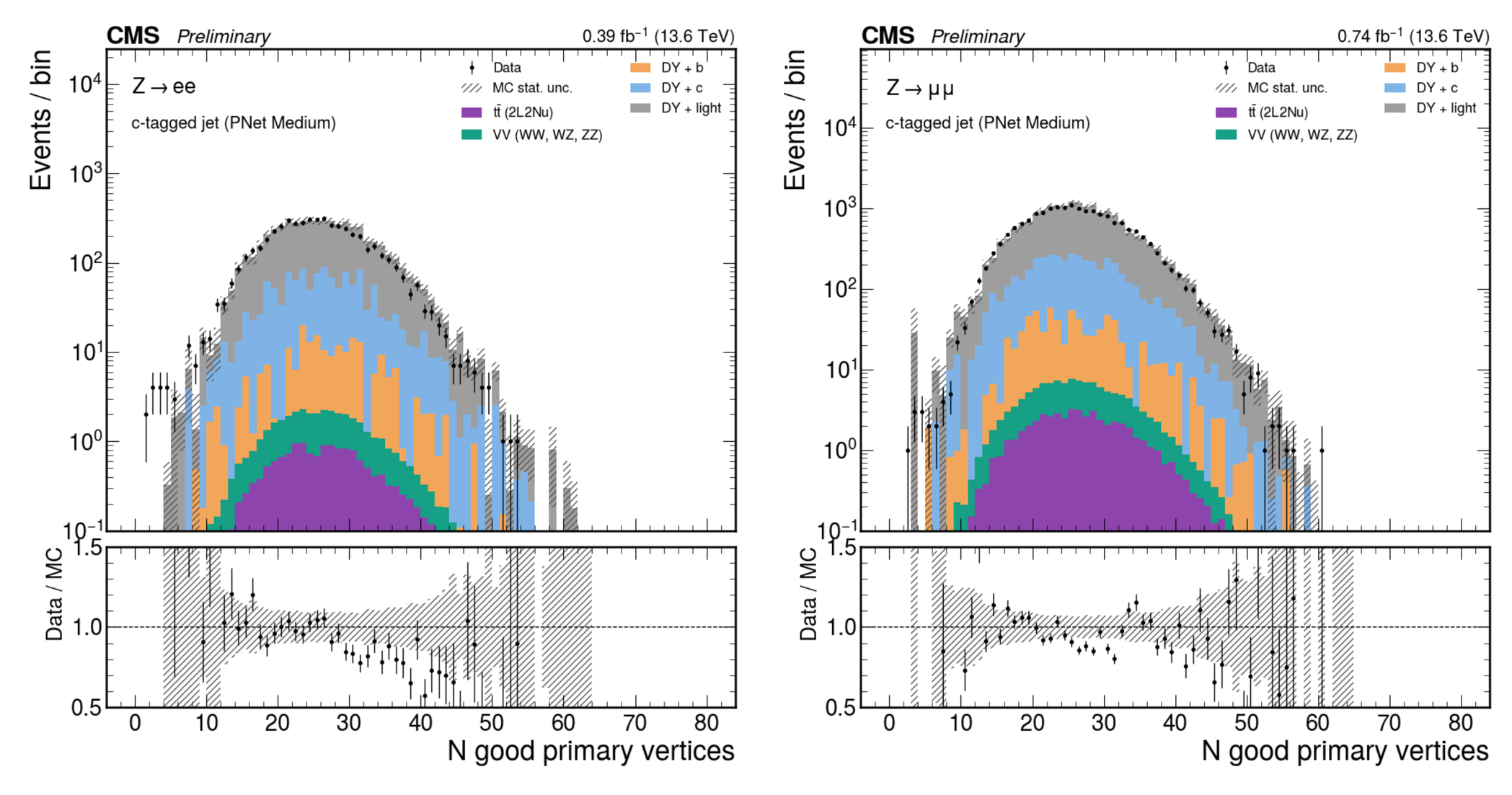
Type-1 corrected; the selection cuts at 50 GeV.



c-tag Medium

Number of good primary vertices

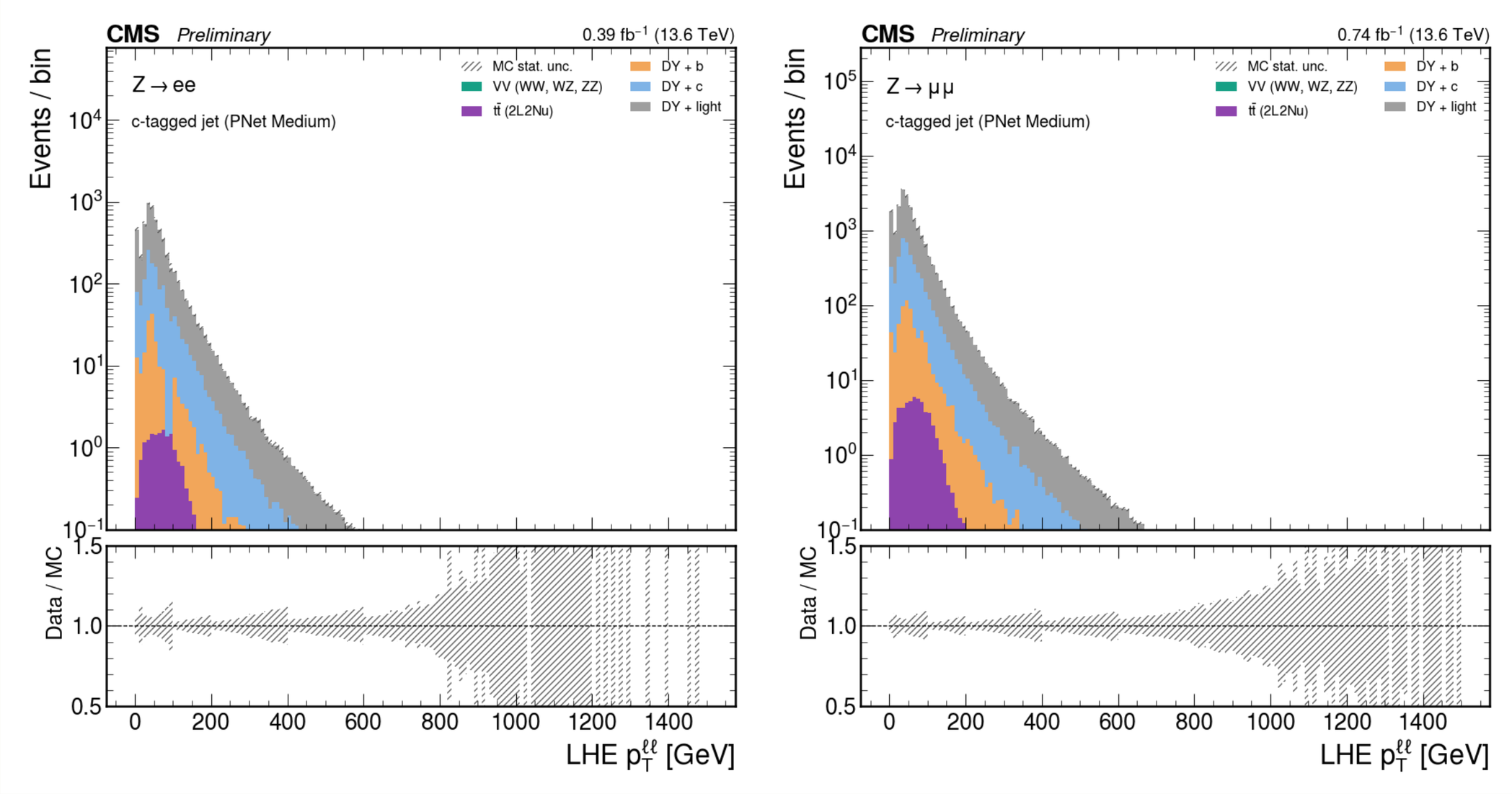
After pileup weights.



c-tag Medium

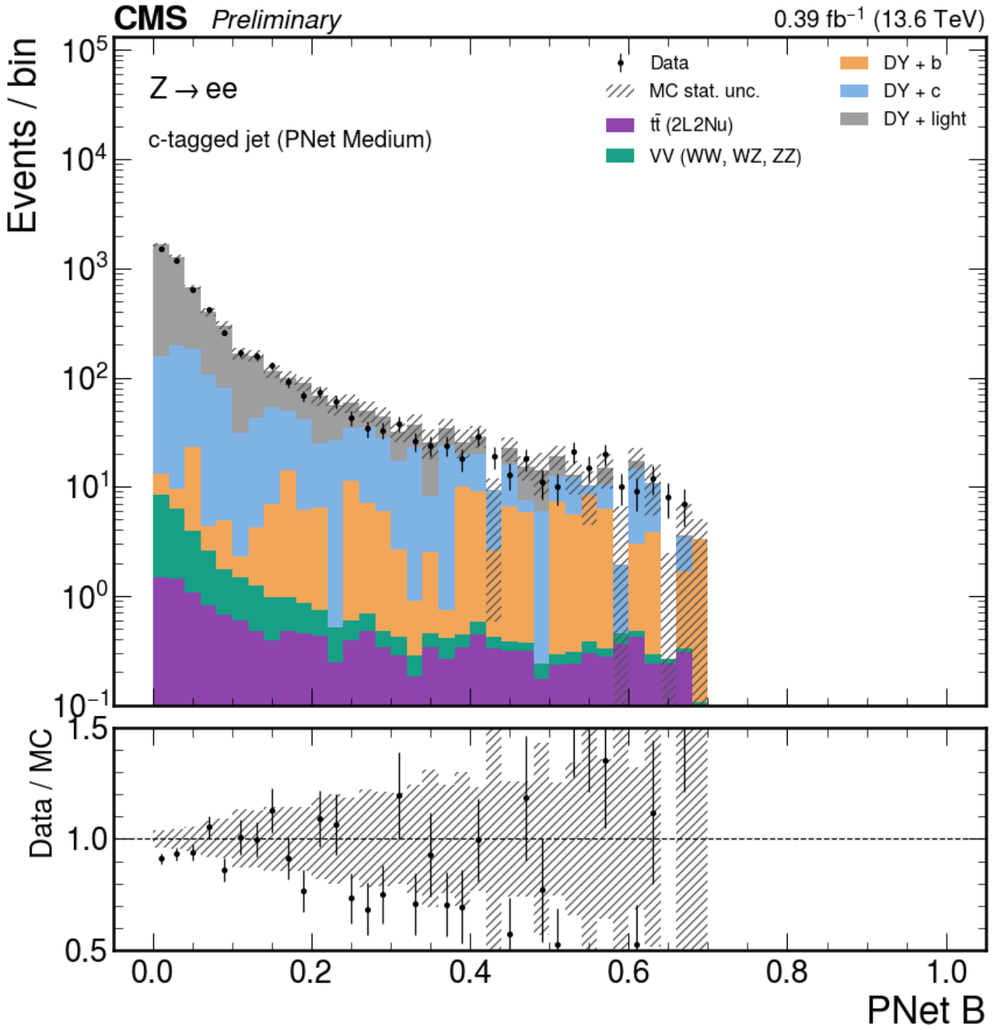
Generator Z pT (LHE)

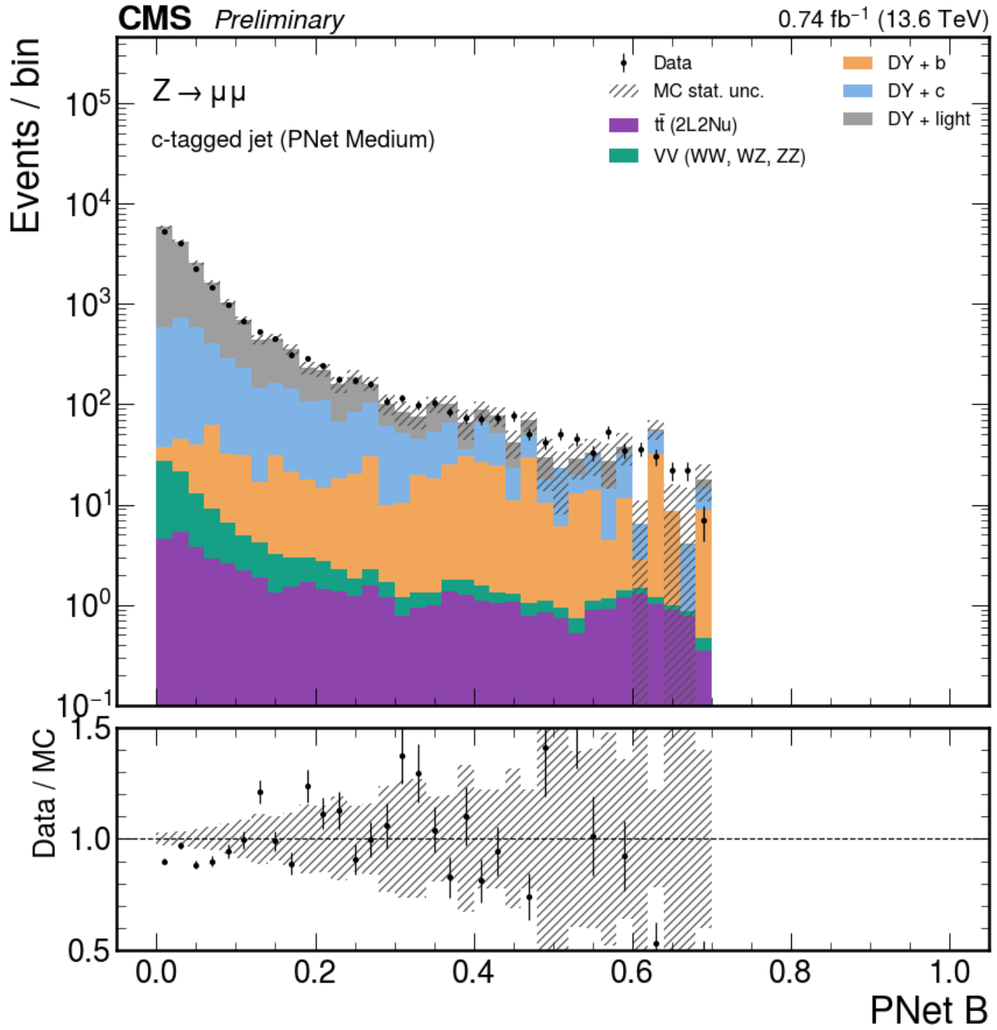
MC only, no data.



c-tag Medium

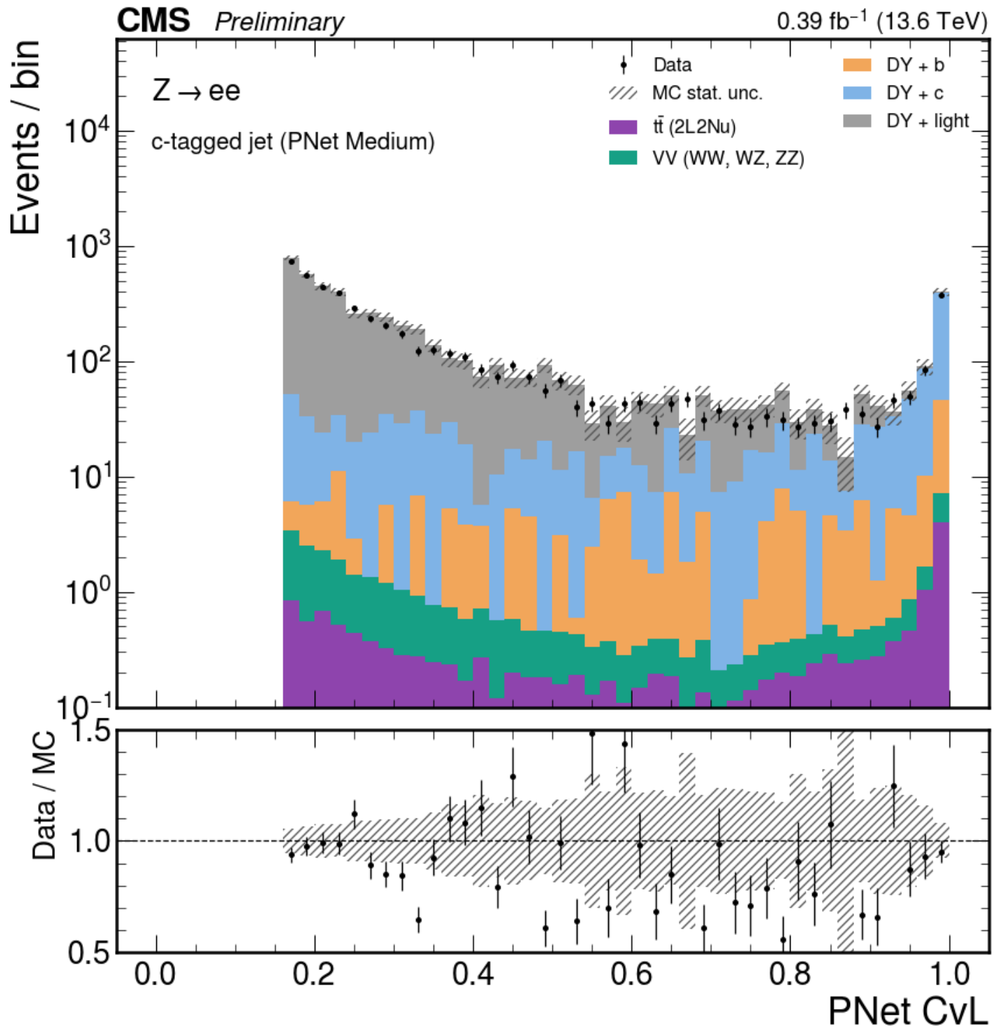
Leading-jet PNet B score

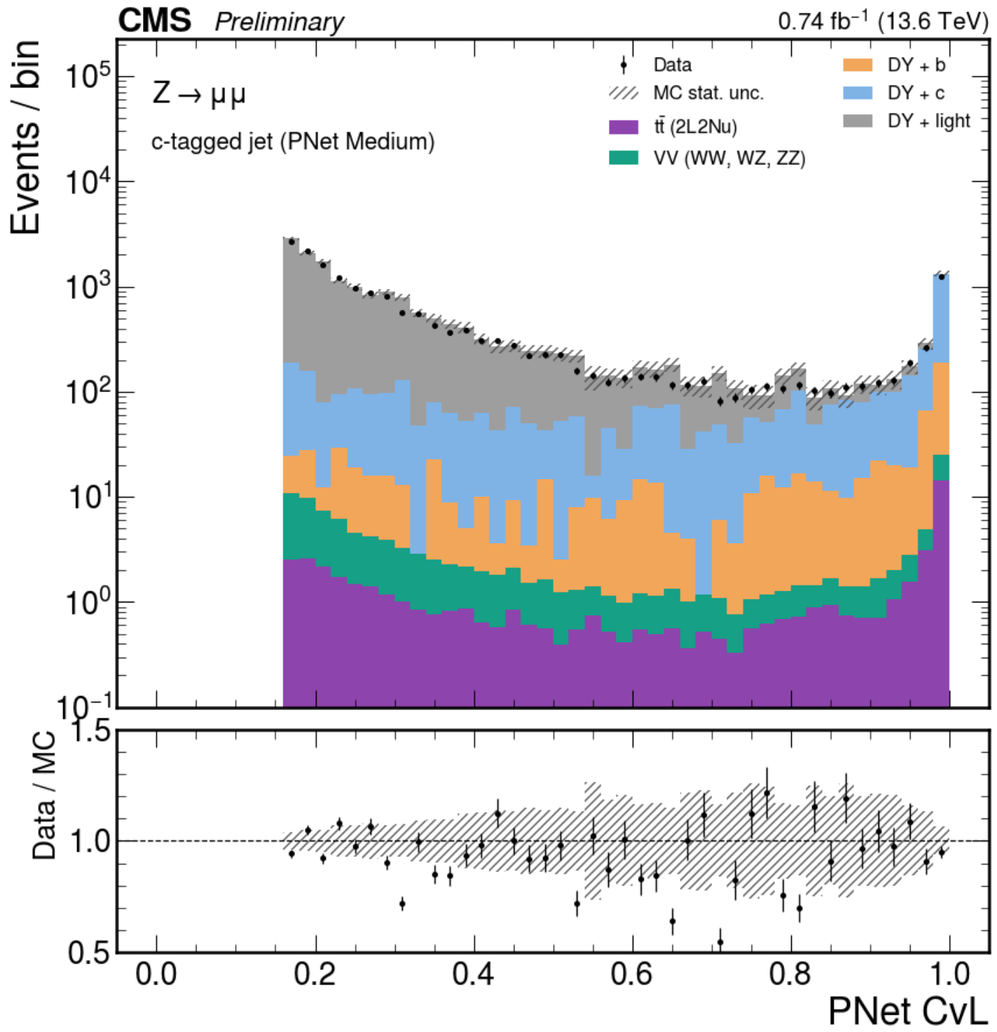




c-tag Medium

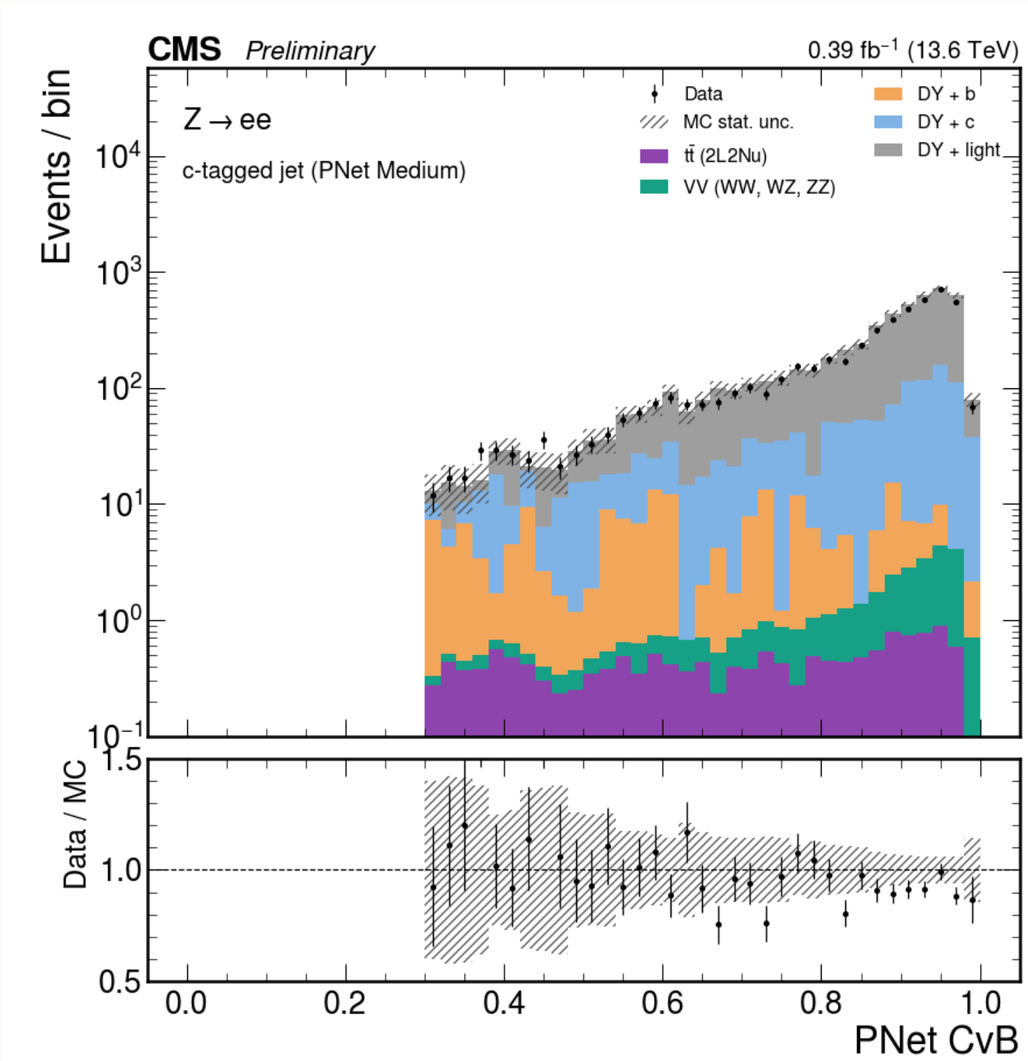
Leading-jet PNet CvL score

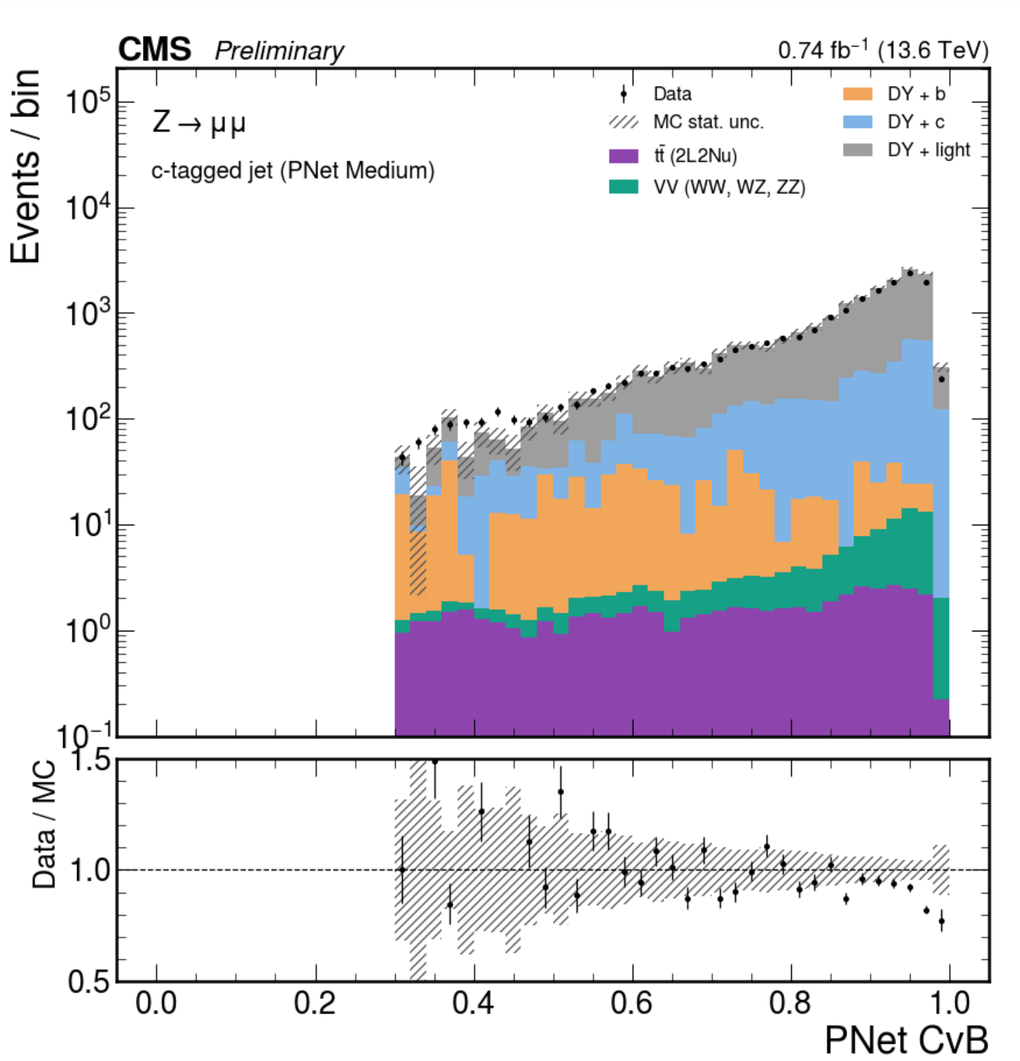




c-tag Medium

Leading-jet PNet CvB score

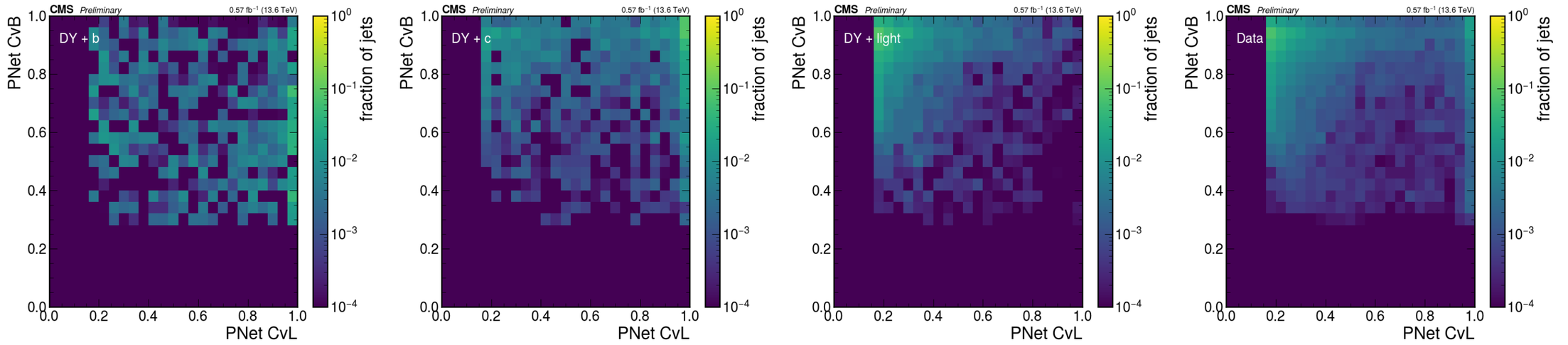




c-tag Medium

PNet CvL vs CvB of the leading jet

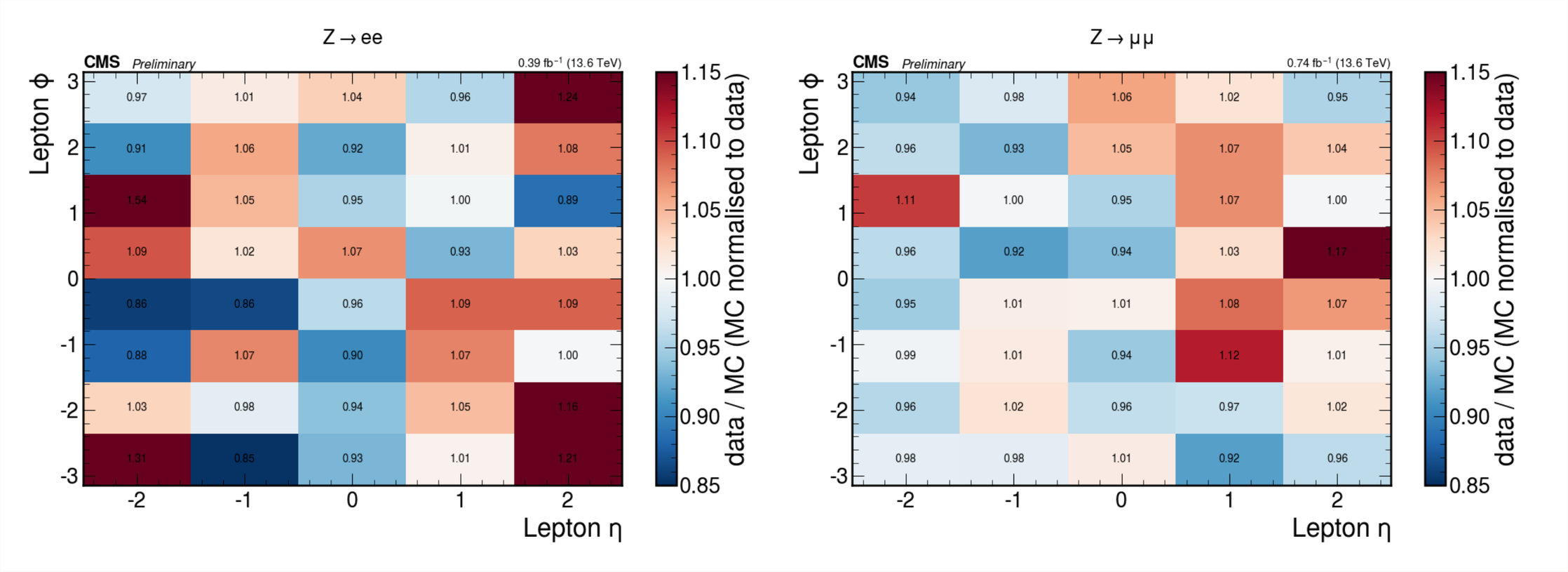
Each panel normalised to 1. Z→ee and Z→μμ added.



c-tag Medium

Lepton η–φ, data / MC

Both Z leptons. MC normalised to data in each map. Colour scale stops at ±15%; in the tagged regions there are few events per cell, so read the numbers.



b-tag Tight

Leading jet PNet B > 0.673

	data/MC	DY+b	DY+c	DY+light	tt	tagged MC	tagged data
Z→ee	1.004 ± 0.074	82.9%	5.4%	4.5%	6.8%	3.2%	3.4%
Z→μμ	0.902 ± 0.045	82.0%	4.8%	6.7%	6.1%	3.5%	3.3%

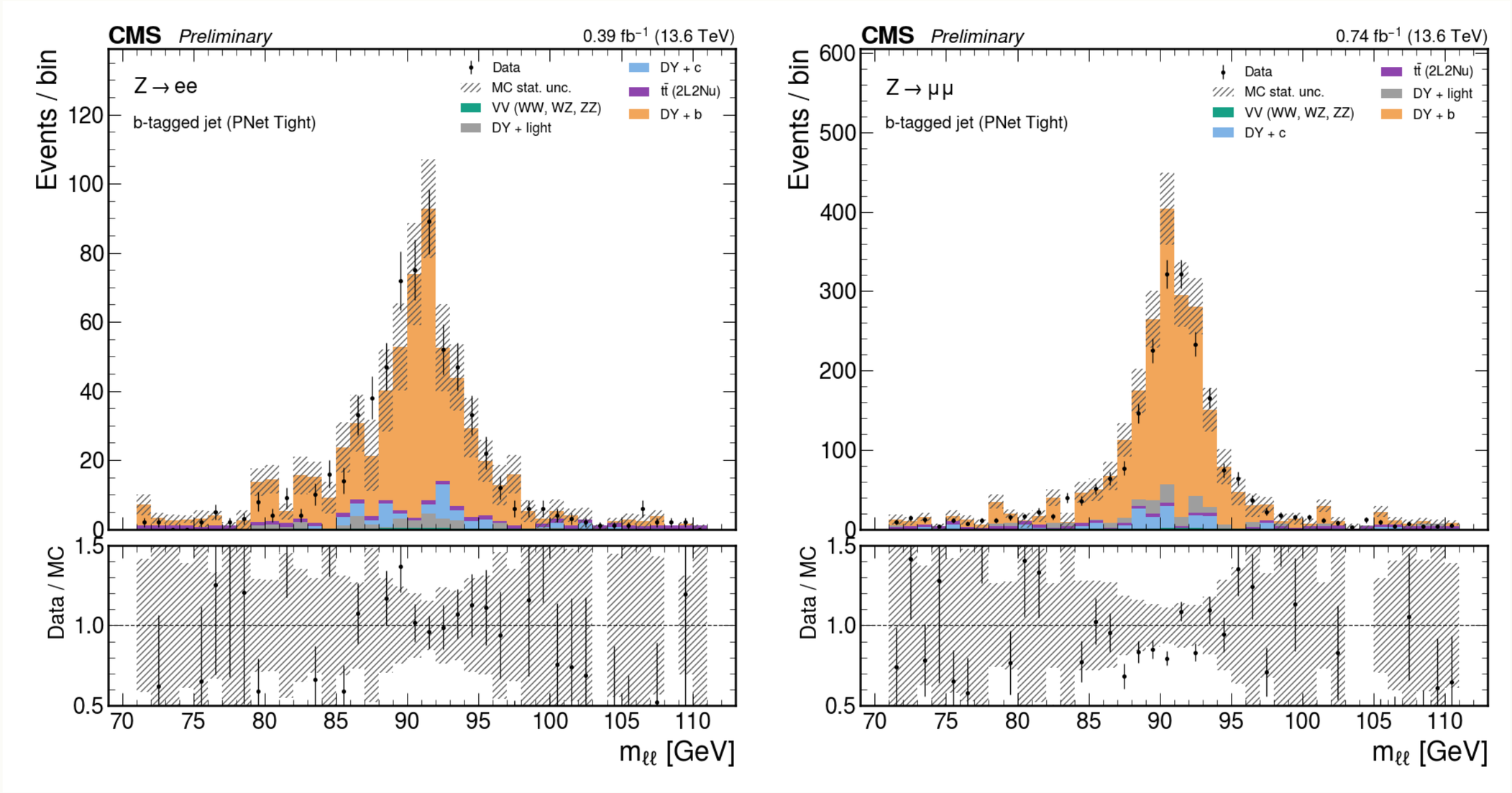
Flavour shares are of all MC. 'Tagged' = fraction of Z + jet events passing the tag. Errors are statistical only.

32 plots follow: Z and leptons, jets, Z–jet, MET and vertices, tagger scores, 2D maps. Left Z→ee, right Z→μμ.

b-tag Tight

Dilepton mass

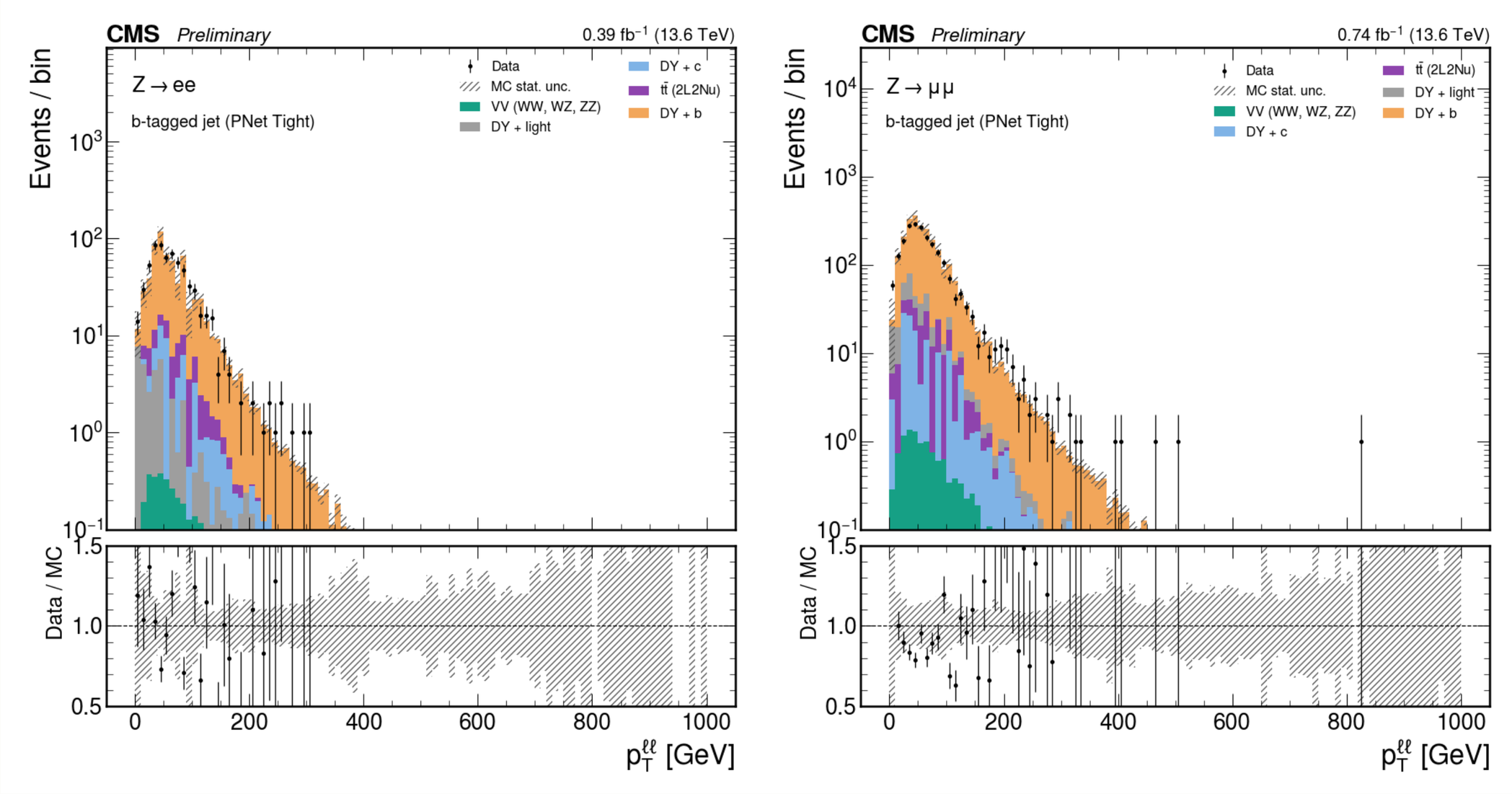
m(ℓℓ), 71–111 GeV window, linear scale.



b-tag Tight

Z pT

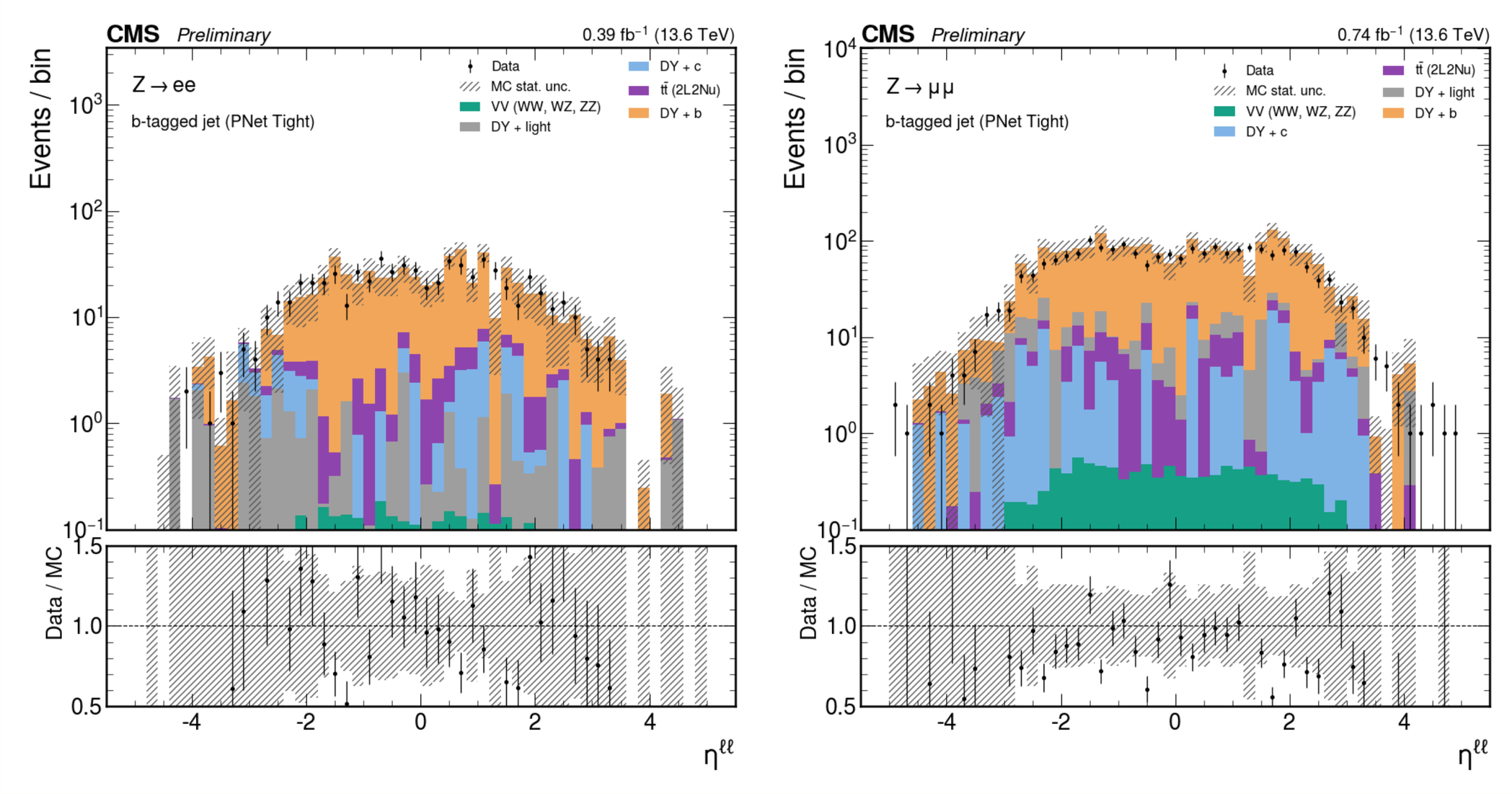
pT of the lepton pair.



b-tag Tight

Z η

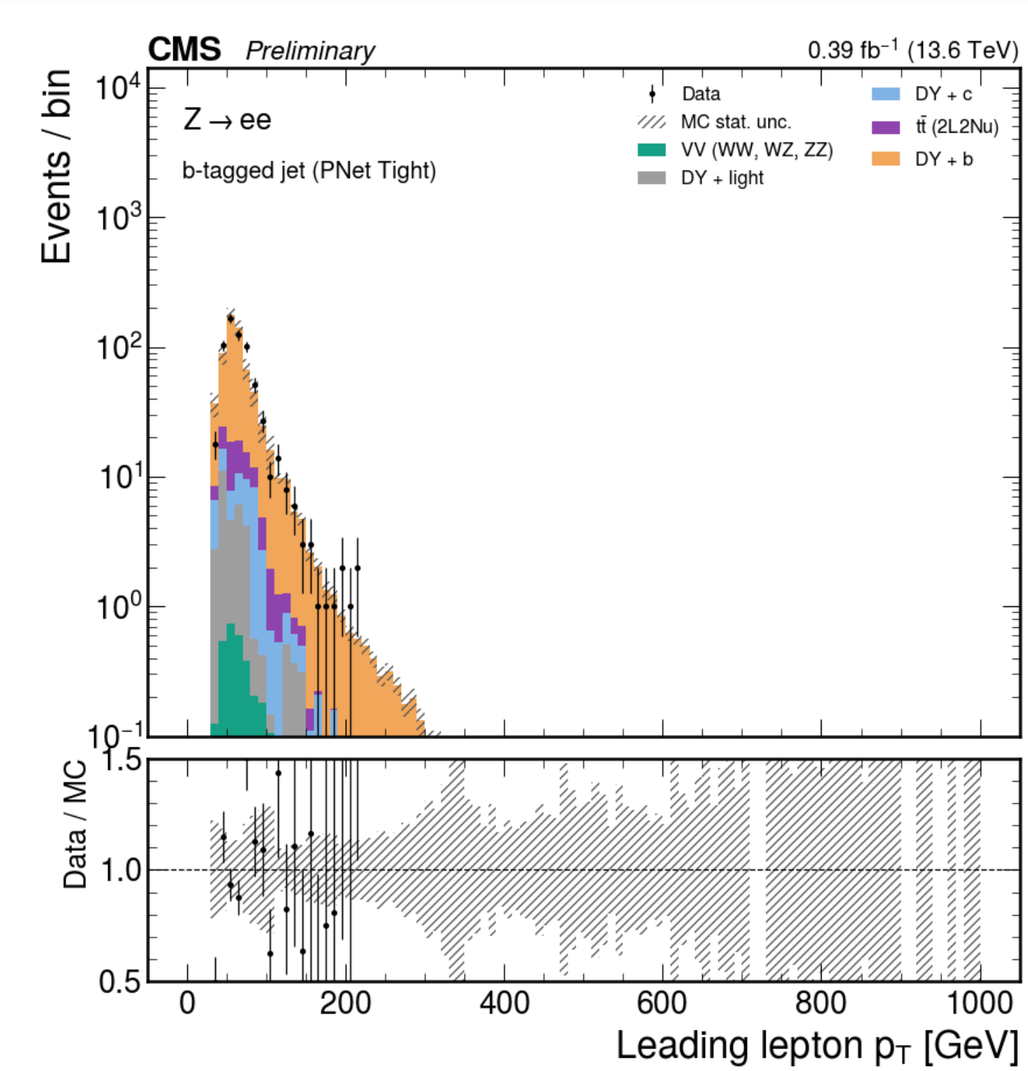
η of the lepton pair.

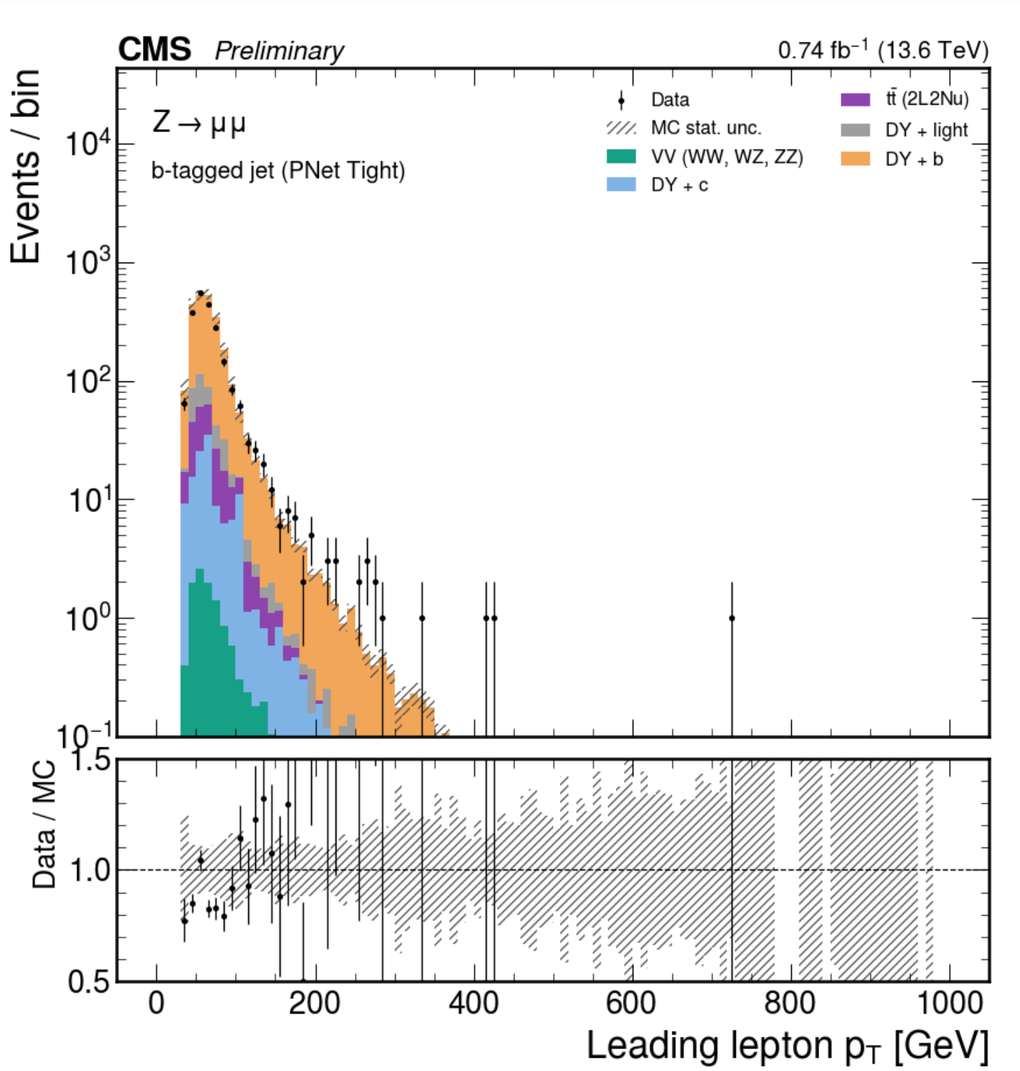


b-tag Tight

Leading-lepton pT

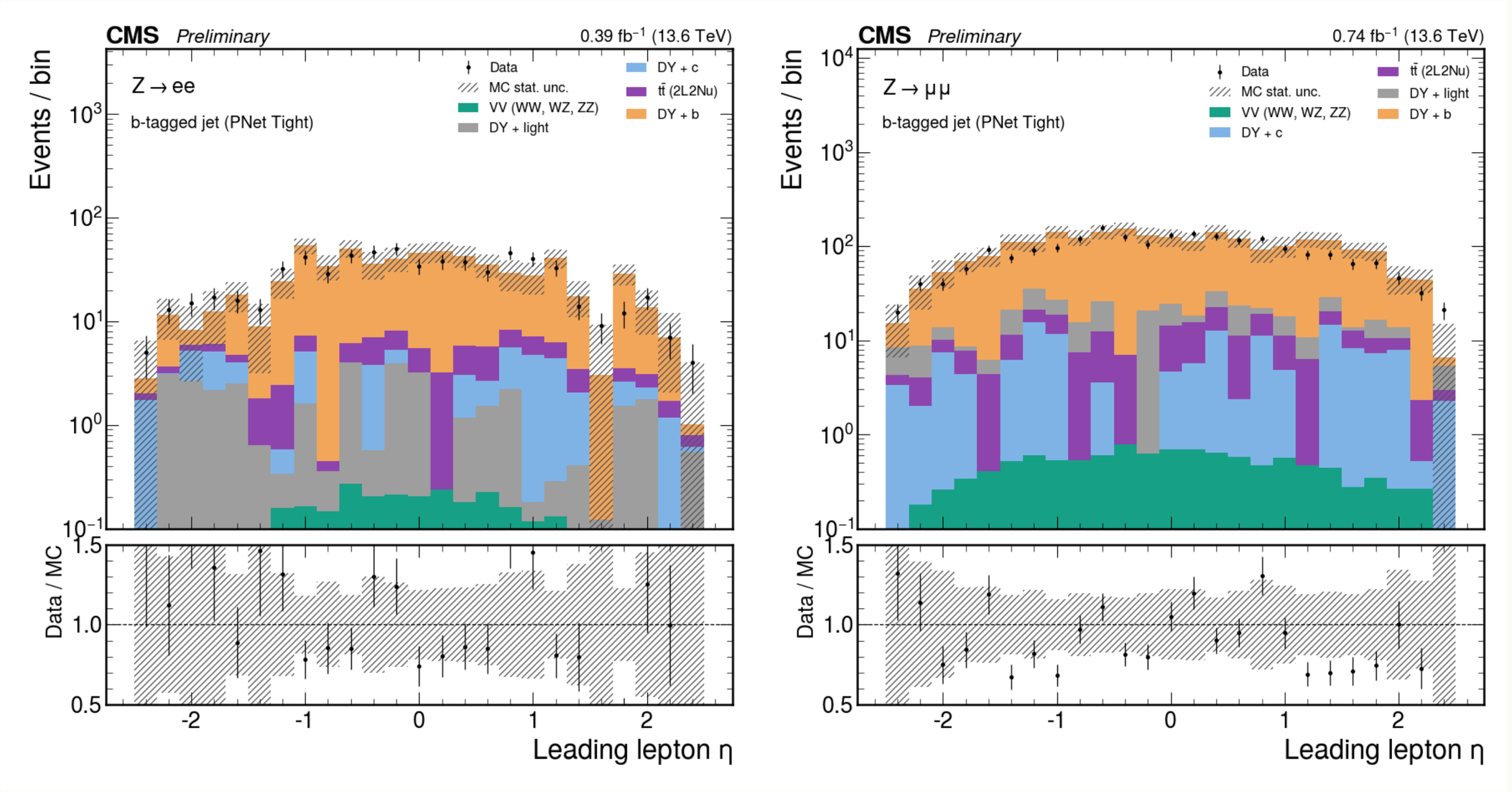
Leading lepton pT > 35 GeV (trigger plateau).





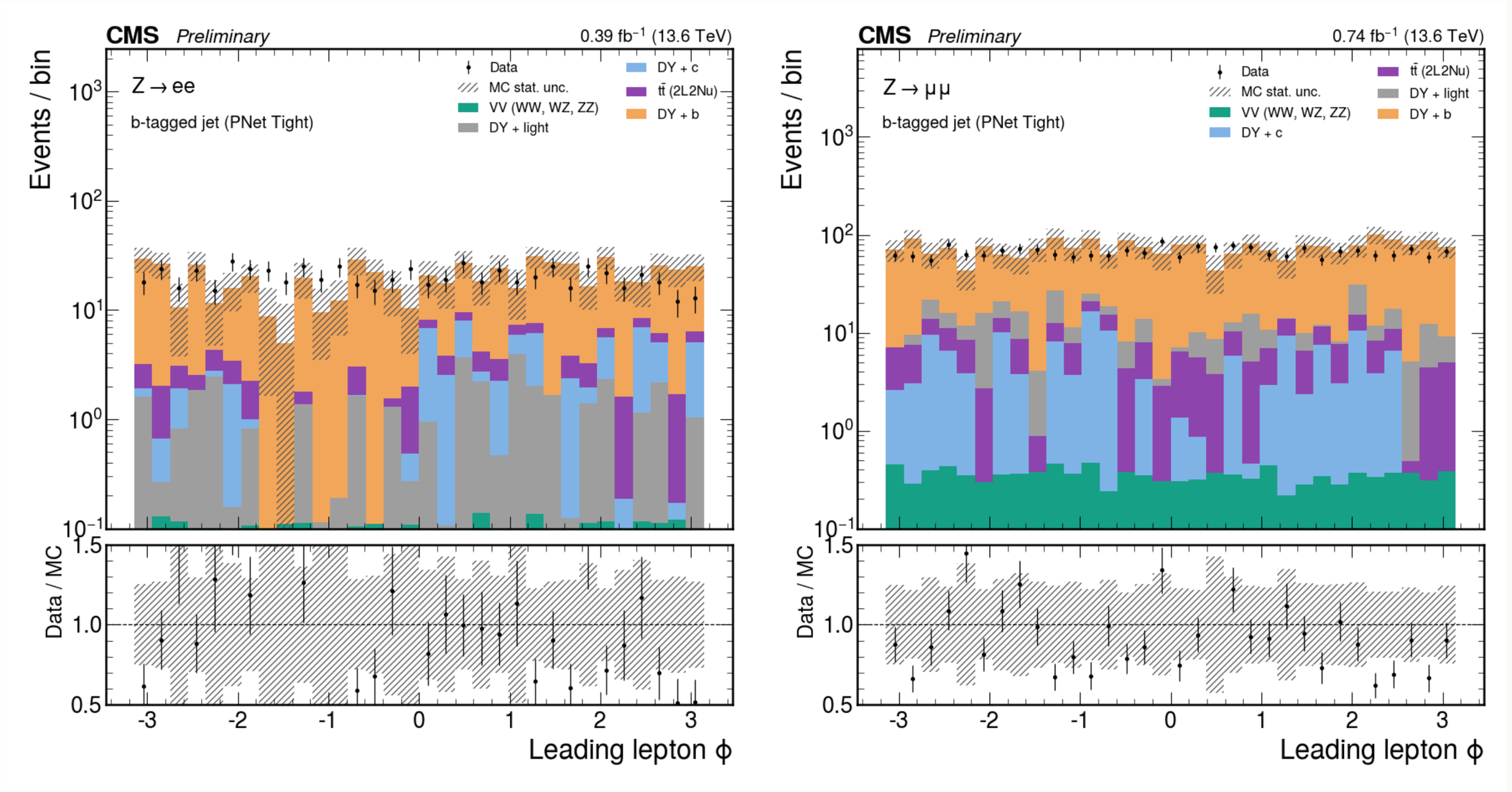
b-tag Tight

Leading-lepton η



b-tag Tight

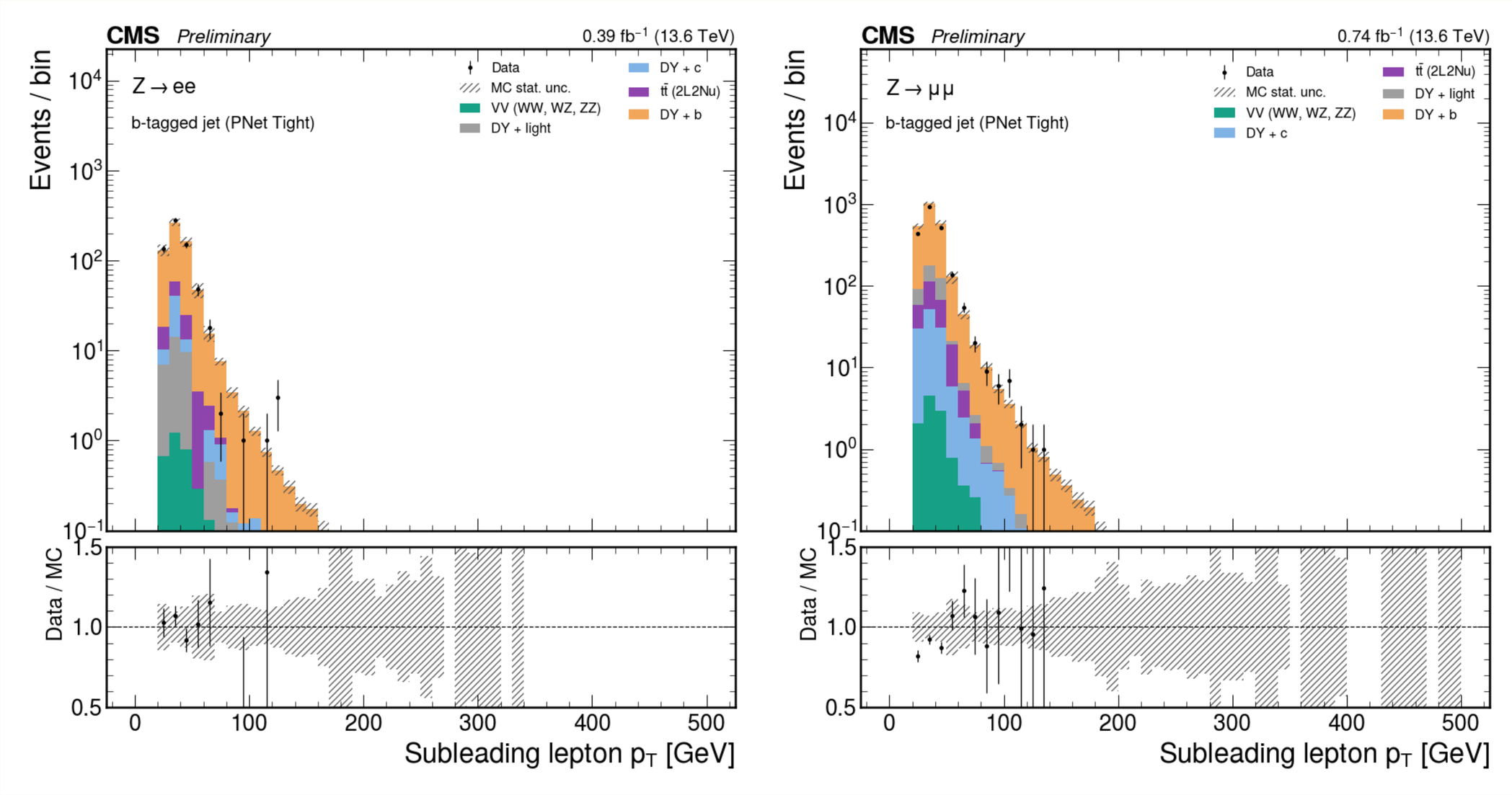
Leading-lepton φ



b-tag Tight

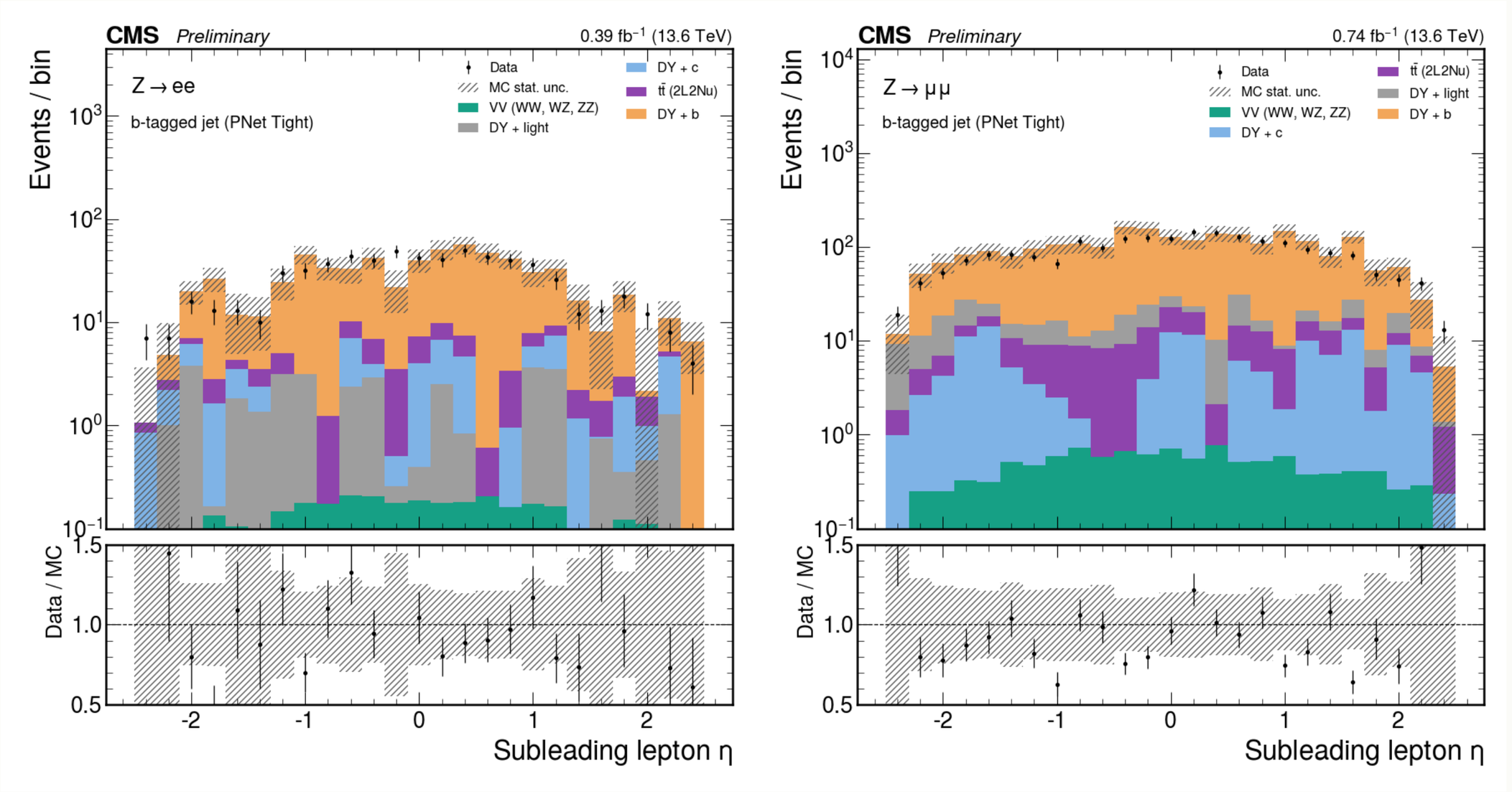
Second-lepton pT

Second lepton pT > 25 GeV.



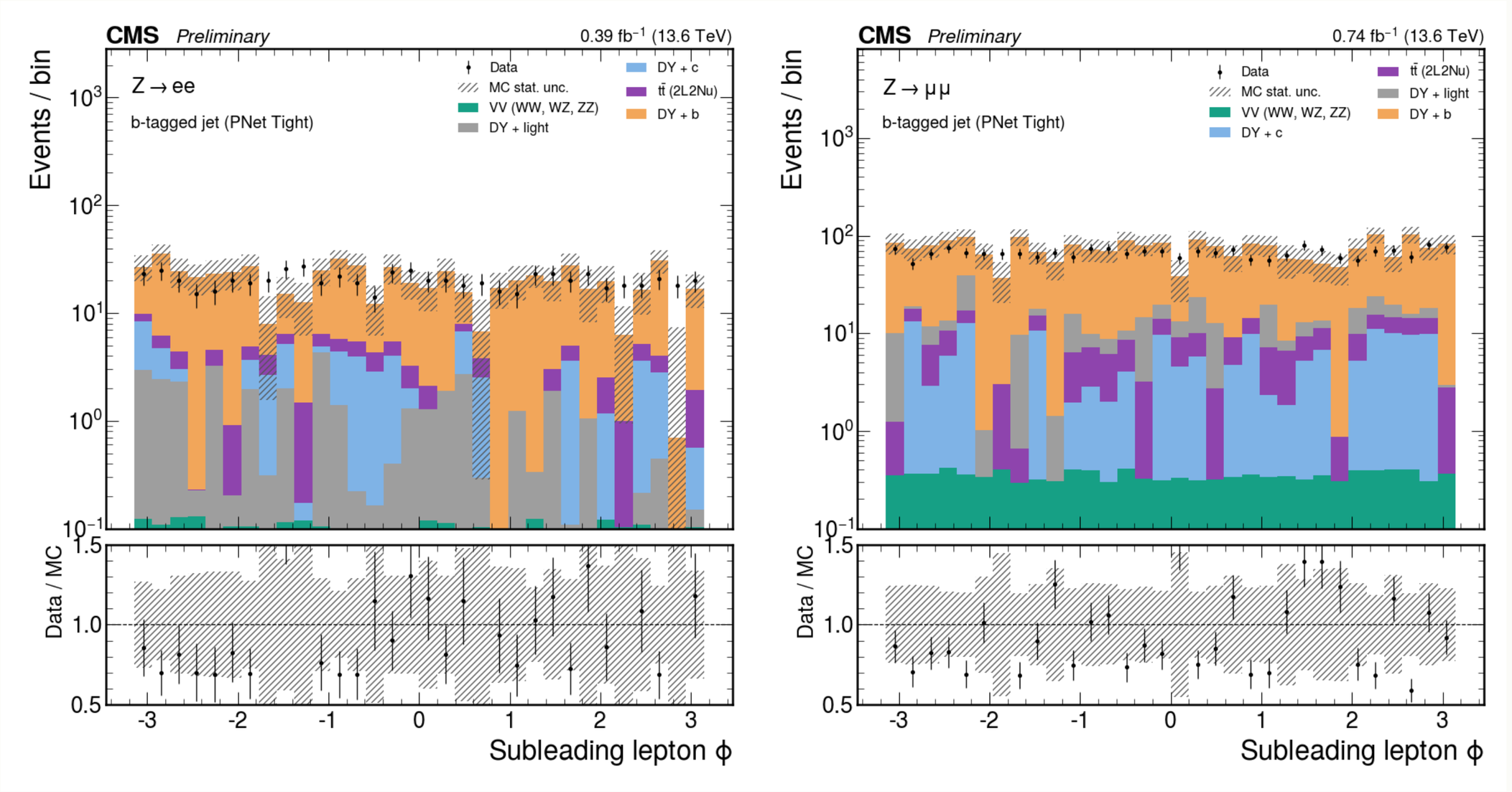
b-tag Tight

Second-lepton η



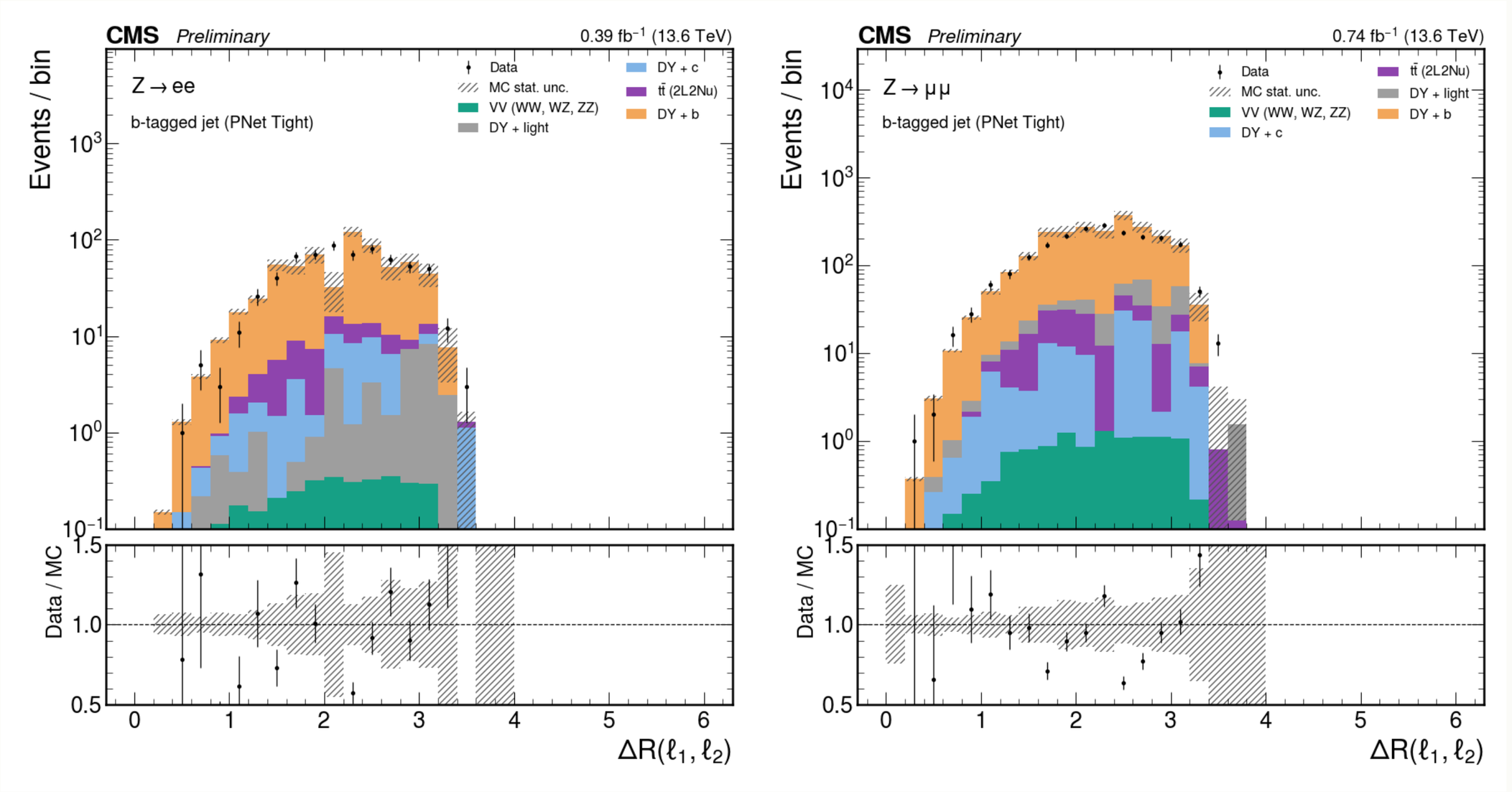
b-tag Tight

Second-lepton φ



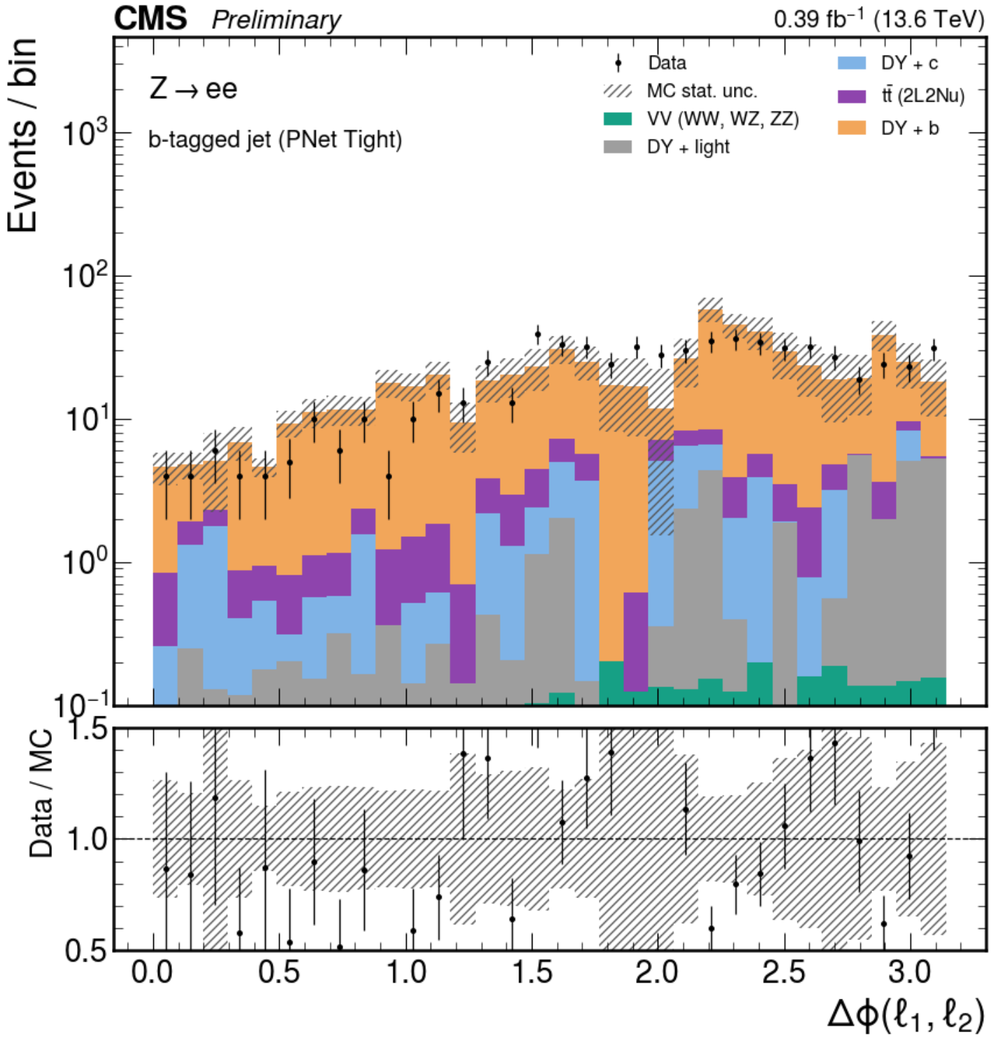
b-tag Tight

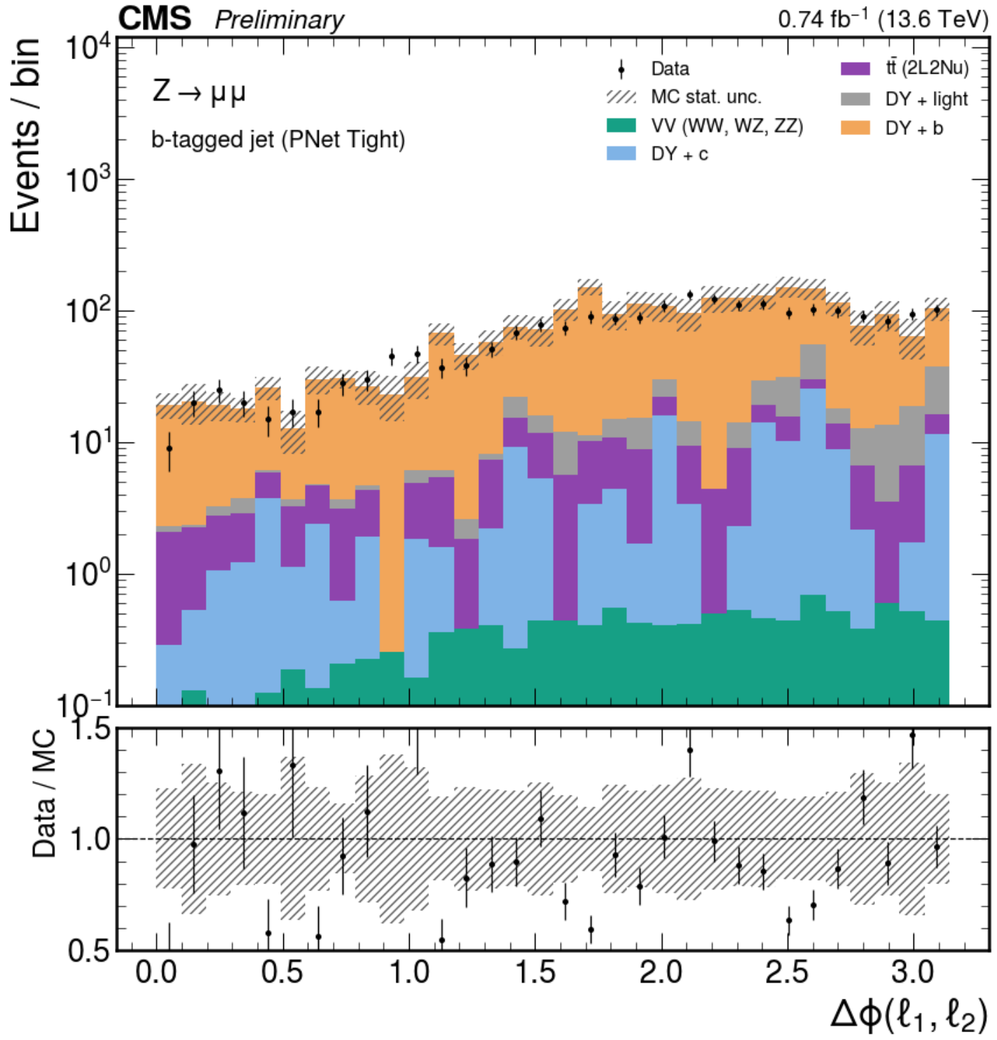
ΔR between the leptons



b-tag Tight

Δφ between the leptons

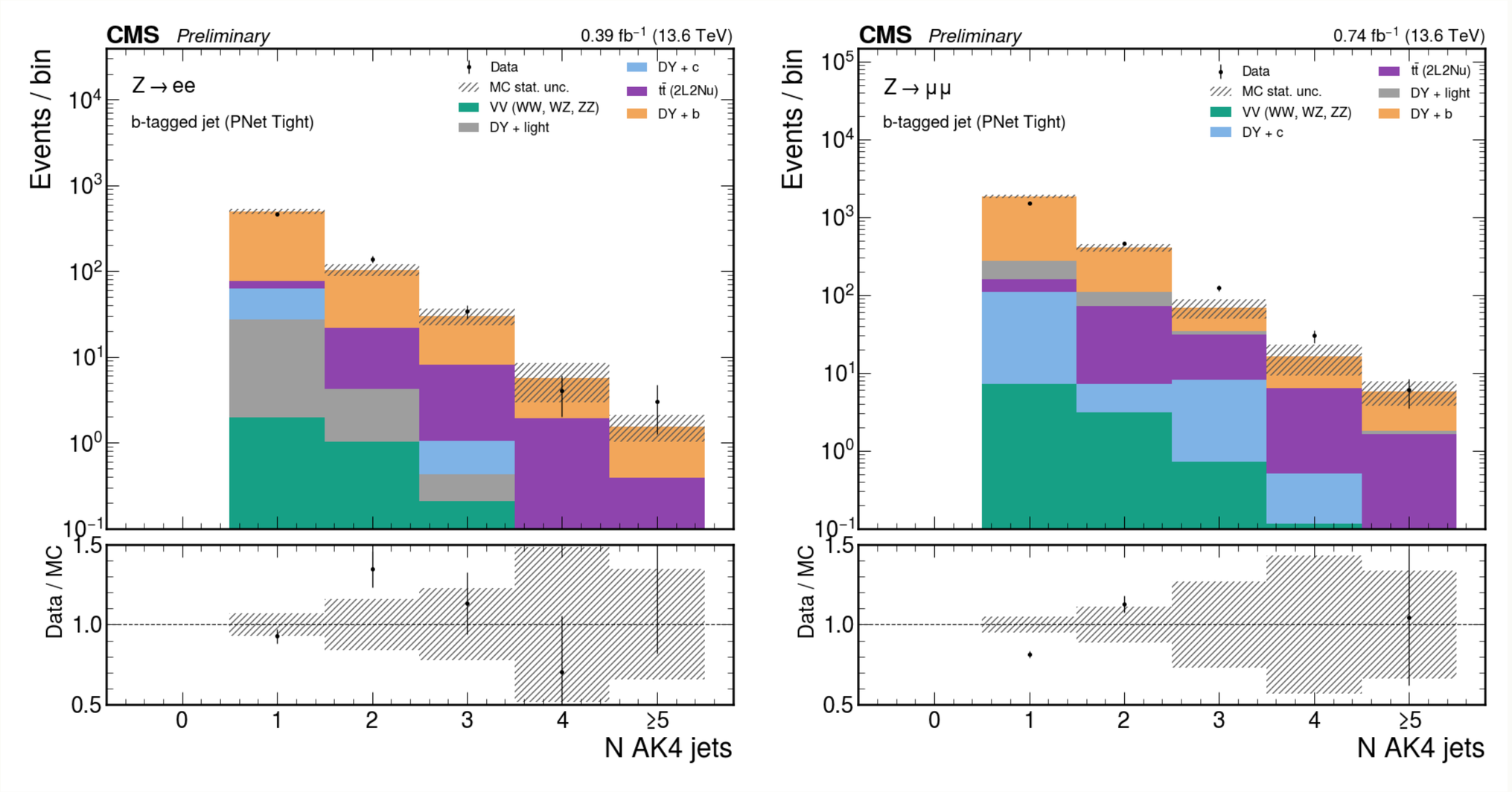




b-tag Tight

Number of jets

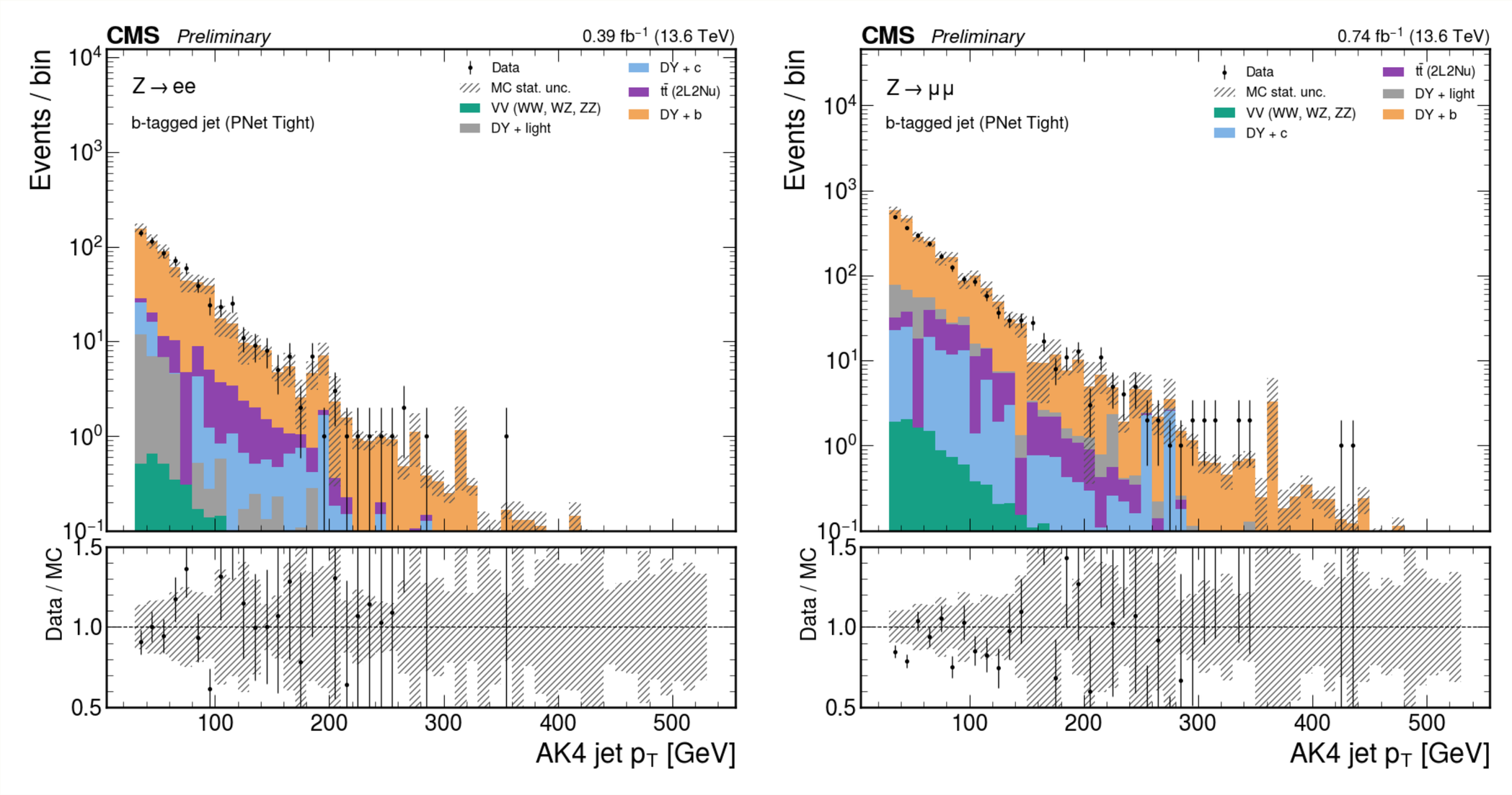
AK4 jets with pT > 30 GeV, |η| < 2.5, away from the leptons. Untagged plot: Z events before the jet cut, so the 0-jet bin is filled.



b-tag Tight

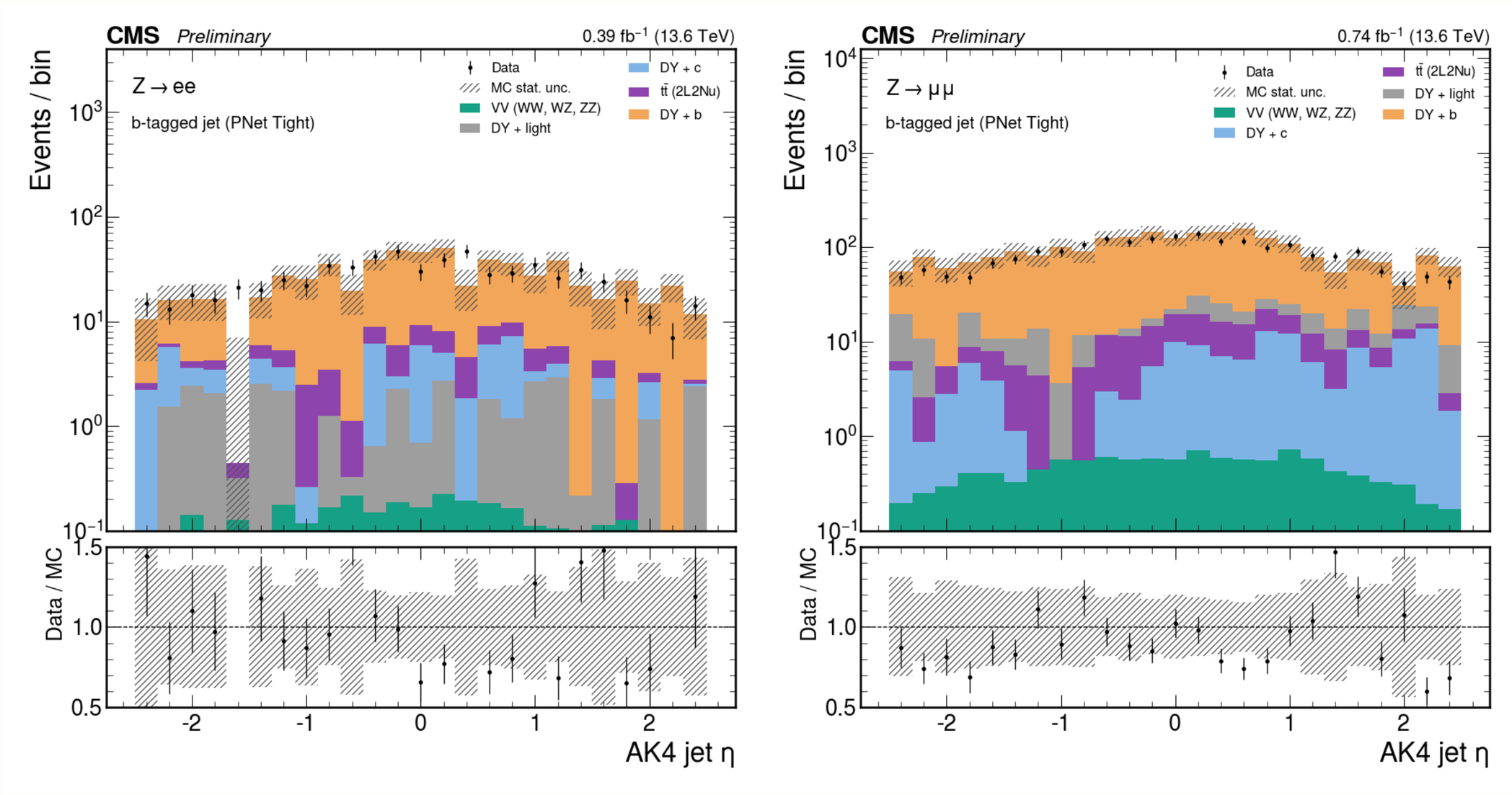
Leading-jet pT

The tag (if any) is on this jet.



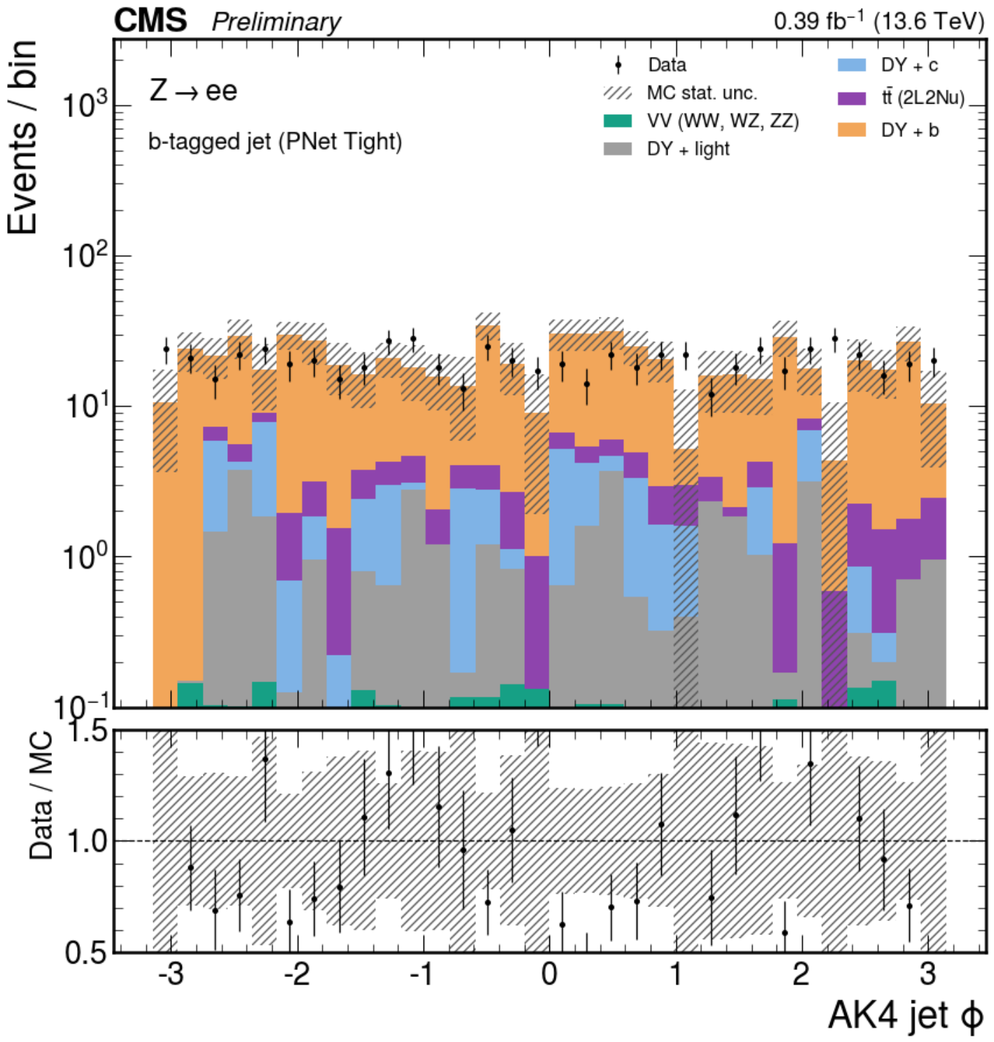
b-tag Tight

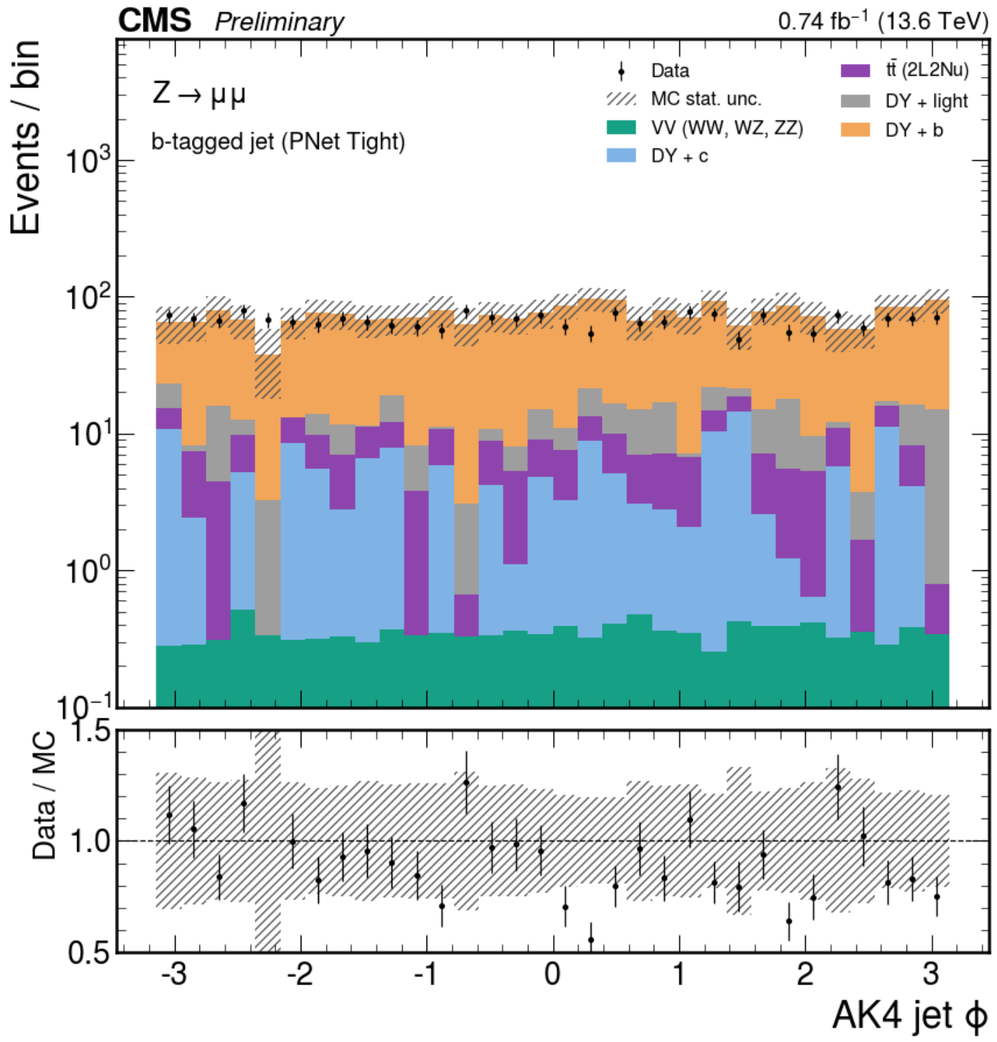
Leading-jet η



b-tag Tight

Leading-jet φ

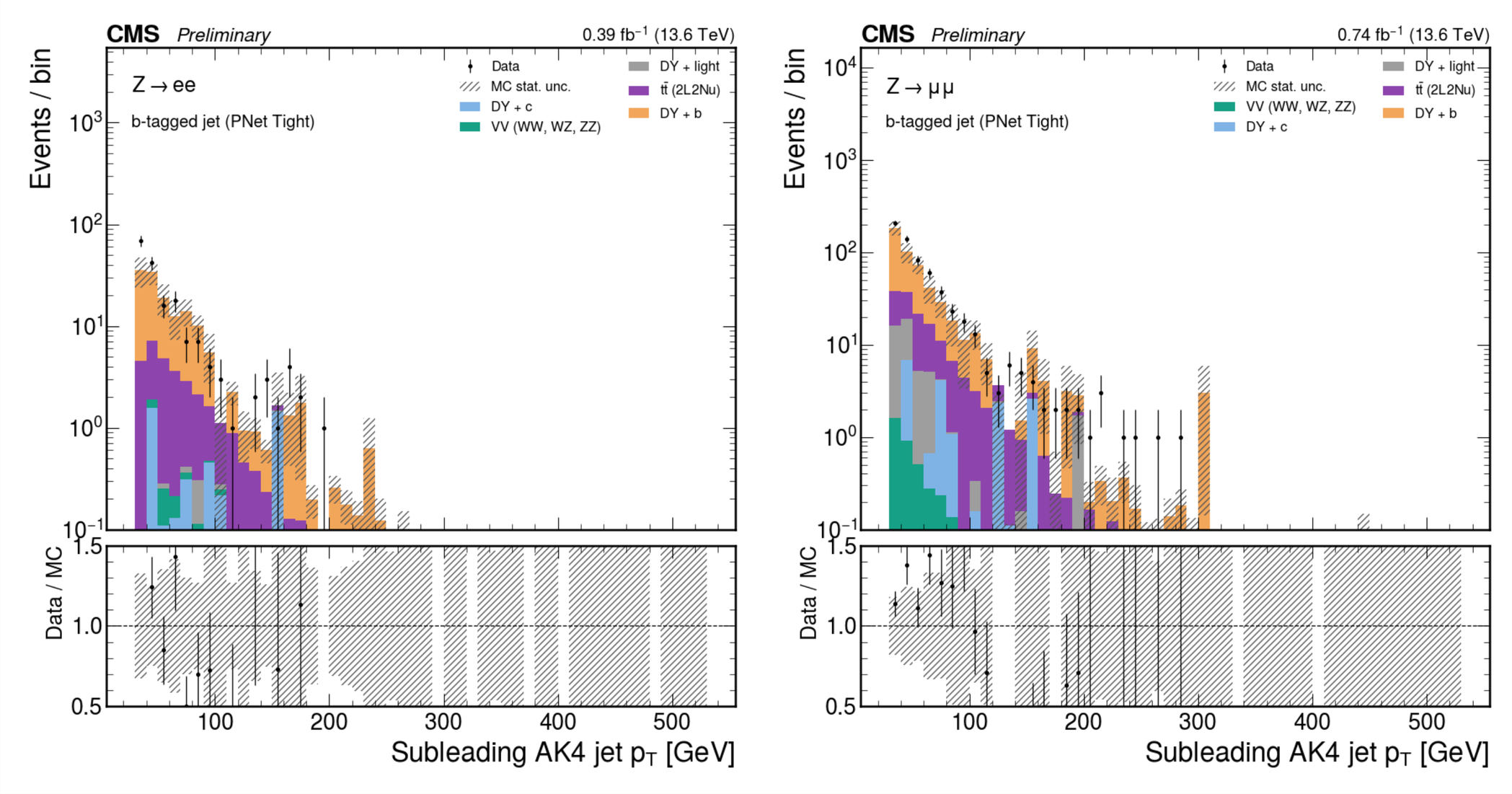




b-tag Tight

Second-jet pT

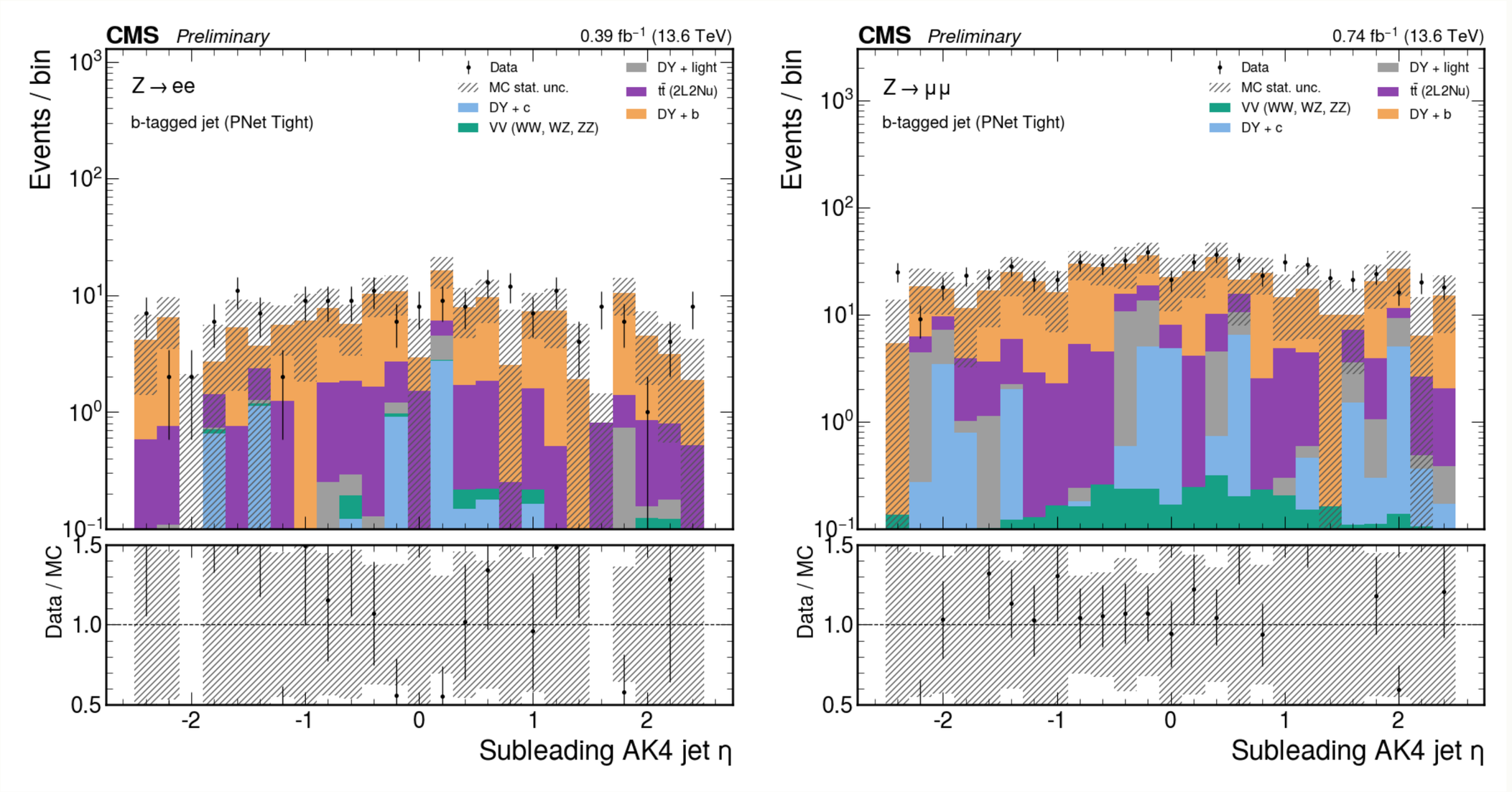
Events with at least 2 jets.



b-tag Tight

Second-jet η

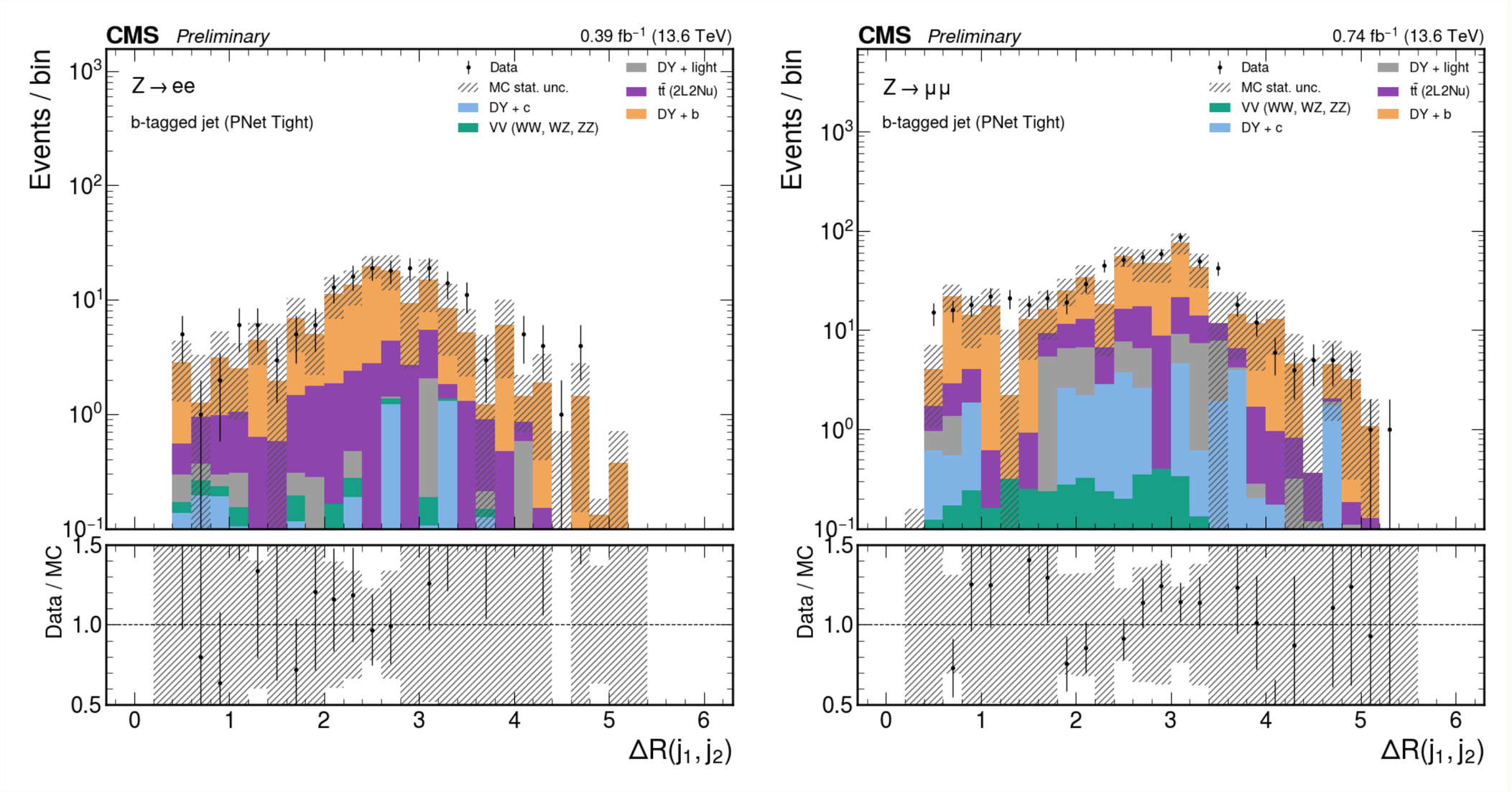
Events with at least 2 jets.



b-tag Tight

ΔR between the two leading jets

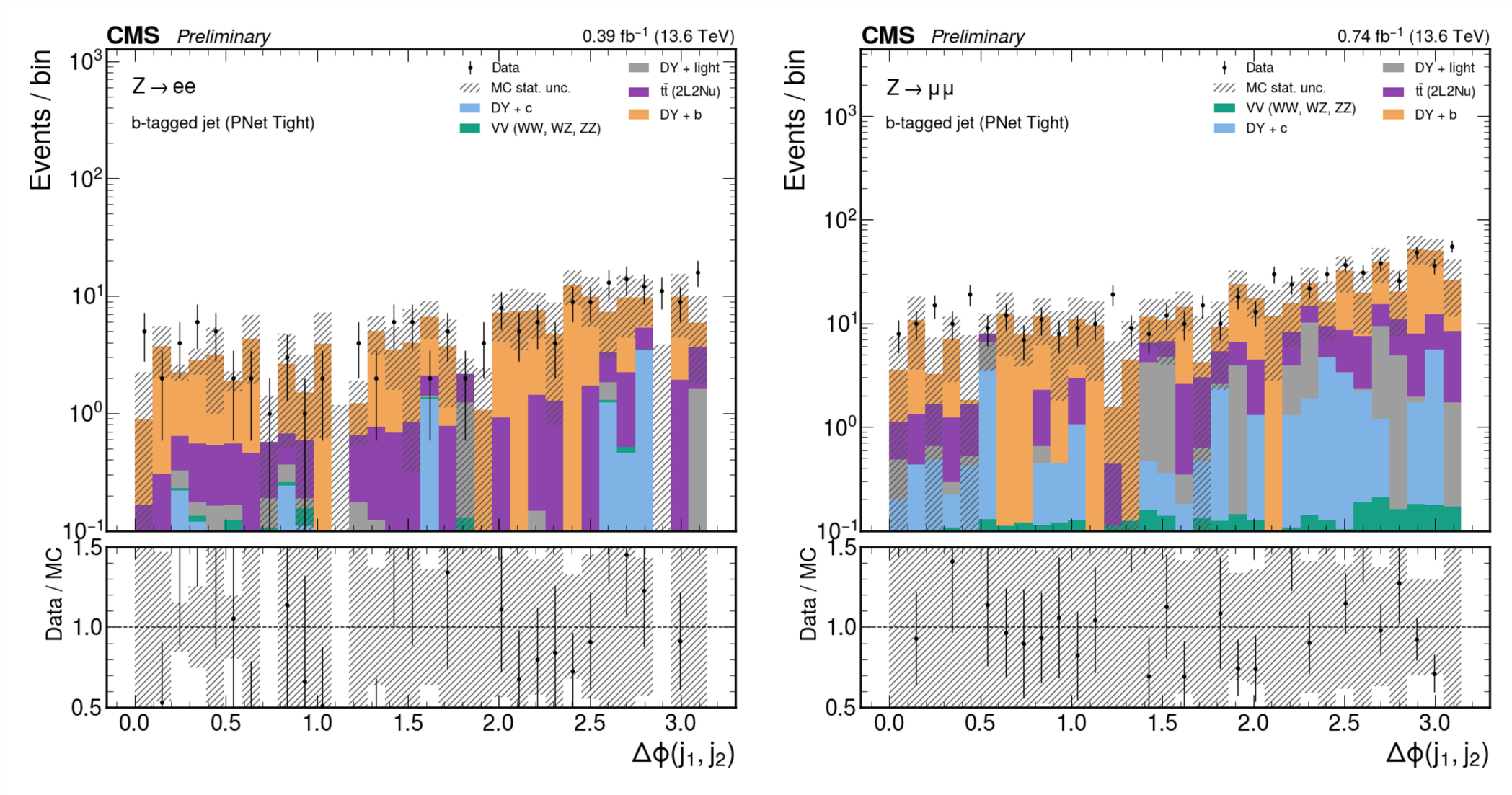
Events with at least 2 jets.



b-tag Tight

Δφ between the two leading jets

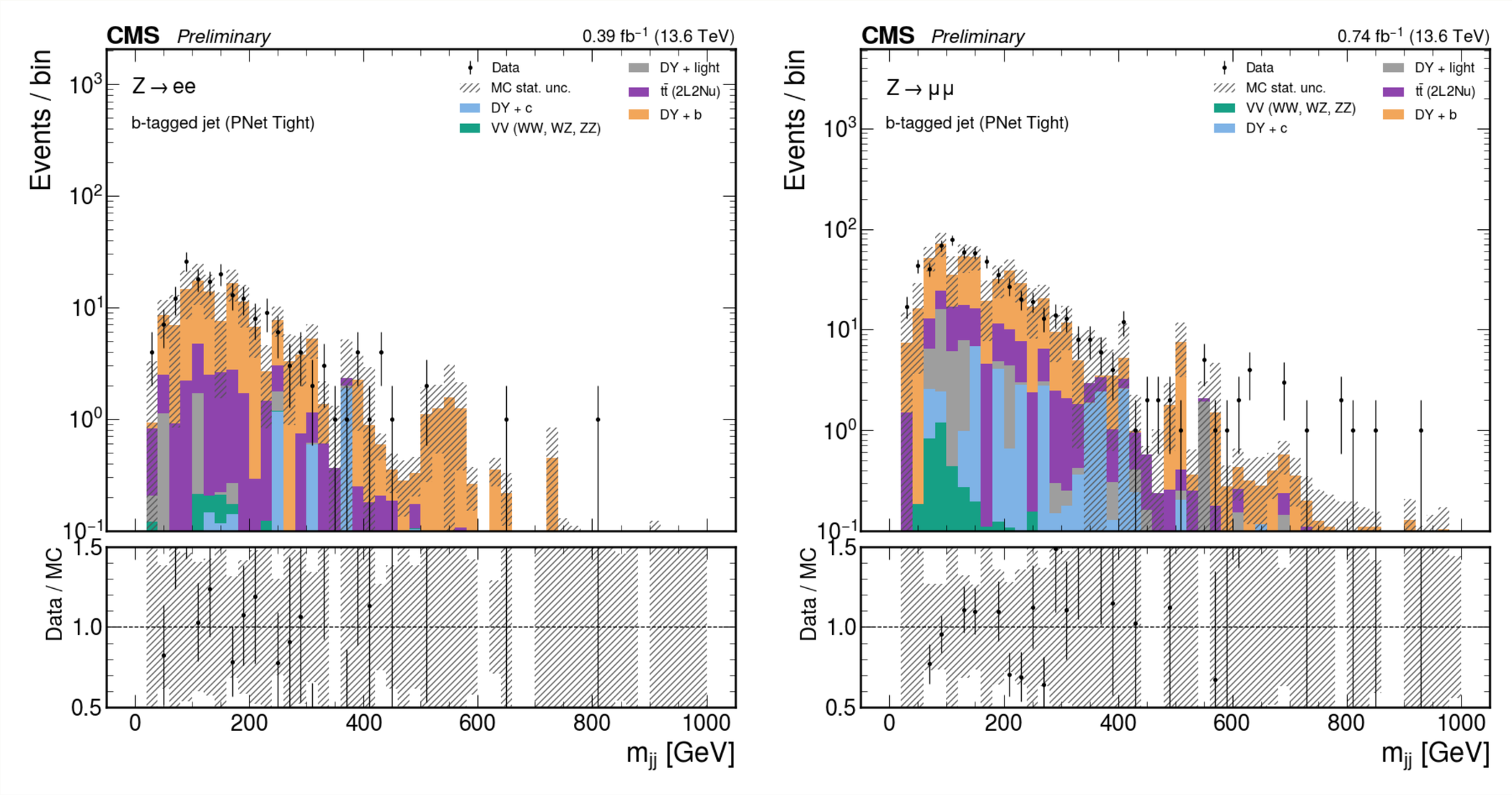
Events with at least 2 jets.



b-tag Tight

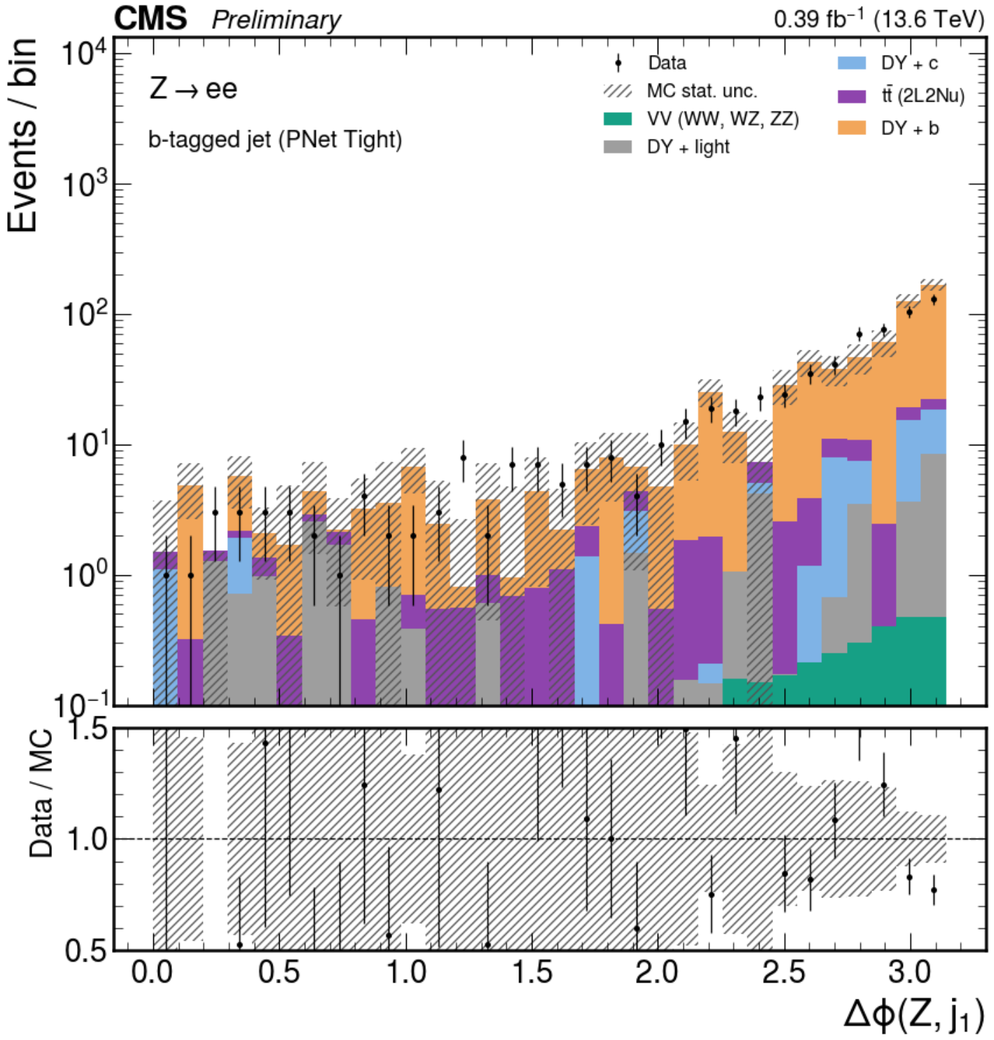
Dijet mass

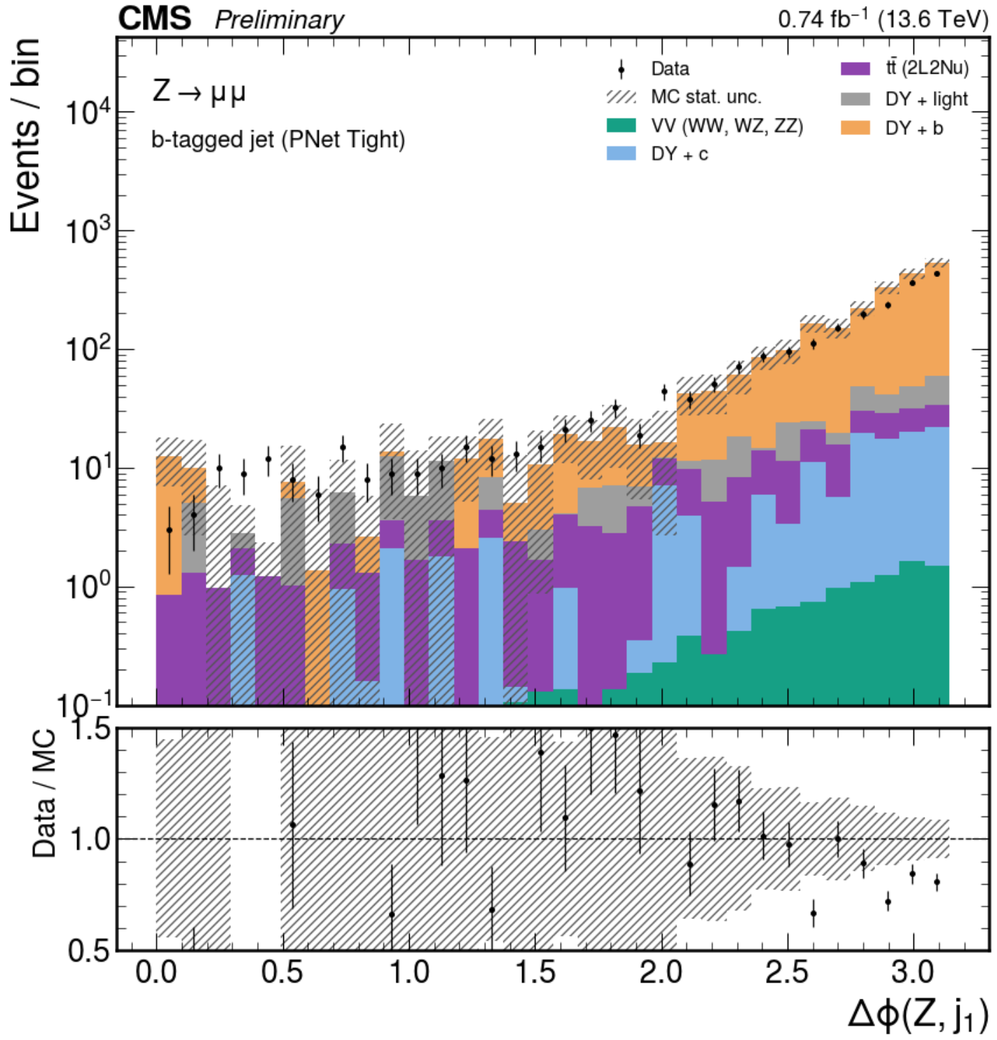
Events with at least 2 jets.



b-tag Tight

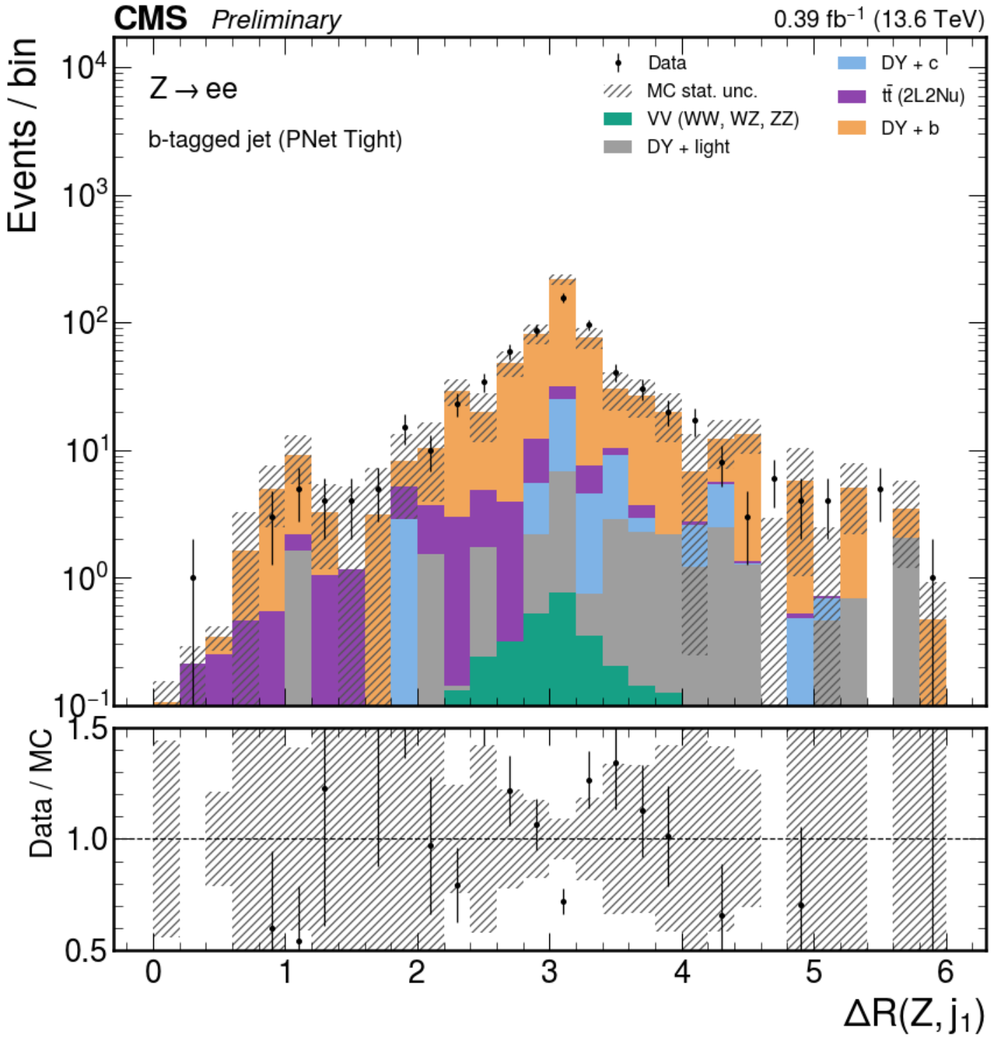
Δφ(Z, leading jet)

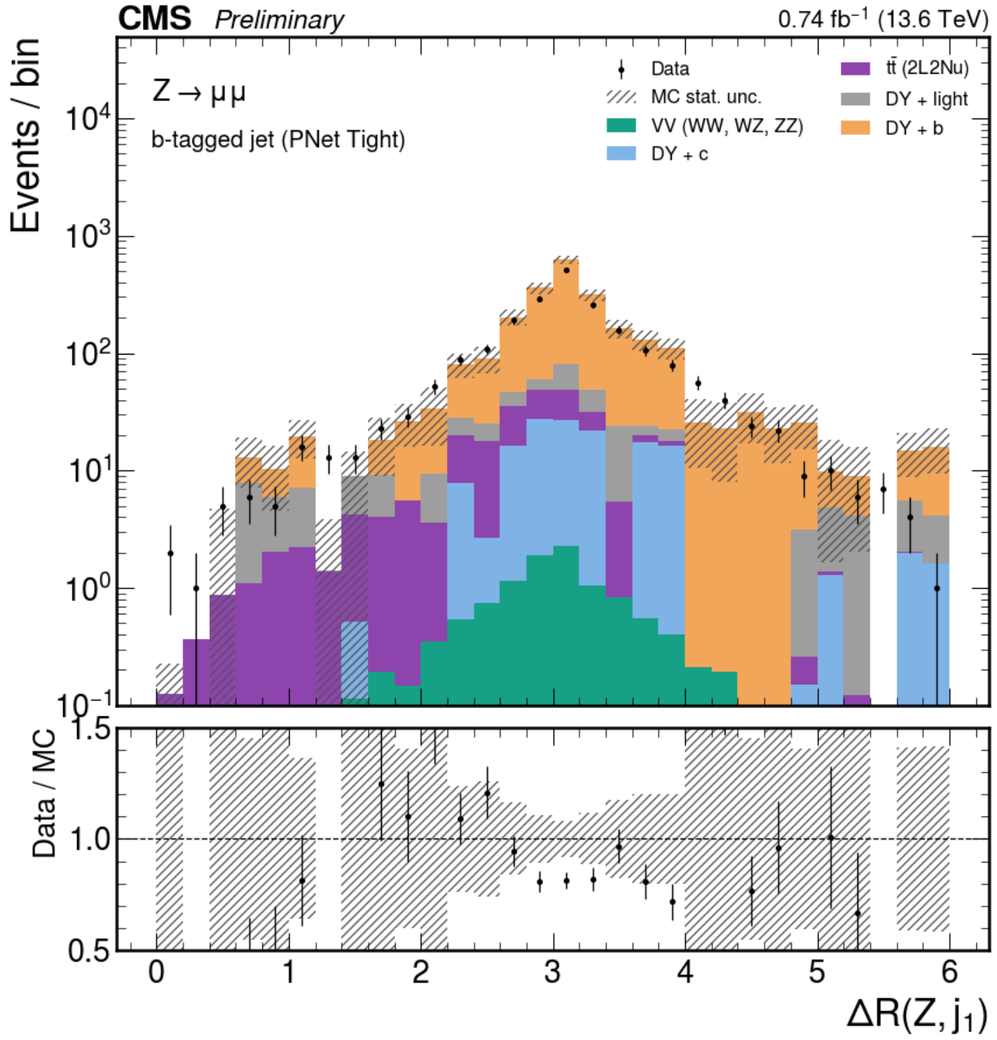




b-tag Tight

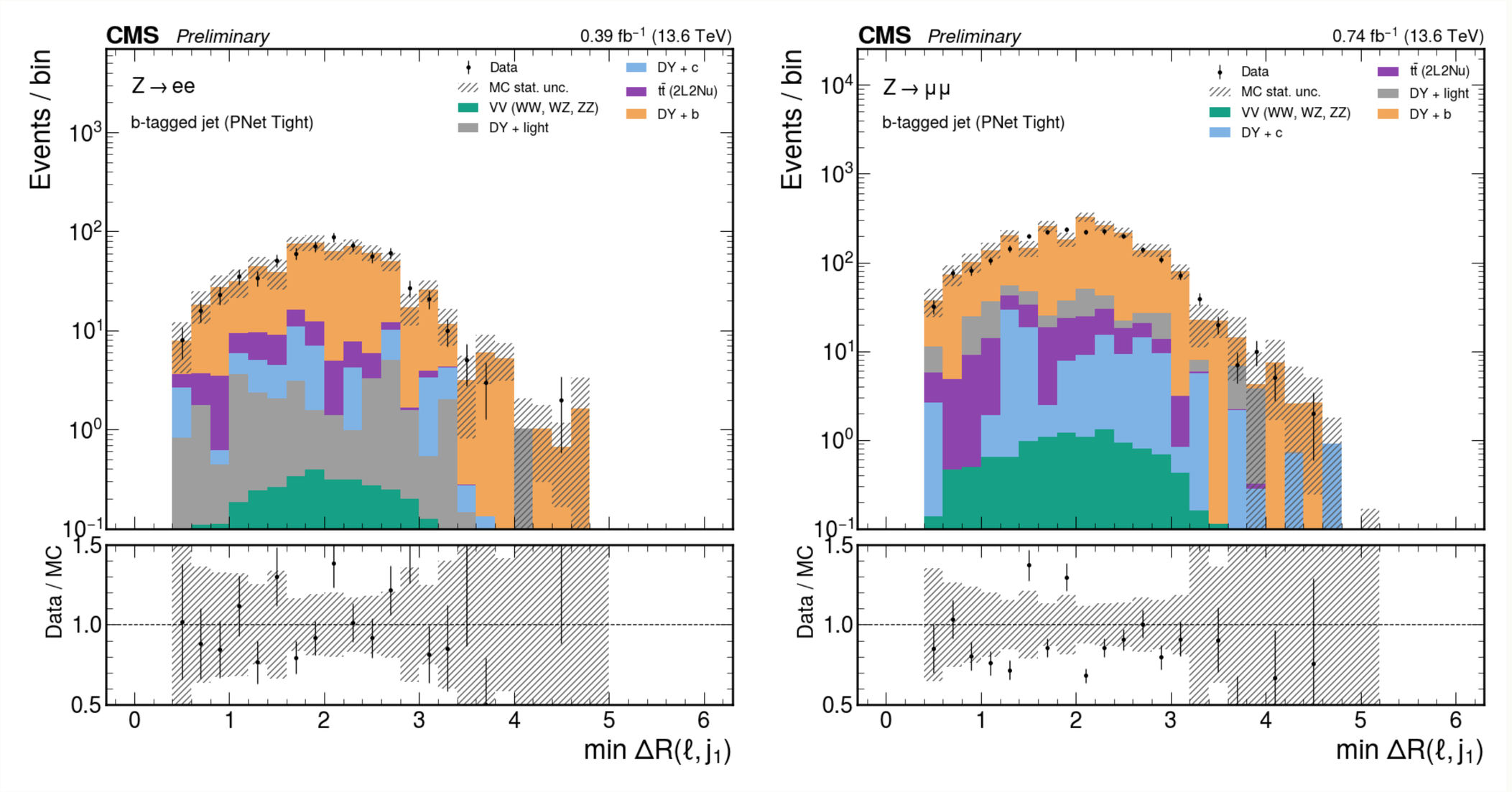
ΔR(Z, leading jet)





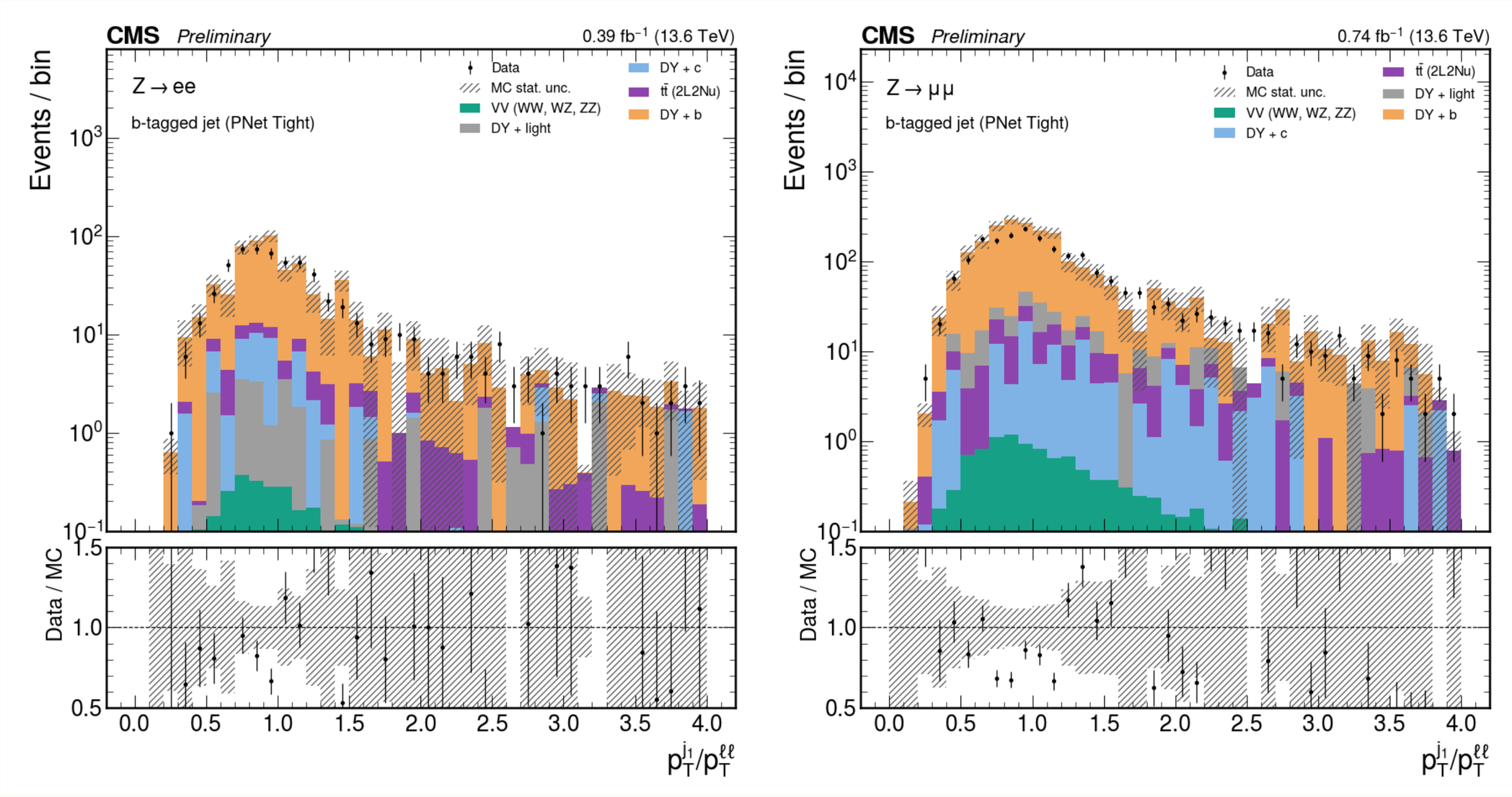
b-tag Tight

Smallest ΔR(lepton, leading jet)



b-tag Tight

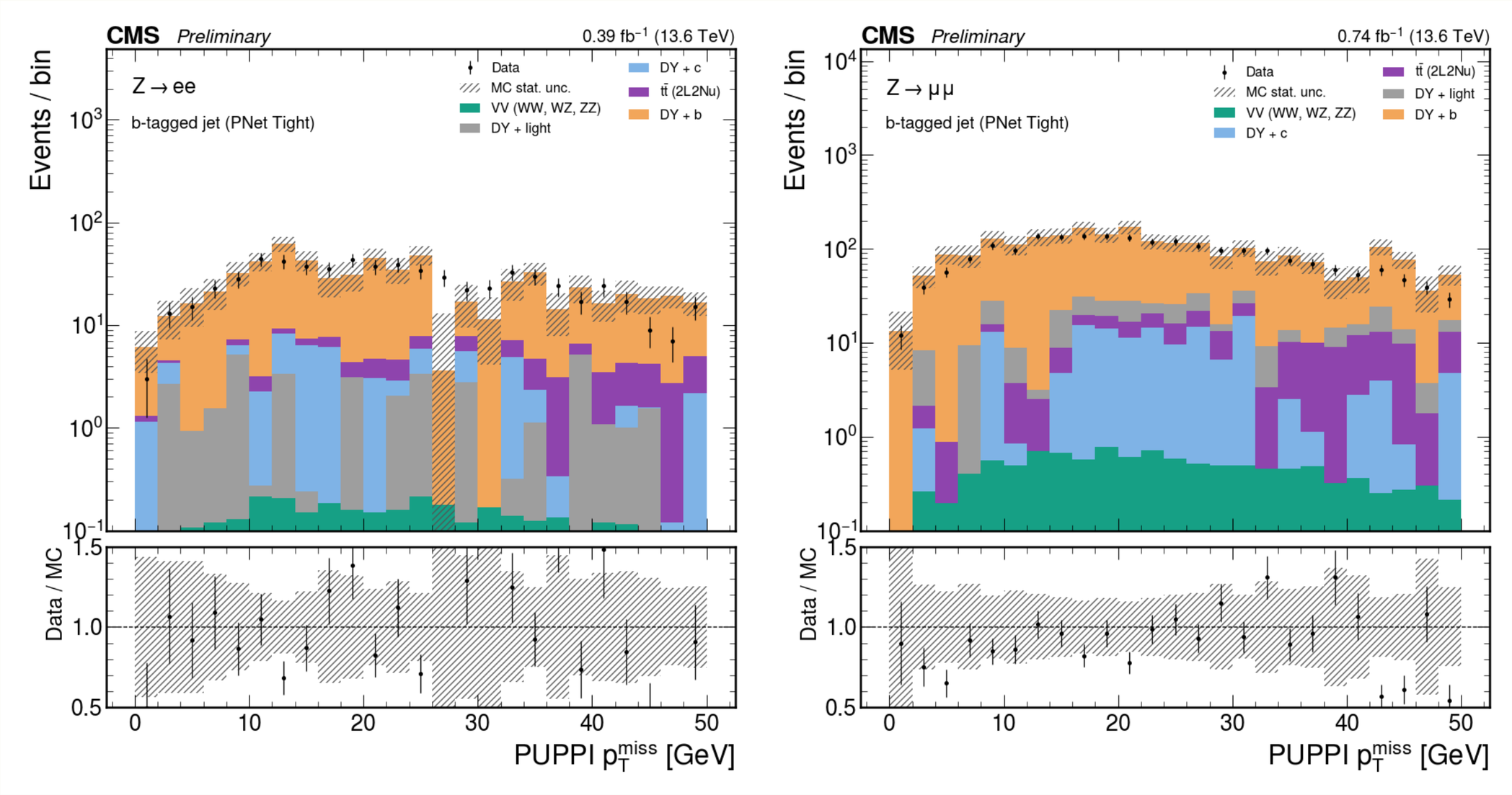
pT balance: leading-jet pT / Z pT



b-tag Tight

PUPPI MET

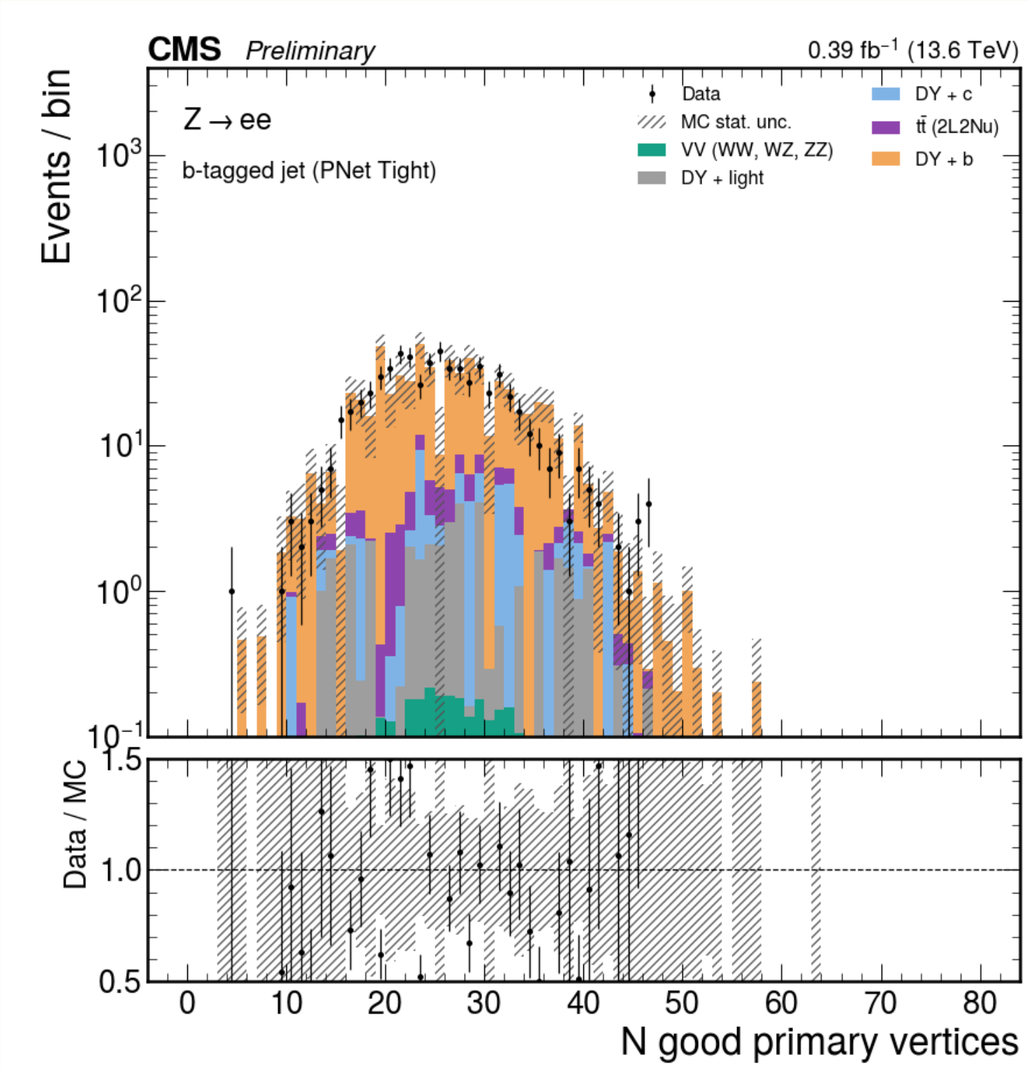
Type-1 corrected; the selection cuts at 50 GeV.

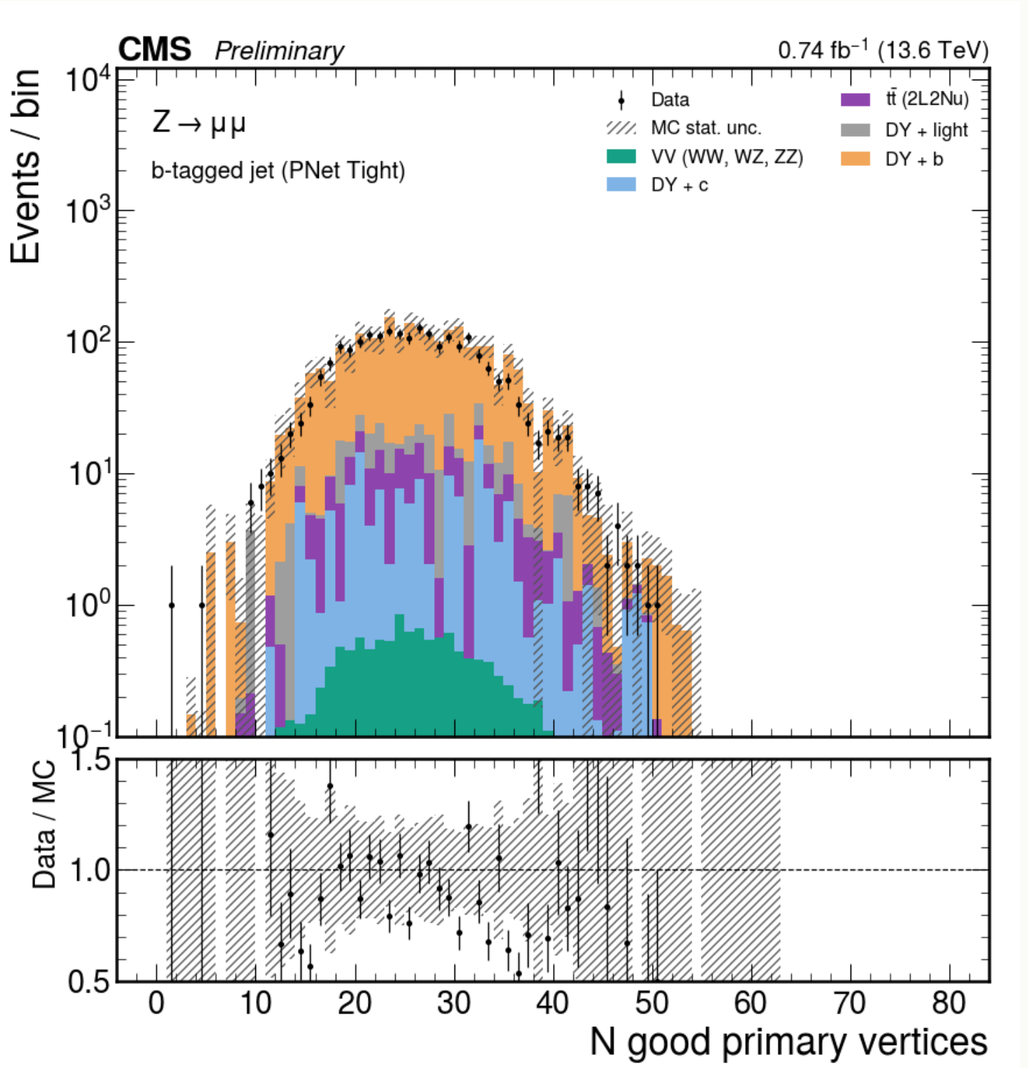


b-tag Tight

Number of good primary vertices

After pileup weights.

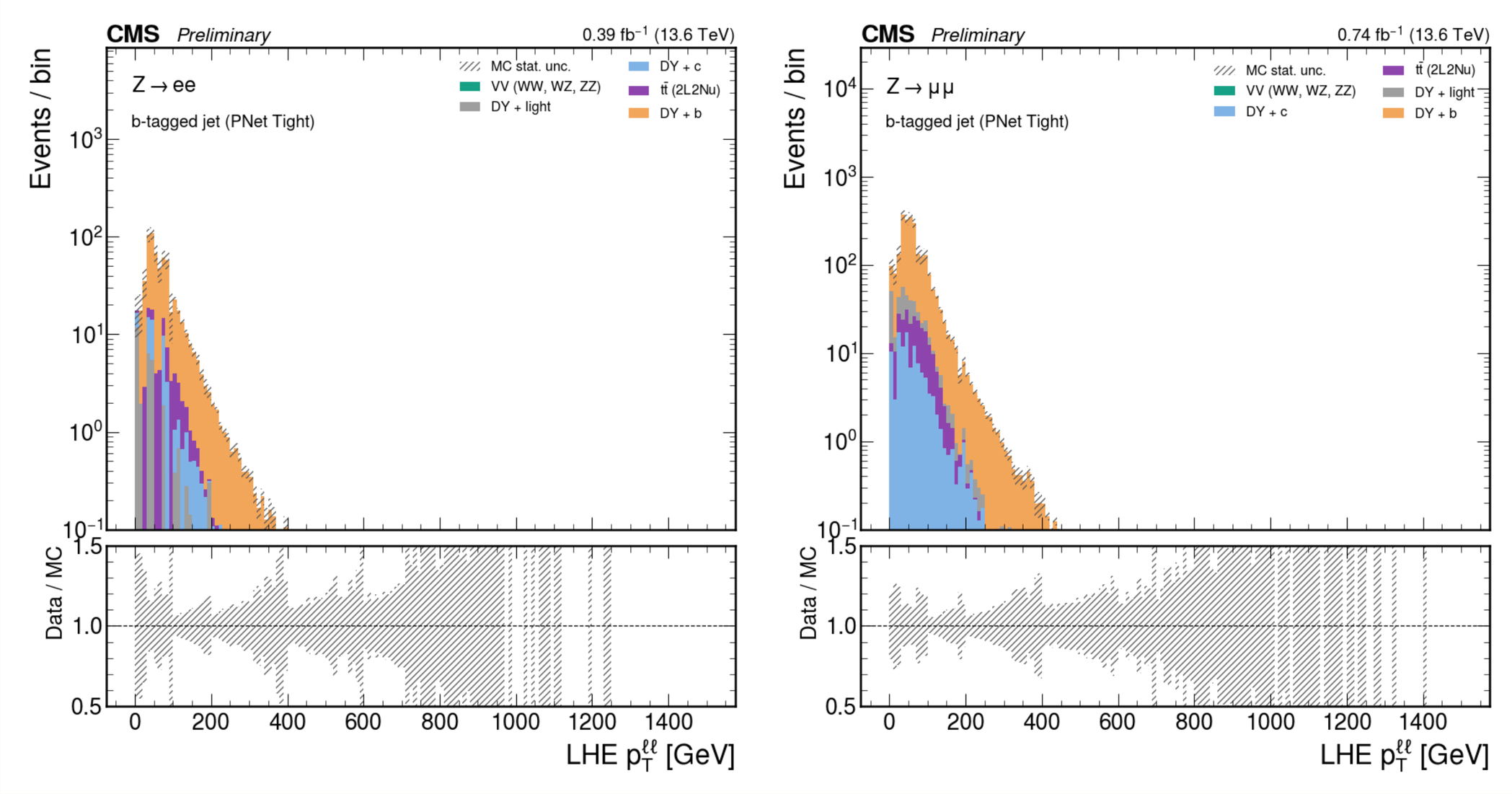




b-tag Tight

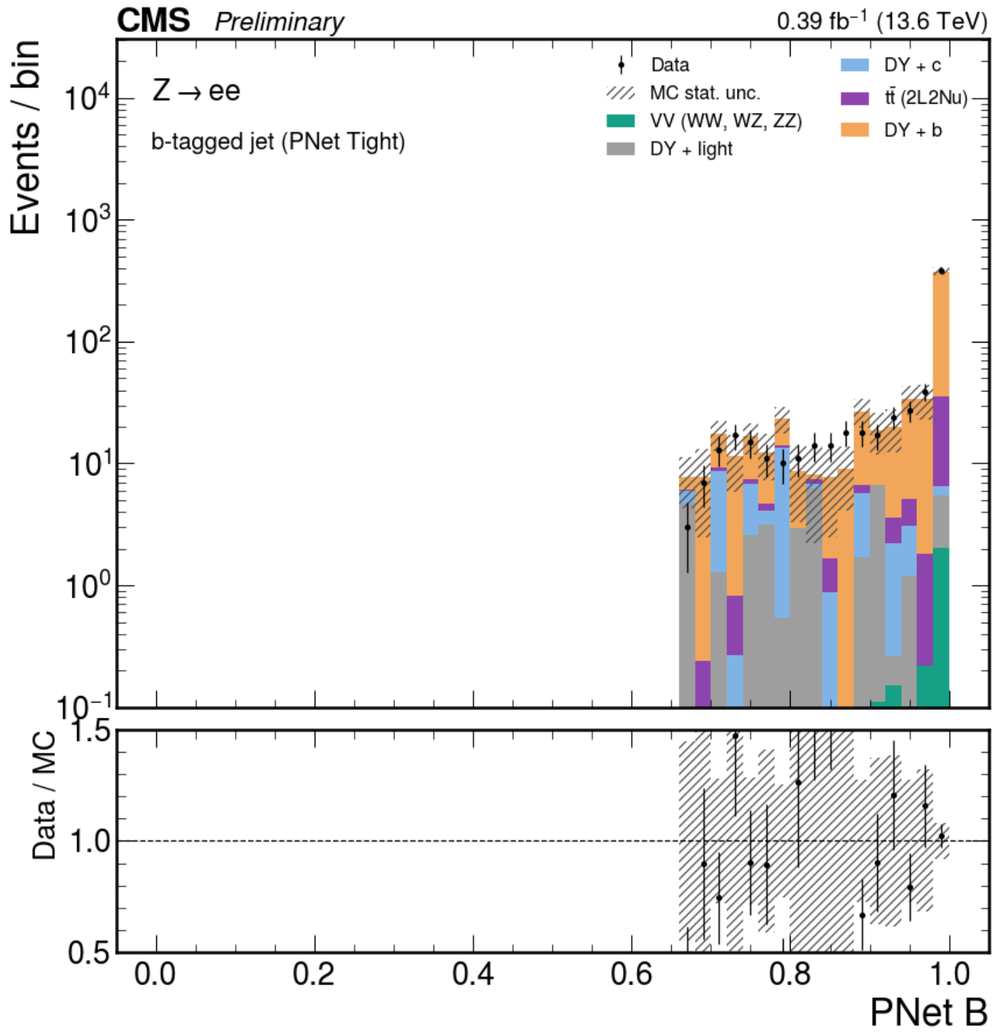
Generator Z pT (LHE)

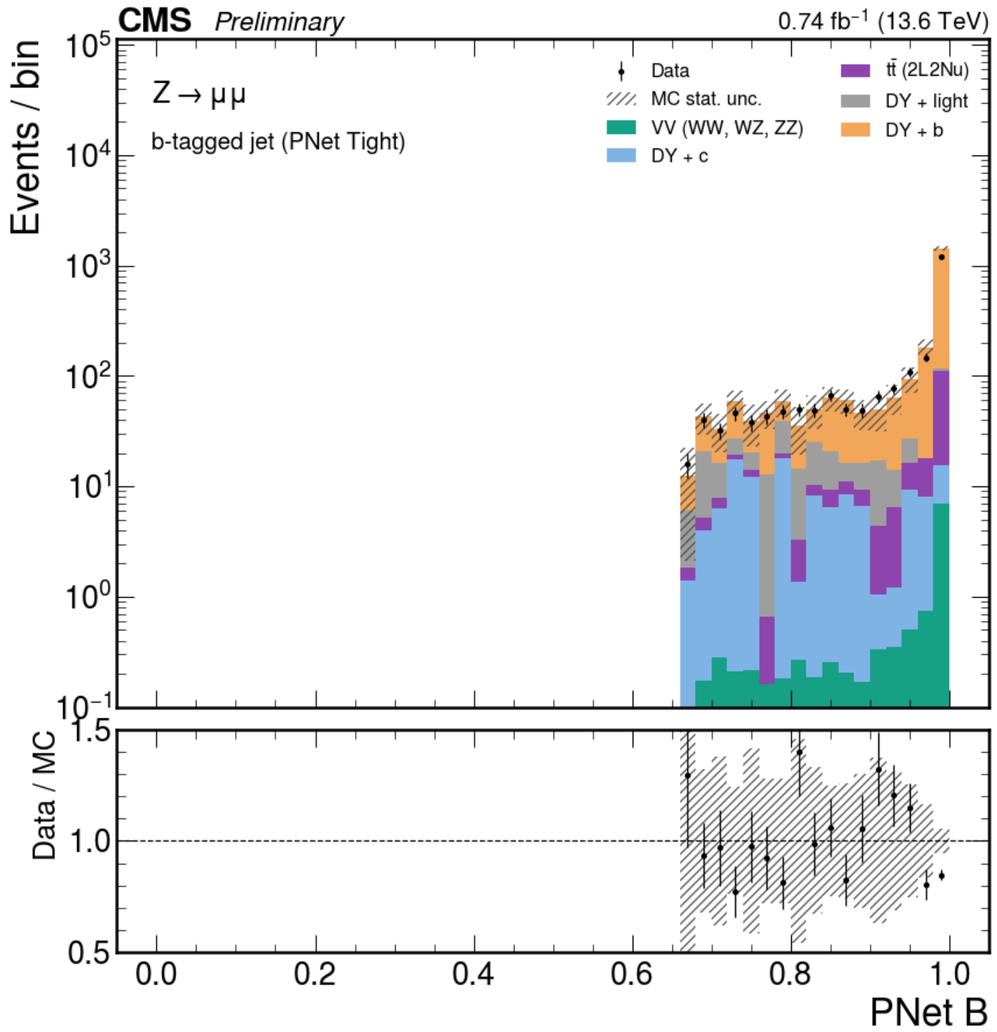
MC only, no data.



b-tag Tight

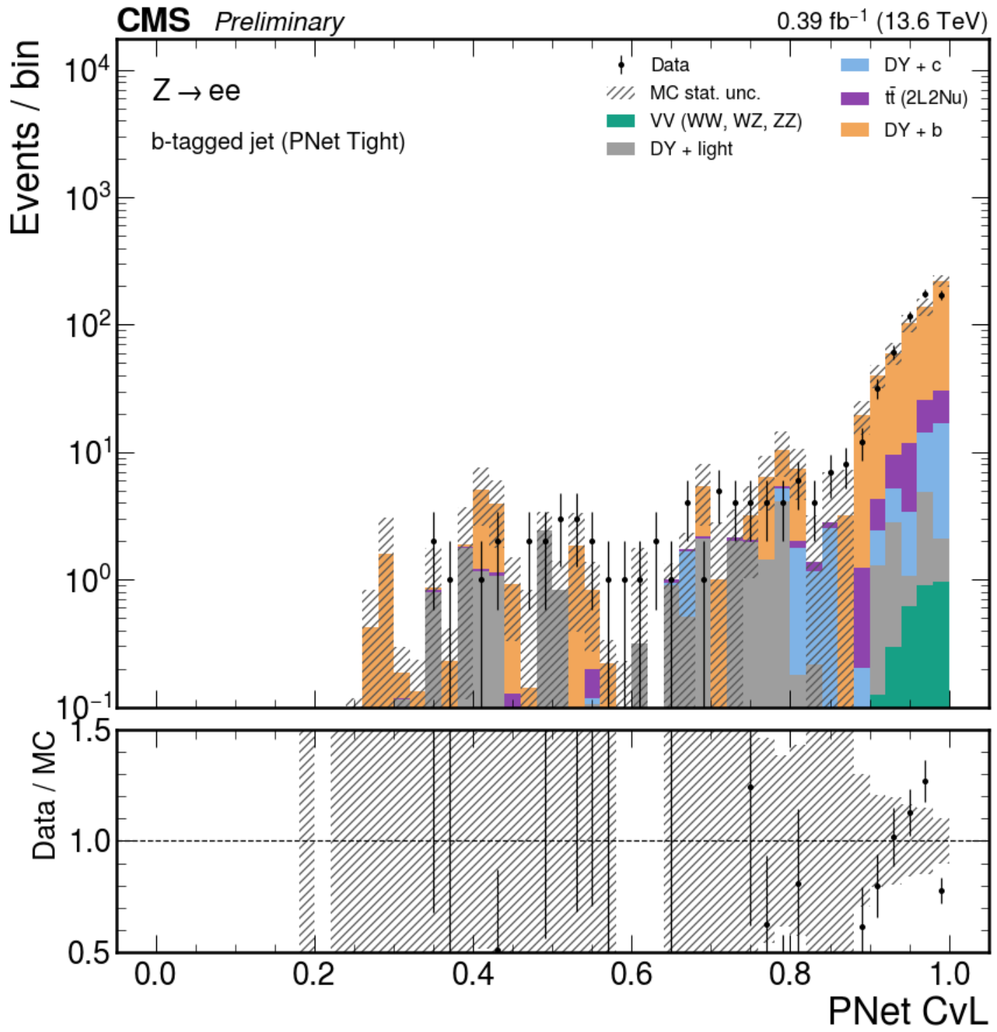
Leading-jet PNet B score

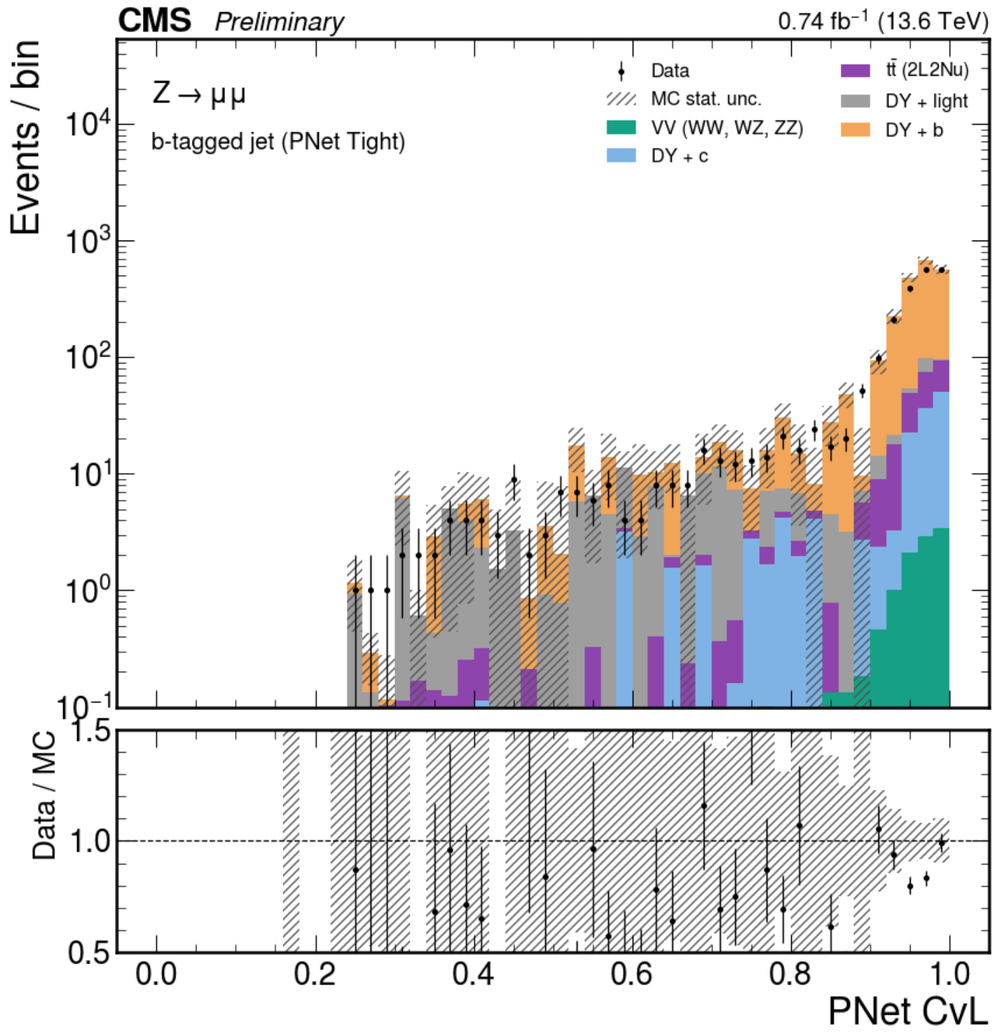




b-tag Tight

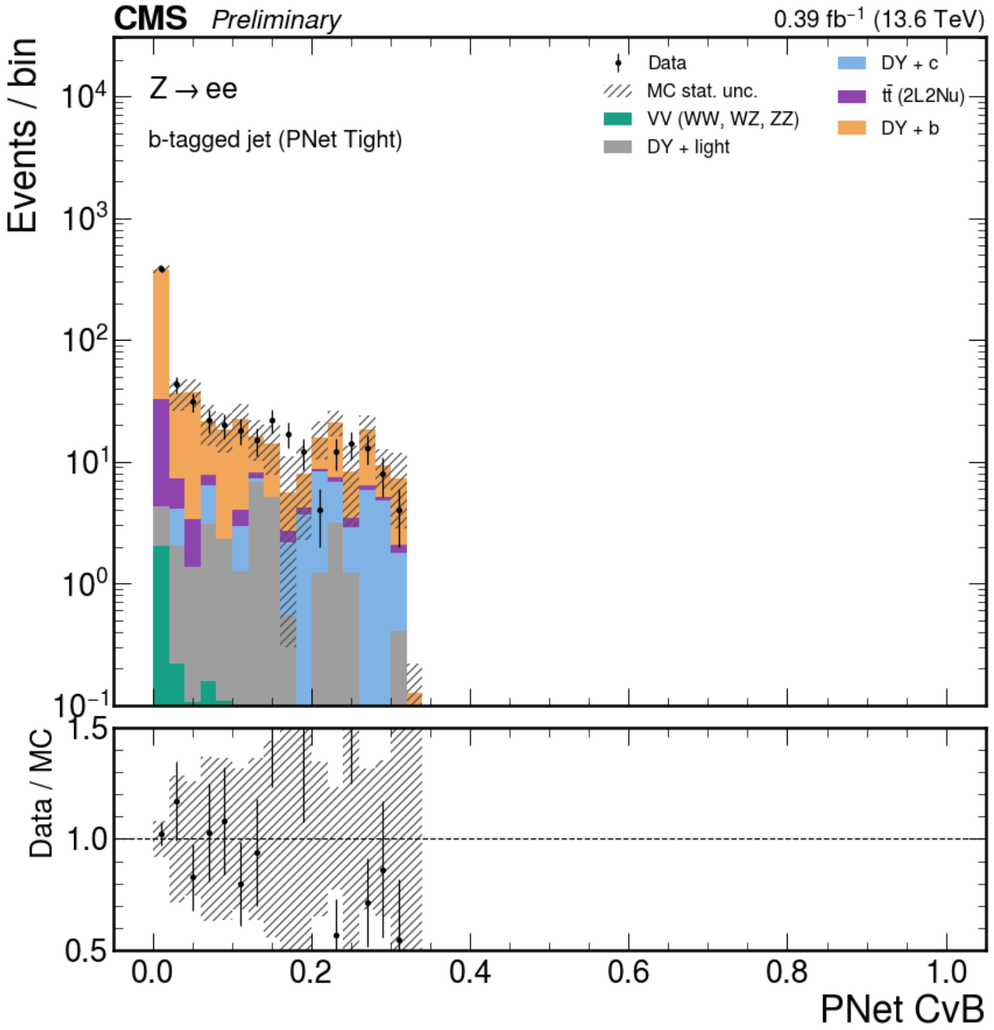
Leading-jet PNet CvL score

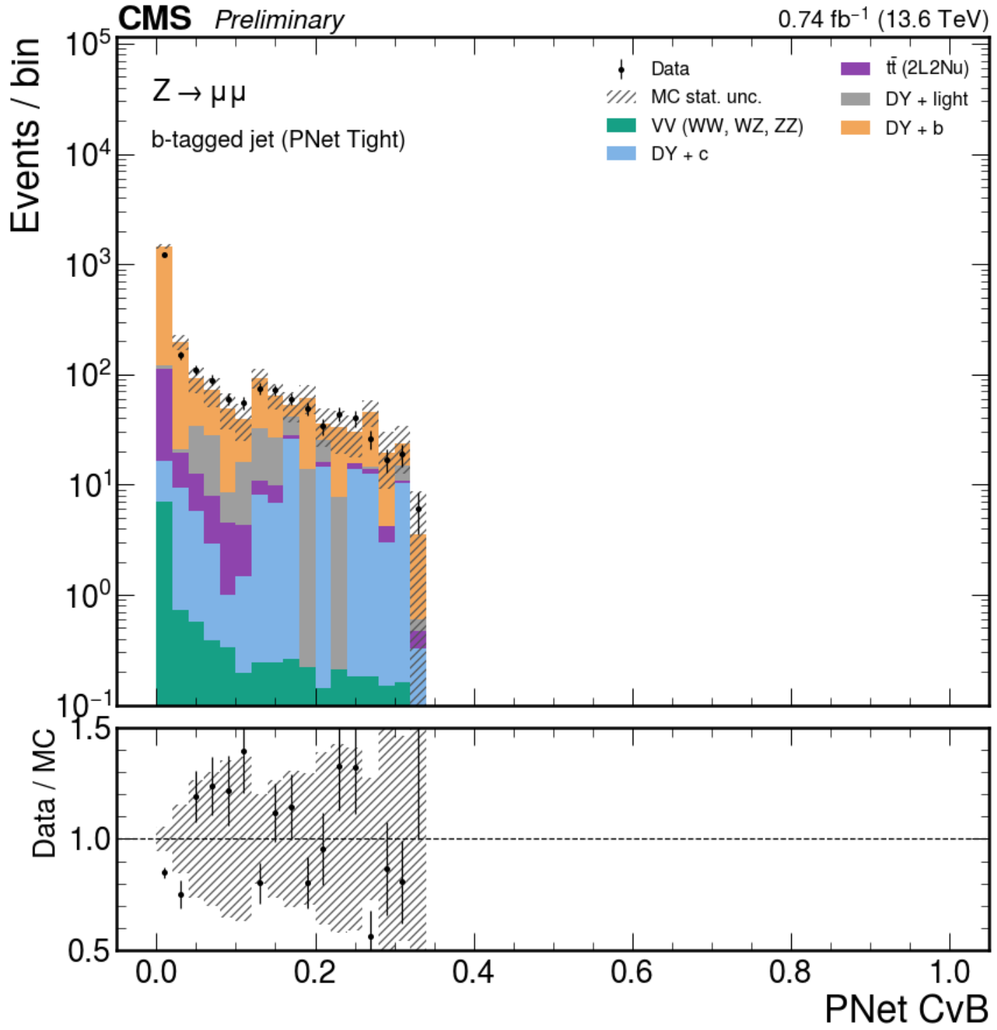




b-tag Tight

Leading-jet PNet CvB score

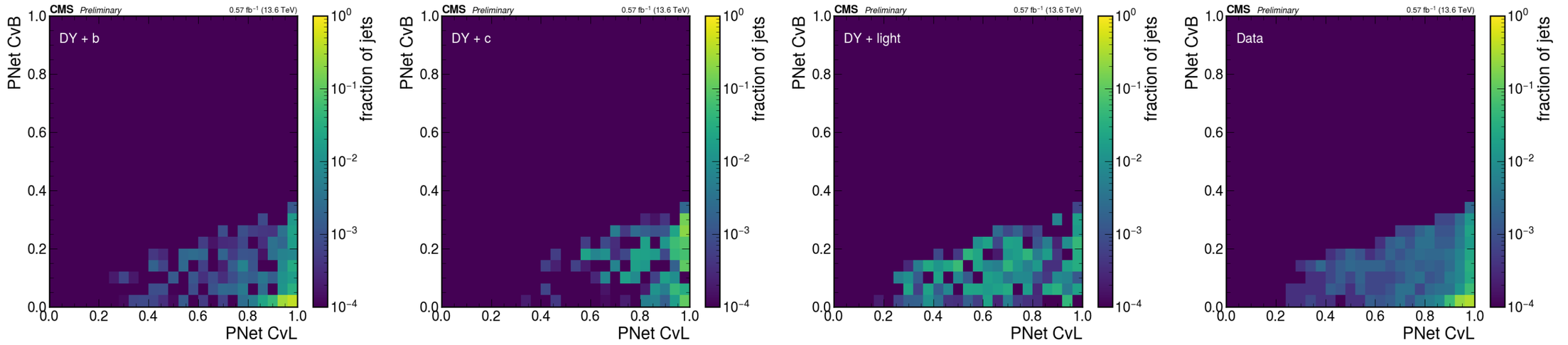




b-tag Tight

PNet CvL vs CvB of the leading jet

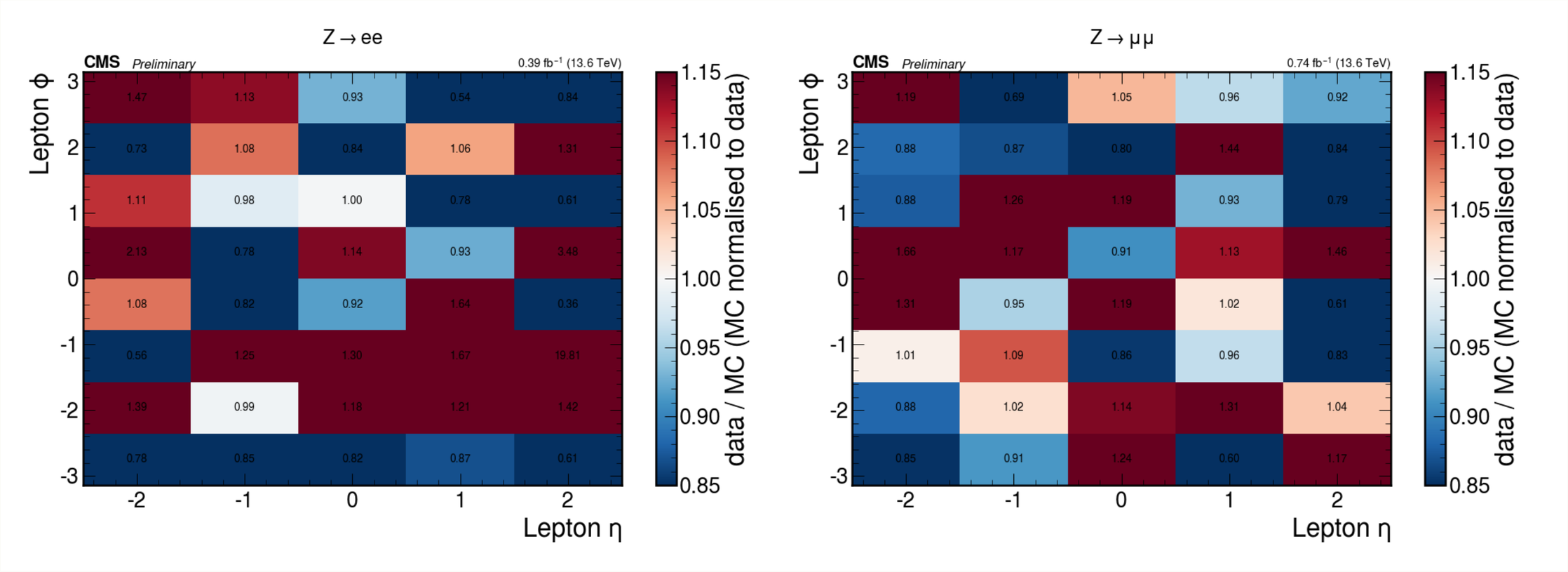
Each panel normalised to 1. Z→ee and Z→μμ added.



b-tag Tight

Lepton η–φ, data / MC

Both Z leptons. MC normalised to data in each map. Colour scale stops at ±15%; in the tagged regions there are few events per cell, so read the numbers.



c-tag Tight

Leading jet PNet CvL > 0.492 and CvB > 0.259

	data/MC	DY+b	DY+c	DY+light	tt	tagged MC	tagged data
Z→ee	0.916 ± 0.043	8.2%	49.4%	41.1%	0.7%	7.6%	7.3%
Z→μμ	0.928 ± 0.033	9.6%	50.2%	38.9%	0.7%	7.3%	7.0%

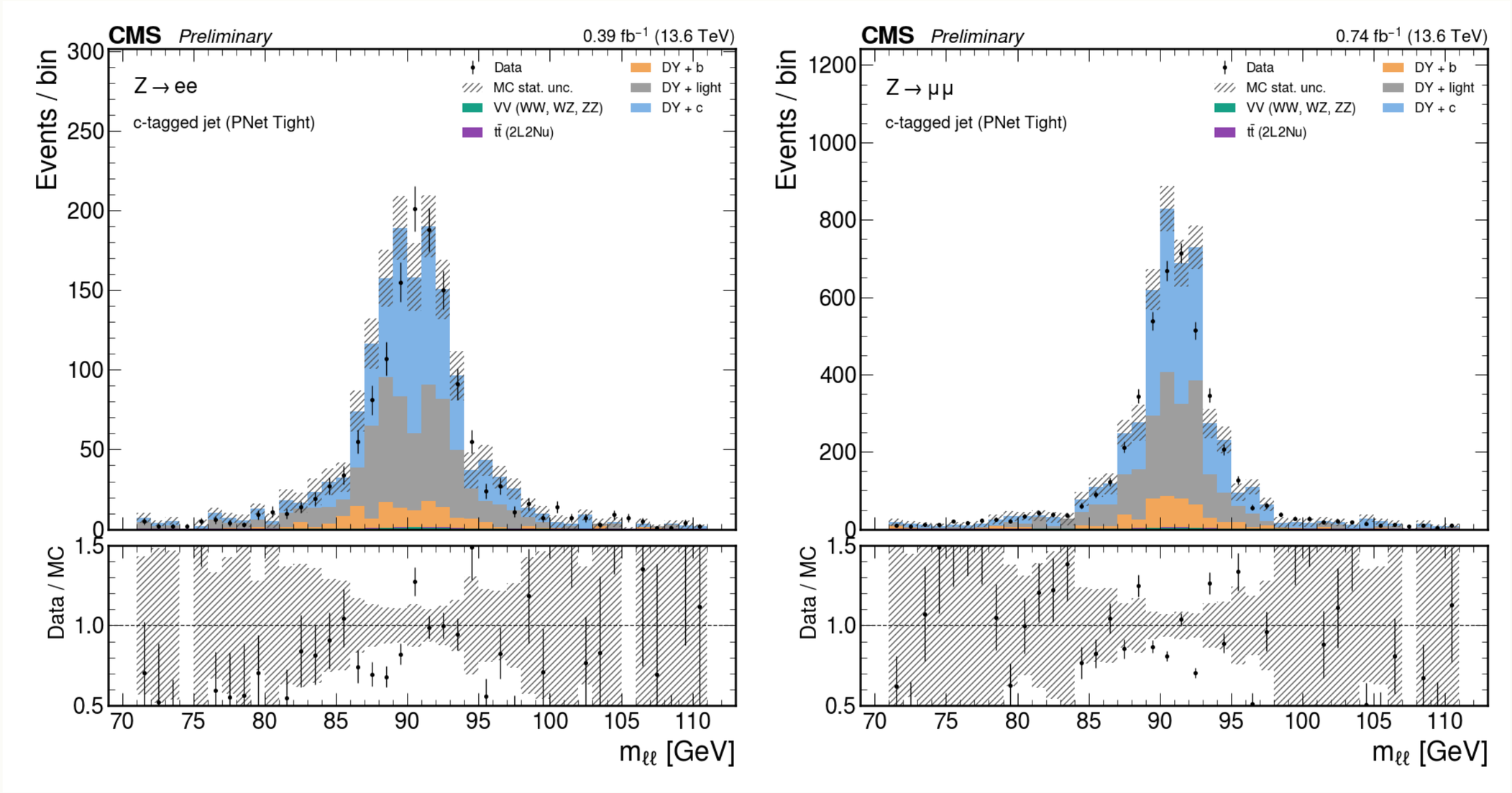
Flavour shares are of all MC. 'Tagged' = fraction of Z + jet events passing the tag. Errors are statistical only.

32 plots follow: Z and leptons, jets, Z–jet, MET and vertices, tagger scores, 2D maps. Left Z→ee, right Z→μμ.

c-tag Tight

Dilepton mass

m(ℓℓ), 71–111 GeV window, linear scale.

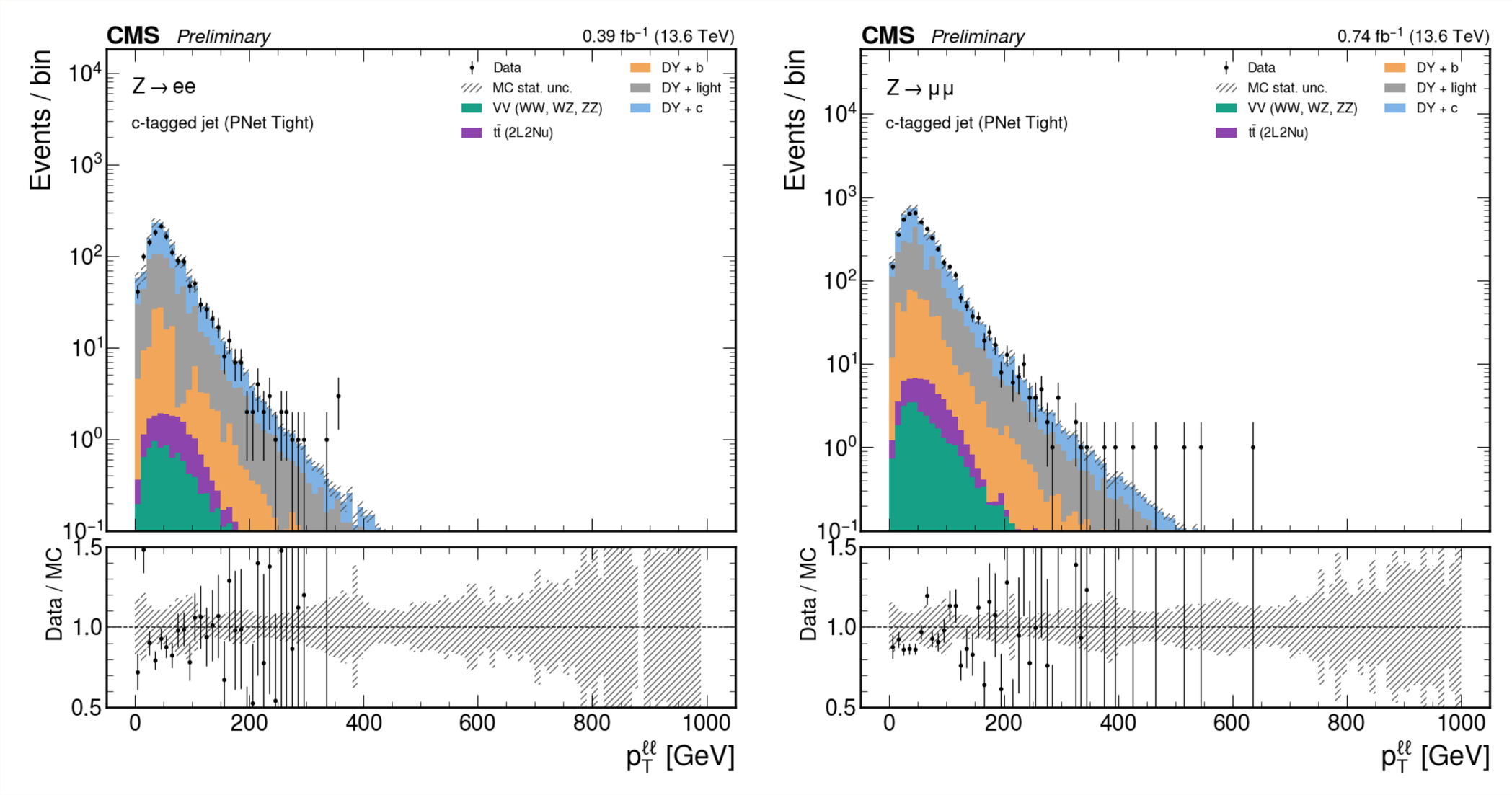


Summer22 (2022 C–D), 13.6 TeV · Z→ℓℓ + jets · 20 files per MC sample and per data channel · AK4 jets · stat. errors only · 205

c-tag Tight

Z pT

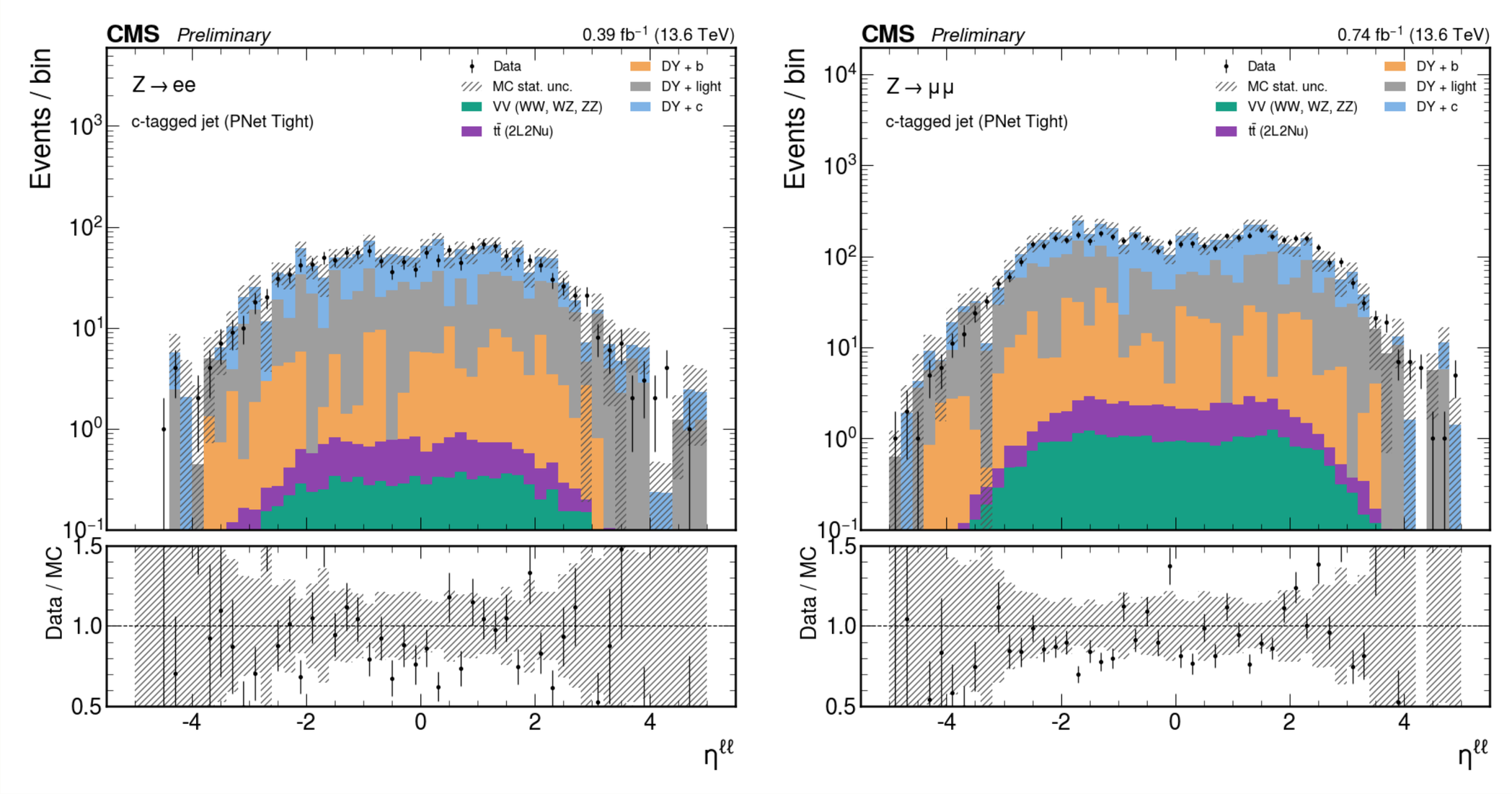
pT of the lepton pair.



c-tag Tight

Z η

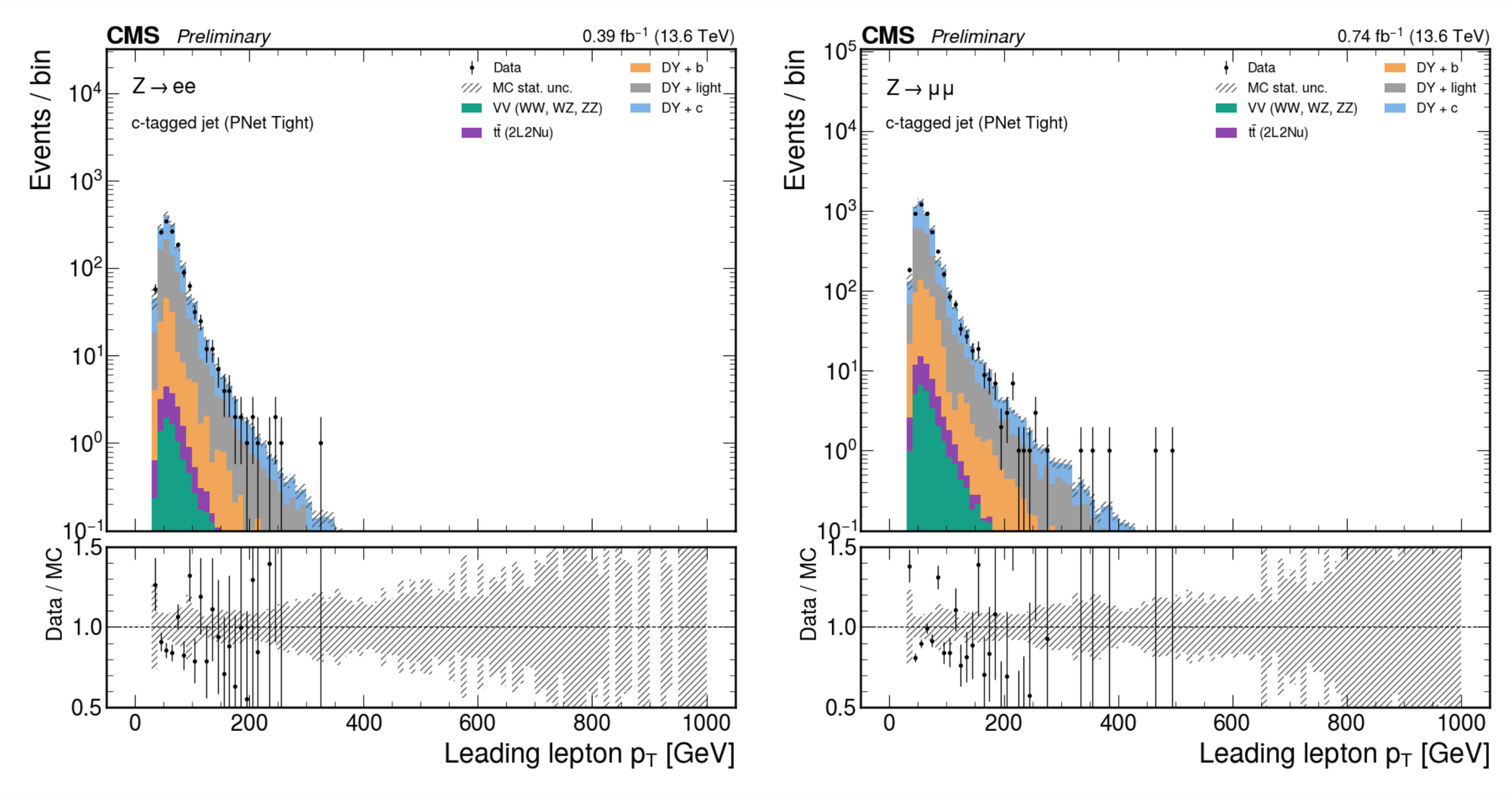
η of the lepton pair.



c-tag Tight

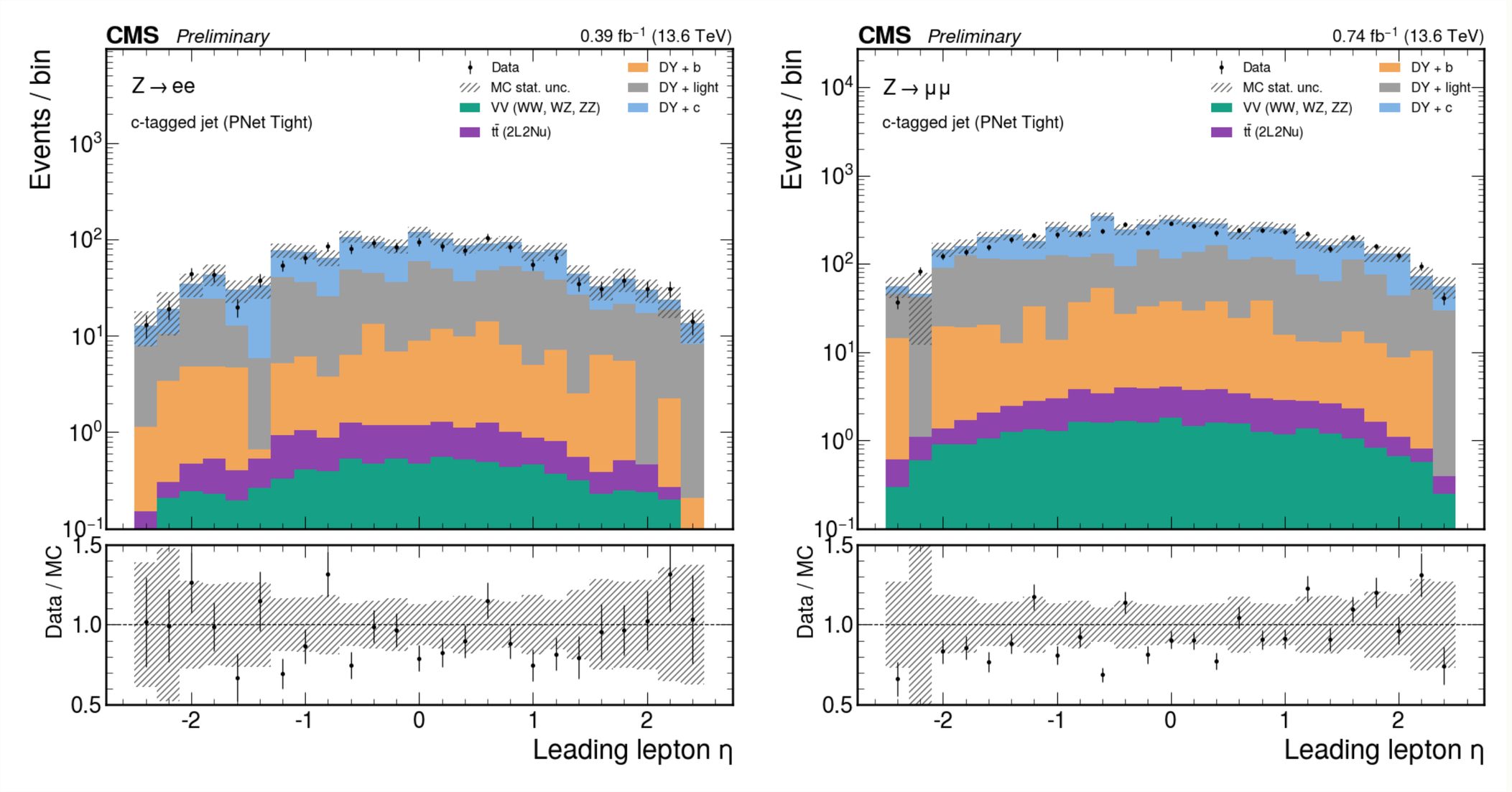
Leading-lepton pT

Leading lepton pT > 35 GeV (trigger plateau).



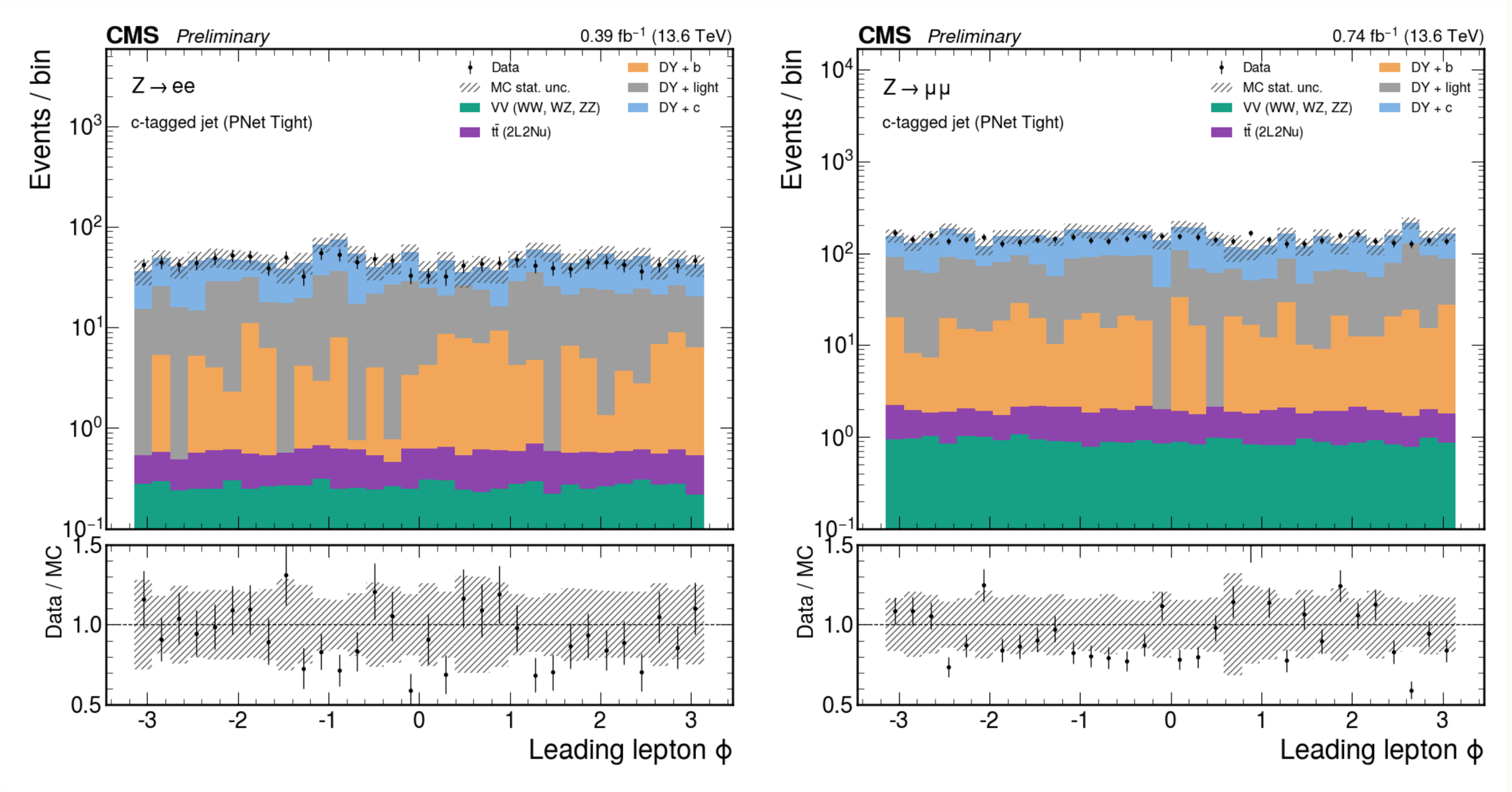
c-tag Tight

Leading-lepton η



c-tag Tight

Leading-lepton φ

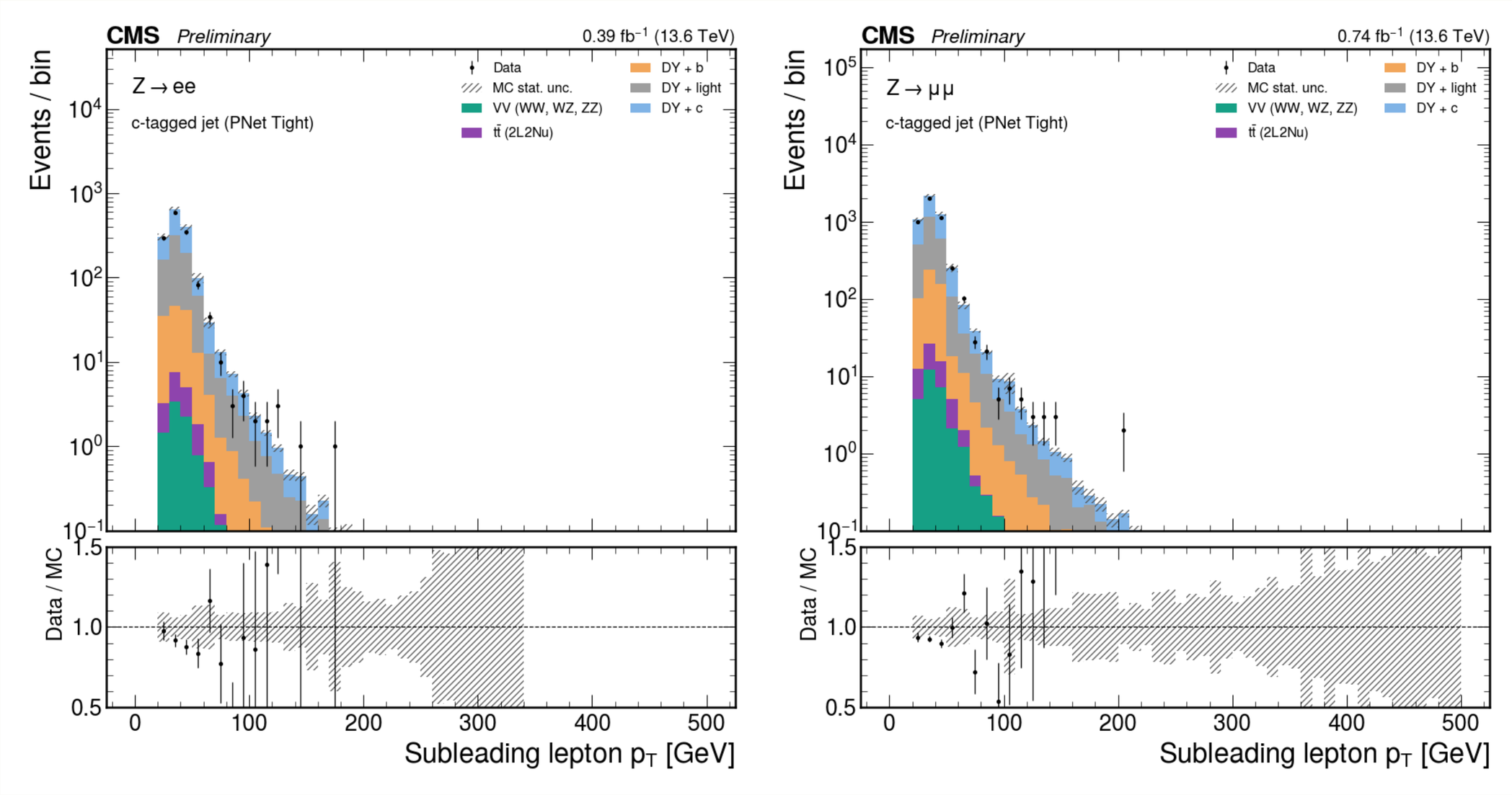


Summer22 (2022 C–D), 13.6 TeV · Z→ℓℓ + jets · 20 files per MC sample and per data channel · AK4 jets · stat. errors only · 210

c-tag Tight

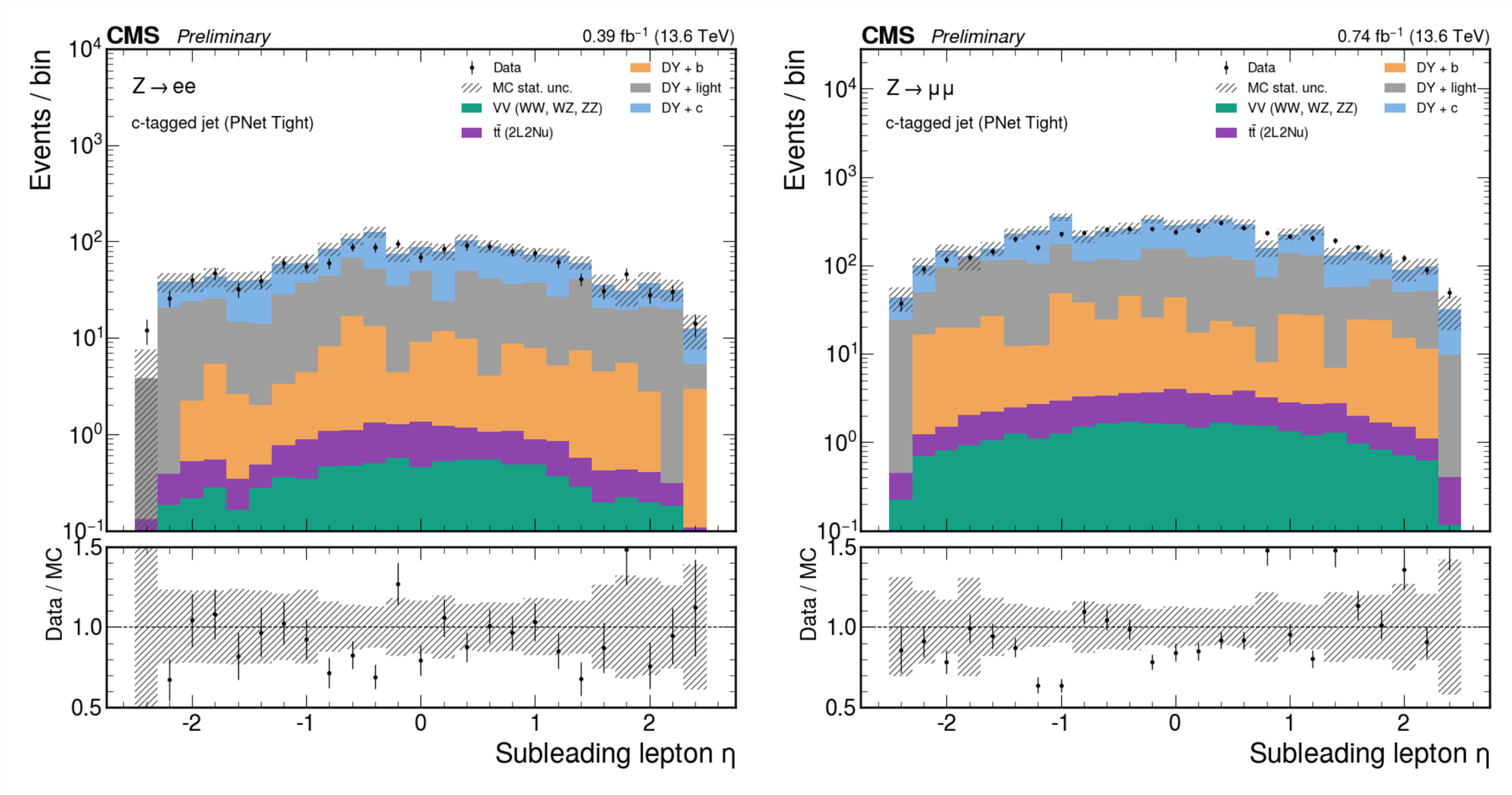
Second-lepton pT

Second lepton pT > 25 GeV.



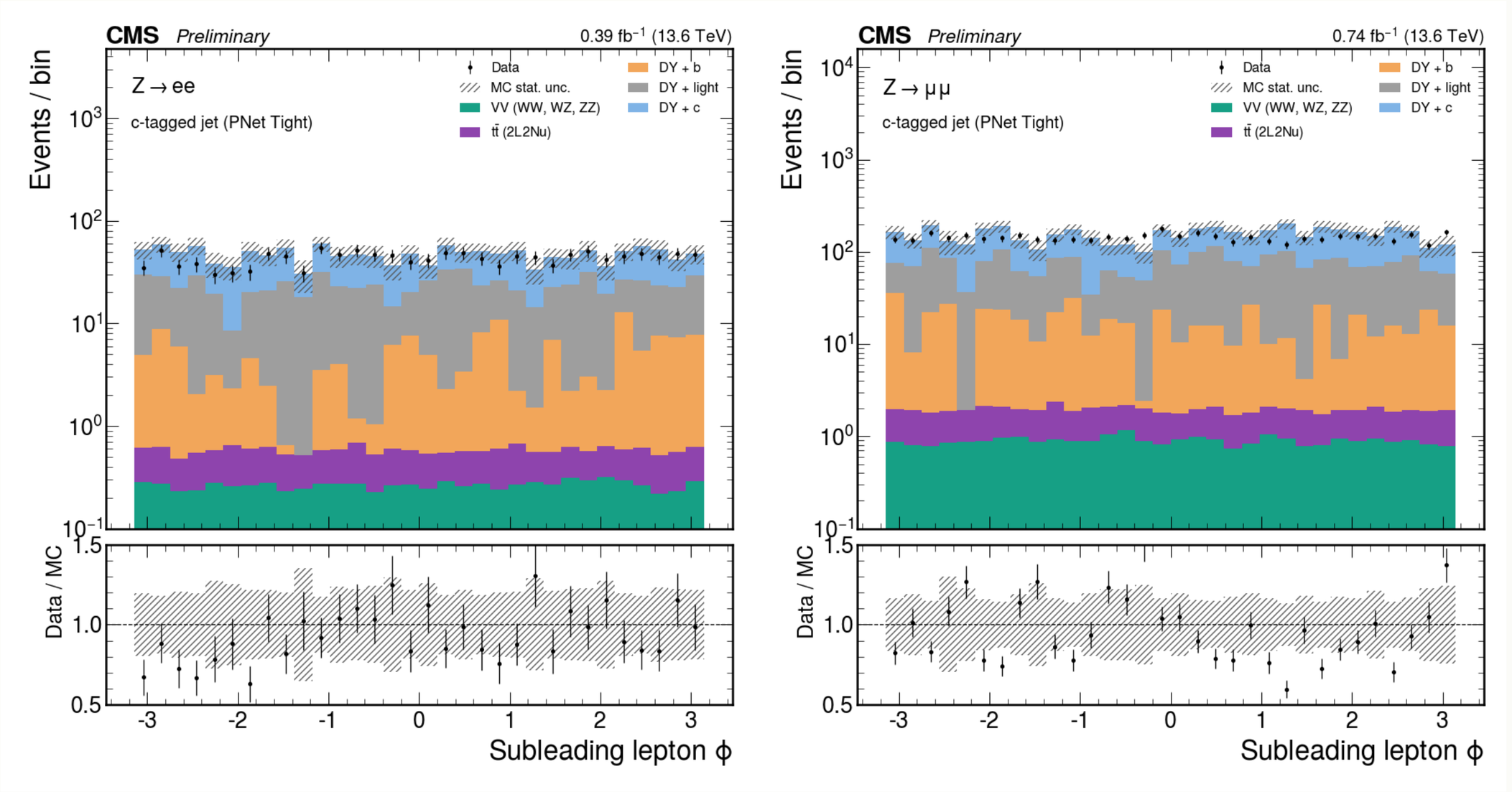
c-tag Tight

Second-lepton η



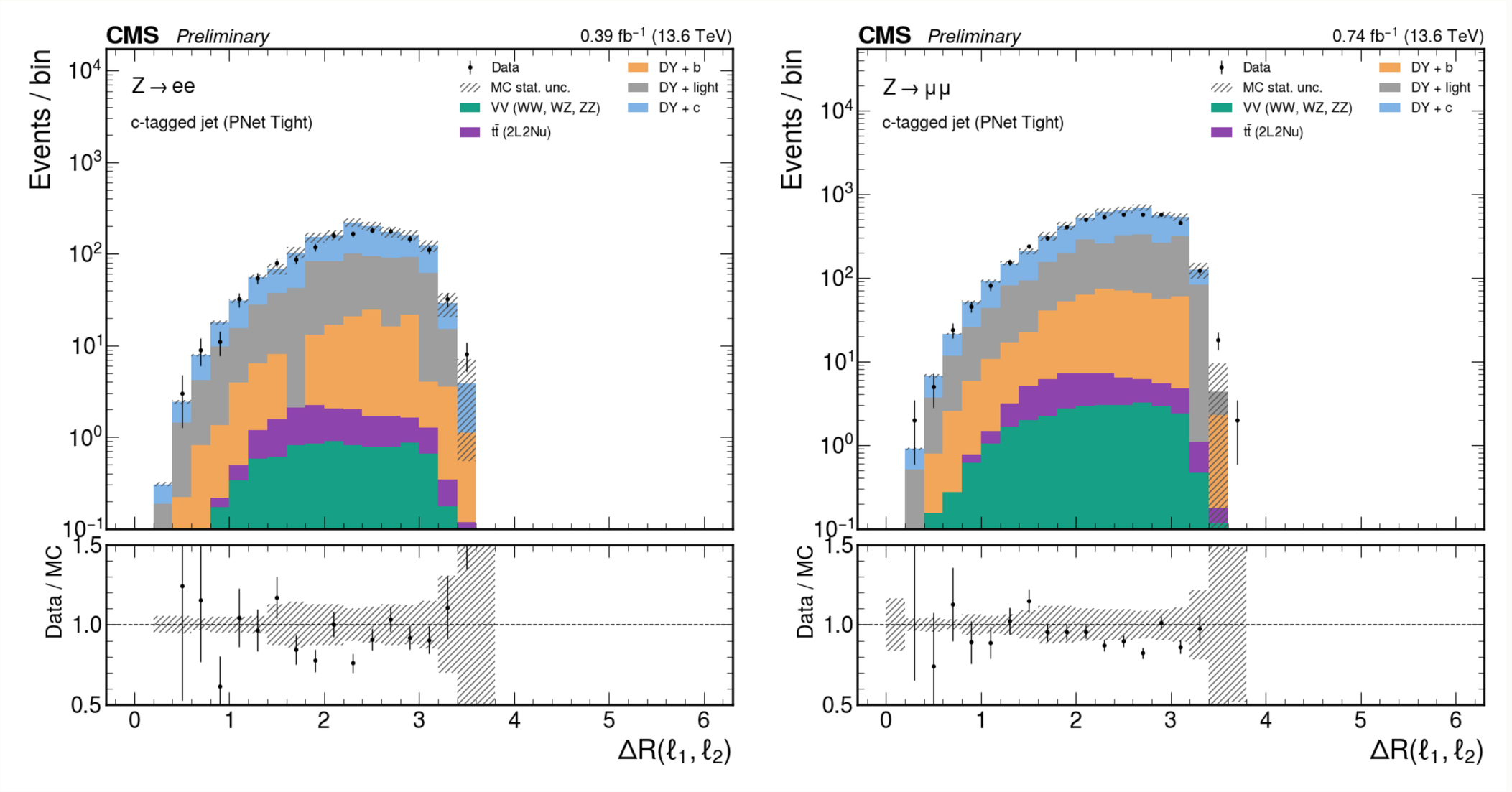
c-tag Tight

Second-lepton φ



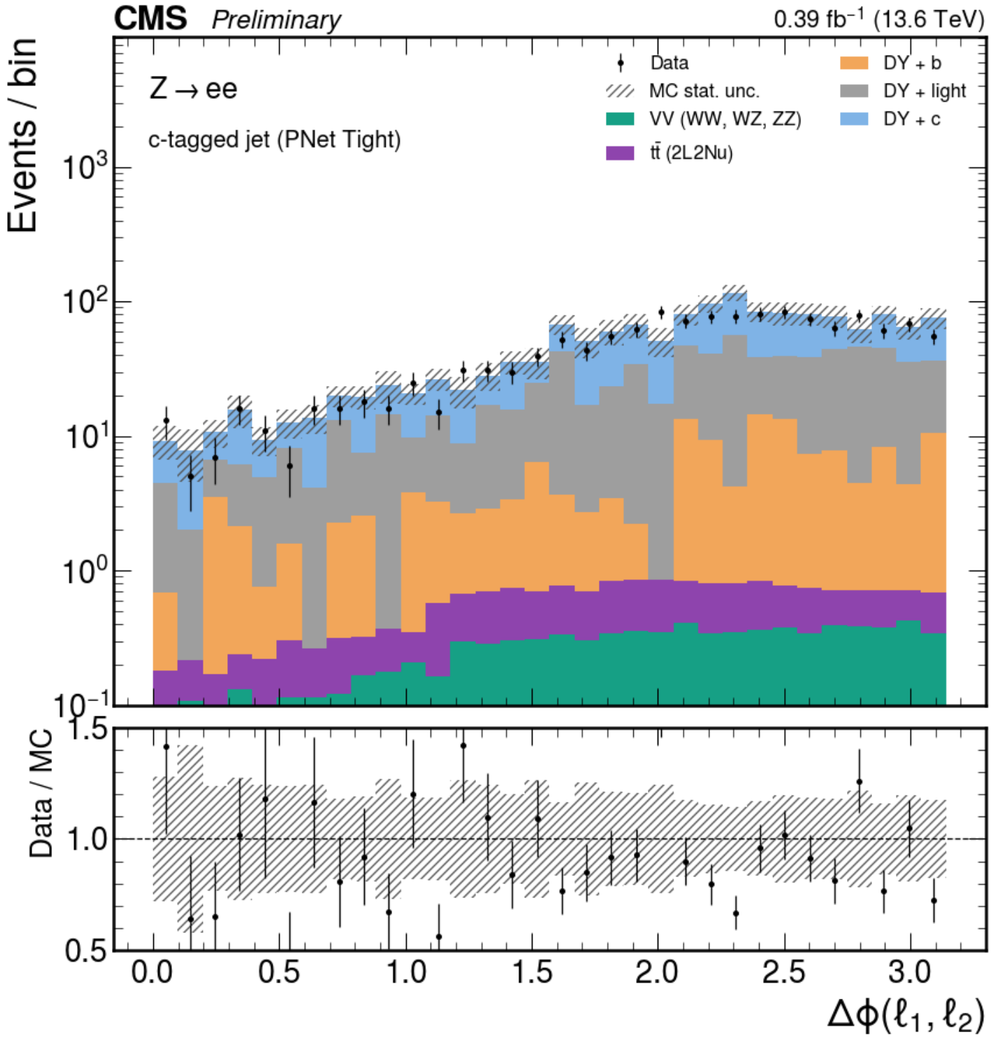
c-tag Tight

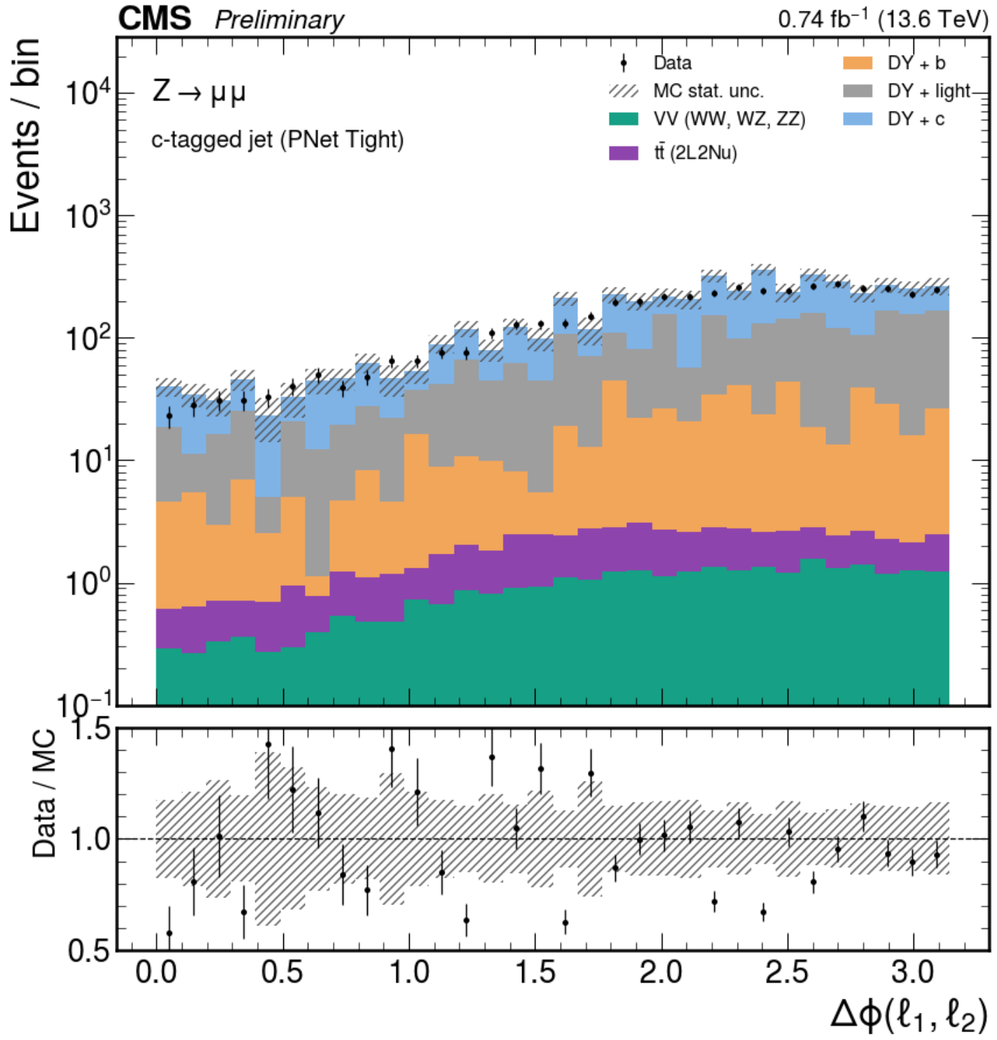
ΔR between the leptons



c-tag Tight

Δφ between the leptons

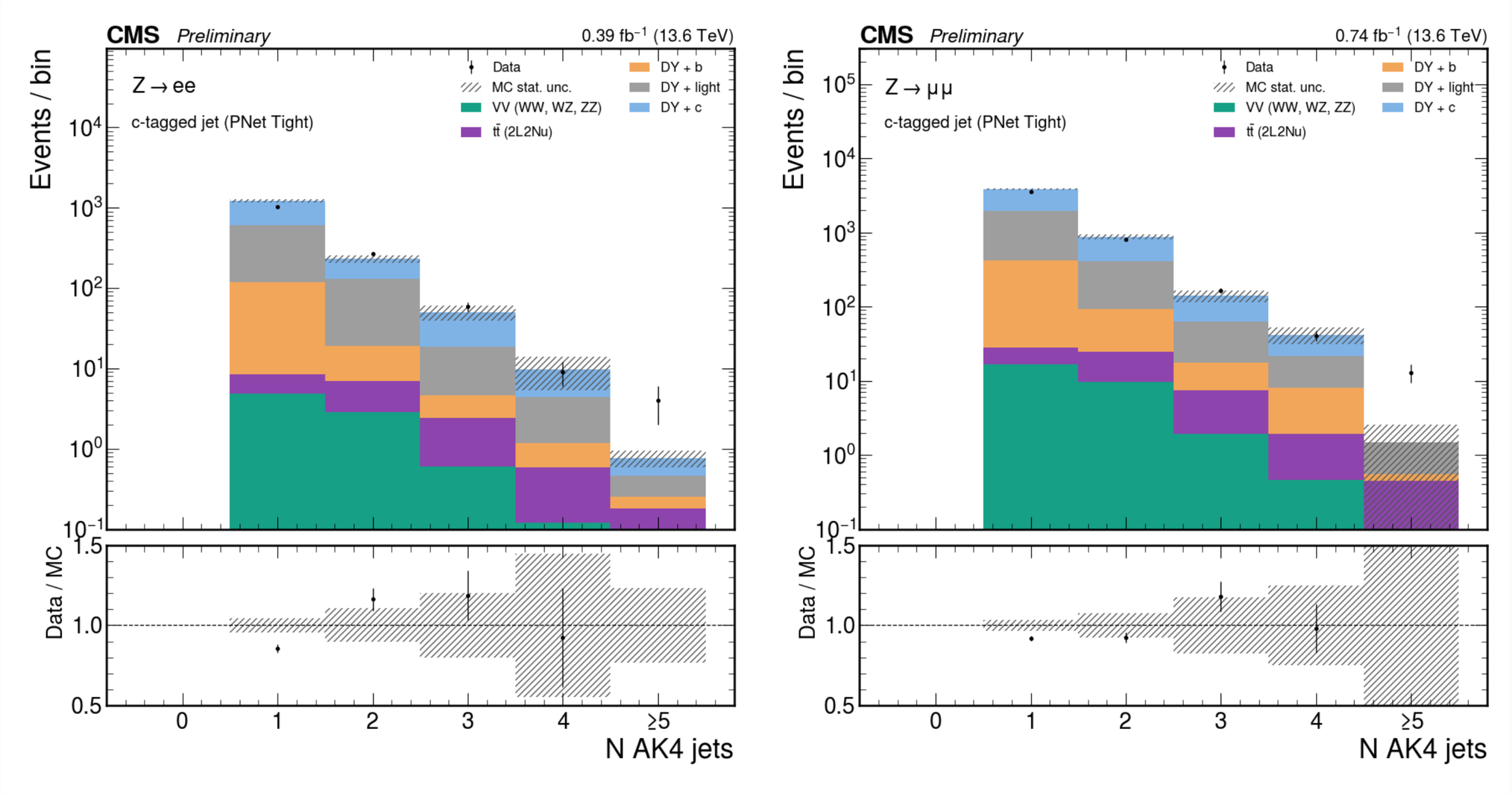




c-tag Tight

Number of jets

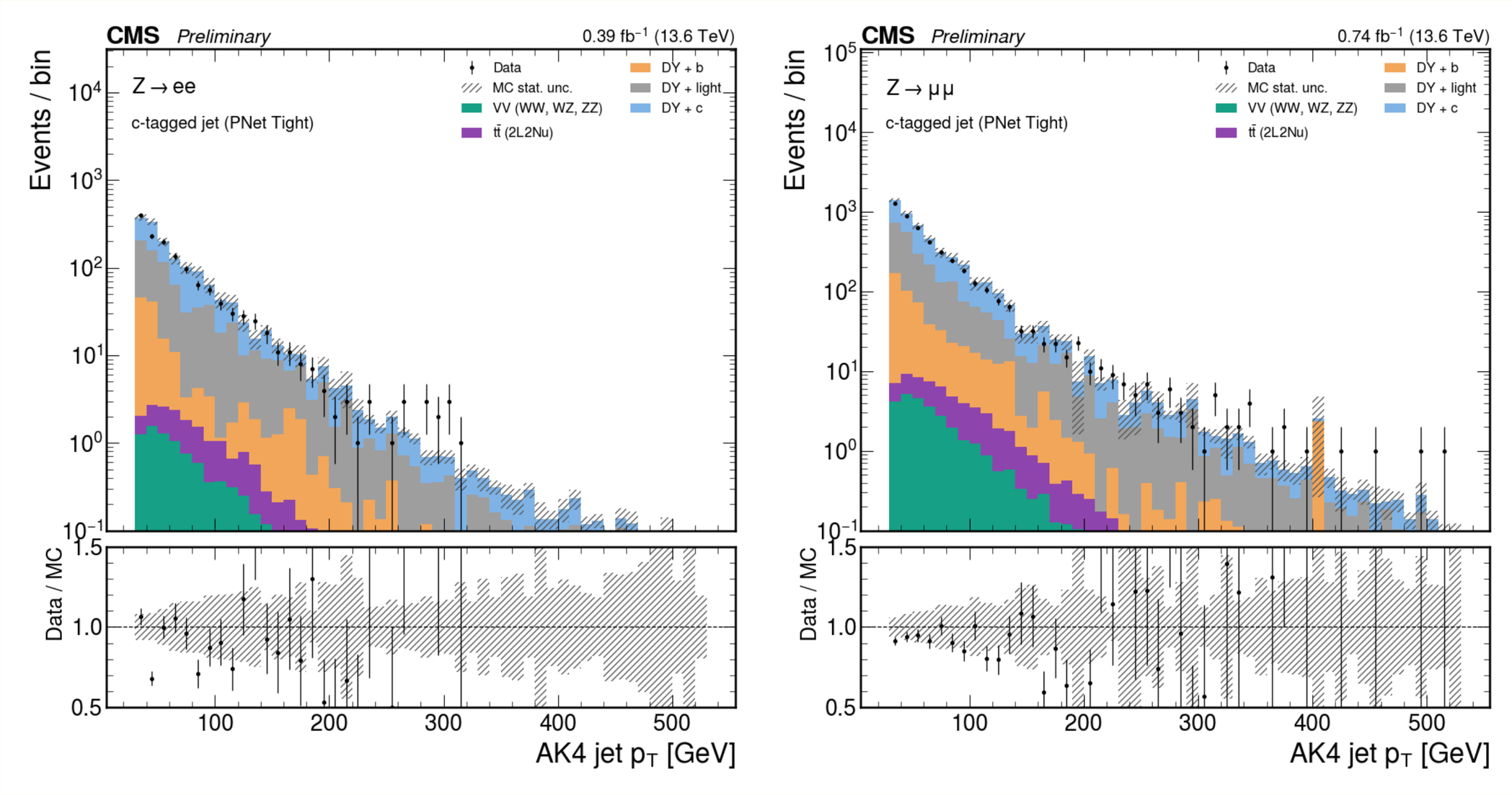
AK4 jets with pT > 30 GeV, |η| < 2.5, away from the leptons. Untagged plot: Z events before the jet cut, so the 0-jet bin is filled.



c-tag Tight

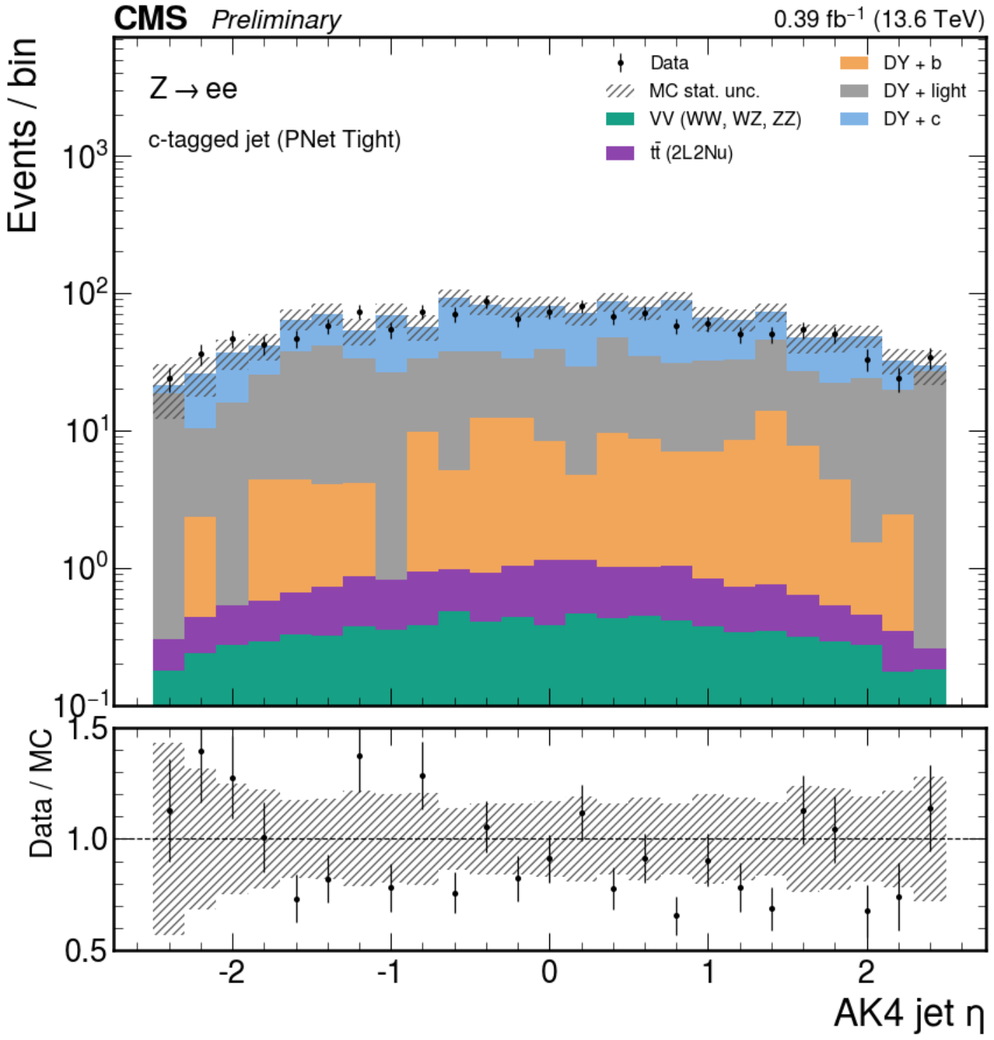
Leading-jet pT

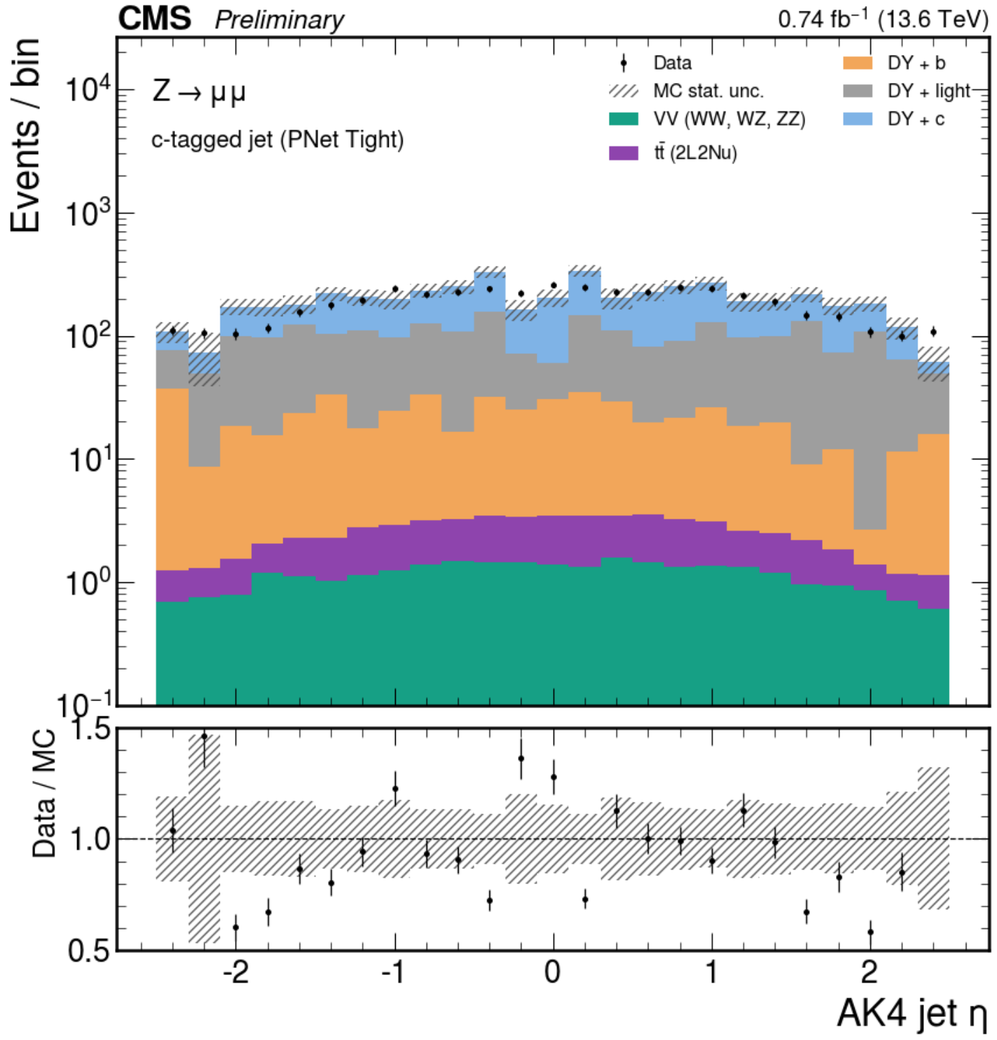
The tag (if any) is on this jet.



c-tag Tight

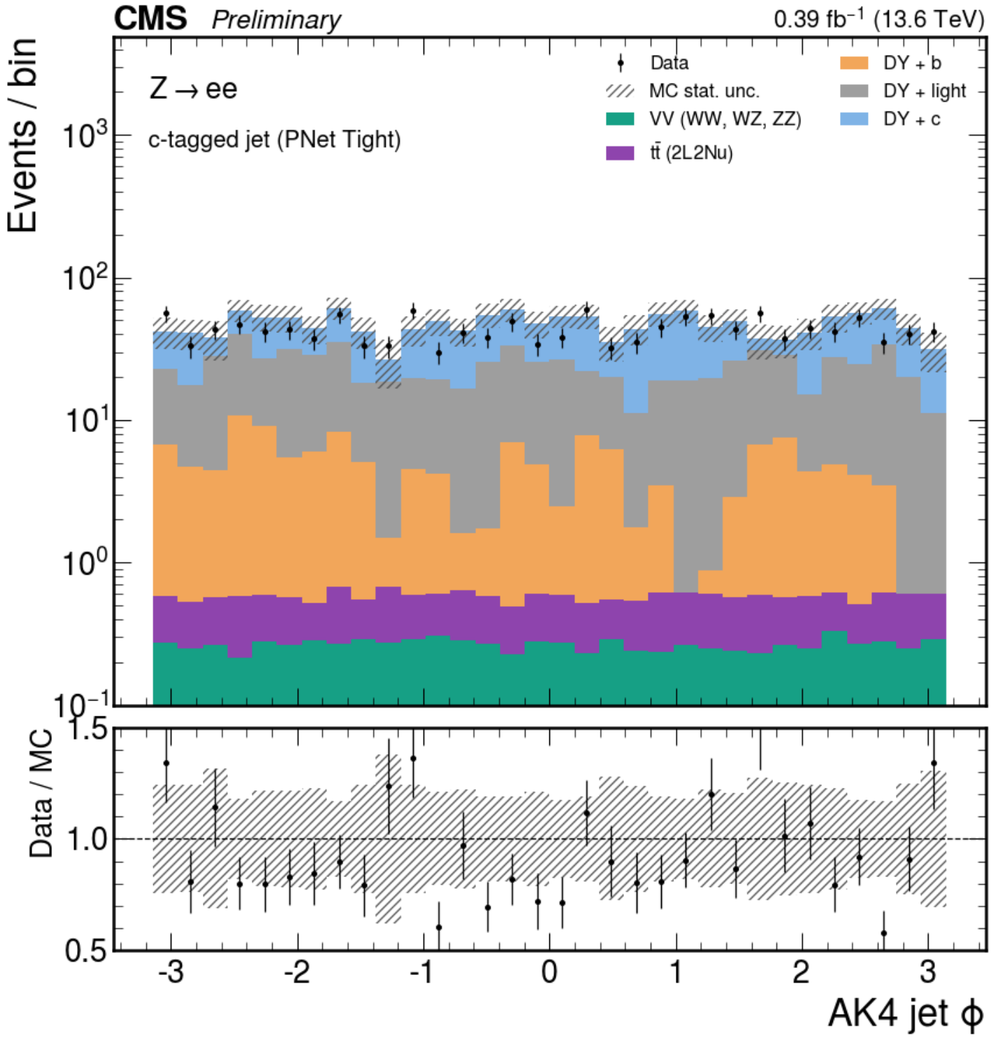
Leading-jet η

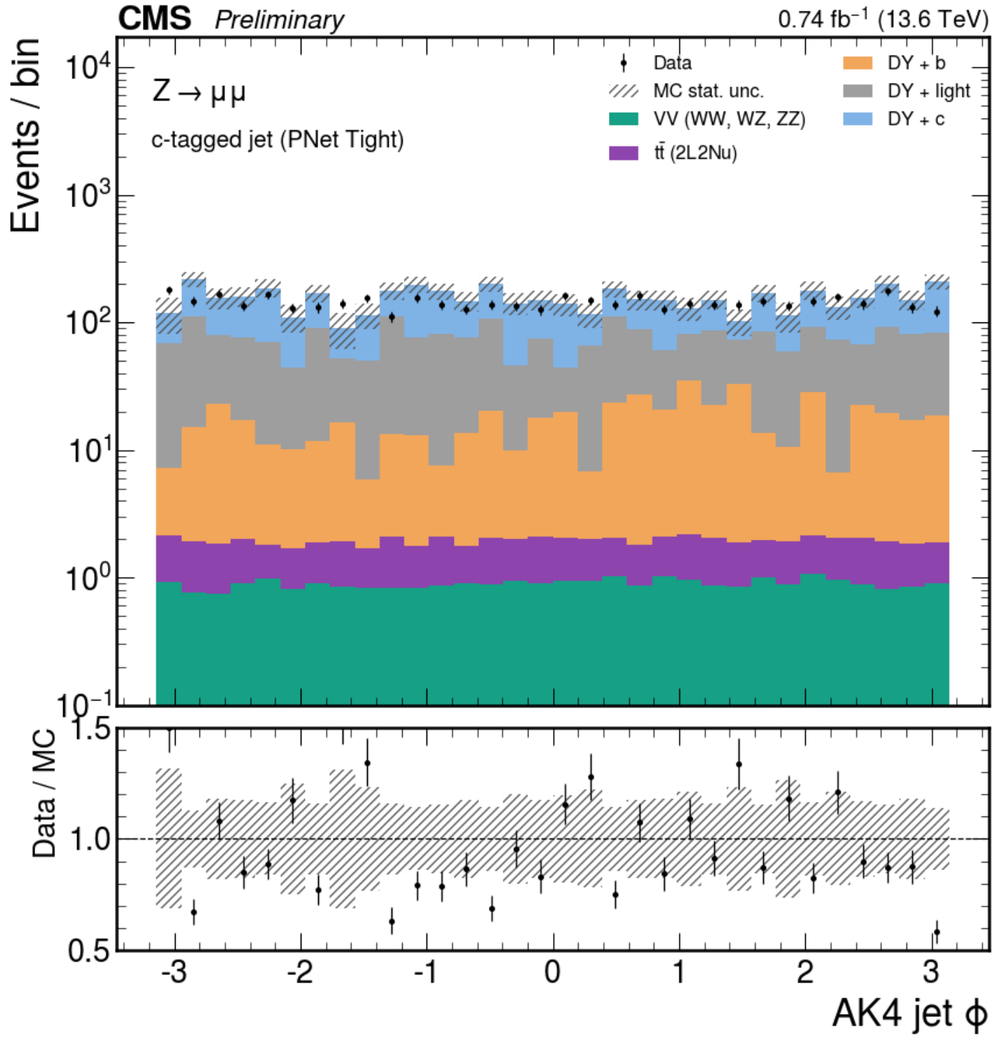




c-tag Tight

Leading-jet φ

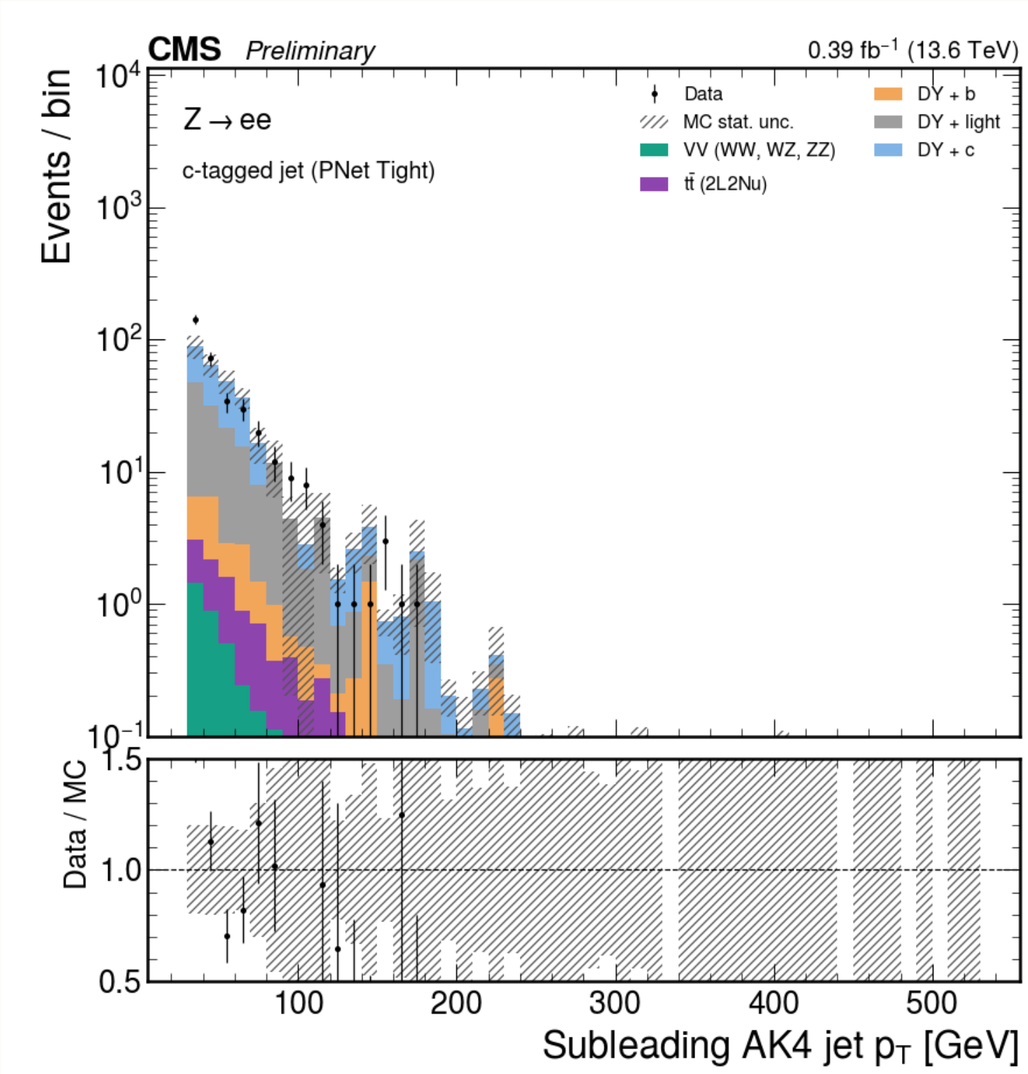


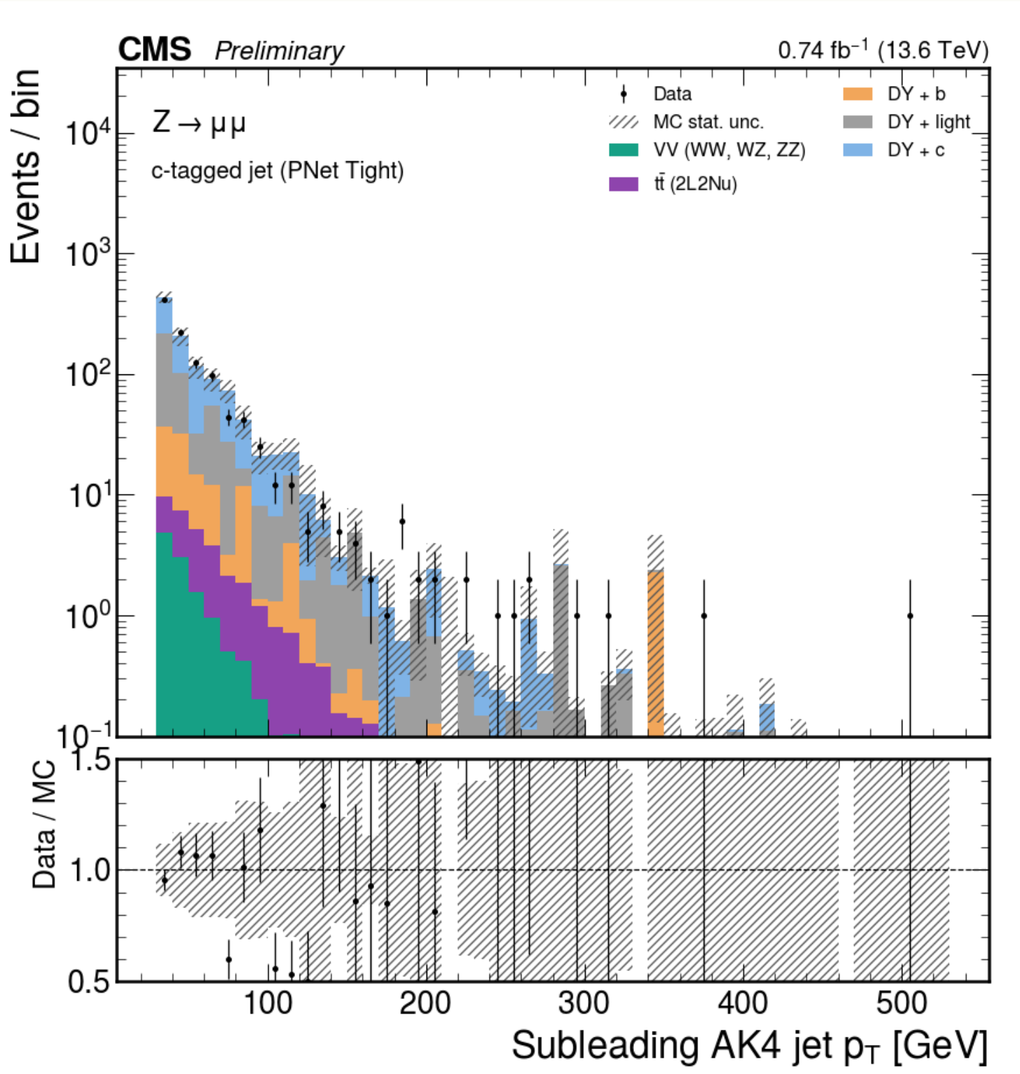


c-tag Tight

Second-jet pT

Events with at least 2 jets.

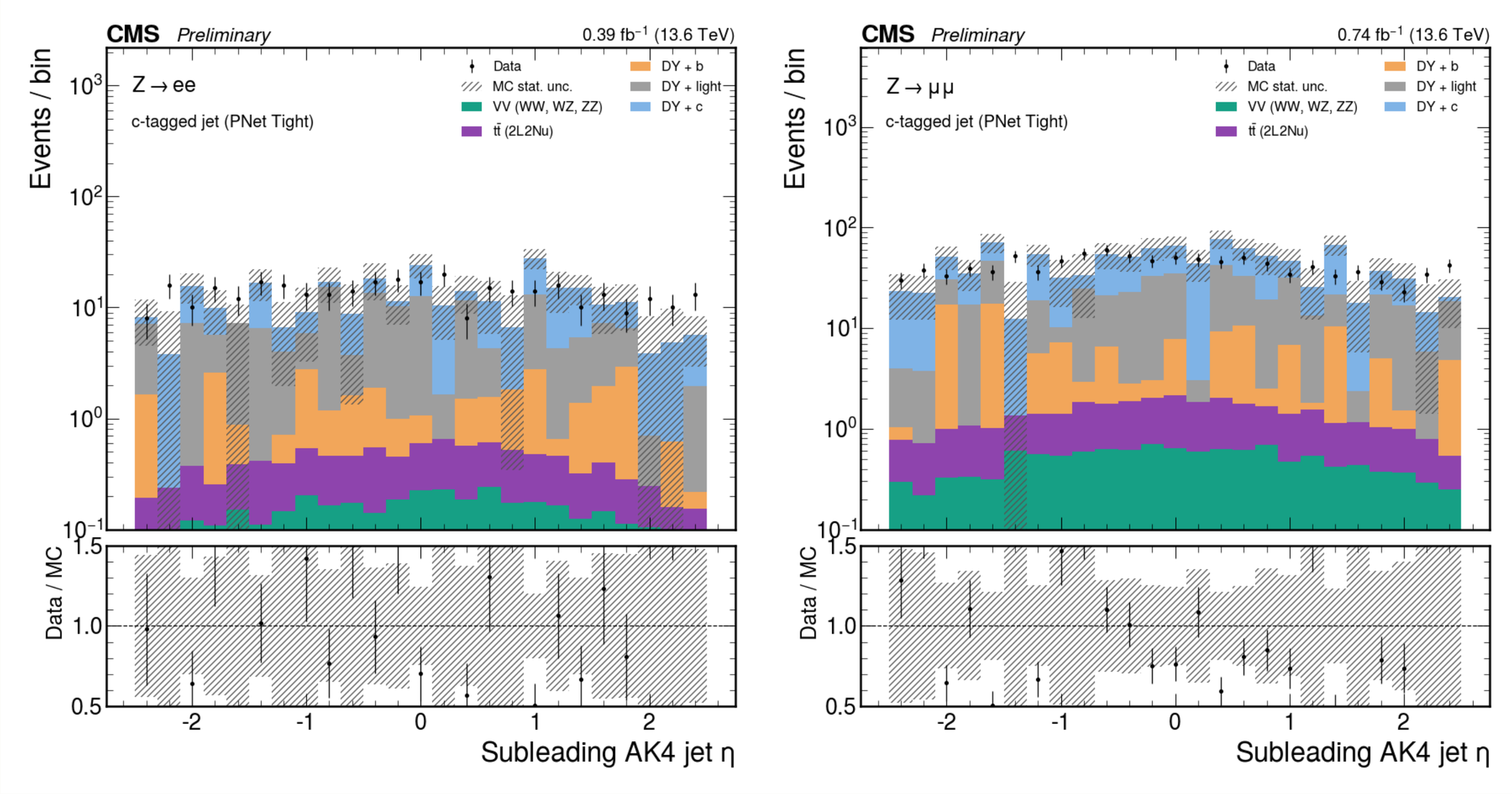




c-tag Tight

Second-jet η

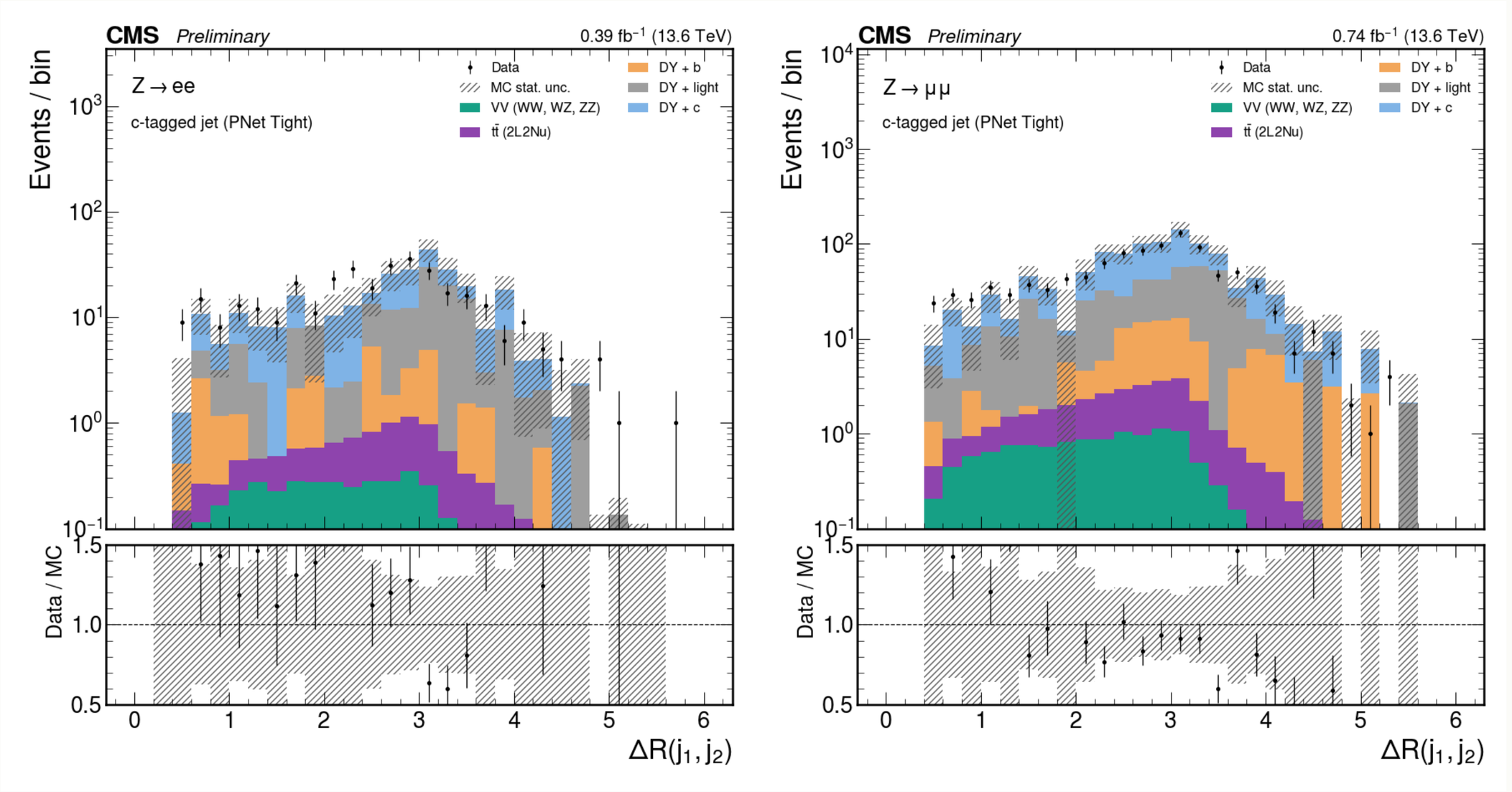
Events with at least 2 jets.



c-tag Tight

ΔR between the two leading jets

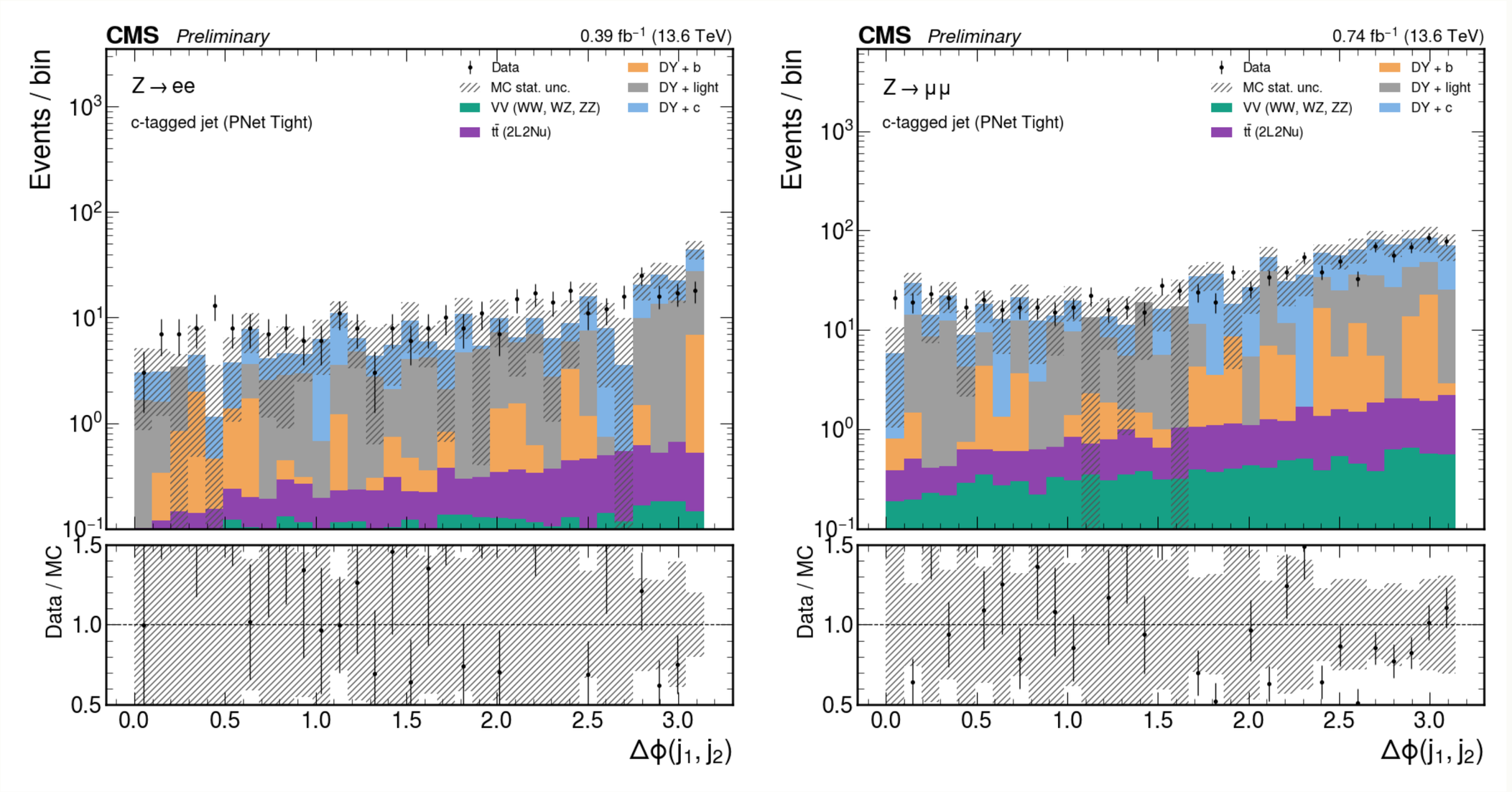
Events with at least 2 jets.



c-tag Tight

Δφ between the two leading jets

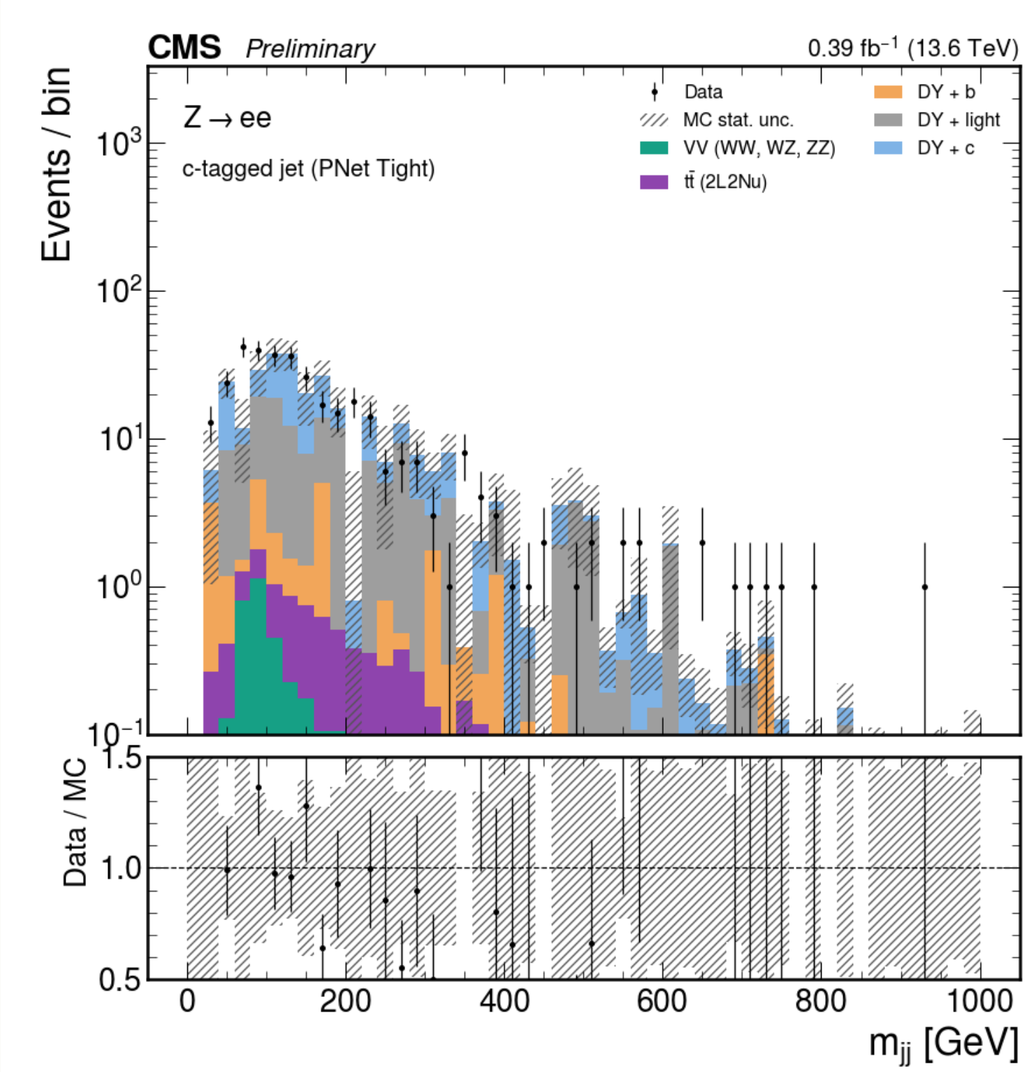
Events with at least 2 jets.

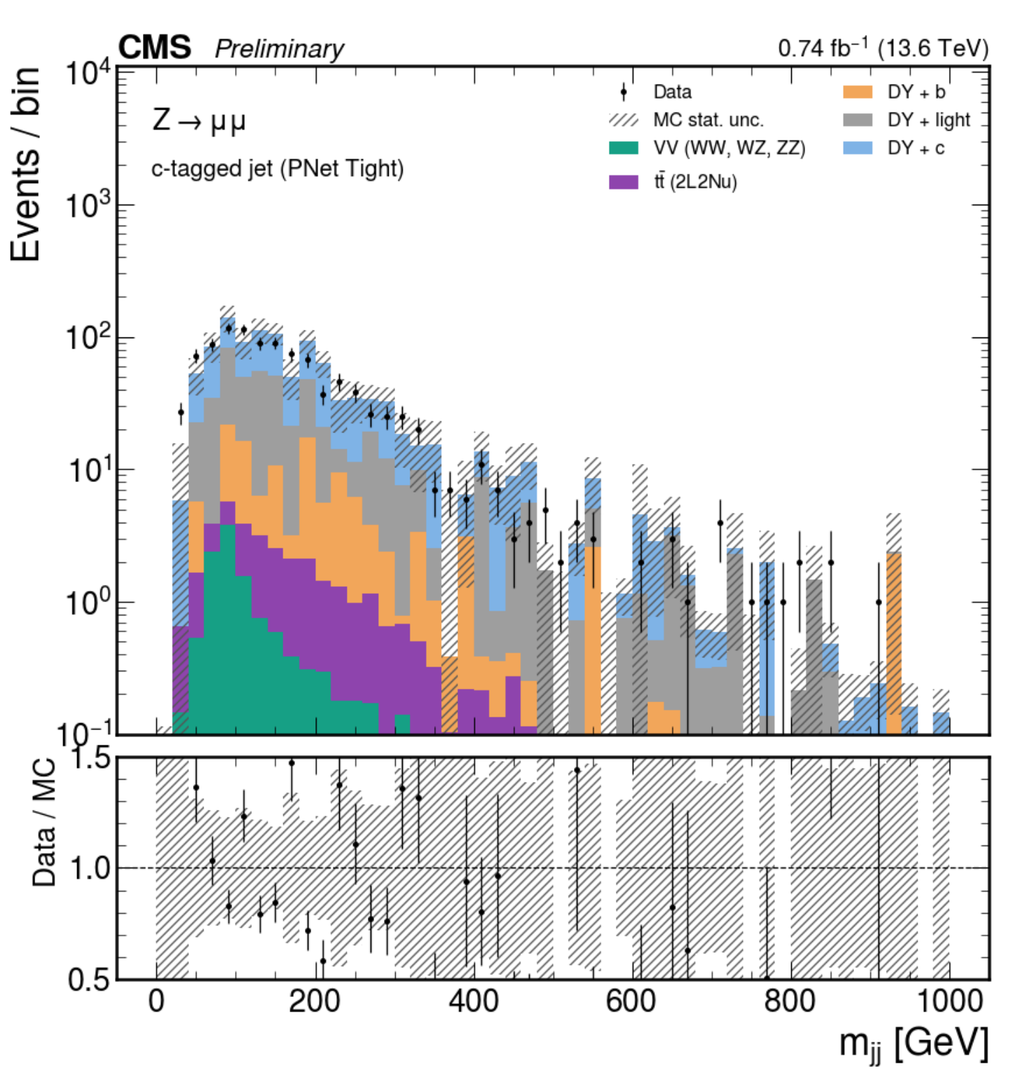


c-tag Tight

Dijet mass

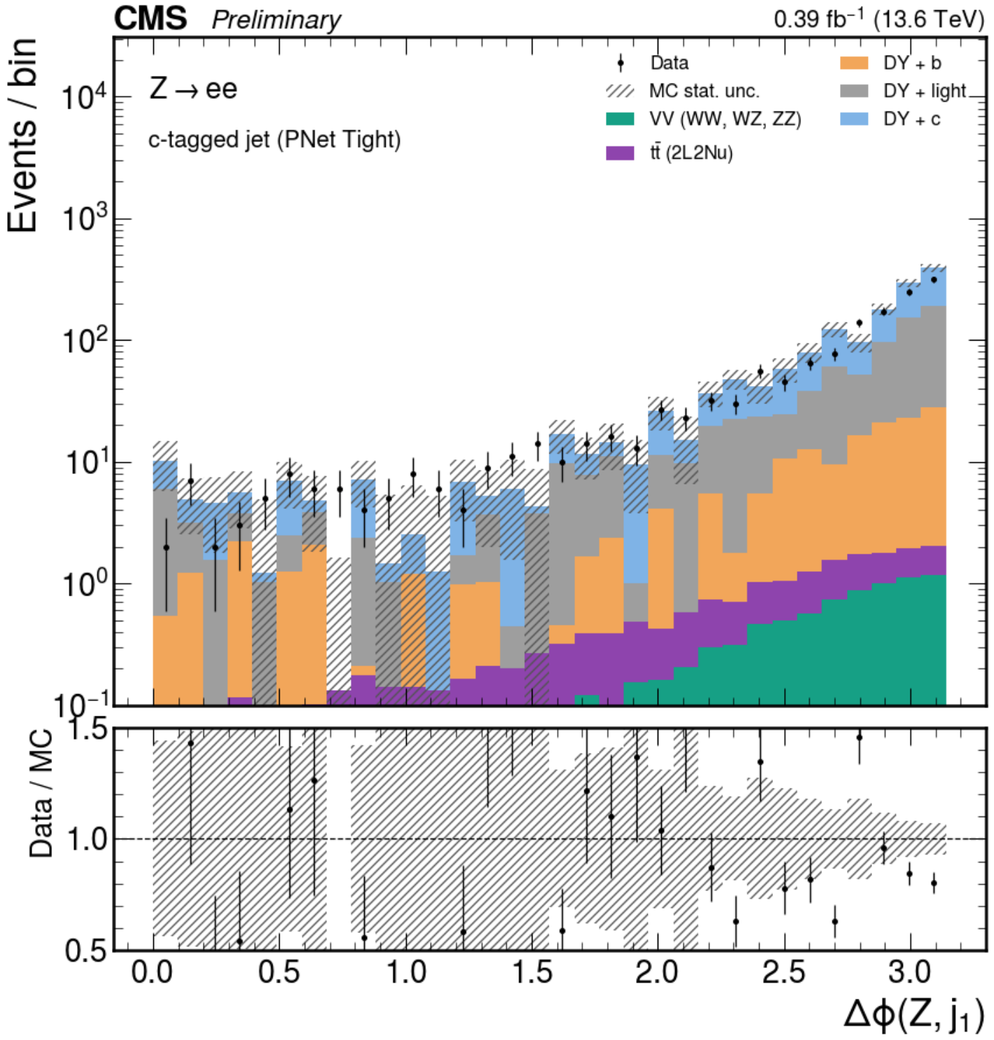
Events with at least 2 jets.

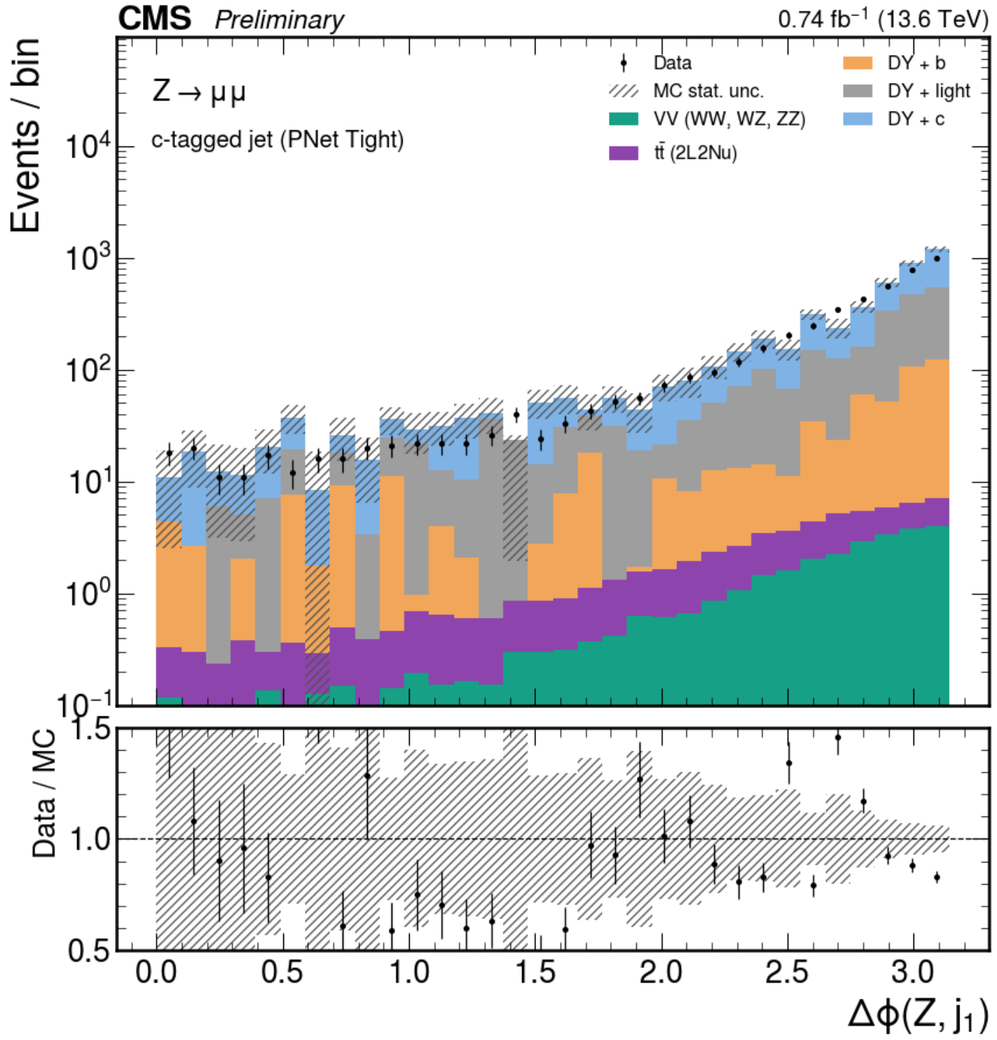




c-tag Tight

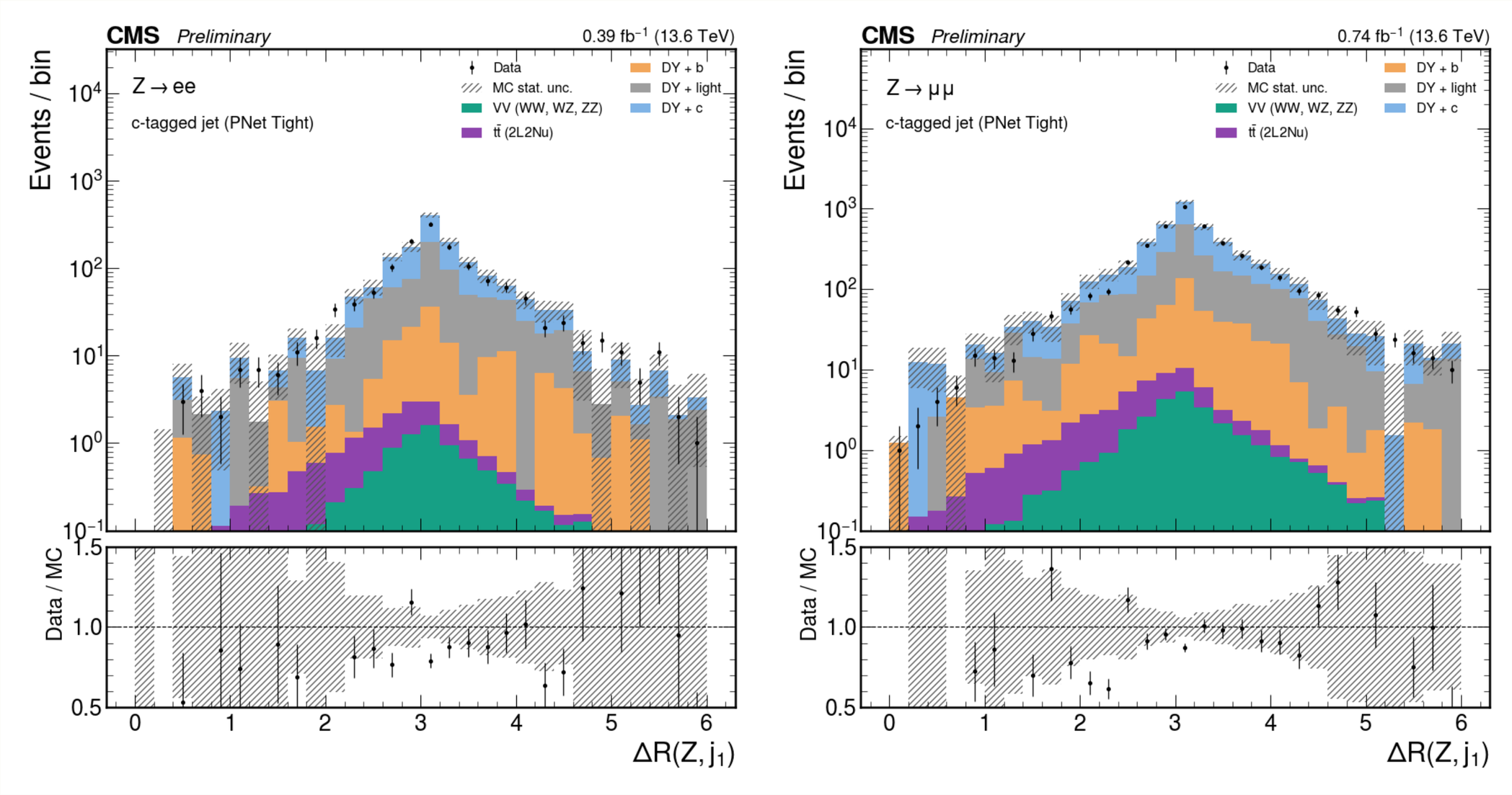
Δφ(Z, leading jet)





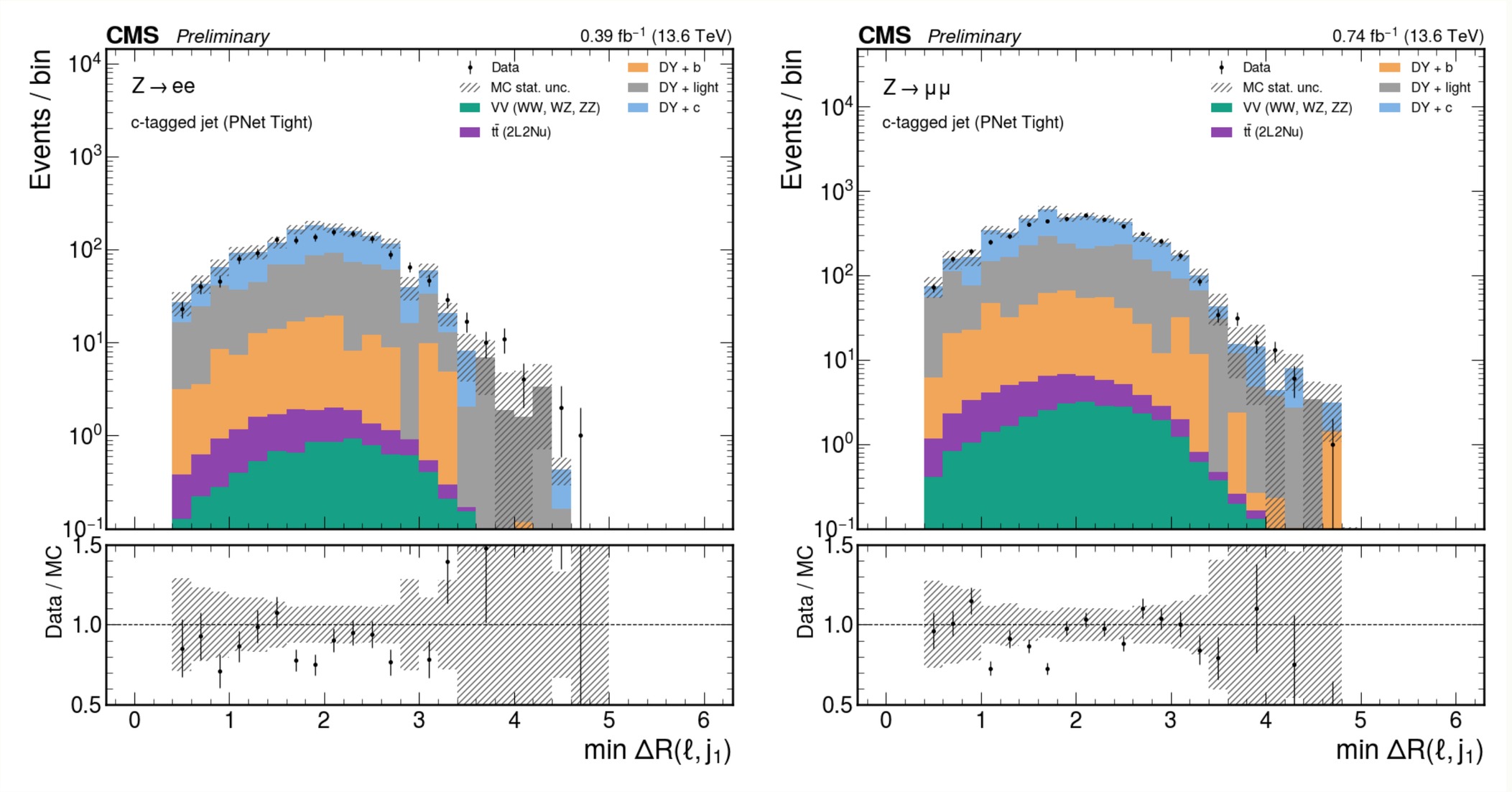
c-tag Tight

ΔR(Z, leading jet)



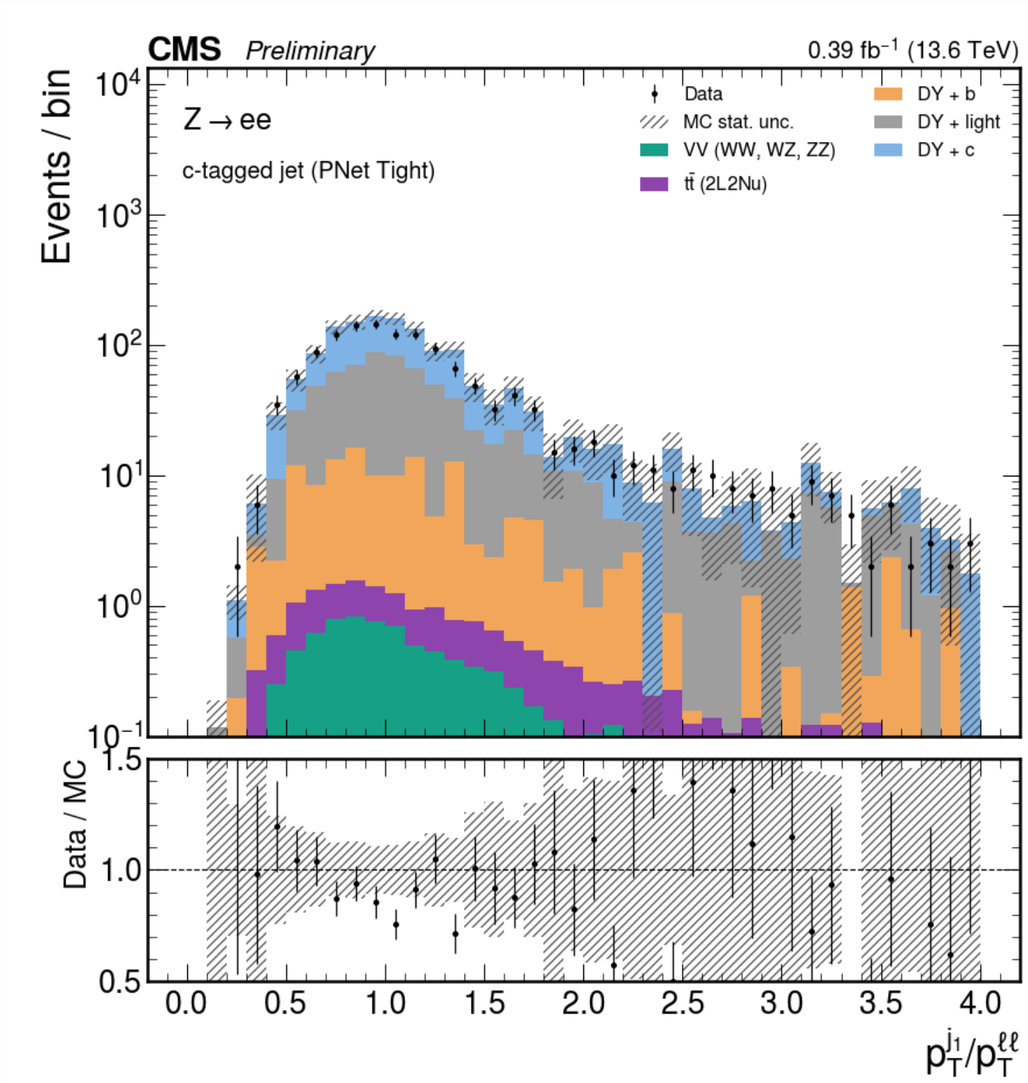
c-tag Tight

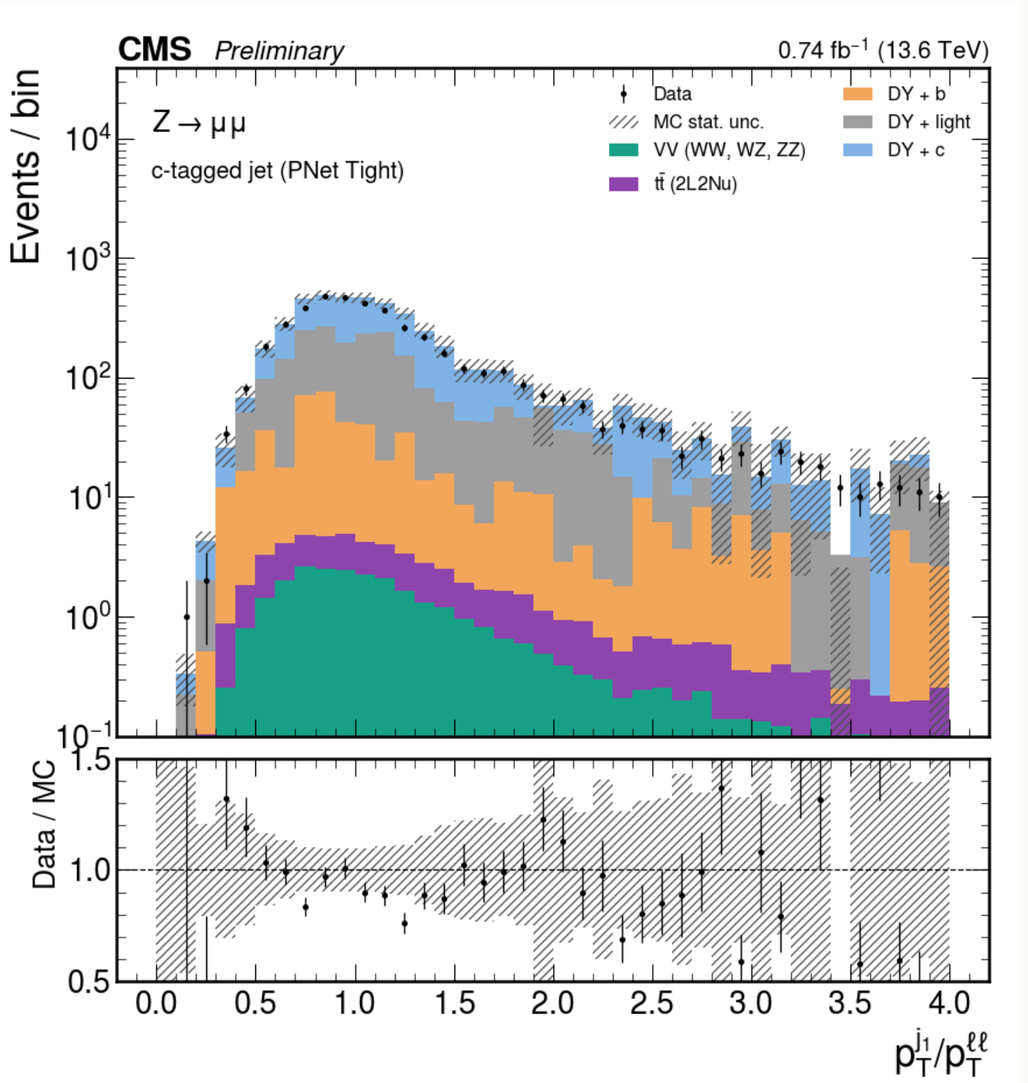
Smallest ΔR(lepton, leading jet)



c-tag Tight

pT balance: leading-jet pT / Z pT

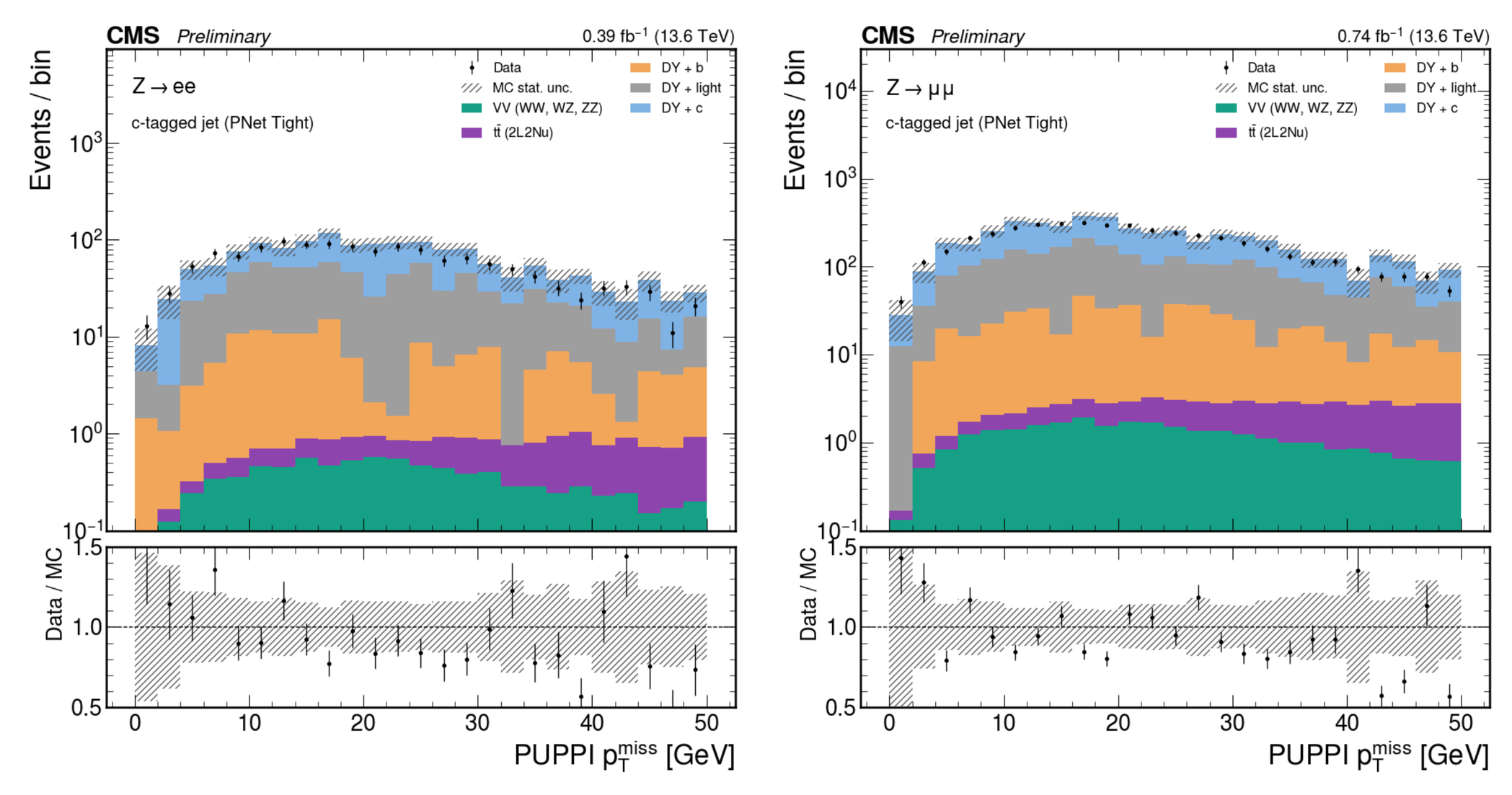




c-tag Tight

PUPPI MET

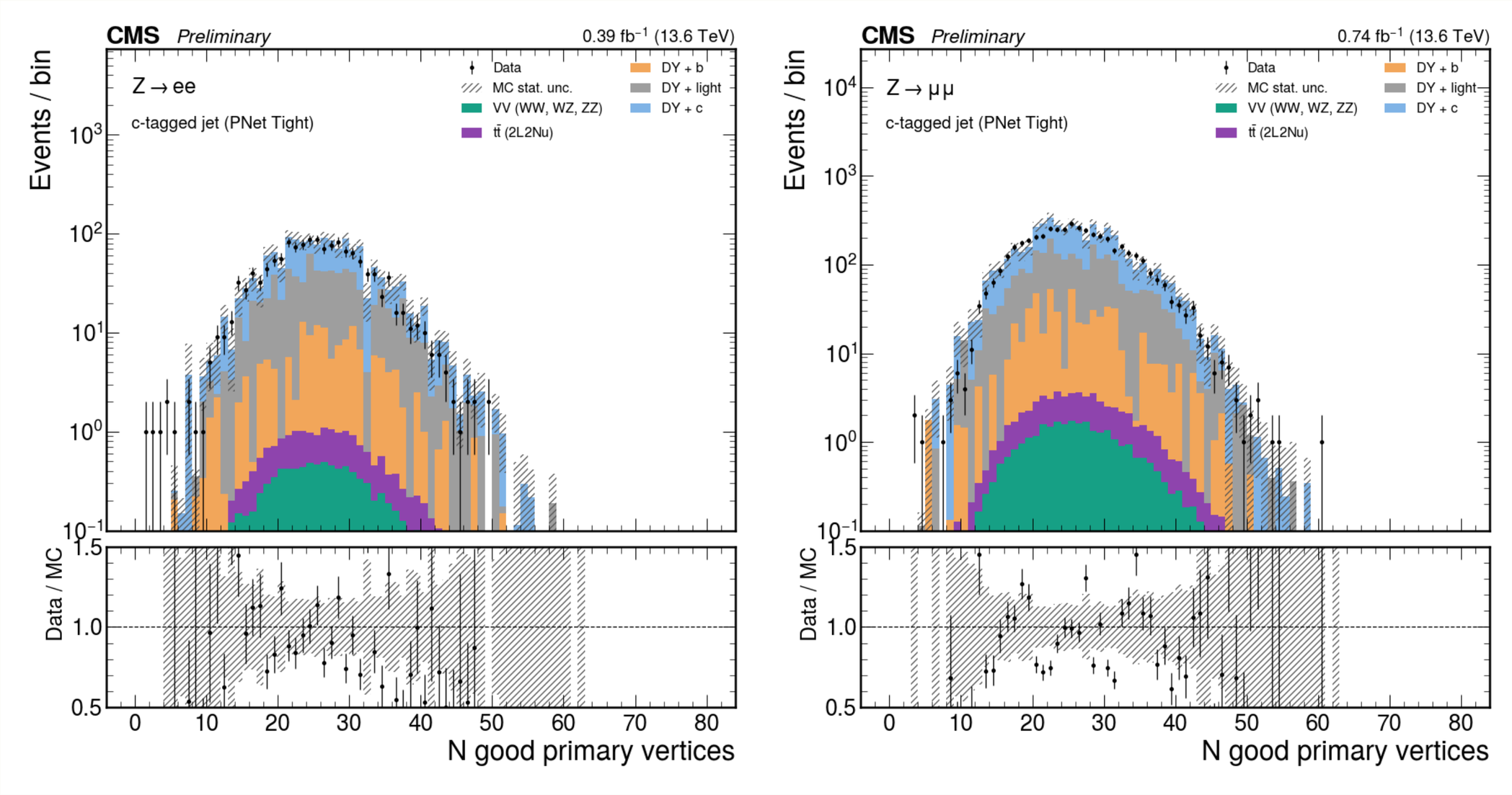
Type-1 corrected; the selection cuts at 50 GeV.



c-tag Tight

Number of good primary vertices

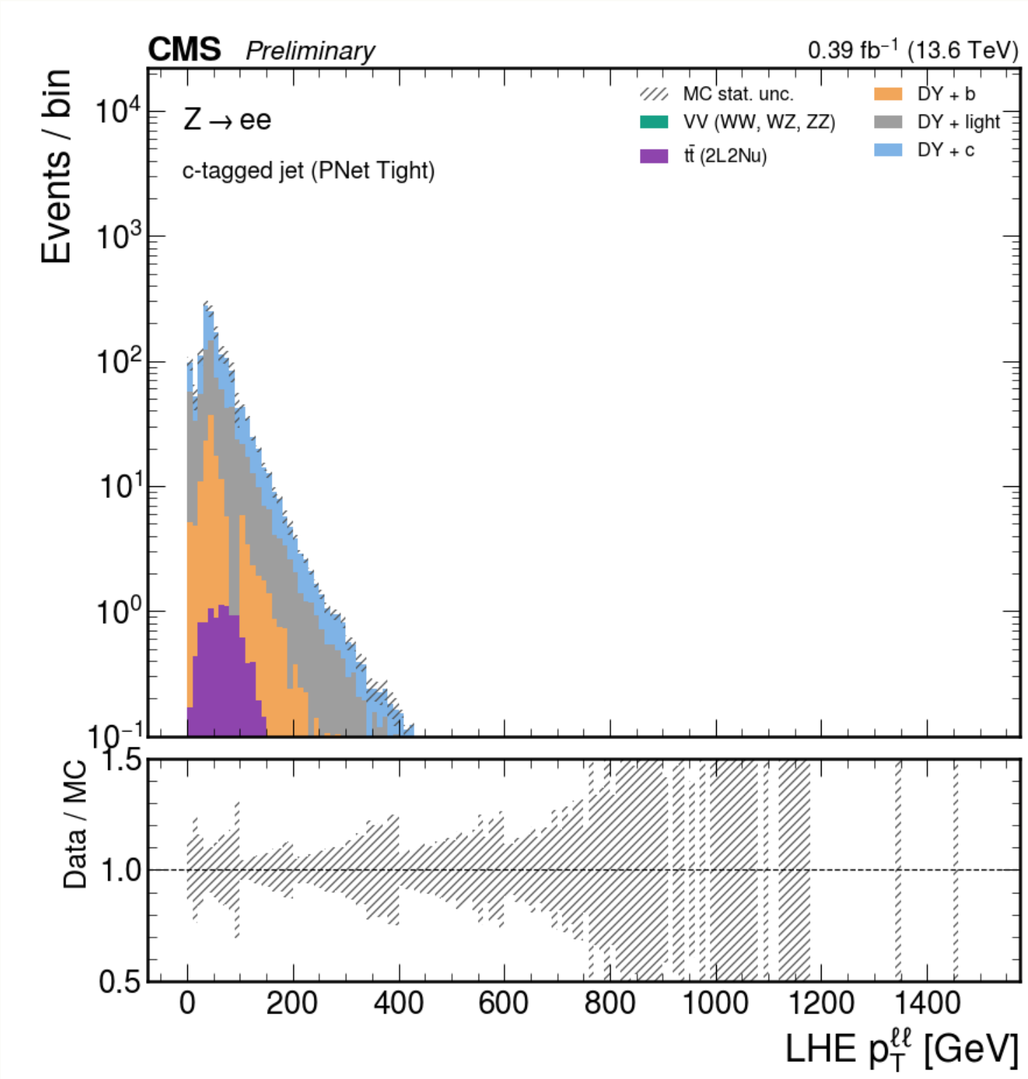
After pileup weights.

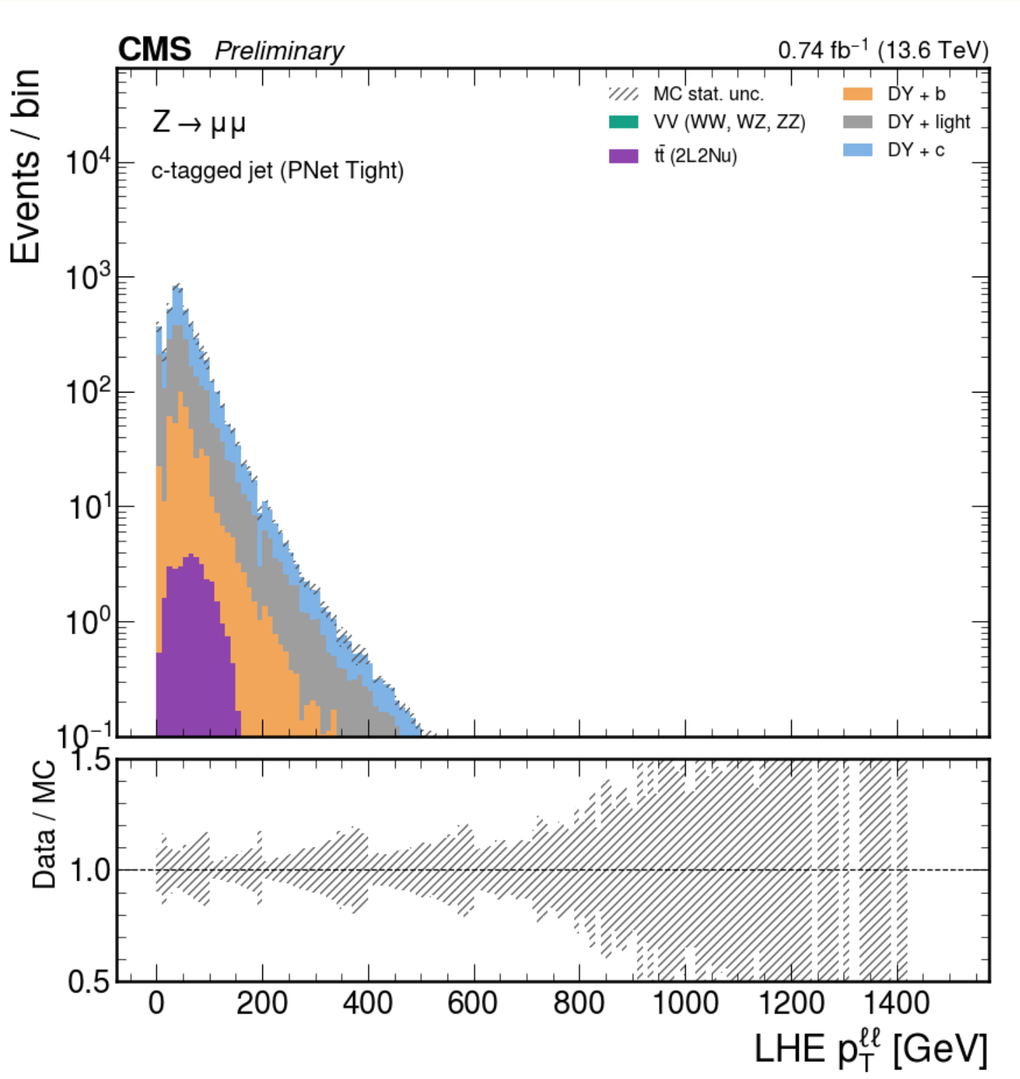


c-tag Tight

Generator Z pT (LHE)

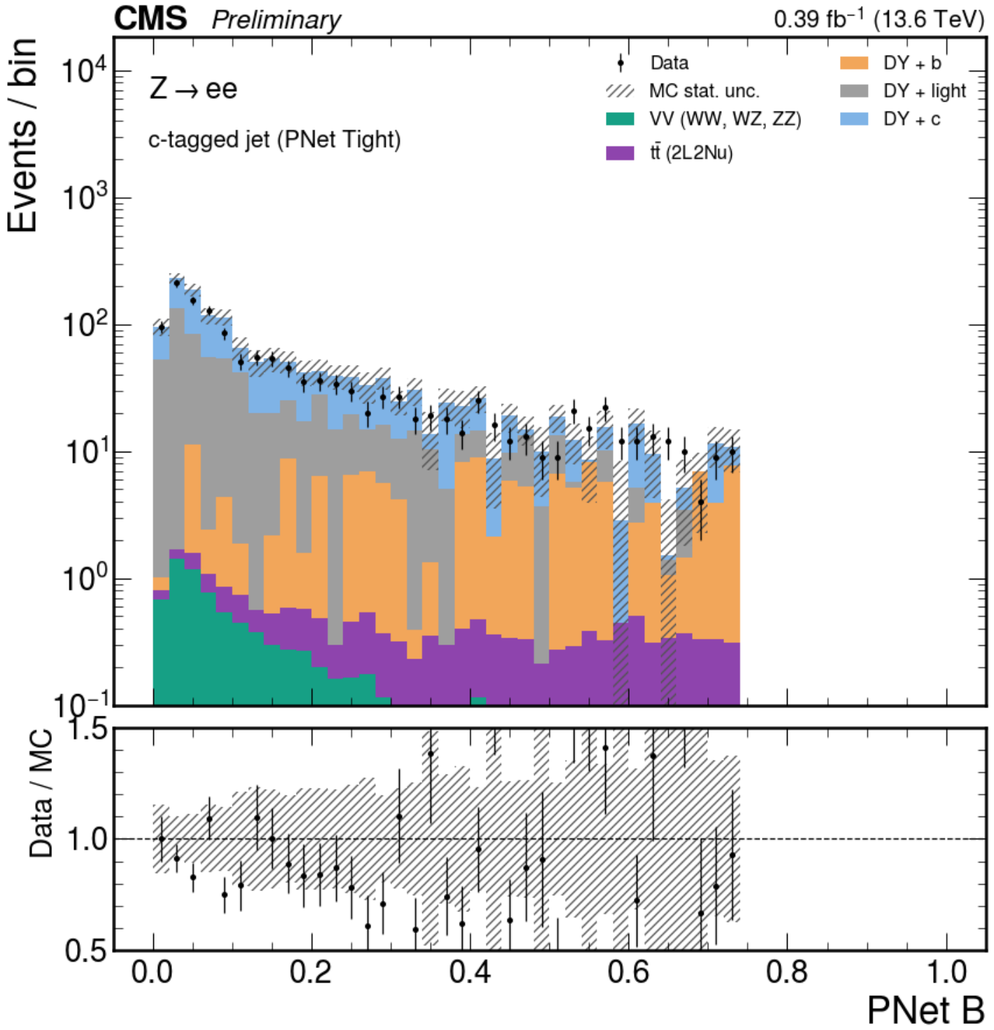
MC only, no data.

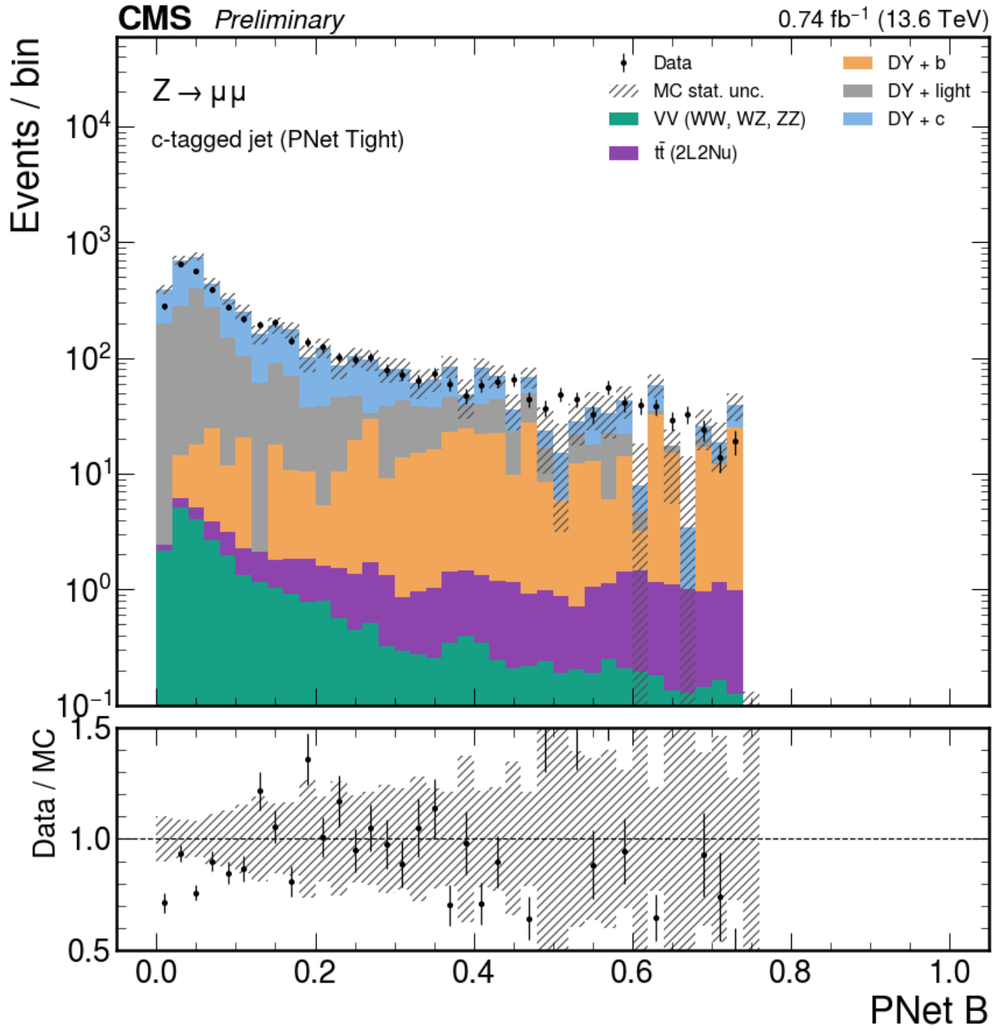




c-tag Tight

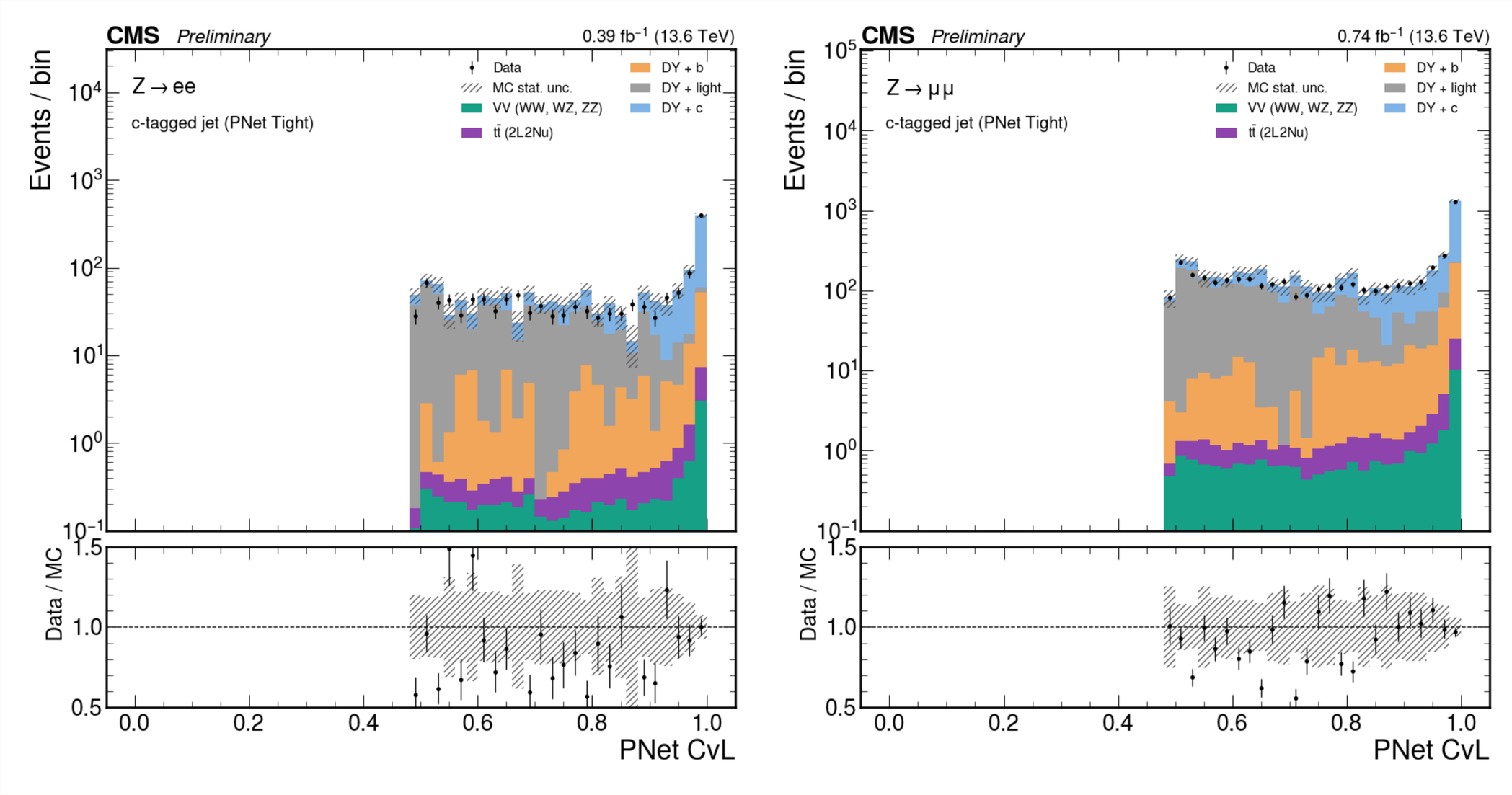
Leading-jet PNet B score





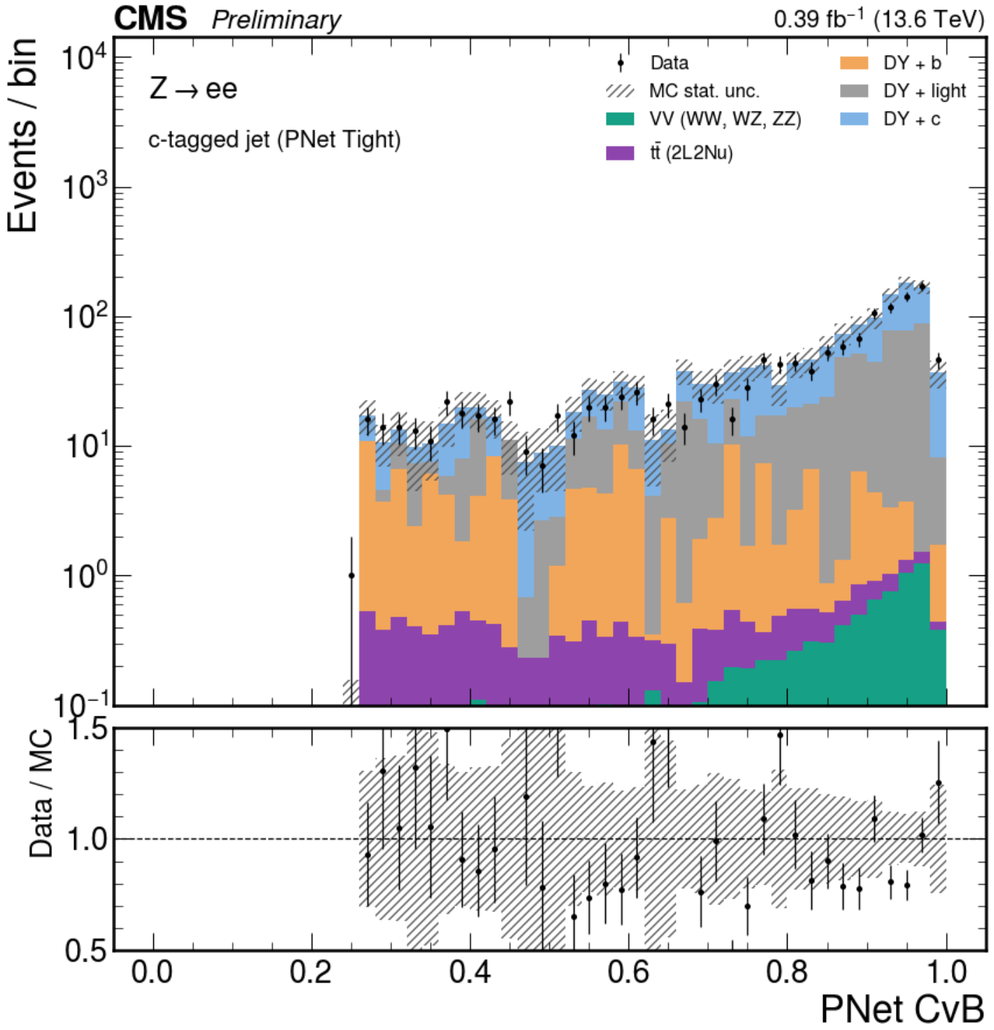
c-tag Tight

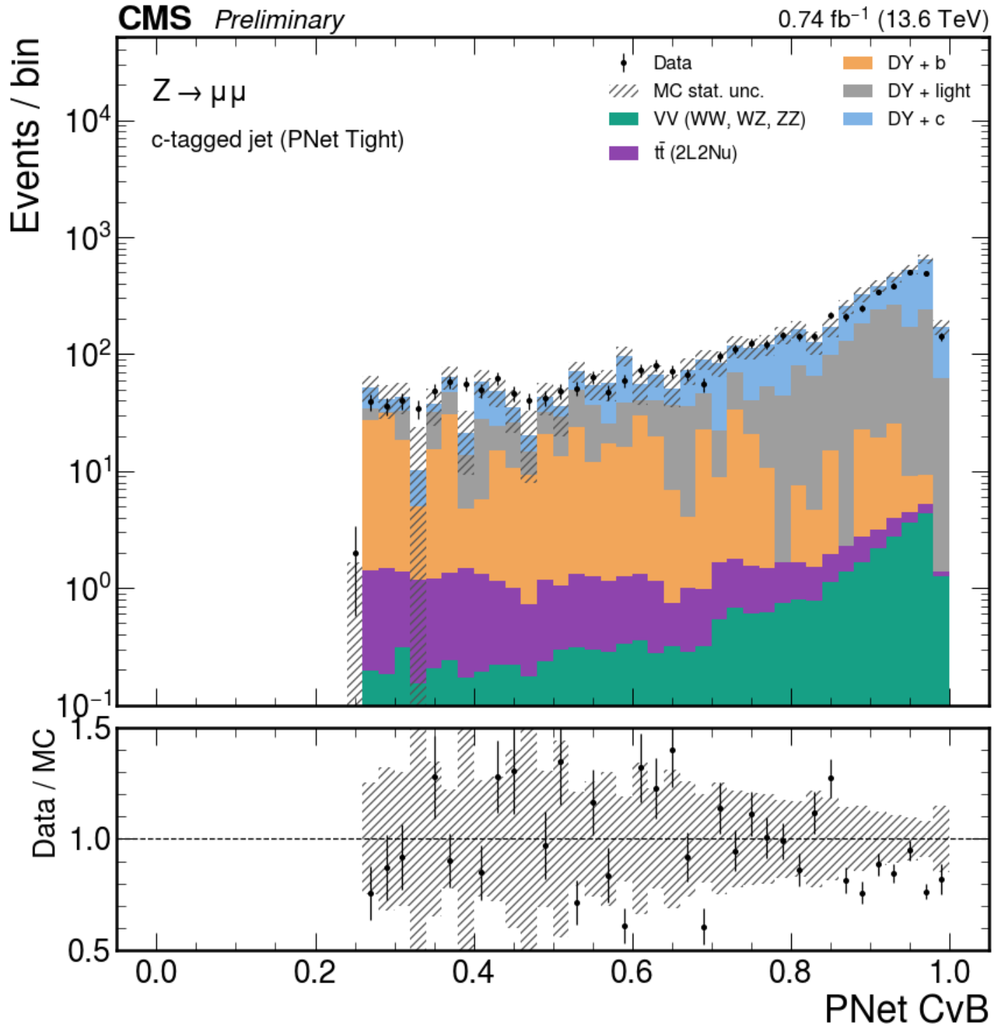
Leading-jet PNet CvL score



c-tag Tight

Leading-jet PNet CvB score

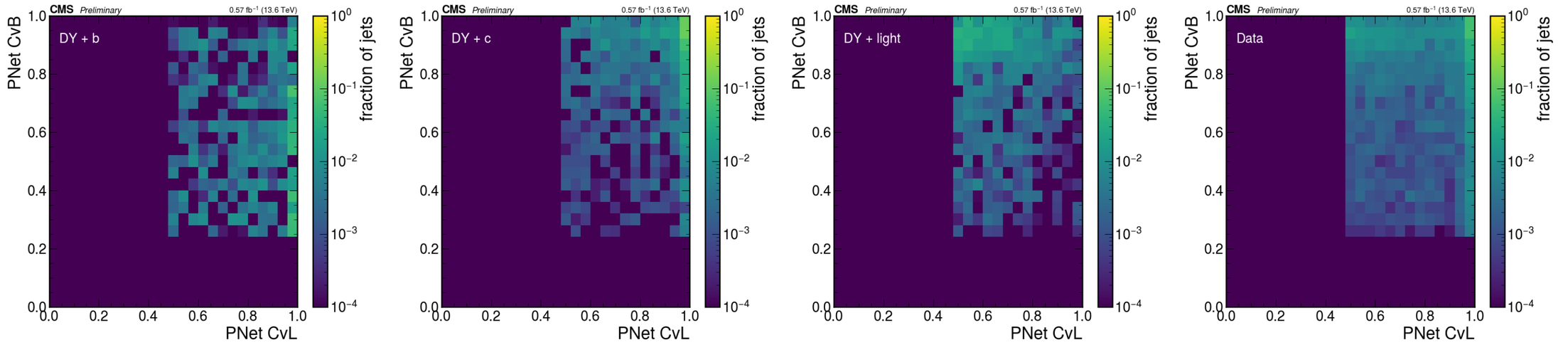




c-tag Tight

PNet CvL vs CvB of the leading jet

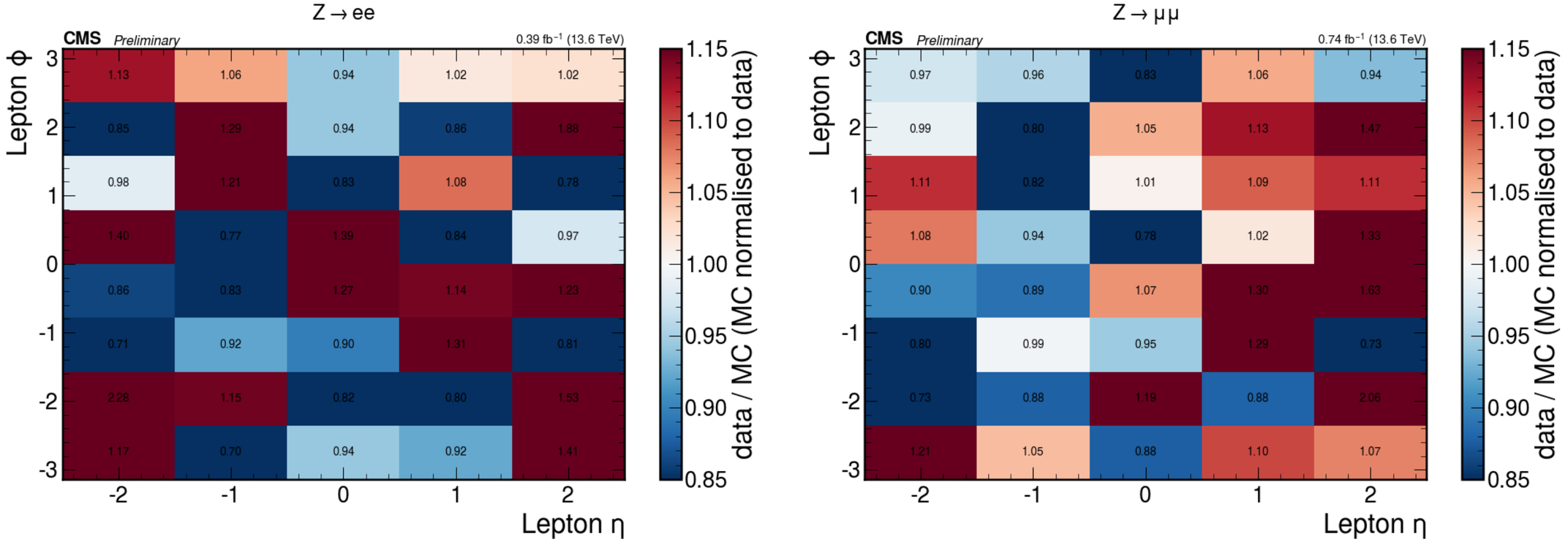
Each panel normalised to 1. Z→ee and Z→μμ added.



c-tag Tight

Lepton η–φ, data / MC

Both Z leptons. MC normalised to data in each map. Colour scale stops at ±15%; in the tagged regions there are few events per cell, so read the numbers.



Summary

All kinematic plots now have every POG correction, plus b- and c-tagging SFs.

32 plots per region in this deck (hadron-flavour stacks); parton-flavour versions in hists_zbzc_run3_tag/Summer22/plots_<region>/. MET filters added; DY Z pT weights off.

Data and MC tag the same fraction of jets, within 1–2%, at every working point.

Data/MC is about 0.95 in all regions, as without tagging. Errors here are statistical only.

Loose c-tag is almost no selection (89% of events, 90% light).

For a charm-enriched sample, Medium (20% c) or Tight (50% c) is better. b-tag Medium gives 55–59% b, Tight 82%.

Next

Add the SF up/down uncertainties as a band. Same plots for 2022 E–G and 2023.